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PERFORMANCE SPECIFICATION
LITTORAL PLANNING CHART (LPC)

This specification is approved for use by all Departments and Agencies of the Department of Defense.



Comments, suggestions or questions on this document should be addressed to the National Geospatial-Intelligence Agency, National Center for Geospatial Intelligence Standards (NCGIS), Mail Stop N32, 7500 GEOINT Dr. Springfield, VA 22150, or emailed to ncgis-mail@nga.mil. Since contact information can change, you may want to verify the currency of this address information using the ASSIST Online database at https://assist.dla.mil.

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1. SCOPE

1.1 Scope. This specification defines hardcopy production requirements for the National Geospatial-Intelligence Agency (NGA) Littoral Planning Chart (LPC), which will replace the existing Combat Chart.

1.2 Purpose. The purpose of this specification is to assure uniformity of treatment among all elements including symbols, accuracy, layout, content, and design needed for the production of the Littoral Planning Chart as related to tactical military operations covering coastal regions. Littoral Planning Charts provide an approximate 50/50 division between land and water, and provide hydrographic and topographic information in those coastal areas.

1.3 Classification. The classification of the Littoral Planning Chart will contain charts that are oriented North-South or East-West. Some Littoral Planning Charts that will have a Skewed orientation.

2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3, 4, or 5 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. Every effort has been made to ensure the completeness of this list. Users are cautioned that they must meet all specified requirements of documents cited in sections 3, 4, or 5 of this specification.

2.2 Government documents.

2.2.1 Specifications, standards, and handbooks. The following specification, standards, and handbooks form part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

INTERNATIONAL STANDARDIZATION AGREEMENTS

STANAG 1022 – Combat Charts, Amphibious Charts and Combat/Landing Charts
S4 – REGULATIONS OF THE IHO FOR INTERNATIONAL (INT) CHARTS AND CHART SPECIFICATIONS OF THE IHO

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DEPARTMENT OF DEFENSE SPECIFICATIONS

MIL-PRF-89201 – Harbor, Approach, Coastal, and General Nautical Charts
MIL-T-89301 – Topographic Line Maps, Scale 1:50,000

DEPARTMENT OF DEFENSE STANDARDS

MIL-STD-2402 – Standard Practice for Mapping, Charting, and Geodesy Symbols for Graphic Products
MIL-STD-2410 - Mapping, Charting, and Geodesy Reproduction and Printing
MIL-STD-2414 - Bar Coding for Geospatial Products
MIL-STD-600001 – Mapping, Charting, and Geodesy Accuracy

(Copies of these documents are available online at https://assist.dla.mil or from the Standardization Document Order Desk, 700 Robbins Avenue, Building 4D, Philadelphia, PA 19111-5094.)

2.2.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

STANDCONTABLE 03 – Standard Conversion Table No. 3
STANDCONTABLE 04 – Standard Conversion Table No. 4

(Copies of the above publications are available from the National Geospatial-Intelligence Agency, National Center for Geospatial Intelligence Standards (NCGIS), Mail Stop N32, 7500 GEOINT Dr. Springfield, VA 22150, or emailed to ncgis-mail@nga.mil)

NGA GEOnet Names Server GNS http://earth-info.nga.mil/gns/html/index.html

NGA Technical Manual, NGA TM 8358.1, National

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Geospatial-Intelligence Agency: Datums, Ellipsoids, Grids, and Grid Reference Systems, Preliminary Edition.
http://earth-info.nga.mil/GandG/publications/index.html

Chart No. 1	-	Nautical Chart Symbols and Abbreviations
Pubs 110-116	-	NGA List of Lights
Pub 117	-	Radio Navigational Aids
NtM	-	US Notice to Mariners
SD Pub	-	Sailing Direction Enroutes and Sailing Direction Planning Guides
NFDD	-	NSG Feature Data Dictionary at https://nsgreg.nga.mil/

(Copies of the above publications are available from National Geospatial-Intelligence Agency, Maritime Safety Information Website. http://msi.nga.mil/NGAPortal/MSI.portal)

National System for Geospatial Intelligence (NSG)
Portrayal Standard for Local Topographic Data Store (LTDS) Data
http://www.gwg.nga.mil/

2.3 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

IHO Special Pub 46	-	Correction of Echo Soundings
IHO S-4	-	Regulations of the IHO for International (INT) Charts and Chart Specifications of the IHO
IH0 Special Pub 44	-	IHO Standards for Hydrographic Surveys

(Copies of the above publication are available on disk or paper format, upon request, from the International Hydrographic Organization- Monaco)

NP139 - Echo Sounding Correction Tables

(Copies of the above publication are available from the British Admiralty, Taunton, U.K.)

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Regulations of the IHO for International (INT) Charts and Chart Specifications of the IHO (Chart #1)

2.4 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First Article. When specified (see 6.2), a sample shall be subjected to first article inspection in accordance with 4.3.

3.2 Accuracy.

3.2.1 Horizontal accuracy. Absolute horizontal accuracy for Littoral Planning Charts is 1.0mm (50 meters) CE at the 90% confidence level. See 6.4 for definition of CE.

3.2.2 Vertical accuracy.

3.2.2.1 Topographic. Absolute vertical accuracy (land) is one contour interval LE at the 90% confidence level.

3.2.2.2 Hydrographic. Absolute vertical accuracy for depths shown on Littoral Planning Charts is 0.3 meters (from 0 to 30 meters), and 1% of depth (greater than 30 meters), linear error (LE), at a 90% confidence level. See 6.4 for definition of LE.

3.2.3 Hydrographic data accuracy. NGA strives to compile Littoral Planning Charts with the most accurate information available. NGA analysts compile charts from a variety of sources, with varying accuracies. Often the metric accuracy (expressed as circular and linear error at a certain level of confidence) of the hydrographic and bathymetric data is unknown, or the information is not available to NGA. In this case, a subjective determination of accuracy is made, based on the survey dates, scale, and originating agency of the source.

3.2.4 Displaced features. Feature symbols, which are displaced for cartographic clarity, are excluded from the accuracy requirement stated above.

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3.3 Datum

3.3.1 Horizontal datum. For new production, and as map/chart sheets are revised or updated for periodic maintenance, the WGS 84 or NAD 83 datum shall be used. Where appropriate, a revised Military Grid system based on WGS 84 or NAD 83 as applicable shall be depicted as the primary grid. The old (local) grid will be depicted as the secondary grid with tick marks along the border of the sheet. A grid conversion note shall also be placed in the marginalia of the chart. Additionally, both the old and new 100,000-meter square two-letter identifiers shall be depicted on the map/chart, if applicable. Appropriate margin notes shall be added to explain the dual lettering.

3.3.2 Vertical datum. In areas where tides are significant (generally greater than 0.3 meter range), Littoral Planning Charts will reference three vertical datums. The contours and heights of topographic features are referenced to Mean Sea Level. Coastal Shoreline is referenced to Mean High Water. Hydrographic features are referenced to a vertical datum based on a low water tide level, called the sounding datum or hydrographic datum. The specific low water datum used depends on the type of tide in the area or the number and magnitude of high and low tides in one tidal cycle. In areas where tidal range is not significant, i.e. less than 0.3 meters, Mean Sea Level (MSL) may be used as the vertical datum for hydrography, shoreline, and elevations.

3.4 Adjoining data set and chart match.

3.4.1 Overlap area. Adjacent Littoral Planning Charts shall have an area of overlap with adjoining charts. This practice makes it easier to transfer plotted positions from one chart to the next.

3.4.2 Chart detail. When adjoining charts are compiled at the same time, chart detail in the overlap area must be identical, except in cases where the newest Littoral Planning Chart is based on later date sources. In these cases, the new Littoral Planning Chart shall show the later information. If the differences in the overlap area are critical for safe

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navigation, the previous published Littoral Planning Chart shall be updated with appropriate Notice to Mariners.

3.4.3 New or new edition. When new or new edition Littoral Planning Charts are produced and the adjacent charts are not updated, chart detail in the overlap area shall be identical to the previously published charts, except as noted in 3.4.2.

3.5 Series. Littoral Planning Charts produced, as part of a series shall be outlined in a location diagram in the marginalia of the chart.

3.6 Scale.

3.6.1 Standard Scale. Littoral Planning Charts are generally constructed at a standard scale of 1:50,000. The linear dimensions (length, width, area, etc.) expressed in Appendix A of this specification are based on a map distance to ground distance ratio of 1:50,000. Linear dimensions for non-standard scale Littoral Planning Charts as described in 3.6.2 shall be modified proportionally.

3.6.2 Non-standard scales. Littoral Planning Charts may be produced at a non-standard scale. The linear dimensions expressed in Appendix A of this specification are based on the standard scale of 1:50,000. All dimensions reference in Appendix A shall be modified proportionally to accommodate non-standard scale charts.

3.6.3 Data density. The standard 1:50,000 scale of the Littoral Planning Chart is adhered to even in areas where the corresponding topographic map coverage is at 1:100,000 scale. The level of topographic detail and density shown on Littoral Planning Charts shall commensurate with 1:50,000 scale.

3.7 Chart design.

a. Littoral Planning Charts at a standard scale of 1:50,000 provide chart coverage for 8 to 15 nautical miles of land distance and 5 to 15 nautical miles of sea distance (total thirty nautical miles); however, the proportion of the land and water may vary for special user requirements. The charts shall be oriented east-west, north-south, or skewed. The long edge of the Littoral Planning Chart runs generally perpendicular to the coastline to provide sufficient coverage inland and out to sea.

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b. Most charts are north oriented but, where the coastline varies from the cardinal direction, charts may be oriented normal to the coast (skewed). When charts are skewed in this manner, more complete coverage of both land and water areas is achieved. Such charts should be clearly marked as "not North oriented".

c. Littoral Planning Charts are rectangular in shape, except where a small island falls just outside the limits of a chart, and is not shown on another chart in the series. An extension is added to the chart to include the island, rather than make a new chart just for the island.

d. An area of overlap is provided for ease of transition from one chart to the next. The width of this overlap is variable but shall not be less than one minute or greater than two minutes.

3.8 Size and neatlines. The standard trim size for charts oriented east-west or north-south is 105.4 cm by 147.3 cm (41.5 by 58 inches). The neatline size for east-west chart is 78.7 cm by 137.2 cm (31 inches by 54 inches). The neatline size for north-south chart is 81.3 cm by 134.6 cm (32 inches by 53 inches). See APPENDICES B and C (Style Sheets).

3.9 Projection. Littoral Planning charts are constructed on the Transverse Mercator projection.

3.9.1 Graticule lines and subdivisions.

a. Parallels and meridians are shown in black (Standard Printing Color (SPC)-58600) at 4 minute intervals, except as noted in 3.9.4.

b. Full minute ticks are shown along the neatline.

c. Minute ticks are shown along a selected central meridian and a selected central parallel. The central meridian and the parallel, which is subdivided, are selected based on clarity of presentation with preference given to open water wherever possible. Another major consideration in the portrayal of subdivisions along projection lines is the location of rows

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of grid labels. See APPENDICES B and C for examples of minute tick subdivisions.

d. Five second ticks subdividing one minute are shown at two selected locations on each neatline, central meridian, and central parallel. Locations are selected based on clarity of presentation with preference given to open water areas.

3.9.2 Graticule labeling.

a. Geographic coordinates showing degrees, minutes, and seconds are shown on all four neatline corners. The appropriate suffix N, S, E, or W is shown with full coordinates values (degrees, minutes, and seconds) at the lower left and upper right corners.

b. Parallels, meridians and minute tick labels are shown along the neatline. The subdivided central parallel and meridian labels show the degree and minute values at the neatline.

c. Every minute tick on the subdivided central parallel and meridian shows a minute label except at the 4 minute graticule, which shows a label with the whole degree and minute value.

d. Five second subdivision tick labels are shown at the 15”, 30”, and 45” ticks.

3.9.3 Additional graticule information. Refer to APPENDICES B and C, Style Sheets for additional information about labeling and subdivision.

3.9.4 Exception to graticule intervals. Although meridians and parallels are usually shown at 4 minute intervals, this interval becomes too dense (approximately 3 inches) at 60 degrees latitude and greater. Whenever the graticule interval measures less than 7.6 cm (3 inches), the interval is increased. For Littoral Planning Charts at 60 degrees latitude and greater, the interval for meridians is 10 minutes and the interval for parallels is 5 minutes.

3.9.5 Graticule and grid accuracy.

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a. The intersections of parallels and meridians shall be within 0.1 mm of computed positions.

b. The overall distance between the first full grid lines adjacent to opposite neatlines shall not vary by more than 0.3 mm from their computed measurements. The distance between adjacent grid lines will not vary by more than 0.1 mm from the computed grid interval.

3.10 Reference systems.

3.10.1 Military grids. The Military Grid System is shown in accordance the National Geospatial-Intelligence Agency Technical Manual 8358.1 "Datums, Ellipsoids, Grids, and Grid Reference Systems". A maximum of three grids are portrayed on Littoral Planning Charts.

3.10.2 Major grid. The major (or primary) grid is normally one or two universal grids, either the Universal Transverse Mercator (UTM), or the Universal Polar Stereographic (UPS). The primary grid is shown in purple (SPC 96532). Normally only one major grid is shown on the chart, unless the area covered by the chart overlaps the junction between the two universal grids as set forth in NGA TM 8358.1.

3.10.3 Additional grids. Up to two additional grids may be portrayed. They are shown in order of precedence first in blue (SPC-48253), and then if needed, in red-brown (SPC-61121). No more than three grids are shown on a Littoral Planning Chart. Additional grids are portrayed in the following order of precedence:

a. Overlapping Grid. If any area covered by the chart is within 40 kilometers of a grid zone junction, or ellipsoid junction, or as set forth in NGA TM 8358.1, the overlapping grid is also shown.

b. Local Grid. In accordance with agreements between NGA and other producers, or other necessity as set forth in NGA TM 8358.1, a local grid (such as a pre-WGS grid) may be shown (see 3.3.1). Examples of these local grids include French Lambert North African Grids, Madagascar Grid, and Ceylon Belt.

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3.10.4 Grid portrayal.

a. The 1000 meter northing grid lines are labeled to the right of each 10,000 easting grid line and the 1000 meter easting grid lines are labeled above each 10,000 meter northing grid line.

b. The dimensions, size and style, and placement of marginal data related to grids and grid formats are contained in the style sheets. For an example of a Littoral Planning Chart with major and overlapping grids, see APPENDIX B. For an example of a Littoral Planning Chart with only a major (primary) grid, see APPENDIX C.

3.11 Margin data.

3.11.1 General. Refer to APPENDIX B – East-West Style Sheet and APPENDIX C – North-South Style Sheet for graphic illustrations of the design, the composition, and location of margin data. Notes and diagrams that will vary from chart to chart are indicated by accompanying notation on the style sheets. Variations of those examples are explained in this section. All Littoral Planning Chart margin notes shown in section 3.11 of this specification are portrayed in a convenient font type and size. The correct fonts (type size and style), color, justification, format, and placement for all margin notes and diagrams are provided in APPENDICES B and C.

3.11.2 Source and Zones of Confidence diagram.

3.11.2.1 Use. The source and zones of confidence diagram is a miniature representation of the chart, which graphically illustrates the location of the various sources shown in the source data list (see 3.11.10) in addition to a zone of confidence (see 3.11.11), which enables mariners to assess the quality of the hydrographic data from which the chart was compiled. Source and zones of confidence diagrams are shown on all charts.

3.11.2.2 Content. The diagram shows the shoreline, the outlines of individual topographic map sheets in the area of the Littoral Planning Chart, corresponding sheet numbers, outlines of hydrographic sources, corresponding letter designators keyed to the source data list, corresponding zone of confidence category keyed to the zone of confidence categories table, and an outline of areas covered by differing contour intervals if more than one interval is used. The source and zones of confidence diagram is

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shown in black (SPC-58600) except for the map sheet outlines and numbers in green (SPC-52813), the topographic contour interval information in red-brown (SPC-61121), the zone of confidence in purple (SPC-96532), and water tint in blue (SPC-48253, 31% screen, 45° angle). The diagram is located in the lower chart margin on the east-west oriented charts and in the right margin on the north-south oriented charts.

3.11.2.3 Topographic map sheet disclaimer. The following note is shown near, or preferably within the land area of the source diagram, if topographic maps do not exist over all or part of the area covered by the Littoral Planning Chart: "For index purposes only – not necessarily an indication of published maps." Type is 6 point mixed case, green (SPC-52813).

3.11.2.4 Contour interval note. If more than one contour interval is shown on the Littoral Planning Chart, due to source materials utilizing different contour intervals, the areas covered by different contour intervals are shown in the Source Diagram. If adjoining topographic sheets utilize different intervals, a dashed red-brown line, lineweight 0.2 mm, dashed length 2.0 mm, dash space 0.5 mm, is shown displaced 0.2 mm away from the green sheet boundary line, and 7 point red-brown type is used to label the contour intervals for the different areas, for example, "Contour interval 10 meters" and "Contour interval 20 meters". If political boundaries, water bodies, or other natural features make it obvious where the boundary between contour intervals is, the dashed line may be omitted, and the labels alone used to indicate the different contour intervals.

3.11.3 Buoyage notes. The buoyage note tells the user that the International Association of Lighthouse Authorities (IALA) Maritime Buoyage System is either in effect or will be in effect in the area covered by the chart. One of four buoyage notes is shown in the margin of the chart, depending on the status of the IALA conversion in the area. The type for the headings – "BUOYAGE" and "CHANGES TO BUOYAGE" is 9 point, upper case black (SPC 58600). The type for the text of the note is 7 point, mixed case black (SPC 58600). The buoyage note is positioned in the chart margin where space is available.

a. When the IALA Buoyage System, Region A, is in effect in the area covered by the chart, the note in FIGURE 1 is shown.

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BUOYAGE

IALA Buoyage System, Region A,
is in effect in the area covered
by this chart. See Chart No. 1.

FIGURE 1. IALA buoyage note- Region A

b. When the IALA Buoyage System, Region B, is in effect in the area covered by the chart, the note in FIGURE 2 is shown.

BUOYAGE

IALA Buoyage System, Region B,
is in effect in the area covered
by this chart. See Chart No. 1.

FIGURE 2. IALA buoyage note- Region B

c. When the aids to navigation in the area covered by the chart are in the process of being converted to the IALA Maritime Buoyage System, Region A, the note in FIGURE 3 is shown.

CHANGES TO BUOYAGE

Certain buoys, lights, and beacons within the area of this chart will be affected by the conversion to IALA Maritime Buoyage System, Region A. For further explanation, see annual Notice to Mariners 1(6) and Chart No. 1.

FIGURE 3. Buoyage note- conversion to Region A

d. When the aids to navigation in the area covered by the chart are in the process of being converted to the IALA Maritime Buoyage System, Region B, the note in FIGURE 4 is shown.

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CHANGES TO BUOYAGE

Certain buoys, lights, and beacons within the area of this chart will be affected by the conversion to IALA Maritime Buoyage System, Region B. for further explanation, see annual Notice to Mariners 1(6) and Chart No. 1.

FIGURE 4. Buoyage note- conversion to Region B

3.11.4 Bar scales. Bar scales are shown on each chart. They are shown in black (SPC-58600) and include statute miles, nautical miles, and meters. Expanded Bar scales indicating 40,000 yards are shown in the chart margins, parallel to the neatlines on opposite sides of the chart and adjacent to the water area.

3.11.5 Catalog number. Catalog numbers are not shown on Littoral Planning Charts. Classified charts may be found in the classified chart catalog, and unclassified charts may be found in the unclassified chart catalog.

3.11.6 Cautions and general notes.

a. Cautions are notes to the mariner warning of specific dangers that exist in specific geographic positions within the chart, or dangerous conditions that apply to the entire area covered by the chart. They are used for situations that are too complex or too unusual to be portrayed by standard symbology.

b. Cautions referring to a specific geographic position require a legend in the general location of the danger to direct the mariner to the text of the caution in the margin. If a single caution appears on the chart, the legend shall read "SEE CAUTION". If multiple cautions are shown on the chart, they are numbered and the legends read "CAUTION NO 1, CAUTION NO 2, CAUTION NO 3," and so on. Note that the "SEE" is not shown for multiple cautions, and that there is no period after the abbreviation "NO".

c. The text of each caution is shown in the margin of the chart. Single cautions are not numbered and the caution title reads "CAUTION". If multiple cautions are shown, each caution is numbered and the caution title reads "CAUTIONS". The caution text in the margin is surrounded by a 0.3 mm lineweight box.

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d. The caution box is positioned in the chart margin where space is available. Type for the caution legends is 12 point upper case. Type for the caution title is 9 point upper case. Type for the text of the caution notes is 7 point mixed case. All caution information is shown in green (SPC-52813).

e. General notes are used to present information specific to the chart, not concerned with dangers to navigation. They are treated the same as cautions, with the exception that they are shown in black instead of green, and the legend "SEE CAUTION, CAUTION NO 1, CAUTIONS, etc." are replaced by the legends "SEE NOTE, NOTE NO 1, NOTES, etc."

3.11.7 Chart numbers. Chart numbers are shown on each chart: outside the upper left, lower left, and lower right neatline in black (SPC-58600). Chart numbers are in 30 point.

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FIGURE 5. Example chart number

3.11.8 Security Classification. The security classification of the products generated by the use of this Specification will be the lowest practicable classification. When it is necessary to assign a higher security classification to the product, it shall be in accordance with established national security procedures.

3.11.9 Classification explanation. Classification explanations are shown on all classified charts in black (SPC-58600) in 8 point upper case. They are located in the lower right chart margin, 3 mm below and centered under the declassification note. For charts whose classification is based exclusively on the location and existence of the chart, the standard note is shown (see style sheets, APPENDICES B and C). If classified information is shown on the chart, the note is tailored to indicate the classified content. See FIGURE 6.

WARNING NOTE - SECURITY CLASSIFICATION IS BASED ON
THE FACT OF EXISTENCE AND AVAILABILITY OF THIS
SHEET. TO ENABLE UNCLASSIFIED REQUISITIONING,
THIS SHEET MUST BE ORDERED BY REFERENCE NUMBER.

FIGURE 6. Example classification explanation note

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3.11.10 Classification notes.

a. Classification notes are shown on each classified Littoral Planning Chart in black (SPC-58600) (see FIGURE 7). Classification will be either SECRET, CONFIDENTIAL, or LIMITED DISTRIBUTION.

b. Classified Littoral Planning Charts carry the classification in the following four places in the chart margin in 30 point upper case.

(1) Between the seal and the chart title block (north-south) or above the chart title block (east-west).

(2) In the upper left margin, next to the chart number.

(3) In the lower right chart margin, next to the subtitle.

(4) In the lower left chart margin, next to the user's note.

CONFIDENTIAL SECRET

FIGURE 7. Classification notes

c. Unclassified Littoral Planning Charts have a limited distribution and shall show the note (see FIGURE 8) in place of classification note, in positions 3.11.9.b.2 and 3, described above. No notes are shown in positions 3.11.9.b.1 and 4.

LIMITED DISTRIBUTION

FIGURE 8. Limited distribution note

d. Type for the limited distribution is 18 point upper case.

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e. The following note (see FIGURE 9) shall be shown under the LIMITED DISTRIBUTION note in the lower right margin (position 3.11.9.b.3 described above). Type for this note is 8 point, mixed case.

> Distribution authorized to DoD, IAW 10 U.S.C. §§ 130 & 455. Release authorized to U.S. DoD contractors, IAW 48 C.F.R. § 252.245-7000. Refer other requests to: Headquarters, NGA, ATTN: Release Officer, Mail Stop S82-OIAD, 7500 GEOINT Dr. Springfield, VA 22150-7500. Destroy IAW DoDI 5030.59. Removal of this caveat is prohibited.

FIGURE 9. Distribution guidance note

3.11.11 Source data list. Source data is shown in black (SPC 58600) on all charts. The source data list is shown to the right of the source diagram on east-west oriented charts and below the source diagram on north-south oriented charts. Source data is shown in 6 point upper and lower case type. Five point type size and/or condensed type may be used for large source data lists, if space is limited.

3.11.11.1 Use. The source data list provides information on the origins, scales, and dates of the hydrographic sources, and the origin and currency of topographic sources, so the user can generally determine their quality. The primary purpose of the hydrographic source data listing, used in conjunction with the source diagram, is to guide navigators and those involved in planning "navigational operations" on the degree of confidence they should have in the adequacy and accuracy of charted depths and positions. As a secondary function, the source listing serves as a readily accessible, but not necessarily comprehensive, record of the sources that were used to compile the chart.

3.11.11.2 Content. The source data list consists of four parts: the topographic source listing, the hydrographic source listing, the note "With additions from other sources", and miscellaneous notes. See FIGURE 10 for an example of a source data list.

3.11.11.2.1 Topographic source listing. The topographic source listing identifies the NGA or foreign topographic map series used as the basis for the Littoral Planning Chart, along with the edition date (printed date) and scale of the maps. Individual sheet numbers are normally shown in the source diagram (see 3.13.2) and in the source listing. The currency

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date of the topographic maps, i.e. the "MAP INFORMATION AS OF (date)", or the currency date of the Littoral Planning Chart, if it has been photo-revised past the currency date of the maps, should be shown. See MIL-T-89301 (latest edition), section 3.11.21.

3.11.11.2.2 Hydrographic source listing. The hydrographic source listing provides data about the hydrographic sources shown in the source and zones of confidence diagram. Each source is identified by a letter, which is keyed to a specific area in the source diagram. If only one source was used, the letter identifier is not shown on either the source data or the source diagram. In all cases, an attempt will be made to cite the original survey data of U.S. and foreign sources being used in each area. Only if the original survey sources are unknown will the charts, which contain information from original surveys, be cited.

a. For hydrographic survey sources (direct or from foreign charts), the country of origin, date, and scale shall be listed. Surveys of similar origin, type, date, and scale may be grouped together to avoid too long a list or too complex a diagram, for example, French Surveys, 1978-1983, 1:20,000-1:30,000. Dates are grouped as follows: prior to 1940 (no sonar), 1940's (sonar but no electronic positioning), and 1950's and later (both sonar and electronic positioning). Lead line and echo-sounder surveys should not be grouped together.

b. For chart sources, the producing country, chart number, edition number (if applicable), edition date, correction date (if different than the edition date), and scale shall be listed. Charts are cited only if no information about the survey(s) used to compile them is known. If the survey(s) used to make a source chart are known, either with geographic limits, or only as general information, the foreign chart is not cited as the source, and the survey information shall be shown (see paragraph 'a.' above).

c. If known, the type of survey, such as "sketch survey" or "reconnaissance survey" shall be shown. These terms imply that there is a significant risk of undetected dangers, even if the survey is of a recent date. "Random track data" (i.e., IHO passage soundings) implies soundings acquired on an uncoordinated basis over a period of years. "Unsurveyed" indicates no data of any kind; it should be written in the

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appropriate area of the diagram, but not shown in the source listing. Qualifying terms such as "leadline" or "no sonar" may be added after the type of survey where the date does not give sufficient indication of the survey methods. Where a charted survey is supplemented by occasional soundings from older or later sources, only the main survey should normally be listed.

d. If unconventional or remotely collected bathymetry was used, it shall be identified in the source listing. An example is shown in FIGURE 10.

e. In listing sources, the only country name that shall be abbreviated is the U.S. All other country names shall be spelled out. "British Admiralty" shall be spelled out when citing chart sources, and "British" shall be used when citing surveys, originating from the Hydrographic Department, Ministry of Defence of the United Kingdom. Surveys made by non-government agencies, such as oil companies, shall be called "Commercial Surveys".

3.11.11.2.3 Other sources note. At the bottom of the source listing is a note "With additions from other sources". This part of the note allows minor sources to be used for enhancement, without listing them in the source data.

3.11.11.2.4 Miscellaneous notes. Notes are sometimes shown below the source list to provide the user with additional information about sources.

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```
Topography
    NGA Maps, Series A1234
     Sheet 5678 I (Ed.1,2010)..................................1:50,000
     Sheet 5678 II (Ed.1,2010).................................1:50,000
    MCE Map, Series A1234.....................................1:50,000
     Sheet 5678 III(Ed.1,2010).................................1:50,000
    Map information as of 2011.

Hydrography
    *A.    U.S. Navy Surveys, 1976-1980..............1:30,000-1:50,000
    *B.    U.S. Navy Surveys, 1946...........................1:25,000
     C.    British Survey, 1980 (Reconnaissance survey).....1:75,000
     D.    Commercial Surveys, 1977-1986.............1:18,000-1:24,000
     E.    Finnish Surveys, 1967-1972................1:20,000-1:40,000
     F.    Finnish Surveys 1929-1950, no other information provided
     G.    British Admiralty Chart 1234.....................1:40,000
           (Ed. 1988, Corr. 1992, sources not identified)
    +H.    Swedish Chart 942................................1:200,000
           (3rd. Ed., 1958, sources not identified)
     I.    Multispectral (LANDSAT), 1988
     J.    Airborne Laser Sounder, 1993
     K.    Random track data
With additions from other sources
* Uncorrected soundings
+ Meters and half meters
```

Source	Example of:
Topo	NGA 1:50,000 scale map sheets, printed in 2010, photo-revised in 2011.
A.	Uncorrected sounding, scales, and dates (all after 1950) grouped together.
B.	Uncorrected soundings, date prior to 1950 shown as separate source.
C.	"British" survey. That is also a special type of survey.
D.	Commercial surveys, grouped into similar scales and ranges.
E.	Finnish surveys identified in the source diagram of a Finnish source chart.
F.	Finnish survey information taken from Finnish chart(s), where only the range of the dates of surveys was provided, i.e., as given in the chart title block.
G.	"British Admiralty" chart, with edition and correction dates, but no edition number, and with no source data provided.
H.	Chart with edition number and date, and no source data provided. Depth units are in meters and half meters.
I.,J.	Examples of unconventional surveys.
K.	Random track data, which is listed after all other hydrographic sources.

FIGURE 10. Example of a source data list

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a. If soundings from a source cannot be corrected for sound velocity the source shall be identified in the source diagram. A note shall be added at the bottom of the list of hydrographic sources. It shall read: "* Uncorrected soundings". An asterisk shall be added to both the source reference line, as shown by sources A and B in FIGURE 10, and to the corresponding source identifier in the source diagram. If all soundings on a chart are either corrected for sound velocity or uncorrected, the "Uncorrected soundings" note is not required in the source diagram and the presence or absence of the corrected sounding note in the vertical datum note (see 3.13.31) shall be used to indicate if the soundings are corrected for sound velocity.

b. Miscellaneous notes are also used to identify sources with different units of soundings than what is shown on the majority of the chart, as shown by source H in FIGURE 10.

3.11.12 Zones of Confidence (ZOC) Category Table. The zones of confidence table provide the data about the zone of confidence shown in the source and zones of confidence diagram. Each source is identified by a zone of confidence letter/number value, which is keyed to a specific area in the source and zones of confidence diagram. The allocation of a ZOC indicates that particular data meets minimum criteria for position and depth accuracy and seafloor coverage defined in this Table. ZOC categories reflect a charting standard and not just a hydrographic survey standard. Depth and position accuracies specified for each ZOC category refer to the errors of the final depicted soundings and include not only survey errors but also other errors introduced in the chart production process.

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ZOC	POSITION ACCURACY	DEPTH ACCURACY	SEAFLOOR COVERAGE
A1	±5m	=0.50m + 1%d	All significant seafloor features detected.
A2	±20m	=1.00m + 2%d	All significant seafloor features detected.
B	±50m	=1.00m + 2%d	Uncharted features hazardous to surface navigation are not expected but may exist.
C	±500m	=2.00m + 5%d	Depth anomalies may be expected.
D	Worse than ZOC C	Worse than ZOC C	Large depth anomalies may be expected.
U	Unassessed – The quality of the bathymetric data has yet to be assessed.		
MDSC	Maintained Depth See Chart		

FIGURE 11. Example Zones of Confidence (ZOC) categories

3.11.12.1 Quality. The quality of the hydrographic source data is assessed according to six categories as in Figure 11 above: five quality categories for assessed data (A1, A2, B, C and D) and a sixth category (U) for data that has not been assessed. If none of the hydrographic sources used on a chart have been assessed, a ZOC diagram indicating only 'U' values should not be added to the chart, as it would not include any information of use to the mariner. The assessment of hydrographic data quality and classification into zones is based on a combination of: a. Position accuracy, b. Depth accuracy, and c. Sea floor coverage (certainty of significant feature detection). Where a charted survey is supplemented by occasional soundings from a less accurate source, only the main survey should normally be categorized. The less accurate depths may be indicated as hairline/upright sounding figures on the chart.

3.11.12.2 Zone of Confidence (ZOC) Categories. The higher ZOC categories, A1 and A2, demand full sea floor ensonification or sweep and require very high accuracy standards, which have only been achievable with the technology available since about 1980. Therefore, many sea lanes, which have been regarded as adequately surveyed, may carry a ZOC B classification. Modern surveys of critical areas can be expected to carry ZOC A2 classification while ZOC A1 will cover only those areas surveyed

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under exceptionally stringent conditions for very special reasons. (See Figure 11.)

3.11.12.2.1 Additional Categories. Additional categories may be added to the Zones of Confidence Diagram. Example: Maintained Depth and Dredged Area. Such areas often do not accurately indicate actual depths, but do indicate minimum depths at the time of dredging. Unsurveyed: this should be evident from the face of the chart, but may also be indicated on the ZOC Diagram.

3.11.12.2.2 Other Considerations. The date of a survey may be important, particularly in areas of mobile or unstable sea floor. The survey date may be inserted in parentheses against the Zone Of Confidence value on the face of the diagram. To avoid too complex a diagram, dates of surveys may be grouped or a suitable note added to the relevant portion of the chart, rather than complicating the diagram.

3.11.12.2.3 Other References. Zones of Confidence are additionally described in the document S4- Regulations for the IHO for International (INT) Charts and Chart Specifications of the IHO. Please refer to this document for other Zones of Confidence Tables and Category condition explanations.

3.11.13 Datum conversion note.

a. Datum conversion notes are shown on Littoral Planning Charts that are not constructed on the World Geodetic System 1984 (WGS-1984) Datum. The datum note indicates the necessary correction required to place the chart on WGS-84. The datum note is based on one of the following conditions.

(1) If a shift can be derived from the chart datum to WGS, the following note shall be shown:

COORDINATE CONVERSION – EUROPEAN DATUM TO WGS

GRID: Subtract 65 m E; Subtract 296 m N
Geographic: Subtract 3.5” Long; Subtract 3.0” Lat.

FIGURE 12. Example of a datum note for shift to WGS

Note that the grid and geographic coordinates and the directions shown above are examples and will vary from chart to chart.

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(2) If the chart cannot be shifted to WGS because of a lack of geodetic data or if the chart is internally inconsistent, the following note shall be shown:

WORLD GEODETIC SYSTEM DATUM ADJUSTMENT

Due to the unavailability of geodetic data, this chart cannot be placed on the World Geodetic System 1984 (WGS-84) Datum.

FIGURE 13. Datum note used when chart cannot be shifted to WGS

(3) When the chart is constructed on the World Geodetic System 1984 (WGS-84), no datum note is required.

b. These notes are shown in black (SPC-58600) in the upper right chart margin on the north-south oriented charts, with the right end aligned with the right chart border, and in the top margin on the east-west oriented charts. Type is 9 point upper case for the title and 7 point mixed case for the text.

3.11.14 Declassification notes. Declassification notes are shown on all classified charts in black (SPC-58600) 8 point upper case type. They provide authority for classification and downgrading instructions for the charts. Declassification notes will be determined by the Regional Command Authority. They are located in the lower margin 3 mm below and centered on the classification note in the lower right margin.

DERIVED FROM:
DECLASSIFIED ON:

FIGURE 14. Declassification note

3.11.15 Depth conversion scale. A standard depth conversion scale showing the relationship between feet, meters and decimeters, and fathoms and feet is shown on each chart in black (SPC 58600). It is located in the lower left chart margin on all charts.

3.11.16 Disclaimer note. A disclaimer note, if applicable, is shown in the chart margin to indicate that boundaries are not necessarily authoritative. This note appears in black (SPC-

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58600). The type is 8 point, mixed case. The disclaimer note is not shown on charts of U.S. domestic areas.

NOTE
Boundary representation is not necessarily authoritative.

FIGURE 15. Boundary disclaimer note

3.11.17 NGA Seal. The NGA seal is shown on each chart. The standard 19mm National Geospatial-Intelligence Agency seal in black (SPC-58600) above the title block on north-south charts and in the top margin on east-west charts.



FIGURE 16. NGA seal

3.11.18 Bar code and stock numbers.

3.11.18.1 National Stock Number. The National Stock Number (NSN), and the edition number, in both bar code and human readable form, is shown on each chart, to uniquely identify the chart in the DoD Logistics Standard System (DLSS). The first four digits of the NSN indicate the Federal Supply Classification (FSC), which is 7642 for hydrographic products. The next two digits indicate the National Codification Bureau that assigned the item identification number to the item of supply. The remaining seven digits are a non-significant serially assigned item identification number. The letters "NSN" are shown in the front of the human readable national stock number to distinguish it from the NGA reference number (see below).

3.11.18.2 NGA Reference number. The NGA reference number, in human readable form only, is shown in addition to the NSN and edition number, for internal NGA use. Reference numbers and bar codes are shown in accordance with MIL-STD-2414. The bar codes and reference numbers are shown in the lower left margin of the chart, 5mm below the chart number. The first five letters of

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the NGA reference number are LPCXX, followed by the six digit chart number.



FIGURE 17. Example reference and bar codes

3.11.19 Edition number and date. The edition number and date are shown in black (SPC-58600). The edition date is the date of the latest unclassified Notice to Mariners checked for changes on source charts. All first and subsequent chart editions will show edition number and date followed by the correction note. This note is placed below the lower left corner on all charts, in 8 point mixed case type. Classified Littoral Planning Charts will be referenced to the Classified Notice to Mariners.

3rd Ed., May 7, 1994, (Correct through NM 19/94)

FIGURE 18. Edition number/date/correction note for unclassified Notices to Mariners

1st Ed., Feb 26, 1994, (Correct through CNM 2/94)

FIGURE 19. Edition number/date/correction note for classified Notices to Mariners

3.11.20 First edition date. The first edition date identifies the earliest date that the chart was published and gives an approximation of the date of the original compilation. The first edition date is shown on each chart in black (SPC-58600) in 7 point type. As only the month and year are shown in the First Edition Date, the months are abbreviated except for May, June and July. The First Edition Date is shown below and centered on the Horizontal Datum Note.

1st Ed., Sept. 1985

FIGURE 20. Example first edition date

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3.11.21 Glossary. The glossary is shown in black (SPC-58600) and located in the right chart margin on the north-south oriented charts, and in the top chart margin on east-west oriented charts. When the native language is other than English, all generic terms appearing on the chart are translated in the glossary. Since one glossary may be utilized for a group or series of charts, all items appearing in the glossary may not apply to any one chart. The heading is shown in 9 point, upper case type and the translations are in 7 point, mixed case type.

GLOSSARY

Ayer.....stream
Batu (Bt).....rock, stone, bank, reef, shoal
Darat.....coast, land
Kepulauan.....islands
Palau (pu).....islet, island
Tanjung (Tg).....cape, point

FIGURE 21. Example glossary

3.11.22 Grid reference box.

a. The grid reference box is shown on each chart. It contains instructions and an example for composing a standard grid reference. The grid reference box also contains diagrams identifying the applicable grid zone designations and grid square identifications. See NGA Technical Manual 8358.1, “Datums, Ellipsoids, Grids, and Grid Reference Systems”.

b. The grid reference box and instructions are shown in purple (SPC-96532). The sample point is shown as it appears inside the neatline. When the primary grid is shown, the grid zone designation and 100,000 meter square identification letters are also shown in purple. When there is a grid zone junction on or near the chart, and the primary and overlapping grids must be shown, the grid zone designation is shown in black (SPC-58600), and the 100,000 meter square identification letters are shown in the grid color. APPENDIX B (East-West Style Sheet) shows an example of a grid reference box when two grids are shown. APPENDIX C (North-South Style Sheet) shows an example of a grid reference box when only the primary grid is shown.

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3.11.23 Heights note.

a. A heights note, indicating the unit of measure for heights, is shown on each chart. The first line of the note "HEIGHTS IN METERS"; is shown in black (SPC-58600) in 14 point, upper case type. Notes indicating the contour interval and reference to supplementary contours, if required, are shown in red-brown (SPC-61121) under the first line. Contour information is in 8 point, upper and lower case and supplementary contour information is in 7 point, mixed case type.

HEIGHTS IN METERS
Contour interval 20 meters
(Supplementary contours in dashed lines at 10 meter interval)

FIGURE 22. Example heights note.

b. The information contained in the heights note is repeated in the vertical datum note and these notes must be in agreement regarding contour interval, and supplementary contour interval if applicable.

c. If more than one contour interval is shown on the chart, the interval covering the greater part of the land area, or the more topographically significant part, is indicated in the margin, and the second contour interval is indicated only in the source diagram (see 3.11.2.4).

3.11.24 Horizontal datum note.

a. Horizontal datum notes are shown on each chart and contain the following information:

(1) Type of projection – Transverse Mercator

(2) Horizontal datum – see below

(3) Grid interval – 1000 meters

(4) Grid zone and ellipsoid – see below

(5) Scale – 1:50,000

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b. The horizontal datums, grid zones, and ellipsoids used on Littoral Planning charts are those specified in the latest edition NGA Technical Manual 8358.1, "Datum, Ellipsoids, Grids and Grid Reference Systems".

c. All type is upper case. The type of projection is in 10 point; the rest of the horizontal datum note is 8 point. The type of projection, datum, and scale are shown in black (SPC-58600). UTM primary grids information is shown in purple (SPC-96532), and any UTM overlapping grid information, if required is shown in blue (SPC-48253). Occasionally a third grid is require; either a local grid or a secondary grid. If the third grid is required, it and all related notes are shown in Red-Brown (SPC-61121).

TRANSVERSE MERCATOR PROJECTION
WORLD GEODETIC SYSTEM-1984 DATUM
PURPLE LINES AND TICKS INDICATE THE 1000 METER
UNIVERSAL TRANSVERSE MERCATOR GRID, ZONE 59N, INTERNATIONAL ELLIPSOID
BLUE LINES AND TICKS INDICATE THE 1000 METER
UNIVERSAL TRANSVERSE MERCATOR GRID, ZONE 60N, INTERNATIONAL ELLIPSOID
SCALE: 1:50,000

FIGURE 23. Example horizontal datum note

3.11.25 Location diagram.

a. A location diagram is shown on each chart. The diagram portrays the location of the chart in relation to other charts in the series by a heavy outline of the chart. Degree graticule lines are shown and labeled. Shoreline and major names (country, city or town and water area identification) are shown. The diagram is black (SPC-58600), except for water area names, which are blue (SPC-48253), and water areas, which are tinted blue (SPC-48253, 31% screen, 45° angle). Location diagrams are not shown for isolated Littoral Planning Charts.

b. The diagram is located below the title block in the right chart margin on the north-south oriented charts and in the top margin on the east-west oriented charts. Country names are in 10 point, upper case, city or town names are 8 point upper and lower case: chart numbers are 8 point type, and water area names are 10 point italic type.

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3.11.26 Main title.

a. The main title is shown on each chart. It includes the following information:

(1) Type of chart – Littoral Planning Chart

(2) Major water name – geographic name of ocean or sea

(3) Country name – name of nation

(4) Chart title – geographic name of specific locality

LITTORAL PLANNING CHART
CARIBBEAN SEA
PUERTO RICO
ISLA DE VIEQUES

FIGURE 24. Example main title

3.11.27 Publishing and copyright note. The NGA publishing note identifies the National Geospatial-Intelligence Agency as the publishing authority, and identifies the product as copyrighted material. The note is shown on each chart in black (SPC-58600) and is centered between the right and left neatlines in the lower chart margin. Type is 8 point, mixed case for the first line, 8 point upper case for the second and mixed case for third lines, and 6 point mixed case for the fourth line.

Prepared and published by the
NATIONAL GEOSPATIAL-INTELLIGENCE AGENCY

©Copyright (year of publication) by the United States Government
No copyright claimed under Title 17 U.S.C.

FIGURE 25. Publishing and copyright note

3.11.28 Sounding note. The unit of measurement used for soundings appears in purple (SPC-96532) and is located in the top and bottom margins on all charts. The type is 30 point upper case.

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SOUNDINGS IN METERS

FIGURE 26. Sounding note

3.11.29 Subtitle.

a. The subtitle is shown on each chart. It identifies the following information:

(1) Type of chart – Littoral Planning Chart

(2) Chart title – geographic name of specific locality. This name must be the same as the one in the main title.

(3) Unit of measure for soundings – Meters

(4) Scale – 1:50,000

b. The subtitle is shown in black (SPC-58600). The type of the chart is in 10 point, upper case type, the chart title is in 14 point, upper and lower case type, and unit of soundings and scale is in 8 point upper case type. The subtitle is located below the lower right chart corner.

LITTORAL PLANNING CHART
Isla de Vieques
SOUNDING IN METERS – SCALE 1:50,000

Figure 27. Example subtitle

3.11.30 Symbol legend.

a. A symbol legend is shown on each chart. The symbol legend defines and illustrates the features most frequently represented in a series of charts. It is the same legend that is used on the equivalent scale topographic maps of the same area (see NSG Portrayal Standard for LTDS Data, latest edition), with additional hydrographic features added. These features are a light, which replaces the topographic symbol for lighthouse, a submerged rock, a rock awash, foreshore, a sunken wreck, an exposed wreck, a hydrographic danger, a reef, a depth curve, and a sounding.

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b. The legend shall be bilingual when the equivalent scale topographic maps are shown with a second language. The English descriptive type shall be vertical and the foreign language type shall be italic.

c. The legend is located below the glossary on north-south oriented charts and in the top margin on the east-west oriented charts. The legend title is shown in 10 point, upper case type. The names of the symbols are in 5 point, upper case type (for major items), or upper and lower case type (for sub-items). Color for this type is black (SPC-58600).

d. The symbols and symbol labels in the legend are shown the same as they appear inside the chart neatlines and are specified in NSG Portrayal Standard for LTDS Data.

e. The red light readability of the chart shall be shown by a note centered under the legend: "THIS CHART IS RED-LIGHT READABLE". Type is 8 point, upper case. The color of this note is Red-Brown (SPC-61121). See APPENDICES B and C.

3.11.31 Tide box.

a. A tide box is shown when sufficient data is available. Tidal information includes the name of the tidal stations located on the chart, or the most representative tidal station within 9 kilometers (5 nautical miles) of the chart limits, and the height of tides in relation to the datum of soundings. The height of tides is indicated by gradients of higher and lower Spring and Neap tides or mean ranges as appropriate. Extreme Low Water may also be shown if sufficient information is available. Place names and tidal heights shown in TABLES 27-29 are examples.

(1) Diurnal tides are tides with a single high and a single low water each day. Tide boxes shall show the tide levels of the mean heights of high and low water, as well as Mean Sea Level. The positions in the tide box for the Mean Lower High Water and the Mean Higher Low Water are left blank. A statement indicating the type of tide shall be included.

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TIDAL INFORMATION

Place	Heights above datum of soundings				
	Mean High Water		Mean Sea Level	Mean Low Water	
	Higher	Lower		Lower	Higher
	meters	meters	meters	meters	meters
Abadan	1.3	-	0.7	0.1	-
Karg Island	1.1	-	0.5	0.0	-

Tide is usually diurnal

FIGURE 28. Example of a tide box for diurnal tides

(2) Semi-Diurnal tides are tides with two high waters and two low waters each day, and the differences between the two high waters and differences between the two low waters are small. The tide box shall show the mean heights of the high and low water at both springs and neaps, and Mean Sea Level.

TIDAL INFORMATION

Place	Lat	Long	Heights above datum of soundings				
			Mean High Water		Mean Sea Level	Mean Low Water	
			Springs	Neaps		Springs	Neaps
			meters	meters	meters	meters	meters
Jutland	55° 42'N	008° 14"E	1.3	1.3	0.8	0.1	0.4
Helgoland	52° 22'N	007° 46'E	1.1	1.1	0.7	0.0	0.2

FIGURE 29. Example of a tide box for semi-diurnal tides

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(3) Mixed tides are tides with two high waters and two low waters each day with a large difference between the heights of the highs and/or a large difference between the lows. Tide boxes shall show the mean heights of the two daily high and low waters, and Mean Sea level.

TIDAL INFORMATION

Place	Heights above datum of soundings				
	Mean High Water		Mean Sea Level	Mean Low Water	
	Higher	Lower		Lower	Higher
	meters	meters	meters	meters	meters
Tokyo Wan	1.3	0.8	0.5	0.1	0.4
Yokosuka	1.1	0.5	0.3	0.0	0.1

FIGURE 30. Example of a tide box for mixed tides

b. If full information is not available, partly complete data, for example, a height for springs only, may be given. Other spaces are left blank.

c. FIGURE 29 shows an example of a tide box with geographic positions. The geographic coordinates of tide stations (to the nearest minute) are given when one or more of the following conditions exist:

(1) The tide station listed in the box falls outside the limits of the chart. Tide stations that are less than nine kilometers (five miles) outside the limits of the chart may be used if no appreciable differences in the tide height will result.

(2) The tide station cannot be readily located on the chart.

d. Tidal information is shown in black (SPC-58600) and located in the lower left chart margin on east-west oriented and in the right chart margin on north-south oriented charts. All

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type is 7 point mixed case, with exception of the title, which is 7 point upper case.

e. When no tidal information is available for a chart, the following note is shown in the space specified above for tidal information:

TIDAL INFORMATION

There is no tidal information available for the area covered by this chart.

FIGURE 31. Note for no tidal information

3.11.32 User's note. The NGA user's note provides instructions for users to provide comments and suggestions for improving the chart. The note appears in black (SPC-58600) and is located in the bottom left margin below the edition number and date on all charts. The type is 6 point mixed case.

USERS SHOULD REFER CORRECTIONS, ADDITIONS, AND COMMENTS TO NGA'S MARITIME OPERATIONS DESK: TOLL FREE, 1-800-362-6289; COMMERCIAL, 571-557-5455; DSN, 574-5455;SEND E-MAIL TO NAVSAFTEY@NGA.MIL; USE CONTACT LINKS FOR DNC AND MARITIME WEBSITES LOCATED AT http://msi.nga.mil/NGAPortal/MSI.portal; OR WRITE TO MARITIME PRODUCTS AND SERVICES DOMAIN, NATIONAL GEOSPATIAL-INTELLIGENCE AGENCY, MAIL STOP N65, 7500 GEOINT DRIVE SPRINGFIELD VA 22150-7500.

FIGURE 32. Users' note

3.11.33 Vertical datum notes.

a. Vertical datum notes are shown on each chart and contain the following information:

(1) Unit of soundings – meters

(2) Unit of soundings to greater precision – identifies the depth to which soundings are shown in meters and decimeters.

(3) Sounding datum – identifies the vertical datum to which depth information is referenced. When the vertical datum for depths is a low water datum, the note is "reduced to the approximate level of (datum name)". When the vertical datum

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for depths is Mean Sea Level, the note is "reduced to approximate Mean Sea Level".

(4) Sounding correction note – indicates that the soundings on the chart have been corrected for sound velocity. The absence of this note indicates that not all or none of the soundings have been corrected for sound velocity.

(5) Units of measure for heights – meters

(6) Contour interval – identifies the contour interval

(7) Supplementary contour note – indicates that the supplementary contours are shown as dashed lines and provides the supplementary contour interval. This note is shown only when supplementary contours are shown on the chart.

b. Unit of soundings is in 10 point, upper case, soundings differing from the main unit soundings are in 8 point, mixed case, sounding datum is in 8 point, mixed case, height units and height datum are in 10 point, upper case, contour intervals are in 8 point, mixed case, and supplementary contours are in 7 point, mixed case. The vertical datum note is shown in black (SPC-58600), except the contour and supplemental contour intervals, which are shown in red-brown (SPC-61121).

SOUNDINGS IN METERS
(Under 21 in meters and decimeters)
reduced to the approximate level of Mean Low Water Springs
Soundings on this chart have been corrected for sound velocity

HEIGHTS IN METERS ABOVE MEAN SEA LEVEL
Contour interval 20 meters
(supplementary contours shown in dashed lines at 10 meter interval)

SOUNDINGS IN METERS
(Under 31 in meters and half meters)
reduced to approximate Mean Sea Level

HEIGHTS IN METERS ABOVE MEAN SEA LEVEL
Contour interval 20 meters

FIGURE 33. Examples of vertical datum notes

3.11.34 Warning note. One of the following warning notes, as appropriate, is shown in the chart margin in purple (SPC-96532).

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The type for the title is 9 point, upper case, and type for the text is 7 point, mixed case:

a. For charts of foreign waters, see FIGURE 34.

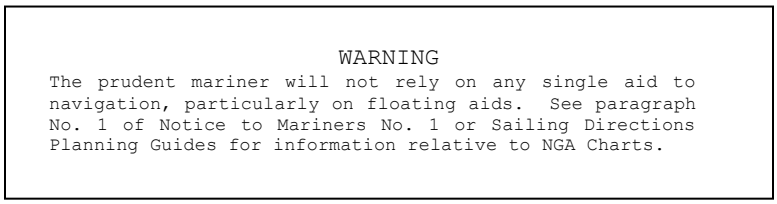


FIGURE 34. Warning Note for charts of foreign waters

b. For charts in U.S. waters see FIGURE 35.

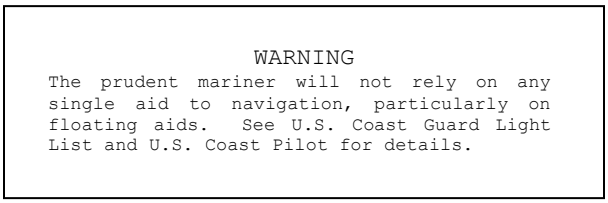


FIGURE 35. Warning Note for charts of U.S. waters

3.12 Culture.

a. Littoral Planning Charts provide a combination of hydrographic chart and topographic map for use in amphibious operations. The cultural information shown on the Littoral Planning Charts is shown as the same on equivalent scale topographic maps, NSG Portrayal Standard for LTDS Data (latest edition), with exception of certain marine aids to navigation.

b. Certain marine aids to navigation are shown on Littoral Planning Charts differently than on topographic line maps. Refer to features in Appendix A of this specification for specific feature requirements.

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(1) Other marine lights and beacons are shown on Littoral Planning Charts even though they are not shown on topographic maps.

3.13 Hydrography. Hydrography is a branch of applied sciences which deals with the measurement and description of the features of the sea and coastal areas for the primary purpose of navigation and all other marine purposes and activities including offshore activities, research, protection of marine environment and prediction services.

a. Inland hydrographic features, which include many rivers, canals, reservoirs, lakes, etc…, are generally derived from topographic sources found within the Local Topographic Data Store LTDS and referenced in Appendix A.

b. The hydrography is compiled from the latest and most reliable hydrographic survey data available. Due to the scarcity of hydrographic surveys and the normal inaccessibility of foreign survey smooth sheets, the most frequently utilized source materials are published nautical charts.

c. The unit of measure for all hydrographic data is meters. Depths under 21 meters are shown in meters and decimeters if source data is available.

d. Soundings shown on Littoral Planning Charts shall be corrected for sound velocity where possible. Soundings shallower than 200 meters shall be corrected whenever bar checks, sound velocity, or temperature/salinity data are available. Soundings 200 meters or deeper shall be corrected using Echo-Sounding Correction Tables, NP139 (latest edition) issued by the British Admiralty Hydrographic Department. If directly observed sound velocity data are available, they should be used if considered to be more reliable than the average correction values for the zones in NP 139.

e. Charts showing only corrected soundings shall show a note with the other vertical datum notes as specified in 3.11.33.a.(4). This note shall be omitted from charts that show soundings that have not been corrected for sound velocity, charts that show a mixture of corrected and uncorrected soundings, or charts that show soundings that could not be determined corrected/uncorrected.

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f. If it is not possible to correct the soundings from sources used on a Littoral Planning Chart, and therefore a mixture of corrected and uncorrected soundings are shown, those sources with uncorrected soundings shall be identified in the compilation data list as described in 3.11.11.

g. Consult IHO Special Publication 46, Correction of Echo Soundings (latest edition) or the producing agency to determine if soundings on foreign charts require sound velocity corrections. Soundings on most foreign charts have already been corrected for sound velocity. If it cannot be determined whether or not a source is corrected, soundings shall be portrayed as shown on the source.

h. Sources in fathoms and feet that have been corrected for sound velocity shall be converted to meters using Standard Conversion Table No. 4. Sources in fathoms and feet that have not been corrected for sound velocity and have an assumed speed of sound in salt water of 1500 meters per second (820 fathoms per second) shall be converted to meters using Conversion Table No. 4 before sound velocity corrections are applied. Sources in fathoms and feet that have not been corrected for sound velocity and have an assumed speed of sound in salt water of 1463 meters per second (800 fathoms per second) shall be converted to meters using Standard Conversion Table No. 3 before sound velocity corrections are applied. See Publication NP 139.

i. Hydrography on Littoral Planning Charts is portrayed primarily by soundings, and supplemented by depth curves. The soundings and depth curves must present a representation of the bottom that will allow a mariner to interpolate the depth of water at places where soundings and depth curves are not shown, with some degree of accuracy. Therefore, the selection of soundings to be shown on the chart is critical to the usability of the chart. Rules for sounding selection cannot cover every conceivable situation. The judgment of an experienced nautical cartographer must be the final authority on the correct selection of soundings.

j. Soundings can be classified as either critical soundings, support soundings, or fill soundings. Critical soundings are those soundings, which identify the shoals, and soundings, which provide essential information required for navigation in non-dangerous areas. Support soundings provide additional information about the shape of the bottom around

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critical soundings and shoals, or are used to provide identifiers to depth curves and to show changes in the bottom slope away from shoals or deeps. Fill soundings are used in flat areas and to fill in deep areas that are not adequately defined by support soundings.

k. Emphasis must be placed on selecting a density of soundings for natural channels, shoals or other hazardous areas that are sufficiently close together so that these areas are properly highlighted (by dense sounding pattern) for quick recognition by the mariner. Additional supportive soundings and fill soundings are selected at a lesser density to complete the bottom description.

l. To support ship to shore movement, the hydrographic detail inside the 10 meter depth curve must be as detailed as possible. Sounding density in this area is greatest, and all dangers and bottom characteristics must be shown. Depth curves at 1, 2, 3, 5, and 7 meters should be shown if sounding data is sufficient to support them.

m. The depth curve interval may be modified to better fit the configuration of the bottom. Depth curve interval may be increased in areas with steeply sloping bottom, and decreased in areas of gently sloping bottom. Depth curve interval is also dependent on the source material. Depth curves can be interpolated at the desired interval from hydrographic surveys but when published nautical charts are used as the primary source, depth curve interval will be dependent on what is shown on the source chart.

n. In certain areas, difficulties for navigators may arise in determining the direction of the lateral buoyage system, i.e., IALA. Examples of this situation might be in a one-way traffic lane where the direction of buoyage is opposed to the traffic direction, the “straight through” buoyage of a strait overrides the “approach from seaward” convention, or where two opposing directions meet, or where the lateral system extends a long way offshore and, at its outer end, has a local direction opposed to the general direction (as occurs in the northern part of the outer Thames Estuary). The mariner’s problem is not that of interpreting charted buoyage, but of knowing which side to pass when confronted with a “new danger” (described by IALA as one that has been marked by buoys, but not yet charted).

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For potentially confusing situations, the following symbol shall be included on the chart to indicate the direction of lateral buoyage. It shall be placed in the water, in the general area of the confusing situation, and point in the direction of buoyage for the area. The note is generally shown in conjunction with the symbol, but the symbol alone may be shown in congested areas.

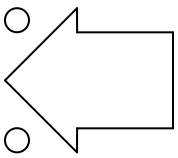


FIGURE 36. Direction of buoyage symbol

3.14 Hypsography/Physiography.

a. The hypsography and physiography are generally portrayed on Littoral Planning Charts the same as on equivalent scale topographic maps, NSG Portrayal Standard for LTDS Data (latest edition).

b. Tints are used to emphasize land and water areas of the chart. A yellow tint (SPC-57377, 21%, 120D, 75° angle screen) is printed over the land areas of the chart, and land areas of all applicable diagrams in the chart margin. Land tint is also printed in offshore features (outside the shoreline) that are above the high water plane of reference, such as breakwaters, piers, cribs, or offshore loading facilities. Land tint is not printed over glaciers or snowfields, which have a white background.

c. Water tint (SPC-48253, 31%, 45° angle screen) is used to emphasize water areas that are less than 10 meters deep. Water tint is shown from the shoreline to the 10 meter depth curve, and overprints the foreshore, features that cover and uncover with rise and fall of the tide, and features that are entirely under water.

3.15 Vegetation. Vegetation is portrayed on Littoral Planning Charts the same as on equivalent scale topographic maps, NSG Portrayal Standard for LTDS Data (latest edition).

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3.16 Demarcation.

a. Boundaries and identification of administrative divisions shall be in accordance with current U.S. Department of State policy.

b. Portrayal of international and first order administrative land boundaries on Littoral Planning Charts are portrayed the same as on equivalent scale topographic maps, NSG Portrayal Standard for LTDS Data (latest edition). The Boundaries Graphic in the margin of topographic maps is not shown on Littoral Planning Charts. Political boundaries shall not be shown in the water.

c. Hydrographic demarcations are generally limited to those features that affect safety to surface navigation. However, boundaries for other regulated areas, such as fishing prohibited areas or traffic separation schemes will also be shown on the Littoral Planning Chart.

d. Hydrographic demarcations on Littoral Planning Charts are symbolized in black or green. Purple symbols are not used because they would visually conflict with the purple UTM grid lines.

3.17 Aeronautical. Aeronautical information is portrayed on Littoral Planning Charts the same as on equivalent scale topographic maps, NSG Portrayal Standard for LTDS Data (latest edition).

3.18 Names and labeling.

a. Geographic names shown on Littoral Planning Charts shall be shown in accordance with the US Board of Geographic Names (BGN). Information on BGN names is available from the NGA GEOnet Names Server at http://earth-info.mil/gns.html.index.html.

b. The following is a list of features, which may not appear in Appendix A, but may be named on the final product. Definitions for the following features may be found in NGA GEOnet Names Server.

Name	Example
Banks	Outer Banks

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Name	Example
Basin	Great Basin
Bay	Chesapeake Bay
Beach	Virginia Beach
Bend	
Bluff	
Bottom	
Break	
Butte	
Canyon	Grand Canyon
Cape	Cape of Good Hope
Channel	English Channel
City	New York City
Cliff	
Corner	Tyson’s Corner
Crossing	
Desert	Sahara Desert
Dispersed Village	
Dome	
Everglade	Florida Everglades
Falls	
Flat	
Flats	
Forest	
Gap	
Gorge	
Gulch	
Gulf	Gulf of Mexico
Gut	
Hamlet	
Harbor	Boston Harbor
Head	
Highland	
Hill	
Hole	
Hollow	
Inlet	Hamilton Inlet
Island Chain	Hawaiian Islands
Junction	
Knob	
Knoll	
Lagoon	
Lake	
Lands	
Lookout	

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Name	Example
Mesa	
Mountain	
Marina	
Mountain Range	Rocky Mountains
Narrows	
Neck	
Ocean	Atlantic Ocean
Park	Yellowstone National Park
Pass	
Passage	
Patch	
Peak	Pikes Peak
Plain	Great Plains
Plateau	Colorado Plateau
Point	
Pool	
Port	
Range	Coastal Range
Ravine	
Region	
Ridge	
Roadstead	
Rock	
Sands	
Scattered Village	Communidads of South America, Streusudlung of Europe
Sea	Caribbean Sea
Sea Mount	
Shelf	
Shoals	
Sink	
Sound	Puget Sound
Spit	
Spring	
Spur	
Strait	Bering Strait
Summit	
Town	
Valley	Death Valley
Village	Greenwich Village
Wood	

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c. The names on Littoral Planning Charts shall be the same names shown on the equivalent scale topographic maps, NSG Portrayal Standard for LTDS Data (latest edition). If possible, names and labels for land features shall not be placed in the water where they might obscure hydrographic detail. Geographic names on offshore features should be the same on hydrographic charts covering the same area. Named features that are described in the Sailing Directions publications should also be named.

3.19 Color separation. The standard for printed colors is the DoD Standard Printing Color (SPC) Catalog for Mapping Charting, and Related Products, March 1972. The standard for printing screens is the DoD Standard Printing Screen (SPS) Catalog for Mapping Charting, and Related Products, April 1982.

3.19.1 LPC Standard Printing Colors (SPC). The standard printed colors for Littoral Planning Charts LPC are black, purple (hydro purple), red-brown, blue (cyan), green, and yellow.

a. Black (SPC 58600)

(1) Projection graticle and values

(2) Cultural planimetry

(3) Geodetic control

(4) Checked elevations and bench marks

(5) Land names, shoal names, and political boundaries

(6) Soundings

(7) Hydrographic symbols

(8) Cables and cable areas

(9) Bottom characteristics

(10) Compass roses

(11) Light legends

(12) Pipelines

(13) Anchorage areas (non-dangerous)

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(14) Shoreline

(15) Foreshore (12% screen)

(16) Depth curves and designators

(17) Marginal information

a) Title block

b) Subtitle

c) NGA Seal

d) Chart numbers

e) Series index

f) Catalog reference

g) Glossary

h) Symbol legend (text and symbols appearing in black on the chart)

i) Depth conversion scale

j) Tidal information

k) Compilation sources text & linework

l) Zone of Confidence text

m) Bar scales

n) Publishing and copyright note

o) Classification notes

p) Heights note

q) Edition date

r) User's note

s) World Geodetic System Datum note

t) Disclaimer note

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u) Skewed chart note

b. Cyan-Blue (SPC 48253)

(1) Drainage

(2) Non-navigable water shoreline

(3) Water names

(4) Miscellaneous water features

(5) Second major or overlapping military grid system and reference notes

(6) Appropriate legend items

(7) Blue (21% screen features)

a) All open water tint to the 10-meter curve

b) Tint over underwater navigational hazards

c) Other features delineated both as drainage and vegetation (nipa or mangrove)

d) Water tint for diagrams

c. Purple (SPC 96532)

(1) First major grid and reference notes

(2) Grid reference box

(3) Light flares

(4) Radio aids

(5) Sounding unit note

(6) Prudent Mariner note

d. Red-Brown (SPC 61121)

(1) Road classification overprint

(2) International and military boundary overprints

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(3) Topographic contours and designators

(4) Built-up areas

(5) Miscellaneous topographic symbols

(6) Contour interval notes

(7) Secondary military grid system

(8) Appropriate legend items

e. Green (SPC 52813)

(1) Vegetation

(2) Prohibited and restricted areas

(3) Warning, firing, operating, anchorage, wire drag areas (swept areas), routes

(4) Maritime limits

(5) Limits and numbers of NGA topographic sources indicating source material in the compilation sources diagram

(6) Appropriate legend items

(7) Anchorage areas (dangerous or restricted)

(8) Traffic Separation Schemes

f. Yellow (31% SPC 57377)

(1) Land tint

(2) Land tint for diagrams

3.20 Symbology. Symbology for Littoral Planning Charts shall be in accordance with National System for Geospatial Intelligence (NSG) Portrayal Standard for Local Topographic Data Store (LTDS), Data and Regulations of the IHO for International (INT) Charts, and Chart Specifications of the IHO (Chart #1).

3.20.1 Symbol color exceptions. The following features in the table below are exceptions to the symbology color as defined by

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the Regulations of the IHO for International (INT) Charts and Chart Specifications of the IHO (Chart #1). For these features, the specification color would be changed to avoid conflict with chart projection and UTM Grid colors.

TABLE I. LPC color exceptions

LPC Color Exceptions		
LPC Feature/ Description	**Chart #1 Symbols**	**LPC Symbol Color**
BB010 Anchorage / Non-Dangerous (Surface, Point)	N11.1, N11.2, N12.1, N12.2, N12.4, N12.5, N13, N14	Black (58600)
AQ070 Ferry Crossing / [Free-moving] or [Ice] (Curve)	M50	Black (58600)
FC021 Maritime Limit and FC037 Maritime Caution Area/ Degaussing Range	N25	Black (58600)
AQ113 Pipeline / All Hydrographic Pipelines (Curve)	L40.1, L41.1, L42, L44	Black (58600)
AT005 Cable/ Hydrographic Cables	L30.1, L31.1, L32	Black (58600)
FC037 Maritime Caution Area/ Cable areas	L30.2, L31.2	Black (58600)
BB010 Anchorage / Dangerous or Restricted (Surface, Point)	N12.6, N12.7, N12.8, N12.9	Green (52813)
FC033 Dumping Ground / Hazardous	N2.1, N23.1, N23.2, N24,	Green (52813)
BD001 Explosive Naval Mine	N23.1	Green (52813)
BB270 Pilot Boarding Place (Point)	T1.1 , T1.4	Green (52813)
FC037 Maritime Caution Area (Surface, Point)	F31, L3, L40.2, L41.2, M14, M18, N1.1, N1.2, N2.1, N20, N21, N22, N23.1, N26, N31, N32, N34, N40.2, N41, N43, N44, N45, N46, N47, N49, N63, N64	Green (52813)
FC021 Maritime Limit	F31, M14. M18, N1.2, N2.1, N22, N26, N30, N31, N32, N33, N34, N45, N49, N63, N64	Green (52813)

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TABLE I. LPC color exceptions - Continued

LPC Color Exceptions		
LPC Feature/ Description	**Chart #1 Symbols**	**LPC Symbol Color**
FC050 Naval Firing and/or Practice Area	N30, N31, N32, N34	Green (52813)
FC055 Naval Operations Area	N33, N34	Green (52813)
FC165 Maritime Route	M3, M4, M5.1, M27.2, M27.3, M28.1, M28.2	Green (52813)
FC177 Swept Area / All Swept (Surface)	I24	Green (52813)
FA000 Administrative Boundary / various hydrographic boundaries	N41, N43, N44, N46, N47, N48	Green (52813)
FC130 Maritime Radar Reference Curve	M32.1	Green (52813)
FC041 Traffic Separation Scheme	M12, M13, M14, M15, M16, M21, M23	Green (52813)

3.21 Reproduction. Reproduction of Littoral Planning Charts can be either by lithography or by utilizing current digital industry standard methods. The final product conforms to the best lithographic and digital standards with respect to clarity, conformance to specified colors and screens, and accuracy of registration.

3.21.1 Paper. Littoral Planning Charts are printed on high wet strength E50 chart paper. The trim size is 105.4 cm (41 ½ inches) by 147.3 cm (58 inches) for all Littoral Planning Charts (north-south and east-west).

3.21.2 Colors and inks. Printing colors and screens shall conform to information and items illustrated in MIL-STD-2410.

3.21.3 Red-light readability. The colors used on Littoral Planning Charts are designed to be readable in both white and red light. The fact that the chart is red light readable is identified in the margin (see 3.11.29). For a better understanding of the impact of various lighting on the readability of Littoral Planning Charts, the following visual efficiency figures are provided:

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SPC Number	Color Name	Visual efficiency (%)		
		Daylight	Redlight	Bluelight
46351	Aero-Blue	85	86	57
48253	Cyan-Blue	62	86	17
52813	Green	60	76	72
57377	Yellow	12	3	84
58600	Black	91	90	91
61121	Red-Brown	82	68	90
96532	Purple	74	62	49

3.22 Magnetic variation.

3.22.1 Compass rose portrayal. Magnetic variation data is shown on Littoral Planning Charts by means of black (SPC-58600) compass roses oriented to true north and a legend in the center of each compass rose indicating the magnetic variation (to the nearest minute) at the position of the compass rose, and annual change (to the nearest second). Refer to the style sheets, APPENDICES B and C, for graphic examples of compass roses.

a. In rounding off to the nearest minutes and seconds, 29 minutes and 29 seconds are rounded down, and 30 minutes and 30 seconds are rounded up.

b. Magnetic variation data is shown for the current year epoch in which the edition date falls. For example, a chart published in 2010 shows magnetic data for the epoch year 2010.

3.22.2 Compass positions. The positions of compass roses are chosen to meet the following criteria:

a. Placed in the water areas of the chart in such a way that they do not obscure dangers, shoal soundings, or aids to navigation.

b. Placed in the vicinity of the meridians, which are central to the areas where the compass roses will most likely be used. This minimizes the error introduced by the convergence of meridians on the Transverse Mercator projection.

c. Clear of chart folds.

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d. Provide coverage over the chart so that no water area, excluding inland hydrography, is farther than 45 cm from a compass rose.

3.22.3 Compass rose specifications.

a. A compass rose consists of 360 ticks radiating out from the center of the compass rose. On a full size rose, the ticks begin at a radial distance of 63 mm from the center of the compass rose. Reduced sizes, with radial distances of 56, 49, or 42 mm, may be shown if space is not available to show the full size rose. Tick lengths are as follows: 10° ticks are 3.0 mm long, 5° ticks are 2.0 mm long, and 1°ticks are 1.0 mm long. All lineweights are 0.15mm.

b. Ten degree ticks (0, 10, 20, 30, 40,....350) are labeled with 7 point type, with the numbers oriented so they are reading from the bottom of the chart (90 and 270 are vertical and reading from the inside of the compass rose). No degree signs are shown. Space between the end of the 10°ticks and the base of the type is 1.0 mm.

c. In addition to these ticks, extended ticks are shown at 0°, 90°, 180°, and 270°. These ticks are 5.0 mm long and begin 1.0 mm outside of the numbers 0, 90, 180, and 270.

d. A five pointed star is shown above the tick over the number 0. The star is 7.5 mm in diameter and is shown by a perimeter line (0.15 mm lineweight). Each point of the star is then divided by another line down the centerline of the point (0.15 mm lineweight). The left half of each point (proceeding clockwise around the star) is shaded by lines, parallel to the center line and spaced at 2.0 mm interval (lines are 0.10 mm lineweight). The star is positioned so one of the points is pointing true north and the end of the tick above the number “0” is touching the perimeter of the star opposite the top point.

e. The cross at the center of the compass rose is 0.15 mm lineweight. Ticks are oriented 90 degrees to each other and are positioned at cardinal directions (north-south, east-west). The overall tick length is 2.0 mm.

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3.23 Feature/Attribution.

3.23.1 General. Appendix A contains LPC feature, feature geometry, inclusion conditions, and specific rules corresponding to Littoral Planning Chart production.

4. VERIFICATION.

4.1 Classification of inspection. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.2)

b. Conformance inspection (see 4.3)

4.2 First article inspection. When a first article inspection is required (see 3.1), the product shall be examined as specified in 4.3.1, and the construction record reviewed for compliance with 4.3.2.

4.3 Conformance inspection. Conformance inspection shall include the examinations of 4.3.1 and 4.3.2.

4.3.1 Visual examination. The map/chart shall be examined for defects and errors as specified by the contract or Government. Any defects or errors detected shall be corrected. Defects detected during the inspection of the printed "catch copy" shall be evaluated by NGA for criticality, and suitable corrective action.

4.3.2 Review of construction records. Records about the construction of the map/chart shall be maintained. The records shall document sources, decisions regarding reconciliation of conflicting data, etc. Chart records/construction histories shall be reviewed concurrently with visual examinations (see 4.3.1) to ensure that proper cartographic procedures have been followed. Digital construction records may also be utilized and maintained.

5. PACKAGING.

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When packaging of material is to be performed by DoD or in-house contractor personnel, these personnel need to contact

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the responsible packaging activity to ascertain packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activities within the Military Service or Defense Agency, or within the military service's system command. Packaging data retrieval is available from the managing Military Department or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

6. NOTES.

6.1 Intended use. The intended use for Littoral Planning Charts is to support amphibious operations, joint air/ground tactical operations, land combat operations, naval gunfire support, and special operations planning. LPC are also used for other non-military missions such as humanitarian relief in littoral waters. Littoral Planning Charts are military-unique because of the classification of the product and sensitivity of the areas of interest identified by LPC coverage.

6.2 Acquisition Requirements. Acquisition documents must specify the following:

a. Title, number and date of this specification.

b. Issue of the DoDISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.2.1 and 2.3).

c. When a first article is required (see 3.1 and 4.2).

6.3 Supersession. These specifications supersede Military Specifications for Combat Charts, MIL-C-89202A, 29 April 1995.

6.4 Definitions.

6.4.1 Circular error (CE). An accuracy figure representing the stated percentage of probability that any point expressed as a function of two linear components (e.g., horizontal position) will be within the given figure.

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6.4.2 Linear error (LE). A one dimensional error (such as an error in elevation) defined by the normal distribution function.

6.5 Subject term (key word) listing.

Amphibious
Bathymetry
Beach
Charting
Hydrography
Landing
Marine
Maritime
MC&G (Mapping, Charting and Geodesy)
National Geospatial-Intelligence Agency
Nautical
Navigation
USMC

6.6 International standardization agreement implementation. This specification implements the NATO standardization agreements shown below. When amendment, revision, or cancellation of this specification is proposed, the preparing activity must coordinate the action with the U.S. National Point of Contact for the international standardization agreement, as identified in the ASSIST database at https://assist.dla.mil.

STANAG 1022 – Combat Chart, Amphibious Charts, and Combat\Landing Charts

STANAG 2211, Geodetic Datums, Spheroids, Grids, and Grid References

STANAG 3673, Identification of Source Data on Nautical and Special Naval Charts

6.7 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.

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6.8 NGA customer help desk. For questions concerning is or other NGA products, services, or specifications, please telephone the NGA Customer Help Desk at 1-800-455-0899, Commercial 314-260-1236, or DSN 490-1236.

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APPENDIX A
Littoral Planning Chart Features

A.1 SCOPE.

A.1.1 Scope. This Appendix is not a mandatory part of the specification. The information contained herein is intended for guidance only.

A.2 Littoral planning chart features. This Appendix provides the majority of feature portrayal and symbol information such as feature description, feature geometry, inclusion conditions, and feature information section. The feature information section includes symbol specification, feature origin, symbol criteria, and symbol reference, label, and color information.

The Appendix information is organized alphabetically based on the Littoral Planning Chart feature name and then further order within each feature grouping by geometry or delineation (i.e. Surface, Curve, and Point).

A.2.1 Feature description. The feature description is the definition of the feature as defined by National System for Geospatial-Intelligence (NSG).

A.2.2 Inclusion condition. The inclusion condition includes the feature information and attributes criteria that define those feature geometry conditions for the Littoral Planning Chart Feature.

A.2.3 Notes. This section contains generic extraction and finishing rules and guidelines for displaying and labeling the LPC features.

A.2.4 Feature information section. The feature information section includes symbol specification, feature origin, symbol criteria, and symbol reference, label, and label color information. A sample Feature Information Section is below.

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Dredged Area (FC034)	
Surface	
Feature Description:	*An area of the bottom of a body of water (for example: a channel) which has been deepened by dredging.*
Inclusion Condition:	*Width >= 50 meters*

Notes

Show Hydrographic Depth values in whole meters if decimeter value is 0; otherwise show the decimeter value showing a comma instead of decimeter point (example: 17,3m)

If Dredged Area is not maintained (Maintained: [False]), the label should include the year if the control surveys and dredging (Example: Dredged to 8,2m (2006)).

If space does not allow for the full legend to be shown, labels for Dredged Channels are condensed in the following order:
(1.) Delete "Dredged to" or "Maintained depth" labels first,
(2.) Delete Date label and parentheses second,
(3.) Delete description of Aids to Navigation label third. Do not delete Hydrographic depth label or the "m" from any Dredged Channel feature

If two line features of the same feature type meet end to end, and have different depths, a short line is shown centered on the point of intersection. It bisects the angle at which the line features meet (i.e., if the lines meet at 180° angle the bisecting line is perpendicular to the meeting line features). The bisecting line is 0.1 mm lineweight, length is 3.0 mm, and it is shown in the same color as the line features.

Dredged channels/areas shall be tinted in accordance with their depths, following the guidance for generation of water tint provided in rules for depth curves and/or open water.

If Aids to Navigation Marked: [True], Description of Aids to Navigation shall be shown if one of the following conditions occur:
(1.) The aids to navigation that mark the feature do not meet the inclusion conditions for aids to navigation.
(2.) The aids to navigation that mark the feature are known to exist, but the details, such as position, type of aid, etc, are not sufficient to chart them as aids to navigation.
If Navaid features are shown, or information about aids to navigation is completely lacking, omit the Description of Aids to Navigation label.

Feature Information					
Specification:	*Chart #1*	**Feature Origin:**	*Hydrographic*	**Symbol:**	I22
Symbol Criteria:	Maintained: [False]			**Label:**	[Hydrographic Depth]"m", Hydrographic Depth: 6 point lower case, Italic
				Color:	Black (58600)

A.2.4.1 Symbol specification. The symbol specification defines whether the feature symbology specification. The feature symbology specifications may be from National System for Geospatial Intelligence (NSG) Portrayal Standard for Local Topographic Data Store (LTDS) Data, Regulations of the IHO for International (INT) Charts and Chart Specifications of the IHO (Chart #1), or from other feature symbology specifications.

A.2.4.2 Feature origin. The feature origin defines whether the feature as defined in the inclusion condition is from Topographic, Hydrographic, Aeronautical, or other origins.

A.2.4.3 Symbol criteria. The symbol criteria define the attribution necessary in determining the feature symbology as defined by the symbol specification.

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A.2.4.4 Symbol reference. The symbol reference as labeled 'Symbol:' in the Feature Information section will be the symbol reference as specified in the symbol specifications. Chart #1 contains a section designation code 'A-X' and a symbol number. LTDS is a concatenation of three items: representation code for area, line, and point entities; symbol color; and entity/feature name.

A.2.4.5 Label. The label information will include the label information as described by the symbol specifications and may include feature attribution information that needs to be included in labeling the feature. The label size, Character case (Upper, Lower, or Mixed) and font style included, must be similar to Swiss742 fonts specified in the MIL-STD-2402 and MIL-STD-2410

(e.g. compatible with the colors, sizes, styles). The TrueType Zurich font has been selected as the most appropriate font to represent the label information for the LPC product.

A.2.4.6 Color. The color will define the color (Standard Printing Color # SPC) of the label information and does not necessarily determine the symbol color. See 3.20 Symbology and 3.20.1 Symbol Color Exceptions for information regarding the color of symbols and exceptions to the specification.

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Access Zone (FA005)

Surface

Feature Description:	*A terrain region between a contact zone and the first passable land transportation route (for example: a road).*
Inclusion Condition:	*All Required*

Notes

Feature Information

Specification:	*Other*	**Feature Origin:**	*Other*	**Symbol:**	TBD
Symbol Criteria:	TBD			**Label:**	TBD
				Color:	TBD

Acoustic Station (BK010)

Point

Feature Description:	*The geographic location at which a set of acoustic observations were taken.*
Inclusion Condition:	*All Required*

Notes

Feature Information

Specification:	*Other*	**Feature Origin:**	*Other*	**Symbol:**	TBD
Symbol Criteria:	TBD			**Label:**	TBD
				Color:	TBD

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Administrative Boundary (FA000)

Curve

Feature Description:	*A boundary between administratively controlled regions.*
Inclusion Condition:	*All Required*

Notes

A Boundary marker will be aligned with the feature.

If the boundary symbol and the projection line have the same line weight, the boundary symbol shall be shown in its entirety 0.25 mm inside the projection line.

Feature must retain all cartographic detail (i.e., not thinned or smoothed).

Label line feature above (preferred) and parallel to the line with a 0.5 mm space between. Above means: readable from south or east Projection neatline.

If a boundary follows a road and the exact location is unknown, label "APPROXIMATE BOUNDARY".

Possession of islands and island groups shall be shown by placing the country name in parentheses below the island name or island group name. If all of the islands in an island group belong to one country, the country name shall be placed under the island group name only. If islands within the same island group belong to different countries, the country name shall be placed under each island name, and not under the island group name. Islands administered jointly by two countries shall show both country names, separated by a dash, e.g., (UK-US). Country names shall be abbreviated in the manner approved by the Board of Geographic Names. Type size for county names shall be 2/3 the size of the island name or island group name, but shall not be less than 5 point.

If a boundary is not recognized by the U.S. Department of State as an official international boundary, but falls under the category of "Other Line of Separation", and the type of boundary is not portrayed by another feature, the text is used to label the line in accordance with Geonames/Boundary guidance; e.g. "Administrative Line", "Provisional administrative Line".

International boundaries and other lines of separation, and their associated labels, are shown in margin diagrams as well as in the body of the map or chart.

If the limits of a lesser Administrative Boundary division is coincident with that of a higher division, the symbol for the higher boundary division shall be shown, in descending order- (International), (Primary/1st Order), (2nd Order), (3rd Order), (City).

Every third unit of the boundary symbol shall be shown when a boundary is coincident within a Road. When the boundary is International, the International Boundary overprint shall be shown as a continuous band.

If a boundary follows an edge of a Road, Track, Trail, or Railroad, every third unit of the boundary symbol shall be shown coincident with the feature. When the boundary is International, the width of the International boundary overprint shall be reduced to one-half of its normal width.

If a boundary follows a Road and the exact location is unknown, the symbol shall be shown in the center of the road with every third unit of the boundary symbol.

If a boundary follows a Shoreline (Coastal or Inland), every third unit of the boundary symbol shall be shown where coincident with the Shoreline.

If a Boundary is coincident with a single-line or narrow double-line stream, (i.e., width <3 mm at map scale) only every third unit of the boundary symbol shall be shown

If a boundary crosses a body of Open Water >= 20 mm width and alignment is known, the complete boundary symbol shall be shown. If the boundary alignment is unknown, the boundary symbol shall be shown in the open water area at the points of entry. If the size of the Open Water permits, show complete units of the boundary symbol.

International boundary symbols shall not be shown crossing bodies of Open water with width >= 20 mm at map scale. The symbol shall terminate at points of entry into the Open water area.

If a boundary location is known and occurs within a double-line stream the complete boundary symbol shall be shown. If the boundary location is unknown, the boundary symbol shall be centered in the Stream and labeled "APPROXIMATE".

Boundaries shown shall be included in legend

In areas where there is no defined boundary between two countries, center Names in the approximate area of their respective sides of the label "NO DEFINED BOUNDARY". Pairs of labels may be repeated if necessary for large areas, but pairs should be positioned far enough apart so that they DO NOT imply a specific division line between the two countries.

Use a point of change for changes in status on an Administrative boundary, Armistice line, Cease-fire line, defacto boundary, demilitarized zone, or zone of occupation, unless change occurs at a symbolized boundary marker.

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Administrative Boundary (FA000) *(Continued)*

Curve

Feature Information					
Specification:	*LTDS*	**Feature Origin:**	*Topographic*	**Symbol:**	LN_Dk-Brown1815_Admin Boundary_International
Symbol Criteria:	Geopolitical Line Type = [Generic International Boundary]			**Label:**	[Geopolitical Entity : Geographic Name Information (1) : Full Name (first side)] "("[Boundary Status]")" [Geopolitical Entity : Geographic Name Information (2) : Full Name (second side)] "("[Boundary Status]")", 10 point Condensed Bold Upper case
				Color:	Black (58600)

Feature Information					
Specification:	*LTDS*	**Feature Origin:**	*Topographic*	**Symbol:**	LN_Dk-Brown1815_Admin Boundary_Armistice
Symbol Criteria:	Geopolitical Line Type = [Armistice Line]			**Label:**	"[Geographic Name Information (1) : Full Name (first side)]/ [Geographic Name Information (2) : Full Name (second side)] ARMISTICE LINE", 10 point Condensed Bold Upper case
				Color:	Black (58600)

Feature Information					
Specification:	*LTDS*	**Feature Origin:**	*Topographic*	**Symbol:**	LN_Black_Admin Boundary_CeaseFire
Symbol Criteria:	Geopolitical Line Type = [Cease Fire Line]			**Label:**	"CEASE-FIRE-LINE", 10 point Condensed Bold Upper case
				Color:	Black (58600)

Feature Information					
Specification:	*LTDS*	**Feature Origin:**	*Topographic*	**Symbol:**	LN_AdminBoundary _NoDefinedBoundary
Symbol Criteria:	Boundary Status = [No Defined Boundary]			**Label:**	"NO DEFINED BOUNDARY", 8 point Condensed Upper case
				Color:	Black (58600)

Feature Information					
Specification:	*LTDS*	**Feature Origin:**	*Topographic*	**Symbol:**	LN_Dk-Brown1815_AdminBoundary _OtherLineOfSeparation
Symbol Criteria:	(Geopolitical Line Type = [Generic Administrative Boundary] or [Provisional Administrative Line] or [Line of Control] or [Demarcation Line] or [Line of Convenience] or [Convention Line] or [Claim Line] or [Intercolonial Line] or [Interentity Line] or [Line of Adjacency] or [Line of Withdrawal] or [Military Disengagement Line] or [Treaty Line] or [Other]) and "[Note : Memorandum]"			**Label:**	"[Geopolitical Entity : Geographic Name Information (1) : Full Name (first side)]/[Geopolitical Entity : Geographic Name Information (2) : Full Name (second side)]" and "[Geopolitical Line Type]" and "[Note : Memorandum]"
				Color:	Black (58600)

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Administrative Boundary (FA000) *(Continued)*

Curve

Feature Information					
Specification:	*Chart #1*	**Feature Origin:**	*Hydrographic*	**Symbol:**	N47
Symbol Criteria:	Maritime Geopolitical Limit Type = [Exclusive Economic Zone Limit]			**Label:**	"[Geographic Name Information (1) : Full Name (first side)]" and "EEZ", 8 point Upper case, Italic
				Color:	Green (52813)

Feature Information					
Specification:	*Chart #1*	**Feature Origin:**	*Hydrographic*	**Symbol:**	N44
Symbol Criteria:	Maritime Geopolitical Limit Type = [Contiguous Zone Limit]			**Label:**	None
				Color:	N/A

Feature Information					
Specification:	*Chart #1*	**Feature Origin:**	*Hydrographic*	**Symbol:**	N48
Symbol Criteria:	Maritime Geopolitical Limit Type= [Customs Boundary]			**Label:**	None
				Color:	N/A

Feature Information					
Specification:	*Chart #1*	**Feature Origin:**	*Hydrographic*	**Symbol:**	N43
Symbol Criteria:	Maritime Geopolitical Limit Type = [Territorial Waters Limit of Sovereignty]			**Label:**	"Twelve Nautical Mile Limit", 8 Point Mixed Case Italic
				Color:	Green (52813)

Feature Information					
Specification:	*Chart #1*	**Feature Origin:**	*Hydrographic*	**Symbol:**	N43
Symbol Criteria:	Maritime Geopolitical Limit Type = [Territorial Sea Baseline]			**Label:**	None
				Color:	N/A

Feature Information					
Specification:	*Chart #1*	**Feature Origin:**	*Hydrographic*	**Symbol:**	N41
Symbol Criteria:	Maritime Geopolitical Limit Type = [International Boundary]			**Label:**	"[Geographic Name Information (1) : Full Name (first side)]/ [Geographic Name Information (2) : Full Name (second side)]" , 8 point Upper case Italic
				Color:	Green (52813)

Feature Information					
Specification:	*Chart #1*	**Feature Origin:**	*Hydrographic*	**Symbol:**	N46
Symbol Criteria:	Maritime Geopolitical Limit Type = [Three Nautical Mile Limit]			**Label:**	"Three Nautical Mile Limit", 8 point Mixed case, Italic
				Color:	Green (52813)

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Administrative Boundary (FA000) *(Continued)*

Curve

Feature Information					
Specification:	*Chart #1*	**Feature Origin:**	*Hydrographic*	**Symbol:**	N46
Symbol Criteria:	Maritime Geopolitical Limit Type = [Continental Shelf Limit]			**Label:**	"[Geographic Name Information (1) : Full Name (first side)]" and "Continental Shelf", 8 point Mixed case, Italic
				Color:	Green (52813)

Feature Information					
Specification:	*Chart #1*	**Feature Origin:**	*Hydrographic*	**Symbol:**	N46
Symbol Criteria:	Maritime Geopolitical Limit Type = [Extended Continental Shelf Limit]			**Label:**	"[Geographic Name Information (1) : Full Name (first side)]" and "Extended Continental Shelf", 8 point Mixed case, Italic
				Color:	Green (52813)

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Administrative Division (FA003)

Surface

Feature Description:	*An administratively subordinate division of a geopolitical entity.*
Inclusion Condition:	*All Required*

Notes

A Boundary marker will be aligned with the feature.

If the boundary symbol and the projection line have the same line weight, the boundary symbol shall be shown in it's entirety 0.25 mm inside the projection line.

Feature must retain all cartographic detail (i.e., not thinned or smoothed).

Label line feature above (preferred) and parallel to the line with a 0.5 mm space between. Above means: readable from south or east Projection neatline.

If a boundary follows a road and the exact location is unknown, label "APPROXIMATE BOUNDARY".

Possession of islands and island groups shall be shown by placing the country name in parentheses below the island name or island group name. If all of the islands in an island group belong to one country, the country name shall be placed under the island group name only. If islands within the same island group belong to different countries, the country name shall be placed under each island name, and not under the island group name. Islands administered jointly by two countries shall show both country names, separated by a dash, e.g., (UK-US). Country names shall be abbreviated in the manner approved by the Board of Geographic Names. Type size for county names shall be 2/3 the size of the island name or island group name, but shall not be less than 5 point.

If a boundary is not recognized by the U.S. Department of State as an official international boundary, but falls under the category of "Other Line of Separation", and the type of boundary is not portrayed by another feature, the text is used to label the line in accordance with Geonames/Boundary guidance; e.g. "Administrative Line", "Provisional administrative Line".

International boundaries and other lines of separation, and their associated labels, are shown in margin diagrams as well as in the body of the map or chart.

If the limits of a lesser Administrative Boundary division is coincident with that of a higher division, the symbol for the higher boundary division shall be shown, in descending order- (International), (Primary/1st Order), (2nd Order), (3rd Order), (City).

Every third unit of the boundary symbol shall be shown when a boundary is coincident within a Road. When the boundary is International, the International Boundary overprint shall be shown as a continuous band.

If a boundary follows an edge of a Road, Track, Trail, or Railroad, every third unit of the boundary symbol shall be shown coincident with the feature. When the boundary is International, the width of the International boundary overprint shall be reduced to one-half of its normal width.

If a boundary follows a Road and the exact location is unknown, the symbol shall be shown in the center of the road with every third unit of the boundary symbol.

If a boundary follows a Shoreline (Coastal or Inland), every third unit of the boundary symbol shall be shown where coincident with the Shoreline.

If a Boundary is coincident with a single-line or narrow double-line stream, (i.e., width <3 mm at map scale) only every third unit of the boundary symbol shall be shown

If a boundary crosses a body of Open Water >= 20 mm width and alignment is known, the complete boundary symbol shall be shown. If the boundary alignment is unknown, the boundary symbol shall be shown in the open water area at the points of entry. If the size of the Open Water permits, show complete units of the boundary symbol.

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Administrative Division (FA003) *(Continued)*

Surface

Notes

International boundary symbols shall not be shown crossing bodies of Open water with width >= 20 mm at map scale. The symbol shall terminate at points of entry into the Open water area.

If a boundary location is known and occurs within a double-line stream the complete boundary symbol shall be shown. If the boundary location is unknown, the boundary symbol shall be centered in the Stream and labeled "APPROXIMATE".

Boundaries shown shall be included in legend

In areas where there is no defined boundary between two countries, center Names in the approximate area of their respective sides of the label "NO DEFINED BOUNDARY". Pairs of labels may be repeated if necessary for large areas, but pairs should be positioned far enough apart so that they DO NOT imply a specific division line between the two countries.

Use a point of change for changes in status on an Administrative boundary, Armistice line, Cease-fire line, defacto boundary, demilitarized zone, or zone of occupation, unless change occurs at a symbolized boundary marker.

Feature Information					
Specification:	*LTDS*	**Feature Origin:**	*Topographic*	**Symbol:**	LN_Black_AdminDivision _Primary
Symbol Criteria:	BGN Administrative Level = [First-order]			**Label:**	"[Geographic Name Information : Full Name]", 8 point Condensed Upper case
				Color:	Black (58600)

Feature Information					
Specification:	*LTDS*	**Feature Origin:**	*Topographic*	**Symbol:**	LN_Black_AdminDivision _Secondary
Symbol Criteria:	BGN Administrative Level = [Second-order]			**Label:**	"[Geographic Name Information : Full Name]", 8 point Condensed Upper case
				Color:	Black (58600)

Feature Information					
Specification:	*LTDS*	**Feature Origin:**	*Topographic*	**Symbol:**	LN_Black_AdminDivision _Tertiary
Symbol Criteria:	BGN Administrative Level = [Third-order]			**Label:**	"[Geographic Name Information : Full Name]", 8 point Condensed Upper case
				Color:	Black (58600)

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Aerial (AT011)

Point

Feature Description:	*A device that is used for emitting and/or sensing electromagnetic energy.*
Inclusion Condition:	*Height Above Surface Level >= 46 m. OR Navigation Landmark: (True)*

Notes

None

Feature Information

Specification:	*LTDS*	**Feature Origin:**	*Topographic*	**Symbol:**	PT_Black_Aerial
Symbol Criteria:	Navigation Landmark: [True] And Height Above Surface Level < 46			**Label:**	None
				Color:	N/A

Feature Information

Specification:	*LTDS*	**Feature Origin:**	*Topographic*	**Symbol:**	PT_Blue072_VO
Symbol Criteria:	Height Above Surface Level >= 46			**Label:**	"Aerial" , 6 point Condensed Mixed case; [Highest Elevation], 6 point Condensed Bold Mixed case; "([Height Above Surface Level])", 6 point Condensed Mixed case
				Color:	Blue (46351)

Aerial Farm (AT012)

Surface

Feature Description:	*A collection of aerials that are collocated and serve a common purpose.*
Inclusion Condition:	*Area >= 15,625 sq m.*

Notes

None

Feature Information

Specification:	*LTDS*	**Feature Origin:**	*Topographic*	**Symbol:**	AR_Black_AerialFarm
Symbol Criteria:	Area >= 15,625			**Label:**	"Aerial farm", 6 point Condensed Mixed case
				Color:	Black (58600)

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Aerial Farm (AT012)

Point

Inclusion Condition:	*Area < 15,625 sq m. AND Navigation Landmark: (True)*

Notes

None

Feature Information					
Specification:	*LTDS*	**Feature Origin:**	*Topographic*	**Symbol:**	PT_Black_AerialFarm
Symbol Criteria:	Area < 15,625 AND Navigation Landmark = [True]			**Label:**	"Aerial farm", 6 point Condensed Mixed case
				Color:	Black (58600)

Aerodrome Beacon (GB013)

Point

Feature Description:	*A beacon used to indicate the location of an aerodrome from the air.*
Inclusion Condition:	*All required*

Notes

Priority for type placement: 1-right center, 2-bottom center, 3-left center, 4-top center.
(A) Minimum distance from symbol - 1 mm.
(B) Maximum distance from symbol before choosing the next highest priority:
(#1) 4 mm measured to the West end;
(#2) 4 mm measured to the North side (top);
(#3) 4 mm measured to the East end;
(#4) 4 mm measured to the South side (bottom)

Light flares (Posicut # 94) shall be oriented in order as follows:
(1) So that it does not overprint on the feature or a legend associated with that feature (except for the 7.1 mm diameter circle representing a radio aid to navigation, which the flare may overprint). ;
(2) So it does not overprint other chart data (i.e., soundings, pipelines, submarine cables, etc).
(3) So that the wide end of the flare is pointed toward the legend of the light, or to seaward along the line, in the case of clearing lines and leading lines.

Feature Information					
Specification:	*Chart #1*	**Feature Origin:**	*Hydrgraphic*	**Symbol:**	P60
Symbol Criteria:	All			**Label:**	"Aero", 7 point Mixed case; [Character of Light], 7 point Mixed case; [Period of Light] and "s", 7 point lower case; [Height Above Surface Level] and "m", 7 point lower case; [Nominal Range], 7 point lower case, "M", 7 point Upper case
				Color:	Black (58600)

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Aerodrome Boundary (GB007)

Curve

Feature Description:	*The geographical limits of an aerodrome property as defined by the local authority.*
Inclusion Condition:	*All required*

Notes

None

Feature Information					
Specification:	*Other*	**Feature Origin:**	*Aeronautical*	**Symbol:**	TBD
Symbol Criteria:	TBD			**Label:**	TBD
				Color:	TBD

Aeronautical Radio Navigation Installation (GA033)

Point

Feature Description:	*A ground based piece of electronic equipment that provides aeronautical navigation guidance information or position data.*
Inclusion Condition:	*Primary Operating Range Category: >= 50 nautical miles and Facility Operational Status: (Continuous Operation) and Distance from Shoreline <= 9,997 meters*

Notes

Priority for type placement: 1-right center, 2-bottom center, 3-left center, 4-top center.
(A) Minimum distance from symbol - 1 mm.
(B) Maximum distance from symbol before choosing the next highest priority:
(#1) 4 mm measured to the West end;
(#2) 4 mm measured to the North side (top);
(#3) 4 mm measured to the East end;
(#4) 4 mm measured to the South side (bottom)

Feature Information					
Specification:	*Chart #1*	**Feature Origin:**	*Hydrographic*	**Symbol:**	S16
Symbol Criteria:	All			**Label:**	"Aero RC", 8 point Mixed case
				Color:	Purple (96532)

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Aircraft Hangar (GB230)

Surface

Feature Description:	*A building for housing aircraft.*
Inclusion Condition:	*Area >= 625 sq m.*

Notes

None

Feature Information

Specification:	*LTDS*	**Feature Origin:**	*Topographic*	**Symbol:**	AR_Black_AircraftHangar
Symbol Criteria:	(Height Above Surface Level < 46) OR (Area >= 5625 and Height Above Surface Level >= 46)			**Label:**	"Hanger", 6 point Condensed Mixed case
				Color:	Black (58600)

Aircraft Hangar (GB230)

Point

Inclusion Condition:	*Area < 625 sq m.*

Notes

None

Feature Information

Specification:	*LTDS*	**Feature Origin:**	*Topographic*	**Symbol:**	PT_Black_AircraftHangar
Symbol Criteria:	Height Above Surface Level < 46			**Label:**	"Hanger", 6 point Condensed Mixed case
				Color:	Black (58600)

Aircraft Revetment (GB050)

Curve

Feature Description:	*A barricade that protects an aircraft, equipment, and/or facility from hostile action.*
Inclusion Condition:	*ALL REQUIRED*

Notes

None

Feature Information

Specification:	*LTDS*	**Feature Origin:**	*Topographic*	**Symbol:**	LN_Blue072_AircraftRevetment
Symbol Criteria:	All			**Label:**	None
				Color:	N/A

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Aircraft Stand (GB016)

Point

Feature Description:	*A designated area on an apron intended to be used for parking an aircraft.*
Inclusion Condition:	*ALL REQUIRED*

Notes

None

Feature Information

Specification:	*Other*	**Feature Origin:**	*Aeronautical*	**Symbol:**	TBD
Symbol Criteria:	TBD			**Label:**	TBD
				Color:	TBD

Amphitheatre (AK164)

Surface

Feature Description:	*A small tract of level ground serving as a stage that is surrounded by rising slopes (either naturally occurring or artificially constructed) supporting tiered seating (for example: benches).*
Inclusion Condition:	*Area >= 625 square meters*

Notes

Area and line features will be generalized to detail compatible with scale.

Type for features on land shall be positioned on land if possible. If type must be placed in the water, it shall be positioned so it does not obscure or overprint hydrographic detail. Type for features in the water shall be positioned in the water. If possible, type shall not be positioned across the shoreline.

Descriptive terms, e.g., "Canal" shall be shown if the name is not known. If the descriptive word appears in the name, for example, "PANAMA CANAL". The descriptive type shall not be shown, i.e. do not show "Panama Canal Canal".

Limiting lines of feature are omitted if it coalesces with a road (AP030).

If a clearing exists inside of the area feature, and the size of the clearing is equal to or greater than the area for the surrounding area feature, the clearing is shown as an open space inside the surround feature. If the area of the clearing is less than the area for the surrounding feature, the clearing is deleted and absorbed into the surrounding area feature.

If two area features with the same feature code do not connect at any point, and have a space between them less than 2.5 mm at map/chart scale, delete the open space that is less than 2.5 mm between the features and combine them into one area feature.

If a portion of an area feature has a minimum width of less than 2.5 mm at map/chart scale, delete that portion of the area feature that is not at least 2.5 mm wide, measured from perimeter to perimeter. If the deletion of the portion of the area based on the above criteria will reduce the area of the remaining portion of the area feature to below the minimum area inclusion, do not delete the narrow portion of the feature that is less than 2.5 mm wide. If the deletion of the portion of the area based on the above criteria will split two larger areas connected by a narrow strip into two separate areas, either of which would be below minimum area inclusion, do not delete the narrow portion of the feature that is less than 2.5 mm wide.

Feature Information

Specification:	*LTDS*	**Feature Origin:**	*Topographic*	**Symbol:**	AR_Black_Amphitheatre
Symbol Criteria:	Area >= 625			**Label:**	[Geographic Name Information : Full Name] if available, 6 point Condensed Mixed case; "amphitheatre", 6 point Condensed Mixed case
				Color:	Black (58600)

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Amphitheatre (AK164)

Point

Inclusion Condition:	*Area < 625 square meters AND Navigation Landmark =* [True]

Notes

None

Feature Information					
Specification:	*LTDS*	**Feature Origin:**	*Topographic*	**Symbol:**	PT_Black_Amphitheatre
Symbol Criteria:	Area < 625 AND Navigation Landmark = [True]			**Label:**	[Geographic Name Information : Full Name] if available, 6 point Condensed Mixed case; "amphitheatre", 6 point Condensed Mixed case
				Color:	Black (58600)

Amusement Park (AK030)

Surface

Feature Description:	*A predominantly man-made facility equipped with recreational devices.*
Inclusion Condition:	*Area >= 15,625 square meters*

Notes

Coincident similar area features will be blended to form a single feature.

Area and line features will be generalized to detail compatible with scale.

Type for features on land shall be positioned on land if possible. If type must be placed in the water, it shall be positioned so it does not obscure or overprint hydrographic detail. Type for features in the water shall be positioned in the water. If possible, type shall not be positioned across the shoreline.

Descriptive terms, e.g., "Canal" shall be shown if the name is not known. If the descriptive word appears in the name, for example, "PANAMA CANAL". The descriptive type shall not be shown, i.e. do not show "Panama Canal Canal".

Limiting lines of feature are omitted if it coalesces with a road (AP030).

If a clearing exists inside of the area feature, and the size of the clearing is equal to or greater than the area for the surrounding area feature, the clearing is shown as an open space inside the surround feature. If the area of the clearing is less than the area for the surrounding feature, the clearing is deleted and absorbed into the surrounding area feature.

If two area features with the same feature code do not connect at any point, and have a space between them less than 2.5 mm at map/chart scale, delete the open space that is less than 2.5 mm between the features and combine them into one area feature.

If a portion of an area feature has a minimum width of less than 2.5 mm at map/chart scale, delete that portion of the area feature that is not at least 2.5 mm wide, measured from perimeter to perimeter. If the deletion of the portion of the area based on the above criteria will reduce the area of the remaining portion of the area feature to below the minimum area inclusion, do not delete the narrow portion of the feature that is less than 2.5 mm wide. If the deletion of the portion of the area based on the above criteria will split two larger areas connected by a narrow strip into two separate areas, either of which would be below minimum area inclusion, do not delete the narrow portion of the feature that is less than 2.5 mm wide.

Feature Information					
Specification:	*LTDS*	**Feature Origin:**	*Topographic*	**Symbol:**	AR_Black_AmusementPark
Symbol Criteria:	Area >= 15,625			**Label:**	[Geographic Name Information : Full Name], 6 point Condensed Mixed case; "Amusement park", 6 point Condensed Mixed case
				Color:	Black (58600)

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Amusement Park (AK030)

Point

Inclusion Condition:	*Area < 15,625 square meters And Navigation Landmark = [True]*

Notes

None

Feature Information					
Specification:	*LTDS*	**Feature Origin:**	*Topographic*	**Symbol:**	PT_Black_AmusementPark
Symbol Criteria:	Area < 15,625 AND Navigation Landmark = [True]			**Label:**	[Geographic Name Information : Full Name], 6 point Condensed Mixed case; "Amusement park", 6 point Condensed Mixed case
				Color:	Black (58600)

Amusement Park Attraction (AK020)

Point

Feature Description:	*A large structure located in an amusement park.*
Inclusion Condition:	*Height Above Surface Level >= 46 m. OR Navigation Landmark = [True]*

Notes

If one symbol coalesces with another symbol for the same type feature, displace symbols to allow a minimum separation of 0.21mm.

Type for features on land shall be positioned on land if possible. If type must be placed in the water, it shall be positioned so it does not obscure or overprint hydrographic detail. Type for features in the water shall be positioned in the water. If possible, type shall not be positioned across the shoreline.

If the existence is Doubtful or Reported, label as such.

When obstructions coalesce at map scale, label with highest obstruction information.

If >= 3 equal symbols would coalesce at map scale, portray with a representative pattern.

Feature Information					
Specification:	*LTDS*	**Feature Origin:**	*Topographic*	**Symbol:**	PT_Black_ParkAttraction
Symbol Criteria:	Height Above Surface Level < 46 AND Navigation Landmark = [True]			**Label:**	"Park attraction", 6 point Condensed mixed case
				Color:	Black (58600)

Feature Information					
Specification:	*LTDS*	**Feature Origin:**	*Topographic*	**Symbol:**	PT_Blue072_VO
Symbol Criteria:	Height Above Surface Level >= 46			**Label:**	"(AMA)", 7 point Condensed Upper case; [Highest Elevation], 6 point Condensed Bold Mixed case; "([Height Above Surface Level])", 6 point Condensed Mixed case
				Color:	Blue (46351)

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Anchor (BB019)

Point

Feature Description:	*A heavy object, kept in place on the seafloor by its weight and possibly with flukes, to which is attached a cable, rope, or chain that keeps a floating structure on the waterbody surface positioned within a narrow region.*
Inclusion Condition:	*All Required*

Notes

None

Feature Information

Specification:	*Chart #1*	**Feature Origin:**	*Hydrographic*	**Symbol:**	L18
Symbol Criteria:	All			**Label:**	"Ground tackle", 8 point Mixed Case
				Color:	Purple (96532)

Anchorage (BB010)

Surface

Feature Description:	*An area of relatively uniform water depth with no cables, pipelines, or hazardous obstructions present on the seafloor in which vessels anchor or may anchor.*
Inclusion Condition:	*Length or Diameter >= 400 m.*

Notes

Labeling areas, in order of preference:
(1.) Centered in area on one line in the area, type is horizontal, reading left to right.
(2.) Centered in area on one line in the area, oriented along the long axis of the feature, reading left to right, or bottom to top if axis is vertical.
(3.) Centered in area on two approximately equal lines, without splitting a word, type is horizontal, reading left to right.
(4.) Centered on two approximately equal lines without splitting a word, type is oriented along the long axis of the feature, reading left to right or bottom to top if axis is vertical.
(5.) Use "Priority for type placement" (below) if the area is too small to place the type inside the area.

Priority for type placement: 1-right center, 2-bottom center, 3-left center, 4-top center.
(A) Minimum distance from symbol - 1 mm.
(B) Maximum distance from symbol before choosing the next highest priority:
(#1) 4 mm measured to the West end;
(#2) 4 mm measured to the North side (top);
(#3) 4 mm measured to the East end;
(#4) 4 mm measured to the South side (bottom)

Type sizes for Maritime Limits and areas:
8 point - < 8 square cm.,
10 point - >= 8 and < 12 square cm,,
12 point - >= 12 and < 24 square cm.,
14 point - >= 24 and < 100 square cm.,
8 point - >= 100 square cm.;

Type placement for areas >= 100 square cm. to < 500 square cm.:
Two labels are shown on approximately opposite sides of the area, preferably top and bottom.;

Type placement for areas >= 500 square cm.:
Labels are placed at approximately 250 mm. interval around the perimeter of the feature.;

Do not place labels around sharp corners (interior angle < 135°), or along chart neatlines. Place type 1 mm away and parallel to area limit line, reading left to right, or bottom to top if limit line is vertical. Type is placed on the INSIDE of the area.

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Anchorage (BB010) *(Continued)*

Surface

Notes

Type placement for areas < 100 square cm. Type shall be center in the area reading from left to right.

If longer of the two axes is North-South +/- 20 degrees or East-West +/- 20 degrees, type is parallel to the south neatline.
(a.) If length < width times two, type shall be placed on two approximately equal lines without splitting words;
(b.) If length >= width times two, and major axis is East-West +/- 20 degrees, type shall be placed on one line;
(c.) If length >= width times two, and major axis is North-South +/- 20 degrees, type shall be placed with each word on a separate line.
If longer of the two axis is more than 20 degrees from either North-South or East-West, and length < width times two, type shall be parallel to south neatline and on two approximately equal lines without splitting words.
If longer of two axis is more than 20 degrees from either North-South or East-West, and length >= width times two, type shall be placed parallel to the major axis on one line. Type to be placed inside area, place type outside area, using "Priority for type placement" (above).

Descriptive terms, e.g., "Canal" shall be shown if the name is not known. If the descriptive word appears in the name, for example, "PANAMA CANAL". The descriptive type shall not be shown, i.e. do not show "Panama Canal Canal".

The name of the anchoring berth (BB010) shall be centered in the circle of Posicut # 78.

If anchorage type category (Unrestricted Anchorage), no Anchorage name label is shown; Otherwise, label Anchorage name using the following labels:
If Anchorage Type = Explosives Anchorage, label "Explosives";
If Anchorage Type = Reserved Anchorage, label "Reserved";
If Anchorage Type = Seaplane Anchorage, label "Seaplane";
If Anchorage Type = Deepwater Anchorage, label "DW";
If Anchorage Type = Tanker Anchorage, label "Tanker".

When the boundary of an areal symbol crosses or overprints the shoreline, the boundary overprinting the shoreline or land is deleted.

If Name type will not fit inside anchoring berth circle, a rectangle or leader line may be used instead. Leader line is 0.1 mm lineweight, Black solid, SPC-58600.

Feature Information					
Specification:	*Chart #1*	**Feature Origin:**	*Hydrographic*	**Symbol:**	N12.1
Symbol Criteria:	Anchorage Type = [Unrestricted Anchorage] or [Small Craft Anchorage] or [Small Craft Mooring Area] or [Other] or [No Information]			**Label:**	[Geographic Name Information : Full Name] if available, 8-14 point Mixed Case, Italic
				Color:	Black (58600)

Feature Information					
Specification:	*Chart #1*	**Feature Origin:**	*Hydrographic*	**Symbol:**	N11.2, N12.2
Symbol Criteria:	Anchorage Type = [Anchorage Berth]			**Label:**	[Geographic Name Information : Full Name] if available, 8-14 point Mixed Case, Italic
				Color:	Black (58600)

Feature Information					
Specification:	*Chart #1*	**Feature Origin:**	*Hydrographic*	**Symbol:**	N12.4
Symbol Criteria:	Anchorage Type = [Deep Water Anchorage]			**Label:**	"DW", 8-14 point Upper Case, Italic
				Color:	Black (58600)

Feature Information					
Specification:	*Chart #1*	**Feature Origin:**	*Hydrographic*	**Symbol:**	N12.7
Symbol Criteria:	Anchorage Type = [Explosives Anchorage]			**Label:**	None
				Color:	N/A

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Anchorage (BB010) *(Continued)*

Surface

Feature Information					
Specification:	*Chart #1*	**Feature Origin:**	*Hydrographic*	**Symbol:**	N12.8
Symbol Criteria:	Anchorage Type = [Quarantine Anchorage]			**Label:**	None
				Color:	N/A

Feature Information					
Specification:	*Chart #1*	**Feature Origin:**	*Hydrographic*	**Symbol:**	N12.9
Symbol Criteria:	Anchorage Type = [Reserved Anchorage] or [Naval Reserve Anchorage]			**Label:**	[Geographic Name Information : Full Name] if available, 8-14 point Mixed Case, Italic; "Reserved See Note", 8-14 point Mixed Case, Italic
				Color:	Green (52813)

Feature Information					
Specification:	*Chart #1*	**Feature Origin:**	*Hydrographic*	**Symbol:**	N12.5
Symbol Criteria:	Anchorage Type = [Tanker Anchorage]			**Label:**	"Tanker", 8-14 point Mixed Case, Italic
				Color:	Black (58600)

Feature Information					
Specification:	*Chart #1*	**Feature Origin:**	*Hydrographic*	**Symbol:**	N12.6
Symbol Criteria:	Anchorage Type = [Time Limited] or [Anchorage for up to 24 hours]			**Label:**	[Activity Duration], 8-14 point lower case, Italic; "h", 8-14 point lower Case, Italic
				Color:	Green (52813)

Feature Information					
Specification:	*Chart #1*	**Feature Origin:**	*Hydrographic*	**Symbol:**	N13
Symbol Criteria:	Anchorage Type = [Seaplane Anchorage]			**Label:**	None
				Color:	N/A

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Anchorage (BB010)

Point

Inclusion Condition:	*Length or Diameter < 400 m.*

Notes

None

Feature Information					
Specification:	*Chart #1*	**Feature Origin:**	*Hydrographic*	**Symbol:**	N12.1
Symbol Criteria:	Anchorage Type = [Unrestricted Anchorage] or [Small Craft Anchorage] or [Small Craft Mooring Area] or [Other] or [No Information]			**Label:**	[Geographic Name Information : Full Name] if available, 8 point Mixed Case, Italic
				Color:	Black (58600)

Feature Information					
Specification:	*Chart #1*	**Feature Origin:**	*Hydrographic*	**Symbol:**	N12.4
Symbol Criteria:	Anchorage Type = [Deep Water Anchorage]			**Label:**	"DW", 8 point Upper Case, Italic
				Color:	Black (58600)

Feature Information					
Specification:	*Chart #1*	**Feature Origin:**	*Hydrographic*	**Symbol:**	N12.7
Symbol Criteria:	Anchorage Type = [Explosives Anchorage]			**Label:**	None
				Color:	N/A

Feature Information					
Specification:	*Chart #1*	**Feature Origin:**	*Hydrographic*	**Symbol:**	N12.8
Symbol Criteria:	Anchorage Type = [Quarantine Anchorage]			**Label:**	None
				Color:	N/A

Feature Information					
Specification:	*Chart #1*	**Feature Origin:**	*Hydrographic*	**Symbol:**	N12.9
Symbol Criteria:	Anchorage Type = [Reserved Anchorage] or [Naval Reserve Anchorage]			**Label:**	[Geographic Name Information : Full Name] if available, 8 point Mixed Case, Italic; "Reserved See Note", 8 point Mixed Case, Italic
				Color:	Green (52813)

Feature Information					
Specification:	*Chart #1*	**Feature Origin:**	*Hydrographic*	**Symbol:**	N12.5
Symbol Criteria:	Anchorage Type = [Tanker Anchorage]			**Label:**	"Tanker", 8 point Mixed Case, Italic
				Color:	Black (58600)

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Anchorage (BB010) *(Continued)*

Point

Feature Information					
Specification:	*Chart #1*	**Feature Origin:**	*Hydrographic*	**Symbol:**	N12.6
Symbol Criteria:	Anchorage Type = [Time Limited] or [Anchorage for up to 24 hours]			**Label:**	[Activity Duration], 8-14 point lower case, Italic; "h", 8-14 point lower Case, Italic
				Color:	Green (52813)

Feature Information					
Specification:	*Chart #1*	**Feature Origin:**	*Hydrographic*	**Symbol:**	N13, N14
Symbol Criteria:	Anchorage Type = [Seaplane Anchorage]			**Label:**	None
				Color:	N/A

Feature Information					
Specification:	*Chart #1*	**Feature Origin:**	*Hydrographic*	**Symbol:**	N11.1
Symbol Criteria:	Anchorage Type = [Anchorage Berth]			**Label:**	[Geographic Name Information : Full Name], 6 point Upper case
				Color:	Black (58600)

Annotated Location (ZD045)

Surface

Feature Description:	*A location at which text pertaining to that location is annotated.*
Inclusion Condition:	*ALL REQUIRED; SURFACE delineation if associated feature is captured as a SURFACE (Area).*

Notes

Type sizes per are sizes at map/chart scale:
Area features only.
06 point - <= 770 mm square area and <= 14 mm width;
07 point - <= 2,296 mm square area and <= 28 mm width;
09 point - <= 5,192 mm square area and 44 mm width;
10 point - <=9,796 mm square area and <= 62 mm width;
12 point - <= 16,632 mm square area and <= 84 mm width;
14 point - <= 24,960 mm square area and <= 104 mm width;
16 point - > 24,960 mm square area. Where area measurements are inconsistent, the larger type size shall be used.
Where the full range of type sizes is not available for a particular label, the closest available type size shall be used.

Label feature as per hierarchy for topo type placement to south neatline corners reading left to right:
1. Position hierarchy: (a.) northeast (preferred position), (b.) southeast (1st alternative), (c.) northwest (2nd alternative, (d.) southwest (3rd alternative), (e.) top-centered (4th alternative), (f.) bottom-centered (5th alternative): (Hierarchy is based on type positioning so as to avoid overprinting other type or obscuring detail.);
2. Minimum space between type placement and feature symbol is 0.5 mm;
3. This method of type placement shall be used for areal features when space does not permit labeling within that feature.

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Annotated Location (ZD045) *(Continued)*

Surface

Feature Information					
Specification:	*LTDS*	**Feature Origin:**	*Topographic*	**Symbol:**	AR_AnnotatedLocation _Culture
Symbol Criteria:	All associated with Topographic Feature			**Label:**	[Note : Memorandum], 6-20 point Condensed Mixed case
				Color:	Black (58600)

Feature Information					
Specification:	*LTDS*	**Feature Origin:**	*Hydrographic*	**Symbol:**	AR_AnnotatedLocation _Hydro
Symbol Criteria:	All associated with Hydrographic Feature			**Label:**	[Note : Memorandum], 6-20 point Condensed Mixed case
				Color:	Blue (48253)

Annotated Location (ZD045)

Curve

Inclusion Condition:	*ALL REQUIRED; CURVE delineation if associated feature is captured as a CURVE.*

Notes

Type sizes for single line features at map/chart scale.
06 point - <= 80 mm length;
07 point - <= 160 mm length;
09 point - > 160 mm length.

Label shall be position above feature, reading from left to right (or to the left of the vertical feature, reading bottom to top), at a 0.5 mm distance and parallel to respective feature. Label shall preferably be positioned at the midpoint of the line segment or symbol; however, it may be displaced laterally along respective feature to avoid overprinting other symbols or labels. If space will not permit placing label parallel to feature, offset the label in accordance with Rule L-4261 below and use a leader line to identify its location along the feature.

Feature name, label, data information holder, and/or symbol shall be positioned, reading left to right, parallel to the tangent of the center of the southern neatline of the Map/Chart.

Feature Information					
Specification:	*LTDS*	**Feature Origin:**	*Topographic*	**Symbol:**	LN_AnnotatedLocation _Culture
Symbol Criteria:	All associated with Topographic Feature			**Label:**	[Note : Memorandum], 6-20 point Condensed Mixed case
				Color:	Black (58600)

Feature Information					
Specification:	*LTDS*	**Feature Origin:**	*Hydrographic*	**Symbol:**	LN_AnnotatedLocation _Hydro
Symbol Criteria:	All associated with Hydrographic Feature			**Label:**	[Note : Memorandum], 6-20 point Condensed Mixed case
				Color:	Blue (48253)

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Annotated Location (ZD045)

Point

Inclusion Condition:	*ALL REQUIRED; POINT delineation if associated feature is captured as a POINT.*

Notes

Label feature as per hierarchy for topo type placement to south neatline corners reading left to right:
1. Position hierarchy: (a.) northeast (preferred position), (b.) southeast (1st alternative), (c.) northwest (2nd alternative, (d.) southwest (3rd alternative), (e.) top-centered (4th alternative), (f.) bottom-centered (5th alternative): (Hierarchy is based on type positioning so as to avoid overprinting other type or obscuring detail.);
2. Minimum space between type placement and feature symbol is 0.5 mm;
3. This method of type placement shall be used for areal features when space does not permit labeling within that feature.

Miscellaneous labels occasionally may be found in association with marine navigational aids. If a text label is shown on hydrographic source material, it should be considered significant for navigation. Examples are:

A fog detection light, label "Fog Det Lt"; A floodlit structure near navigable water, label "(Illuminated)";

A daytime light, if character of light in the day is different from the character shown at night. Show daytime character, followed by "Day" in parentheses, for example: (F 37m 11M Day);

Unwatched light, with no standby or emergency arrangements, label "(U)";

A temporary light or buoy, label "(temp)". If seasonal, include months, for example: "(Apr-Oct)";

A fog light, if light is only shown in fog, or the light during fog is different from the character of light shown at other times, show character in fog, followed by "(in fog)", in parentheses. For example: Fl 5s (in fog);
A privately maintained light or buoy, label "(priv)";

RACONs occasionally will show a morse code identification, or an operating frequency, for example, "Racon (Z) (10 cm)", "Racon (Z) (3&10 cm)". A RACON responding on a fixed frequency outside the marine band is labeled with an "F" in front of the label "RACON".

Feature Information					
Specification:	*LTDS*	**Feature Origin:**	*Topographic*	**Symbol:**	PT_AnnotatedLocation _Culture
Symbol Criteria:	All associated with Topographic Feature			**Label:**	[Note : Memorandum], 6-20 point Condensed Mixed case
				Color:	Black (58600)

Feature Information					
Specification:	*LTDS*	**Feature Origin:**	*Hydrographic*	**Symbol:**	PT_AnnotatedLocation _Hydro
Symbol Criteria:	All associated with Hydrographic Feature			**Label:**	[Note : Memorandum], 6-20 point Condensed Mixed case
				Color:	Blue (48253)

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Anti-shipping Activity (ZV010)

Point

Feature Description:	*An occurrence of criminal action against a ship.*
Inclusion Condition:	*ALL REQUIRED*

Notes

None

Feature Information					
Specification:	*Other*	**Feature Origin:**	*Hydrographic*	**Symbol:**	TBD
Symbol Criteria:	TBD			**Label:**	TBD
				Color:	TBD

Approach Lighting System (GB485)

Surface

Feature Description:	*An airport lighting facility which provides visual guidance to landing aircraft by radiating light beams in a directional pattern by which the pilot aligns the aircraft with the final approach path for landing.*
Inclusion Condition:	*TBD*

Notes

None

Feature Information					
Specification:	*Other*	**Feature Origin:**	*Aeronautical*	**Symbol:**	TBD
Symbol Criteria:	TBD			**Label:**	TBD
				Color:	TBD

Approach Lighting System (GB485)

Curve

Inclusion Condition:	*TBD*

Notes

None

Feature Information					
Specification:	*Other*	**Feature Origin:**	*Aeronautical*	**Symbol:**	TBD
Symbol Criteria:	TBD			**Label:**	TBD
				Color:	TBD

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Apron (GB015)

Surface

Feature Description:	*A defined area, on a land aerodrome/heliport, intended to accommodate aircraft/helicopters for purposes of loading and unloading passengers, mail or cargo, and for fuelling, parking or maintenance.*
Inclusion Condition:	*ALL REQUIRED*

Notes

Contours will be adjusted to planimetric features.

When 2 or more similar area features are separated by less than 0.5 mm at chart scale, the feature will be agglomerated.

Area and line features will be generalized to detail compatible with scale.

Feature Information

Specification:	*LTDS*	**Feature Origin:**	*Topographic*	**Symbol:**	AR_Blue072_Apron
Symbol Criteria:	All			**Label:**	None
				Color:	N/A

Aquatic Vegetation (BD061)

Surface

Feature Description:	*A region of waterborne cellular or vascular plants (for example: algae, grasses, reeds, and water hyacinths).*
Inclusion Condition:	*Length or Diameter >= 250 meters*

Notes

Point features, or individual posicuts of an area symbol, may be displaced <= 5 mm, at chart scale, to avoid overprints.

Features with areas greater than 100 square cm shall be represented by a legend, e.g., Wd, Kelp, Eddies, Overfalls, rather than by the graphic symbol. Type is 6 point, Black SPC-58600, and label is scattered over area at approximately 50 mm intervals. Position is horizontal and shall not overprint other features.

Feature Information

Specification:	*Chart #1*	**Feature Origin:**	*Hydrographic*	**Symbol:**	J13.1, J13.2
Symbol Criteria:	All			**Label:**	"Wd" if applicable, 6 point Mixed Case Italic
				Color:	Black (58600)

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Aquatic Vegetation (BD061)

Point

Inclusion Condition:	*Length or Diameter < 250 meters*

Notes

Point features, or individual posicuts of an area symbol, may be displaced <= 5 mm, at chart scale, to avoid overprints.

Feature Information

Specification:	*Chart #1*	**Feature Origin:**	*Hydrographic*	**Symbol:**	J13.2
Symbol Criteria:	All			**Label:**	None
				Color:	N/A

Aqueduct (BH010)

Surface

Feature Description:	*A pipe or artificial channel that is designed to transport water from a remote source, usually by gravity, for freshwater supply, agricultural, and/or industrial use.*
Inclusion Condition:	*Length or Diameter >= 75 m. AND Width >= 25 m.*

Notes

Label area features with upper case type within its limits and centered between the sides with a proportional size if the width of the feature will allow its inclusion. However, should the feature be too narrow, then place the type 0.5 mm above and parallel to the feature. When the feature is continuous, repeat label approximately every 30 to 40 cm for either situation, or at least two items, length permitting. In either condition (in or above the feature), curve the type necessary to the curvature of the feature. Should the feature change back and forth between the area and a line feature, the type style will change from upper case type for the area portions, to upper and lower case type for the linear portions. The repeat dimensions remain the same.

When feature continues for a long distance (>25mm), the label shall be repeated at 152 mm intervals, and is not to overprint any type or symbology.

Feature Information

Specification:	*LTDS*	**Feature Origin:**	*Topographic*	**Symbol:**	AR_Cyan_Aqueduct
Symbol Criteria:	Aqueduct Type = [Surface] AND Condition of Facility <> [Under Construction] or [Destroyed]			**Label:**	[Geographic Name Information : Full Name], 6-12 point Condensed Italics Upper case if available; "AQUEDUCT", 6-12 point Condensed Italics Upper case
				Color:	Blue (48253)

Feature Information

Specification:	*LTDS*	**Feature Origin:**	*Topographic*	**Symbol:**	AR_Cyan_Aqueduct_ UnderConstruction_Destroyed
Symbol Criteria:	Aqueduct Type = [Surface] AND Condition of Facility = [Under Construction] or [Destroyed]			**Label:**	[Geographic Name Information : Full Name], 6-12 point Condensed Italics Upper case if available; "AQUEDUCT (CONSTRUCTION)" if Condition of Facility: [Under Construction], or "AQUEDUCT (DESTROYED)" if Condition of Facility = [Destroyed], 6-12 point Condensed Italics Upper case
				Color:	Blue (48253)

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Aqueduct (BH010)

Curve

Inclusion Condition:	*Length or Diameter >= 75 m. AND Width < 25 m.*

Notes

Label area features with upper case type within its limits and centered between the sides with a proportional size if the width of the feature will allow its inclusion. However, should the feature be too narrow, then place the type 0.5 mm above and parallel to the feature. When the feature is continuous, repeat label approximately every 30 to 40 cm for either situation, or at least two items, length permitting. In either condition (in or above the feature), curve the type necessary to the curvature of the feature. Should the feature change back and forth between the area and a line feature, the type style will change from upper case type for the area portions, to upper and lower case type for the linear portions. The repeat dimensions remain the same.

When feature continues for a long distance (>25mm), the label shall be repeated at 152 mm intervals, and is not to overprint any type or symbology.

Feature Information

Specification:	*LTDS*	Feature Origin:	*Topographic*	Symbol:	LN_Cyan_Aqueduct_OnSurface
Symbol Criteria:	Aqueduct Type = [Surface]			Label:	[Geographic Name Information : Full Name] "Aqueduct", 6 point Condensed Italics Mixed case if Condition of Facility = [Abandoned] or [Fully Functional] or [Partially Functional] or [Partially Destroyed] or [Damaged] or [Disused] or [Other] or [No Information]; [Geographic Name Information : Full Name] "Aqueduct (construction)", 6 point Condensed Italics Mixed case if Condition of Facility = [Under Construction]; [Geographic Name Information : Full Name] "Aqueduct (destroyed)", 6 point Condensed Italics Mixed case if Condition of Facility = [Destroyed]
				Color:	Blue (48253)

Feature Information

Specification:	*LTDS*	Feature Origin:	*Topographic*	Symbol:	LN_Cyan_Aqueduct_ AboveSurface
Symbol Criteria:	(Relative Level = [Raised]) or (Supported by Bridge Span = [True])			Label:	[Geographic Name Information : Full Name] "Aqueduct", 6 point Condensed Italics Mixed case if Condition of Facility = [Abandoned] or [Fully Functional] or [Partially Functional] or [Partially Destroyed] or [Damaged] or [Disused] or [Other] or [No Information] ; [Geographic Name Information : Full Name] "Aqueduct (construction)", 6 point Condensed Italics Mixed case if Condition of Facility = [Under Construction]; [Geographic Name Information : Full Name] "Aqueduct (destroyed)", 6 point Condensed Italics Mixed case if Condition of Facility = [Destroyed]
				Color:	Blue (48253)

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Aqueduct (BH010) *(Continued)*

Curve

Feature Information					
Specification:	*LTDS*	**Feature Origin:**	*Topographic*	**Symbol:**	LN_Cyan_Aqueduct_ BelowGround
Symbol Criteria:	Aqueduct Type = [Underground] and Aqueduct Access: <> [Qanat Shaft]			**Label:**	[Geographic Name Information : Full Name] "Aqueduct", 6 point Condensed Italics Mixed case if Condition of Facility = [Abandoned] or [Fully Functional] or [Partially Functional] or [Partially Destroyed] or [Damaged] or [Disused] or [Other] or [No Information] ; [Geographic Name Information : Full Name] "Aqueduct (construction)", 6 point Condensed Italics Mixed case if Condition of Facility = [Under Construction]; [Geographic Name Information : Full Name] "Aqueduct (destroyed)", 6 point Condensed Italics Mixed case if Condition of Facility = [Destroyed]
				Color:	Blue (48253)

Feature Information					
Specification:	*LTDS*	**Feature Origin:**	*Topographic*	**Symbol:**	LN_Cyan_Aqueduct_ WithQanats
Symbol Criteria:	Aqueduct Type = [Underground] and Aqueduct Access = [Qanat Shaft]			**Label:**	[Geographic Name Information : Full Name], 6 point Condensed Italics Mixed case if available
				Color:	Blue (48253)

Archeological Site (AL012)

Surface

Feature Description:	*A site where remains of past civilizations or human activity have been discovered.*
Inclusion Condition:	*Area >= 15,625 square meters*

Notes

None

Feature Information					
Specification:	*LTDS*	**Feature Origin:**	*Topographic*	**Symbol:**	AR_Black_Archeological Site
Symbol Criteria:	All			**Label:**	[Geographic Name Information : Full Name] "Archeological site", 6 point Condensed Mixed case
				Color:	Black (58600)

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Archeological Site (AL012)

Point

Inclusion Condition:	*Area < 15,625 square meters*

Notes

None

Feature Information					
Specification:	*LTDS*	**Feature Origin:**	*Topographic*	**Symbol:**	PT_Black_ArcheologicalSite
Symbol Criteria:	Height above Surface < 46 meters			**Label:**	[Geographic Name Information : Full Name] "Archeological site", 6 point Condensed Mixed case
				Color:	Black (58600)

Feature Information					
Specification:	*LTDS*	**Feature Origin:**	*Topographic*	**Symbol:**	PT_Blue072_VO
Symbol Criteria:	Height above Surface >= 46 meters			**Label:**	[Geographic Name Information : Full Name] "Archeological site", 6 point Condensed Mixed case; [Highest Elevation], 6 point Condensed Bold Mixed case; "([Height Above Surface Level])", 6 point Condensed Mixed case
				Color:	Blue (46351)

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Asphalt Lake (DA005)

Surface

Feature Description:	*A natural accumulation of liquid asphalt.*
Inclusion Condition:	*Area >= 15,625 sq m.*

Notes

Coincident similar area features will be blended to form a single feature.

Area and line features will be generalized to detail compatible with scale.

Feature will be generalized to provide a more aesthetic contoured feature (i.e., smoothed).

Label feature as per hierarchy for topo type placement to south neatline corners reading left to right:
1. Position hierarchy: (a.) northeast (preferred position), (b.) southeast (1st alternative), (c.) northwest (2nd alternative, (d.) southwest (3rd alternative), (e.) top-centered (4th alternative), (f.) bottom-centered (5th alternative): (Hierarchy is based on type positioning so as to avoid overprinting other type or obscuring detail.);
2. Minimum space between type placement and feature symbol is 0.5 mm;
3. This method of type placement shall be used for areal features when space does not permit labeling within that feature.

If a clearing exists inside of an area feature, and the size of the clearing is equal to or greater than the area inclusion condition for the surrounding feature, the clearing is shown as an open space inside the surrounding feature. If the area of the clearing is less than the area inclusion condition for the surrounding feature, the clearing is deleted and absorbed into the surrounding area feature.

If two area features with the same feature code do not connect at any point, and have a space between them of less than 2.5 mm at map/chart scale, delete the open space that is less than 2.5 mm wide between the features and combine them into one area feature.

If a portion of an area feature has a minimum width of less than 2.5 mm at map/chart scale, delete that portion of the area feature that is not at least 2.5 mm wide, measured from perimeter to perimeter. If the deletion of the portion of the area based on the above criteria will reduce the area of the remaining portion of the area feature to below the minimum Area Inclusion, do not delete the narrow portion of the feature that is less than 2.5 mm wide. If the deletion of the portion of the area based on the above criteria will split two larger areas connected by a narrow strip into two separate areas, either of which would be below minimum Area inclusion, do not delete the narrow portion of the feature that is less than 2.5 mm wide.

Feature Information

Specification:	*LTDS*	**Feature Origin:**	*Topographic*	**Symbol:**	AR_Black_AsphaltLake
Symbol Criteria:	All			**Label:**	"Asphalt", 6 point Condensed Mixed case
				Color:	Black (58600)

Astronomical Observatory (AL142)

Surface

Feature Description:	*A building designed and equipped (for example: with a telescope) for making observations of celestial objects (including the earth in relation to them), of space, and of the universe as a whole.*
Inclusion Condition:	*Length or Diameter >= 25 m.*

Notes

None

Feature Information

Specification:	*LTDS*	**Feature Origin:**	*Topographic*	**Symbol:**	AR_Black_Astronomical Observatory
Symbol Criteria:	(Height Above Surface Level < 46) OR (Area >= 5625 and Height Above Surface Level >= 46)			**Label:**	"Astronomical observatory", 6 point Condensed Mixed case
				Color:	Black (58600)

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Astronomical Observatory (AL142)

Point

Inclusion Condition:	*Length or Diameter < 25 m.*

Notes

None

Feature Information					
Specification:	*LTDS*	**Feature Origin:**	*Topographic*	**Symbol:**	PT_Black_Astronomical Observatory
Symbol Criteria:	Height Above Surface Level < 46 meters			**Label:**	"Astronomical observatory", 6 point Condensed Mixed case
				Color:	Black (58600)

Feature Information					
Specification:	*LTDS*	**Feature Origin:**	*Topographic*	**Symbol:**	PT_Blue072_VO
Symbol Criteria:	Height Above Surface Level >= 46 meters			**Label:**	"Astronomical observatory", 6 point Condensed Mixed case; [Highest Elevation], 6 point Condensed Bold Mixed case; "([Height Above Surface Level])", 6 point Condensed Mixed case
				Color:	Blue (46351)

Backshore (BA022)

Surface

Feature Description:	*That part of a shore or beach which is usually dry, being reached only by the highest tides.*
Inclusion Condition:	*All Required*

Notes

None

Feature Information					
Specification:	*Other*	**Feature Origin:**	*Other*	**Symbol:**	TBD
Symbol Criteria:	TBD			**Label:**	TBD
				Color:	TBD

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Bank Vegetation Zone (EE060)

Surface

Feature Description:	*A region along the edge of a body of water or watercourse generally dominated by riparian vegetation.*
Inclusion Condition:	*All Required*

Notes

None

Feature Information

Specification:	*Chart #1*	**Feature Origin:**	*Hydrographic*	**Symbol:**	C33
Symbol Criteria:	All			**Label:**	[Bank Vegetation Character], 8 point Mixed case Italic
				Color:	Black (58600)

Basin Gate (BI045)

Curve

Feature Description:	*A gate that impounds water within a basin or chamber that is used by watercraft.*
Inclusion Condition:	*Length >= 20*

Notes

None

Feature Information

Specification:	*LTDS*	**Feature Origin:**	*Topographic*	**Symbol:**	Not symbolized
Symbol Criteria:	All			**Label:**	None
				Color:	N/A

Basin Gate (BI045)

Point

Inclusion Condition:	*Length < 20*

Notes

None

Feature Information

Specification:	*LTDS*	**Feature Origin:**	*Topographic*	**Symbol:**	Not symbolized
Symbol Criteria:	All			**Label:**	None
				Color:	N/A

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Beach (BA050)

Surface

Feature Description:	*On a shore, the area on which the waves break and over which shore debris (for example: sand, shingle, and/or pebbles) accumulate.*
Inclusion Condition:	*TBD*

Notes

None

Feature Information					
Specification:	*Other*	**Feature Origin:**	*Hydrographic*	**Symbol:**	TBD
Symbol Criteria:	TBD			**Label:**	TBD
				Color:	TBD

Beach (BA050)

Curve

Inclusion Condition:	*TBD*

Notes

None

Feature Information					
Specification:	*Other*	**Feature Origin:**	*Hydrographic*	**Symbol:**	TBD
Symbol Criteria:	TBD			**Label:**	TBD
				Color:	TBD

Beach (BA050)

Point

Inclusion Condition:	*TBD*

Notes

None

Feature Information					
Specification:	*Other*	**Feature Origin:**	*Hydrographic*	**Symbol:**	TBD
Symbol Criteria:	TBD			**Label:**	TBD
				Color:	TBD

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Beach Landing Site (BB150)

Surface

Feature Description:	*A location on a beach suitable for the landing of troops and vehicles*
Inclusion Condition:	*TBD*

Notes

None

Feature Information					
Specification:	*Other*	**Feature Origin:**	*Hydrographic*	**Symbol:**	TBD
Symbol Criteria:	TBD			**Label:**	TBD
				Color:	TBD

Beach Landing Site (BB150)

Point

Inclusion Condition:	*TBD*

Notes

None

Feature Information					
Specification:	*Other*	**Feature Origin:**	*Hydrographic*	**Symbol:**	TBD
Symbol Criteria:	TBD			**Label:**	TBD
				Color:	TBD

Beach Profile (BE050)

Curve

Feature Description:	*A representation of the three dimensional relief of the beach along a line or series of connected lines.*
Inclusion Condition:	*TBD*

Notes

None

Feature Information					
Specification:	*Other*	**Feature Origin:**	*Hydrographic*	**Symbol:**	TBD
Symbol Criteria:	TBD			**Label:**	TBD
				Color:	TBD

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Berth (BB020)

Point

Feature Description:	*A named or numbered place where a vessel is moored at a wharf.*
Inclusion Condition:	*ALL REQUIRED*

Notes

The Berth Identifier of the Berth (BB020) shall be centered in the circle.

If Berth Identifier type will not fit inside berth circle, a rectangle or leader line may be used instead. Leader line is 0.1 mm lineweight, Black solid, SPC-58600.

Feature Information

Specification:	*Chart #1*	**Feature Origin:**	*Hydrographic*	**Symbol:**	F19
Symbol Criteria:	All			**Label:**	[Berth Identifier], 6 point Upper Case
				Color:	Black (58600)

Billboard (AG050)

Point

Feature Description:	*A large outdoor board for advertisements.*
Inclusion Condition:	*ALL REQUIRED*

Notes

None

Feature Information

Specification:	*LTSD*	**Feature Origin:**	*Toprographic*	**Symbol:**	PT_Blue072_VO
Symbol Criteria:	Height above Surface Level >= 46			**Label:**	"Billboard", 6 point Condensed Mixed case; [Highest Elevation], 6 point Condensed Bold Mixed case; "([Height Above Surface Level])", 6 point Condensed Mixed case
				Color:	Blue (46351)

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Blast Barrier (GB025)

Curve

Feature Description:	*A barrier used to divert and/or dissipate jet and/or propeller blast.*
Inclusion Condition:	*TBD*

Notes

None

Feature Information					
Specification:	*Other*	**Feature Origin:**	*Aeronautical*	**Symbol:**	TBD
Symbol Criteria:	TBD			**Label:**	TBD
				Color:	TBD

Blast Barrier (GB025)

Point

Inclusion Condition:	*TBD*

Notes

None

Feature Information					
Specification:	*Other*	**Feature Origin:**	*Aeronautical*	**Symbol:**	TBD
Symbol Criteria:	TBD			**Label:**	TBD
				Color:	TBD

Blast-furnace (AC010)

Point

Feature Description:	*A smelting furnace in which a blast of air is used, especially one for iron-smelting using a compressed hot air blast.*
Inclusion Condition:	*Height Above Surface Level: >= 46 m.*

Notes

None

Feature Information					
Specification:	*LTDS*	**Feature Origin:**	*Topographic*	**Symbol:**	PT_Blue072_VO
Symbol Criteria:	Height Above Surface Level: >= 46 m.			**Label:**	"Blast-furnace", 6 point Condensed Mixed case; [Highest Elevation], 6 point Condensed Bold Mixed Case; "([Height Above Surface Level])", 6 point Condensed Mixed case
				Color:	Blue (46351)

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Boat Landing (BB149)

Surface

Feature Description:	*A place where boats receive or discharge passengers and freight, but lacking most port facilities.*
Inclusion Condition:	*Length or Diameter >= 100 meters and Width >= 20 meters*

Notes

Label shall be placed on land, 1 mm from shoreline and actual location of boat landing (BB149), orient label perpendicular to shoreline.

When the area symbol or cased line symbol overprints the shoreline, shoreline is deleted.

Show land tint inside of symbol.

Feature Information

Specification:	*Chart #1*	**Feature Origin:**	*Hydrographic*	**Symbol:**	F17
Symbol Criteria:	All			**Label:**	"Lndg", 6 point Mixed Case, Italic
				Color:	Black (58600)

Boat Landing (BB149)

Point

Inclusion Condition:	*Length or Diameter < 100 meters*

Notes

Label shall be placed on land, 1 mm from shoreline and actual location of boat landing (BB149), orient label perpendicular to shoreline.

Feature Information

Specification:	*Chart #1*	**Feature Origin:**	*Hydrographic*	**Symbol:**	Not Symbolized
Symbol Criteria:	All			**Label:**	"Lndg", 6 point Mixed Case, Italic
				Color:	Black (58600)

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Bog (BH015)

Surface

Feature Description:	*A permanently wet area of land consisting of incompletely decayed organic material and mainly stagnant fresh water.*
Inclusion Condition:	*Area >= 15,625 sq m.*

Notes

Coincident similar area features will be blended to form a single feature.

Area and line features will be generalized to detail compatible with scale.

Feature will be generalized to provide a more aesthetic contoured feature (i.e., smoothed).

Where area measurements are inconsistent, the larger type size shall be used. Where the full range of type sizes is not available for a particular label, the closest available type size shall be used.

Label feature as per hierarchy for topo type placement to south neatline corners reading left to right:
1. Position hierarchy: (a.) northeast (preferred position), (b.) southeast (1st alternative), (c.) northwest (2nd alternative, (d.) southwest (3rd alternative), (e.) top-centered (4th alternative), (f.) bottom-centered (5th alternative): (Hierarchy is based on type positioning so as to avoid overprinting other type or obscuring detail.);
2. Minimum space between type placement and feature symbol is 0.5 mm; 3. This method of type placement shall be used for areal features when space does not permit labeling within that feature.

Names placement shall be oriented to the longest axis of the feature reading left to right and placed within the area outline and centered. If the longest axis is perpendicular to the south neatline, the type shall be placed outside of the area outline, preferred position is northeast of the feature, but may be placed at any position around the feature so as not to overprint any other feature type and reading left to right.

Symbols and associated area patterns of underpassing features (except drainage shorelines) are broken for all bridges, except footbridges. This rule does not apply to land tint on Littoral Planning Charts.

If a clearing exists inside of the area feature, and the size of the clearing is equal to or greater than the area for the surrounding area feature, the clearing is shown as an open space inside the surround feature. If the area of the clearing is less than the area for the surrounding feature, the clearing is deleted and absorbed into the surrounding area feature.
If two area features with the same feature code do not connect at any point, and have a space between them less than 2.5 mm at map/chart scale, delete the open space that is less than 2.5 mm between the features and combine them into one area feature.

If a portion of an area feature has a minimum width of less than 2.5 mm at map/chart scale, delete that portion of the area feature that is not at least 2.5 mm wide, measured from perimeter to perimeter. If the deletion of the portion of the area based on the above criteria will reduce the area of the remaining portion of the area feature to below the minimum area inclusion, do not delete the narrow portion of the feature that is less than 2.5 mm wide. If the deletion of the portion of the area based on the above criteria will split two larger areas connected by a narrow strip into two separate areas, either of which would be below minimum area inclusion, do not delete the narrow portion of the feature that is less than 2.5 mm wide.

Feature Information

Specification:	*LTDS*	**Feature Origin:**	*Topographic*	**Symbol:**	AR_Cyan_Bog
Symbol Criteria:	Bog Type: (Cranberry Bog) or (Peat Bog)			**Label:**	[Bog Type], 6 point Condensed Italics Mixed case
				Color:	Blue (48253)

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Bollard (BB030)

Point

Feature Description:	*A small shaped post, mounted on the ground or pavement surface.*
Inclusion Condition:	*ALL REQUIRED*

Notes

None

Feature Information					
Specification:	*Chart #1*	**Feature Origin:**	*Hydrographic*	**Symbol:**	F22
Symbol Criteria:	All			**Label:**	"Bollard", 6 point Mixed case
				Color:	Black (58600)

Booming Ground (BD076)

Surface

Feature Description:	*An area in a shallow portion of a body of water, usually at the mouth of a river, where logs are collected, staged, and formed into log booms for further transportation and delivery to a logging mill.*
Inclusion Condition:	*ALL Required*

Notes

None

Feature Information					
Specification:	*Chart #1*	**Feature Origin:**	*Hydrographic*	**Symbol:**	F29.1
Symbol Criteria:	All			**Label:**	Geographic Name Information : Full Name], 6 Point Mixed case, Italic; [Note : Memorandum], 6 Point Mixed case, Italic
				Color:	Black (58600)

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Bottom Characteristic Region (BF010)

Surface

Feature Description:	*A region of a waterbody bottom that is homogeneous with respect to a measurement (for example: consistency, color, and/or composition).*
Inclusion Condition:	*Area >= 15,625 square meters*

Notes

Abbreviations for Bottom Characteristics are:
For the Maritime Bottom Characteristic (first material type) and Maritime Bottom Characteristic (second material type): Ash (Ash), Boulders (Bo), Chalk (Ck), Cinders (Cn), Cirripedia (Cir), Clay (Cy), Cobble (Cb), Coral (Co), Coral Head (Co Hd), Diatoms (Di), Foraminifera (Fr), Fucus (Fu), Globigerina (Gl), Grass (Grs), Gravel (G), Ground (Gd), Lava (Lv), Madrepores (Md), Manganese (Mn), Marl (Ml), Mattes (Ma), Mud (M), Mussels (Ms), Ooze (Oz), Oysters (Oy), Pebbles (P), Polyzoa (Po), Pteropods (Pt), Pumice (Pm), Quartz (Qz), Radiolaria (Rd), Rock (R), Sand (S), Schist (Sch), Scoria (Sc), Sea Tangle (Stg), Seaweed (Wd), Shells (Sh), Shingles (Sn), Silt (Si), Spicules (Spi), Sponge (Sp), Stones (St), Tufa (T).

For characteristics of the Maritime Bottom Characteristic (first material quality) and Maritime Bottom Characteristic (second material quality) : broken (bk), calcareous (ca), coarse (c), decayed (dec), fine (f), flinty (fly), glacial (ga), gritty (gty), ground (grd), hard (h), large (l), rocky (rky), rotten (rt), small (sm), soft (so), speckled (spk), sticky (sy), stiff (sf), streaky (str), tenacious (ten), uneven (unev), varied (vard), volcanic (v), medium (m).

The mobile bottom sand wave symbol should be used primarily in close association with the most significant soundings, usually the shallowest sounding in each area of mobile bottom/sand waves area. The use of the sandwave symbol draws attention to the most significant depths, and also indicates the degree of unreliability of the figure chart.

When frequently repeated surveys show some variation in least depth soundings within areas of sandwaves, the shallowest one found over a period of years should be charted. This blending of details of surveys from different dates must be carried out with care; In particular, long term deepening over time must not be overlooked.

The extent of mobile bottom/sandwave areas, if known and considered navigationally significant, may be indicated by the legend "Sandwaves". The legend should be placed over areas where the depths may be critical to surface navigation, and used in conjunction with the sandwave symbol associated with the most significant soundings. Type style for the legend is 6 point Mixed case italic. Color is Black SPC-58600 solid.

Areas of sandwave/mobile bottom shown on the chart are further explained by the following Caution, shown in the margin. See Notes and Cautions section of product specifications.

Caution: Sandwaves build up during particular states of weather and tide. Surveys may not have been made in those conditions, so the chart may not show the minimum depth possible. The Caution title will be centered and stacked over the note.

Where the secondary material is known to differ from the first material layer, the predominant material [First Material Quality and First Material Type] and the symbol of the underlayer [Second Material Quality and Second Material Type] shall be written in that order, on one line, separated by a slash "/". If Second Material Quality or Second Material Type is unknown, delete the slash.

Where mixtures of materials occur, the predominant material [First Material Quality and First Material Type] shall be shown first, followed by the secondary material [Second Material Quality and Second Material Type], on one line, separated by a space. If no secondary material is present in the mixture, do not show [Second Material Quality and Second Material Type. If a third characteristic/material is present in the mixture, this is shown by the [Note: Memorandum] label, using the standard abbreviations above; otherwise [Note : Memorandum] is not shown.

In water deeper than 100 meters, only show [First Material Type] of bottom characteristics. Bottom characteristics shall be shown, if known, an all shoal and in anchorage areas. Elsewhere they shall be selected to show variations in the composition of the seabed. In uniform areas, bottom characteristics shall be shown at approximately 50 mm interval, if data is available.

Feature Information

Specification:	*Chart #1*	**Feature Origin:**	*Hydrographic*	**Symbol:**	J12.1, J12.2
Symbol Criteria:	[Maritime Bottom Characteristic (first material quality)] or [Maritime Bottom Characteristic (second material quality)]: <> (Mobile Bottom) OR Maritime Bottom Characteristic (first material quality) or [Maritime Bottom Characteristic (second material quality)]: (Other) and [Note : Memorandum]: <> "Springs in Seabed"			**Label:**	[Maritime Bottom Characteristic (first material quality)] If available, 8 point Lower case Italic; [Maritime Bottom Characteristic (first material type)], 8 point Mixed case, Italic; [Maritime Bottom Characteristic (second material quality)] if available, 8 point Lower case, Italic; [Maritime Bottom Characteristic (second material type)] if available, 8 point Mixed case, Italic; [Note : Memorandum] if available, 8 point Mixed case Italic
				Color:	Black (58600)

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Bottom Characteristic Region (BF010)

Point

Inclusion Condition:	*Area < 15,625 square meters*

Notes

Abbreviations for Bottom Characteristics are:
For the Maritime Bottom Characteristic (first material type) and Maritime Bottom Characteristic (second material type): Ash (Ash), Boulders (Bo), Chalk (Ck), Cinders (Cn), Cirripedia (Cir), Clay (Cy), Cobble (Cb), Coral (Co), Coral Head (Co Hd), Diatoms (Di), Foraminifera (Fr), Fucus (Fu), Globigerina (Gl), Grass (Grs), Gravel (G), Ground (Gd), Lava (Lv), Madrepores (Md), Manganese (Mn), Marl (Ml), Mattes (Ma), Mud (M), Mussels (Ms), Ooze (Oz), Oysters (Oy), Pebbles (P), Polyzoa (Po), Pteropods (Pt), Pumice (Pm), Quartz (Qz), Radiolaria (Rd), Rock (R), Sand (S), Schist (Sch), Scoria (Sc), Sea Tangle (Stg), Seaweed (Wd), Shells (Sh), Shingles (Sn), Silt (Si), Spicules (Spi), Sponge (Sp), Stones (St), Tufa (T).

For characteristics of the Maritime Bottom Characteristic (first material quality) and Maritime Bottom Characteristic (second material quality) : broken (bk), calcareous (ca), coarse (c), decayed (dec), fine (f), flinty (fly), glacial (ga), gritty (gty), ground (grd), hard (h), large (l), rocky (rky), rotten (rt), small (sm), soft (so), speckled (spk), sticky (sy), stiff (sf), streaky (str), tenacious (ten), uneven (unev), varied (vard), volcanic (v), medium (m).

The mobile bottom sand wave symbol should be used primarily in close association with the most significant soundings, usually the shallowest sounding in each area of mobile bottom/sand waves area. The use of the sandwave symbol draws attention to the most significant depths, and also indicates the degree of unreliability of the figure chart.

When frequently repeated surveys show some variation in least depth soundings within areas of sandwaves, the shallowest one found over a period of years should be charted. This blending of details of surveys from different dates must be carried out with care; In particular, long term deepening over time must not be overlooked.

The extent of mobile bottom/sandwave areas, if known and considered navigationally significant, may be indicated by the legend "Sandwaves". The legend should be placed over areas where the depths may be critical to surface navigation, and used in conjunction with the sandwave symbol associated with the most significant soundings. Type style for the legend is 6 point Mixed case italic. Color is Black SPC-58600 solid.

Areas of sandwave/mobile bottom shown on the chart are further explained by the following Caution, shown in the margin. See Notes and Cautions section of product specifications.

Caution: Sandwaves build up during particular states of weather and tide. Surveys may not have been made in those conditions, so the chart may not show the minimum depth possible. The Caution title will be centered and stacked over the note.

Where the secondary material is known to differ from the first material layer, the predominant material [First Material Quality and First Material Type] and the symbol of the underlayer [Second Material Quality and Second Material Type] shall be written in that order, on one line, separated by a slash "/". If Second Material Quality or Second Material Type is unknown, delete the slash.

Where mixtures of materials occur, the predominant material [First Material Quality and First Material Type] shall be shown first, followed by the secondary material [Second Material Quality and Second Material Type], on one line, separated by a space. If no secondary material is present in the mixture, do not show [Second Material Quality and Second Material Type. If a third characteristic/material is present in the mixture, this is shown by the [Note: Memorandum] label, using the standard abbreviations above; otherwise [Note : Memorandum] is not shown.

In water deeper than 100 meters, only show [First Material Type] of bottom characteristics. Bottom characteristics shall be shown, if known, an all shoal and in anchorage areas. Elsewhere they shall be selected to show variations in the composition of the seabed. In uniform areas, bottom characteristics shall be shown at approximately 50 mm interval, if data is available.

Feature Information

Specification:	*Chart #1*	Feature Origin:	*Hydrographic*	Symbol:	J12.1, J12.2
Symbol Criteria:	[Maritime Bottom Characteristic (first material quality)] or [Maritime Bottom Characteristic (second material quality)]: <> (Mobile Bottom) OR Maritime Bottom Characteristic (first material quality) or [Maritime Bottom Characteristic (second material quality)]: (Other) and [Note : Memorandum]: <> "Springs in Seabed"			Label:	[Maritime Bottom Characteristic (first material quality)] If available, 8 point Lower case Italic; [Maritime Bottom Characteristic (first material type)], 8 point Mixed case, Italic; [Maritime Bottom Characteristic (second material quality)] if available, 8 point Lower case, Italic; [Maritime Bottom Characteristic (second material type)] if available, 8 point Mixed case, Italic; [Note : Memorandum] if available, 8 point Mixed case Italic
				Color:	Black (58600)

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Bottom Characteristic Region (BF010) *(Continued)*

Point

Feature Information					
Specification:	*Chart #1*	**Feature Origin:**	*Hydrographic*	**Symbol:**	J14
Symbol Criteria:	[Maritime Bottom Characteristic (first material quality)] or [Maritime Bottom Characteristic (second material quality)]: (Mobile Bottom)			**Label:**	None
				Color:	N/A

Feature Information					
Specification:	*Chart #1*	**Feature Origin:**	*Hydrographic*	**Symbol:**	J15
Symbol Criteria:	Maritime Bottom Characteristic (first material quality) or [Maritime Bottom Characteristic (second material quality)]: (Other) and [Note : Memorandum]: <> "Springs in Seabed"			**Label:**	None
				Color:	N/A

Boundary Monument (ZB030)

Point

Feature Description:	*A marker identifying the location of a surveyed boundary line.*
Inclusion Condition:	*ALL REQUIRED*

Notes

Label feature as per hierarchy for topo type placement to south neatline corners reading left to right:
1. Position hierarchy: (a.) northeast (preferred position), (b.) southeast (1st alternative), (c.) northwest (2nd alternative, (d.) southwest (3rd alternative), (e.) top-centered (4th alternative), (f.) bottom-centered (5th alternative): (Hierarchy is based on type positioning so as to avoid overprinting other type or obscuring detail.);
2. Minimum space between type placement and feature symbol is 0.5 mm;
3. This method of type placement shall be used for areal features when space does not permit labeling within that feature.

Feature Information					
Specification:	*LTDS*	**Feature Origin:**	*Topographic*	**Symbol:**	PT_Black_BoundaryMonument
Symbol Criteria:	All			**Label:**	[Geographic Name Information : Full Name], 6 point Condensed Mixed case
				Color:	Black (58600)

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Bridge (AQ040)

Curve

Feature Description:	*A structure that connects two locations and provides for the passage of a transportation route (for example: a road or a railway) over a terrain obstacle (for example: a waterbody, a gully, and/or a road).*
Inclusion Condition:	*Length or Diameter >= 75 m.*

Notes

The sides of a linear bridge which is stacked under a road shall have sides of the bridges abutted up against the sides of the road.

Label feature as per hierarchy for topo type placement to south neatline corners reading left to right:
1. Position hierarchy:
(a.) northeast (preferred position),
(b.) southeast (1st alternative),
(c.) northwest (2nd alternative,
(d.) southwest (3rd alternative),
(e.) top-centered (4th alternative),
(f.) bottom-centered (5th alternative)
(Hierarchy is based on type positioning so as to avoid overprinting other type or obscuring detail.);
2. Minimum space between type placement and feature symbol is 0.5 mm;
3. This method of type placement shall be used for areal features when space does not permit labeling within that feature.

If a bridge feature satisfies vertical obstruction criteria, then symbolize the bridge, and overprint with obstruction symbol (PT_Blue072_VO) and label.

Show at least a 0.50 mm symbol overlap on shore for each terminus (end).

Symbols and associated area patterns of underpassing features (except drainage shorelines) are broken for all bridges, except footbridges. This rule does not apply to land tint on Littoral Planning Charts.

Show land tint inside of symbol.

The Vertical Clearance, Safe and the associated bars shall be positioned reading horizontal, in the following priority: (a.) In the water, alongside the feature, without overprinting other hydrographic detail. (b.) If necessary to avoid overprints, label may be placed on land, adjacent to the feature.

The Safe Horizontal Clearance figure, if shown, shall be placed next to the Safe Overhead Clearance figure, with the number reading horizontal, i.e., the Safe Horizontal Clearance bars vertical. If placed in the water, it shall not overprint other hydrographic detail.

The datum above which Vertical Clearance, Safe are given shall be a high water datum, preferably Mean High Water Springs where the tide is appreciable. If the tide is not appreciable, i.e., less than 0.3 meter, Mean Sea Level may be used. Vertical Clearance, Safe shall be rounded down to the nearest whole meter, unless under 10 meters, where it is rounded down to the nearest meter and decimeter value.

Vertical Clearance, Safe is the sounding datum to which a hydrographic depth, drying height, or the high water datum to which a Vertical Clearance, Safe is referenced. If the hydrographic depth, drying height, or Vertical Clearance, Safe is unknown or not applicable, the Vertical Clearance, Safe is not applicable.

Feature Information

Specification:	*LTDS*	**Feature Origin:**	*Topographic*	**Symbol:**	LN_Black_Bridge_DrawLift
Symbol Criteria:	Bridge Opening Type = [Bascule] or [Swing-bridge] or [Lift-bridge] or [Retractable] or [Drawbridge] or [Opening] and Transportation System Type [1] <> [Pedestrian] and (Length or Diameter >= 75 m.)			**Label:**	[Geographic Name Information : Full Name], 6 point Regular Mixed case if available and [Bridge Opening Type] if Bridge Opening Type = [Bascule] or [Swing-bridge] or [Lift-bridge] or [Retractable] or [Drawbridge], 6 point Regular Mixed case and "(open [Maximum Vertical Clearance])" if Bridge Opening Type = [Lift-bridge], 6 point Regular Mixed case and [Underbridge Clearance] if available , 6 point Regular Mixed case and "(construction)", 6 point Regular Lower case if Condition of Facility = [Under Construction], and "(destroyed)" if Condition of Facility = [Destroyed] or [Ruined]
				Color:	Black (58600)

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Bridge (AQ040) *(Continued)*

Curve

Feature Information					
Specification:	*LTDS*	**Feature Origin:**	*Topographic*	**Symbol:**	LN_Black_Bridge
Symbol Criteria:	Bridge Opening Type = [Fixed] or [Submersible] or [Other] or [No Information] and Transportation System Type [1] <> [Pedestrian] and (Length or Diameter >= 75)			**Label:**	[Geographic Name Information : Full Name], 6 point Regular Mixed case if available and [Underbridge Clearance] if available , 6 point Regular Mixed case and "Pontoon Bridge" if Bridge Structure Type = [Floating], 6 point Regular Mixed case and "Submersible bridge, [Vertical Clearance, Safe]m below datum when lowered" if Bridge Opening Type = [Submersible], 6 point Regular Mixed case and "(construction)", 6 point Regular Lower case if Condition of Facility = [Under Construction] and "(destroyed)" if Condition of Facility = [Destroyed] or [Ruined]
				Color:	Black (58600)

Feature Information					
Specification:	*LTDS*	**Feature Origin:**	*Topographic*	**Symbol:**	LN_Black_Bridge_Pedestrian
Symbol Criteria:	Transportation System Type [1] = [Pedestrian] and (Length or Diameter >= 75 m.)			**Label:**	None
				Color:	N/A

Bridge (AQ040)

Point

Inclusion Condition:	*Length or Diameter < 75 m.*

Notes

A point bridge that is stack under a road shall have the sides of the bridge abutted up against the sides of the road, and the bridge oriented so that the bridge is aligned with the road.

The supporting feature shall be aligned with a Cart Track, Trail, RR Track, and RR Siding/RR Spur.

Label feature as per hierarchy for topo type placement to south neatline corners reading left to right:
1. Position hierarchy:
(a.) northeast (preferred position),
(b.) southeast (1st alternative),
(c.) northwest (2nd alternative,
(d.) southwest (3rd alternative),
(e.) top-centered (4th alternative),
(f.) bottom-centered (5th alternative)
(Hierarchy is based on type positioning so as to avoid overprinting other type or obscuring detail.);
2. Minimum space between type placement and feature symbol is 0.5 mm;
3. This method of type placement shall be used for areal features when space does not permit labeling within that feature.

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Bridge (AQ040) *(Continued)*

Point

Feature Information					
Specification:	*LTDS*	**Feature Origin:**	*Topographic*	**Symbol:**	PT_Black_Bridge
Symbol Criteria:	Transportation System Type [1] <> [Pedestrian] and Length or Diameter < 75 m.			**Label:**	[Geographic Name Information : Full Name], 6 point Regular Mixed case if available and "(construction)", 6 point Regular Lower case if Condition of Facility = [Under Construction], "(destroyed)" if Condition of Facility = [Destroyed] or [Ruined]
				Color:	Black (58600)

Feature Information					
Specification:	*LTDS*	**Feature Origin:**	*Topographic*	**Symbol:**	PT_Black_Bridge_Pedestrian
Symbol Criteria:	Transportation System Type [1] = [Pedestrian] and Length or Diameter < 75 m.			**Label:**	None
				Color:	N/A

Feature Information					
Specification:	*LTDS*	**Feature Origin:**	*Topographic*	**Symbol:**	PT_Black_Bridge_DrawLift
Symbol Criteria:	Bridge Opening Type = [Bascule] or [Swing-bridge] or [Lift-bridge] or [Retractable] or [Drawbridge] or [Opening] and Transportation System Type [1] <> [Pedestrian] and (Length or Diameter < 75 m.)			**Label:**	[Geographic Name Information : Full Name], 6 point Regular Mixed case if available and "(construction)", 6 point Regular Lower case if Condition of Facility = [Under Construction], "(destroyed)" if Condition of Facility = [Destroyed] or [Ruined]
				Color:	Black (58600)

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Bridge Span (AQ045)

Curve

Feature Description:	*A component of the deck of a bridge spanning successive bridge piers.*
Inclusion Condition:	*ALL REQUIRED*

Notes

None

Feature Information					
Specification:	*LTDS*	**Feature Origin:**	*Topographic*	**Symbol:**	No Symbology needed
Symbol Criteria:	All			**Label:**	None
				Color:	N/A

Bridge Superstructure (AQ050)

Point

Feature Description:	*A superstructure of a bridge, above the lowest deck, not including pylons or towers.*
Inclusion Condition:	*Height Above Surface Level >= 46 m.*

Notes

None

Feature Information					
Specification:	*LTDS*	**Feature Origin:**	*Topographic*	**Symbol:**	PT_Blue072_LinearObstruction
Symbol Criteria:	Height Above Surface Level >= 46 m.			**Label:**	"Bridge superstructure", 6 point Condensed Mixed case; [Highest Elevation], 6 point Condensed Bold Mixed case; "([Height Above Surface Level])", 6 point Condensed Mixed case
				Color:	Blue (48253)

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Bridge Tower (AQ055)

Point

Feature Description:	*A tower and/or pylon from which the deck of a bridge is suspended.*
Inclusion Condition:	*Height Above Surface Level >= 46 m.*

Notes

None

Feature Information					
Specification:	*LTDS*	**Feature Origin:**	*Topographic*	**Symbol:**	PT_Blue072_VO
Symbol Criteria:	Height Above Surface Level >= 46 m.			**Label:**	"Bridge tower", 6 point Condensed Mixed case; [Highest Elevation], 6 point Condensed Bold Mixed case; "([Height Above Surface Level])", 6 point Condensed Mixed case
				Color:	Blue (48253)

Brush (EB070)

Surface

Feature Description:	*A tract covered mainly by short, uncultured, woody plants.*
Inclusion Condition:	*Area >= 15,625 sq m.*

Notes

Coincident similar area features will be blended to form a single feature.

Area and line features will be generalized to detail compatible with scale.

Feature will be generalized to provide a more aesthetic contoured feature (i.e., smoothed).

Symbols and associated area patterns of underpassing features (except drainage shorelines) are broken for all bridges, except footbridges. This rule does not apply to land tint on Littoral Planning Charts.

If a clearing exists inside of the area feature, and the size of the clearing is equal to or greater than the area for the surrounding area feature, the clearing is shown as an open space inside the surround feature. If the area of the clearing is less than the area for the surrounding feature, the clearing is deleted and absorbed into the surrounding area feature.

If two area features with the same feature code do not connect at any point, and have a space between them less than 2.5 mm at map/chart scale, delete the open space that is less than 2.5 mm between the features and combine them into one area feature.

If a portion of an area feature has a minimum width of less than 2.5 mm at map/chart scale, delete that portion of the area feature that is not at least 2.5 mm wide, measured from perimeter to perimeter. If the deletion of the portion of the area based on the above criteria will reduce the area of the remaining portion of the area feature to below the minimum area inclusion, do not delete the narrow portion of the feature that is less than 2.5 mm wide. If the deletion of the portion of the area based on the above criteria will split two larger areas connected by a narrow strip into two separate areas, either of which would be below minimum area inclusion, do not delete the narrow portion of the feature that is less than 2.5 mm wide.

Feature Information					
Specification:	*LTDS*	**Feature Origin:**	*Topographic*	**Symbol:**	AR_Green362_Brush
Symbol Criteria:	All			**Label:**	None
				Color:	N/A

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Bucket Elevator (AF021)

Point

Feature Description:	*Equipment, usually a series of buckets, for raising materials or liquids to a higher level.*
Inclusion Condition:	*Height Above Surface Level >= 46 m.*

Notes

None

Feature Information

Specification:	*LTDS*	**Feature Origin:**	*Topographic*	**Symbol:**	PT_Blue072_VO
Symbol Criteria:	Height Above Surface Level: >= 46 meters			**Label:**	"Bucket elevator", 6 point Condensed Mixed case; [Highest Elevation], 6 point Condensed Bold Mixed case; "([Height Above Surface Level])": 6 point Condensed Mixed case
				Color:	Blue (46351)

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Building (AL013)

Surface

Feature Description:	*A free-standing self-supporting construction that is roofed, usually walled, and is intended for human occupancy (for example: a place of work or recreation) and/or habitation.*
Inclusion Condition:	*Length or Diameter >= 25 m.*

Notes

If feature coalesce at map scale, when shown in their true positions, they shall be displaced 0.2 mm from one another.

When symbolized feature is < 0.2 mm from a line feature, displace to 0.2 mm (map scale).

Area and line features will be generalized to detail compatible with scale.

Type for features on land shall be positioned on land if possible. If type must be placed in the water, it shall be positioned so it does not obscure or overprint hydrographic detail. Type for features in the water shall be positioned in the water. If possible, type shall not be positioned across the shoreline.

Passenger terminal shall not be labeled, unless they are identified with a proper name.

Passenger terminal in Built-up Areas shall not be named if the name is the same as the populated place name.

If Building Function (Other), identify building's function using Text Description.

The generic part of a name is not shown when the building has a posicut identification (i.e., ST PATRICKS CATHEDRAL is shortened to ST. PATRICKS).

If Height Above Surface Level >= 46 meters, then depict as an obstruction symbol.

If coalescing features being thinned are a mix of heights, with some < 46 meters and some >= 46 meters, then only the obstruction symbol shall be shown.

Omit within Built-up Area AL020 unless it is a landmark or Height Above Surface Level >= 46 meters.

When obstructions coalesce at map scale, label with highest obstruction information.

Building symbols with distinguishing characteristic attached shall show the staff of the symbol at right angles to the Road. If the symbol coalesces with another symbol, the staff shall be repositioned to an unobstructed side of the building.

Spacing between Building symbols shall not be less than 0.2 mm.

The Building symbol shall be shown in its true position if a space of >=0.2 mm (map scale) exists between the Building and Road symbols on map.

A space of not less than 0.2 mm shall be shown between Building symbols and Tracks and Trails.

Symbolize apron/hardstands and buildings inside areal Aircraft Facilities (Airport), or (Seaplane)

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Building (AL013) *(Continued)*

Surface

Feature Information					
Specification:	*LTDS*	**Feature Origin:**	*Topographic*	**Symbol:**	AR_Black_Building_Important
Symbol Criteria:	Feature Function: ([Forest Warden] or [Railway Transport] or [Railway Passenger Transport] or [Customs Checkpoint] or [Inspection Station] or [Administration] or [Government] or [Local Government] or [Subnational Government] or [Palace] or [Capitol] or [Civil Activities] or [Maritime Defense] or [Defence Activities] or [Armory] or [Military Reserve Activities] or [Judicial Activities] or [Law Enforcement] or [Immigration Control] or [Imprisonment] or [Juvenile Corrections] or [Firefighting] or [Rescue and Paramedical Services] or [Opera House] or [Museum]) and (Condition of Facility: [Under Construction] or [Abandoned] or [Dismantled] or [Fully Functional] or [Partially Functional] or [Partially Destroyed] or [Damaged] or [Disused] or [Planned] or [Other] or [No Information]) and (Height Above Surface Level < 46)			**Label:**	[Geographic Name Information : Full Name]if available, 6 point Mixed Case Condensed; or "Ranger" if (Feature Function: [Forest Warden]); "Station" if (Feature Function: [Railway Transport] or [Railway Passenger Transport]); "Checkpoint" if (Feature Function: [Customs Checkpoint] or [Inspection Station] or [Immigration Control]); "Government" if (Feature Function: [Administration] or [Government] or [Local Government] or [Subnational Government] or [Palace] or [Capitol] or [Civil Activities]); "Military" if (Feature Function: [Maritime Defense] or [Defence Activities] or [Armory] or [Military Reserve Activities]); "Courthouse"if (Feature Function: [Judicial Activities]); "Police" if (Feature Function: [Law Enforcement]); "Prison" if (Feature Function: [Imprisonment] or [Juvenile Corrections]); "Fire/Rescue" if (Feature Function: [Firefighting] or [Rescue and Paramedical Services]); "Opera House" if (Feature Function: [Opera House]); "Museum" if (Feature Function: [Museum]), 6 point Condensed Mixed case
				Color:	Black (58600)

Feature Information					
Specification:	*LTDS*	**Feature Origin:**	*Topographic*	**Symbol:**	AR_Black_Building_Generic
Symbol Criteria:	Feature Function: <> ([Forest Warden] or [Railway Transport] or [Railway Passenger Transport] or [Customs Checkpoint] or [Inspection Station] or [Administration] or [Government] or [Local Government] or [Subnational Government] or [Palace] or [Capitol] or [Civil Activities] or [Maritime Defense] or [Defence Activities] or [Armory] or [Military Reserve Activities] or [Judicial Activities] or [Law Enforcement] or [Immigration Control] or [Imprisonment] or [Juvenile Corrections] or [Firefighting] or [Rescue and Paramedical Services] or [Opera House] or [Museum]) and (Condition of Facility: [Under Construction] or [Abandoned] or [Dismantled] or [Fully Functional] or [Partially Functional] or [Partially Destroyed] or [Damaged] or [Disused] or [Planned] or [Other] or [No Information]) and (Height Above Surface Level < 46)			**Label:**	None
				Color:	N/A

Feature Information					
Specification:	*LTDS*	**Feature Origin:**	*Topographic*	**Symbol:**	AR_Black_Building_Destroyed
Symbol Criteria:	(Condition of Facility: [Destroyed] or [Ruined])			**Label:**	None
				Color:	N/A

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Building (AL013) *(Continued)*

Surface

Feature Information					
Specification:	*LTDS*	**Feature Origin:**	*Topographic*	**Symbol:**	AR_Black_Building_Diplomatic
Symbol Criteria:	(Feature Function: [Diplomacy] or [Diplomatic Mission] or [Embassy] or [Consul]) and (Condition of Facility: [Under Construction] or [Abandoned] or [Dismantled] or [Fully Functional] or [Partially Functional] or [Partially Destroyed] or [Damaged] or [Disused] or [Other] or [No Information]) and (Height Above Surface Level < 46)			**Label:**	[Geographic Name Information: Full Name], 6 point Condensed Mixed case if available
				Color:	Black (58600)

Feature Information					
Specification:	*LTDS*	**Feature Origin:**	*Topographic*	**Symbol:**	AR_Black_Building_Hospital
Symbol Criteria:	(Area >= 5625) and (Condition of Facility = [Under Construction] or [Abandoned] or [Dismantled] or [Fully Functional] or [Partially Functional] or [Partially Destroyed] or [Damaged] or [Disused] or [Other] or [No Information]) and (Feature Function = [In-patient Care] or [Urgent Medical Care]) and (Height Above Surface Level < 46)			**Label:**	[Geographic Name Information: Full Name], 6 point Condensed Mixed case if available
				Color:	Black (58600)

Feature Information					
Specification:	*LTDS*	**Feature Origin:**	*Topographic*	**Symbol:**	AR_Black_School
Symbol Criteria:	((Feature Function: [Education] or [Primary Education] or [Secondary Education] or [Higher Education] or [Vocational Education] or [Institution]) or (Religious Information : Religious Facility Type: [Seminary]) and (Condition of Facility: [Under Construction] or [Abandoned] or [Dismantled] or [Fully Functional] or [Partially Functional] or [Partially Destroyed] or [Damaged] or [Disused] or [Other] or [No Information]) and (Height Above Surface Level < 46)			**Label:**	[Geographic Name Information: Full Name], 6 point Condensed Mixed case if available
				Color:	Black (58600)

Feature Information					
Specification:	*LTDS*	**Feature Origin:**	*Topographic*	**Symbol:**	AR_Black_Mosque
Symbol Criteria:	((Feature Function: [Religious Activities] or [Place of Worship] or [Islamic Prayer Hall]) and (Religious Information : Religious Facility Type: [Marabout] or [Minaret] or [Mosque]) or Religious Information : Religious Designation: [Islam] or [Shia] or [Sunni])) or (Feature Function: [Islamic Prayer Hall]) and (Condition of Facility: [Under Construction] or [Abandoned] or [Dismantled] or [Fully Functional] or [Partially Functional] or [Partially Destroyed] or [Damaged] or [Disused] or [Other] or [No Information]) and (Height Above Surface Level < 46)			**Label:**	[Geographic Name Information: Full Name], 6 point Condensed Mixed case if available
				Color:	Black (58600)

Feature Information					
Specification:	*LTDS*	**Feature Origin:**	*Topographic*	**Symbol:**	AR_Black_Synagogue
Symbol Criteria:	(Feature Function: [Religious Activities] or [Place of Worship]) and (Religious Information : Religious Facility Type: [Synagogue]) or Religious Information : Religious Designation: [Judaism] and (Condition of Facility: [Under Construction] or [Abandoned] or [Dismantled] or [Fully Functional] or [Partially Functional] or [Partially Destroyed] or [Damaged] or [Disused] or [Other] or [No Information]) and (Height Above Surface Level < 46)			**Label:**	[Geographic Name Information: Full Name], 6 point Condensed Mixed case if available
				Color:	Black (58600)

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Building (AL013) *(Continued)*

Surface

Feature Information					
Specification:	*LTDS*	**Feature Origin:**	*Topographic*	**Symbol:**	AR_Black_Church
Symbol Criteria:	(Feature Function: [Religious Activities] or [Place of Worship]) and (Religious Information : Religious Facility Type: [Cathedral] or [Chapel] or [Church] or [Religious Community] or [Shrine] or [Tabernacle]) or Religious Information : Religious Designation: [Christian] or [Orthodox] or [Protestant] or [Nestorian] or [Chaldean] and (Condition of Facility: [Under Construction] or [Abandoned] or [Dismantled] or [Fully Functional] or [Partially Functional] or [Partially Destroyed] or [Damaged] or [Disused] or [Other] or [No Information]) and (Height Above Surface Level < 46)			**Label:**	[Geographic Name Information: Full Name], 6 point Condensed Mixed case if available
				Color:	Black (58600)

Feature Information					
Specification:	*LTDS*	**Feature Origin:**	*Topographic*	**Symbol:**	AR_Black_Pagoda
Symbol Criteria:	(Feature Function: [Religious Activities] or [Place of Worship]) and (Religious Information : Religious Facility Type: [Pagoda]) or Religious Information : Religious Designation: [Hinduism] and (Condition of Facility: [Under Construction] or [Abandoned] or [Dismantled] or [Fully Functional] or [Partially Functional] or [Partially Destroyed] or [Damaged] or [Disused] or [Other] or [No Information]) and (Height Above Surface Level < 46)			**Label:**	[Geographic Name Information: Full Name], 6 point Condensed Mixed case if available
				Color:	Black (58600)

Feature Information					
Specification:	*LTDS*	**Feature Origin:**	*Topographic*	**Symbol:**	AR_Black_Temple
Symbol Criteria:	(Feature Function: [Religious Activities] or [Place of Worship]) and (Religious Information : Religious Facility Type: [Temple]) and Religious Information : Religious Designation: [Buddhism] or [Shinto] or [Hinduism] and (Condition of Facility: [Under Construction] or [Abandoned] or [Dismantled] or [Fully Functional] or [Partially Functional] or [Partially Destroyed] or [Damaged] or [Disused] or [Other] or [No Information]) and (Height Above Surface Level < 46)			**Label:**	[Geographic Name Information: Full Name], 6 point Condensed Mixed case if available
				Color:	Black (58600)

Feature Information					
Specification:	*LTDS*	**Feature Origin:**	*Topographic*	**Symbol:**	AR_Black_Stupa
Symbol Criteria:	(Feature Function: [Religious Activities] or [Place of Worship]) and (Religious Information : Religious Facility Type: [Stupa]) or Religious Information : Religious Designation: [Buddhism] and (Condition of Facility: [Under Construction] or [Abandoned] or [Dismantled] or [Fully Functional] or [Partially Functional] or [Partially Destroyed] or [Damaged] or [Disused] or [Other] or [No Information]) and (Height Above Surface Level < 46)			**Label:**	[Geographic Name Information: Full Name], 6 point Condensed Mixed case if available
				Color:	Black (58600)

Feature Information					
Specification:	*LTDS*	**Feature Origin:**	*Topographic*	**Symbol:**	AR_Black_Building_HouseOf Worship_Generic
Symbol Criteria:	(Feature Function: [Religious Activities] or [Place of Worship]) and (Religious Information : Religious Facility Type: [Other] or [No Information]) and (Religious Information : Religious Designation: [Other] or [No Information]) and (Height Above Surface Level < 46)			**Label:**	"House of Worship", 6 point Mixed Case Condensed
				Color:	Black (58600)

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Building (AL013) *(Continued)*

Surface

Feature Information					
Specification:	*LTDS*	**Feature Origin:**	*Topographic*	**Symbol:**	AR_Black_Building_HealthOffice
Symbol Criteria:	(Area >= 5625) and (Condition of Facility = [Under Construction] or [Abandoned] or [Dismantled] or [Fully Functional] or [Partially Functional] or [Partially Destroyed] or [Damaged] or [Disused] or [Other] or [No Information]) and (Feature Function [1] = [Human Health Activities] or [Out-patient Care] or [Public Health Activities] or [Leprosy Care] or [Intermediate Care] or [Psychiatric In-patient Care] or [Residential Care]) and (Height Above Surface Level < 46)			**Label:**	None
				Color:	N/A

Feature Information					
Specification:	*LTDS*	**Feature Origin:**	*Topographic*	**Symbol:**	PT_Blue072_VO
Symbol Criteria:	Height Above Surface Level: >= 46 meters AND Area: < 5,625 Sq m			**Label:**	"Building", 6 point Condensed Mixed case; [Highest Elevation], 6 point Condensed Bold Mixed case; "([Height Above Surface Level])", 6 point Condensed Mixed case
				Color:	Blue (46351)

Feature Information					
Specification:	*LTDS*	**Feature Origin:**	*Topographic*	**Symbol:**	AR_Black_Building_LargeArea VO
Symbol Criteria:	Height Above Surface Level >= 46 meters AND Area >= 5,625 Sq m			**Label:**	[Highest Elevation], 6 point Condensed Bold Mixed case; "([Height Above Surface Level])", 6 point Condensed Mixed case
				Color:	Blue (46351)

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Building (AL013)

Point

Inclusion Condition:	*Length or Diameter < 25 m.*

Notes

The feature (when Height Above Surface Level <= 46 meters or when height is not known) shall be oriented perpendicular (90 degrees) to the nearby road (AP030), cart track (AP010), trail (AP050), or railway (AN010).

If feature coalesce at map scale, when shown in their true positions, they shall be displaced 0.2 mm from one another.

When symbolized feature is < 0.2 mm from a line feature, displace to 0.2 mm (map scale).

Like point features which coalesce in clusters of 3 or more will be thinned to form a representative pattern.

Type for features on land shall be positioned on land if possible. If type must be placed in the water, it shall be positioned so it does not obscure or overprint hydrographic detail. Type for features in the water shall be positioned in the water. If possible, type shall not be positioned across the shoreline.

Passenger terminal shall not be labeled, unless they are identified with a proper name.

Passenger terminal in Built-up Areas shall not be named if the name is the same as the populated place name.

If Building Function (Other), identify building's function using Text Description.

The generic part of a name is not shown when the building has a posicut identification (i.e., ST PATRICKS CATHEDRAL is shortened to ST. PATRICKS).

Descriptive terms, e.g., "Canal" shall be shown if the name is not known. If the descriptive word appears in the name, for example, "PANAMA CANAL". The descriptive type shall not be shown, i.e. do not show "Panama Canal Canal".

If the existence is Doubtful or Reported, label as such.

If coalescing features being thinned are a mix of heights, with some < 46 meters and some >= 46 meters, then only the obstruction symbol shall be shown.

Omit within Built-up Area AL020 unless it is a landmark or Height Above Surface Level >= 46 meters.

When obstructions coalesce at map scale, label with highest obstruction information.

Building symbols with distinguishing characteristic attached shall show the staff of the symbol at right angles to the Road. If the symbol coalesces with another symbol, the staff shall be repositioned to an unobstructed side of the building.

Spacing between Building symbols shall not be less than 0.2 mm.

The Building symbol shall be shown in its true position if a space of >=0.2 mm (map scale) exists between the Building and Road symbols on map.

A space of not less than 0.2 mm shall be shown between Building symbols and Tracks and Trails.

Symbolize apron/hardstands and buildings inside areal Aircraft Facilities (Airport), or (Seaplane)

Buildings (AL013), single or occurring in rows or clusters, shall be shown in their true orientation except when falling <= 0.2 mm of the following linear features: Road AP030, Railway (AN010), Cart Track (AP010), Trail (AP050), Aqueduct (BH010), Canal (BH020), and Ditch (BH030). In these cases, Buildings (AL013 point), single (or occurring in rows or clusters with <= 0.2 mm separation between buildings) shall be collectively oriented parallel to those linear features at a distance of 0.2 mm.

Feature Information

Specification:	*LTDS*	Feature Origin:	*Topographic*	Symbol:	PT_Black_Building_Generic
Symbol Criteria:	(Feature Function: <> [Forest Warden] or [Administration] or [Government] or [Local Government] or [Subnational Government] or [Palace] or [Capitol] or [Civil Activities] or [Maritime Defense] or [Defence Activities] or [Armory] or [Military Reserve Activities] or [Judicial Activities] or [Law Enforcement] or [Imprisonment] or [Juvenile Corrections] or [Firefighting] or [Rescue and Paramedical Services] or [Opera House] or [Museum]) and (Condition of Facility: [Under Construction] or [Abandoned] or [Dismantled] or [Fully Functional] or [Partially Functional] or [Partially Destroyed] or [Damaged] or [Disused] or [Other] or [No Information]) and (Height Above Surface Level < 46)			Label:	None
				Color:	N/A

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Building (AL013) *(Continued)*

Point

Feature Information					
Specification:	*LTDS*	**Feature Origin:**	*Topographic*	**Symbol:**	PT_Black_Building_Important_Non-landmark
Symbol Criteria:	(Feature Function: [Forest Warden] or [Administration] or [Government] or [Local Government] or [Subnational Government] or [Palace] or [Capitol] or [Civil Activities] or [Maritime Defense] or [Defence Activities] or [Armory] or [Military Reserve Activities] or [Judicial Activities] or [Law Enforcement] or [Imprisonment] or [Juvenile Corrections] or [Firefighting] or [Rescue and Paramedical Services] or [Opera House] or [Museum]) and (Condition of Facility: [Under Construction] or [Abandoned] or [Dismantled] or [Fully Functional] or [Partially Functional] or [Partially Destroyed] or [Damaged] or [Disused] or [Other] or [No Information]) and (Navigation Landmark: [False]) and (Height Above Surface Level < 46)			**Label:**	None
				Color:	N/A

Feature Information					
Specification:	*LTDS*	**Feature Origin:**	*Topographic*	**Symbol:**	PT_Black_Building_Important_Landmark
Symbol Criteria:	(Feature Function: [Forest Warden] or [Administration] or [Government] or [Local Government] or [Subnational Government] or [Palace] or [Capitol] or [Civil Activities] or [Maritime Defense] or [Defence Activities] or [Armory] or [Military Reserve Activities] or [Judicial Activities] or [Law Enforcement] or [Imprisonment] or [Juvenile Corrections] or [Firefighting] or [Rescue and Paramedical Services] or [Opera House] or [Museum]) and (Condition of Facility: [Under Construction] or [Abandoned] or [Dismantled] or [Fully Functional] or [Partially Functional] or [Partially Destroyed] or [Damaged] or [Disused] or [Other] or [No Information]) and (Navigation Landmark: [True]) and (Height Above Surface Level < 46)			**Label:**	[Geographic Name Information : Full Name] if available; or "Ranger" if (Feature Function: [Forest Warden]); "Government" if (Feature Function: [Administration] or [Government] or [Local Government] or [Subnational Government] or [Palace] or [Capitol] or [Civil Activities]); "Military" if (Feature Function: [Maritime Defense] or [Defence Activities] or [Armory] or [Military Reserve Activities]); "Courthouse" if (Feature Function: [Judicial Activities]); "Police" if (Feature Function: [Law Enforcement]); "Prison" if (Feature Function: [Imprisonment] or [Juvenile Corrections]); "Fire/Rescue" if (Feature Function: [Firefighting] or [Rescue and Paramedical Services]); "Opera House" if (Feature Function: [Opera House]); "Museum" if (Feature Function: [Museum]) , 6 Point Mixed case Condensed
				Color:	Black (58600)

Feature Information					
Specification:	*LTDS*	**Feature Origin:**	*Topographic*	**Symbol:**	PT_Blue072_VO
Symbol Criteria:	Height Above Surface Level: >= 46 meters			**Label:**	"Building", 6 point Condensed Mixed case; [Highest Elevation], 6 point Condensed Bold Mixed case; "([Height Above Surface Level])", 6 point Condensed Mixed case
				Color:	Blue (46351)

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Building (AL013) *(Continued)*

Point

Feature Information					
Specification:	*LTDS*	**Feature Origin:**	*Topographic*	**Symbol:**	PT_Black_Building_Diplomatic
Symbol Criteria:	(Feature Function: [Diplomacy] or [Diplomatic Mission] or [Embassy] or [Consul]) and (Condition of Facility: [Under Construction] or [Abandoned] or [Dismantled] or [Fully Functional] or [Partially Functional] or [Partially Destroyed] or [Damaged] or [Disused] or [Other] or [No Information]) and (Height Above Surface Level < 46)			**Label:**	[Geographic Name Information: Full Name], 6 point Condensed Mixed case if available
				Color:	Black (58600)

Feature Information					
Specification:	*LTDS*	**Feature Origin:**	*Topographic*	**Symbol:**	PT_Black_Hospital
Symbol Criteria:	(Area < 5625) and (Condition of Facility = [Under Construction] or [Abandoned] or [Dismantled] or [Fully Functional] or [Partially Functional] or [Partially Destroyed] or [Damaged] or [Disused] or [Other] or [No Information]) and (Feature Function = [In-patient Care] or [Urgent Medical Care]) and (Height Above Surface Level < 46)			**Label:**	[Geographic Name Information: Full Name], 6 point Condensed Mixed case if available
				Color:	Black (58600)

Feature Information					
Specification:	*LTDS*	**Feature Origin:**	*Topographic*	**Symbol:**	PT_Black_School
Symbol Criteria:	((Feature Function: [Education] or [Primary Education] or [Secondary Education] or [Higher Education] or [Vocational Education] or [Institution]) or (Religious Information : Religious Facility Type: [Seminary]) and (Condition of Facility: [Under Construction] or [Abandoned] or [Dismantled] or [Fully Functional] or [Partially Functional] or [Partially Destroyed] or [Damaged] or [Disused] or [Other] or [No information]) and (Height Above Surface Level < 46)			**Label:**	[Geographic Name Information: Full Name], 6 point Condensed Mixed case if available
				Color:	Black (58600)

Feature Information					
Specification:	*LTDS*	**Feature Origin:**	*Topographic*	**Symbol:**	PT_Black_Customs_Checkpoint
Symbol Criteria:	(Feature Function: [Customs Checkpoint] or [Inspection Station] or [Immigration Control]) and (Condition of Facility: [Under Construction] or [Abandoned] or [Dismantled] or [Fully Functional] or [Partially Functional] or [Partially Destroyed] or [Damaged] or [Disused] or [Other] or [No Information]) and (Height Above Surface Level < 46)			**Label:**	[Geographic Name Information: Full Name], 6 point Condensed Mixed case if available
				Color:	Black (58600)

Feature Information					
Specification:	*LTDS*	**Feature Origin:**	*Topographic*	**Symbol:**	PT_Black_Health_Office
Symbol Criteria:	(Area < 5625) and (Condition of Facility = [Under Construction] or [Abandoned] or [Dismantled] or [Fully Functional] or [Partially Functional] or [Partially Destroyed] or [Damaged] or [Disused] or [Other] or [No Information]) and (Feature Function [1] = [Human Health Activities] or [Out-patient Care] or [Public Health Activities] or [Leprosy Care] or [Intermediate Care] or [Psychiatric In-patient Care] or [Residential Care]) and (Height Above Surface Level < 46)			**Label:**	None
				Color:	N/A

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Building (AL013) *(Continued)*

Point

Feature Information					
Specification:	*LTDS*	**Feature Origin:**	*Topographic*	**Symbol:**	PT_Black_TransportationStation
Symbol Criteria:	(Feature Function: [Railway Transport] or [Railway Passenger Transport]) and (Condition of Facility: [Under Construction] or [Abandoned] or [Dismantled] or [Fully Functional] or [Partially Functional] or [Partially Destroyed] or [Damaged] or [Disused] or [Other] or [No Information]) and (Height Above Surface Level < 46)			**Label:**	[Geographic Name Information: Full Name], 6 point Condensed Mixed case if available; [Geographic Name Information: Full Name] "(construction)" if Condition of Facility: (Under Construction), 6 point Condensed Mixed case ; [Geographic Name Information: Full Name] "(not usable)" if Condition of Facility: (Abandoned), 6 point Condensed Mixed case
				Color:	Black (58600)

Feature Information					
Specification:	*LTDS*	**Feature Origin:**	*Topographic*	**Symbol:**	PT_Black_Building_Landmark _Destroyed
Symbol Criteria:	(Condition of Facility: [Destroyed] or [Ruined]) and (Navigation Landmark: [True])			**Label:**	None
				Color:	N/A

Feature Information					
Specification:	*LTDS*	**Feature Origin:**	*Topographic*	**Symbol:**	PT_Black_Marabout_Muslim Shrine
Symbol Criteria:	(Feature Function: [Religious Activities] or [Place of Worship]) and (Religious Information : Religious Facility Type: [Marabout] or (Religious Information : Religious Facility Type: [Shrine] and Religious Information : Religious Designation: [Islam] or [Shia] or [Sunni]) and (Condition of Facility: [Under Construction] or [Abandoned] or [Dismantled] or [Fully Functional] or [Partially Functional] or [Partially Destroyed] or [Damaged] or [Disused] or [Other] or [No Information]) and (Height Above Surface Level < 46)			**Label:**	[Geographic Name Information: Full Name], 6 point Condensed Mixed case if available
				Color:	Black (58600)

Feature Information					
Specification:	*LTDS*	**Feature Origin:**	*Topographic*	**Symbol:**	PT_Black_Synagogue
Symbol Criteria:	(Feature Function: [Religious Activities] or [Place of Worship]) and (Religious Information : Religious Facility Type: [Synagogue] or Religious Information : Religious Designation: [Judaism]) and (Condition of Facility: [Under Construction] or [Abandoned] or [Dismantled] or [Fully Functional] or [Partially Functional] or [Partially Destroyed] or [Damaged] or [Disused]or [Other] or [No Information]) and (Height Above Surface Level < 46)			**Label:**	[Geographic Name Information: Full Name], 6 point Condensed Mixed case if available
				Color:	Black (58600)

Feature Information					
Specification:	*LTDS*	**Feature Origin:**	*Topographic*	**Symbol:**	PT_Black_Church
Symbol Criteria:	(Feature Function: [Religious Activities] or [Place of Worship]) and (Religious Information : Religious Facility Type: [Cathedral] or [Chapel] or [Church] or [Religious Community] or [Tabernacle]) or (Religious Information : Religious Designation: [Roman Catholic] or [Christian] or [Orthodox] or [Protestant] or [Nestorian] or [Chaldean]) and (Condition of Facility: [Under Construction] or [Abandoned] or [Dismantled] or [Fully Functional] or [Partially Functional] or [Partially Destroyed] or [Damaged] or [Disused] or [Other] or [No Information]) and (Height Above Surface Level < 46)			**Label:**	[Geographic Name Information: Full Name], 6 point Condensed Mixed case if available
				Color:	Black (58600)

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Building (AL013) *(Continued)*

Point

Feature Information					
Specification:	*LTDS*	**Feature Origin:**	*Topographic*	**Symbol:**	PT_Black_Pagoda
Symbol Criteria:	(Feature Function: [Religious Activities] or [Place of Worship]) and (Religious Information : Religious Facility Type: [Pagoda]) or (Religious Information : Religious Designation: [Hinduism]) and (Condition of Facility: [Under Construction] or [Abandoned] or [Dismantled] or [Fully Functional] or [Partially Functional] or [Partially Destroyed] or [Damaged] or [Disused] or [Other] or [No Information]) and (Height Above Surface Level < 46)			**Label:**	[Geographic Name Information: Full Name], 6 point Condensed Mixed case if available
				Color:	Black (58600)

Feature Information					
Specification:	*LTDS*	**Feature Origin:**	*Topographic*	**Symbol:**	PT_Black_Temple
Symbol Criteria:	(Feature Function: [Religious Activities] or [Place of Worship]) and (Religious Information : Religious Facility Type: [Temple]) or (Religious Information : Religious Designation: [Buddhism] or [Shinto] or [Hinduism]) and (Condition of Facility: [Under Construction] or [Abandoned] or [Dismantled] or [Fully Functional] or [Partially Functional] or [Partially Destroyed] or [Damaged] or [Disused] or [Other] or [No Information]) and (Height Above Surface Level < 46)			**Label:**	[Geographic Name Information: Full Name], 6 point Condensed Mixed case if available
				Color:	Black (58600)

Feature Information					
Specification:	*LTDS*	**Feature Origin:**	*Topographic*	**Symbol:**	PT_Black_Stupa
Symbol Criteria:	(Feature Function: [Religious Activities] or [Place of Worship]) and (Religious Information : Religious Facility Type: [Stupa]) or (Religious Information : Religious Designation: [Buddhism]) and (Condition of Facility: [Under Construction] or [Abandoned] or [Dismantled] or [Fully Functional] or [Partially Functional] or [Partially Destroyed] or [Damaged] or [Disused] or [Other] or [No Information]) and (Height Above Surface Level < 46)			**Label:**	[Geographic Name Information: Full Name], 6 point Condensed Mixed case if available
				Color:	Black (58600)

Feature Information					
Specification:	*LTDS*	**Feature Origin:**	*Topographic*	**Symbol:**	PT_Black_Minaret
Symbol Criteria:	(Feature Function: [Religious Activities] or [Place of Worship]) and (Religious Information : Religious Facility Type: [Minaret]) and (Condition of Facility: [Under Construction] or [Abandoned] or [Dismantled] or [Fully Functional] or [Partially Functional] or [Partially Destroyed] or [Damaged] or [Disused] or [Other] or [No Information]) and (Height Above Surface Level < 46)			**Label:**	[Geographic Name Information: Full Name], 6 point Condensed Mixed case if available
				Color:	Black (58600)

Feature Information					
Specification:	*LTDS*	**Feature Origin:**	*Topographic*	**Symbol:**	PT_Black_Mosque
Symbol Criteria:	(Feature Function: [Religious Activities] or [Place of Worship] or [Islamic Prayer Hall]) and (Religious Information : Religious Facility Type: [Mosque]) or (Religious Information : Religious Designation: [Islam] or [Shia] or [Sunni]) and (Condition of Facility: [Under Construction] or [Abandoned] or [Dismantled] or [Fully Functional] or [Partially Functional] or [Partially Destroyed] or [Damaged] or [Disused] or [Other] or [No Information]) and (Height Above Surface Level < 46)			**Label:**	[Geographic Name Information: Full Name], 6 point Condensed Mixed case if available
				Color:	Black (58600)

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Building (AL013) *(Continued)*

Point

Feature Information					
Specification:	*LTDS*	**Feature Origin:**	*Topographic*	**Symbol:**	PT_Black_Building_HouseOf Worship_Generic
Symbol Criteria:	(Feature Function: [Religious Activities] or [Place of Worship]) and (Religious Information : Religious Facility Type: [Other] or [No Information] and Religious Information : Religious Designation: [Other] or [No Information] and (Height Above Surface Level < 46)			**Label:**	"House of Worship", 6 point Mixed Case Condensed
				Color:	Black (58600)

Feature Information					
Specification:	*LTDS*	**Feature Origin:**	*Topographic*	**Symbol:**	PT_Black_Shrine_Generic
Symbol Criteria:	(Feature Function: [Religious Activities] or [Place of Worship] and Religious Information : Religious Facility Type: [Shrine] and Religious Information : Religious Designation: [Buddhism] or [Judaism] or [Shinto] or [Hinduism] or [Other] or [No Information]) and (Condition of Facility: [Under Construction] or [Abandoned] or [Dismantled] or [Fully Functional] or [Partially Functional] or [Partially Destroyed] or [Damaged] or [Disused] or [Other] or [No Information]) and (Height Above Surface Level < 46)			**Label:**	[Geographic Name Information: Full Name], 6 point Condensed Mixed case if available; “Shrine", 6 point Condensed Mixed case
				Color:	Black (58600)

Building Superstructure (AL018)

Surface

Feature Description:	*A supplemental portion of a building which rises from the roof but is not considered to be a portion of the roof.*
Inclusion Condition:	*Length or Diameter >= 25 meters AND Height Above Surface Level >= 46 meters*

Notes

None

Feature Information					
Specification:	*LTDS*	**Feature Origin:**	*Topographic*	**Symbol:**	PT_Blue072_VO
Symbol Criteria:	Height Above Surface Level >= 46 meters			**Label:**	"([Building Superstructure Type])", 6 point Condensed Mixed case; [Highest Elevation], 6 point Condensed Bold Mixed case; "([Height Above Surface Level])", 6 point Condensed Mixed case
				Color:	Blue (46351)

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Building Superstructure (AL018)

Point

Inclusion Condition:	*Length or Diameter < 25 meters And Height Above Surface Level >= 46 meters*

Notes

A Feature "Landmark (False)" shall not displace a Feature "Landmark (True)"

Priority for type placement: 1-right center, 2-bottom center, 3-left center, 4-top center.
(A) Minimum distance from symbol - 1 mm.
(B) Maximum distance from symbol before choosing the next highest priority:
(#1) 4 mm measured to the West end;
(#2) 4 mm measured to the North side (top);
(#3) 4 mm measured to the East end;
(#4) 4 mm measured to the South side (bottom)

Feature Information

Specification:	*LTDS*	Feature Origin:	*Topographic*	Symbol:	PT_Blue072_VO
Symbol Criteria:	Height Above Surface Level >= 46 meters			Label:	"([Building Superstructure Type])", 6 point Condensed Mixed case; [Highest Elevation], 6 point Condensed Bold Mixed case; "([Height Above Surface Level])", 6 point Condensed Mixed case
				Color:	Blue (46351)

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Built-up Area (AL020)

Surface

Feature Description:	*A tract containing a concentration of buildings and/or other structures.*
Inclusion Condition:	*Area >= 15,625 sq m.*

Notes

Variable Type Specification:
1st Class: 14 point Condensed Bold Upper case
2nd Class: 10 point Condensed Bold Upper case
3rd Class: 10 point Condensed Bold Mixed case
4th Class: 10 point Condensed Mixed case
5th Class: 8 point Condensed Mixed case

When 2 or more similar area features are separated by less than 0.5 mm at chart scale, the feature will be agglomerated.

Coincident similar area features will be blended to form a single feature.

Area and line features will be generalized to detail compatible with scale.

Name label shall be positioned 0.5 mm from respective side of feature symbol so that wording may read from left to right except perpendicular wording which shall read bottom to top (east side) of feature.

When a Wall symbol coalesces with Built-Up Area outline, or Shantytown outline, omit Built-Up Area outline or Shantytown outline, and show Wall with Built-Up Area tint only.

Where Wall is around a populated place that is not symbolized as Built-Up Area or Shantytown, the Wall symbol shall be omitted but "(Walled)" will be labeled in parentheses below the place name when place name is known.

The Built-Up Area tint shall be cleared from all through Routes and streets.

The limiting outline of the Built-up Area tint shall be dropped when it overprints linear features (Streams, Roads, and Railroads, etc), or if the space between the symbols is < 0.5mm.

Areal features (parks, railways, factory complexes, port facilities, fabrication complexes, hospital complexes, cemeteries, and other similar complexes) within the Built-Up Area tint shall be void of the built-up tint if >= 2.5 mm X 2.5 mm.

If a clearing exists inside of the area feature, and the size of the clearing is equal to or greater than the area for the surrounding area feature, the clearing is shown as an open space inside the surround feature. If the area of the clearing is less than the area for the surrounding feature, the clearing is deleted and absorbed into the surrounding area feature.

If two area features with the same feature code do not connect at any point, and have a space between them less than 2.5 mm at map/chart scale, delete the open space that is less than 2.5 mm between the features and combine them into one area feature.

If a portion of an area feature has a minimum width of less than 2.5 mm at map/chart scale, delete that portion of the area feature that is not at least 2.5 mm wide, measured from perimeter to perimeter. If the deletion of the portion of the area based on the above criteria will reduce the area of the remaining portion of the area feature to below the minimum area inclusion, do not delete the narrow portion of the feature that is less than 2.5 mm wide. If the deletion of the portion of the area based on the above criteria will split two larger areas connected by a narrow strip into two separate areas, either of which would be below minimum area inclusion, do not delete the narrow portion of the feature that is less than 2.5 mm wide.

Feature Information

Specification:	*LTDS*	**Feature Origin:**	*Topographic*	**Symbol:**	AR_Dk-Brown1815-42_BUA_Dense
Symbol Criteria:	(Built-up Area Density Category: [Dense]) and (Condition of Facility: <> [Destroyed] or [Ruined])			**Label:**	[Geographic Name Information: Full Name], 8-14 point Condensed Bold Mixed case if available
				Color:	Black (58600)

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Built-up Area (AL020) *(Continued)*

Surface

Feature Information					
Specification:	*LTDS*	**Feature Origin:**	*Topographic*	**Symbol:**	AR_Dk-Brown1815-12_BUA_Sparse
Symbol Criteria:	(Built-up Area Density Category: [Sparse] or [Moderate]) and (Condition of Facility: <> [Destroyed] or [Ruined])			**Label:**	[Geographic Name Information: Full Name], 8-14 point Condensed Bold Mixed case if available
				Color:	Black (58600)

Feature Information					
Specification:	*LTDS*	**Feature Origin:**	*Topographic*	**Symbol:**	AR_Dk-Brown1815_BUA_Destroyed
Symbol Criteria:	(Condition of Facility: [Destroyed] or [Ruined])			**Label:**	[Geographic Name Information: Full Name], 8 point Condensed Mixed case if available
				Color:	Black (58600)

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Buoy (BC020)

Point

Feature Description:	*A floating object moored to the bottom in a particular place, as an aid to navigation or for other specific purposes.*
Inclusion Condition:	*All Required*

Notes

Buoys and additional co-located Maritime Features:

Buoys may contain additional Maritime features that co-located as the same coordinate. These additional features can be Maritime Navigation Light (BC040), Fog Signal (BC101), Sound Signal Device (BC103), and Maritime Radio Beacon (BC034). These other Maritime features in conjunction with the Buoy feature will provide all of the necessary components of a Buoy that will be located at the same coordinate, but collected as separate features. Refer to rules pertaining to those additional features.

Cardinal Displacement Rules for hazards:
If an IALA cardinal buoy must be displaced off of a point hydrographic danger, it shall be displaced the minimum distance required to resolve the overprint of the central danger symbol (excluding danger circle and type), in the safe direction of the IALA cardinal buoy. The following are IALA cardinal buoys:

(1.) If (Buoy Type: [North Cardinal], buoy is displaced north;

(2.) If (Buoy Type: [East Cardinal], buoy is displaced east;

(3.) If (Buoy Type: [South Cardinal], buoy is displaced south;

(4.) If (Buoy Type: [West Cardinal], buoy is displaced west;

Displacement Rules for NAVAIDS overprinting hydrographic dangers or depth curves:

When aids to navigation overprint hydrographic dangers, or depth curves, the following displacement criteria applies:

(a.) If the aid to navigation is a fixed aid (Maritime Navigation Beacon and Maritime Navigation Light), and the hydrographic danger symbol is a point symbol, delete the hydrographic danger point symbol. Show both if the hydrographic danger symbol is an area symbol.

(b.) If the aid to navigation is a buoy, the hydrographic danger symbol's danger curve, i.e., the dotted perimeter line, if present, is broken for the buoy symbol (excluding type).

(c.) If a point danger symbol and a buoy are overprinting, refer to Cardinal Displacement Rules for hazards and displacement shall be as little as possible to resolve the overprint.

(d.) The above criteria refer to the graphic portion of symbols only. All type and labels are movable around the graphic elements to avoid overprints. If the hydrographic danger symbol's hydrographic depth type must be moved, it shall be placed outside the danger circle, and enclosed in parentheses.

(e.) Depth curves are broken for aids to navigation and dangers, and the type associated with these symbols.

Lateral Marks Guidance: (Chart #1 Q130.1)

Lateral marks may exhibit red or green lights of any rhythm (but not fixed), including Long Flashing lights. A composite group flashing red or green light (eg Fl(2+1)R) is exhibited only from modified lateral (preferred channel) buoys.

IALA Region A:
- *Port-hand marks are red with cylindrical topmarks (if any). Lights are red and have any rhythm except Fl(2+1)R*
- *Preferred channel to port Buoy will show Maritime Navigation Navigation Lights as Fl(2+1)G, if lit*
- *Starboard-hand marks are green with conical topmarks (if any). Lights are green and have any rhythm except Fl(2+1)G*
- *Preferred channel to starboard Buoy will show Maritime Navigation Navigation Lights as Fl(2+1)R, if lit*
- *Port-hand Edge Buoys (Buoy Type:[Port-hand Edge]) may be shown Can and Pillar and will be colored red with cylinder Topmarks.*
- *Starboard-hand Edge Buoys (Buoy Type:[Starboard-hand Edge]) may be shown Conical and Pillar and will be colored green with cone Topmarks.*

IALA Region B:
- *Port-hand marks are green with cylindrical topmarks (if any). Lights are green and have any rhythm except Fl(2+1)G*
- *Preferred channel to port will show Maritime Navigation Navigation Lights as Fl(2+1)R, if lit*
- *Starboard-hand marks are red with conical topmarks (if any). Lights are red and have any rhythm except Fl(2+1)R*
- *Preferred channel to starboard will show Maritime Navigation Navigation Lights as Fl(2+1)G, if lit*
- *Port-hand Edge Buoys (Buoy Type:[Port-hand Edge]) may be shown Can and Pillar and will be colored green with cylinder Topmarks.*
- *Starboard-hand Edge Buoys (Buoy Type:[Starboard-hand Edge]) may be shown Conical and Pillar and will be colored red with cone Topmarks.*

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Buoy (BC020) *(Continued)*

Point

Notes

Feature name/label shall be positioned parallel to lines of latitude and readable left to right.

Yellow, abbreviated "Y", shall be substituted for Orange "Or" or Amber "Am" when describing light color in the character of light.

If the only light color is white, omit color from light. E.g., "Fl 6s 12m 8M" is a white light.

The name of the buoy shall be shown in quotes (I.e., "Heron").

The Period of Light label for a Maritime Navigation Light that is associated with a buoy may be omitted if that buoy is shown on a chart at a larger scale.

In areas of congested type, if it is necessary to abridge a Maritime Navigation Light legend due to clutter, the period may be omitted.

A topmark as depicted in Chart #1 Q9 must be in the same orientation as the symbol to which it is attached, ie, a buoy topmark must slope at the same angle as the rest of the buoy and a beacon topmark must normally be upright.

With the lower end of the topmark touching the feature, use the following IALA Maritime Buoyage System - Topmarks guidance.

IALA Maritime Buoyage System - Topmarks guidance

Lateral marks may have a single can (cylindrical) topmark (Topmark Shape: [Cylinder]) on the port hand and a single conical topmark (point up) (Topmark Shape: [Cone Point Upward]) on the starboard hand, coloured red or green as appropriate for Region A and Region B. Can and conical buoys indicate by their shape which is the correct side to pass, so these buoys may not be fitted with topmarks. If the buoy does not have a distinctive shape, a topmark will usually be fitted. Region A: red to port, green to starboard, Region B: green to port, red to starboard.

Cardinal marks have two black cone topmarks, one above the other. There must be a clear separation between each cone; in particular, two cones base to base must not be shown as a diamond shape.

North Cardinal - (Buoy Type: [North Cardinal]) will use Topmark Shape: [2 Cones Points Upward]

South Cardinal - (Buoy Type: [South Cardinal]) will use Topmark Shape: [2 Cones Points Downward]

East Cardinal - (Buoy Type: [East Cardinal]) will use Topmark Shape: [2 Cones Base to Base]

West Cardinal - (Buoy Type: [West Cardinal]) will use Topmark Shape: [2 Cones Point to Point]

The point of the cone reflects the position of the black band(s) on the body of the buoy (or beacon), eg for a West cardinal mark, the black band is in the middle.

Isolated danger marks, which indicate the location of an isolated danger of limited size that is completely surrounded by navigable water, have two black sphere topmarks, one above the other (Topmark Shape: [2 Spheres One over Other].

Safe water marks, as used for centre-lines of channels or as landfall marks, may have a single red sphere topmark (Topmark Shape: [Sphere]. Spherical buoys indicate by their shape that there is navigable water all around them, so such buoys may not be fitted with topmarks.

Special marks, not primarily intended to assist navigation but indicating a special area or feature, may have a single yellow diagonal 'X' shaped topmark (St Andrew's cross) (Topmark Shape: [X-shaped].

Special marks may also be used to mark Traffic Separation Schemes, or channels within channels (eg a Deep Water route within a wider navigation channel marked by standard lateral buoys) or special purpose channels (eg for small craft).

Emergency wreck marking buoys, intended for the temporary marking of a new wreck, may have a yellow standing/upright '+' shaped topmark (St George's cross) (Topmark Shape: [Upright Cross].

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Buoy (BC020) *(Continued)*

Point

Notes

Sound Signal Device Guidance:

To avoid clutter, where a Sound Signal Device is co-located with another aid to navigation, the [Fog Signal Type] legend or the Chart #1 symbol R1 should be shown, but not both.

If a Sound Signal Device (Fog Signal Type: [No Information]) or not shown, label the sound signal type with appropriate legend and show Chart #1 (R1) symbol.

If Fog Signal Type: [Bell], use label "Bell", Chart #1 (R14)

If Fog Signal Type: [Diaphone], use legend "Dia", Chart #1 (R11)

If Fog Signal Type: [Explosive], use legend "Explos", Chart #1 (R10)

If Fog Signal Type: [Gong], use legend "Gong", Chart #1 (R16)

If Fog Signal Type: [Horn], use legend "Horn", Chart #1 (R13)

If Fog Signal Type: [Siren], use legend "Siren", Chart #1 (R12)

A Sound Signal Device with Fog Signal Type: [Other] or [No Information] the Chart #1 (R1) symbol shall be shown and can be used alone or in conjunction with other aids to navigation such as Buoys, Maritime Navigational Lights, or Maritime Radiobeacons.

Color of Navigational Aids Guidance:

The international abbreviations for colour are:
The International Abbreviations for Navigation Mark Colours for labeling color of Navigational Aids is:

Navigation Mark Colour: [White], abbreviate "W";

Navigation Mark Colour: [Red], abbreviate "R";

Navigation Mark Colour: [Green], abbreviate "G";

Navigation Mark Colour: [Blue], abbreviate "Bu";

Navigation Mark Colour: [Violet], abbreviate "Vi";

Navigation Mark Colour: [Yellow], abbreviate "Y";

Navigation Mark Colour: [Orange], abbreviate "Or"; (may be charted as "Y")

Navigation Mark Colour: [Amber], abbreviate "Am"; (may be charted as "Y")

Navigation Mark Colour: [Black], abbreviate "B";

Colour abbreviations must be in capital letters in all cases except for the second letter of two-letter abbreviations. These abbreviations must be used for the colours of lights and structures.

Multicoloured structures. Where the colours are in bands, the sequence of colour abbreviations must be from top to bottom. Where the colours are in stripes (vertical or diagonal) or the sequence of horizontal bands is not known, the darker colour must be given first. Buoys and Beacons which consist of multiple colors (Bands or Stripes) should contain multiple Navigation Mark Colour attributes collected for each color and in the order top to bottom for Bands and darker color first for Stripes.

The Navigation Mark Colour label for Buoys and Maritime Navigation Beacon shall be centered immediately below the small circle marking the position of the feature.

Within the IALA Maritime Buoyage System:
• red and green are used for lateral marks,
• black and yellow bands are used for cardinal marks,
• black and red bands are used for isolated danger marks,
• red and white stripes are used for safe-water buoys,
• yellow is used for special marks,
• blue and yellow stripes are used for emergency wreck marking buoys (on trial from 2006).

Red and green on lateral marks have different meanings in IALA Regions A and B, therefore the applicable region must be stated on charts.

Colour representation is effective in the case of filled and open (ie unfilled outline) symbols. The old scheme of lines, dots and chequers to represent colour(s) is obsolete and should no longer be used, as it cannot satisfactorily be used for topmarks and some types of buoy symbols (eg spar buoys and most multicoloured buoys). On multicoloured charts, buoys may be shown in their actual colour, or follow the rules for 'standard' coloured charts, which are:

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Buoy (BC020) *(Continued)*

Point

Notes

a. A black or green buoy (Navigation Mark Colour: [Black] or [Green]), must be represented by a filled symbol, Chart #1 (Q2)
Note: a spar buoy is always charted filled, irrespective of its actual colour, Chart #1 (Q5)

b. Any other colour of buoy, or multicoloured buoy (except spar buoys, see (a) above, and some preferred channel buoys, see (e) below) must be represented by an open symbol (ie unfilled outline) Chart #1 (Q3),

c. A striped buoy (Example: Buoy Type: [Safe Water]; Colour Pattern: [Vertical Stripes]) (except spar buoys, see (a) above) must be represented by an open buoy symbol with a single line from top to bottom, Chart #1 (Q5)

d. A banded buoy (Example: Colour Pattern: [Horizontal Stripes]) (except spar buoys, see (a) above, and predominantly green preferred channel buoys, see (e) below) must be represented by an open buoy symbol. No change is made to the buoy symbol to represent bands. This can be deduced from the multiple letter abbreviations, the topmark and lack of vertical line, Chart #1 (Q4)

e. Preferred channel (or bifurcation) buoys (Example Buoy Type: [Bifurcation]) are modified lateral marks within the IALA Maritime Buoyage System (ie green with red band, or red with green band). The symbol used should follow the lateral convention, ie a filled symbol must represent a predominantly green mark, and an open symbol must represent a predominantly red mark, Chart #1 (Q130.1)

Where there is insufficient space on charts for abbreviations, the topmarks alone (for cardinal buoys and beacons) or the filled and open symbols (for lateral buoys) may be considered adequate to indicate colours, without abbreviations.

Abbreviations for multiple colours on buoys (and beacons) must be shown in accordance with the following conventions:

a. Where the colours are in bands the sequence of colour abbreviations must be from top to bottom, eg in the IALA System: Chart #1 (Q4)
- *a north cardinal mark (black above yellow): BY*
- *an east cardinal mark (black with single broad horizontal yellow band): BYB*
- *a south cardinal mark (yellow above black): YB*
- *a west cardinal mark (yellow with a single broad horizontal black band): YBY*
- *a preferred channel lateral mark: GRG or RGR*
- *an isolated danger mark (black with one or more broad horizontal red bands): BRB*

Note: It helps to remember that the points of the topmark cones (for cardinal marks) reflect the position of the black band(s) on the body of the buoy (or beacon), eg for a north cardinal mark, the black band is at the top.

b. Where the colours are in stripes (vertical or diagonal) or the sequence of horizontal bands is not known, the darker colour must be given first, eg in the IALA System: Chart #1 (Q5)
- *a safe water buoy (red and white vertical stripes): RW*
- *an emergency wreck marking buoy (blue and yellow vertical stripes): BuY*

RADAR REFLECTORS ON BUOYS Guidance:

Areas where radar reflectors are fitted to most buoys (Example: Radar Reflector Present: [True]). In many areas of the world, radar reflectors are fitted to nearly all major buoys and to many minor ones. In such areas, the symbol for a radar reflector (Chart #1 (S4)) should not be shown on buoy symbols in order to reduce the complexity of buoy symbols and associated legends. In these areas, nations wishing to show the radar reflector symbol on unlit buoys may, exceptionally, do so but should include a chart note explaining why they are not shown on light buoys.

In other areas where radar reflectors are not widely fitted to buoys, the existence of a radar reflector should be indicated by the symbol (Chart #1 (S4) in black), Example: Chart #1 (Q11)

If Radar Reflector Present:[False] do not show Radar Reflector (Chart #1 (S4)).

Miscellaneous Buoy Guidance:

Miscellaneous labels occasionally may *be found in association with marine navigational aids. If a text label is shown on hydrographic source material, it should be considered significant for navigation. Examples are:*

A fog detection light, label "Fog Det Lt";

A floodlit structure near navigable water, label "(Illuminated)"; (Floodlit: [True])

A daytime light, if character of light in the day is different from the character shown at night. Show daytime character, followed by "Day" in parentheses, for example: (F 37m 11M Day);

Unwatched light, with no standby or emergency arrangements, label "(U)";

A temporary light or buoy, label "(temp)". If seasonal, include months, for example: "(Apr-Oct)";

A fog light, if light is only shown in fog, or the light during fog is different from the character of light shown at other times, show character in fog, followed by "(in fog)", in parentheses. For example: Fl 5s (in fog);

A privately maintained light or buoy, label "(priv)"; (Buoy Type: [Private])

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Buoy (BC020) *(Continued)*

Point

Notes

Seasonal buoys shall be shown without mention of their seasonal nature.

The slope of the buoy, which is normally 25° from vertical, may be varied from 5° to 45° from vertical to avoid overprints.

Reserve fog signals shall not be shown on product.
If a channel is marked by a feature but because of the product scale individual features overprint or are closer than 3 mm, replace the features with the legend "Buoyed channel" or "Channel marked by beacons". Legend will be aligned with the channel.

If thinning of the buoys is necessary due to clutter because of product scale, first thin buoys with the same Navigation Mark Colour, then those with the same Buoy Shape.

Feature Information					
Specification:	*Chart #1*	**Feature Origin:**	*Hydrographic*	**Symbol:**	Q23
Symbol Criteria:	(Buoy Shape: [Pillar] or [Diamond])			**Label:**	"[Geographic Name Information : Full Name]" if available, 7 point Mixed case Italic; [Navigation Mark Colour], 5 point Upper case Italic
				Color:	Black (58600)

Feature Information					
Specification:	*Chart #1*	**Feature Origin:**	*Hydrographic*	**Symbol:**	Q25
Symbol Criteria:	(Buoy Shape: [Barrel])			**Label:**	"[Geographic Name Information : Full Name]" if available, 7 point Mixed case Italic; [Navigation Mark Colour], 5 point Upper case Italic
				Color:	Black (58600)

Feature Information					
Specification:	*Chart #1*	**Feature Origin:**	*Hydrographic*	**Symbol:**	Q20
Symbol Criteria:	(Buoy Shape: [Conical])			**Label:**	"[Geographic Name Information : Full Name]" if available, 7 point Mixed case Italic; [Navigation Mark Colour], 5 point Upper case Italic
				Color:	Black (58600)

Feature Information					
Specification:	*Chart #1*	**Feature Origin:**	*Hydrographic*	**Symbol:**	Q21
Symbol Criteria:	(Buoy Shape: [Can])			**Label:**	"[Geographic Name Information : Full Name]" if available, 7 point Mixed case Italic; [Navigation Mark Colour], 5 point Upper case Italic
				Color:	Black (58600)

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Buoy (BC020) *(Continued)*

Point

Feature Information					
Specification:	*Chart #1*	**Feature Origin:**	*Hydrographic*	**Symbol:**	Q24
Symbol Criteria:	(Buoy Shape: [Spar] or [Ice buoy]) OR (Buoy Type: [Ice Buoy])			**Label:**	"[Geographic Name Information : Full Name]" if available, 7 point Mixed case Italic; [Navigation Mark Colour], 5 point Upper case Italic
				Color:	Black (58600)

Feature Information					
Specification:	*Chart #1*	**Feature Origin:**	*Hydrographic*	**Symbol:**	Q22
Symbol Criteria:	(Buoy Shape: [Spherical])			**Label:**	"[Geographic Name Information : Full Name]" if available, 7 point Mixed case Italic; [Navigation Mark Colour], 5 point Upper case Italic
				Color:	Black (58600)

Feature Information					
Specification:	*Chart #1*	**Feature Origin:**	*Hydrographic*	**Symbol:**	Q26
Symbol Criteria:	(Buoy Shape: [Super-buoy])			**Label:**	"[Geographic Name Information : Full Name]" if available, 7 point Mixed case Italic;
				Color:	Black (58600)

Feature Information					
Specification:	*Chart #1*	**Feature Origin:**	*Hydrographic*	**Symbol:**	Q58
Symbol Criteria:	(Buoy Type: [Ocean Data Acquisition System (ODAS) Buoy])			**Label:**	"ODAS", 7 point Upper case, Italic
				Color:	Black (58600)

Feature Information					
Specification:	*Chart #1*	**Feature Origin:**	*Hydrographic*	**Symbol:**	Q40
Symbol Criteria:	Buoy Type: [Mooring]			**Label:**	"[Geographic Name Information : Full Name]" if available, 7 point Mixed case Italic; [Navigation Mark Colour], 5 point Upper case Italic
				Color:	Black (58600)

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Buoy (BC020) *(Continued)*

Point

Feature Information					
Specification:	*Chart #1*	**Feature Origin:**	*Hydrographic*	**Symbol:**	P6
Symbol Criteria:	Buoy Type: [Large Automatic Navigation Buoy]			**Label:**	"[Geographic Name Information : Full Name]" if available, 7 point Mixed case Italic; [Navigation Mark Colour], 5 point Upper case Italic
				Color:	Black (58600)

Feature Information					
Specification:	*Chart #1*	**Feature Origin:**	*Hydrographic*	**Symbol:**	P5
Symbol Criteria:	(Buoy Type: [Articulated Light])			**Label:**	"[Geographic Name Information : Full Name]" if available, 7 point Mixed case Italic; [Navigation Mark Colour], 5 point Upper case
				Color:	Black (58600)

Feature Information					
Specification:	*Chart #1*	**Feature Origin:**	*Hydrographic*	**Symbol:**	Q30
Symbol Criteria:	(Buoy Type: [Light Float])			**Label:**	"[Geographic Name Information : Full Name]" if available, 7 point Mixed case Italic; [Navigation Mark Colour], 5 point Upper case Italic
				Color:	Black (58600)

Feature Information					
Specification:	*Chart #1*	**Feature Origin:**	*Hydrographic*	**Symbol:**	Q130.4
Symbol Criteria:	(Buoy Type: [Isolated Danger]) and (Buoy Shape: [Pillar] or [Spar]) and (Colour Pattern: [Horizontal Stripes]) and (Navigation Mark Colour: [Black] and [Red] and [Black]) and (Topmark Shape: [2 Spheres One over Other])			**Label:**	"[Geographic Name Information : Full Name]" if available, 7 point Mixed case Italic; [Navigation Mark Colour], 5 point Upper case Italic
				Color:	Black (58600)

Feature Information					
Specification:	*Chart #1*	**Feature Origin:**	*Hydrographic*	**Symbol:**	Q130.5
Symbol Criteria:	(Buoy Type: [Landfall] or [Safe Water] or [Fairway] or [Mid-Channel]) and (Buoy Shape: [Pillar] or [Spar] or [Spherical]) and (Colour Pattern: [Vertical Stripes]) and (Navigation Mark Colour: [Red] and [White]) and (Topmark Shape: [Sphere])			**Label:**	"[Geographic Name Information : Full Name]" if available, 7 point Mixed case Italic; [Navigation Mark Colour], 5 point Upper case Italic
				Color:	Black (58600)

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Buoy (BC020) *(Continued)*

Point

Feature Information					
Specification:	*Chart #1*	**Feature Origin:**	*Hydrographic*	**Symbol:**	Q130.6
Symbol Criteria:	(Buoy Type: [Special] or [Anchorage] or [Warping] or [Quarantine] or [Exercise Area] or [Explosive Anchorage] or [Aeronautical Anchorage] or [Compass Adjustment] or [Fish Trap] or [Diving] or [Notice] or [APEX Float] or [General Warning] or [Private] or [Swim] or [Control] or [Entry Prohibited] or [Accelerometer] or [Wavemeter] or [Navigation, communication and control buoy (NCCB)] or [Target] or [Marker Ship] or [Barge] or [Recording] or [Recreation Zone] or [Anchoring Prohibited] or [Berthing Prohibited] or [Overtaking Prohibited] or [Two-Way Traffic Prohibited] or [Reduced Wake] or [Speed Limit] or [Stop] or [Sound Ship's Siren] or [Restricted Vertical Clearance] or [Maximum Vessel's Draft] or [Restricted Horizontal Clearance] or [Strong Current Warning] or [Berthing Permitted] or [Overhead Power Cable] or [Channel Edge Gradient] or [Ferry Crossing] or [Pipeline] or [Clearing Line] or [Refuge] or [Foul Ground] or [Yachting] or [Heliport] or [GPS Mark] or [Work In Progress] or [Artificial Reef] or [Marine Farm] or [Seaplane Landing] or [Wellhead] or [Water Column Sensor] or [Other] or [No Information]) and (Buoy Shape: (Any)) and (Colour Pattern: [Vertical Stripes] or [Horizontal Stripes]) and (Navigation Mark Colour: ([White] and [Orange]) or [Yellow]) and (Topmark Shape: [X-shaped])			**Label:**	"[Geographic Name Information : Full Name]" if available, 7 point Mixed case Italic; [Navigation Mark Colour], 5 point Upper case Italic; if (Buoy Type: [Private]), Label: "Priv", 7 point Mixed case Italic; if (Buoy Type: [Wavemeter]), Label: "Wave", 7 point Mixed case Italic; if (Buoy Type: [Navigation, communication and control buoy (NCCB)]), Label: "(NCCB)", 7 point Mixed case Italic; Label: "Target" if (Buoy Type: [Target]), 7 point Mixed case Italic; Label: "Marker Ship" if (Buoy Type: [Marker Ship]), 7 point Mixed case Italic; Label: "Barge" if (Buoy Type: [Barge]), 7 point Mixed case Italic; if (Buoy Type: [Outfall]), Label: "Outfall", 7 point Mixed case Italic;
				Color:	Black (58600)

Feature Information					
Specification:	*Chart #1*	**Feature Origin:**	*Hydrographic*	**Symbol:**	Q130.1
Symbol Criteria:	(Buoy Type: [Bifurcation] or [Preferred Channel to Port] or [Preferred Channel to Starboard]) and (Buoy Shape: [Can] or [Conical]) and (Navigation Mark Colour: ([Green] and [Red] and [Green]) or ([Red] and [Green] and [Red]))			**Label:**	"[Geographic Name Information : Full Name]" if available, 7 point Mixed case Italic; [Navigation Mark Colour], 5 point Upper case Italic
				Color:	Black (58600)

Feature Information					
Specification:	*Chart #1*	**Feature Origin:**	*Hydrographic*	**Symbol:**	Q43
Symbol Criteria:	(Buoy Type: [Telegraph Cable] or [Telephone])			**Label:**	"[Geographic Name Information : Full Name]" if available, 7 point Mixed case Italic;
				Color:	Black (58600)

Feature Information					
Specification:	*Chart #1*	**Feature Origin:**	*Hydrographic*	**Symbol:**	Q50
Symbol Criteria:	(Buoy Type: [Firing Danger Area])			**Label:**	"[Geographic Name Information : Full Name]" if available, 7 point Mixed case Italic; Label: "DZ", 7 point Upper case Italic
				Color:	Black (58600)

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Buoy (BC020) *(Continued)*

Point

Feature Information					
Specification:	*Chart #1*	**Feature Origin:**	*Hydrographic*	**Symbol:**	Q56
Symbol Criteria:	(Buoy Type: [Spoil Ground])			**Label:**	"[Geographic Name Information : Full Name]" if available, 7 point Mixed case Italic;
				Color:	Black (58600)

Feature Information					
Specification:	*Chart #1*	**Feature Origin:**	*Hydrographic*	**Symbol:**	Q42
Symbol Criteria:	(Buoy Type: [Trot])			**Label:**	"[Geographic Name Information : Full Name]" if available, 7 point Mixed case Italic;
				Color:	Black (58600)

Feature Information					
Specification:	*Chart #1*	**Feature Origin:**	*Hydrographic*	**Symbol:**	Q130.3
Symbol Criteria:	(Buoy Type: [West Cardinal])			**Label:**	"[Geographic Name Information : Full Name]" if available, 7 point Mixed case Italic; [Navigation Mark Colour], 5 point Upper case Italic; Label: "West Mark", 7 point Mixed case Italic
				Color:	Black (58600)

Feature Information					
Specification:	*Chart #1*	**Feature Origin:**	*Hydrographic*	**Symbol:**	Q130.3
Symbol Criteria:	(Buoy Type: [South Cardinal])			**Label:**	"[Geographic Name Information : Full Name]" if available, 7 point Mixed case Italic; [Navigation Mark Colour], 5 point Upper case Italic; Label: "South Mark", 7 point Mixed case Italic
				Color:	Black (58600)

Feature Information					
Specification:	*Chart #1*	**Feature Origin:**	*Hydrographic*	**Symbol:**	Q130.3
Symbol Criteria:	(Buoy Type: [East Cardinal])			**Label:**	"[Geographic Name Information : Full Name]" if available, 7 point Mixed case Italic; [Navigation Mark Colour], 5 point Upper case Italic; Label: "East Mark", 7 point Mixed case Italic
				Color:	Black (58600)

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Buoy (BC020) *(Continued)*

Point

Feature Information					
Specification:	*Chart #1*	**Feature Origin:**	*Hydrographic*	**Symbol:**	Q130.3
Symbol Criteria:	(Buoy Type: [North Cardinal])			**Label:**	"[Geographic Name Information : Full Name]" if available, 7 point Mixed case Italic; [Navigation Mark Colour], 5 point Upper case Italic; Label: "North Mark", 7 point Mixed case Italic
				Color:	Black (58600)

Feature Information					
Specification:	*Chart #1*	**Feature Origin:**	*Hydrographic*	**Symbol:**	Q54
Symbol Criteria:	(Buoy Type: [Degaussing Range])			**Label:**	"[Geographic Name Information : Full Name]" if available, 7 point Mixed case Italic; [Navigation Mark Colour], 5 point Upper case Italic; Label: "DG", 7 point Upper case Italic
				Color:	Black (58600)

Feature Information					
Specification:	*Chart #1*	**Feature Origin:**	*Hydrographic*	**Symbol:**	L16
Symbol Criteria:	Buoy Type: [Installation]			**Label:**	"[Geographic Name Information : Full Name]" if available, 7 point Mixed case Italic;
				Color:	Black (58600)

Feature Information					
Specification:	*Chart #1*	**Feature Origin:**	*Hydrographic*	**Symbol:**	Q130.1
Symbol Criteria:	(Buoy Type: [Junction])			**Label:**	"[Geographic Name Information : Full Name]" if available, 7 point Mixed case Italic; [Navigation Mark Colour], 5 point Upper case Italic
				Color:	Black (58600)

Feature Information					
Specification:	*Chart #1*	**Feature Origin:**	*Hydrographic*	**Symbol:**	Q130.1
Symbol Criteria:	(Buoy Type: [Lateral] or [Starboard-hand Edge] or [Port-hand Edge])			**Label:**	"[Geographic Name Information : Full Name]" if available, 7 point Mixed case Italic; [Navigation Mark Colour], 5 point Upper case Italic
				Color:	Black (58600)

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Buoy (BC020) *(Continued)*

Point

Feature Information					
Specification:	*Chart #1*	**Feature Origin:**	*Hydrographic*	**Symbol:**	Q55
Symbol Criteria:	(Buoy Type: [Cable])			**Label:**	"[Geographic Name Information : Full Name]" if available, 7 point Mixed case Italic; [Navigation Mark Colour], 5 point Upper case Italic; Label : "Cable", 7 point Mixed Case Italic
				Color:	Black (58600)

Feature Information					
Specification:	*Chart #1*	**Feature Origin:**	*Hydrographic*	**Symbol:**	Q57
Symbol Criteria:	(Buoy Type: [Outfall])			**Label:**	"[Geographic Name Information : Full Name]" if available, 7 point Mixed case Italic
				Color:	Black (58600)

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Cable (AT005)

Curve

Feature Description:	*A single continuous rope-like bundle consisting of multiple strands.*
Inclusion Condition:	*Length or Diameter >= 125 m.*

Notes

Type for features on land shall be positioned on land if possible. If type must be placed in the water, it shall be positioned so it does not obscure or overprint hydrographic detail. Type for features in the water shall be positioned in the water. If possible, type shall not be positioned across the shoreline.

Where (Waterbody Overhead Obstruction: [True]) and (Navigation Landmark: [True]), the [Height Above Surface Level] label shall be placed in the water below the Cable feature but not to obscure or overprint hydrographic detail.

Label shall be position above feature, reading from left to right (or to the left of the vertical feature, reading bottom to top), at a 0.5 mm distance and parallel to respective feature. Label shall preferably be positioned at the midpoint of the line segment or symbol; however, it may be displaced laterally along respective feature to avoid overprinting other symbols or labels. If space will not permit placing label parallel to feature, offset the label (reading left to right, parallel to the tangent of the center of the southern neatline of the Map/Chart) and use a leader line to identify its location along the feature.

Feature shall not be shown within Built-up Area.

If feature parallels a Road, or Railroad at a distance of <= 5.0 mm at map scale, then do not portray. Show only the segments that run across country.

When powerlines run through an area of Trees, the area tree symbol is masked for 1.0 mm on each side of the powerline symbol, to represent the cleared way through the powerlines run.

Place Pylon symbols at 12.5 mm intervals along line feature, and also at points of the line feature directional change.

Where several cables within the waterbody converge at the same point, the symbols may be terminated before they reach the coast or inshore water on smaller scale paper charts, to avoid obscuring more important detail. In oil and gasfields, where pipelines and cables are often laid on the same route, the cable may be omitted.

The power 'flash' of Cable feature Chart #1 (L31.1) (Cable Type: [Power Line] or [Transmission Line]) should break the cable symbol at intervals of about 50mm.

In the case of power cables across narrow channels, where it is considered that notice boards give adequate warning of the danger, the 'flash' symbol may be omitted.

If Pipelines or cables in the water meet the following criteria, they are represented as pipeline areas (FC037) or cable areas (FC037) respectively, rather than shown as separate pipelines or cables. If pipelines and cables together meet the following criteria, they are represented as "cables and pipelines area" (FC037). Criteria: (a.) more than two linear features, AND (b.) space between any two linear features is less than 8 mm at chart scale, AND (c.) space between the outermost linear features in the group is greater than 3 mm at chart scale. If more than two cables or pipelines are <= 3 mm apart at chart scale, show only the outermost linear features. If cable symbols overprint other cable symbols, show only one cable. If pipeline symbols overprint other pipeline symbols, show only one pipeline. If cable symbols and pipeline symbols overprint, show a cable and pipeline area (FC037). The outermost limits of the cable, pipeline, or cables and pipeline area (FC037) area feature are displaced for 2 mm past the outermost cables and pipelines, so that the area within which anchoring, trawling, and dredging are prohibited or inadvisable includes a safety margin beyond the outermost cables and/or pipelines.

In certain circumstances, high voltage power cables may cause a deviation in a ship's magnetic compass; in these cases, where reports have been received, they should be treated as local magnetic anomalies (ZC040) and the legend "Magnetic Anomaly (see Note)" should be added in magenta at appropriate points along the cable.

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Cable (AT005) *(Continued)*

Curve

Feature Information					
Specification:	*LTDS*	**Feature Origin:**	*Topographic*	**Symbol:**	LN_Blue072_Cable_Powerline
Symbol Criteria:	Cable Type [1]: (Power Line) or (Transmission Line) and Height Above Surface Level < 46 and (Vertical Relative Location: [Above Surface])			**Label:**	"([Height Above Surface Level])", 6 point Condensed Mixed case if (Waterbody Overhead Obstruction: [True]) and (Navigation Landmark: [True])
				Color:	Blue (48251)

Feature Information					
Specification:	*LTDS*	**Feature Origin:**	*Topographic*	**Symbol:**	LN_Black_Cable_ CommunicationLine
Symbol Criteria:	(Cable Type [1]: [Telephone] or [Telegraph] or [Digital Communication Line] or [Communication Line] or [Fibre-optic] or [Tethering] or [Load Bearing] or [Guide] or [Barrier] or [Restraining] or [Towing] or [Other] or [No Information]) and (Height Above Surface Level < 46) and (Vertical Relative Location: [Above Surface])			**Label:**	"Tel", 6 point Condensed Mixed case if Cable Type [1]: (Telephone) or (Telegraph) or (Digital Communication Line) or (Communication Line) or (Fibre-optic); "Cable", 6 point Condensed Mixed case if Cable Type [1]: (Tethering) or (Load Bearing) or (Guide) or (Barrier) or (Restraining) or (Towing) or (Other) or (No Information); "([Height Above Surface Level])", 6 point Condensed Mixed case if (Waterbody Overhead Obstruction: [True]) and (Navigation Landmark: [True])
				Color:	Black (58600)

Feature Information					
Specification:	*LTDS*	**Feature Origin:**	*Topographic*	**Symbol:**	LN_Blue072_Cable_Obstruction
Symbol Criteria:	Length or Diameter >= 125 and Height Above Surface Level >= 46 and (Vertical Relative Location: [Above Surface])			**Label:**	"Cable" , 6 point Condensed Mixed case; [Highest Elevation], 6 point Condensed Bold Mixed case; "([Height Above Surface Level])", 6 point Condensed Mixed case
				Color:	Blue (48251)

Feature Information					
Specification:	*Chart #1*	**Feature Origin:**	*Hydrographic*	**Symbol:**	L30.1
Symbol Criteria:	(Cable Type: <> [Power Line] or [Transmission Line]) and (Condition of Facility: [Fully Functional] or [Partially Functional] or [Under Construction]) and (Vertical Relative Location: [On Waterbody Bottom] or [Below Waterbody Bottom] or [Above Waterbody Bottom] or [Below Waterbody Surface])			**Label:**	"Buried", 7 Point Mixed Case, Italic if (Vertical Relative Location: [Below Waterbody Bottom])
				Color:	Black (58600)

Feature Information					
Specification:	*Chart #1*	**Feature Origin:**	*Hydrographic*	**Symbol:**	L31.1
Symbol Criteria:	(Cable Type: [Power Line] or [Transmission Line]) and (Condition of Facility: [Fully Functional] or [Partially Functional] or [Under Construction]) and (Vertical Relative Location: [On Waterbody Bottom] or [Below Waterbody Bottom] or [Above Waterbody Bottom] or [Below Waterbody Surface])			**Label:**	None
				Color:	N/A

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Cable (AT005) *(Continued)*

Curve

Feature Information					
Specification:	*Chart #1*	**Feature Origin:**	*Hydrographic*	**Symbol:**	L32
Symbol Criteria:	(Vertical Relative Location: [On Waterbody Bottom] or [Below Waterbody Bottom] or [Above Waterbody Bottom] or [Below Waterbody Surface]) and (Condition of Facility: [Abandoned] or [Destroyed] or [Partially Destroyed] or [Disused] or [Other])			**Label:**	None
				Color:	N/A

Cable (AT005)

Point

Inclusion Condition:	*Length or Diameter < 125 m.*

Notes

None

Feature Information					
Specification:	*LTDS*	**Feature Origin:**	*Topographic*	**Symbol:**	PT_Blue072_CableVO
Symbol Criteria:	Length or Diameter < 125 AND Height Above Surface Level >= 46 and (Vertical Relative Location: [Above Surface])			**Label:**	"Cable" , 6 point Condensed Mixed case; [Highest Elevation], 6 point Condensed Bold Mixed case; "([Height Above Surface Level])", 6 point Condensed Mixed case
				Color:	Blue (48251)

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Cableway (AT041)

Curve

Feature Description:	*A suspended transportation system consisting of one or more load cables, supporting pylons, carrier units (for example: cars or buckets intended to transport people, material, and/or equipment) and usually stations.*
Inclusion Condition:	*Length or Diameter >= 125 m.*

Notes

Area and line features will be generalized to detail compatible with scale.

Type for features on land shall be positioned on land if possible. If type must be placed in the water, it shall be positioned so it does not obscure or overprint hydrographic detail. Type for features in the water shall be positioned in the water. If possible, type shall not be positioned across the shoreline.

Label shall be position above feature, reading from left to right (or to the left of the vertical feature, reading bottom to top), at a 0.5 mm distance and parallel to respective feature. Label shall preferably be positioned at the midpoint of the line segment or symbol; however, it may be displaced laterally along respective feature to avoid overprinting other symbols or labels. If space will not permit placing label parallel to feature, offset the label to be positioned, (reading left to right, parallel to the tangent of the center of the southern neatline of the Map/Chart) and use a leader line to identify its location along the feature.

Feature Information

Specification:	*LTDS*	**Feature Origin:**	*Topographic*	**Symbol:**	LN_Black_Cableway
Symbol Criteria:	Height Above Surface Level < 46 m			**Label:**	"[Cableway Type]", 6 point Condensed Mixed case if (Cableway Type: [Chair-lift] or [Ski Tow] or [Aerial Tramway] or [Gondola Lift] or [T-bar Lift] or [Industrial Ropeway] or [Material Tramway]); "Cableway", 6 point Condensed Mixed case if (Cableway Type: [Other] or [No Information])
				Color:	Black (58600)

Feature Information

Specification:	*LTDS*	**Feature Origin:**	*Topographic*	**Symbol:**	LN_Blue072_Cableway_ Obstruction
Symbol Criteria:	Length or Diameter >= 125 and Height Above Surface Level >= 46			**Label:**	"Cable" , 6 point Condensed Mixed case; [Highest Elevation], 6 point Condensed Bold Mixed case; "([Height Above Surface Level])": 6 point Condensed Mixed case
				Color:	Blue (48251)

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Cableway (AT041)

Point

Inclusion Condition:	*Length or Diameter < 125 m.*

Notes

None

Feature Information					
Specification:	*LTDS*	**Feature Origin:**	*Topographic*	**Symbol:**	PT_Blue072_CablewayVO
Symbol Criteria:	Length or Diameter < 125 and Height Above Surface Level >= 46			**Label:**	"Cable" , 6 point Condensed Mixed case; [Highest Elevation], 6 point Condensed Bold Mixed case; "([Height Above Surface Level])": 6 point Condensed Mixed case
				Color:	Blue (48251)

Cairn (AL025)

Point

Feature Description:	*A heap of stones piled up as a memorial or a landmark.*
Inclusion Condition:	*Navigation Landmark: (True)*

Notes

None

Feature Information					
Specification:	*LTDS*	**Feature Origin:**	*Topographic*	**Symbol:**	PT_Black_Cairn
Symbol Criteria:	All			**Label:**	None
				Color:	N/A

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Calling-in Point (BB050)

Point

Feature Description:	*The location at which vessels are required to report to a traffic control center.*
Inclusion Condition:	*ALL REQUIRED*

Notes

The symbol must be shown in the centre of the fairway if a position is not otherwise specified, and oriented to denote the direction(s) of vessel movement. A single arrowhead indicates that a report is required only when a vessel is bound in that direction. At route junctions, reporting points may have more than two arrowheads, each pointing in the general direction as denoted in [Direction of Traffic - 1} or [Direction of Traffic - 2] or [Direction of Traffic - 3] or [Direction of Traffic - 4] of vessel movement.

If the radio reporting point has an alphanumeric designator [Geographic Name Information : Full Name], it must be shown in magenta within the circle where possible. Alphanumeric type shall be CAPS. If the designator is too large to fit in the circle, it may be shown adjacent to it.

Point features may be displaced <= 5 mm, at chart scale, to avoid overprints.

Feature Information

Specification:	*Chart #1*	**Feature Origin:**	*Hydrographic*	**Symbol:**	M40.1
Symbol Criteria:	All			**Label:**	[Geographic Name Information : Full Name] if available, 7 point upper case
				Color:	Purple (96532)

Camp (AI030)

Surface

Feature Description:	*An encampment where tents and/or other easily moveable structures (for example: yurts) serve as full-time, temporary, or seasonal residences.*
Inclusion Condition:	*Area >= 15,625 sq m.*

Notes

None

Feature Information

Specification:	*LTDS*	**Feature Origin:**	*Topographic*	**Symbol:**	AR_Black_Camp
Symbol Criteria:	All			**Label:**	[Geographic Name Information : Full Name] if available: 6 point Condensed Mixed case; "[Feature Function]", 6 point Condensed Mixed case if <> (Other) or (No information); "Camp", 6 point Condensed Mixed case if Feature Function: (Other) or (No information)
				Color:	Black (58600)

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Camp (AI030)

Point

Inclusion Condition:	*Area < 15,625 sq m.*

Notes

None

Feature Information					
Specification:	*LTDS*	**Feature Origin:**	*Topographic*	**Symbol:**	PT_Black_Camp
Symbol Criteria:	All			**Label:**	[Geographic Name Information : Full Name] if available: 6 point Condensed Mixed case; "[Feature Function]", 6 point Condensed Mixed case if <> (Other) or (No information)
				Color:	Black (58600)

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Camp-site (AK060)

Surface

Feature Description:	*A designated place for recreational camping.*
Inclusion Condition:	*Area >= 15,625 square meters*

Notes

Area and line features will be generalized to detail compatible with scale.

Type for features on land shall be positioned on land if possible. If type must be placed in the water, it shall be positioned so it does not obscure or overprint hydrographic detail. Type for features in the water shall be positioned in the water. If possible, type shall not be positioned across the shoreline.

Descriptive terms, e.g., "Canal" shall be shown if the name is not known. If the descriptive word appears in the name, for example, "PANAMA CANAL". The descriptive type shall not be shown, i.e. do not show "Panama Canal Canal".

If area feature symbol is <= 2.5 mm at map scale (125 meters on ground at 1:50,000 scale) from another (area) feature with unlike attributes, the larger of the feature outline may be extended to touch the other's outline. They would have one dividing line between them and where the outlines would coalesce, one of the features would omit that portion.

Limiting lines of feature are omitted if it coalesces with a road (AP030).

If a clearing exists inside of the area feature, and the size of the clearing is equal to or greater than the area for the surrounding area feature, the clearing is shown as an open space inside the surround feature. If the area of the clearing is less than the area for the surrounding feature, the clearing is deleted and absorbed into the surrounding area feature.

If two area features with the same feature code do not connect at any point, and have a space between them less than 2.5 mm at map/chart scale, delete the open space that is less than 2.5 mm between the features and combine them into one area feature.

If a portion of an area feature has a minimum width of less than 2.5 mm at map/chart scale, delete that portion of the area feature that is not at least 2.5 mm wide, measured from perimeter to perimeter. If the deletion of the portion of the area based on the above criteria will reduce the area of the remaining portion of the area feature to below the minimum area inclusion, do not delete the narrow portion of the feature that is less than 2.5 mm wide. If the deletion of the portion of the area based on the above criteria will split two larger areas connected by a narrow strip into two separate areas, either of which would be below minimum area inclusion, do not delete the narrow portion of the feature that is less than 2.5 mm wide.

Feature Information

Specification:	*LTDS*	**Feature Origin:**	*Topographic*	**Symbol:**	AR_Black_CampSite
Symbol Criteria:	Area >= 15,625			**Label:**	[Geographic Name Information : Full Name] "Campsite", 6 point Condensed Mixed case
				Color:	Black (58600)

Camp-site (AK060)

Point

Inclusion Condition:	*Area < 15,625 square meters*

Notes

None

Feature Information

Specification:	*LTDS*	**Feature Origin:**	*Topographic*	**Symbol:**	PT_Black_CampSite
Symbol Criteria:	(Area < 15,625) and (Navigation Landmark: [True])			**Label:**	[Geographic Name Information : Full Name] if available, 6 point Condensed Mixed case;
				Color:	Black (58600)

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Canal (BH020)

Surface

Feature Description:	*An artificial waterway with no flow, or a controlled flow, usable or built for navigation.*
Inclusion Condition:	*Width >= 25 m.*

Notes

Rivers, canals, and ditches will be partially collapsed when area and line delineations are supported on the product and the area feature does not meet the minimum geometric inclusion condition.

Coincident similar area features will be blended to form a single feature.

Area and line features will be generalized to detail compatible with scale.

Feature will be generalized to provide a more aesthetic contoured feature (i.e., smoothed).

Label area features with upper case type within its limits and centered between the sides with a proportional size if the width of the feature will allow its inclusion. However, should the feature be too narrow, then place the type 0.5 mm above and parallel to the feature. When the feature is continuous, repeat label approximately every 30 to 40 cm for either situation, or at least two items, length permitting. In either condition (in or above the feature), curve the type necessary to the curvature of the feature. Should the feature change back and forth between the area and a line feature, the type style will change from upper case type for the area portions, to upper and lower case type for the linear portions. The repeat dimensions remain the same.

Label shall be position above feature, reading from left to right (or to the left of the vertical feature, reading bottom to top), at a 0.5 mm distance and parallel to respective feature. Label shall preferably be positioned at the midpoint of the line segment or symbol; however, it may be displaced laterally along respective feature to avoid overprinting other symbols or labels. If space will not permit placing label parallel to feature, offset the label (reading left to right, parallel to the tangent of the center of the southern neatline of the Map/Chart) and use a leader line to identify its location along the feature.

Show Hydrographic Depth or Hydrographic Height values in whole meters if decimeter value is 0; otherwise show the decimeter value as a subscript to the meter value.

Descriptive terms, e.g., "Canal" shall be shown if the name is not known. If the descriptive word appears in the name, for example, "PANAMA CANAL". The descriptive type shall not be shown, i.e. do not show "Panama Canal Canal".

Symbols and associated area patterns of underpassing features (except drainage shorelines) are broken for all bridges, except footbridges. This rule does not apply to land tint on Littoral Planning Charts.

Do not show land tint in the symbol. Do not show land tint if (Perennial).

Symbolize the casement portions (Left Bank/Right Bank) of the feature using individual river or canal banks in conjunction with the inland shoreline symbology. The inland shoreline shall correspond to the associated water body i.e., Perennial river = Perennial Inland Shoreline.

Feature Information

Specification:	*LTDS*	**Feature Origin:**	*Topographic*	**Symbol:**	AR_Cyan_Canal_Perennial
Symbol Criteria:	(Water Resource Information : Hydrologic Persistence [Perennial] or [No Information]) and (Condition of Facility: [Abandoned] or [Fully Functional] or [Partially Functional] or [Partially Destroyed] or [Damaged] or [Disused] or [Other] or [No Information])			**Label:**	[Geographic Name Information : Full Name]if available "CANAL", 7-16 point Condensed Italics Upper case
				Color:	Blue (48253)

Feature Information

Specification:	*LTDS*	**Feature Origin:**	*Topographic*	**Symbol:**	AR_Cyan_Canal_Non-perennial
Symbol Criteria:	(Water Resource Information : Hydrologic Persistence: [Intermittent] or [Dry]) and (Condition of Facility: [Abandoned] or [Fully Functional] or [Partially Functional] or [Partially Destroyed] or [Damaged] or [Disused] or [Other] or [No Information])			**Label:**	[Geographic Name Information : Full Name]if available "CANAL", 7-16 point Condensed Italics Upper case
				Color:	Blue (48253)

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Canal (BH020) *(Continued)*

Surface

Feature Information					
Specification:	*LTDS*	**Feature Origin:**	*Topographic*	**Symbol:**	AR_Cyan_Canal_ UnderConstruction_Destroyed
Symbol Criteria:	(Condition of Facility: [Under Construction] or [Destroyed])			**Label:**	[Geographic Name Information : Full Name]if available "CANAL (CONSTRUCTION)", 7-16 point Condensed Italics Mixed case if Condition of Facility: [Under Construction]; [Geographic Name Information : Full Name]if available "CANAL (DESTROYED)", 7-16 point Condensed Italics Upper case if Condition of Facility: [Destroyed]
				Color:	Blue (48253)

Canal (BH020)

Curve

Inclusion Condition:	*Width < 25 m.*

Notes

Rivers, canals, and ditches will be partially collapsed when area and line delineations are supported on the product and the area feature does not meet the minimum geometric inclusion condition.

Coincident similar area features will be blended to form a single feature.

Area and line features will be generalized to detail compatible with scale.

Feature will be generalized to provide a more aesthetic contoured feature (i.e., smoothed).

Label area features with upper case type within its limits and centered between the sides with a proportional size if the width of the feature will allow its inclusion. However, should the feature be too narrow, then place the type 0.5 mm above and parallel to the feature. When the feature is continuous, repeat label approximately every 30 to 40 cm for either situation, or at least two items, length permitting. In either condition (in or above the feature), curve the type necessary to the curvature of the feature. Should the feature change back and forth between the area and a line feature, the type style will change from upper case type for the area portions, to upper and lower case type for the linear portions. The repeat dimensions remain the same.

Label shall be position above feature, reading from left to right (or to the left of the vertical feature, reading bottom to top), at a 0.5 mm distance and parallel to respective feature. Label shall preferably be positioned at the midpoint of the line segment or symbol; however, it may be displaced laterally along respective feature to avoid overprinting other symbols or labels. If space will not permit placing label parallel to feature, offset the label (reading left to right, parallel to the tangent of the center of the southern neatline of the Map/Chart) and use a leader line to identify its location along the feature.

Show Hydrographic Depth or Hydrographic Height values in whole meters if decimeter value is 0; otherwise show the decimeter value as a subscript to the meter value.

Descriptive terms, e.g., "Canal" shall be shown if the name is not known. If the descriptive word appears in the name, for example, "PANAMA CANAL". The descriptive type shall not be shown, i.e. do not show "Panama Canal Canal".

Symbols and associated area patterns of underpassing features (except drainage shorelines) are broken for all bridges, except footbridges. This rule does not apply to land tint on Littoral Planning Charts.

Do not show land tint in the symbol. Do not show land tint if (Perennial).

Symbolize the casement portions (Left Bank/Right Bank) of the feature using individual river or canal banks in conjunction with the inland shoreline symbology. The inland shoreline shall correspond to the associated water body i.e., Perennial river = Perennial Inland Shoreline.

Incorporate shorter Canals and Ditches <= 320 meters length as a connector feature and incorporate spacing of > 200 meters. Always retain the outermost limits of these features before generalization takes place.

Omit from Built-up Area.

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Canal (BH020) *(Continued)*

Curve

Feature Information					
Specification:	*LTDS*	**Feature Origin:**	*Topographic*	**Symbol:**	LN_Cyan_Canal_Perennial
Symbol Criteria:	(Water Resource Information : Hydrologic Persistence: [Perennial] or [No Information]) and (Condition of Facility: [Abandoned] or [Fully Functional] or [Partially Functional] or [Partially Destroyed] or [Damaged] or [Disused] or [Other] or [No Information])			**Label:**	[Geographic Name Information : Full Name] if available "Canal", 6 point Condensed Italics Mixed case
				Color:	Blue (48253)

Feature Information					
Specification:	*LTDS*	**Feature Origin:**	*Topographic*	**Symbol:**	LN_Cyan_Canal_Dry
Symbol Criteria:	(Water Resource Information : Hydrologic Persistence: [Intermittent] or [Dry]) or (Condition of Facility: [Under Construction] or [Destroyed])			**Label:**	[Geographic Name Information : Full Name] if available "Canal (construction)", 6 point Condensed Italics Mixed case if Condition of Facility: [Under Construction]; [Geographic Name Information : Full Name] if available "Canal (destroyed)" if Condition of Facility: [Destroyed]
				Color:	Blue (48253)

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Cane (EC010)

Surface

Feature Description:	*A tract covered mainly by large treelike grasses.*
Inclusion Condition:	*Area >= 15,625 sq m.*

Notes

Coincident similar area features will be blended to form a single feature.

Area and line features will be generalized to detail compatible with scale.

Feature will be generalized to provide a more aesthetic contoured feature (i.e., smoothed).

Symbols and associated area patterns of underpassing features (except drainage shorelines) are broken for all bridges, except footbridges. This rule does not apply to land tint on Littoral Planning Charts.

If a clearing exists inside of the area feature, and the size of the clearing is equal to or greater than the area for the surrounding area feature, the clearing is shown as an open space inside the surround feature. If the area of the clearing is less than the area for the surrounding feature, the clearing is deleted and absorbed into the surrounding area feature.

If two area features with the same feature code do not connect at any point, and have a space between them less than 2.5 mm at map/chart scale, delete the open space that is less than 2.5 mm between the features and combine them into one area feature.

If a portion of an area feature has a minimum width of less than 2.5 mm at map/chart scale, delete that portion of the area feature that is not at least 2.5 mm wide, measured from perimeter to perimeter. If the deletion of the portion of the area based on the above criteria will reduce the area of the remaining portion of the area feature to below the minimum area inclusion, do not delete the narrow portion of the feature that is less than 2.5 mm wide. If the deletion of the portion of the area based on the above criteria will split two larger areas connected by a narrow strip into two separate areas, either of which would be below minimum area inclusion, do not delete the narrow portion of the feature that is less than 2.5 mm wide.

Feature Information

Specification:	*LTDS*	**Feature Origin:**	*Topographic*	**Symbol:**	AR_Green362_Cane
Symbol Criteria:	All			**Label:**	None
				Color:	N/A

Caravan Park (AI020)

Surface

Feature Description:	*A prepared site, typically including facilities, used for holiday accommodations where caravans and/or motor homes are parked.*
Inclusion Condition:	*Area >= 15,625 sq m.*

Notes

None

Feature Information

Specification:	*LTDS*	**Feature Origin:**	*Topographic*	**Symbol:**	AR_Black_CaravanPark
Symbol Criteria:	Area >= 15,625			**Label:**	[Geographic Name Information : Full Name] if available "Caravan park", 6 point Condensed Mixed case
				Color:	Black (58600)

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Caravan Park (AI020)

Point

Inclusion Condition:	*Area < 15,625 sq m.*

Notes

None

Feature Information					
Specification:	*LTDS*	**Feature Origin:**	*Topographic*	**Symbol:**	PT_Black_CaravanPark
Symbol Criteria:	Area < 15,625 and Navigation Landmark: [True}			**Label:**	[Geographic Name Information : Full Name] if available "Caravan park", 6 point Condensed Mixed case
				Color:	Black (58600)

Cart Track (AP010)

Curve

Feature Description:	*An unimproved road.*
Inclusion Condition:	*Road Weather Restriction: (Fair Weather) or (Winter Only) AND Length or Diameter > 600 m. AND (in moderate-to-dense areas are spaced >= 2000 m. apart) OR (in sparse-to-moderate areas are spaced >= 600 m. apart) OR (needed to complete the component network or to establish connectivity)*

Notes

If feature coalesce at map scale, when shown in their true positions, they shall be displaced 0.2 mm from one another.

Area and line features will be generalized to detail compatible with scale.

For Road, Cart track, and Trail within Built-up Area; Symbolize the portion of the feature within the Built-up Area as white.

When the project area or sheet falls within an area defined as having sparse culture; the inclusion condition defaults to all required.

A space of not less than 0.2 mm shall be shown between Building symbols and Tracks and Trails.

Collect Cart Tracks to complete the transportation network and provide connectivity to areas of habitation/utilization. Cart Tracks and Trail (AP050) need to be extracted to portray an inter-connective framework linking towns, villages, or other features, but lose their importance in denser feature portrayals. In sparse areas, use them to connect Road networks or where no other significant features have been captured. In sparsely populated areas the network of Cart Tracks and Trail (AP050) would provide the main transportation routes in the area, and would therefore become very important. If the Road (AP030) network is dense, the need to capture Cart Tracks and Trail (AP050) will lessen and could be negligible.

Feature Information					
Specification:	*LTDS*	**Feature Origin:**	*Topographic*	**Symbol:**	LN_Black_CartTrack
Symbol Criteria:	All			**Label:**	None
				Color:	N/A

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Castle (AL375)

Surface

Feature Description:	*A single large fortified building that has thick walls, battlements, and often the presence of a moat, and is commonly of some historical significance.*
Inclusion Condition:	*Length or Diameter >= 25 m.*

Notes

None

Feature Information

Specification:	*LTDS*	**Feature Origin:**	*Topographic*	**Symbol:**	AR_Black_Castle
Symbol Criteria:	(Area >= 5625)			**Label:**	[Geographic Name Information: Full Name] if available "Castle", 6 point Condensed Mixed case
				Color:	Black (58600)

Castle (AL375)

Point

Inclusion Condition:	*Length or Diameter < 25 m.*

Notes

None

Feature Information

Specification:	*LTDS*	**Feature Origin:**	*Topographic*	**Symbol:**	PT_Black_Castle
Symbol Criteria:	((Height Above Surface Level < 46 m.) and (Area < 5625)) OR (Height Above Surface Level < 46 m.)			**Label:**	[Geographic Name Information: Full Name] if available, 6 point Condensed Mixed case
				Color:	Black (58600)

Feature Information

Specification:	*LTDS*	**Feature Origin:**	*Topographic*	**Symbol:**	PT_Blue072_VO
Symbol Criteria:	Height Above Surface Level >= 46 m.			**Label:**	"Castle", 6 point Condensed Mixed case; [Highest Elevation], 6 point Condensed Bold Mixed case; "([Height Above Surface Level])", 6 point Condensed Mixed case
				Color:	Blue (46351)

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Castle Complex (AL376)

Surface

Feature Description:	*A fortified complex of buildings and related structures that consists principally of a central keep with surrounding thick walls, battlements, and often the presence of a moat, and is commonly of some historical significance.*
Inclusion Condition:	*Area >= 15,625 sq m.*

Notes

None

Feature Information

Specification:	*LTDS*	**Feature Origin:**	*Topographic*	**Symbol:**	AR_Black_CastleComplex
Symbol Criteria:	All			**Label:**	[Geographic Name Information: Full Name] if available or (([Geographic Name Information: Full Name] <> "UNK") and ([Geographic Name Information: Full Name] is not null)), 6 point Condensed Mixed case; "Castle" if (([Geographic Name Information: Full Name] = "UNK" or [Geographic Name Information: Full Name] is null)), 6 point Condensed Mixed case
				Color:	Black (58600)

Castle Complex (AL376)

Point

Inclusion Condition:	*Area < 15,625 sq m.*

Notes

None

Feature Information

Specification:	*LTDS*	**Feature Origin:**	*Topographic*	**Symbol:**	PT_Black_CastleComplex
Symbol Criteria:	All			**Label:**	[Geographic Name Information: Full Name] if available, 6 point Condensed Mixed case
				Color:	Black (58600)

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Catalytic Cracker (AC020)

Point

Feature Description:	*A device in which the separation of petroleum is carried out in the presence of a catalyst.*
Inclusion Condition:	*Height Above Surface Level: >= 46 m. OR Navigation Landmark: (True)*

Notes

The feature (when height < 46 meters or when height is not known) shall be oriented perpendicular (90 degrees) to the nearby road (AP030), cart track (AP010), trail (AP050), or railroad (AN010).

If one symbol coalesces with another symbol for the same type feature, displace symbols to allow a minimum separation of 0.21mm.

Label feature as per hierarchy for topo type placement to south neatline corners reading left to right:
1. Position hierarchy:
(a.) northeast (preferred position
(b.) southeast (1st alternative)
(c.) northwest (2nd alternative
(d.) southwest (3rd alternative)
(e.) top-centered (4th alternative)
(f.) bottom-centered (5th alternative)
(Hierarchy is based on type positioning so as to avoid overprinting other type or obscuring detail.)
2. Minimum space between type placement and feature symbol is 0.5 mm
3. This method of type placement shall be used for areal features when space does not permit labeling within that feature.

Type for features on land shall be positioned on land if possible. If type must be placed in the water, it shall be positioned so it does not obscure or overprint hydrographic detail. Type for features in the water shall be positioned in the water. If possible, type shall not be positioned across the shoreline.

Feature Information

Specification:	*LTDS*	**Feature Origin:**	*Topographic*	**Symbol:**	PT_Black_CatalyticCracker
Symbol Criteria:	(Height Above Surface Level < 46) and (Navigation Landmark = [True])			**Label:**	"Catalytic cracker", 6 point Condensed Mixed case
				Color:	Black (58600)

Feature Information

Specification:	*LTDS*	**Feature Origin:**	*Topographic*	**Symbol:**	PT_Blue072_VO
Symbol Criteria:	Height Above Surface Level >= 46 meters			**Label:**	"Catalytic cracker", 6 point Condensed Mixed case; [Highest Elevation], 6 point Condensed Bold Mixed case; "[Height Above Surface Level]", 6 point Condensed Mixed case
				Color:	Blue (46351)

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Causeway Structure (AQ063)

Surface

Feature Description:	*A solid raised way across a terrain obstacle (for example: a wetland or a body of shallow water) that is intended to support a transportation route (for example: a road or a railway).*
Inclusion Condition:	*Width >= 25 m.*

Notes

When 2 or more similar area features are separated by less than 0.5 mm at chart scale, the feature will be agglomerated.

Area and line features will be generalized to detail compatible with scale.

Label feature as per hierarchy for topo type placement to south neatline corners reading left to right:
1. Position hierarchy:
(a.) northeast (preferred position),
(b.) southeast (1st alternative),
(c.) northwest (2nd alternative,
(d.) southwest (3rd alternative),
(e.) top-centered (4th alternative),
(f.) bottom-centered (5th alternative)
(Hierarchy is based on type positioning so as to avoid overprinting other type or obscuring detail.);
2. Minimum space between type placement and feature symbol is 0.5 mm;
3. This method of type placement shall be used for areal features when space does not permit labeling within that feature.

Names placement shall be oriented to the longest axis of the feature reading left to right and placed within the area outline and centered. If the longest axis is perpendicular to the south neatline, the type shall be placed outside of the area outline, preferred position is northeast of the feature (Rule L-3505), but may be placed at any position around the feature so as not to overprint any other feature type and reading left to right.

Where a Cut Line or Fill coincides with a contour, the contour shall be suppressed. The cut Lines ticks shall point downhill towards the bottom of the cut.

Where symbols with the same feature code but different values of Water Level Effect (i.e. (Covers and Uncovers) or (Always Dry)) are adjacent to each other, the boundary of the feature that is the highest in vertical plane shall be shown, and the boundary of the feature above water is shown whole, while the feature that covers and uncovers has a portion of its symbol perimeter that overprints the higher feature deleted.

Show land tint inside the symbol. If the Water Level Effect: (Always Dry), show land tint only. Show land tint and water tint if Water Level Effect: (Covers and Uncovers)

A blue 31% tint shall not overprint other blue 31% tints. If two blue tinted symbols, or one tinted symbol and water tint overprint, only one 31% tint shall be shown in the area.

Feature Information

Specification:	*LTDS*	**Feature Origin:**	*Topographic*	**Symbol:**	AR_Cyan_CausewayStructure_Dry
Symbol Criteria:	Water Level Effect = [No Information] or [Always Dry] or [Other]			**Label:**	[Geographic Name Information : Full Name] if available "CAUSEWAY", 6 point Light Condensed Italic Upper case
				Color:	Black (58600)

Feature Information

Specification:	*LTDS*	**Feature Origin:**	*Topographic*	**Symbol:**	AR_BlackCyan_Causeway Structure_NotDry
Symbol Criteria:	Water Level Effect = [Covers and Uncovers]			**Label:**	[Geographic Name Information : Full Name] if available "CAUSEWAY", 6 point Light Condensed Italic Upper case
				Color:	Black (58600)

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Causeway Structure (AQ063)

Point

Inclusion Condition:	*Width < 25 m.*

Notes

Area and line features will be generalized to detail compatible with scale.

Label line feature above (preferred) and parallel to the line with a 0.5 mm space between. Above means: readable from south or east Projection neatline.

Where a Cut Line or Fill coincides with a contour, the contour shall be suppressed. The cut Lines ticks shall point downhill towards the bottom of the cut.

Omit from Built-up Area.

Where symbols with the same feature code but different values of Water Level Effect (i.e. (Covers and Uncovers) or (Always Dry)) are adjacent to each other, the boundary of the feature that is the highest in vertical plane shall be shown, and the boundary of the feature above water is shown whole, while the feature that covers and uncovers has a portion of its symbol perimeter that overprints the higher feature deleted.

Show land tint inside the symbol. If the Water Level Effect: (Always Dry), show land tint only. Show land tint and water tint if Water Level Effect: (Covers and Uncovers)

A blue 31% tint shall not overprint other blue 31% tints. If two blue tinted symbols, or one tinted symbol and water tint overprint, only one 31% tint shall be shown in the area.

Feature Information

Specification:	*LTDS*	Feature Origin:	*Topographic*	Symbol:	LN_Causeway
Symbol Criteria:	All			Label:	[Geographic Name Information : Full Name] if available "CAUSEWAY", 6 point Light Condensed Italic Upper case
				Color:	Black (58600)

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Cave Mouth (DB029)

Point

Feature Description:	*The entrance to an interconnected series of naturally occurring subterranean chambers.*
Inclusion Condition:	*ALL REQUIRED*

Notes

Like point features which coalesce in clusters of 3 or more will be thinned to form a representative pattern.

Label feature as per hierarchy for topo type placement to south neatline corners reading left to right:
1. Position hierarchy: (a.) northeast (preferred position), (b.) southeast (1st alternative), (c.) northwest (2nd alternative, (d.) southwest (3rd alternative), (e.) top-centered (4th alternative), (f.) bottom-centered (5th alternative): (Hierarchy is based on type positioning so as to avoid overprinting other type or obscuring detail.);
2. Minimum space between type placement and feature symbol is 0.5 mm;
3. This method of type placement shall be used for areal features when space does not permit labeling within that feature.

Type for features on land shall be positioned on land if possible. If type must be placed in the water, it shall be positioned so it does not obscure or overprint hydrographic detail. Type for features in the water shall be positioned in the water. If possible, type shall not be positioned across the shoreline.

Descriptive terms, e.g., "Canal" shall be shown if the name is not known. If the descriptive word appears in the name, for example, "PANAMA CANAL". The descriptive type shall not be shown, i.e. do not show "Panama Canal Canal".

The triangle part of the symbol (PT_Black_CaveMouth) shall mark the location of the entrance, and the stem of the symbol shall extend in the same direction as the Cave.

Feature Information

Specification:	*LTDS*	**Feature Origin:**	*Topographic*	**Symbol:**	PT_Black_CaveMouth
Symbol Criteria:	All			**Label:**	[Geographic Name Information : Full Name], 6 point Condensed Mixed case if available
				Color:	Black (58600)

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Cemetery (AL030)

Surface

Feature Description:	*A site and associated structures devoted to the burial of the dead.*
Inclusion Condition:	*Area >= 15,625 sq m.*

Notes

Coincident similar area features will be blended to form a single feature.

Area and line features will be generalized to detail compatible with scale.

Type for features on land shall be positioned on land if possible. If type must be placed in the water, it shall be positioned so it does not obscure or overprint hydrographic detail. Type for features in the water shall be positioned in the water. If possible, type shall not be positioned across the shoreline.

Descriptive terms, e.g., "Canal" shall be shown if the name is not known. If the descriptive word appears in the name, for example, "PANAMA CANAL". The descriptive type shall not be shown, i.e. do not show "Panama Canal Canal".

The limiting outline of the Built-up Area tint shall be dropped when it overprints linear features (Streams, Roads, and Railroads, etc), or if the space between the symbols is < 0.5mm.

If a clearing exists inside of the area feature, and the size of the clearing is equal to or greater than the area for the surrounding area feature, the clearing is shown as an open space inside the surround feature. If the area of the clearing is less than the area for the surrounding feature, the clearing is deleted and absorbed into the surrounding area feature.

If two area features with the same feature code do not connect at any point, and have a space between them less than 2.5 mm at map/chart scale, delete the open space that is less than 2.5 mm between the features and combine them into one area feature.

If a portion of an area feature has a minimum width of less than 2.5 mm at map/chart scale, delete that portion of the area feature that is not at least 2.5 mm wide, measured from perimeter to perimeter. If the deletion of the portion of the area based on the above criteria will reduce the area of the remaining portion of the area feature to below the minimum area inclusion, do not delete the narrow portion of the feature that is less than 2.5 mm wide. If the deletion of the portion of the area based on the above criteria will split two larger areas connected by a narrow strip into two separate areas, either of which would be below minimum area inclusion, do not delete the narrow portion of the feature that is less than 2.5 mm wide.

Feature Information

Specification:	*LTDS*	**Feature Origin:**	*Topographic*	**Symbol:**	AR_Black_Cemetery
Symbol Criteria:	(Religious Information : Religious Designation: [Hinduism] or [Other] or [No Information])			**Label:**	[Geographic Name Information: Full Name], 6 point Condensed Mixed case if available
				Color:	Black (58600)

Feature Information

Specification:	*LTDS*	**Feature Origin:**	*Topographic*	**Symbol:**	AR_Black_Cemetery_Christian
Symbol Criteria:	(Religious Information : Religious Designation: [Roman Catholic] or [Christian] or [Orthodox] or [Protestant] or [Nestorian] or [Chaldean])			**Label:**	[Geographic Name Information: Full Name], 6 point Condensed Mixed case if available
				Color:	Black (58600)

Feature Information

Specification:	*LTDS*	**Feature Origin:**	*Topographic*	**Symbol:**	AR_Black_Cemetery_Buddhism
Symbol Criteria:	(Religious Information : Religious Designation: [Buddhism])			**Label:**	[Geographic Name Information: Full Name], 6 point Condensed Mixed case if available
				Color:	Black (58600)

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Cemetery (AL030) *(Continued)*

Surface

Feature Information					
Specification:	*LTDS*	**Feature Origin:**	*Topographic*	**Symbol:**	AR_Black_Cemetery_Islam
Symbol Criteria:	(Religious Information : Religious Designation: [Islam] or [Shia] or [Sunni])			**Label:**	[Geographic Name Information: Full Name], 6 point Condensed Mixed case if available
				Color:	Black (58600)

Feature Information					
Specification:	*LTDS*	**Feature Origin:**	*Topographic*	**Symbol:**	AR_Black_Cemetery_Judaism
Symbol Criteria:	(Religious Information : Religious Designation: [Judaism])			**Label:**	[Geographic Name Information: Full Name], 6 point Condensed Mixed case if available
				Color:	Black (58600)

Feature Information					
Specification:	*LTDS*	**Feature Origin:**	*Topographic*	**Symbol:**	AR_Black_Cemetery_Shinto
Symbol Criteria:	(Religious Information : Religious Designation: [Shinto])			**Label:**	[Geographic Name Information: Full Name], 6 point Condensed Mixed case if available
				Color:	Black (58600)

Cemetery (AL030)

Point

Inclusion Condition:	*Area < 15,625 sq m.*

Notes

If one symbol coalesces with another symbol for the same type feature, displace symbols to allow a minimum separation of 0.21mm.

A cluster of 3 or more coalescing similar point features will be aggregated when an area delineation is supported by the product.

Feature Information					
Specification:	*LTDS*	**Feature Origin:**	*Topographic*	**Symbol:**	PT_Black_Cemetery_Unknown
Symbol Criteria:	(Religious Information : Religious Designation: [Hinduism] or [Other] or [No Information])			**Label:**	[Geographic Name Information: Full Name], 6 point Condensed Mixed case if available
				Color:	Black (58600)

Feature Information					
Specification:	*LTDS*	**Feature Origin:**	*Topographic*	**Symbol:**	PT_Black_Cemetery_Christian
Symbol Criteria:	(Religious Information : Religious Designation: [Roman Catholic] or [Christian] or [Orthodox] or [Protestant] or [Nestorian] or [Chaldean])			**Label:**	[Geographic Name Information: Full Name], 6 point Condensed Mixed case if available
				Color:	Black (58600)

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Cemetery (AL030) *(Continued)*

Point

Feature Information					
Specification:	*LTDS*	**Feature Origin:**	*Topographic*	**Symbol:**	PT_Black_Cemetery_Buddhism
Symbol Criteria:	(Religious Information : Religious Designation: [Buddhism])			**Label:**	[Geographic Name Information: Full Name], 6 point Condensed Mixed case if available
				Color:	Black (58600)

Feature Information					
Specification:	*LTDS*	**Feature Origin:**	*Topographic*	**Symbol:**	PT_Black_Cemetery_Islam
Symbol Criteria:	(Religious Information : Religious Designation: [Islam] or [Shia] or [Sunni[)			**Label:**	[Geographic Name Information: Full Name], 6 point Condensed Mixed case if available
				Color:	Black (58600)

Feature Information					
Specification:	*LTDS*	**Feature Origin:**	*Topographic*	**Symbol:**	PT_Black_Cemetery_Judaism
Symbol Criteria:	(Religious Information : Religious Designation: [Judaism])			**Label:**	[Geographic Name Information: Full Name], 6 point Condensed Mixed case if available
				Color:	Black (58600)

Feature Information					
Specification:	*LTDS*	**Feature Origin:**	*Topographic*	**Symbol:**	PT_Black_Cemetery_Shinto
Symbol Criteria:	(Religious Information : Religious Designation: [Shinto])			**Label:**	[Geographic Name Information: Full Name], 6 point Condensed Mixed case if available
				Color:	Black (58600)

Channel (BH191)

Surface

Feature Description:	*That part of a body of water, either ocean or inland, (sometimes dredged) deep enough for navigation through an area otherwise not navigable.*
Inclusion Condition:	*Width >= 50 meters*

Notes

Intermittent Channel Rules:
Intermittent rivers are those that are dry most of the time. The symbol shall be a dashed line. Where both banks can be shown, or where, in the case of 'braided' rivers, the normal flow does not fill the river bed but is carried in a number of small channels, the bands and intermediate channels are each to be shown by dashed lines. Land tint is to be carried across such rivers.

Channel General Rules:
Channels may be entirely natural features such as passages between islands or sandbanks; apart from naming them, no other chart action should be required. Channels may also be artificially marked, and sometimes deepened. Such channels may have their outer limits shown, usually in part only, by:

- *transits or 'clearing lines';*
- *light sectors or direction lights;*
- *lateral buoys or beacons;*
- *dredged area or swept area limits.*

It should not usually be necessary to chart the outer limits of a channel, as these should be defined by the appropriate symbols above. However, if the outer limits of a channel are not clear from natural or artificial features, but a regulatory authority requires or recommends certain vessels to navigate within defined limits, they should be charted by the appropriate Routeing Measure symbol.

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Channel (BH191) *(Continued)*

Surface

Notes

Fairway Rules:
A Fairway designated by a regulatory authority must be delimited by bold magenta dashed lines (M15). The ends of the fairway should be closed. Sections of different minimum depth or maximum draught should be separated by the same limit symbol (M15). The M18 symbol is the equivalent Surface (Area) feature as M15. The magenta legend FAIRWAY (or equivalent) or its name should be inserted if possible, in the area and parallel with the channel's limits. Soundings and depth contours should be included as appropriate. Fairways may consist partly or entirely of dredged or maintained areas. If the limits of the Fairway coincide with the limits of a dredged or maintained area, or light sector limit, the usual cartographic principles apply, see below. A Fairway may be highlighted with grey tint. Regulated or recommended direction of traffic flow arrows may be added, as with a two-way route. A note may be added to provide further guidance, eg: classes of vessel recommended to use the fairway, survey quality and date, an explanation of the maximum draught symbol. Alternatively, where a leading line lies within a fairway, the maximum authorized (or recommended) draught may be shown on the leading line (M6).

Cartographic Principles:
For coincident limits, the limit symbol (line) portraying the area which is considered to be potentially the most dangerous to navigation (usually bounding permanent physical obstructions) has priority. In general, restrictive limits have precedence over non-restrictive limits, other limits being broken accordingly. However, where an important area feature includes restrictions, the restrictions may be shown by an associated note or one or more point symbols within the area (eg for a designated fairway, an anchoring prohibited symbol could be inserted within the fairway without its associated limit). The colour hierarchy is generally: black, magenta, green, other.

Dredged Channel Rules:
If space does not allow for the full legend to be shown, labels for Dredged Channels are condensed in the following order:
(1.) Delete "Dredged to" or "Maintained depth" labels first,
(2.) Delete Date label and parentheses second,
(3.) Delete description of Aids to Navigation label third. Do not delete Hydrographic depth label or the "m" from any Dredged Channel feature

Dredged channels/areas shall be tinted in accordance with their depths, following the guidance for generation of water tint provided in rules for depth curves and/or open water.

Show Hydrographic Depth or Hydrographic Height values in whole meters if decimeter value is 0; otherwise show the decimeter value as a subscript to the meter value.

When the boundary of an areal symbol crosses or overprints the shoreline, the boundary overprinting the shoreline or land is deleted.

If (Maritime Navigation Marked: [True]), Description of Aids to Navigation shall be shown if one of the following conditions occur:
(1.) The aids to navigation that mark the feature do not meet the inclusion conditions for aids to navigation.
(2.) The aids to navigation that mark the feature are known to exist, but the details, such as position, type of aid, etc, are not sufficient to chart them as aids to navigation

Symbol perimeter shall be broken where ship traffic enters and exits the feature. Feature boundary is symbolized only on those edges where ship traffic does not enter or exit.

Feature Information					
Specification:	*Chart #1*	**Feature Origin:**	*Hydrographic*	**Symbol:**	C21
Symbol Criteria:	(Channel Type: [River Channel]) and (Hydrologic Persistence: [Intermittent] or [Ephemeral] or [Dry])			**Label:**	None
				Color:	N/A

Feature Information					
Specification:	*Chart #1*	**Feature Origin:**	*Hydrographic*	**Symbol:**	C20
Symbol Criteria:	(Channel Type: [River Channel]) and (Hydrologic Persistence: [Perennial])			**Label:**	[Geographic Name Information : Full Name], 8 point Mixed case Italic if available
				Color:	Black (58600)

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Channel (BH191) *(Continued)*

Surface

Feature Information					
Specification:	*Chart #1*	**Feature Origin:**	*Hydrographic*	**Symbol:**	M18
Symbol Criteria:	(Channel Type: [Marine Channel]) and ((Maintained: [True]) or (Dredged: [True])			**Label:**	"FAIRWAY [Depth Range (lower value)]m (see Note)", 8 point Mixed Case Italic if ([Depth Range (lower value)]) <> Null; "FAIRWAY <[Depth Range (upper value)]m> (see Note)", 8 point Mixed Case Italic if ([Depth Range (upper value)]) <> Null; "SAFETY FAIRWAY", 8 point upper case Italic if ([Depth Range (upper value)] = Null) and ([Depth Range (lower value)] = Null)
				Color:	Purple (96532)

Channel (BH191)

Curve

Inclusion Condition:	*Width < 50 meters*

Notes

Intermittent Channel Rules:
Intermittent rivers are those that are dry most of the time. The symbol shall be a dashed line. Where both banks can be shown, or where, in the case of 'braided' rivers, the normal flow does not fill the river bed but is carried in a number of small channels, the bands and intermediate channels are each to be shown by dashed lines. Land tint is to be carried across such rivers.

Feature Information					
Specification:	*Chart #1*	**Feature Origin:**	*Hydrographic*	**Symbol:**	C21
Symbol Criteria:	(Channel Type: [River Channel]) and (Hydrologic Persistence: [Intermittent] or [Ephemeral] or [Dry])			**Label:**	None
				Color:	N/A

Feature Information					
Specification:	*Chart #1*	**Feature Origin:**	*Hydrographic*	**Symbol:**	C20
Symbol Criteria:	(Channel Type: [River Channel]) and (Hydrologic Persistence: [Perennial])			**Label:**	[Geographic Name Information : Full Name], 8 point Mixed case Italic if available
				Color:	Black (58600)

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Checkpoint (AH070)

Point

Feature Description:	*An official place to register, declare and/or inspect goods and/or people.*
Inclusion Condition:	*All required*

Notes

None

Feature Information					
Specification:	*LTDS*	**Feature Origin:**	*Topographic*	**Symbol:**	PT_Black_Checkpoint
Symbol Criteria:	All			**Label:**	[Geographic Name Information : Full Name], 6 point Condensed Mixed case if available
				Color:	Black (58600)

Cistern (BI010)

Point

Feature Description:	*A man-made container used for the collection and/or storage of water.*
Inclusion Condition:	*All required*

Notes

The feature (when Height Above Surface Level <= 46 meters or when height is not known) shall be oriented perpendicular (90 degrees) to the nearby road (AP030), cart track (AP010), trail (AP050), or railroad (AN010).

Like point features which coalesce in clusters of 3 or more will be thinned to form a representative pattern.

Feature Information					
Specification:	*LTDS*	**Feature Origin:**	*Topographic*	**Symbol:**	PT_Cyan_Cistern
Symbol Criteria:	All			**Label:**	None
				Color:	N/A

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Cleared Way (EC040)

Surface

Feature Description:	*A man-made cleared strip through a vegetated region.*
Inclusion Condition:	*All required*

Notes

None

Feature Information					
Specification:	*LTDS*	**Feature Origin:**	*Topographic*	**Symbol:**	No Symbology needed
Symbol Criteria:	All			**Label:**	None
				Color:	N/A

Conservation Area (FA210)

Surface

Feature Description:	*An area of land and/or sea especially dedicated to the protection and maintenance of biological diversity, and of natural and associated cultural resources, and managed through legal or other effective means.*
Inclusion Condition:	*All required*

Notes

None

Feature Information					
Specification:	*LTDS*	**Feature Origin:**	*Topographic*	**Symbol:**	AR_Black_ConservationArea
Symbol Criteria:	All			**Label:**	[Geographic Name Information : Full Name] [Conservation Area Management Category], 6-16 point Condensed Mixed case if (Conservation Area Management Category: [Strict Nature Reserve] or [Wilderness Area] or [National Park] or [Natural Monument] or [Habitat and/or Species Management Area] or [Habitat Management Area] or [Species Management Area] or [Breeding Ground] or [Protected Landscape or Seascape] or [Managed Resource Protected Area] or [Managed Forest Protected Area]); [Geographic Name Information : Full Name] "Conservation area", 6-16 point Condensed Mixed case if (Conservation Area Management Category: [Other] or [No Information])
				Color:	Black (58600)

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Contact Zone (FA041)

Surface

Feature Description:	*A terrain strip 3 metres wide limited on the water side by the contact between land and water (under average water level conditions).*
Inclusion Condition:	*All required*

Notes

None

Feature Information					
Specification:	*Other*	**Feature Origin:**	*Other*	**Symbol:**	TBD
Symbol Criteria:	TBD			**Label:**	TBD
				Color:	TBD

Contaminated Region (FA012)

Surface

Feature Description:	*A region whose prevailing natural conditions have been degraded through contamination by harmful or objectionable substances.*
Inclusion Condition:	*Area >= 15,625 sq m*

Notes

None

Feature Information					
Specification:	*Other*	**Feature Origin:**	*Other*	**Symbol:**	TBD
Symbol Criteria:	TBD			**Label:**	TBD
				Color:	TBD

Contaminated Region (FA012)

Point

Inclusion Condition:	*Area < 15,625 sq m*

Notes

None

Feature Information					
Specification:	*Other*	**Feature Origin:**	*Other*	**Symbol:**	TBD
Symbol Criteria:	TBD			**Label:**	TBD
				Color:	TBD

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Control Point (ZB035)

Point

Feature Description:	*A terrain object (for example: a marker) with a known location (for example: horizontal position, elevation, or both) intended to serve as a survey control.*
Inclusion Condition:	*ALL REQUIRED*

Notes

The preferred position of elevation values for horizontal control points are as indicated below, in descending order of preference: Preferred: Bottom right side (southeast) corner of triangle symbol.
Second: Right side (east) of triangle adjacent to and centered on dot center point.
Third: Top left side (northwest) corner, adjacent to triangle symbol.
Fourth: Bottom left side (southwest) corner of triangle symbol.

When control points and bench marks have a name or number to identify them (such as Station 16, or STA 116), and an elevation value, the name or number and elevation are positioned in the following order of precedence.
Preferred: Station name on top left side (northwest) corner of triangle symbol, and elevation value on bottom right (southeast) corner, adjacent to the apex of the triangle symbol.
Second: Stacked, name and values, centered on the right side of the triangle symbol.
Third: Same as second, except to the left side of the triangle symbol.
Fourth: Station name is centered to the left side on the dot of the triangle symbol, with the elevation value centered on the same line as the station name, but to the right side of the triangle symbol.

Control Points shall not be shown < 75 mm apart. In areas of high concentration of points, (more than one every 75 mm), the points of the higher order of preference will be shown no less than 75 mm or more than 125 mm apart.
The order of preference is –
(1.) trig stations,
(2.) Bench marks,
(3.) spot heights.

Feature Information

Specification:	*LTDS*	**Feature Origin:**	*Topographic*	**Symbol:**	PT_Black_ControlPoint_ Horizontal
Symbol Criteria:	(Survey Control Point Type: [Horizontal Position] or [Horizontal Position and Benchmark])			**Label:**	[Geographic Name Information : Full Name], 6 point Condensed Mixed case if available "BM", 6 point Condensed Upper case "([Base Elevation])", 6 point Condensed Lower case if (Survey Control Point Type: [Horizontal Position and Benchmark]); [Geographic Name Information : Full Name], 6 point Condensed Mixed case if available if (Survey Control Point Type: [Horizontal Position])
				Color:	Black (58600)

Feature Information

Specification:	*LTDS*	**Feature Origin:**	*Topographic*	**Symbol:**	PT_Black_ControlPoint_Vertical
Symbol Criteria:	(Survey Control Point Type: [Vertical Position])			**Label:**	[Geographic Name Information : Full Name], 6 point Condensed Mixed case if available "BM", 6 point Condensed Upper case "([Base Elevation])", 6 point Condensed Lower case
				Color:	Black (58600)

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Control Tower (AQ060)

Surface

Feature Description:	*A structure that houses personnel and equipment used to control the flow of traffic within a specified range of an installation.*
Inclusion Condition:	*Length or Diameter >=25 m.*

Notes

Type for features on land shall be positioned on land if possible. If type must be placed in the water, it shall be positioned so it does not obscure or overprint hydrographic detail. Type for features in the water shall be positioned in the water. If possible, type shall not be positioned across the shoreline.

If coalescing features being thinned are a mix of heights, with some < 46 meters and some >= 46 meters, then only the obstruction symbol shall be shown.

When obstructions coalesce at map scale, label with highest obstruction information.

Feature Information

Specification:	*LTDS*	**Feature Origin:**	*Topographic*	**Symbol:**	AR_Black_ControlTower
Symbol Criteria:	(Height Above Surface Level < 46) OR (Area >= 5625 and Height Above Surface Level >= 46)			**Label:**	"Control tower", 6 point Condensed Mixed case
				Color:	Black (58600)

Control Tower (AQ060)

Point

Inclusion Condition:	*Length or Diameter < 25 m.*

Notes

Type for features on land shall be positioned on land if possible. If type must be placed in the water, it shall be positioned so it does not obscure or overprint hydrographic detail. Type for features in the water shall be positioned in the water. If possible, type shall not be positioned across the shoreline.

If coalescing features being thinned are a mix of heights, with some < 46 meters and some >= 46 meters, then only the obstruction symbol shall be shown.

When obstructions coalesce at map scale, label with highest obstruction information.

Feature Information

Specification:	*LTDS*	**Feature Origin:**	*Topographic*	**Symbol:**	PT_Black_ControlTower
Symbol Criteria:	Height Above Surface Level < 46 m.			**Label:**	"Control tower", 6 point Condensed Mixed case
				Color:	Black (58600)

Feature Information

Specification:	*LTDS*	**Feature Origin:**	*Topographic*	**Symbol:**	PT_Blue072_VO
Symbol Criteria:	Height Above Surface Level >= 46 m.			**Label:**	"Control tower", 6 point Condensed Mixed case; [Highest Elevation], 6 point Condensed Bold Mixed case; "([Height Above Surface Level])", 6 point Condensed Mixed case
				Color:	Blue (46351)

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Conveyor (AF020)

Curve

Feature Description:	*A mechanical device for conveying articles or materials during manufacture or processing using an endless moving belt or series of rollers.*
Inclusion Condition:	*Length or Diameter >= 375 m.*

Notes

Area and line features will be generalized to detail compatible with scale.

Type for features on land shall be positioned on land if possible. If type must be placed in the water, it shall be positioned so it does not obscure or overprint hydrographic detail. Type for features in the water shall be positioned in the water. If possible, type shall not be positioned across the shoreline.

The Conveyor symbol shall only be shown outside of a Built-up area tint, and begin and end at another symbolized feature.

Feature Information

Specification:	*LTDS*	**Feature Origin:**	*Topographic*	**Symbol:**	LN_Black_Conveyor
Symbol Criteria:	All			**Label:**	"Conveyor", 6 point Condensed Mixed case
				Color:	Black (58600)

Cooling Facility (AD055)

Surface

Feature Description:	*A facility for the removal of thermal energy (for example: by generating and circulating chilled water) for cooling purposes.*
Inclusion Condition:	*Area: >= 15,625 sq m.*

Notes

None

Feature Information

Specification:	*LTDS*	**Feature Origin:**	*Topographic*	**Symbol:**	AR_Black_CoolingFacility
Symbol Criteria:	All			**Label:**	[Geographic Name Information : Full Name] "Cooling facility", 6 point Condensed Mixed case
				Color:	Black (58600)

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Cooling Facility (AD055)

Point

Inclusion Condition:	*Area < 15,625sq.m*

Notes

None

Feature Information					
Specification:	*LTDS*	**Feature Origin:**	*Topographic*	**Symbol:**	PT_Black_CoolingFacility
Symbol Criteria:	All			**Label:**	[Geographic Name Information : Full Name] "Cooling facility", 6 point Condensed Mixed case
				Color:	Black (58600)

Cooling Tower (AF030)

Point

Feature Description:	*A tall tower for cooling hot water from an industrial process before reuse.*
Inclusion Condition:	*All Required*

Notes

If one symbol coalesces with another symbol for the same type feature, displace symbols to allow a minimum separation of 0.21mm.

Type for features on land shall be positioned on land if possible. If type must be placed in the water, it shall be positioned so it does not obscure or overprint hydrographic detail. Type for features in the water shall be positioned in the water. If possible, type shall not be positioned across the shoreline.

When obstructions coalesce at map scale, label with highest obstruction information.

If >= 3 equal symbols would coalesce at map scale, portray with a representative pattern.

Feature Information					
Specification:	*LTDS*	**Feature Origin:**	*Topographic*	**Symbol:**	PT_Black_CoolingTower
Symbol Criteria:	Height Above Surface Level: < 46 meters			**Label:**	"Cooling tower", 6 point Condensed Mixed case
				Color:	Black (58600)

Feature Information					
Specification:	*LTDS*	**Feature Origin:**	*Topographic*	**Symbol:**	PT_Blue072_VO
Symbol Criteria:	Height Above Surface Level: >= 46 meters			**Label:**	"Cooling tower", 6 point Condensed Mixed case; [Highest Elevation], 6 point Condensed Bold Mixed case; "([Height Above Surface Level])", 6 point Condensed Mixed case
				Color:	Blue (46351)

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Crane (AF040)

Point

Feature Description:	*Equipment for lifting, shifting, and lowering objects or materials by means of a swinging boom or with the lifting apparatus supported on an overhead track.*
Inclusion Condition:	*All Required*

Notes

If one symbol coalesces with another symbol for the same type feature, displace symbols to allow a minimum separation of 0.21mm.

Type for features on land shall be positioned on land if possible. If type must be placed in the water, it shall be positioned so it does not obscure or overprint hydrographic detail. Type for features in the water shall be positioned in the water. If possible, type shall not be positioned across the shoreline.

When obstructions coalesce at map scale, label with highest obstruction information.

If >= 3 equal symbols would coalesce at map scale, portray with a representative pattern.

Feature Information

Specification:	*LTDS*	**Feature Origin:**	*Topographic*	**Symbol:**	PT_Black_Crane
Symbol Criteria:	(Height Above Surface Level < 46) and (Location Referenced to Shoreline: [Inland] or [At Shoreline] or [No Information])			**Label:**	None
				Color:	N/A

Feature Information

Specification:	*LTDS*	**Feature Origin:**	*Topographic*	**Symbol:**	PT_Black_Crane_Offshore
Symbol Criteria:	(Height Above Surface Level < 46) and (Location Referenced to Shoreline: [Offshore])			**Label:**	"Crane", 6 point Condensed Mixed case; "([Height Above Surface Level])", 6 point Condensed Mixed case
				Color:	Black (58600)

Feature Information

Specification:	*LTDS*	**Feature Origin:**	*Topographic*	**Symbol:**	PT_Blue072_VO
Symbol Criteria:	Height Above Surface Level: >= 46 meters			**Label:**	"Crane", 6 point Condensed Mixed case; [Highest Elevation], 6 point Condensed Bold Mixed case; "([Height Above Surface Level])", 6 point Condensed Mixed case
				Color:	Blue (46351)

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Crevasse (BJ031)

Surface

Feature Description:	*A deep crack or fissure in a glacier that results from differential movement of ice.*
Inclusion Condition:	*Length or Diameter >= 250 m. AND Width >= 50 m.*

Notes

None

Feature Information

Specification:	*LTDS*	**Feature Origin:**	*Topographic*	**Symbol:**	AR_Cyan_Crevasse
Symbol Criteria:	All			**Label:**	"Crevasse", 6-10 point Condensed Mixed case Italic
				Color:	Blue (46351)

Crevasse (BJ031)

Curve

Inclusion Condition:	*Length or Diameter >= 250 m. AND Width < 50 m.*

Notes

None

Feature Information

Specification:	*LTDS*	**Feature Origin:**	*Topographic*	**Symbol:**	LN_Cyan_Crevasse
Symbol Criteria:	All			**Label:**	"Crevasse", 6 point Condensed Mixed case Italic
				Color:	Blue (46351)

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Crevice (DB061)

Surface

Feature Description:	*A narrow opening or fissure produced by a crack in the land, especially in rock.*
Inclusion Condition:	*Length or Diameter >= 250 m. AND Width >= 50 m.*

Notes

When any portion of the area feature does not meet the minimum geometric inclusion condition and line delineation for the feature is supported on the product, the area feature will be partially collapsed.

Coincident similar area features will be blended to form a single feature.

Area and line features will be generalized to detail compatible with scale.

Feature will be generalized to provide a more aesthetic contoured feature (i.e., smoothed).

Label feature as per hierarchy for topo type placement to south neatline corners reading left to right:
1. Position hierarchy: (a.) northeast (preferred position), (b.) southeast (1st alternative), (c.) northwest (2nd alternative, (d.) southwest (3rd alternative), (e.) top-centered (4th alternative), (f.) bottom-centered (5th alternative): (Hierarchy is based on type positioning so as to avoid overprinting other type or obscuring detail.);
2. Minimum space between type placement and feature symbol is 0.5 mm;
3. This method of type placement shall be used for areal features when space does not permit labeling within that feature.

Feature Information

Specification:	*LTDS*	**Feature Origin:**	*Topographic*	**Symbol:**	AR_Dk-Brown1815_Crevice
Symbol Criteria:	All			**Label:**	"Crevice", 6-10 point Condensed Italics Mixed case
				Color:	Black (58600)

Crevice (DB061)

Curve

Inclusion Condition:	*Length or Diameter >= 250 m. AND Width < 50 m.*

Notes

Area and line features will be generalized to detail compatible with scale.

Feature will be generalized to provide a more aesthetic contoured feature (i.e., smoothed).

Label line feature above (preferred) and parallel to the line with a 0.5 mm space between. Above means: readable from south or east Projection neatline.

Feature Information

Specification:	*LTDS*	**Feature Origin:**	*Topographic*	**Symbol:**	LN_Dk-Brown1815_Crevice
Symbol Criteria:	All			**Label:**	"Crevice", 6 point Condensed Italics Mixed case
				Color:	Black (58600)

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Crib (BD020)

Surface

Feature Description:	*A permanent marine structure usually designed to support or elevate pipelines; especially a structure enclosing a screening device at the offshore end of a potable water intake pipe.*
Inclusion Condition:	*Length >= 50 meters*

Notes

Diffusers and cribs at the end of pipes, and templates, manifolds and other underwater installations associated with pipelines should be charted in the same way as other obstructions, either with the abbreviation 'obstn' , 'Subm Crib', or an appropriate legend, eg 'Diffuser', 'Manifold'. Crib (BD020) shall be labeled "Subm Crib".

Use the following abbreviations for Horizontal Accuracy and Existence values:
If Position Approximate, label "PA";
If Position Doubtful, label "PD";
If Existence Doubtful, label "ED";
If Reported, label "Rep".

Date shall only be shown if Existence = Reported. Date shall be shown in parentheses if available. Always show "Rep" and date on the same line.
Unless specifically indicated on source material as inaccurate or doubtful, etc., or modified by other override rules, features shall be assumed to be accurate and definite. This rule applied to accuracy and existence of unique features, and is not related to the accuracy of the source as a whole.

Following IHO practice, features whose positions are approximate are labeled "PA" on the source charts. Features whose positions are doubtful are labeled "PD" on the source charts. Features whose existence is doubtful are labeled "ED" on the source charts. Features that have only been reported are labeled "Rep" on the source charts, and usually show the date of report in parentheses.

Definitions:
PA - Position Approximate = The position has not been accurately determined, or does not remain fixed.;
PD - Position Doubtful = The feature has been reported in various positions, but not definitely determined to be in any.;
ED - Existence Doubtful = Indicates the possible existence of the feature, the actual existence of which has not been established.

Show Hydrographic Depth or Hydrographic Height values in whole meters if decimeter value is 0; otherwise show the decimeter value as a subscript to the meter value.

Date shall only be shown if Existence = Reported. Date shall be shown in parentheses if available. Always show "Rep" and date on the same line.

Priority for type placement Rules:
1-right center,
2-bottom center,
3-left center,
4-top center.
(A) Minimum distance from symbol - 1 mm.
(B) Maximum distance from symbol before choosing the next highest priority:
(#1) 4 mm measured to the West end;
(#2) 4 mm measured to the North side (top);
(#3) 4 mm measured to the East end;
(#4) 4 mm measured to the South side (bottom)

If symbols overprint each other, labels are condensed as follows:
(1.) If the labels are identical, only one is retained.;
(2,) If the labels are not identical, they shall be condensed in to one legend, e.g., "Fishhaven and Well". If multiple depths are shown, only the shallowest is retained.

Labels may be stacked, readable top to bottom to avoid overprints.

Type Preference Placement Rules:
(a.) Placed on one horizontal line centered in feature.;
(b.) Broken into multiple horizontal lines, centered in feature, with a 1 mm space between lines.;
(c.) Placed on one line outside feature. Use Priority for type placement Rules for placement, Hydrographic Depth shall always be positioned inside area.;
(d.) Placed on two lines to avoid overprints. Use Priority for type placement Rules. If present, Hydrographic Depth shall always be placed inside area.

Labels shall be placed in water area, and shall not overprint the shoreline or be placed on land or in the foreshore.

When two or more identical features fall within 10 mm of each other, one legend with plural description shall be shown, with type placed as close as possible to the center of the group without overprinting features.
Where symbols with the same feature code but different values of Hydrographic Vertical Positioning Information: Water Level Effect (i.e. Always Dry , Always Submerged, Covers and Uncovers)) are adjacent to each other, the boundary of the feature that is the highest in vertical plane shall be shown, and the boundary of the feature above water is shown whole, while the feature that covers and uncovers has a portion of its symbol perimeter that overprints the higher feature deleted.

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Crib (BD020) *(Continued)*

Surface

Notes

When two or more identical symbols overprint each other, delete the portions that overlap and show as one symbol with an outline of the combined area of the symbols.

Show land tint inside the symbol. If the Hydrographic Vertical Positioning Information: Water Level Effect: [Always Dry] show land tint only.

A blue 31% tint shall not overprint other blue 31% tints. If two blue tinted symbols, or one tinted symbol and water tint overprint, only one 31% tint shall be shown in the area.

Feature Information					
Specification:	*Chart #1*	**Feature Origin:**	*Hydrographic*	**Symbol:**	K40
Symbol Criteria:	(Hydrographic Vertical Positioning Information : Water Level Effect: [Always Submerged]) and (Hydrographic Vertical Positioning Information : Sounding Metadata : Bathymetric Measurement Quality Category: [Depth Unknown])			**Label:**	"Subm Crib", 6 point Mixed case, Italic
				Color:	Black (58600)

Crib (BD020)

Point

Inclusion Condition:	*Length < 50 meters*

Notes

Diffusers and cribs at the end of pipes, and templates, manifolds and other underwater installations associated with pipelines should be charted in the same way as other obstructions, either with the abbreviation 'obstn' , 'Subm Crib', or an appropriate legend, eg 'Diffuser', 'Manifold'. Crib (BD020) shall be labeled "Subm Crib".

Use the following abbreviations for Horizontal Accuracy and Existence values:
If Position Approximate, label "PA";
If Position Doubtful, label "PD";
If Existence Doubtful, label "ED";
If Reported, label "Rep".

Date shall only be shown if Existence = Reported. Date shall be shown in parentheses if available. Always show "Rep" and date on the same line.
Unless specifically indicated on source material as inaccurate or doubtful, etc., or modified by other override rules, features shall be assumed to be accurate and definite. This rule applied to accuracy and existence of unique features, and is not related to the accuracy of the source as a whole.

Following IHO practice, features whose positions are approximate are labeled "PA" on the source charts. Features whose positions are doubtful are labeled "PD" on the source charts. Features whose existence is doubtful are labeled "ED" on the source charts. Features that have only been reported are labeled "Rep" on the source charts, and usually show the date of report in parentheses.

Definitions:
PA - Position Approximate = The position has not been accurately determined, or does not remain fixed.;
PD - Position Doubtful = The feature has been reported in various positions, but not definitely determined to be in any.;
ED - Existence Doubtful = Indicates the possible existence of the feature, the actual existence of which has not been established.

Show Hydrographic Depth or Hydrographic Height values in whole meters if decimeter value is 0; otherwise show the decimeter value as a subscript to the meter value.

Date shall only be shown if Existence = Reported. Date shall be shown in parentheses if available. Always show "Rep" and date on the same line.

Priority for type placement Rules:
1-right center,
2-bottom center,
3-left center,
4-top center.
(A) Minimum distance from symbol - 1 mm.
(B) Maximum distance from symbol before choosing the next highest priority:
(#1) 4 mm measured to the West end;
(#2) 4 mm measured to the North side (top);
(#3) 4 mm measured to the East end;
(#4) 4 mm measured to the South side (bottom)

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Crib (BD020) ***(Continued)***

Point

Notes

If symbols overprint each other, labels are condensed as follows:
(1.) If the labels are identical, only one is retained.;
(2,) If the labels are not identical, they shall be condensed in to one legend, e.g., "Fishhaven and Well". If multiple depths are shown, only the shallowest is retained.

Labels may be stacked, readable top to bottom to avoid overprints.

Type Preference Placement Rules:
(a.) Placed on one horizontal line centered in feature.;
(b.) Broken into multiple horizontal lines, centered in feature, with a 1 mm space between lines.;
(c.) Placed on one line outside feature. Use Priority for type placement Rules for placement, Hydrographic Depth shall always be positioned inside area.;
(d.) Placed on two lines to avoid overprints. Use Priority for type placement Rules. If present, Hydrographic Depth shall always be placed inside area.

Labels shall be placed in water area, and shall not overprint the shoreline or be placed on land or in the foreshore.

When two or more identical features fall within 10 mm of each other, one legend with plural description shall be shown, with type placed as close as possible to the center of the group without overprinting features.
Where symbols with the same feature code but different values of Hydrographic Vertical Positioning Information: Water Level Effect (i.e. Always Dry , Always Submerged, Covers and Uncovers)) are adjacent to each other, the boundary of the feature that is the highest in vertical plane shall be shown, and the boundary of the feature above water is shown whole, while the feature that covers and uncovers has a portion of its symbol perimeter that overprints the higher feature deleted.

Feature Information					
Specification:	*Chart #1*	**Feature Origin:**	*Hydrographic*	**Symbol:**	K41
Symbol Criteria:	(Hydrographic Vertical Positioning Information : Water Level Effect: [Always Submerged]) and (Hydrographic Vertical Positioning Information : Sounding Metadata : Bathymetric Measurement Quality Category: [Depth Known])			**Label:**	[Hydrographic Vertical Positioning Information : Hydrographic Depth], 7 point lower case/ 5 point subscript; "Subm Crib", 6 point Mixed case, Italic
				Color:	Black (58600)

Feature Information					
Specification:	*Chart #1*	**Feature Origin:**	*Hydrographic*	**Symbol:**	K40
Symbol Criteria:	(Hydrographic Vertical Positioning Information : Water Level Effect: [Always Submerged]) and (Hydrographic Vertical Positioning Information : Sounding Metadata : Bathymetric Measurement Quality Category: [Depth Unknown])			**Label:**	"Subm Crib", 6 point Mixed case, Italic
				Color:	Black (58600)

Feature Information					
Specification:	*Chart #1*	**Feature Origin:**	*Hydrographic*	**Symbol:**	K2, K42
Symbol Criteria:	(Hydrographic Vertical Positioning Information : Water Level Effect: [Always Submerged]) and (Hydrographic Vertical Positioning Information : Sounding Metadata : Bathymetric Measurement Quality Category: [Depth Known]) and (Hydrographic Vertical Positioning Information : Bathymetric Measurement Technique: [Diver] or [Area Swept by Wire-drag])			**Label:**	[Hydrographic Vertical Positioning Information : Hydrographic Depth], 7 point lower case/ 5 point subscript; "Subm Crib", 6 point Mixed case, Italic
				Color:	Black (58600)

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Crib (BD020) ***(Continued)***

Point

Feature Information					
Specification:	*Chart #1*	**Feature Origin:**	*Hydrographic*	**Symbol:**	K3
Symbol Criteria:	(Hydrographic Vertical Positioning Information : Water Level Effect: [Always Submerged]) and (Hydrographic Vertical Positioning Information : Sounding Metadata : Bathymetric Measurement Quality Category: [Safe Clearance Known])			**Label:**	[Hydrographic Vertical Positioning Information : Hydrographic Depth], 7 point lower case/ 5 point subscript; "Subm Crib", 6 point Mixed case, Italic
				Color:	Black (58600)

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Crop Land (EA010)

Surface

Feature Description:	*An area that has been tilled for the planting of crops.*
Inclusion Condition:	*Area >= 15,625 sq m.*

Notes

Coincident similar area features will be blended to form a single feature.

Area and line features will be generalized to detail compatible with scale.

Feature will be generalized to provide a more aesthetic contoured feature (i.e., smoothed).

Label feature as per hierarchy for topo type placement to south neatline corners reading left to right:
1. Position hierarchy: (a.) northeast (preferred position), (b.) southeast (1st alternative), (c.) northwest (2nd alternative, (d.) southwest (3rd alternative), (e.) top-centered (4th alternative), (f.) bottom-centered (5th alternative): (Hierarchy is based on type positioning so as to avoid overprinting other type or obscuring detail.);
2. Minimum space between type placement and feature symbol is 0.5 mm;
3. This method of type placement shall be used for areal features when space does not permit labeling within that feature.

Names placement shall be oriented to the longest axis of the feature reading left to right and placed within the area outline and centered. If the longest axis is perpendicular to the south neatline, the type shall be placed outside of the area outline, preferred position is northeast of the feature (Rule L-3505), but may be placed at any position around the feature so as not to overprint any other feature type and reading left to right.

Type for features on land shall be positioned on land if possible. If type must be placed in the water, it shall be positioned so it does not obscure or overprint hydrographic detail. Type for features in the water shall be positioned in the water. If possible, type shall not be positioned across the shoreline.

Symbols and associated area patterns of underpassing features (except drainage shorelines) are broken for all bridges, except footbridges. This rule does not apply to land tint on Littoral Planning Charts.

If a clearing exists inside of the area feature, and the size of the clearing is equal to or greater than the area for the surrounding area feature, the clearing is shown as an open space inside the surround feature. If the area of the clearing is less than the area for the surrounding feature, the clearing is deleted and absorbed into the surrounding area feature.

If two area features with the same feature code do not connect at any point, and have a space between them less than 2.5 mm at map/chart scale, delete the open space that is less than 2.5 mm between the features and combine them into one area feature.

If a portion of an area feature has a minimum width of less than 2.5 mm at map/chart scale, delete that portion of the area feature that is not at least 2.5 mm wide, measured from perimeter to perimeter. If the deletion of the portion of the area based on the above criteria will reduce the area of the remaining portion of the area feature to below the minimum area inclusion, do not delete the narrow portion of the feature that is less than 2.5 mm wide. If the deletion of the portion of the area based on the above criteria will split two larger areas connected by a narrow strip into two separate areas, either of which would be below minimum area inclusion, do not delete the narrow portion of the feature that is less than 2.5 mm wide.

Apply the inclusion condition to cultivated land only when the project area or sheet is nearly devoid (<= 10%) of vegetation, or by special instruction.

Feature Information

Specification:	*LTDS*	**Feature Origin:**	*Topographic*	**Symbol:**	AR_Dk-Brown1815-7_Cropland
Symbol Criteria:	All			**Label:**	"Terraces", 6 point Condensed Mixed case if Crop Information : Farming Pattern [1] = [Terraced]
				Color:	Black (58600)

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Culvert (AQ065)

Curve

Feature Description:	*An enclosed channel for carrying a watercourse (for example: a stream, a sewer, or a drain) under a route (for example: a road, a railway, or an embankment).*
Inclusion Condition:	*ALL REQUIRED*

Notes

The supporting feature shall be aligned with a Cart Track, Trail, RR Track, and RR Siding/RR Spur.

Orientation of the culvert symbol is with the headline parallel with the over passing feature, and centered on the drain if possible.

Omit from Built-up Area.

Feature Information

Specification:	*LTDS*	**Feature Origin:**	*Topographic*	**Symbol:**	LN_Black_Culvert
Symbol Criteria:	All			**Label:**	None
				Color:	N/A

Cut (DB070)

Curve

Feature Description:	*An excavation in the terrain to provide passage for a land or water transportation route (for example: a road, a railway, and/or a canal).*
Inclusion Condition:	*Length or Diameter >= 125 m. AND Predominant Feature Depth >= 3 m.*

Notes

This is a non-symbolized feature. It is extracted down the centerline of the overall "Cut" and can be used to help orient the Cut Lines. The individual Cut Lines will represent the required symbol.

Feature Information

Specification:	*LTDS*	**Feature Origin:**	*Topographic*	**Symbol:**	No Symbology
Symbol Criteria:	All			**Label:**	None
				Color:	N/A

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Cut Line (DB071)

Curve

Feature Description:	*The demarcation line between a cut and the surrounding land surface.*
Inclusion Condition:	*Length or Diameter >= 125 m.*

Notes

Area and line features will be generalized to detail compatible with scale.

Feature will be generalized to provide a more aesthetic contoured feature (i.e., smoothed).

Where a Cut Line or Fill coincides with a contour, the contour shall be suppressed. The cut Lines ticks shall point downhill towards the bottom of the cut.

Omit from Built-up Area.

When a Contour coalesces with a Bluff/Cliff, Escarpment, Crevice, Crevasse, Esker, Fault, or Rock Formation, the coalescing portion of the contour shall be omitted.

Show longest of line feature in ground truth position.

Feature Information					
Specification:	*LTDS*	**Feature Origin:**	*Topographic*	**Symbol:**	LN_Dk-Brown1815_Cutline_GreaterthanContour
Symbol Criteria:	Predominant Feature Height >= contour interval			**Label:**	None
				Color:	N/A

Feature Information					
Specification:	*LTDS*	**Feature Origin:**	*Topographic*	**Symbol:**	LN_Dk-Brown1815_Cutline_LessthanContour
Symbol Criteria:	Predominant Feature Height < contour interval			**Label:**	None
				Color:	N/A

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Dam (BI020)

Surface

Feature Description:	*A barrier constructed to hold back water and raise its level to form a reservoir or to prevent flooding.*
Inclusion Condition:	*ALL REQUIRED*

Notes

Contours will be adjusted to planimetric features.

Area and line features will be generalized to detail compatible with scale.

Label feature as per hierarchy for topo type placement to south neatline corners reading left to right:
1. Position hierarchy: (a.) northeast (preferred position), (b.) southeast (1st alternative), (c.) northwest (2nd alternative, (d.) southwest (3rd alternative), (e.) top-centered (4th alternative), (f.) bottom-centered (5th alternative): (Hierarchy is based on type positioning so as to avoid overprinting other type or obscuring detail.);
2. Minimum space between type placement and feature symbol is 0.5 mm;
3. This method of type placement shall be used for areal features when space does not permit labeling within that feature.

Descriptive terms, e.g., "Canal" shall be shown if the name is not known. If the descriptive word appears in the name, for example, "PANAMA CANAL". The descriptive type shall not be shown, i.e. do not show "Panama Canal Canal".

Dams across single line Streams without a back-up Lake/Pond shall not be shown.

Omit if not shown in conjunction with drainage feature.

If a feature is not associated with (touching) a river, omit the feature.

Feature Information

Specification:	*LTDS*	**Feature Origin:**	*Topographic*	**Symbol:**	AR_Black_Dam
Symbol Criteria:	Width >= 100 AND (Structural Material Type [1]: <> [Soil] or [Other] or [No information])			**Label:**	[Geographic Name Information : Full Name] "Dam" if available and if (Condition of Facility: [Abandoned] or [Dismantled] or [Proposed] or [Fully Functional] or [Partially Functional] or [Partially Destroyed] or [Damaged] or [Disused] or [Planned] or [Other] or [No Information]), 6 point Condensed Mixed case; [Geographic Name Information : Full Name] "Dam (construction)"if available and if (Condition of Facility: [Under Construction]), 6 point Condensed Mixed case; [Geographic Name Information : Full Name] "Dam (destroyed)" if available and if (Condition of Facility: [Destroyed] or [Ruined]), 6 point Condensed Mixed case
				Color:	Black (58600)

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Dam (BI020) *(Continued)*

Surface

Feature Information					
Specification:	*LTDS*	**Feature Origin:**	*Topographic*	**Symbol:**	AR_Dk-Brown1815_Dam
Symbol Criteria:	Width >= 100 AND (Structural Material Type [1]: [Soil] or [Other] or [No Information])			**Label:**	[Geographic Name Information : Full Name] "Dam" if available and if (Condition of Facility: [Abandoned] or [Dismantled] or [Proposed] or [Fully Functional] or [Partially Functional] or [Partially Destroyed] or [Damaged] or [Disused] or [Planned] or [Other] or [No Information]), 6 point Condensed Mixed case; [Geographic Name Information : Full Name] "Dam (construction)"if available and if (Condition of Facility: [Under Construction]), 6 point Condensed Mixed case; [Geographic Name Information : Full Name] "Dam (destroyed)" if available and if (Condition of Facility: [Destroyed] or [Ruined]), 6 point Condensed Mixed case
				Color:	Black (58600)

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Dam (BI020)

Curve

Inclusion Condition:	*ALL REQUIRED*

Notes

Contours will be adjusted to planimetric features.

Area and line features will be generalized to detail compatible with scale.

Label feature as per hierarchy for topo type placement to south neatline corners reading left to right:
1. Position hierarchy: (a.) northeast (preferred position), (b.) southeast (1st alternative), (c.) northwest (2nd alternative, (d.) southwest (3rd alternative), (e.) top-centered (4th alternative), (f.) bottom-centered (5th alternative): (Hierarchy is based on type positioning so as to avoid overprinting other type or obscuring detail.);
2. Minimum space between type placement and feature symbol is 0.5 mm;
3. This method of type placement shall be used for areal features when space does not permit labeling within that feature.

Descriptive terms, e.g., "Canal" shall be shown if the name is not known. If the descriptive word appears in the name, for example, "PANAMA CANAL". The descriptive type shall not be shown, i.e. do not show "Panama Canal Canal".

Dams across single line Streams without a back-up Lake/Pond shall not be shown.

Omit if not shown in conjunction with drainage feature.

If a feature is not associated with (touching) a river, omit the feature.

Feature Information

Specification:	*LTDS*	Feature Origin:	*Topographic*	Symbol:	LN_Black_Dam
Symbol Criteria:	Width < 100 AND Structural Material Type [1] <> [Soil]or [Other] or [No information] And (Transportation System Type = [No Transportation System] or [Pedestrian] or [Other] or [No Information])			Label:	"[Geographic Name Information : Full Name]" if available and "([Structural Material Type])" if Structural Material Type <> [Concrete] or [Masonry], 6 point Condensed Mixed case; "[Geographic Name Information : Full Name]" if available and "(construction)" if Condition of Facility = [Under Construction], 6 point Condensed Mixed case; "[Geographic Name Information : Full Name]" if available and "(destroyed)" if Condition of Facility = [Destroyed] or [Ruined], 6 point Condensed Mixed case
				Color:	Black (58600)

Feature Information

Specification:	*LTDS*	Feature Origin:	*Topographic*	Symbol:	LN_Dk-Brown1815_Dam
Symbol Criteria:	Width < 100 AND (Structural Material Type [1]: [Soil] or [Other] or [No Information]) And (Transportation System Type = [No Transportation System] or [Other] or [No Information])			Label:	"[Geographic Name Information : Full Name]" if available and "(construction)" if Condition of Facility = [Under Construction], 6 point Condensed Mixed case; "[Geographic Name Information : Full Name]" if available and "(destroyed)" if Condition of Facility = [Destroyed] or [Ruined], 6 point Condensed Mixed case
				Color:	Black (58600)

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Dam (BI020) *(Continued)*

Curve

Feature Information					
Specification:	*LTDS*	**Feature Origin:**	*Topographic*	**Symbol:**	LN_Black_Dam_Carrying Transportation
Symbol Criteria:	Width < 100 AND (Structural Material Type [1] <> [Soil] or [Other] or [No information]) AND (Transportation System Type [1] = [Railway] or [Road])			**Label:**	"[Geographic Name Information : Full Name]" if available and "([Structural Material Type])" if Structural Material Type <> [Concrete] or [Masonry], 6 point Condensed Mixed case; "[Geographic Name Information : Full Name]" if available and "(construction)" if Condition of Facility = [Under Construction], 6 point Condensed Mixed case; "[Geographic Name Information : Full Name]" if available and "(destroyed)" if Condition of Facility = [Destroyed] or [Ruined], 6 point Condensed Mixed case
				Color:	Black (58600)

Feature Information					
Specification:	*LTDS*	**Feature Origin:**	*Topographic*	**Symbol:**	LN_Dk-Brown1815_Dam_Carrying Transportation
Symbol Criteria:	Width < 100 AND (Structural Material Type [1] = [Soil] or [Other] or [No Information]) AND (Transportation System Type [1] = [Railway] or [Road])			**Label:**	"[Geographic Name Information : Full Name]" if available and "(construction)" if Condition of Facility = [Under Construction], 6 point Condensed Mixed case; "[Geographic Name Information : Full Name]" if available and "(destroyed)" if Condition of Facility = [Destroyed] or [Ruined], 6 point Condensed Mixed case
				Color:	Black (58600)

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Dam (BI020)

Point

Inclusion Condition:	*ALL REQUIRED*

Notes

The feature shall be oriented perpendicular (90 degrees) with respect to area drainage features (Ditch, River/Stream).

Contours will be adjusted to planimetric features.

The feature symbology shall be positioned such that the longest axis of the symbol is aligned coincident with the centerline of the associated road, railroad track, or RR siding/RR spur feature.

Label feature as per hierarchy for topo type placement to south neatline corners reading left to right:
1. Position hierarchy: (a.) northeast (preferred position), (b.) southeast (1st alternative), (c.) northwest (2nd alternative, (d.) southwest (3rd alternative), (e.) top-centered (4th alternative), (f.) bottom-centered (5th alternative): (Hierarchy is based on type positioning so as to avoid overprinting other type or obscuring detail.);
2. Minimum space between type placement and feature symbol is 0.5 mm;
3. This method of type placement shall be used for areal features when space does not permit labeling within that feature.

Descriptive terms, e.g., "Canal" shall be shown if the name is not known. If the descriptive word appears in the name, for example, "PANAMA CANAL". The descriptive type shall not be shown, i.e. do not show "Panama Canal Canal".

Omit if not shown in conjunction with drainage feature.

If a feature is not associated with (touching) a river, omit the feature.

Feature Information

Specification:	*LTDS*	Feature Origin:	*Topographic*	Symbol:	PT_Black_Dam
Symbol Criteria:	Length or Diameter < 75 AND (Structural Material Type [1]: <> [Soil] or [Other] or [No information])			Label:	"[Geographic Name Information : Full Name]" if available and "([Structural Material Type])" if Structural Material Type <> [Concrete] or [Masonry], 6 point Condensed Mixed case; "[Geographic Name Information : Full Name]" if available and "(construction)" if Condition of Facility = [Under Construction], 6 point Condensed Mixed case; "[Geographic Name Information : Full Name]" if available and "(destroyed)" if Condition of Facility = [Destroyed] or [Ruined], 6 point Condensed Mixed case
				Color:	Black (58600)

Feature Information

Specification:	*LTDS*	Feature Origin:	*Topographic*	Symbol:	PT_Dk-Brown1815_Dam
Symbol Criteria:	Length or Diameter < 75 AND (Structural Material Type [1] = [Soil])			Label:	"[Geographic Name Information : Full Name]" if available and "(construction)" if Condition of Facility = [Under Construction], 6 point Condensed Mixed case; "[Geographic Name Information : Full Name]" if available and "(destroyed)" if Condition of Facility = [Destroyed] or [Ruined], 6 point Condensed Mixed case
				Color:	Black (58600)

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Dam (BI020) *(Continued)*

Point

Feature Information					
Specification:	*LTDS*	**Feature Origin:**	*Topographic*	**Symbol:**	PT_Blue072_VO
Symbol Criteria:	Height Above Surface Level >= 46			**Label:**	"Dam", 6 point Condensed mixed case AND [Highest Elevation], 6 point Condensed Bold Mixed case AND "([Height Above Surface Level])", 6 point Condensed Mixed case
				Color:	Blue (46351)

Depth Area (BE019)

Surface

Feature Description:	*A water area whose depth is within a defined range of values.*
Inclusion Condition:	*ALL REQUIRED*

Notes

Depth Areas will either be shown as no tint or a Blue Tint depending on the depth curve area designated per specific LPC Chart.

Generally, if Depth Range (lower value) > 0 and Depth Range (upper value) <= 20 the blut tint will be shown.

Feature Information					
Specification:	*Chart #1*	**Feature Origin:**	*Hydrographic*	**Symbol:**	I30
Symbol Criteria:	All			**Label:**	None
				Color:	N/A

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Depth Curve (BE010)

Curve

Feature Description:	*A navigational safety line indicating that no sounding of a lesser depth exists seaward of the line, but greater depths may occur on the shallow side of the line.*
Inclusion Condition:	*ALL REQUIRED*

Notes

Depth curves/contours shall be labeled with the numeral in the same unit of measurement as the soundings. The term "meters" shall not be part of the label.

Depth Curve/Contour labels:
(1.) Break curve the width of the label plus 1 mm on each side. Orient label parallel to curve, readable left to right, or bottom to top, if curve is vertical.
(2.) Start labels at the middle of the curve/contour, space every 12 cm. Labels may be moved any distance to avoid overprints, except on a closed curve where an overprint cannot be avoided. If the overprint is another Depth Curve/Contour, break the curve/contour. Label every curve at least once if length of curve is 10 mm greater than window and does not close

Depth curves which surround a single sounding shall not be labeled if the length of the depth curve is less than 20 mm.
Approximate Curve/Contour symbology:
Where it is necessary to draw the navigator's attention to inadequacy in survey data, depth contours should be indicated as approximate by breaking them into lengths of 4mm, with spaces of 2mm. Longer sections may be used where there are extensive areas containing approximate contours. For short lengths of approximate contours, such as around a small isolated shoal, an alternative presentation breaking them into shorter lengths of 2mm with spaces of 1mm may be used. This is represented in Chart #1 symbol I31.

When accurate depth curves/contours taken directly from source charts are shown on a product that has been enlarged by a factor greater than two, compared to the scale of the source chart, e.g., 1:50,000 source on a product larger than 1:25,000, the depth curve's/contour's accuracy shall be approximate. When approximate depth curves/contours taken directly from source charts are shown on a product that has been reduced by a factor greater than two compared to the source chart, e.g., 1:50,000 source on a product smaller than 1:100,000, the depth curve/contour accuracy shall be accurate. When depth curves taken from source charts are enlarged or reduced by a factor equal to or less than two, they shall retain the same accuracy as the source chart.

If other curves/contours are selected for portrayal (based on published source material), use the curve/contour values that are equal to the value of the curves/contours on the source material.

The depth curve/contour to which water tint is shown may be interpolated from soundings shown on nautical chart sources in order to develop the required open water blue tint. If this is done the curve/contour shall be approximate.

In delineating depth curves, the line shall be positioned as near as possible to the deeper side of the sounding value without touching. The depth curve shall not be broken for the sounding value.

Depth curves/contours that coalesce on steep slopes shall show only the deepest and shoalest curves/contours. Where space is limited in "steep to channels" portrayal of the deepest curve/contour is preferred. In other general areas where space is limited because of scale, the shoalest curve/contour shall be shown with the deep curves/contours broken. Where a blue tint invades a steep slope, the curve/contour delimiting the blue tint must be shown.

Small depressions within shallow areas shall not be surrounded by a depth curve/contour if less than three soundings fall within the curve/contour.

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Depth Curve (BE010) *(Continued)*

Curve

Notes

When published nautical charts in meters are used as source material for NGA charts, the NGA charts shall retain the depth curves/contours shown on the published chart source. Occasionally foreign sources will portray soundings which are the same value as the depth curve/contour seaward of the depth curve/contour. In this event, the depth curve/contour is broken and a dashed approximate curve is extended seaward around the sounding(s).

Foreign charts showing soundings and depth curves in fathoms, that are used as source material for nautical charts, shall have the sounding converted to meters, and have the depth curves converted to meters as follows:
(a.) 1 fathom curve/contour shall be retained, and labeled 2;
(b.) 3 fathom curve/contour shall be retained, and labeled 5. This policy shall be followed only when the soundings seaward of the curve/contour are greater than the value of the curve/contour. When the above conversion is not practical, an approximate depth curve/contour shall be delineated and dashed lines displayed. Indefinite (approximate) depth curves/contours shall replace fathom curves/contours of depths other than above.

Show water tint (Blue SPC-48253), 31% screen, at 45°) from the shoreline (BA010), to the 10 meter depth curve/contour (BE010) and all offshore areas shallower than 10 meters (inside the 10 meter curve/contour). Blue tint is deleted from inland hydrographic features (BH), in those areas that are deeper than 10 meters (outside the 10 meter depth curve/contour).

If the shoal sounding and selected depth curves/contours will adequately portray a danger, it is not necessary for the complete sequence of depth curves/contours to be shown around an isolated pinnacle.

Accurate depth curves/contours shall be shown when the sounding data from which they are interpolated has a density of <= 10 mm maximum spacing, at the product chart scale before soundings have been thinned. When this sounding density is > 10 mm maximum spacing, any interpolated depth curves/contours shall be approximate.

In areas of the chart where the primary source of the hydrographic data is a foreign nautical chart, and that foreign chart does not show depth curves/contours, depth curves/contours shall not be interpolated, and soundings alone shall be used to depict the bottom topography.

In rapidly changing areas where surveys with different dates adjoin but do not agree, gaps in depth curves/contours shall be left to indicate data discrepancy to the user. Gap width shall be commensurate with chart scale and area covered by the sources.

Feature Information					
Specification:	*Chart #1*	**Feature Origin:**	*Hydrographic*	**Symbol:**	I30
Symbol Criteria:	All			**Label:**	[Depth Curve : Depth Curve or Contour Value], 7 point lower case, Italic
				Color:	Black (58600)

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Direction of Buoyage Indicator (FC010)

Point

Feature Description:	*A location that marks a change in the direction of buoyage.*
Inclusion Condition:	*ALL REQUIRED*

Notes

The Note "GENERAL DIRECTION OF BUOYAGE ON THIS CHART" is generally shown, reading horizontally, near the stem of the arrow, but it may be omitted in congested areas.

The standard size "Direction of Buoyage" symbol may be reduced in size to 75% or 50% for use in congested areas.

Where two opposing local directions meet, two arrows (size at cartographer's discretion) should be inserted, point to point but with a gap of at least 5mm between them. Additionally, a dashed magenta line (N1.2) may be inserted perpendicular to the arrows' direction, across the channel, with a magenta explanatory legend, eg: 'Change of buoyage direction', 8 point Mixed case Purple (96532) shall be placed parallel to the dashed line.

Feature Information

Specification:	*Chart #1*	**Feature Origin:**	*Hydrographic*	**Symbol:**	Q130.2
Symbol Criteria:	All			**Label:**	"Change in buoyage direction", 7 point Mixed Case
				Color:	Purple (96532)

Discoloured Water (BD030)

Surface

Feature Description:	*A region of water having a colour distinctly different from that of the surrounding region.*
Inclusion Condition:	*Length >= 200 meters*

Notes

The labeling for Discoloured Water using Chart #1 symbol K40 shall be "Discolored Water", 6 point Mixed case, Italic, Black (58600).

Use the following abbreviations for Horizontal Accuracy and Existence values:
If Position Approximate, label "PA";
If Position Doubtful, label "PD";
If Existence Doubtful, label "ED";
If Reported, label "Rep".

Date shall only be shown if Existence = Reported. Date shall be shown in parentheses if available. Always show "Rep" and date on the same line.

Unless specifically indicated on source material as inaccurate or doubtful, etc., or modified by other override rules, features shall be assumed to be accurate and definite. This rule applied to accuracy and existence of unique features, and is not related to the accuracy of the source as a whole.

Following IHO practice, features whose positions are approximate are labeled "PA" on the source charts. Features whose positions are doubtful are labeled "PD" on the source charts. Features whose existence is doubtful are labeled "ED" on the source charts. Features that have only been reported are labeled "Rep" on the source charts, and usually show the date of report in parentheses.

Definitions:
PA - Position Approximate = The position has not been accurately determined, or does not remain fixed.;
PD - Position Doubtful = The feature has been reported in various positions, but not definitely determined to be in any.;
ED - Existence Doubtful = Indicates the possible existence of the feature, the actual existence of which has not been established.

Show Hydrographic Depth or Hydrographic Height values in whole meters if decimeter value is 0; otherwise show the decimeter value as a subscript to the meter value.

Date shall only be shown if Existence = Reported. Date shall be shown in parentheses if available. Always show "Rep" and date on the same line.

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Discoloured Water (BD030) *(Continued)*

Surface

Notes

Priority for type placement Rules:
1-right center,
2-bottom center,
3-left center,
4-top center.
(A) Minimum distance from symbol - 1 mm.
(B) Maximum distance from symbol before choosing the next highest priority:
(#1) 4 mm measured to the West end;
(#2) 4 mm measured to the North side (top);
(#3) 4 mm measured to the East end;
(#4) 4 mm measured to the South side (bottom)

If symbols overprint each other, labels are condensed as follows:
(1.) If the labels are identical, only one is retained.;
(2,) If the labels are not identical, they shall be condensed in to one legend, e.g., "Fishhaven and Well". If multiple depths are shown, only the shallowest is retained.

Labels may be stacked, readable top to bottom to avoid overprints.

Type Preference Placement Rules:
(a.) Placed on one horizontal line centered in feature.;
(b.) Broken into multiple horizontal lines, centered in feature, with a 1 mm space between lines.;
(c.) Placed on one line outside feature. Use Priority for type placement Rules for placement, Hydrographic Depth shall always be positioned inside area.;
(d.) Placed on two lines to avoid overprints. Use Priority for type placement Rules. If present, Hydrographic Depth shall always be placed inside area.

Labels shall be placed in water area, and shall not overprint the shoreline or be placed on land or in the foreshore.

When two or more identical symbols overprint each other, delete the portions that overlap and show as one symbol with an outline of the combined area of the symbols.

A blue 31% tint shall not overprint other blue 31% tints. If two blue tinted symbols, or one tinted symbol and water tint overprint, only one 31% tint shall be shown in the area.

Feature Information

Specification:	*Chart #1*	**Feature Origin:**	*Hydrographic*	**Symbol:**	K40
Symbol Criteria:	All			**Label:**	"Discoloured Water", 6 point Mixed case, Italic
				Color:	Black (58600)

Discoloured Water (BD030)

Point

Notes

The labeling for Discoloured Water using Chart #1 symbol K40 shall be "Discolored Water", 6 point Mixed case, Italic, Black (58600).

Use the following abbreviations for Horizontal Accuracy and Existence values:
If Position Approximate, label "PA";
If Position Doubtful, label "PD";
If Existence Doubtful, label "ED";
If Reported, label "Rep".

Date shall only be shown if Existence = Reported. Date shall be shown in parentheses if available. Always show "Rep" and date on the same line.

Unless specifically indicated on source material as inaccurate or doubtful, etc., or modified by other override rules, features shall be assumed to be accurate and definite. This rule applied to accuracy and existence of unique features, and is not related to the accuracy of the source as a whole.

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Discoloured Water (BD030) *(Continued)*

Point

Notes

Following IHO practice, features whose positions are approximate are labeled "PA" on the source charts. Features whose positions are doubtful are labeled "PD" on the source charts. Features whose existence is doubtful are labeled "ED" on the source charts. Features that have only been reported are labeled "Rep" on the source charts, and usually show the date of report in parentheses.

Definitions:
PA - Position Approximate = The position has not been accurately determined, or does not remain fixed.;
PD - Position Doubtful = The feature has been reported in various positions, but not definitely determined to be in any.;
ED - Existence Doubtful = Indicates the possible existence of the feature, the actual existence of which has not been established.

Show Hydrographic Depth or Hydrographic Height values in whole meters if decimeter value is 0; otherwise show the decimeter value as a subscript to the meter value.

Date shall only be shown if Existence = Reported. Date shall be shown in parentheses if available. Always show "Rep" and date on the same line.

Priority for type placement Rules:
1-right center,
2-bottom center,
3-left center,
4-top center.
(A) Minimum distance from symbol - 1 mm.
(B) Maximum distance from symbol before choosing the next highest priority:
(#1) 4 mm measured to the West end;
(#2) 4 mm measured to the North side (top);
(#3) 4 mm measured to the East end;
(#4) 4 mm measured to the South side (bottom)

If symbols overprint each other, labels are condensed as follows:
(1.) If the labels are identical, only one is retained.;
(2,) If the labels are not identical, they shall be condensed in to one legend, e.g., "Fishhaven and Well". If multiple depths are shown, only the shallowest is retained.

Labels may be stacked, readable top to bottom to avoid overprints.

Type Preference Placement Rules:
(a.) Placed on one horizontal line centered in feature.;
(b.) Broken into multiple horizontal lines, centered in feature, with a 1 mm space between lines.;
(c.) Placed on one line outside feature. Use Priority for type placement Rules for placement, Hydrographic Depth shall always be positioned inside area.;
(d.) Placed on two lines to avoid overprints. Use Priority for type placement Rules. If present, Hydrographic Depth shall always be placed inside area.

Labels shall be placed in water area, and shall not overprint the shoreline or be placed on land or in the foreshore.

When two or more identical symbols overprint each other, delete the portions that overlap and show as one symbol with an outline of the combined area of the symbols.

A blue 31% tint shall not overprint other blue 31% tints. If two blue tinted symbols, or one tinted symbol and water tint overprint, only one 31% tint shall be shown in the area.

Feature Information

Specification:	*Chart #1*	**Feature Origin:**	*Hydrographic*	**Symbol:**	K40
Symbol Criteria:	All			**Label:**	"Discoloured Water", 6 point Mixed case, Italic
				Color:	Black (58600)

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Dish Aerial (AT010)

Point

Feature Description:	*A concave-shaped aerial that is used for emitting and/or sensing electromagnetic energy.*
Inclusion Condition:	*Length or Diameter >= 3 m.*

Notes

Type for features on land shall be positioned on land if possible. If type must be placed in the water, it shall be positioned so it does not obscure or overprint hydrographic detail. Type for features in the water shall be positioned in the water. If possible, type shall not be positioned across the shoreline.

If the existence is Doubtful or Reported, label as such.

When obstructions coalesce at map scale, label with highest obstruction information.

Feature Information

Specification:	*LTDS*	**Feature Origin:**	*Topographic*	**Symbol:**	PT_Black_DishAerial
Symbol Criteria:	Height Above Surface Level < 46			**Label:**	None
				Color:	N/A

Feature Information

Specification:	*LTDS*	**Feature Origin:**	*Topographic*	**Symbol:**	PT_Blue072_VO
Symbol Criteria:	Height Above Surface Level >= 46			**Label:**	"Dish" , 6 point Condensed Mixed case; [Highest Elevation], 6 point Condensed Bold Mixed case; "([Height Above Surface Level])", 6 point Condensed Mixed case
				Color:	Blue (46351)

Display Sign (AL050)

Point

Feature Description:	*An upright panel or similar structure used to convey visual information.*
Inclusion Condition:	*All Required*

Notes

None

Feature Information

Specification:	*Chart #1*	**Feature Origin:**	*Hydrographic*	**Symbol:**	Q126
Symbol Criteria:	All			**Label:**	None
				Color:	N/A

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Disposal Site (AB000)

Surface

Feature Description:	*A site on land for the collection and/or deposition of refuse or discarded material.*
Inclusion Condition:	*Area: >= 15,625 sq m. and Height Above Surface Level: < 46 m.*

Notes

When 2 or more similar area features are separated by less than 0.5 mm at chart scale, the feature will be agglomerated.

Coincident similar area features will be blended to form a single feature.

Area and line features will be generalized to detail compatible with scale.

Type for features on land shall be positioned on land if possible. If type must be placed in the water, it shall be positioned so it does not obscure or overprint hydrographic detail. Type for features in the water shall be positioned in the water. If possible, type shall not be positioned across the shoreline.

Limiting lines of feature are omitted if it coalesces with a road (AP030).

Feature Information

Specification:	*LTDS*	**Feature Origin:**	*Topographic*	**Symbol:**	AR_Black_DisposalSite_Refuse
Symbol Criteria:	All			**Label:**	"Disposal site", 6 point Condensed Mixed case
				Color:	Black (58600)

Disposal Site (AB000)

Point

Inclusion Condition:	*Area: < 15,625 sq m. OR Height Above Surface Level: >= 46 m.*

Notes

When 2 or more similar area features are separated by less than 0.5 mm at chart scale, the feature will be agglomerated.

Coincident similar area features will be blended to form a single feature.

Area and line features will be generalized to detail compatible with scale.

Type for features on land shall be positioned on land if possible. If type must be placed in the water, it shall be positioned so it does not obscure or overprint hydrographic detail. Type for features in the water shall be positioned in the water. If possible, type shall not be positioned across the shoreline.

Limiting lines of feature are omitted if it coalesces with a road (AP030).

Feature Information

Specification:	*LTDS*	**Feature Origin:**	*Topographic*	**Symbol:**	PT_Blue072_VO
Symbol Criteria:	Height Above Surface Level: >= 46 m.			**Label:**	"Disposal site", 6 point Condensed Mixed case; [Highest Elevation], 6 point Condensed Bold Mixed Case; "([Height Above Surface Level])", 6 point Condensed Mixed case
				Color:	Blue (46351)

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Distance Mark (ZB036)

Point

Feature Description:	*A marker that indicates the value of distance measured from a starting location.*
Inclusion Condition:	*Distance Mark : Maritime Navigation Marked = [True] and (Distance Mark : Transportation System Type = [Inland Waterway])*

Notes

Marks which indicate distances along a channel in either nautical miles, kilometres or some other unit, shall be shown, where considered useful. The symbol shall be shown either ashore or in the channel, and will be chosen at the discretion of each country where it represents visible marks. The unit of measurement (M, km, etc) should be shown.

Where there are no visible marks, the distance figures, with units, should be shown in magenta with no symbols.

Feature Information

Specification:	*Chart #1*	**Feature Origin:**	*Hydrographic*	**Symbol:**	B25
Symbol Criteria:	Distance Mark : Maritime Navigation Marked = [True] and (Distance Mark : Transportation System Type = [Inland Waterway])			**Label:**	"km [distance at point]", 6 point lower case Italic if (Distance Mark : Transportation System Type =[Inland Waterway])
				Color:	Purple (96532)

Ditch (BH030)

Surface

Feature Description:	*An artificial waterway with no flow, or a controlled flow, usually unlined, used for draining or irrigating land.*
Inclusion Condition:	*Width >= 25 m.*

Notes

If one symbol coalesces with another symbol for the same type feature, displace symbols to allow a minimum separation of 0.21mm.

Rivers, canals, and ditches will be partially collapsed when area and line delineations are supported on the product and the area feature does not meet the minimum geometric inclusion condition.

Coincident similar area features will be blended to form a single feature.

Area and line features will be generalized to detail compatible with scale.

Feature will be generalized to provide a more aesthetic contoured feature (i.e., smoothed).

Omit from Built-up Area.

Symbols and associated area patterns of underpassing features (except drainage shorelines) are broken for all bridges, except footbridges. This rule does not apply to land tint on Littoral Planning Charts.

Do not show land tint in the symbol. Do not show land tint if (Perennial).

Symbolize the casement portions (Left Bank/Right Bank) of the feature using individual river or canal banks in conjunction with the inland shoreline symbology. The inland shoreline shall correspond to the associated water body i.e., Perennial river = Perennial Inland Shoreline.

Feature Information

Specification:	*LTDS*	**Feature Origin:**	*Topographic*	**Symbol:**	AR_Cyan_Ditch_Perennial
Symbol Criteria:	Water Resource Information : Hydrologic Persistence = (Perennial) or (No Information)			**Label:**	None
				Color:	N/A

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Ditch (BH030) *(Continued)*

Surface

Feature Information					
Specification:	*LTDS*	**Feature Origin:**	*Topographic*	**Symbol:**	AR_Cyan_Ditch_Intermittent
Symbol Criteria:	Water Resource Information : Hydrologic Persistence = (Intermittent) or (Dry)			**Label:**	None
				Color:	N/A

Ditch (BH030)

Curve

Inclusion Condition:	*Width < 25 m.*

Notes

If one symbol coalesces with another symbol for the same type feature, displace symbols to allow a minimum separation of 0.21mm.

Area and line features will be generalized to detail compatible with scale.

Feature will be generalized to provide a more aesthetic contoured feature (i.e., smoothed).

Incorporate shorter Canals and Ditches <= 320 meters length as a connector feature and incorporate spacing of > 200 meters. Always retain the outermost limits of these features before generalization takes place.

Omit from Built-up Area.

Ditches to drain swamps and areas subject to natural inundation shall be shown as perennial Ditches.

Feature Information					
Specification:	*LTDS*	**Feature Origin:**	*Topographic*	**Symbol:**	LN_Cyan_Ditch_Perennial
Symbol Criteria:	Water Resource Information : Hydrologic Persistence = (Perennial) or (No Information)			**Label:**	None
				Color:	N/A

Feature Information					
Specification:	*LTDS*	**Feature Origin:**	*Topographic*	**Symbol:**	LN_Cyan_Ditch_Intermittent
Symbol Criteria:	Water Resource Information : Hydrologic Persistence = (Intermittent) or (Dry)			**Label:**	None
				Color:	N/A

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Dock (BB095)

Surface

Feature Description:	*An artificially enclosed body of water within which vessels may moor and which may have gates used to regulate the interior water level.*
Inclusion Condition:	*ALL REQUIRED*

Notes

None

Feature Information					
Specification:	*Chart #1*	**Feature Origin:**	*Hydrographic*	**Symbol:**	F27
Symbol Criteria:	All			**Label:**	None
				Color:	N/A

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Dolphin (BB080)

Point

Feature Description:	*A post or group of posts used for mooring, warping a ship or as an aid to navigation.*
Inclusion Condition:	*ALL REQUIRED*

Notes

Where dolphins are very large, eg on either side of the pierhead of a deep-water tanker terminal, their outlines should be shown true to scale (possibly with small light stars where appropriate). Land tint should be superimposed. Small dolphins (or large ones on smaller scale charts) shall be shown symbolically by a small square, two of the sides being aligned with the centre line of any moored vessel. (Single dolphins to which vessels may secure in any direction shall be shown with the symbol having two sides horizontal). No land tint need be shown on the small square symbol.

Priority for type placement: 1-right center, 2-bottom center, 3-left center, 4-top center.
(A) Minimum distance from symbol - 1 mm.
(B) Maximum distance from symbol before choosing the next highest priority:
(#1) 4 mm measured to the West end;
(#2) 4 mm measured to the North side (top);
(#3) 4 mm measured to the East end;
(#4) 4 mm measured to the South side (bottom)

Feature name/label shall be positioned parallel to lines of latitude and readable left to right.

If two or more like features are in a group, label once in plural form, e.g., Dols, Fish Traps, etc. If single features are separated by 5 mm or more, they shall be labeled individually.

The label 'Dn' or 'Dns', or Deviation Dn (if necessary), 6 point Mixed case Italic Black (58600) should be inserted only if the nature of the feature is not self-evident.

Feature Information					
Specification:	*Chart #1*	**Feature Origin:**	*Hydrographic*	**Symbol:**	F20
Symbol Criteria:	Mooring and/or Warping Facility Type: [Dolphin]			**Label:**	None
				Color:	N/A

Feature Information					
Specification:	*Chart #1*	**Feature Origin:**	*Hydrographic*	**Symbol:**	F21
Symbol Criteria:	Mooring and/or Warping Facility Type: [Deviation Dolphin]			**Label:**	None
				Color:	N/A

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Dragon's Teeth (AL060)

Surface

Feature Description:	*Upward-pointing obstacles laid in the ground to slow or stop the movement of vehicles.*
Inclusion Condition:	*Width >= 25 meters*

Notes

None

Feature Information					
Specification:	*LTDS*	**Feature Origin:**	*Topographic*	**Symbol:**	AR_Black_DragonsTeeth
Symbol Criteria:	All			**Label:**	None
				Color:	N/A

Dragon's Teeth (AL060)

Curve

Inclusion Condition:	*Width < 25 meters*

Notes

None

Feature Information					
Specification:	*LTDS*	**Feature Origin:**	*Topographic*	**Symbol:**	LN_Black_DragonsTeeth
Symbol Criteria:	All			**Label:**	None
				Color:	N/A

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Dredged Area (FC034)

Surface

Feature Description:	*An area of the bottom of a body of water (for example: a channel) which has been deepened by dredging.*
Inclusion Condition:	*Width >= 50 meters*

Notes

Show Hydrographic Depth values in whole meters if decimeter value is 0; otherwise show the decimeter value showing a comma instead of decimeter point (example: 17,3m)

If Dredged Area is not maintained (Maintained: [False]), the label should include the year if the control surveys and dredging (Example: Dredged to 8,2m (2006)).

If space does not allow for the full legend to be shown, labels for Dredged Channels are condensed in the following order:
(1.) Delete "Dredged to" or "Maintained depth" labels first,
(2.) Delete Date label and parentheses second,
(3.) Delete description of Aids to Navigation label third. Do not delete Hydrographic depth label or the "m" from any Dredged Channel feature

If two line features of the same feature type meet end to end, and have different depths, a short line is shown centered on the point of intersection. It bisects the angle at which the line features meet (i.e., if the lines meet at 180° angle the bisecting line is perpendicular to the meeting line features). The bisecting line is 0.1 mm lineweight, length is 3.0 mm, and it is shown in the same color as the line features.

Dredged channels/areas shall be tinted in accordance with their depths, following the guidance for generation of water tint provided in rules for depth curves and/or open water.

If Aids to Navigation Marked: [True], Description of Aids to Navigation shall be shown if one of the following conditions occur:
(1.) The aids to navigation that mark the feature do not meet the inclusion conditions for aids to navigation.
(2.) The aids to navigation that mark the feature are known to exist, but the details, such as position, type of aid, etc, are not sufficient to chart them as aids to navigation.
If Navaid features are shown, or information about aids to navigation is completely lacking, omit the Description of Aids to Navigation label.

Feature Information

Specification:	*Chart #1*	**Feature Origin:**	*Hydrographic*	**Symbol:**	I22
Symbol Criteria:	Maintained: [False]			**Label:**	[Hydrographic Depth]"m", Hydrographic Depth: 6 point lower case, Italic
				Color:	Black (58600)

Feature Information

Specification:	*Chart #1*	**Feature Origin:**	*Hydrographic*	**Symbol:**	I23
Symbol Criteria:	Maintained: [True]			**Label:**	[Hydrographic Depth]"m", Hydrographic Depth: 6 point lower case, Italic
				Color:	Black (58600)

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Dredged Area (FC034)

Curve

Inclusion Condition:	*Width < 50 meters*

Notes

If two line features of the same feature type meet end to end, and have different depths, a short line is shown centered on the point of intersection. It bisects the angle at which the line features meet (i.e., if the lines meet at 180° angle the bisecting line is perpendicular to the meeting line features). The bisecting line is 0.1 mm lineweight, length is 3.0 mm, and it is shown in the same color as the line features.

Dredged channels/areas shall be tinted in accordance with their depths, following the guidance for generation of water tint provided in rules for depth curves and/or open water.

If Aids to Navigation Marked: [True], Description of Aids to Navigation shall be shown if one of the following conditions occur:
(1.) The aids to navigation that mark the feature do not meet the inclusion conditions for aids to navigation.
(2.) The aids to navigation that mark the feature are known to exist, but the details, such as position, type of aid, etc, are not sufficient to chart them as aids to navigation.
If Navaid features are shown, or information about aids to navigation is completely lacking, omit the Description of Aids to Navigation label.

Feature Information					
Specification:	*Chart #1*	**Feature Origin:**	*Hydrographic*	**Symbol:**	I20
Symbol Criteria:	All			**Label:**	None
				Color:	N/A

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Drive-in Theatre (AK070)

Surface

Feature Description:	*A place where motion pictures are shown while viewers remain in their vehicles.*
Inclusion Condition:	*Area >= 15,625 square meters*

Notes

Coincident similar area features will be blended to form a single feature.

Area and line features will be generalized to detail compatible with scale.

Type for features on land shall be positioned on land if possible. If type must be placed in the water, it shall be positioned so it does not obscure or overprint hydrographic detail. Type for features in the water shall be positioned in the water. If possible, type shall not be positioned across the shoreline.

Limiting lines of feature are omitted if it coalesces with a road (AP030).

If a clearing exists inside of the area feature, and the size of the clearing is equal to or greater than the area for the surrounding area feature, the clearing is shown as an open space inside the surround feature. If the area of the clearing is less than the area for the surrounding feature, the clearing is deleted and absorbed into the surrounding area feature.

If two area features with the same feature code do not connect at any point, and have a space between them less than 2.5 mm at map/chart scale, delete the open space that is less than 2.5 mm between the features and combine them into one area feature.

If a portion of an area feature has a minimum width of less than 2.5 mm at map/chart scale, delete that portion of the area feature that is not at least 2.5 mm wide, measured from perimeter to perimeter. If the deletion of the portion of the area based on the above criteria will reduce the area of the remaining portion of the area feature to below the minimum area inclusion, do not delete the narrow portion of the feature that is less than 2.5 mm wide. If the deletion of the portion of the area based on the above criteria will split two larger areas connected by a narrow strip into two separate areas, either of which would be below minimum area inclusion, do not delete the narrow portion of the feature that is less than 2.5 mm wide.

Feature Information

Specification:	*LTDS*	**Feature Origin:**	*Topographic*	**Symbol:**	AR_Black_DriveInTheatre
Symbol Criteria:	Area >= 15,625			**Label:**	"Drive-in theater", 6 point Condensed Mixed case
				Color:	Black (58600)

Drive-in Theatre (AK070)

Point

Inclusion Condition:	*Area < 15,625 square meters AND Navigation Landmark = [True]*

Notes

None

Feature Information

Specification:	*LTDS*	**Feature Origin:**	*Topographic*	**Symbol:**	PT_Black_DriveInTheatre
Symbol Criteria:	Area < 15,625 AND Navigation Landmark = [True]			**Label:**	"Drive-in theater", 6 point Condensed Mixed case
				Color:	Black (58600)

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Dry Dock (BB090)

Surface

Feature Description:	*An artificial basin fitted with a gate or caisson into which vessels can be floated and the water pumped out to expose the bottom of the vessel.*
Inclusion Condition:	*Width >= 20 m.*

Notes

Priority for type placement: 1-right center, 2-bottom center, 3-left center, 4-top center.
(A) Minimum distance from symbol - 1 mm.
(B) Maximum distance from symbol before choosing the next highest priority:
(#1) 4 mm measured to the West end;
(#2) 4 mm measured to the North side (top);
(#3) 4 mm measured to the East end;
(#4) 4 mm measured to the South side (bottom)

When the area symbol or cased line symbol overprints the shoreline, shoreline is deleted.

Land tint shall be shown over dry docks to distinguish them from wet docks

Feature Information

Specification:	*LTDS*	**Feature Origin:**	*Topographic*	**Symbol:**	AR_Black_DryDock
Symbol Criteria:	All			**Label:**	[Geographic Name Information : Full Name] if available, 6 point Condensed Mixed case; "Dry dock", 6 point Condensed Mixed case
				Color:	Black (58600)

Dumping Ground (FC033)

Surface

Feature Description:	*A sea area where dredged material or other potentially more harmful material (for example: explosives or chemical waste) is deliberately deposited.*
Inclusion Condition:	*Area >= 15,625*

Notes

Type sizes for Maritime Limits and areas:
8 point - < 8 square cm.,
10 point - >= 8 and < 12 square cm,,
12 point - >= 12 and < 24 square cm.,
14 point - >= 24 and < 100 square cm.,
8 point - >= 100 square cm.;
Type placement for areas >= 100 square cm. to < 500 square cm.: Two labels are shown on approximately opposite sides of the area, preferably top and bottom.;
Type placement for areas >= 500 square cm.: Labels are placed at approximately 250 mm. interval around the perimeter of the feature.;
Do not place labels around sharp corners (interior angle < 135°), or along chart neatlines. Place type 1 mm away and parallel to area limit line, reading left to right, or bottom to top if limit line is vertical. Type is placed on the INSIDE of the area.

Priority for type placement: 1-right center, 2-bottom center, 3-left center, 4-top center.
(A) Minimum distance from symbol - 1 mm.
(B) Maximum distance from symbol before choosing the next highest priority:
(#1) 4 mm measured to the West end;
(#2) 4 mm measured to the North side (top);
(#3) 4 mm measured to the East end;
(#4) 4 mm measured to the South side (bottom)

Maritime Limit type "Unsurveyed Area" shall be labeled with legends spaced every 50 mm along the boundary line, with type positioned 1 mm away from the line. Labels should be on the inside of the area reading from left to right, or from bottom to top if boundary is vertical. Do not place text around sharp corners.

When the boundary of an areal symbol crosses or overprints the shoreline, the boundary overprinting the shoreline or land is deleted.

Minimum width for maritime limit symbols, other than pilot boarding areas, shall be 4 mm at chart scale.

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Dumping Ground (FC033) *(Continued)*

Surface

Feature Information					
Specification:	*Chart #1*	**Feature Origin:**	*Hydrographic*	**Symbol:**	N23.1
Symbol Criteria:	Dumping Ground Type = [Explosives] and Dumping Ground : Operational = [True]			**Label:**	"Explosives Dumping Ground", 8-14 point Mixed case, Italic
				Color:	Green (52813)

Feature Information					
Specification:	*Chart #1*	**Feature Origin:**	*Hydrographic*	**Symbol:**	N23.2
Symbol Criteria:	Dumping Ground Type = [Explosives] and (Dumping Ground : Operational = [False] or Dumping Ground : Maintained = [False])			**Label:**	"Explosives Dumping Ground" "(disused - [Date])", 8-14 point Mixed case, Italic
				Color:	Green (52813)

Feature Information					
Specification:	*Chart #1*	**Feature Origin:**	*Hydrographic*	**Symbol:**	N24
Symbol Criteria:	Dumping Ground Type = [Chemical Waste] and Dumping Ground : Operational = [True]			**Label:**	"Dumping Ground for Chemicals", 8-14 point Mixed case, Italic
				Color:	Green (52813)

Feature Information					
Specification:	*Chart #1*	**Feature Origin:**	*Hydrographic*	**Symbol:**	N2.1
Symbol Criteria:	(Dumping Ground Type = [Hazardous Material] or [Nuclear Waste] or [Spoil] or [Vessels])			**Label:**	"Dumping Ground for" [Dumping Ground Type]"if (Dumping Ground : Operational = [True] or Dumping Ground : Maintained: [True]), 8-14 point Mixed case, Italic; "Dumping Ground for" [Dumping Ground Type]""(disused - [Date])" if (Dumping Ground : Operational = [False] or Dumping Ground : Maintained = [False]), 8-14 point Mixed case, Italic;
				Color:	Green (52813)

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Dumping Ground (FC033)

Point

Inclusion Condition:	*Area < 15,625*

Notes

None

Feature Information					
Specification:	*Chart #1*	**Feature Origin:**	*Hydrographic*	**Symbol:**	N23.1
Symbol Criteria:	Dumping Ground Type = [Explosives]			**Label:**	None
				Color:	N/A

Elevation Contour (CA010)

Curve

Feature Description:	*A line connecting points having the same elevation value relative to a vertical datum.*
Inclusion Condition:	*ALL REQUIRED*

Notes

Label only the index contours unless the area has < 5% rise.

Contour values shall be labeled on the 1/2 and 1/4 interval auxiliaries, at the ends, and where necessary, every 100 to 150 mm.

Contours that are coincident with the datum plane shall be labeled "ZERO", and those Contours below the datum plane are labeled with numerals prefixed by the label "MINUS".

The Contour values shall be positioned so that they progress in smooth-flowing curves, reading uphill towards the higher elevation. Contour values shall not be positioned upside down.

Contour values shall be centered on the axis of the Contour line.

Sets of contour values shall be repeated at distances of from >= 100 mm to <= 150 mm.

Contour values shall not be shown < 20 mm from a control point, bench mark or spot elevation.

Contours shall intersect and cross linear feature at a right angle to that feature with a right angle contour length of 0.25 mm out from each side of the crossed feature. Features are: road, railroad, and all Inland water area and line features except for Lake/Pond.

For aesthetics reasons, contours crossing the following features shall be portrayed perpendicular to the linear or areal feature symbols, and extending 0.2 mm beyond the symbol on either side: Railroad Track (line), Railroad Siding (line), Interchange (line), Road (line), Apron/Hardstand (area), Overrun/Stopway (area), Runway (area), Taxiway (area).

For aesthetics reasons, contours crossing the following features shall be portrayed perpendicular to the areal feature's symbol: Aqueduct (area), Canal (area), Ditch (area), River/Stream (area).

Where index contours begin to coalesce (<0.5 mm from adjacent contours for any interval) the following hierarchy shall apply for dropping intermediate contours: (a.) The two inner-most intermediate contours shall be dropped first. (b.) The two outer-most intermediate contours shall be last to be dropped. All index contours shall remain unless they coalesce, then apply Rule below.

Index contours shall be drawn continuously throughout the sheet graphic. When they coalesce, this condition shall be represented by a single index contour for the length of the coalescing condition.

The ticks of the depression contour shall be shortened by one-half if distance between contours are <= 0.40 mm at map scale.

Where a Cut Line or Fill coincides with a contour, the contour shall be suppressed. The cut Lines ticks shall point downhill towards the bottom of the cut.

Contour values shall be haloed 0.2 mm from the contour line.

When a Contour coalesces with a Bluff/Cliff, Escarpment, Crevice, Crevasse, Esker, Fault, or Rock Formation, the coalescing portion of the contour shall be omitted.

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Elevation Contour (CA010) *(Continued)*

Curve

Notes

Auxiliary contours shall be shown to indicate summits or tops when feature cannot be shown by normal contour intervals.

Auxiliary contours need not be continuous. They shall be any length > 25 mm. When shown in sections, they must start and end at interpolated points between contours.

Auxiliary contours shall be shown at one-half of the prescribed contour interval when: (a.) the % of slope is > 2 <= 5, or (b.) isolated relief formations need to be shown.

Auxiliary contours shall be shown at one-quarter of the prescribed contour interval if the % of slope is <= 2.

Form lines shall not be shown as continuations of other contours. A space of 1.3 mm shall be between other contours and form lines.

Contours shall be broken for Ravines/Gorges/Canyons, etc., represented by limiting lines, or the appropriate feature symbol.

Sand and gravel areas shall be contoured.

Contours shall be broken for Sand Dunes at limits of area patterns.

Feature Information					
Specification:	*LTDS*	**Feature Origin:**	*Topographic*	**Symbol:**	LN_Dk-Brown1815_Contour_LandIndexAccurate
Symbol Criteria:	(Hypsography Portrayal Type = [Index Contour] or [Index Carrying Contour]) and (Elevation Surface Category = [Land] or [No Information])			**Label:**	[Highest Elevation], 7 point Condensed Italics lower case
				Color:	Red/Brown (61121)

Feature Information					
Specification:	*LTDS*	**Feature Origin:**	*Topographic*	**Symbol:**	LN_Dk-Brown1815_Contour_LandIndexApproximate
Symbol Criteria:	(Hypsography Portrayal Type = [Approximate Index Contour]) and (Elevation Surface Category = [Land] or [No Information])			**Label:**	[Highest Elevation], 7 point Condensed Italics lower case
				Color:	Red/Brown (61121)

Feature Information					
Specification:	*LTDS*	**Feature Origin:**	*Topographic*	**Symbol:**	LN_Dk-Brown1815_Contour_LandHalfAuxillary
Symbol Criteria:	(Hypsography Portrayal Type = [Half Auxiliary Contour] or [Auxiliary Carrying Contour]) and (Elevation Surface Category = [Land] or [No Information])			**Label:**	[Highest Elevation], 7 point Condensed Italics lower case
				Color:	Red/Brown (61121)

Feature Information					
Specification:	*LTDS*	**Feature Origin:**	*Topographic*	**Symbol:**	LN_Dk-Brown1815_Contour_Land_QuarterAuxillary
Symbol Criteria:	(Hypsography Portrayal Type = [Quarter Auxiliary Contour]) and (Elevation Surface Category = [Land] or [No Information])			**Label:**	[Highest Elevation], 7 point Condensed Italics lower case
				Color:	Red/Brown (61121)

Feature Information					
Specification:	*LTDS*	**Feature Origin:**	*Topographic*	**Symbol:**	LN_Dk-Brown1815_Contour_LandIntermediateAccurate
Symbol Criteria:	(Hypsography Portrayal Type = [Intermediate Contour] or [Intermediate Carrying Contour]) and (Elevation Surface Category = [Land] or [No Information])			**Label:**	[Highest Elevation] if needed, 7 point Condensed Italics lower case
				Color:	Red/Brown (61121)

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Elevation Contour (CA010) *(Continued)*

Curve

Feature Information					
Specification:	*LTDS*	**Feature Origin:**	*Topographic*	**Symbol:**	LN_Dk-Brown1815_Contour_LandIntermediateApproximate
Symbol Criteria:	(Hypsography Portrayal Type = [Approximate Intermediate Contour]) and (Elevation Surface Category = [Land] or [No Information])			**Label:**	[Highest Elevation] if needed, 7 point Condensed Italics lower case
				Color:	Red/Brown (61121)

Feature Information					
Specification:	*LTDS*	**Feature Origin:**	*Topographic*	**Symbol:**	LN_Dk-Brown1815_Contour_LandFormLine
Symbol Criteria:	(Hypsography Portrayal Type = [Form Line]) and (Elevation Surface Category = [Land] or [No Information])			**Label:**	None
				Color:	N/A

Feature Information					
Specification:	*LTDS*	**Feature Origin:**	*Topographic*	**Symbol:**	LN_Dk-Brown1815_Contour_Land_MoundIndex
Symbol Criteria:	(Hypsography Portrayal Type = [Mound Index Contour]) and (Elevation Surface Category = [Land] or [No Information])			**Label:**	[Highest Elevation], 7 point Condensed Italics lower case
				Color:	Red/Brown (61121)

Feature Information					
Specification:	*LTDS*	**Feature Origin:**	*Topographic*	**Symbol:**	LN_Dk-Brown1815_Contour_Land_MoundIntermediate
Symbol Criteria:	(Hypsography Portrayal Type = [Mound Intermediate Contour]) and (Elevation Surface Category = [Land] or [No Information])			**Label:**	[Highest Elevation] if needed, 7 point Condensed Italics lower case
				Color:	Red/Brown (61121)

Feature Information					
Specification:	*LTDS*	**Feature Origin:**	*Topographic*	**Symbol:**	LN_Dk-Brown1815_Contour_Land_DepressionIndex
Symbol Criteria:	(Hypsography Portrayal Type = [Depression Index Contour]) and (Elevation Surface Category = [Land] or [No Information])			**Label:**	[Highest Elevation], 7 point Condensed Italics lower case
				Color:	Red/Brown (61121)

Feature Information					
Specification:	*LTDS*	**Feature Origin:**	*Topographic*	**Symbol:**	LN_Dk-Brown1815_Contour_Land_DepressionAuxillary
Symbol Criteria:	(Hypsography Portrayal Type = [Depression Auxiliary Contour]) and (Elevation Surface Category = [Land] or [No Information])			**Label:**	[Highest Elevation], 7 point Condensed Italics lower case
				Color:	Red/Brown (61121)

Feature Information					
Specification:	*LTDS*	**Feature Origin:**	*Topographic*	**Symbol:**	LN_Dk-Brown1815_Contour_Land_DepressionIntermediate
Symbol Criteria:	(Hypsography Portrayal Type = [Depression Intermediate Contour]) and (Elevation Surface Category = [Land] or [No Information])			**Label:**	[Highest Elevation] if needed, 7 point Condensed Italics lower case
				Color:	Red/Brown (61121)

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Elevation Contour (CA010) *(Continued)*

Curve

Feature Information					
Specification:	*LTDS*	**Feature Origin:**	*Topographic*	**Symbol:**	LN_Cyan_Contour_ SnowIceIndexAccurate
Symbol Criteria:	(Hypsography Portrayal Type = [Index Contour] or [Index Carrying Contour]) and (Elevation Surface Category = [Snow Field and/or Ice-field])			**Label:**	[Highest Elevation], 7 point Condensed Italics lower case
				Color:	Blue (48253)

Feature Information					
Specification:	*LTDS*	**Feature Origin:**	*Topographic*	**Symbol:**	LN_Cyan_Contour_ SnowIceIndexApproximate
Symbol Criteria:	(Hypsography Portrayal Type = [Approximate Index Contour]) and (Elevation Surface Category = [Snow Field and/or Ice-field])			**Label:**	[Highest Elevation], 7 point Condensed Italics lower case
				Color:	Blue (48253)

Feature Information					
Specification:	*LTDS*	**Feature Origin:**	*Topographic*	**Symbol:**	LN_Cyan_Contour_ SnowIceHalfAuxillary
Symbol Criteria:	(Hypsography Portrayal Type = [Half Auxiliary Contour] or [Auxiliary Carrying Contour]) and (Elevation Surface Category = [Snow Field and/or Ice-field])			**Label:**	[Highest Elevation], 7 point Condensed Italics lower case
				Color:	Blue (48253)

Feature Information					
Specification:	*LTDS*	**Feature Origin:**	*Topographic*	**Symbol:**	LN_Cyan_Contour_ SnowIce_QuarterAuxillary
Symbol Criteria:	(Hypsography Portrayal Type = [Quarter Auxiliary Contour]) and (Elevation Surface Category = [Snow Field and/or Ice-field])			**Label:**	[Highest Elevation], 7 point Condensed Italics lower case
				Color:	Blue (48253)

Feature Information					
Specification:	*LTDS*	**Feature Origin:**	*Topographic*	**Symbol:**	LN_Cyan_Contour_ SnowIceIntermediateAccurate
Symbol Criteria:	(Hypsography Portrayal Type = [Intermediate Contour] or [Intermediate Carrying Contour]) and (Elevation Surface Category = [Snow Field and/or Ice-field])			**Label:**	[Highest Elevation] if needed, 7 point Condensed Italics lower case
				Color:	Blue (48253)

Feature Information					
Specification:	*LTDS*	**Feature Origin:**	*Topographic*	**Symbol:**	LN_Cyan_Contour_ SnowIceIntermediateApproximate
Symbol Criteria:	(Hypsography Portrayal Type = [Approximate Intermediate Contour]) and (Elevation Surface Category = [Snow Field and/or Ice-field])			**Label:**	[Highest Elevation] if needed, 7 point Condensed Italics lower case
				Color:	Blue (48253)

Feature Information					
Specification:	*LTDS*	**Feature Origin:**	*Topographic*	**Symbol:**	LN_Cyan_Contour_ SnowIceFormLine
Symbol Criteria:	(Hypsography Portrayal Type = [Form Line]) and (Elevation Surface Category = [Snow Field and/or Ice-field])			**Label:**	None
				Color:	N/A

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Elevation Contour (CA010) *(Continued)*

Curve

Feature Information					
Specification:	*LTDS*	**Feature Origin:**	*Topographic*	**Symbol:**	LN_Cyan_Contour_ SnowIce_DepressionIndex
Symbol Criteria:	(Hypsography Portrayal Type = [Depression Index Contour]) and (Elevation Surface Category = [Snow Field and/or Ice-field])			**Label:**	[Highest Elevation], 7 point Condensed Italics lower case
				Color:	Blue (48253)

Feature Information					
Specification:	*LTDS*	**Feature Origin:**	*Topographic*	**Symbol:**	LN_Cyan_Contour_ SnowIce_DepressionAuxillary
Symbol Criteria:	(Hypsography Portrayal Type = [Depression Auxiliary Contour]) and (Elevation Surface Category = [Snow Field and/or Ice-field])			**Label:**	[Highest Elevation], 7 point Condensed Italics lower case
				Color:	Blue (48253)

Feature Information					
Specification:	*LTDS*	**Feature Origin:**	*Topographic*	**Symbol:**	LN_Cyan_Contour_ SnowIce_DepressionIntermediate
Symbol Criteria:	(Hypsography Portrayal Type = [Depression Intermediate Contour]) and (Elevation Surface Category = [Snow Field and/or Ice-field])			**Label:**	[Highest Elevation] if needed, 7 point Condensed Italics lower case
				Color:	Blue (48253)

Embankment (DB090)

Surface

Feature Description:	*A man-made raised long mound of earth or other material.*
Inclusion Condition:	*Width >= 25 m.*

Notes

When 2 or more similar area features are separated by less than 0.5 mm at chart scale, the feature will be agglomerated.

Area and line features will be generalized to detail compatible with scale.

Where a Cut Line or Fill coincides with a contour, the contour shall be suppressed. The cut Lines ticks shall point downhill towards the bottom of the cut.

When a Contour coalesces with a Bluff/Cliff, Escarpment, Crevice, Crevasse, Esker, Fault, or Rock Formation, the coalescing portion of the contour shall be omitted.

Show land tint inside the symbol. If the vertical reference category is (above surface /does not cover at High Water), show land tint only.

Feature Information					
Specification:	*LTDS*	**Feature Origin:**	*Topographic*	**Symbol:**	AR_Dk-Brown1815_Embankment_ GreaterthanContourInterval
Symbol Criteria:	Predominant Feature Height >= contour interval			**Label:**	None
				Color:	N/A

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Embankment (DB090) *(Continued)*

Surface

Feature Information					
Specification:	*LTDS*	**Feature Origin:**	*Topographic*	**Symbol:**	AR_Dk-Brown1815_Embankment_LessthanContourInterval
Symbol Criteria:	Predominant Feature Height < contour interval			**Label:**	None
				Color:	N/A

Embankment (DB090)

Curve

Inclusion Condition:	*Width < 25 m.*

Notes

Area and line features will be generalized to detail compatible with scale.

Label line feature above (preferred) and parallel to the line with a 0.5 mm space between. Above means: readable from south or east Projection neatline.

Where a Cut Line or Fill coincides with a contour, the contour shall be suppressed. The cut Lines ticks shall point downhill towards the bottom of the cut.

Omit from Built-up Area.

When a Contour coalesces with a Bluff/Cliff, Escarpment, Crevice, Crevasse, Esker, Fault, or Rock Formation, the coalescing portion of the contour shall be omitted.

Where symbols with the same feature code but different values of vertical reference categories (i.e. cover/uncovers, submerged, above surface (at high water)) are adjacent to each other, the boundary of the feature that is the highest in vertical plane shall be shown, and the boundary of the feature above water is shown whole, while the feature that covers and uncovers has a portion of its symbol perimeter that overprints the higher feature deleted.

Show land tint inside the symbol. If the vertical reference category is (above surface /does not cover at High Water), show land tint only.

Feature Information					
Specification:	*LTDS*	**Feature Origin:**	*Topographic*	**Symbol:**	LN_Dk-Brown1815_Embankment_GreaterthanContourInterval
Symbol Criteria:	Embankment Type = [Fill] AND Predominant Feature Height >= contour interval			**Label:**	None
				Color:	N/A

Feature Information					
Specification:	*LTDS*	**Feature Origin:**	*Topographic*	**Symbol:**	LN_Dk-Brown1815_Embankment_LessthanContourInterval
Symbol Criteria:	Embankment Type = [Fill] AND Predominant Feature Height < contour interval			**Label:**	None
				Color:	N/A

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Embankment (DB090) *(Continued)*

Curve

Feature Information					
Specification:	*LTDS*	**Feature Origin:**	*Topographic*	**Symbol:**	LN_Dk-Brown1815_Embankment_DykeWithTransportation
Symbol Criteria:	Embankment Type <> [Fill] AND (Transportation System Type [1] = [Aqueduct] or [Canal] or [Pipeline] or [Railway] or [Road])			**Label:**	None
				Color:	N/A

Feature Information					
Specification:	*LTDS*	**Feature Origin:**	*Topographic*	**Symbol:**	LN_Dk-Brown1815_Embankment_DykeWithoutTransportation
Symbol Criteria:	Embankment Type <> [Fill] AND (Transportation System Type [1] = [No Transportation System] or [Other] or [No Information])			**Label:**	None
				Color:	N/A

Engine Test Cell (AF060)

Surface

Feature Description:	*A structure wherein aircraft or rocket engines are tested.*
Inclusion Condition:	*Area >= 625 sq m.*

Notes

None

Feature Information					
Specification:	*LTDS*	**Feature Origin:**	*Topographic*	**Symbol:**	AR_Black_EngineTestCell
Symbol Criteria:	(Height Above Surface Level < 46) OR (Area >= 5625 and Height Above Surface Level >= 46)			**Label:**	"Engine test cell", 6 point Condensed Mixed case
				Color:	Black (58600)

Engine Test Cell (AF060)

Point

Inclusion Condition:	*Area < 625 sq m*

Notes

None

Feature Information					
Specification:	*LTDS*	**Feature Origin:**	*Topographic*	**Symbol:**	PT_Black_EngineTestCell
Symbol Criteria:	Height Above Surface Level: < 46 meters			**Label:**	"Engine test cell", 6 point Condensed Mixed case
				Color:	Black (58600)

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Engine Test Cell (AF060) *(Continued)*

Point

Feature Information					
Specification:	*LTDS*	**Feature Origin:**	*Topographic*	**Symbol:**	PT_Blue072_VO
Symbol Criteria:	Height Above Surface Level: >= 46 meters			**Label:**	"Engine test cell", 6 point Condensed Mixed case; [Highest Elevation], 6 point Condensed Bold Mixed case; "([Height Above Surface Level])", 6 point Condensed Mixed case
				Color:	Blue (46351)

Engineered Earthwork (AH025)

Surface

Feature Description:	*An excavation and/or embankment created by remolding the natural configuration of the terrain for the purpose of enhancing the defense of a site from armed attack.*
Inclusion Condition:	*Width >= 25 m.*

Notes

None

Feature Information					
Specification:	*LTDS*	**Feature Origin:**	*Topographic*	**Symbol:**	AR_Black_Engineered Earthwork
Symbol Criteria:	All			**Label:**	"Trench" if (Engineered Earthwork Type = [No Information] or [Other] or [Military Trench]); "[Engineered Earthwork Type]" if (Engineered Earthwork Type = [Battery] or [Rampart] or [Redoubt]); "Parapet" if (Engineered Earthwork Type = [Military Parapet]), 6 point Condensed Mixed case
				Color:	Black (58600)

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Engineered Earthwork (AH025)

Point

Inclusion Condition:	*Width < 25 m.*

Notes

None

Feature Information					
Specification:	*LTDS*	**Feature Origin:**	*Topographic*	**Symbol:**	LN_Black_EngineeredEarthwork
Symbol Criteria:	All			**Label:**	"Trench" if (Engineered Earthwork Type = [No Information] or [Other] or [Military Trench]); "[Engineered Earthwork Type]" if (Engineered Earthwork Type = [Battery] or [Rampart] or [Redoubt]); "Parapet" if (Engineered Earthwork Type = [Military Parapet]), 6 point Condensed Mixed case
				Color:	Black (58600)

Esker (DB100)

Curve

Feature Description:	*A long, narrow ridge of sand and gravel deposited by a glacial stream.*
Inclusion Condition:	*Length or Diameter >= 125 m.*

Notes

None

Feature Information					
Specification:	*LTDS*	**Feature Origin:**	*Topographic*	**Symbol:**	LN_Dk-Brown1815_Esker
Symbol Criteria:	All			**Label:**	"Esker", 6 point Condensed Mixed case
				Color:	Black (58600)

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Excavating Machine (AF050)

Curve

Feature Description:	*A mechanical device for removing materials from the ground.*
Inclusion Condition:	*Height Above Surface Level >= 46 m.*

Notes

None

Feature Information

Specification:	*LTDS*	**Feature Origin:**	*Topographic*	**Symbol:**	PT_Blue072_VO
Symbol Criteria:	Height Above Surface Level >= 46			**Label:**	"Excavating machine", 6 point Condensed Mixed case; and [Highest Elevation], 6 point Condensed Bold Mixed Case; and "([Height Above Surface Level])", 6 point Condensed Mixed Case
				Color:	Blue (46351)

Explosive Naval Mine (BD001)

Point

Feature Description:	*An explosive device used in naval warfare deployed on or below the sea surface.*
Inclusion Condition:	*All Required*

Notes

None

Feature Information

Specification:	*Chart #1*	**Feature Origin:**	*Hydrographic*	**Symbol:**	N23.1
Symbol Criteria:	All			**Label:**	"Naval mine", 7 Point Mixed case Italic
				Color:	Green (52813)

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Extraction Mine (AA010)

Surface

Feature Description:	*An excavation made in the terrain for the purpose of extracting and/or exploiting natural resources.*
Inclusion Condition:	*Area >= 15,625 square meters*

Notes

When 2 or more similar area features are separated by less than 0.5 mm at chart scale, the feature will be agglomerated to form an area multiple feature outline.

Coincident similar area features will be blended to form a single feature.

Area and line features will be generalized to detail compatible with scale.

Feature will be generalized to provide a more aesthetic contoured feature (i.e., smoothed).

Type for features on land shall be positioned on land if possible. If type must be placed in the water, it shall be positioned so it does not obscure or overprint hydrographic detail. Type for features in the water shall be positioned in the water. If possible, type shall not be positioned across the shoreline.

Limiting lines of feature are omitted if it coalesces with a road (AP030).

Feature Information

Specification:	*LTDS*	**Feature Origin:**	*Topographic*	**Symbol:**	AR_Black_Mine_Operational
Symbol Criteria:	(Condition of Facility = [Fully functional] or [Partially Functional] or [Other] or [No Information]) AND (Feature Function [1] <> [Peat Extraction]) AND (Mine Extraction Type <> [Peatery])			**Label:**	[Geographic Name Information : Full Name] if available: 6 point Condensed lower case; "([Product [1]])", 6 point Condensed lower case
				Color:	Black (58600)

Feature Information

Specification:	*LTDS*	**Feature Origin:**	*Topographic*	**Symbol:**	AR_Black_Mine_Abandoned
Symbol Criteria:	(Condition of Facility <> [Fully functional] or [Partially Functional] or [Other] or [No Information]) AND (Feature Function [1] <> [Peat Extraction]) AND (Mine Extraction Type <> [Peatery])			**Label:**	[Geographic Name Information : Full Name] if available: 6 point Condensed lower case; "([Product [1]])", 6 point Condensed lower case
				Color:	Black (58600)

Feature Information

Specification:	*LTDS*	**Feature Origin:**	*Topographic*	**Symbol:**	AR_Black_Mine_PeatCuttings
Symbol Criteria:	(Condition of Facility = [Fully functional] or [Partially Functional] or [Other] or [No Information]) and (Feature Function [1] = [Peat Extraction]) or (Mine Extraction Type = [Peatery])			**Label:**	"Peat cuttings", 6 point Regular Mixed case
				Color:	Black (58600)

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Extraction Mine (AA010)

Point

Inclusion Condition:	*Area < 15,625 square meters*

Notes

If one symbol coalesces with another symbol for the same type feature, displace symbols to allow a minimum separation of 0.21mm.

A cluster of 3 or more coalescing similar point features will be aggregated to form an area multiple feature outline.

Type for features on land shall be positioned on land if possible. If type must be placed in the water, it shall be positioned so it does not obscure or overprint hydrographic detail. Type for features in the water shall be positioned in the water. If possible, type shall not be positioned across the shoreline.

If >= 3 equal symbols would coalesce at map scale, portray with a representative pattern.

Feature Information					
Specification:	*LTDS*	**Feature Origin:**	*Topographic*	**Symbol:**	PT_Black_Mine_Operational
Symbol Criteria:	(Condition of Facility = [Fully functional] or [Partially Functional] or [Other] or [No Information]) AND (Feature Function [1] <> [Peat Extraction]) AND (Mine Extraction Type <> [Peatery])			**Label:**	[Geographic Name Information : Full Name] if available: 6 point Condensed lower case; "([Product [1]])", 6 point Condensed lower case
				Color:	Black (58600)

Feature Information					
Specification:	*LTDS*	**Feature Origin:**	*Topographic*	**Symbol:**	PT_Black_Mine_Abandoned
Symbol Criteria:	(Condition of Facility <> [Fully functional] or [Partially Functional] or [Other] or [No Information]) AND (Feature Function [1] <> [Peat Extraction]) AND (Mine Extraction Type <> [Peatery])			**Label:**	[Geographic Name Information : Full Name] if available: 6 point Condensed lower case; "([Product [1]])", 6 point Condensed lower case
				Color:	Black (58600)

Feature Information					
Specification:	*LTDS*	**Feature Origin:**	*Topographic*	**Symbol:**	PT_Black_Mine_PeatCutting
Symbol Criteria:	(Condition of Facility = [Fully functional] or [Partially Functional] or [Other] or [No Information]) and (Feature Function [1] = [Peat Extraction]) or (Mine Extraction Type = [Peatery]			**Label:**	None
				Color:	N/A

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Facility (AL010)

Surface

Feature Description:	*An area that has been developed to perform a specific principal function, consisting of one or more vertical constructions (for example: structures or buildings), horizontal constructions (for example: pavements, roads, rail tracks, or bridges), and/or supporting utilities (for example: power lines, water supply, or sewerage), plus the underlying land.*
Inclusion Condition:	*Area >= 15,625 square meters*

Notes

Coincident similar area features will be blended to form a single feature.

Area and line features will be generalized to detail compatible with scale.

Type for features on land shall be positioned on land if possible. If type must be placed in the water, it shall be positioned so it does not obscure or overprint hydrographic detail. Type for features in the water shall be positioned in the water. If possible, type shall not be positioned across the shoreline.

Descriptive terms, e.g., "Canal" shall be shown if the name is not known. If the descriptive word appears in the name, for example, "PANAMA CANAL". The descriptive type shall not be shown, i.e. do not show "Panama Canal Canal".

Limiting lines of feature are omitted if it coalesces with a road (AP030).

Feature Information

Specification:	*LTDS*	**Feature Origin:**	*Topographic*	**Symbol:**	AR_Black_Facility
Symbol Criteria:	Feature Function [1] = [Gas Oil Separation] or [Ore Dressing] or [Sugar Refining] or [Petroleum Refining] or [Chemical Manufacture] or [Metal Refining]			**Label:**	[Geographic Name Information : Full Name] if available, 6 point Condensed Mixed case, "([Manufacturing Information : Product [1]])" and "Refinery", 6 point Condensed Mixed case, if <> [No Product] or [Other] or [No Information];"Refinery" if Manufacturing Information : Product [1] = [No Product] or [Other] or [No Information] , "(destroyed)" if Condition of Facility = [Destroyed] or [Ruined], 6 point Condensed lower case
				Color:	Black (58600)

Feature Information

Specification:	*LTDS*	**Feature Origin:**	*Topographic*	**Symbol:**	AR_Black_Facility
Symbol Criteria:	Feature Function [1] = [Grain Milling] or [Sawmilling] or [Paper-mill] or [Cement Mill] or [Steel Mill]			**Label:**	[Geographic Name Information : Full Name] if available, 6 point Condensed Mixed case, "([Manufacturing Information : Product [1]])" and "Mill", 6 point Condensed Mixed case, if <> [No Product] or [Other] or [No Information]; "Mill" if [Manufacturing Information : Product [1] = [No Product] or [Other] or [No Information], "(destroyed)" if Condition of Facility = [Destroyed] or [Ruined], 6 point Condensed lower case
				Color:	Black (58600)

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Facility (AL010) *(Continued)*

Surface

Feature Information					
Specification:	*LTDS*	**Feature Origin:**	*Topographic*	**Symbol:**	AR_Black_Facility
Symbol Criteria:	Feature Function [1] = [Manufacturing] or [Food Product Manufacture] or [Food Processing] or [Meat Processing] or [Seafood Processing] or [Fruit and/or Vegetable Processing] or [Oil-mill] or [Dairying] or [Grain Milling] or [Baking] or [Sugar Manufacture] or [Sugar Milling] or [Confection Manufacture] or [Pasta Manufacture] or [Prepared Meal Manufacture] or [Animal Feed Manufacture] or [Spirit Distillery] or [Winery] or [Brewing] or [Soft Drink Manufacture] or [Tobacco Product Manufacture] or [Textile Manufacture] or [Apparel Manufacture] or [Leather Product Manufacture] or [Footwear Manufacturing] or [Sawmilling] or [Wooden Construction Product Manufacture] or [Paper-mill] or [Printing] or [Coke Manufacture] or [Medicinal Product Manufacture] or [Rubber Product Manufacture] or [Plastic Product Manufacture] or [Glass Product Manufacture] or [Refractory Product Manufacture] or [Clay Product Manufacture] or [Ceramic Product Manufacture] or [Cement Mill] or [Cement Product Manufacture] or [Stone Product Manufacture] or [Steel Mill] or [Foundry] or [Structural Metal Product Manufacture] or [Munitions Manufacture] or [Fabricated Metal Product Manufacture] or [Electronic Equipment Manufacture] or [Electrical Equipment Manufacture] or [Machinery Manufacture] or [Motor Vehicle Manufacture] or [Ship Construction] or [Railway Vehicle Manufacture] or [Aircraft Manufacture] or [Military Vehicle Manufacture] or [Furniture Manufacture] or [Jewellery Manufacture] or [Musical Instrument Manufacture] or [Sports Goods Manufacture] or [Game and/or Toy Manufacture] or [Medical and/or Dental Equipment Manufacture] or [Sewerage] or [Materials Recovery]			**Label:**	[[Geographic Name Information : Full Name] if available, 6 point Condensed Mixed case, "([Manufacturing Information : Product [1]])" and "Plant", 6 point Condensed Mixed case, if <> [No Product] or [Other] or [No Information]; "Plant" if [Manufacturing Information : Product [1] = [No Product] or [Other] or [No Information], "(destroyed)" if Condition of Facility = [Destroyed] or [Ruined], 6 point Condensed lower case
				Color:	Black (58600)

Feature Information					
Specification:	*LTDS*	**Feature Origin:**	*Topographic*	**Symbol:**	AR_Black_Facility
Symbol Criteria:	Feature Function [1] = [Radio Broadcasting] or [Television Broadcasting] or [Telecommunications] or [Retail Telecommunications] or [Wired Telecommunications] or [Main Telephone Exchange] or [Branch Telephone Exchange] or [Wireless Telecommunications] or [Mobile Phone Service] or [Satellite Telecommunications] or [Satellite Ground Control]			**Label:**	[Geographic Name Information : Full Name] if available, 6 point Condensed Mixed case, "Commmunications facility", "(destroyed)" if Condition of Facility = [Destroyed] or [Ruined], 6 point Condensed lower case
				Color:	Black (58600)

Feature Information					
Specification:	*LTDS*	**Feature Origin:**	*Topographic*	**Symbol:**	AR_Black_Facility
Symbol Criteria:	Feature Function [1] = [Nuclear Research Centre] or [Other] or [No information]			**Label:**	[Geographic Name Information : Full Name] if available, 6 point Condensed Mixed case, "Nuclear research center" if Feature Function [1] = (Nuclear Research Centre), 6 point Condensed Mixed case, "Facility" if Feature Function [1] = [Other] or [No information], 6 point Condensed Mixed case,"(destroyed)" if Condition of Facility = [Destroyed] or [Ruined], 6 point Condensed lower case
				Color:	Black (58600)

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Facility (AL010)

Point

Inclusion Condition:	*Area < 15,625 square meters*

Notes

The feature (when height <= 46 meters or when height is not known) shall be oriented perpendicular (90 degrees) to the nearby road (AP030), cart track (AP010), trail (AP050), or railroad (AN010).

If one symbol coalesces with another symbol for the same type feature, displace symbols to allow a minimum separation of 0.21mm.

Type for features on land shall be positioned on land if possible. If type must be placed in the water, it shall be positioned so it does not obscure or overprint hydrographic detail. Type for features in the water shall be positioned in the water. If possible, type shall not be positioned across the shoreline.

Descriptive terms, e.g., "Canal" shall be shown if the name is not known. If the descriptive word appears in the name, for example, "PANAMA CANAL". The descriptive type shall not be shown, i.e. do not show "Panama Canal Canal".

If >= 3 equal symbols would coalesce at map scale, portray with a representative pattern.

Feature Information

Specification:	*LTDS*	Feature Origin:	*Topographic*	Symbol:	PT_Black_Facility
Symbol Criteria:	(Feature Function [1] = [Gas Oil Separation] or [Ore Dressing] or [Sugar Refining] or [Petroleum Refining] or [Chemical Manufacture] or [Metal Refining]) and (Condition of Facility <> [Destroyed] or [Proposed] or [Ruined] or [Planned])			Label:	[Geographic Name Information : Full Name] if available, 6 point Condensed Mixed case, "([Manufacturing Information : Product [1]])" and "Refinery", 6 point Condensed Mixed case, if <> [No Product] or [Other] or [No Information];"Refinery" if Manufacturing Information : Product [1] = [No Product] or [Other] or [No Information] , "(destroyed)" if Condition of Facility = [Destroyed] or [Ruined], 6 point Condensed lower case
				Color:	Black (58600)

Feature Information

Specification:	*LTDS*	Feature Origin:	*Topographic*	Symbol:	PT_Black_Facility
Symbol Criteria:	(Feature Function [1] = [Grain Milling] or [Sawmilling] or [Paper-mill] or [Cement Mill] or [Steel Mill]) and (Condition of Facility <> [Destroyed] or [Proposed] or [Ruined] or [Planned])			Label:	[Geographic Name Information : Full Name] if available, 6 point Condensed Mixed case, "([Manufacturing Information : Product [1]])" and "Mill", 6 point Condensed Mixed case, if <> [No Product] or [Other] or [No Information]; "Mill" if [Manufacturing Information : Product [1] = [No Product] or [Other] or [No Information], "(destroyed)" if Condition of Facility = [Destroyed] or [Ruined], 6 point Condensed lower case
				Color:	Black (58600)

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Facility (AL010) *(Continued)*

Point

Feature Information					
Specification:	*LTDS*	**Feature Origin:**	*Topographic*	**Symbol:**	PT_Black_Facility
Symbol Criteria:	(Feature Function [1] = [Radio Broadcasting] or [Television Broadcasting] or [Telecommunications] or [Retail Telecommunications] or [Wired Telecommunications] or [Main Telephone Exchange] or [Branch Telephone Exchange] or [Wireless Telecommunications] or [Mobile Phone Service] or [Satellite Telecommunications] or [Satellite Ground Control]) and (Condition of Facility <> [Destroyed] or [Proposed] or [Ruined] or [Planned])			**Label:**	[Geographic Name Information : Full Name] if available, 6 point Condensed Mixed case; "Commmunication facility", 6 point Condensed Mixed case
				Color:	Black (58600)

Feature Information					
Specification:	*LTDS*	**Feature Origin:**	*Topographic*	**Symbol:**	PT_Black_Facility
Symbol Criteria:	(Feature Function [1]= [Manufacturing] or [Food Product Manufacture] or [Food Processing] or [Meat Processing] or [Seafood Processing] or [Fruit and/or Vegetable Processing] or [Oil-mill] or [Dairying] or [Grain Milling] or [Baking] or [Sugar Manufacture] or [Sugar Milling] or [Confection Manufacture] or [Pasta Manufacture] or [Prepared Meal Manufacture] or [Animal Feed Manufacture] or [Spirit Distillery] or [Winery] or [Brewing] or [Soft Drink Manufacture] or [Tobacco Product Manufacture] or [Textile Manufacture] or [Apparel Manufacture] or [Leather Product Manufacture] or [Footwear Manufacturing] or [Sawmilling] or [Wooden Construction Product Manufacture] or [Paper-mill] or [Printing] or [Coke Manufacture] or [Medicinal Product Manufacture] or [Rubber Product Manufacture] or [Plastic Product Manufacture] or [Glass Product Manufacture] or [Refractory Product Manufacture] or [Clay Product Manufacture] or [Ceramic Product Manufacture] or [Cement Mill] or [Cement Product Manufacture] or [Stone Product Manufacture] or [Steel Mill] or [Foundry] or [Structural Metal Product Manufacture] or [Munitions Manufacture] or [Fabricated Metal Product Manufacture] or [Electronic Equipment Manufacture] or [Electrical Equipment Manufacture] or [Machinery Manufacture] or [Motor Vehicle Manufacture] or [Ship Construction] or [Railway Vehicle Manufacture] or [Aircraft Manufacture] or [Military Vehicle Manufacture] or [Furniture Manufacture] or [Jewellery Manufacture] or [Musical Instrument Manufacture] or [Sports Goods Manufacture] or [Game and/or Toy Manufacture] or [Medical and/or Dental Equipment Manufacture] or [Sewerage] or [Materials Recovery]) and (Condition of Facility <> [Destroyed] or [Proposed] or [Ruined] or [Planned])			**Label:**	[[Geographic Name Information : Full Name] if available, 6 point Condensed Mixed case, "([Manufacturing Information : Product [1]])" and "Plant", 6 point Condensed Mixed case, if <> [No Product] or [Other] or [No Information]; "Plant" if [Manufacturing Information : Product [1] = [No Product] or [Other] or [No Information], "(destroyed)" if Condition of Facility = [Destroyed] or [Ruined], 6 point Condensed lower case
				Color:	Black (58600)

Feature Information					
Specification:	*LTDS*	**Feature Origin:**	*Topographic*	**Symbol:**	PT_Black_Facility
Symbol Criteria:	Feature Function [1] = [Nuclear Research Centre] or [Other]) and (Condition of Facility <> [Destroyed] or [Proposed] or [Ruined] or [Planned]) and (Height Above Surface level < 46)			**Label:**	[Geographic Name Information : Full Name] if available, 6 point Condensed Mixed case, "Nuclear research center" if Feature Function [1] = [Nuclear Research Centre], 6 point Condensed Mixed case, "Facility" if Feature Function [1] = [Other] or [No information], 6 point Condensed Mixed case
				Color:	Black (58600)

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Facility (AL010) *(Continued)*

Point

Feature Information					
Specification:	*LTDS*	**Feature Origin:**	*Topographic*	**Symbol:**	PT_Black_Facility_Destroyed
Symbol Criteria:	Condition of Facility = [Destroyed] or [Ruined]			**Label:**	None
				Color:	N/A

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Fairground (AK090)

Surface

Feature Description:	*An area where permanent facilities exist to hold outdoor fairs, circuses or exhibitions.*
Inclusion Condition:	*Area >= 15,625 square meters*

Notes

Coincident similar area features will be blended to form a single feature.

Area and line features will be generalized to detail compatible with scale.

Type for features on land shall be positioned on land if possible. If type must be placed in the water, it shall be positioned so it does not obscure or overprint hydrographic detail. Type for features in the water shall be positioned in the water. If possible, type shall not be positioned across the shoreline.

Descriptive terms, e.g., "Canal" shall be shown if the name is not known. If the descriptive word appears in the name, for example, "PANAMA CANAL". The descriptive type shall not be shown, i.e. do not show "Panama Canal Canal".

Limiting lines of feature are omitted if it coalesces with a road (AP030).

If a clearing exists inside of the area feature, and the size of the clearing is equal to or greater than the area for the surrounding area feature, the clearing is shown as an open space inside the surround feature. If the area of the clearing is less than the area for the surrounding feature, the clearing is deleted and absorbed into the surrounding area feature.

If two area features with the same feature name do not connect at any point, and have a space between them less than 2.5 mm at map/chart scale, delete the open space that is less than 2.5 mm between the features and combine them into one area feature.

If a portion of an area feature has a minimum width of less than 2.5 mm at map/chart scale, delete that portion of the area feature that is not at least 2.5 mm wide, measured from perimeter to perimeter. If the deletion of the portion of the area based on the above criteria will reduce the area of the remaining portion of the area feature to below the minimum area inclusion, do not delete the narrow portion of the feature that is less than 2.5 mm wide. If the deletion of the portion of the area based on the above criteria will split two larger areas connected by a narrow strip into two separate areas, either of which would be below minimum area inclusion, do not delete the narrow portion of the feature that is less than 2.5 mm wide.

Feature Information

Specification:	*LTDS*	**Feature Origin:**	*Topographic*	**Symbol:**	AR_Black_Fairground
Symbol Criteria:	Area >= 15,625			**Label:**	[Geographic Name Information : Full Name] if available, "Fairgrounds", 6-16 point Condensed Mixed case
				Color:	Black (58600)

Fairground (AK090)

Point

Inclusion Condition:	*Area < 15,625 square meters AND Navigation Landmark: [True]*

Notes

None

Feature Information

Specification:	*LTDS*	**Feature Origin:**	*Topographic*	**Symbol:**	PT_Black_Fairground
Symbol Criteria:	Area < 15,625 square meters AND Navigation Landmark = [True]			**Label:**	[Geographic Name Information : Full Name] if available, "Fairgrounds", 6 point Condensed Mixed case
				Color:	Black (58600)

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Fence (AL070)

Curve

Feature Description:	*A man-made barrier of relatively light structure used as an enclosure or boundary.*
Inclusion Condition:	*Length or Diameter >= 125 m. AND Predominant Feature Height >= 2 m.*

Notes

Area and line features will be generalized to detail compatible with scale.

Fences shall not be shown if parallel to and < 25 meters from any linear feature.

Walls or Fences which enclose the following areal features shall not be shown: Mobile Home park, Amusement Park, Athletic Field, Campground, Drive-In Theater, Fairgrounds, Golf Course, Stadium, Zoo, and Cemetery.

Feature Information

Specification:	*LTDS*	**Feature Origin:**	*Topographic*	**Symbol:**	LN_Black_Fence
Symbol Criteria:	All			**Label:**	None
				Color:	N/A

Ferry Crossing (AQ070)

Curve

Feature Description:	*A route where a ferry crosses from one shore to another.*
Inclusion Condition:	*ALL REQUIRED*

Notes

Area and line features will be generalized to detail compatible with scale.

Ferries may be abbreviated to "Fy" when the label coalesces with other detail.

Label shall be position above feature, reading from left to right (or to the left of the vertical feature, reading bottom to top), at a 0.5 mm distance and parallel to respective feature. Label shall preferably be positioned at the midpoint of the line segment or symbol; however, it may be displaced laterally along respective feature to avoid overprinting other symbols or labels. If space will not permit placing label parallel to feature, offset the label in accordance with Rule L-4261 below and use a leader line to identify its location along the feature.

Feature name, label, data information holder, and/or symbol shall be positioned, reading left to right, parallel to the tangent of the center of the southern neatline of the Map/Chart.

Descriptive terms, e.g., "Canal" shall be shown if the name is not known. If the descriptive word appears in the name, for example, "PANAMA CANAL". The descriptive type shall not be shown, i.e. do not show "Panama Canal Canal".

Omit if not shown in conjunction with drainage feature.

Pedestrian Ferry crossings are shown only when there is no Road, Bridge, Causeway, Vehicular Ferry Crossing, Railroad Ferry Crossing, or Both Road and Railroad Ferry Crossing the water body within 635 meters of the pedestrian ferry.

Feature Information

Specification:	*Chart #1*	**Feature Origin:**	*Hydrographic*	**Symbol:**	M51
Symbol Criteria:	Ferry Crossing Type = [Cable]			**Label:**	"Cable Ferry", 6 point Mixed case;
				Color:	Black (58600)

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Ferry Crossing (AQ070) *(Continued)*

Curve

Feature Information					
Specification:	*Chart #1*	**Feature Origin:**	*Hydrographic*	**Symbol:**	M50
Symbol Criteria:	Ferry Crossing Type = [Free-moving] or [Ice] or [Other] or [No Information]			**Label:**	None
				Color:	N/A

Ferry Station (AQ080)

Surface

Feature Description:	*A location where a ferry takes on or discharges its load.*
Inclusion Condition:	*Length or Diameter >= 25 m.*

Notes

None

Feature Information					
Specification:	*LTDS*	**Feature Origin:**	*Topographic*	**Symbol:**	AR_Black_FerryStation
Symbol Criteria:	All			**Label:**	[Geographic Name Information : Full Name] if available, "Ferry station", 6 point Condensed Mixed case
				Color:	Black (58600)

Ferry Station (AQ080)

Point

Inclusion Condition:	*Length or Diameter < 25 m.*

Notes

None

Feature Information					
Specification:	*LTDS*	**Feature Origin:**	*Topographic*	**Symbol:**	PT_Black_FerryStation
Symbol Criteria:	Height Above Surface level < 46			**Label:**	[Geographic Name Information : Full Name] if available, "Ferry station", 6 point Condensed Mixed case
				Color:	Black (58600)

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Firing Range (FA015)

Surface

Feature Description:	*A site designated for the purpose of discharging firearms or detonating munitions.*
Inclusion Condition:	*Area >= 15,625 sq m.*

Notes

None

Feature Information					
Specification:	*LTDS*	**Feature Origin:**	*Topographic*	**Symbol:**	AR_Black_FiringRange
Symbol Criteria:	All			**Label:**	[Geographic Name Information : Full Name] if available and "Firing range", 6 point Condensed Mixed case
				Color:	Black (58600)

Fish Farm Facility (BH051)

Surface

Feature Description:	*A facility involved in the breeding (hatching and associated activities) and cultivation (raising for release or harvesting) of fish in tanks or landlocked enclosures.*
Inclusion Condition:	*Area >= 15,625 sq m.*

Notes

When 2 or more similar area features are separated by less than 0.5 mm at chart scale, the feature will be agglomerated.

Area and line features will be generalized to detail compatible with scale.

Label feature as per hierarchy for topo type placement to south neatline corners reading left to right:
1. Position hierarchy: (a.) northeast (preferred position), (b.) southeast (1st alternative), (c.) northwest (2nd alternative, (d.) southwest (3rd alternative), (e.) top-centered (4th alternative), (f.) bottom-centered (5th alternative): (Hierarchy is based on type positioning so as to avoid overprinting other type or obscuring detail.);
2. Minimum space between type placement and feature symbol is 0.5 mm;
3. This method of type placement shall be used for areal features when space does not permit labeling within that feature.

Names placement shall be oriented to the longest axis of the feature reading left to right and placed within the area outline and centered. If the longest axis is perpendicular to the south neatline, the type shall be placed outside of the area outline, preferred position is northeast of the feature (Rule L-3505), but may be placed at any position around the feature so as not to overprint any other feature type and reading left to right.

Omit from Built-up Area.

Do not show land tint inside symbol.

Feature Information					
Specification:	*LTDS*	**Feature Origin:**	*Topographic*	**Symbol:**	AR_Cyan-31_MarineFarm
Symbol Criteria:	All			**Label:**	[Geographic Name Information : Full Name] if available, 6 point Condensed Italics Mixed case; "Fish farm", 6 point Condensed Italics Mixed case
				Color:	Black (58600)

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Fish Farm Facility (BH051)

Point

Inclusion Condition:	*Area < 15,625 sq m.*

Notes

When 2 or more similar area features are separated by less than 0.5 mm at chart scale, the feature will be agglomerated.

Area and line features will be generalized to detail compatible with scale.

Label feature as per hierarchy for topo type placement to south neatline corners reading left to right:
1. Position hierarchy: (a.) northeast (preferred position), (b.) southeast (1st alternative), (c.) northwest (2nd alternative, (d.) southwest (3rd alternative), (e.) top-centered (4th alternative), (f.) bottom-centered (5th alternative): (Hierarchy is based on type positioning so as to avoid overprinting other type or obscuring detail.);
2. Minimum space between type placement and feature symbol is 0.5 mm;
3. This method of type placement shall be used for areal features when space does not permit labeling within that feature.

Names placement shall be oriented to the longest axis of the feature reading left to right and placed within the area outline and centered. If the longest axis is perpendicular to the south neatline, the type shall be placed outside of the area outline, preferred position is northeast of the feature (Rule L-3505), but may be placed at any position around the feature so as not to overprint any other feature type and reading left to right.

Omit from Built-up Area.

Feature Information

Specification:	*LTDS*	**Feature Origin:**	*Topographic*	**Symbol:**	PT_Cyan_FishFarm
Symbol Criteria:	All			**Label:**	[Geographic Name Information : Full Name] if available, 6 point Condensed Italics Mixed case
				Color:	Black (58600)

Fish Haven (BD125)

Surface

Feature Description:	*An area on a water body bottom established to simulate a natural reef and intended to enhance fishery resources and commercial and/or recreational fishing opportunities.*
Inclusion Condition:	*Length >= 200 meters*

Notes

Use the following abbreviations for Horizontal Accuracy and Existence values:
If Position Approximate, label "PA";
If Position Doubtful, label "PD";
If Existence Doubtful, label "ED";
If Reported, label "Rep".

Date shall only be shown if Existence = Reported. Date shall be shown in parentheses if available. Always show "Rep" and date on the same line.
Unless specifically indicated on source material as inaccurate or doubtful, etc., or modified by other override rules, features shall be assumed to be accurate and definite. This rule applied to accuracy and existence of unique features, and is not related to the accuracy of the source as a whole.

Following IHO practice, features whose positions are approximate are labeled "PA" on the source charts. Features whose positions are doubtful are labeled "PD" on the source charts. Features whose existence is doubtful are labeled "ED" on the source charts. Features that have only been reported are labeled "Rep" on the source charts, and usually show the date of report in parentheses.

Definitions:
PA - Position Approximate = The position has not been accurately determined, or does not remain fixed.;
PD - Position Doubtful = The feature has been reported in various positions, but not definitely determined to be in any.;
ED - Existence Doubtful = Indicates the possible existence of the feature, the actual existence of which has not been established.

Show Hydrographic Depth or Hydrographic Height values in whole meters if decimeter value is 0; otherwise show the decimeter value as a subscript to the meter value.

Date shall only be shown if Existence = Reported. Date shall be shown in parentheses if available. Always show "Rep" and date on the same line.

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Fish Haven (BD125) *(Continued)*

Surface

Notes

Priority for type placement:
1-right center,
2-bottom center,
3-left center,
4-top center.
(A) Minimum distance from symbol - 1 mm.
(B) Maximum distance from symbol before choosing the next highest priority:
(#1) 4 mm measured to the West end;
(#2) 4 mm measured to the North side (top);
(#3) 4 mm measured to the East end;
(#4) 4 mm measured to the South side (bottom)

If symbols overprint each other, labels are condensed as follows:
(1.) If the labels are identical, only one is retained.;
(2,) If the labels are not identical, they shall be condensed in to one legend, e.g., "Fishhaven and Well".
If multiple depths are shown, only the shallowest is retained.

Labels may be stacked, readable top to bottom to avoid overprints.

Type shall be placed in the following preference:
(a.) Placed on one horizontal line centered in feature.;
(b.) Broken into multiple horizontal lines, centered in feature, with a 1 mm space between lines.;
(c.) Placed on one line outside feature. Hydrographic Depth shall always be positioned inside area.;
(d.) Placed on two lines to avoid overprints. If present, Hydrographic Depth shall always be placed inside area.

Labels shall be placed in water area, and shall not overprint the shoreline or be placed on land or in the foreshore.

Unless specifically indicated on source material as inaccurate or doubtful, etc., or modified by other override rules, features shall be assumed to be accurate and definite. This rule applied to accuracy and existence of unique features, and is not related to the accuracy of the source as a whole.

The height of a hydrographic feature is that above the surface of the water at High Water. Hydrographic Depth/Height information is not applicable at High Water. Hydrographic Depth/Height information values (Depth Known), (Depth Known by Wire Drag), and (Depth Unknown) are used to describe knowledge about a depth of a feature when the feature is below the water surface/ submerged.
Hydrographic Depth/Height information values (Uncovering Height Known) and (Uncovering Height Unknown) are used to describe knowledge about the drying height of a feature when it covers and uncovers, i.e., between high and low water. Hydrographic depth is used to record the depth of a feature when Hydrographic Depth/Height information values are (Depth Known) and (Depth Known by Wire Drag). Hydrographic depth is not applicable when the depth is unknown or when the feature covers and uncovers.
Hydrographic Drying Height records the drying height of a feature when Hydrographic Depth/Height information values are known. Hydrographic Drying Height is not applicable when the drying height is unknown, or the feature is below water.

Vertical Datum category is the sounding datum to which a hydrographic depth, drying height, or the high water datum to which a Safe Overhead Clearance is referenced. If the hydrographic depth, drying height, or Safe Overhead Clearance is unknown or not applicable, the vertical datum category is not applicable.

When the boundary of an areal symbol crosses or overprints the shoreline, the boundary overprinting the shoreline or land is deleted.

If areas where separate dangers coalesce because of scale, the danger curves (dotted lines) are deleted except for those on the outermost perimeter of the aggregate danger hazard. A smooth dotted line is drawn around the area.

Dangers with danger curves (dotted lines) around them which fall inside other danger areas with danger curves around them, shall have the inner dotted line deleted.

A blue 31% tint shall not overprint other blue 31% tints. If two blue tinted symbols, or one tinted symbol and water tint overprint, only one 31% tint shall be shown in the area.

Feature Information					
Specification:	*Chart #1*	**Feature Origin:**	*Hydrographic*	**Symbol:**	K46.1
Symbol Criteria:	Hydrographic Vertical Positioning Information : Hydrographic Depth = (0 or null)			**Label:**	None
				Color:	N/A

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Fish Haven (BD125) *(Continued)*

Surface

Feature Information					
Specification:	*Chart #1*	**Feature Origin:**	*Hydrographic*	**Symbol:**	K46.2
Symbol Criteria:	Hydrographic Vertical Positioning Information : Hydrographic Depth > 0			**Label:**	[Hydrographic Vertical Positioning Information : Hydrographic Depth], 7 point lower case Italic/ 5 point subscript
				Color:	Black (58600)

Fish Haven (BD125)

Point

Inclusion Condition:	*Length < 200 meters*

Notes

Use the following abbreviations for Horizontal Accuracy and Existence values:
If Position Approximate, label "PA";
If Position Doubtful, label "PD";
If Existence Doubtful, label "ED";
If Reported, label "Rep".

Date shall only be shown if Existence = Reported. Date shall be shown in parentheses if available. Always show "Rep" and date on the same line.
Unless specifically indicated on source material as inaccurate or doubtful, etc., or modified by other override rules, features shall be assumed to be accurate and definite. This rule applied to accuracy and existence of unique features, and is not related to the accuracy of the source as a whole.

Following IHO practice, features whose positions are approximate are labeled "PA" on the source charts. Features whose positions are doubtful are labeled "PD" on the source charts. Features whose existence is doubtful are labeled "ED" on the source charts. Features that have only been reported are labeled "Rep" on the source charts, and usually show the date of report in parentheses.

Definitions:
PA - Position Approximate = The position has not been accurately determined, or does not remain fixed.;
PD - Position Doubtful = The feature has been reported in various positions, but not definitely determined to be in any.;
ED - Existence Doubtful = Indicates the possible existence of the feature, the actual existence of which has not been established.

Show Hydrographic Depth or Hydrographic Height values in whole meters if decimeter value is 0; otherwise show the decimeter value as a subscript to the meter value.

Date shall only be shown if Existence = Reported. Date shall be shown in parentheses if available. Always show "Rep" and date on the same line.

If point symbol overprints shoreline, the Hydrographic Depth type, posicut, or graphic in the center of the symbol shall be displaced seaward until they no longer overprint the shoreline. If the danger curve (dotted perimeter line) overprints the shoreline, that portion of the dotted perimeter line falling on land shall be deleted.

Priority for type placement:
1-right center,
2-bottom center,
3-left center,
4-top center.
(A) Minimum distance from symbol - 1 mm.
(B) Maximum distance from symbol before choosing the next highest priority:
(#1) 4 mm measured to the West end;
(#2) 4 mm measured to the North side (top);
(#3) 4 mm measured to the East end;
(#4) 4 mm measured to the South side (bottom)

If symbols overprint each other, labels are condensed as follows:
(1.) If the labels are identical, only one is retained.;
(2,) If the labels are not identical, they shall be condensed in to one legend, e.g., "Fishhaven and Well".
If multiple depths are shown, only the shallowest is retained.

If symbols overprint each other, labels are condensed as follows: (1.) If the labels are identical, only one is retained.; (2,) If the labels are not identical, they shall be condensed in to one legend, e.g., "Fishhaven and Well". If multiple depths are shown, only the shallowest is retained.

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Fish Haven (BD125) *(Continued)*

Point

Notes

Labels may be stacked, readable top to bottom to avoid overprints.

Labels shall be placed in water area, and shall not overprint the shoreline or be placed on land or in the foreshore.

Hydrographic Depth or minimum draft depth label shall be placed adjacent to the symbol. The minimum draft depth shall be labeled as "<[minimum draft depth]>", 7 point lower Italic case/ 5 point subscript.

Variable type size for Hydrographic Depth values enclosed by danger curves (dotted circles): If Hydrographic Depth < 10, (a single digit principle digit), apply 7 point type to the principle digit, and 5 point type to the subscript, if there is one. If Hydrographic Depth >= 10 (a double digit principle digit), apply 6 point type to the principle digit, and 5 point type to the subscript, if there is one.

Unless specifically indicated on source material as inaccurate or doubtful, etc., or modified by other override rules, features shall be assumed to be accurate and definite. This rule applied to accuracy and existence of unique features, and is not related to the accuracy of the source as a whole.

Following IHO practice, features whose positions are approximate are labeled "PA" on the source charts. Features whose positions are doubtful are labeled "PD" on the source charts. Features whose existence is doubtful are labeled "ED" on the source charts. Features that have only been reported are labeled "Rep" on the source charts, and usually show the date of report in parentheses.

Definitions:
PA - Position Approximate = The position has not been accurately determined, or does not remain fixed.;
PD - Position Doubtful = The feature has been reported in various positions, but not definitely determined to be in any.;
ED - Existence Doubtful = Indicates the possible existence of the feature, the actual existence of which has not been established.

The height of a hydrographic feature is that above the surface of the water at High Water. Hydrographic Depth/Height information is not applicable at High Water. Hydrographic Depth/Height information values (Depth Known), (Depth Known by Wire Drag), and (Depth Unknown) are used to describe knowledge about a depth of a feature when the feature is below the water surface/ submerged. Hydrographic Depth/Height information values (Uncovering Height Known) and (Uncovering Height Unknown) are used to describe knowledge about the drying height of a feature when it covers and uncovers, i.e., between high and low water.
Hydrographic depth is used to record the depth of a feature when Hydrographic Depth/Height information values are (Depth Known) and (Depth Known by Wire Drag). Hydrographic depth is not applicable when the depth is unknown or when the feature covers and uncovers.
Hydrographic Drying Height records the drying height of a feature when Hydrographic Depth/Height information values are known. Hydrographic Drying Height is not applicable when the drying height is unknown, or the feature is below water.

Vertical Datum category is the sounding datum to which a hydrographic depth, drying height, or the high water datum to which a Safe Overhead Clearance is referenced. If the hydrographic depth, drying height, or Safe Overhead Clearance is unknown or not applicable, the vertical datum category is not applicable.

If areas where separate dangers coalesce because of scale, the danger curves (dotted lines) are deleted except for those on the outermost perimeter of the aggregate danger hazard. A smooth dotted line is drawn around the area.

Dangers with danger curves (dotted lines) around them which fall inside other danger areas with danger curves around them, shall have the inner dotted line deleted.

Hydrographic depth information (Depth Known by Wire Drag) is not applicable when Seafloor Category is (Fish Haven).

A blue 31% tint shall not overprint other blue 31% tints. If two blue tinted symbols, or one tinted symbol and water tint overprint, only one 31% tint shall be shown in the area.

The diameter of the symbol perimeter line may be reduced in congested area to avoid overprinting other symbols, but must be at least 2.0 mm diameter, and cannot come closer than 0.2 mm to any type or graphic symbol shown inside the circle. In non-congested areas the normal symbol diameter shall be shown.

When two or more point obstruction symbols, (dotted lines) overprint, and the attribution values are identical, one symbol shall be placed in the center of the group and shall be labeled with the number of obstructions in the group, e.g., 2 Obstr's, etc. Type is 6 point italic, in color Black SPC-58600.

Feature Information					
Specification:	*Chart #1*	**Feature Origin:**	*Hydrographic*	**Symbol:**	K46.1
Symbol Criteria:	Hydrographic Vertical Positioning Information : Hydrographic Dept = (0 or null)			**Label:**	None
				Color:	N/A

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Fish Haven (BD125) *(Continued)*

Point

Feature Information					
Specification:	*Chart #1*	**Feature Origin:**	*Hydrographic*	**Symbol:**	K46.2
Symbol Criteria:	Hydrographic Vertical Positioning Information : Hydrographic Depth > 0			**Label:**	[Hydrographic Vertical Positioning Information : Hydrographic Depth], 7 point lower case Italic/ 5 point subscript
				Color:	Black (58600)

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Fish Trap Area (BB112)

Surface

Feature Description:	*An area containing structures (usually portable) for catching fish.*
Inclusion Condition:	*Length >= 100 meters and Width >= 20 meters*

Notes

Type sizes for Maritime Limits and areas:
8 point - < 8 square cm.,
10 point - >= 8 and < 12 square cm,,
12 point - >= 12 and < 24 square cm.,
14 point - >= 24 and < 100 square cm.,
8 point - >= 100 square cm.;

Type placement for areas >= 100 square cm. to < 500 square cm.: Two labels are shown on approximately opposite sides of the area, preferably top and bottom.;

Type placement for areas >= 500 square cm.: Labels are placed at approximately 250 mm. interval around the perimeter of the feature.;

Do not place labels around sharp corners (interior angle < 135°), or along chart neatlines. Place type 1 mm away and parallel to area limit line, reading left to right, or bottom to top if limit line is vertical. Type is placed on the INSIDE of the area.

Priority for type placement: 1-right center, 2-bottom center, 3-left center, 4-top center.
(A) Minimum distance from symbol - 1 mm.
(B) Maximum distance from symbol before choosing the next highest priority:
(#1) 4 mm measured to the West end;
(#2) 4 mm measured to the North side (top);
(#3) 4 mm measured to the East end;
(#4) 4 mm measured to the South side (bottom)

Type placement for areas < 100 square cm. Type shall be center in the area reading from left to right.
If longer of the two axes is North-South +/- 20 degrees or East-West +/- 20 degrees, type is parallel to the south neatline.
(a.) If length < width times two, type shall be placed on two approximately equal lines without splitting words;
(b.) If length >= width times two, and major axis is East-West +/- 20 degrees, type shall be placed on one line;
(c.) If length >= width times two, and major axis is North-South +/- 20 degrees, type shall be placed with each word on a separate line.
If longer of the two axis is more than 20 degrees from either North-South or East-West, and length < width times two, type shall be parallel to south neatline and on two approximately equal lines without splitting words.
If longer of two axis is more than 20 degrees from either North-South or East-West, and length >= width times two, type shall be placed parallel to the major axis on one line. Type to be placed inside area, place type outside area, using Rule L-4722.

If two or more like features are in a group, label once in plural form, e.g., Dols, Fish Traps, etc. If single features are separated by 5 mm or more, they shall be labeled individually.

When the boundary of an areal symbol crosses or overprints the shoreline, the boundary overprinting the shoreline or land is deleted.

Minimum width for maritime limit symbols, other than pilot boarding areas, shall be 4 mm at chart scale.

Feature Information

Specification:	*Chart #1*	**Feature Origin:**	*Hydrographic*	**Symbol:**	K45
Symbol Criteria:	All			**Label:**	"Fish traps", 8-14 point Mixed case, Italic
				Color:	Black (58600)

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Fish Trap Area (BB112)

Point

Inclusion Condition:	*Length < 100 meters and Width < 20 meters*

Notes

If two or more like features are in a group, label once in plural form, e.g., Dols, Fish Traps, etc. If single features are separated by 5 mm or more, they shall be labeled individually.

When the boundary of an areal symbol crosses or overprints the shoreline, the boundary overprinting the shoreline or land is deleted.

Minimum width for maritime limit symbols, other than pilot boarding areas, shall be 4 mm at chart scale.

Feature Information

Specification:	*Chart #1*	**Feature Origin:**	*Hydrographic*	**Symbol:**	K44.2
Symbol Criteria:	All			**Label:**	None
				Color:	N/A

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Fish Weir (BB110)

Surface

Feature Description:	*A fence of stakes or wall of stones set in a river or along the shore to direct fish towards fish traps or nets.*
Inclusion Condition:	*Length >= 100 meters*

Notes

Type sizes for Maritime Limits and areas:
8 point - < 8 square cm.,
10 point - >= 8 and < 12 square cm,,
12 point - >= 12 and < 24 square cm.,
14 point - >= 24 and < 100 square cm.,
8 point - >= 100 square cm.;

Type placement for areas >= 100 square cm. to < 500 square cm.: Two labels are shown on approximately opposite sides of the area, preferably top and bottom.;

Type placement for areas >= 500 square cm.: Labels are placed at approximately 250 mm. interval around the perimeter of the feature.;

Do not place labels around sharp corners (interior angle < 135°), or along chart neatlines. Place type 1 mm away and parallel to area limit line, reading left to right, or bottom to top if limit line is vertical. Type is placed on the INSIDE of the area.

Priority for type placement: 1-right center, 2-bottom center, 3-left center, 4-top center.
(A) Minimum distance from symbol - 1 mm.
(B) Maximum distance from symbol before choosing the next highest priority:
(#1) 4 mm measured to the West end;
(#2) 4 mm measured to the North side (top);
(#3) 4 mm measured to the East end;
(#4) 4 mm measured to the South side (bottom)

Type placement for areas < 100 square cm. Type shall be center in the area reading from left to right.
If longer of the two axes is North-South +/- 20 degrees or East-West +/- 20 degrees, type is parallel to the south neatline.
(a.) If length < width times two, type shall be placed on two approximately equal lines without splitting words;
(b.) If length >= width times two, and major axis is East-West +/- 20 degrees, type shall be placed on one line;
(c.) If length >= width times two, and major axis is North-South +/- 20 degrees, type shall be placed with each word on a separate line.
If longer of the two axis is more than 20 degrees from either North-South or East-West, and length < width times two, type shall be parallel to south neatline and on two approximately equal lines without splitting words.
If longer of two axis is more than 20 degrees from either North-South or East-West, and length >= width times two, type shall be placed parallel to the major axis on one line. Type to be placed inside area, place type outside area, using Rule L-4722.

If two or more like features are in a group, label once in plural form, e.g., Dols, Fish Traps, etc. If single features are separated by 5 mm or more, they shall be labeled individually.

When the boundary of an areal symbol crosses or overprints the shoreline, the boundary overprinting the shoreline or land is deleted.

Minimum width for maritime limit symbols, other than pilot boarding areas, shall be 4 mm at chart scale.

Feature Information

Specification:	*Chart #1*	**Feature Origin:**	*Hydrographic*	**Symbol:**	K45
Symbol Criteria:	All			**Label:**	"Fish traps", 8-14 point Mixed case, Italic
				Color:	Black (58600)

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Fish Weir (BB110)

Point

Inclusion Condition:	*Length < 100 meters*

Notes

If two or more like features are in a group, label once in plural form, e.g., Dols, Fish Traps, etc. If single features are separated by 5 mm or more, they shall be labeled individually.

When the boundary of an areal symbol crosses or overprints the shoreline, the boundary overprinting the shoreline or land is deleted.

Minimum width for maritime limit symbols, other than pilot boarding areas, shall be 4 mm at chart scale.

Feature Information

Specification:	*Chart #1*	**Feature Origin:**	*Hydrographic*	**Symbol:**	K44.2
Symbol Criteria:	All			**Label:**	None
				Color:	N/A

Fishing Stakes (BB100)

Curve

Feature Description:	*Poles or stakes placed in shallow water to outline fishing grounds or to catch fish.*
Inclusion Condition:	*Length >= 100 meters and Width < 20 meters*

Notes

None

Feature Information

Specification:	*Chart #1*	**Feature Origin:**	*Hydrographic*	**Symbol:**	K44.1
Symbol Criteria:	All			**Label:**	None
				Color:	N/A

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Flagpole (AL073)

Point

Feature Description:	*A staff or pole on which a flag is raised.*
Inclusion Condition:	*Height Above Surface Level >= 46 m. OR Navigation Landmark: (True)*

Notes

A Feature (Not Conspicuous) shall not displace a Feature (Conspicuous)

Priority for type placement: 1-right center, 2-bottom center, 3-left center, 4-top center.
(A) Minimum distance from symbol - 1 mm.
(B) Maximum distance from symbol before choosing the next highest priority:
(#1) 4 mm measured to the West end;
(#2) 4 mm measured to the North side (top);
(#3) 4 mm measured to the East end;
(#4) 4 mm measured to the South side (bottom)

If the feature is symbolized by one of the following Posicuts, and that posicut overprints the shoreline, or other point symbols when oriented vertical, replace the posicut and label as indicated below.

Feature Information

Specification:	*LTDS*	**Feature Origin:**	*Topographic*	**Symbol:**	PT_Blue072_VO
Symbol Criteria:	Height Above Surface Level >= 46			**Label:**	"Flagpole", 6 point Condensed Mixed case; [Highest Elevation], 6 point Condensed Bold Mixed case; "([Height Above Surface Level])", 6 point Condensed Mixed case
				Color:	Blue (46351)

Flare Pipe (AF070)

Point

Feature Description:	*An open-ended pipe at which waste gases are burned.*
Inclusion Condition:	*Location Referenced to Shoreline = [Offshore] OR Height Above Surface Level >= 46 m. OR Navigation Landmark = [True]*

Notes

If one symbol coalesces with another symbol for the same type feature, displace symbols to allow a minimum separation of 0.21mm.

Type for features on land shall be positioned on land if possible. If type must be placed in the water, it shall be positioned so it does not obscure or overprint hydrographic detail. Type for features in the water shall be positioned in the water. If possible, type shall not be positioned across the shoreline.

When obstructions coalesce at map scale, label with highest obstruction information.

If >= 3 equal symbols would coalesce at map scale, portray with a representative pattern.

Feature Information

Specification:	*LTDS*	**Feature Origin:**	*Topographic*	**Symbol:**	PT_Black_FlarePipe
Symbol Criteria:	(Height above Surface Level < 46) and (Location Referenced to Shoreline <> [Offshore])			**Label:**	None
				Color:	N/A

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Flare Pipe (AF070) *(Continued)*

Point

Feature Information					
Specification:	*LTDS*	**Feature Origin:**	*Topographic*	**Symbol:**	PT_Blue072_VO
Symbol Criteria:	Height Above Surface Level >= 46 meters and (Location Referenced to Shoreline <> [Offshore])			**Label:**	"Flare pipe", 6 point Condensed Mixed case; [Highest Elevation], 6 point Condensed Bold Mixed case; "([Height Above Surface Level])", 6 point Condensed Mixed case
				Color:	Blue (46351)

Feature Information					
Specification:	*Chart #1*	**Feature Origin:**	*Hydrographic*	**Symbol:**	L11
Symbol Criteria:	Location Referenced to Shoreline = [Offshore]			**Label:**	"Fla", 6 point Condensed Mixed case
				Color:	Black (58600)

Floating Barrier (BD123)

Curve

Feature Description:	*A floating barrier used to protect a region of a body of water or to create a sheltered area on a body of water for storage purposes.*
Inclusion Condition:	*All Required*

Notes

Type shall be placed on the inside along the border of the barrier of whatever the barrier is containing or protecting.

Feature Information					
Specification:	*Chart #1*	**Feature Origin:**	*Hydrographic*	**Symbol:**	F29.1
Symbol Criteria:	Floating Barrier Type <> [Log]			**Label:**	"Floating Barrier", 6 point Mixed Case, Italic
				Color:	Black (58600)

Feature Information					
Specification:	*Chart #1*	**Feature Origin:**	*Hydrographic*	**Symbol:**	N61
Symbol Criteria:	Floating Barrier Type = [Log]			**Label:**	"Log Pond", 6 point Mixed Case, Italic
				Color:	Black (58600)

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Floating Dry Dock (BB199)

Surface

Feature Description:	*A form of dry dock consisting of a floating structure of one or more sections which can be partly submerged by controlled flooding to receive a vessel, then raised by pumping out the water so that the vessel's bottom can be exposed.*
Inclusion Condition:	*Length or Diameter >= 20 m.*

Notes

Priority for type placement: 1-right center, 2-bottom center, 3-left center, 4-top center.
(A) Minimum distance from symbol - 1 mm.
(B) Maximum distance from symbol before choosing the next highest priority:
(#1) 4 mm measured to the West end;
(#2) 4 mm measured to the North side (top);
(#3) 4 mm measured to the East end;
(#4) 4 mm measured to the South side (bottom)

When the area symbol or cased line symbol overprints the shoreline, shoreline is deleted.

If feature is floating and width < 2 mm, show only the perimeter.

If a drydock is floating, show land tint inside the symbol. If the drydock is non-floating, delete land tint from inside the symbol.

Feature Information

Specification:	*LTDS*	**Feature Origin:**	*Topographic*	**Symbol:**	AR_Black_FloatingDryDock
Symbol Criteria:	All			**Label:**	[Geographic Name Information : Full Name] if available; "Dry dock", 6 point Regular Mixed case
				Color:	Black (58600)

Flood Control Structure (BI044)

Surface

Feature Description:	*An artificial structure or gate that is utilized as a defense against flooding or storm surges.*
Inclusion Condition:	*Width >= 25 meters*

Notes

None

Feature Information

Specification:	*LTDS*	**Feature Origin:**	*Topographic*	**Symbol:**	AR_Black_FloodControl Structure
Symbol Criteria:	All			**Label:**	"Gate", 6 point Condensed mixed case if Flood Control Structure Type = [Dyke Gate] or [Emergency Gate] or [Flood Gate]; "Barrage", 6 point Condensed mixed case if Flood Control Structure Type = [Fixed Barrage] or [Movable Barrage]
				Color:	Black (58600)

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Flood Control Structure (BI044)

Curve

Inclusion Condition:	*Width < 25 meters*

Notes

None

Feature Information					
Specification:	*LTDS*	**Feature Origin:**	*Topographic*	**Symbol:**	LN_Black_FloodControl Structure
Symbol Criteria:	All			**Label:**	"Gate", 6 point Condensed mixed case if Flood Control Structure Type = [Dyke Gate] or [Emergency Gate] or [Flood Gate]; "Barrage", 6 point Condensed mixed case if Flood Control Structure Type = [Fixed Barrage] or [Movable Barrage]
				Color:	Black (58600)

Fog Signal (BC101)

Point

Feature Description:	*A device (for example: located on a vessel or an aid to navigation) that transmits a warning signal during periods of low visibility, especially due to fog.*
Inclusion Condition:	*All Required*

Notes

Whether to chart a fog signal:
A fog signal may be shown on charts on which vessels may navigate within range. Therefore its inclusion depends, in part, on some definition of its probable range.
IALA defines the 'usual' range of a fog signal as: 'the distance at which, in foggy weather, an observer has a 50% probability of hearing a sound signal when he is situated on the wing of a ship's bridge [on a vessel with an average ambient noise level]… in relatively calm weather, with no intervening obstacles'. Although not precise enough to chart, for the guidance of cartographers, the following 'usual' ranges are assumed;
• Powerful diaphone: 4 to 5 miles,
• Horn: up to 3 miles (but signals at harbour entrances are usually much weaker).
• Wave actuated bell or whistle: about 1/2 mile or less.
As the importance of fog signals has declined, they should usually be the first aid to navigation omitted in areas where chart clutter becomes a problem.

Types of Fog Signal:
It is impossible to indicate on charts all the variations in the sounds emitted but some major differences can be conveyed to the mariner by distinguishing the following types of fog signal.
If it is required to include the type of fog signal, the following international abbreviations or legends must be used. To avoid clutter, where a fog signal is co-located with another aid to navigation, the fog signal legend or the symbol R 1 should be shown, but not both.

Type of signal not labeled:
In these cases, the magenta symbol R1 (oriented and placed as necessary for clarity) must be shown on the appropriate scale charts. Examples of its use, alone or in conjunction with other aids to navigation are: Fog Signal (Alone), Floating Maritime Aids to Navigation (Buoys), Maritime Navigation Lights, and Maritime Radiobeacons.

Fog Signal Labels:
All Fog Signal labels associated with Floating Maritime Aids to Navigation (Buoys) will be shown in 7 point Mixed case Italic, Black (58600). All Fog Signal labels associated with Maritime Navigation Lights and Maritime Radiobeacons or are non-floating aids to navigation will be shown in 7 point Mixed case, Black (58600).
Label the following Fog Signal types if:
Fog Signal Type: [Explosive], label "Explos" (R10)
Fog Signal Type: [Diaphone], label "Dia" (R11)
Fog Signal Type: [Siren], label "Siren" (R12)
Fog Signal Type: [Horn], label "Horn" (R13)
Fog Signal Type: [Bell], label "Bell" (R14)
Fog Signal Type: [Whistle], label "Whis" (R15)
Fog Signal Type: [Gong], label "Gong" (R16)
Fog Signal : Characters Emitted:

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Fog Signal (BC101) *(Continued)*

Point

Notes

The characteristic rhythm of fog signals (other than those actuated by waves, which are irregular) may be more important than their type when mariners are attempting to identify them. The number of sound emissions (eg blasts, strokes) and the period may be charted, as described below:

A single sound (blast) repeated at intervals should be shown by '(1)' following the type of signal. (Example: Horn (1))

Multiple sounds (blasts) (other than Morse or composite signals) repeated at intervals must be shown by '(2)', '(3)', etc, following the type of signal. (Example: Horn (3))

Morse code rhythms must be shown by 'Mo' followed by the Morse letter in brackets. (Example: Siren Mo(N))

Composite rhythms (other than Morse) where groups of blasts are sounded must be shown as the full sequence. (Example: Siren (2+3))

The period of a fog signal is the time taken for a complete sequence of sound emissions. Where space permits, it must be charted for major signals (and on the largest scale charts for minor signals where considered useful) following the number of blasts. The period must be given in seconds, even for periods of one minute or longer. (Example: Dia(1)30s, Horn(2+3)90s)

FOG SIGNALS ON BUOYS:
The existence of fog signals on buoys must be indicated, on appropriate scales, by legends such as 'Bell', 'Whis' or 'Gong' or the symbol R1 if there is no requirement to define the type of fog signal.

Wave-actuated fog signals have no regular rhythm and must be charted by a legend indicating the type of signal eg 'Bell', 'Whis', 'Gong' against the buoy symbol R21.

Fog signals operated automatically should be charted by a legend which includes the number of blasts (or strokes) and the period.

Wave-actuated signals in conjunction with automatic signals should be charted as in the following example: Q(6)+LFl.15s Horn(1)15s Whis Symbol R22. Note: there is no R1 symbol included as the type of fog signal is shown.

Feature Information					
Specification:	*Chart #1*	**Feature Origin:**	*Hydrographic*	**Symbol:**	R1
Symbol Criteria:	All			**Label:**	None
				Color:	N/A

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Ford (BH070)

Curve

Feature Description:	*A shallow place in a body of water used as a crossing.*
Inclusion Condition:	*All Required*

Notes

Area and line features will be generalized to detail compatible with scale.

Label shall be position above feature, reading from left to right (or to the left of the vertical feature, reading bottom to top), at a 0.5 mm distance and parallel to respective feature. Label shall preferably be positioned at the midpoint of the line segment or symbol; however, it may be displaced laterally along respective feature to avoid overprinting other symbols or labels. If space will not permit placing label parallel to feature, offset the label and use a leader line to identify its location along the feature.

Feature name, label, data information holder, and/or symbol shall be positioned, reading left to right, parallel to the tangent of the center of the southern neatline of the Map/Chart.

Omit if not shown in conjunction with drainage feature.

Fords are shown where they relate to Roads, Cart Track, or Trail.

Retain feature only when associated with Cart Track, Road. Or Trail

Feature Information

Specification:	*LTDS*	**Feature Origin:**	*Topographic*	**Symbol:**	LN_Black_Ford
Symbol Criteria:	All			**Label:**	None
				Color:	N/A

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Foreshore (BA023)

Surface

Feature Description:	*The part of the shore or beach which lies between the low water mark and the upper limit of normal wave action.*
Inclusion Condition:	*All Required*

Notes

For [Maritime Bottom Characteristic : Maritime Bottom Material Type or Reason] is abbreviations.

Abbreviations for [Maritime Bottom Material Type or Reason] are:
Ash (Ash), Boulders (Bo), Chalk (Ck), Cinders (Cn), Cirripedia (Cir), Clay (Cy), Cobble (Cb), Coral (Co), Coral Head (Co Hd), Diatoms (Di), Foraminifera (Fr), Fucus (Fu), Globigerina (Gl), Grass (Grs), Gravel (G), Ground (Gd), Lava (Lv), Madrepores (Md), Manganese (Mn), Marl (Ml), Mattes (Ma), Mud (M), Mussels (Ms), Ooze (Oz), Oysters (Oy), Pebbles (P), Polyzoa (Po), Pteropods (Pt), Pumice (Pm), Quartz (Qz), Radiolaria (Rd), Rock (R), Sand (S), Schist (Sch), Scoria (Sc), Sea Tangle (Stg), Seaweed (Wd), Shells (Sh), Shingles (Sn), Silt (Si), Spicules (Spi), Sponge (Sp), Stones (St), Tufa (T).

Labeling areas, in order of preference:
(1.) Centered in area on one line in the area, type is horizontal, reading left to right.
(2.) Centered in area on one line in the area, oriented along the long axis of the feature, reading left to right, or bottom to top if axis is vertical.
(3.) Centered in area on two approximately equal lines, without splitting a word, type is horizontal, reading left to right.
(4.) Centered on two approximately equal lines without splitting a word, type is oriented along the long axis of the feature, reading left to right or bottom to top if axis is vertical.
(5.) Use priority for type placement if the area is too small to place the type inside the area.

Priority for type placement: 1-right center, 2-bottom center, 3-left center, 4-top center.
(A) Minimum distance from symbol - 1 mm.
(B) Maximum distance from symbol before choosing the next highest priority:
(#1) 4 mm measured to the West end;
(#2) 4 mm measured to the North side (top);
(#3) 4 mm measured to the East end;
(#4) 4 mm measured to the South side (bottom)

Symbols and associated area patterns of underpassing features (except drainage shorelines) are broken for all bridges, except footbridges. This rule does not apply to land tint on Littoral Planning Charts.

Delete dot portion of the symbol that is within 0.5 mm, at chart scale, of the shoreline (BA010)

Features with the same code, separated by less than 2 mm at chart scale, shall be combined into one areal feature.

A blue 31% tint shall not overprint other blue 31% tints. If two blue tinted symbols, or one tinted symbol and water tint overprint, only one 31% tint shall be shown in the area.

Feature Information

Specification:	*LTDS*	**Feature Origin:**	*Topographic*	**Symbol:**	AR_BlackCyan_Foreshore
Symbol Criteria:	All			**Label:**	[Maritime Bottom Characteristic : Maritime Bottom Material Type or Reason], 6 point Mixed Case, Italic
				Color:	Black (58600)

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Forest (EC015)

Surface

Feature Description:	*A large tract of land covered with trees and undergrowth sometimes mixed with pasture.*
Inclusion Condition:	*Width >= 65 m.*

Notes

Coincident similar area features will be blended to form a single feature.

Area and line features will be generalized to detail compatible with scale.

Feature will be generalized to provide a more aesthetic contoured feature (i.e., smoothed).

Type sizes per are sizes at map/chart scale: Area features only.
06 point - <= 770 mm square area and <= 14 mm width;
07 point - <= 2,296 mm square area and <= 28 mm width;
09 point - <= 5,192 mm square area and 44 mm width;
10 point - <=9,796 mm square area and <= 62 mm width;
12 point - <= 16,632 mm square area and <= 84 mm width;
14 point - <= 24,960 mm square area and <= 104 mm width;
16 point - > 24,960 mm square area. Where area measurements are inconsistent, the larger type size shall be used.
Where the full range of type sizes is not available for a particular label, the closest available type size shall be used.

Label feature as per hierarchy for topo type placement to south neatline corners reading left to right:
1. Position hierarchy: (a.) northeast (preferred position), (b.) southeast (1st alternative), (c.) northwest (2nd alternative, (d.) southwest (3rd alternative), (e.) top-centered (4th alternative), (f.) bottom-centered (5th alternative): (Hierarchy is based on type positioning so as to avoid overprinting other type or obscuring detail.);
2. Minimum space between type placement and feature symbol is 0.5 mm;
3. This method of type placement shall be used for areal features when space does not permit labeling within that feature.

Names placement shall be oriented to the longest axis of the feature reading left to right and placed within the area outline and centered. If the longest axis is perpendicular to the south neatline, the type shall be placed outside of the area outline, preferred position is northeast of the feature, but may be placed at any position around the feature so as not to overprint any other feature type and reading left to right.

Type for features on land shall be positioned on land if possible. If type must be placed in the water, it shall be positioned so it does not obscure or overprint hydrographic detail. Type for features in the water shall be positioned in the water. If possible, type shall not be positioned across the shoreline.

Symbols and associated area patterns of underpassing features (except drainage shorelines) are broken for all bridges, except footbridges. This rule does not apply to land tint on Littoral Planning Charts.

If a clearing exists inside of the area feature, and the size of the clearing is equal to or greater than the area for the surrounding area feature, the clearing is shown as an open space inside the surround feature. If the area of the clearing is less than the area for the surrounding feature, the clearing is deleted and absorbed into the surrounding area feature.

If two area features with the same feature code do not connect at any point, and have a space between them less than 2.5 mm at map/chart scale, delete the open space that is less than 2.5 mm between the features and combine them into one area feature.

If a portion of an area feature has a minimum width of less than 2.5 mm at map/chart scale, delete that portion of the area feature that is not at least 2.5 mm wide, measured from perimeter to perimeter. If the deletion of the portion of the area based on the above criteria will reduce the area of the remaining portion of the area feature to below the minimum area inclusion, do not delete the narrow portion of the feature that is less than 2.5 mm wide. If the deletion of the portion of the area based on the above criteria will split two larger areas connected by a narrow strip into two separate areas, either of which would be below minimum area inclusion, do not delete the narrow portion of the feature that is less than 2.5 mm wide.

Feature Information

Specification:	*LTDS*	**Feature Origin:**	*Topographic*	**Symbol:**	AR_Green362-31_Forest_Evergreen
Symbol Criteria:	(Area >= 15,625) and (Foliage Type = [Evergreen]) AND (Canopy Cover > = 51) AND (Predominant Vegetation Height > = 3)			**Label:**	[Geographic Name Information : Full Name], 6-16 point Condensed Upper case if available
				Color:	Black (58600)

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Forest (EC015) *(Continued)*

Surface

Feature Information					
Specification:	*LTDS*	**Feature Origin:**	*Topographic*	**Symbol:**	AR_Green362-31_Forest_Deciduous
Symbol Criteria:	(Area >= 15,625) and (Foliage Type: [Deciduous]) AND (Canopy Cover > = 51) AND (Predominant Vegetation Height > = 3)			**Label:**	[Geographic Name Information : Full Name], 6-16 point Condensed Upper case if available
				Color:	Black (58600)

Feature Information					
Specification:	*LTDS*	**Feature Origin:**	*Topographic*	**Symbol:**	AR_Green362-31_Forest_Mixed
Symbol Criteria:	(Area >= 15,625) and (Foliage Type = [Mixed]) AND (Canopy Cover > = 51) AND (Predominant Vegetation Height > = 3)			**Label:**	[Geographic Name Information : Full Name], 6-16 point Condensed Upper case if available
				Color:	Black (58600)

Feature Information					
Specification:	*LTDS*	**Feature Origin:**	*Topographic*	**Symbol:**	AR_Green362_Forest_ ScatteredTrees
Symbol Criteria:	(Area >= 15,625) and (Canopy Cover > =25 and Canopy Cover < =50) AND (Predominant Vegetation Height > = 3)			**Label:**	[Geographic Name Information : Full Name], 6-16 point Condensed Upper case if available
				Color:	Black (58600)

Feature Information					
Specification:	*LTDS*	**Feature Origin:**	*Topographic*	**Symbol:**	AR_Green362-31_Forest_Unknown
Symbol Criteria:	(Area >= 15,625) and (Foliage Type = [Other] or [No Information]) AND (Canopy Cover > = 51) AND (Predominant Vegetation Height > = 3)			**Label:**	[Geographic Name Information : Full Name], 6-16 point Condensed Upper case if available
				Color:	Black (58600)

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Forest (EC015)

Curve

Inclusion Condition:	*Width < 65 m.*

Notes

Area and line features will be generalized to detail compatible with scale.

Feature will be generalized to provide a more aesthetic contoured feature (i.e., smoothed).

Symbols and associated area patterns of underpassing features (except drainage shorelines) are broken for all bridges, except footbridges. This rule does not apply to land tint on Littoral Planning Charts.

If a portion of an area feature has a minimum width of less than 2.5 mm at map/chart scale, delete that portion of the area feature that is not at least 2.5 mm wide, measured from perimeter to perimeter. If the deletion of the portion of the area based on the above criteria will reduce the area of the remaining portion of the area feature to below the minimum area inclusion, do not delete the narrow portion of the feature that is less than 2.5 mm wide. If the deletion of the portion of the area based on the above criteria will split two larger areas connected by a narrow strip into two separate areas, either of which would be below minimum area inclusion, do not delete the narrow portion of the feature that is less than 2.5 mm wide.

Feature Information

Specification:	*LTDS*	Feature Origin:	*Topographic*	Symbol:	LN_Green362_RowOfTrees
Symbol Criteria:	Predominant Vegetation Height >= 3			Label:	None
				Color:	N/A

Forest (EC015)

Point

Inclusion Condition:	*(Area < 15,625) and (Navigation Landmark: [True]) and (Predominant Vegetation Height >= 3)*

Notes

None

Feature Information

Specification:	*LTDS*	Feature Origin:	*Topographic*	Symbol:	PT_Green362_Tree
Symbol Criteria:	*(Area < 15,625) and (Navigation Landmark = [True]) and (Predominant Vegetation Height >= 3)*			Label:	None
				Color:	N/A

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Fortified Building (AH055)

Surface

Feature Description:	*A building that is specifically designed or reinforced to provide for defense from armed attack.*
Inclusion Condition:	*Length or Diameter >= 25 m. OR Height Above Surface Level >= 46 m.*

Notes

None

Feature Information

Specification:	*LTDS*	**Feature Origin:**	*Topographic*	**Symbol:**	AR_Black_Fortification
Symbol Criteria:	(Area >= 5625)			**Label:**	[Geographic Name Information : Full Name] if available, 6 point Condensed Mixed case AND "Fort", 6 point Condensed Mixed case and "(destroyed)", 6 point Condensed Lower case if (Condition of Facility = [Destroyed] or [Ruined])
				Color:	Black (58600)

Fortified Building (AH055)

Point

Inclusion Condition:	*Length or Diameter < 25 m. OR Height Above Surface Level >= 46 m.*

Notes

None

Feature Information

Specification:	*LTDS*	**Feature Origin:**	*Topographic*	**Symbol:**	PT_Black_Fortification
Symbol Criteria:	((Height Above Surface Level < 46 meters) and (Area < 5625)) OR (Height Above Surface Level < 46 meters)			**Label:**	[Geographic Name Information : Full Name] if available: 6 point Condensed Mixed case AND "Fort", 6 point Condensed Mixed case and "(destroyed)", 6 point Condensed Lower case if (Condition of Facility = [Destroyed] or [Ruined])
				Color:	Black (58600)

Feature Information

Specification:	*LTDS*	**Feature Origin:**	*Topographic*	**Symbol:**	PT_Blue072_VO
Symbol Criteria:	Height Above Surface Level >= 46 meters			**Label:**	"Fort", 6 point Condensed Mixed case; [Highest Elevation], 6 point Condensed Bold Mixed case; "([Height Above Surface Level])", 6 point Condensed Mixed case
				Color:	Blue (46351)

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Foul Ground (BD050)

Surface

Feature Description:	*A region over which it is safe to navigate but which should be avoided for anchoring, intentional grounding, or ground fishing.*
Inclusion Condition:	*Length >= 200 meters*

Notes

Use the following abbreviations for Horizontal Accuracy and Existence values:
If Position Approximate, label "PA";
If Position Doubtful, label "PD";
If Existence Doubtful, label "ED";
If Reported, label "Rep".

Date shall only be shown if Existence = Reported. Date shall be shown in parentheses if available. Always show "Rep" and date on the same line.
Unless specifically indicated on source material as inaccurate or doubtful, etc., or modified by other override rules, features shall be assumed to be accurate and definite. This rule applied to accuracy and existence of unique features, and is not related to the accuracy of the source as a whole.
Following IHO practice, features whose positions are approximate are labeled "PA" on the source charts. Features whose positions are doubtful are labeled "PD" on the source charts. Features whose existence is doubtful are labeled "ED" on the source charts. Features that have only been reported are labeled "Rep" on the source charts, and usually show the date of report in parentheses.

Definitions:
PA - Position Approximate = The position has not been accurately determined, or does not remain fixed.;
PD - Position Doubtful = The feature has been reported in various positions, but not definitely determined to be in any.;
ED - Existence Doubtful = Indicates the possible existence of the feature, the actual existence of which has not been established.

Date shall only be shown if Existence = Reported. Date shall be shown in parentheses if available. Always show "Rep" and date on the same line.

Priority for type placement Rules:
1-right center,
2-bottom center,
3-left center,
4-top center.
(A) Minimum distance from symbol - 1 mm.
(B) Maximum distance from symbol before choosing the next highest priority:
(#1) 4 mm measured to the West end;
(#2) 4 mm measured to the North side (top);
(#3) 4 mm measured to the East end;
(#4) 4 mm measured to the South side (bottom)

If symbols overprint each other, labels are condensed as follows:
(1.) If the labels are identical, only one is retained.;
(2,) If the labels are not identical, they shall be condensed in to one legend, e.g., "Fishhaven and Well". If multiple depths are shown, only the shallowest is retained.

Labels may be stacked, readable top to bottom to avoid overprints.

Type Preference Placement Rules:
(a.) Placed on one horizontal line centered in feature.;
(b.) Broken into multiple horizontal lines, centered in feature, with a 1 mm space between lines.;
(c.) Placed on one line outside feature. Use Priority for type placement Rules for placement, Hydrographic Depth shall always be positioned inside area.;
(d.) Placed on two lines to avoid overprints. Use Priority for type placement Rules. If present, Hydrographic Depth shall always be placed inside area.

Labels shall be placed in water area, and shall not overprint the shoreline or be placed on land or in the foreshore.

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Foul Ground (BD050) *(Continued)*

Surface

Notes

When two or more identical features fall within 10 mm of each other, one legend with plural description shall be shown, with type placed as close as possible to the center of the group without overprinting features.
Where symbols with the same feature code but different values of Hydrographic Vertical Positioning Information: Water Level Effect (i.e. Always Dry , Always Submerged, Covers and Uncovers)) are adjacent to each other, the boundary of the feature that is the highest in vertical plane shall be shown, and the boundary of the feature above water is shown whole, while the feature that covers and uncovers has a portion of its symbol perimeter that overprints the higher feature deleted.

When two or more identical symbols overprint each other, delete the portions that overlap and show as one symbol with an outline of the combined area of the symbols.

Show land tint inside the symbol. If the Hydrographic Vertical Positioning Information: Water Level Effect: [Always Dry] show land tint only.

A blue 31% tint shall not overprint other blue 31% tints. If two blue tinted symbols, or one tinted symbol and water tint overprint, only one 31% tint shall be shown in the area.

Unless specifically indicated on source material as inaccurate or doubtful, etc., or modified by other override rules, features shall be assumed to be accurate and definite. This rule applied to accuracy and existence of unique features, and is not related to the accuracy of the source as a whole.

When the boundary of an areal symbol crosses or overprints the shoreline, the boundary overprinting the shoreline or land is deleted.

If areas where separate dangers coalesce because of scale, the danger curves (dotted lines) are deleted except for those on the outermost perimeter of the aggregate danger hazard. A smooth dotted line is drawn around the area.

Feature Information					
Specification:	*Chart #1*	**Feature Origin:**	*Hydrographic*	**Symbol:**	K31
Symbol Criteria:	All			**Label:**	"Foul", 6 point Mixed case, Italic
				Color:	Black (58600)

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Foul Ground (BD050)

Point

Inclusion Condition:	*Length < 200 meters*

Notes

Use the following abbreviations for Horizontal Accuracy and Existence values:
If Position Approximate, label "PA";
If Position Doubtful, label "PD";
If Existence Doubtful, label "ED";
If Reported, label "Rep".

Date shall only be shown if Existence = Reported. Date shall be shown in parentheses if available. Always show "Rep" and date on the same line.
Unless specifically indicated on source material as inaccurate or doubtful, etc., or modified by other override rules, features shall be assumed to be accurate and definite. This rule applied to accuracy and existence of unique features, and is not related to the accuracy of the source as a whole.

Following IHO practice, features whose positions are approximate are labeled "PA" on the source charts. Features whose positions are doubtful are labeled "PD" on the source charts. Features whose existence is doubtful are labeled "ED" on the source charts. Features that have only been reported are labeled "Rep" on the source charts, and usually show the date of report in parentheses.

Definitions:
PA - Position Approximate = The position has not been accurately determined, or does not remain fixed.;
PD - Position Doubtful = The feature has been reported in various positions, but not definitely determined to be in any.;
ED - Existence Doubtful = Indicates the possible existence of the feature, the actual existence of which has not been established.

Date shall only be shown if Existence = Reported. Date shall be shown in parentheses if available. Always show "Rep" and date on the same line.

If point symbol overprints shoreline, the Hydrographic Depth type, posicut, or graphic in the center of the symbol shall be displaced seaward until they no longer overprint the shoreline. If the danger curve (dotted perimeter line) overprints the shoreline, that portion of the dotted perimeter line falling on land shall be deleted.

Unless specifically indicated on source material as inaccurate or doubtful, etc., or modified by other override rules, features shall be assumed to be accurate and definite. This rule applied to accuracy and existence of unique features, and is not related to the accuracy of the source as a whole.

Feature Information					
Specification:	*Chart #1*	**Feature Origin:**	*Hydrographic*	**Symbol:**	K31, L22
Symbol Criteria:	All			**Label:**	None
				Color:	N/A

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Fountain (BH075)

Point

Feature Description:	*A monumental and/or ornamental structure containing moving water, often including jets, falls, or other decorative features.*
Inclusion Condition:	*Navigation Landmark = [True]*

Notes

None

Feature Information

Specification:	*LTDS*	**Feature Origin:**	*Topographic*	**Symbol:**	PT_Black_Fountain
Symbol Criteria:	Navigation Landmark = [True]			**Label:**	Geographic Name Information : Full Name if available, 6 point Condensed Mixed case; "Fountain", 6 point Condensed Mixed case
				Color:	Black (58600)

Fuel Storage Facility (AM075)

Surface

Feature Description:	*A facility for the storage of fuel (for example: jet fuel, gasoline, or diesel oil).*
Inclusion Condition:	*Area >= 15,625 sq m.*

Notes

None

Feature Information

Specification:	*LTDS*	**Feature Origin:**	*Topographic*	**Symbol:**	AR_Black_FuelStorageFacility
Symbol Criteria:	All			**Label:**	[Geographic Name Information: Full Name], 6 point Condensed Mixed case if available AND "Fuel storage", 6 point Condensed Mixed case AND "(destroyed)", 6 point Condensed Mixed case if (Condition of Facility = [Destroyed] or [Ruined])
				Color:	Black (58600)

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Fuel Storage Facility (AM075)

Point

Inclusion Condition:	*Area < 15,625 sq m.*

Notes

None

Feature Information					
Specification:	*LTDS*	**Feature Origin:**	*Topographic*	**Symbol:**	PT_Black_FuelStorageFacility
Symbol Criteria:	All			**Label:**	[Geographic Name Information: Full Name], 6 point Condensed Mixed case if available AND "Fuel storage", 6 point Condensed Mixed case AND "(destroyed)", 6 point Condensed Mixed case if (Condition of Facility = [Destroyed] or [Ruined])
				Color:	Black (58600)

Gantry (AL080)

Point

Feature Description:	*A permanent raised structure used to support equipment (for example: cranes, signal lights, or signs) while spanning over or around an object (for example: over a road or railroad, or around a ship hull or rocket).*
Inclusion Condition:	*Height Above Surface Level >= 46 m..*

Notes

None

Feature Information					
Specification:	*LTDS*	**Feature Origin:**	*Topographic*	**Symbol:**	PT_Blue072_VO
Symbol Criteria:	Height Above Surface Level >= 46			**Label:**	"Gantry", 6 point Condensed Mixed case; [Highest Elevation], 6 point Condensed Bold Mixed case; "([Height Above Surface Level])", 6 point Condensed Mixed case
				Color:	Blue (46351)

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Gate (AP040)

Curve

Feature Description:	*A barrier on a transportation route (for example: a road, a railway, a tunnel, or a bridge) that controls passage (may be opened and closed).*
Inclusion Condition:	*Length or Diameter >= 10 m.*

Notes

None

Feature Information					
Specification:	*LTDS*	**Feature Origin:**	*Topographic*	**Symbol:**	LN_Black_Gate
Symbol Criteria:	Gate Use: [Toll-gate] or [Border Crossing]			**Label:**	[Geographic Name Information : Full Name] if available, 6 point Condensed Mixed case
				Color:	Black (58600)

Gate (AP040)

Point

Inclusion Condition:	*Length or Diameter < 10 m.*

Notes

None

Feature Information					
Specification:	*LTDS*	**Feature Origin:**	*Topographic*	**Symbol:**	PT_Black_Gate
Symbol Criteria:	Gate Use: [Toll-gate] or [Border Crossing]			**Label:**	[Geographic Name Information : Full Name] if available, 6 point Condensed Mixed case
				Color:	Black (58600)

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Gauging Station (BI070)

Surface

Feature Description:	*A device that monitors river flow and/or depth, and any associated support structures.*
Inclusion Condition:	*TBD.*

Notes

None

Feature Information					
Specification:	*Other*	**Feature Origin:**	*Other*	**Symbol:**	TBD
Symbol Criteria:	TBD			**Label:**	TBD
				Color:	TBD

Gauging Station (BI070)

Point

Inclusion Condition:	*TBD.*

Notes

None

Feature Information					
Specification:	*Other*	**Feature Origin:**	*Other*	**Symbol:**	TBD
Symbol Criteria:	TBD			**Label:**	TBD
				Color:	TBD

Geodetic Point (ZB060)

Point

Feature Description:	*A physical point on the Earth's surface having a known position (for example: as determined by survey) and established as part of a geodetic network.*
Inclusion Condition:	*TBD.*

Notes

None

Feature Information					
Specification:	*Other*	**Feature Origin:**	*Other*	**Symbol:**	TBD
Symbol Criteria:	TBD			**Label:**	TBD
				Color:	TBD

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Geologic Fault (DB110)

Curve

Feature Description:	*A fracture in a rock formation, marked by the relative displacement on either side of the plane of the fracture.*
Inclusion Condition:	*Length or Diameter >= 125 m.*

Notes

Area and line features will be generalized to detail compatible with scale.

Feature will be generalized to provide a more aesthetic contoured feature (i.e., smoothed).

The names of Geologic Faults shall be shown along the fault line when known.

Label shall be position above feature, reading from left to right (or to the left of the vertical feature, reading bottom to top), at a 0.5 mm distance and parallel to respective feature. Label shall preferably be positioned at the midpoint of the line segment or symbol; however, it may be displaced laterally along respective feature to avoid overprinting other symbols or labels. If space will not permit placing label parallel to feature, offset the label and use a leader line to identify its location along the feature.

Feature Information

Specification:	*LTDS*	**Feature Origin:**	*Topographic*	**Symbol:**	LN_Dk-Brown1815_GeologicFault
Symbol Criteria:	All			**Label:**	[Geographic Name Information : Full Name], 6 point Condensed Mixed case if available and "Fault", 6 point Condensed Mixed case
				Color:	Black (58600)

Geopolitical Entity (FA002)

Surface

Feature Description:	*A region controlled by a political community having an organized government and possessing internal and external sovereignity, most often as a State but sometimes having a dependent relationship on another political authority or a special sovereignity status.*
Inclusion Condition:	*ALL REQUIRED*

Notes

If the boundary symbol and the projection line have the same line weight, the boundary symbol shall be shown in it's entirety 0.25 mm inside the projection line.

Feature must retain all cartographic detail (i.e., not thinned or smoothed).

Label line feature above (preferred) and parallel to the line with a 0.5 mm space between. Above means: readable from south or east Projection neatline.

If a boundary follows a road and the exact location is unknown, label "APPROXIMATE BOUNDARY".

If a boundary follows an edge of a Road, Track, Trail, or Railroad, every third unit of the boundary symbol shall be shown coincident with the feature. When the boundary is International, the width of the International boundary overprint shall be reduced to one-half of its normal width.

If a boundary follows a Road and the exact location is unknown, the symbol shall be shown in the center of the road with every third unit of the boundary symbol.

If a boundary follows a Shoreline (Coastal or Inland), every third unit of the boundary symbol shall be shown where coincident with the Shoreline.

If a Boundary is coincident with a single-line or narrow double-line stream, (i.e., width <3 mm at map scale) only every third unit of the boundary symbol shall be shown

If a boundary crosses a body of Open Water >= 20 mm width and alignment is known, the complete boundary symbol shall be shown. If the boundary alignment is unknown, the boundary symbol shall be shown in the open water area at the points of entry. If the size of the Open Water permits, show complete units of the boundary symbol.

Boundaries shown shall be included in legend.

Use a point of change for changes in status on an Administrative boundary, Armistice line, Cease-fire line, defacto boundary, demilitarized zone, or zone of occupation, unless change occurs at a symbolized boundary marker.

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Geopolitical Entity (FA002) *(Continued)*

Surface

Feature Information					
Specification:	*LTDS*	**Feature Origin:**	*Topographic*	**Symbol:**	AR_Black_GeopoliticalEntity _DMZ
Symbol Criteria:	Geopolitical Entity Type: [Demilitarized Zone (DMZ]			**Label:**	"DEMILITARIZED ZONE", 10 point Condensed Bold Upper case
				Color:	Black (58600)

Feature Information					
Specification:	*LTDS*	**Feature Origin:**	*Topographic*	**Symbol:**	LN_Black_GeopoliticalEntity_ ZoneOfOccupation
Symbol Criteria:	Geopolitical Entity Type: [Zone of Occupation]			**Label:**	[Geographic Name Information : Full Name] and "OCCUPIED", 10 point Condensed Bold Upper case
				Color:	Black (58600)

Geothermal Outlet (DB115)

Surface

Feature Description:	*A terrain surface feature controlled by or derived from the heat of the Earth's interior.*
Inclusion Condition:	*Area >= 625 sq m.*

Notes

Label feature as per hierarchy for topo type placement to south neatline corners reading left to right:
1. Position hierarchy: (a.) northeast (preferred position), (b.) southeast (1st alternative), (c.) northwest (2nd alternative, (d.) southwest (3rd alternative), (e.) top-centered (4th alternative), (f.) bottom-centered (5th alternative): (Hierarchy is based on type positioning so as to avoid overprinting other type or obscuring detail.);
2. Minimum space between type placement and feature symbol is 0.5 mm;
3. This method of type placement shall be used for areal features when space does not permit labeling within that feature.

In areas where fumaroles, geysers, and hot springs, are too numerous to symbolize, an area feature shall be symbolized and labeled.

Feature Information					
Specification:	*LTDS*	**Feature Origin:**	*Topographic*	**Symbol:**	AR_Dk-Brown1815_GeothermalOutlet
Symbol Criteria:	All			**Label:**	[Geographic Name Information : Full Name], 6 point Condensed Mixed case if available; "[Geothermal Outlet Type]", 6 point Condensed Mixed case if Geothermal Outlet Type: [Fissure] or [Fumarole] or [Hot Spring] or [Sulphur Spring] or [Geyser]
				Color:	Black (58600)

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Geothermal Outlet (DB115)

Point

Inclusion Condition:	*Area < 625 sq m.*

Notes

Label feature as per hierarchy for topo type placement to south neatline corners reading left to right:
1. Position hierarchy: (a.) northeast (preferred position), (b.) southeast (1st alternative), (c.) northwest (2nd alternative, (d.) southwest (3rd alternative), (e.) top-centered (4th alternative), (f.) bottom-centered (5th alternative): (Hierarchy is based on type positioning so as to avoid overprinting other type or obscuring detail.);
2. Minimum space between type placement and feature symbol is 0.5 mm;
3. This method of type placement shall be used for areal features when space does not permit labeling within that feature.

Squiggly tail of symbol to point downhill to align with the Direction Of Flow (DOF). If DOF cannot be determined, then orient the tail to the bottom of the sheet.

In areas where fumaroles, geysers, and hot springs, are too numerous to symbolize, an area feature shall be symbolized and labeled.

Feature Information

Specification:	*LTDS*	Feature Origin:	*Topographic*	Symbol:	PT_Dk-Brown1815_GeothermalOutlet
Symbol Criteria:	All			Label:	[Geographic Name Information : Full Name], 6 point Condensed Mixed case if available; "[Geothermal Outlet Type]", 6 point Condensed Mixed case if Geothermal Outlet Type: [Fissure] or [Fumarole] or [Hot Spring] or [Sulphur Spring] or [Geyser]
				Color:	Black (58600)

Glacier (BJ030)

Surface

Feature Description:	*A large mass or river of ice formed by accumulation and compaction of snow on higher ground that is moving slowly down a slope or valley from above the snowline.*
Inclusion Condition:	*Area >= 15,625 sq m.*

Notes

Coincident similar area features will be blended to form a single feature.

Area and line features will be generalized to detail compatible with scale.

Feature will be generalized to provide a more aesthetic contoured feature (i.e., smoothed).

Symbols and associated area patterns of underpassing features (except drainage shorelines) are broken for all bridges, except footbridges. This rule does not apply to land tint on Littoral Planning Charts.

If a clearing exists inside of the area feature, and the size of the clearing is equal to or greater than the area for the surrounding area feature, the clearing is shown as an open space inside the surround feature. If the area of the clearing is less than the area for the surrounding feature, the clearing is deleted and absorbed into the surrounding area feature.

If two area features with the same feature name do not connect at any point, and have a space between them less than 2.5 mm at map/chart scale, delete the open space that is less than 2.5 mm between the features and combine them into one area feature.

If a portion of an area feature has a minimum width of less than 2.5 mm at map/chart scale, delete that portion of the area feature that is not at least 2.5 mm wide, measured from perimeter to perimeter. If the deletion of the portion of the area based on the above criteria will reduce the area of the remaining portion of the area feature to below the minimum area inclusion, do not delete the narrow portion of the feature that is less than 2.5 mm wide. If the deletion of the portion of the area based on the above criteria will split two larger areas connected by a narrow strip into two separate areas, either of which would be below minimum area inclusion, do not delete the narrow portion of the feature that is less than 2.5 mm wide.

Do not show land tint inside symbol.

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Glacier (BJ030) *(Continued)*

Surface

Feature Information					
Specification:	*LTDS*	**Feature Origin:**	*Topographic*	**Symbol:**	AR_Cyan_Glacier
Symbol Criteria:	All			**Label:**	[Geographic Name Information : Full Name] if available, 6-16 point Condensed Mixed case
				Color:	Blue (46351)

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Golf Course (AK100)

Surface

Feature Description:	*A tract of land on which golf is played.*
Inclusion Condition:	*Area >= 15,625 square meters*

Notes

Coincident similar area features will be blended to form a single feature.

Area and line features will be generalized to detail compatible with scale.

Type sizes per are sizes at map/chart scale: Area features only.
06 point - <= 770 mm square area and <= 14 mm width
07 point - <= 2,296 mm square area and <= 28 mm width
09 point - <= 5,192 mm square area and 44 mm width
10 point - <=9,796 mm square area and <= 62 mm width
12 point - <= 16,632 mm square area and <= 84 mm width
14 point - <= 24,960 mm square area and <= 104 mm width
16 point - > 24,960 mm square area.
Where area measurements are inconsistent, the larger type size shall be used. Where the full range of type sizes is not available for a particular label, the closest available type size shall be used.

Type for features on land shall be positioned on land if possible. If type must be placed in the water, it shall be positioned so it does not obscure or overprint hydrographic detail. Type for features in the water shall be positioned in the water. If possible, type shall not be positioned across the shoreline.

Descriptive terms, e.g., "Canal" shall be shown if the name is not known. If the descriptive word appears in the name, for example, "PANAMA CANAL". The descriptive type shall not be shown, i.e. do not show "Panama Canal Canal".

Limiting lines of feature are omitted if it coalesces with a road (AP030).

If a clearing exists inside of the area feature, and the size of the clearing is equal to or greater than the area for the surrounding area feature, the clearing is shown as an open space inside the surround feature. If the area of the clearing is less than the area for the surrounding feature, the clearing is deleted and absorbed into the surrounding area feature.

If two area features with the same feature name do not connect at any point, and have a space between them less than 2.5 mm at map/chart scale, delete the open space that is less than 2.5 mm between the features and combine them into one area feature.

If a portion of an area feature has a minimum width of less than 2.5 mm at map/chart scale, delete that portion of the area feature that is not at least 2.5 mm wide, measured from perimeter to perimeter. If the deletion of the portion of the area based on the above criteria will reduce the area of the remaining portion of the area feature to below the minimum area inclusion, do not delete the narrow portion of the feature that is less than 2.5 mm wide. If the deletion of the portion of the area based on the above criteria will split two larger areas connected by a narrow strip into two separate areas, either of which would be below minimum area inclusion, do not delete the narrow portion of the feature that is less than 2.5 mm wide.

Feature Information

Specification:	*LTDS*	**Feature Origin:**	*Topographic*	**Symbol:**	AR_Black_GolfCourse
Symbol Criteria:	Area >= 15,625			**Label:**	[Geographic Name Information : Full Name], 6 point Condensed Mixed case; "Golf course", 6 point Condensed Mixed case
				Color:	Black (58600)

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Golf Course (AK100)

Point

Inclusion Condition:	*Area < 15,625 square meters AND Navigation Landmark: [True]*

Notes

None

Feature Information

Specification:	*LTDS*	Feature Origin:	*Topographic*	Symbol:	PT_Green362_GolfCourse
Symbol Criteria:	Area < 15,625 AND Navigation Landmark: [True]			Label:	[Geographic Name Information : Full Name], 6 point Condensed Mixed case; "Golf course", 6 point Condensed Mixed case
				Color:	Black (58600)

Grain Elevator (AM030)

Surface

Feature Description:	*A tall structure, equipped for loading, unloading, processing, and/or storing grain.*
Inclusion Condition:	*Area >= 625 sq m.*

Notes

When 2 or more similar area features are separated by less than 0.5 mm at chart scale, the feature will be agglomerated.

Area and line features will be generalized to detail compatible with scale.

Type for features on land shall be positioned on land if possible. If type must be placed in the water, it shall be positioned so it does not obscure or overprint hydrographic detail. Type for features in the water shall be positioned in the water. If possible, type shall not be positioned across the shoreline.

If Height >= 46 meters, then depict as an obstruction symbol.

Feature Information

Specification:	*LTDS*	Feature Origin:	*Topographic*	Symbol:	AR_Black_GrainElevator
Symbol Criteria:	(Height Above Surface Level < 46) OR (Area >= 5625 and Height Above Surface Level >= 46)			Label:	"Grain elevator", 6 point Condensed Mixed case
				Color:	Black (58600)

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Grain Elevator (AM030)

Point

Inclusion Condition:	*Area < 625 sq m.*

Notes

Label feature as per hierarchy for topo type placement to south neatline corners reading left to right:
1. Position hierarchy:
(a.) northeast (preferred position
(b.) southeast (1st alternative)
(c.) northwest (2nd alternative
(d.) southwest (3rd alternative)
(e.) top-centered (4th alternative)
(f.) bottom-centered (5th alternative)
(Hierarchy is based on type positioning so as to avoid overprinting other type or obscuring detail.)
2. Minimum space between type placement and feature symbol is 0.5 mm
3. This method of type placement shall be used for areal features when space does not permit labeling within that feature.

Type for features on land shall be positioned on land if possible. If type must be placed in the water, it shall be positioned so it does not obscure or overprint hydrographic detail. Type for features in the water shall be positioned in the water. If possible, type shall not be positioned across the shoreline.

When obstructions coalesce at map scale, label with highest obstruction information.

Feature Information

Specification:	*LTDS*	**Feature Origin:**	*Topographic*	**Symbol:**	PT_Black_GrainElevator
Symbol Criteria:	Height Above Surface Level < 46			**Label:**	"Grain elevator", 6 point Condensed Mixed case
				Color:	Black (58600)

Feature Information

Specification:	*LTDS*	**Feature Origin:**	*Topographic*	**Symbol:**	PT_Blue072_VO
Symbol Criteria:	Height Above Surface Level >= 46			**Label:**	"Grain elevator", 6 point Condensed Mixed case; [Highest Elevation], 6 point Condensed Bold Mixed case; "([Height Above Surface Level])", 6 point Condensed Mixed case
				Color:	Blue (46351)

Grain Storage Structure (AM020)

Surface

Feature Description:	*An enclosed container, used for storing grain or fodder.*
Inclusion Condition:	*Area >= 625 sq m. OR Height Above Surface Level >= 46 m.*

Notes

The feature (when height <= 46 meters or when height is not known) shall be oriented perpendicular (90 degrees) to the nearby road (AP030), cart track (AP010), trail (AP050), or railroad (AN010).

When 2 or more similar area features are separated by less than 0.5 mm at chart scale, the feature will be agglomerated.

Area and line features will be generalized to detail compatible with scale.

Type for features on land shall be positioned on land if possible. If type must be placed in the water, it shall be positioned so it does not obscure or overprint hydrographic detail. Type for features in the water shall be positioned in the water. If possible, type shall not be positioned across the shoreline.

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Grain Storage Structure (AM020) *(Continued)*

Surface

Feature Information					
Specification:	*LTDS*	**Feature Origin:**	*Topographic*	**Symbol:**	AR_Black_GrainStorage
Symbol Criteria:	(Height Above Surface Level < 46) OR (Area >= 5625 and Height Above Surface Level >= 46)			**Label:**	"Silo", 6 point Condensed Mixed case
				Color:	Black (58600)

Grain Storage Structure (AM020)

Point

Inclusion Condition:	*Area < 625 sq m. OR Height Above Surface Level >= 46 m.*

Notes

The feature (when height <= 46 meters or when height is not known) shall be oriented perpendicular (90 degrees) to the nearby road (AP030), cart track (AP010), trail (AP050), or railroad (AN010).

When symbolized feature is < 0.2 mm from a line feature, displace to 0.2 mm (map scale).

A cluster of 3 or more coalescing similar point features will be aggregated to form an area multiple feature outline.

Label feature as per hierarchy for topo type placement to south neatline corners reading left to right:
1. Position hierarchy:
(a.) northeast (preferred position
(b.) southeast (1st alternative)
(c.) northwest (2nd alternative
(d.) southwest (3rd alternative)
(e.) top-centered (4th alternative)
(f.) bottom-centered (5th alternative)
(Hierarchy is based on type positioning so as to avoid overprinting other type or obscuring detail.)
2. Minimum space between type placement and feature symbol is 0.5 mm
3. This method of type placement shall be used for areal features when space does not permit labeling within that feature.

Type for features on land shall be positioned on land if possible. If type must be placed in the water, it shall be positioned so it does not obscure or overprint hydrographic detail. Type for features in the water shall be positioned in the water. If possible, type shall not be positioned across the shoreline.

Feature Information					
Specification:	*LTDS*	**Feature Origin:**	*Topographic*	**Symbol:**	PT_Black_GrainStorage
Symbol Criteria:	Height Above Surface Level < 46			**Label:**	"Silo", 6 point Condensed Mixed case
				Color:	Black (58600)

Feature Information					
Specification:	*LTDS*	**Feature Origin:**	*Topographic*	**Symbol:**	PT_Blue072_VO
Symbol Criteria:	Height Above Surface Level >= 46			**Label:**	"Silo", 6 point Condensed Mixed case; [Highest Elevation], 6 point Condensed Bold Mixed case; "([Height Above Surface Level])", 6 point Condensed Mixed case
				Color:	Blue (46351)

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Grandstand (AK110)

Surface

Feature Description:	*A structure for special viewing of outdoor events, usually roofed, that has tiers of seats or standing room for spectators.*
Inclusion Condition:	*Length or Diameter: >= 125 m. AND Width: >= 25 m.*

Notes

None

Feature Information

Specification:	*LTDS*	**Feature Origin:**	*Topographic*	**Symbol:**	AR_Black_Grandstand
Symbol Criteria:	All			**Label:**	"Grandstand", 6 point Condensed Mixed case
				Color:	Black (58600)

Grandstand (AK110)

Point

Inclusion Condition:	*Length or Diameter: < 125 m.OR Width: < 25 m. AND* Height Above Surface Level >= 46 m.

Notes

None

Feature Information

Specification:	*LTDS*	**Feature Origin:**	*Topographic*	**Symbol:**	PT_Blue072_VO
Symbol Criteria:	Height Above Surface Level >= 46			**Label:**	"Grandstand" , 7 point Condensed Mixed case; [Highest Elevation], 6 point Condensed Bold Mixed case; "([Height Above Surface Level])", 6 point Condensed Mixed case
				Color:	Blue (46351)

Grassland (EB010)

Surface

Feature Description:	*A tract covered mainly by grasses that have little or no woody tissue.*
Inclusion Condition:	*Area >= 15,625 sq m.*

Notes

Coincident similar area features will be blended to form a single feature.

Area and line features will be generalized to detail compatible with scale.

Feature will be generalized to provide a more aesthetic contoured feature (i.e., smoothed).

Symbols and associated area patterns of underpassing features (except drainage shorelines) are broken for all bridges, except footbridges. This rule does not apply to land tint on Littoral Planning Charts.

If a clearing exists inside of the area feature, and the size of the clearing is equal to or greater than the area for the surrounding area feature, the clearing is shown as an open space inside the surround feature. If the area of the clearing is less than the area for the surrounding feature, the clearing is deleted and absorbed into the surrounding area feature.

If two area features with the same feature code do not connect at any point, and have a space between them less than 2.5 mm at map/chart scale, delete the open space that is less than 2.5 mm between the features and combine them into one area feature.

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Grassland (EB010) *(Continued)*

Surface

Notes

If a portion of an area feature has a minimum width of less than 2.5 mm at map/chart scale, delete that portion of the area feature that is not at least 2.5 mm wide, measured from perimeter to perimeter. If the deletion of the portion of the area based on the above criteria will reduce the area of the remaining portion of the area feature to below the minimum area inclusion, do not delete the narrow portion of the feature that is less than 2.5 mm wide. If the deletion of the portion of the area based on the above criteria will split two larger areas connected by a narrow strip into two separate areas, either of which would be below minimum area inclusion, do not delete the narrow portion of the feature that is less than 2.5 mm wide.

Feature Information					
Specification:	*LTDS*	**Feature Origin:**	*Topographic*	**Symbol:**	AR_Green362_Grassland
Symbol Criteria:	Vegetation Characteristic: [Grassland]			**Label:**	[Geographic Name Information : Full Name], 6 point Condensed Mixed case if available
				Color:	Black (58600)

Greenhouse (AJ110)

Surface

Feature Description:	*A building or structure constructed primarily of transparent material (for example: glass or plastic), in which temperature and humidity can be controlled for the cultivation and/or protection of plants.*
Inclusion Condition:	*Length or Diameter >= 25 m.*

Notes

If feature coalesce at map scale, when shown in their true positions, they shall be displaced 0.2 mm from one another.

When symbolized feature is < 0.2 mm from a line feature, displace to 0.2 mm (map scale).

Area and line features will be generalized to detail compatible with scale.

Omit within Built-up Area AL020 unless it is a landmark or height >= 46 meters.

Greenhouse symbols with distinguishing characteristic attached shall show the staff of the symbol at right angles to the Road. If the symbol coalesces with another symbol, the staff shall be repositioned to an unobstructed side of the building.

Spacing between Greenhouse symbols shall not be less than 0.2 mm.

The Greenhouse symbol shall be shown in its true position if a space of >=0.2 mm (map scale) exists between the Building and Road symbols on map.

A space of not less than 0.2 mm shall be shown between Greenhouse symbols and Tracks and Trails.

Feature Information					
Specification:	*LTDS*	**Feature Origin:**	*Topographic*	**Symbol:**	AR_Green362_Greenhouse
Symbol Criteria:	Length or Diameter >= 50			**Label:**	None
				Color:	N/A

Feature Information					
Specification:	*LTDS*	**Feature Origin:**	*Topographic*	**Symbol:**	PT_Green362_Greenhouse
Symbol Criteria:	Length or Diameter < 50			**Label:**	None
				Color:	N/A

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Greenhouse (AJ110)

Point

Inclusion Condition:	*Length or Diameter >= 25 m.*

Notes

The feature (when height <= 46 meters or when height is not known) shall be oriented perpendicular (90 degrees) to the nearby road (AP030), cart track (AP010), trail (AP050), or railroad (AN010).

If feature coalesce at map scale, when shown in their true positions, they shall be displaced 0.2 mm from one another.

When symbolized feature is 0.2 mm from a line feature, displace to 0.2 mm (map scale).

Like point features which coalesce in clusters of 3 or more will be thinned to form a representative pattern.

If coalescing features being thinned are a mix of heights, with some < 46 meters and some >= 46 meters, then only the obstruction symbol shall be shown.

Omit within Built-up Area AL020 unless it is a landmark or height >= 46 meters.

Greenhouse symbols with distinguishing characteristic attached shall show the staff of the symbol at right angles to the Road. If the symbol coalesces with another symbol, the staff shall be repositioned to an unobstructed side of the building.

Spacing between Greenhouse symbols shall not be less than 0.2 mm.

The Greenhouse symbol shall be shown in its true position if a space of >=0.2 mm (map scale) exists between the Building and Road symbols on map.

A space of not less than 0.2 mm shall be shown between Building symbols and Tracks and Trails.

Greenhouses, single or occurring in rows or clusters, shall be shown in their true orientation except when falling <= 0.2 mm of the following linear features: Road AP030, Railroad Track (AN010), Cart Track (AP010), Trail (AP050), Aqueduct (BH010), Canal (BH020), and Ditch (BH030). In these cases, Greenhouses, single (or occurring in rows or clusters with <= 0.2 mm separation between buildings) shall be collectively oriented parallel to those linear features at a distance of 0.2 mm.

Feature Information

Specification:	*LTDS*	**Feature Origin:**	*Topographic*	**Symbol:**	PT_Green362_Greenhouse
Symbol Criteria:	All			**Label:**	None
				Color:	N/A

Gridiron (BB115)

Surface

Feature Description:	*A timber structure in the intertidal zone serving as a support for vessels at low stages of the tide to permit work on the exposed portion of the vessel's hull.*
Inclusion Condition:	*Width >= 20 meters*

Notes

None

Feature Information

Specification:	*Chart #1*	**Feature Origin:**	*Hydrographic*	**Symbol:**	F24
Symbol Criteria:	All			**Label:**	"Gridiron", 6 point Mixed case Italic
				Color:	Black (58600)

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Harbour (BB005)

Surface

Feature Description:	*A natural or artificial improved body of water providing protection for vessels and generally anchorage and docking facilities.*
Inclusion Condition:	*Area >= 15,625*

Notes

None

Feature Information					
Specification:	*Chart #1*	**Feature Origin:**	*Hydrographic*	**Symbol:**	No Symbol needed
Symbol Criteria:	Navigation Landmark = [True]			**Label:**	None
				Color:	N/A

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Hardened Aircraft Shelter (GB250)

Surface

Feature Description:	*A hardened structure built above or partially above ground that encloses aircraft to provide protection from enemy attack.*
Inclusion Condition:	*Area >= 625 sq m.*

Notes

None

Feature Information

Specification:	*LTDS*	**Feature Origin:**	*Topographic*	**Symbol:**	AR_Black_Hardened AircraftShelter
Symbol Criteria:	(Height Above Surface Level < 46) OR (Area >= 5625 and Height Above Surface Level >= 46)			**Label:**	"Aircraft shelter", 6 point Condensed Mixed case
				Color:	Black (58600)

Hardened Aircraft Shelter (GB250)

Point

Inclusion Condition:	*Area < 625 sq m.*

Notes

None

Feature Information

Specification:	*LTDS*	**Feature Origin:**	*Topographic*	**Symbol:**	PT_Black_Hardened AircraftShelter
Symbol Criteria:	Height Above Surface Level < 46			**Label:**	"Aircraft shelter", 6 point Condensed Mixed case
				Color:	Black (58600)

Hazardous Rock (BD130)

Surface

Feature Description:	*An isolated rocky formation or a single large stone or coral, usually one constituting a danger to navigation.*
Inclusion Condition:	*Width >= 50 m.*

Notes

Rocks (or large boulders) which do not cover must be shown as islets, ie using the coastline symbol and, where the size permits, land tint. Where the height is shown, it must be in metres, or metres and decimetres for heights of less than 5m, above the height datum for the chart as stated in the explanatory notes. The same style of numeral as used for land spot heights must be used. If there is not sufficient space to insert the numeral within the rock it must be inserted adjacent to it, in brackets. Islets too small to be shown in their true (scale) size, must not be reduced to a width less than the width of the coastline symbol (to avoid confusion with display imperfections). Islets may be landmarks; for the charting of landmarks and conspicuous objects.

*Rocks, boulders and rocky areas which cover and uncover must be shown either by being outlined with the low water rock edge symbol, with intertidal tint superimposed, or, where isolated, by the symbol for a pinnacle * . Drying heights must be shown above Chart Datum and enclosed in brackets where out of position.*

Rocks which are always underwater must be shown as follows, according to their depth:
a. Where the depth is unknown but the rock is considered to be dangerous to some surface vessels capable of navigating in the vicinity, by the symbol + with danger line and blue tint as shown on symbol K13 (Chart #1).
b. Where the depth is known, by either:
- *the symbol + with the depth, in metres and decimetres, alongside it in brackets, or*
- *by a sounding with the abbreviation for a rocky seabed beneath it*

Numerals for the depth must be shown in the normal style for soundings. Blue tint should be added as appropriate to the depth. If the rock is considered to be dangerous to some surface vessels capable of navigating in the vicinity, because the rock is significantly shoaler than the general depth in the vicinity, the symbol + or the sounding should be enclosed in a danger line. as shown on symbol K15 (Chart #1).

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Hazardous Rock (BD130) *(Continued)*

Surface

Notes

Submerged coral reefs and pinnacles must be charted by the same symbols as isolated rocks, using the abbreviation for coral as appropriate as shown on symbol K16 (Chart #1).

If point symbol overprints shoreline, the Hydrographic Depth type, posicut, or graphic in the center of the symbol shall be displaced seaward until they no longer overprint the shoreline. If the danger curve (dotted perimeter line) overprints the shoreline, that portion of the dotted perimeter line falling on land shall be deleted.

Rocks and isolated coral heads within submerged reefs shall be charted using the appropriate rock symbol. When depths over selected rocks are shown, an overall depth over the reef is not required, since the depth over the reef is shown by the depth of the shallowest rock. Where it is possible to chart depth information for separate rocks, the shallowest depth over the reef shall be shown by (Hydrographic Depth) or (Hydrographic Drying Height) on reef.

Submerged rocks with known depths of 30.0 meters or less are considered dangerous if the depth of the rock is shallower than the corresponding depth area, as defined by the adjacent depth curves. They are considered not dangerous if the depth of the rock falls within the corresponding depth area. For example, on a chart showing 10, 20, and 30 meter depth curves, a rock with a depth of 12.0 meters would be considered dangerous if it fell between the 20 and 30 meter curves, but would be considered not dangerous if it fell between the 10 and 20 meter depth curves.

If areas where separate dangers coalesce because of scale, the danger curves (dotted lines) are deleted except for those on the outermost perimeter of the aggregate danger hazard. A smooth dotted line is drawn around the area.

Dangers with danger curves (dotted lines) around them which fall inside other danger areas with danger curves around them, shall have the inner dotted line deleted.

If an uncovering rock falls inside the foreshore, show the rock center symbol without the blue tint or dotted circle.

When central graphic symbols of hydrographic dangers, excluding the danger curve (dotted line) overprint or coalesce, they shall be thinned, with preference given to retaining those dangers with the shallower depth, if it is known. Danger curves shall not be affected by this rule.

Use the following abbreviations for Horizontal Accuracy and Existence values:
If Position Approximate, label "PA";
If Position Doubtful, label "PD";
If Existence Doubtful, label "ED";
If Reported, label "Rep".

Date shall only be shown if Existence = Reported. Date shall be shown in parentheses if available. Always show "Rep" and date on the same line.
Unless specifically indicated on source material as inaccurate or doubtful, etc., or modified by other override rules, features shall be assumed to be accurate and definite. This rule applied to accuracy and existence of unique features, and is not related to the accuracy of the source as a whole.

Following IHO practice, features whose positions are approximate are labeled "PA" on the source charts. Features whose positions are doubtful are labeled "PD" on the source charts. Features whose existence is doubtful are labeled "ED" on the source charts. Features that have only been reported are labeled "Rep" on the source charts, and usually show the date of report in parentheses.

Definitions:
PA - Position Approximate = The position has not been accurately determined, or does not remain fixed.;
PD - Position Doubtful = The feature has been reported in various positions, but not definitely determined to be in any.;
ED - Existence Doubtful = Indicates the possible existence of the feature, the actual existence of which has not been established.

Show Hydrographic Depth or Hydrographic Height values in whole meters if decimeter value is 0; otherwise show the decimeter value as a subscript to the meter value.

Date shall only be shown if Existence = Reported. Date shall be shown in parentheses if available. Always show "Rep" and date on the same line.

Priority for type placement Rules:
1-right center,
2-bottom center,
3-left center,
4-top center.
(A) Minimum distance from symbol - 1 mm.
(B) Maximum distance from symbol before choosing the next highest priority:
(#1) 4 mm measured to the West end;
(#2) 4 mm measured to the North side (top);
(#3) 4 mm measured to the East end;
(#4) 4 mm measured to the South side (bottom)

If symbols overprint each other, labels are condensed as follows:
(1.) If the labels are identical, only one is retained.;
(2,) If the labels are not identical, they shall be condensed in to one legend, e.g., "Fishhaven and Well". If multiple depths are shown, only the shallowest is retained.

Labels may be stacked, readable top to bottom to avoid overprints.

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Hazardous Rock (BD130) *(Continued)*

Surface

Notes

Type Preference Placement Rules:
(a.) Placed on one horizontal line centered in feature.;
(b.) Broken into multiple horizontal lines, centered in feature, with a 1 mm space between lines.;
(c.) Placed on one line outside feature. Use Priority for type placement Rules for placement, Hydrographic Depth shall always be positioned inside area.;
(d.) Placed on two lines to avoid overprints. Use Priority for type placement Rules. If present, Hydrographic Depth shall always be placed inside area.

Labels shall be placed in water area, and shall not overprint the shoreline or be placed on land or in the foreshore.

When two or more identical features fall within 10 mm of each other, one legend with plural description shall be shown, with type placed as close as possible to the center of the group without overprinting features.
Where symbols with the same feature code but different values of Hydrographic Vertical Positioning Information: Water Level Effect (i.e. Always Dry , Always Submerged, Covers and Uncovers)) are adjacent to each other, the boundary of the feature that is the highest in vertical plane shall be shown, and the boundary of the feature above water is shown whole, while the feature that covers and uncovers has a portion of its symbol perimeter that overprints the higher feature deleted.

When two or more identical symbols overprint each other, delete the portions that overlap and show as one symbol with an outline of the combined area of the symbols.

Show land tint inside the symbol. If the Hydrographic Vertical Positioning Information: Water Level Effect: [Always Dry] show land tint only.

A blue 31% tint shall not overprint other blue 31% tints. If two blue tinted symbols, or one tinted symbol and water tint overprint, only one 31% tint shall be shown in the area.

Feature Information

Specification:	*Chart #1*	**Feature Origin:**	*Hydrographic*	**Symbol:**	K10
Symbol Criteria:	Hydrographic Vertical Positioning Information : Water Level Effect = [Always Dry]			**Label:**	"([Hydrographic Drying Height])" if available, 7 point lower case/ 5 point subscript
				Color:	Black (58600)

Feature Information

Specification:	*Chart #1*	**Feature Origin:**	*Hydrographic*	**Symbol:**	K16
Symbol Criteria:	(Hydrographic Vertical Positioning Information : Water Level Effect: [Always Submerged]) AND (Dangerous to Navigation: [True]) AND (Hydrographic Depth: (Null) or [0])			**Label:**	"Co" if Structure Material Type =[Coral] or [Coral Head]), 6 point Mixed Case Italic; "R" if Structure Material Type = [Rock], 6 point Upper Case Italic
				Color:	Black (58600)

Hazardous Rock (BD130)

Point

Inclusion Condition:	*Width < 50 m.*

Notes

<u>Rocks (or large boulders) which do not cover</u> must be shown as islets, ie using the coastline symbol and, where the size permits, land tint. Where the height is shown, it must be in metres, or metres and decimetres for heights of less than 5m, above the height datum for the chart as stated in the explanatory notes. The same style of numeral as used for land spot heights must be used. If there is not sufficient space to insert the numeral within the rock it must be inserted adjacent to it, in brackets. Islets too small to be shown in their true (scale) size, must not be reduced to a width less than the width of the coastline symbol (to avoid confusion with display imperfections). Islets may be landmarks; for the charting of landmarks and conspicuous objects.

*<u>Rocks, boulders and rocky areas which cover and uncover</u> must be shown either by being outlined with the low water rock edge symbol, with intertidal tint superimposed, or, where isolated, by the symbol for a pinnacle * . Drying heights must be shown above Chart Datum and enclosed in brackets where out of position.*

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Hazardous Rock (BD130) *(Continued)*

Point

Notes

Rocks which are always underwater must be shown as follows, according to their depth:
a. Where the depth is unknown but the rock is considered to be dangerous to some surface vessels capable of navigating in the vicinity, by the symbol + with danger line and blue tint as shown on symbol K13 (Chart #1).
b. Where the depth is known, by either:
- *the symbol + with the depth, in metres and decimetres, alongside it in brackets, or*
- *by a sounding with the abbreviation for a rocky seabed beneath it*

Numerals for the depth must be shown in the normal style for soundings. Blue tint should be added as appropriate to the depth. If the rock is considered to be dangerous to some surface vessels capable of navigating in the vicinity, because the rock is significantly shoaler than the general depth in the vicinity, the symbol + or the sounding should be enclosed in a danger line. as shown on symbol K15 (Chart #1).

Submerged coral reefs and pinnacles must be charted by the same symbols as isolated rocks, using the abbreviation for coral as appropriate as shown on symbol K16 (Chart #1).

If point symbol overprints shoreline, the Hydrographic Depth type, posicut, or graphic in the center of the symbol shall be displaced seaward until they no longer overprint the shoreline. If the danger curve (dotted perimeter line) overprints the shoreline, that portion of the dotted perimeter line falling on land shall be deleted.

Rocks and isolated coral heads within submerged reefs shall be charted using the appropriate rock symbol. When depths over selected rocks are shown, an overall depth over the reef is not required, since the depth over the reef is shown by the depth of the shallowest rock. Where it is possible to chart depth information for separate rocks, the shallowest depth over the reef shall be shown by (Hydrographic Depth) or (Hydrographic Drying Height) on reef.

Submerged rocks with known depths of 30.0 meters or less are considered dangerous if the depth of the rock is shallower than the corresponding depth area, as defined by the adjacent depth curves. They are considered not dangerous if the depth of the rock falls within the corresponding depth area. For example, on a chart showing 10, 20, and 30 meter depth curves, a rock with a depth of 12.0 meters would be considered dangerous if it fell between the 20 and 30 meter curves, but would be considered not dangerous if it fell between the 10 and 20 meter depth curves.

If areas where separate dangers coalesce because of scale, the danger curves (dotted lines) are deleted except for those on the outermost perimeter of the aggregate danger hazard. A smooth dotted line is drawn around the area.

Dangers with danger curves (dotted lines) around them which fall inside other danger areas with danger curves around them, shall have the inner dotted line deleted.

If an uncovering rock falls inside the foreshore, show the rock center symbol without the blue tint or dotted circle.

When central graphic symbols of hydrographic dangers, excluding the danger curve (dotted line) overprint or coalesce, they shall be thinned, with preference given to retaining those dangers with the shallower depth, if it is known. Danger curves shall not be affected by this rule.

Use the following abbreviations for Horizontal Accuracy and Existence values:
If Position Approximate, label "PA";
If Position Doubtful, label "PD";
If Existence Doubtful, label "ED";
If Reported, label "Rep".

Date shall only be shown if Existence = Reported. Date shall be shown in parentheses if available. Always show "Rep" and date on the same line.
Unless specifically indicated on source material as inaccurate or doubtful, etc., or modified by other override rules, features shall be assumed to be accurate and definite. This rule applied to accuracy and existence of unique features, and is not related to the accuracy of the source as a whole.

Following IHO practice, features whose positions are approximate are labeled "PA" on the source charts. Features whose positions are doubtful are labeled "PD" on the source charts. Features whose existence is doubtful are labeled "ED" on the source charts. Features that have only been reported are labeled "Rep" on the source charts, and usually show the date of report in parentheses.

Definitions:
PA - Position Approximate = The position has not been accurately determined, or does not remain fixed.;
PD - Position Doubtful = The feature has been reported in various positions, but not definitely determined to be in any.;
ED - Existence Doubtful = Indicates the possible existence of the feature, the actual existence of which has not been established.

Show Hydrographic Depth or Hydrographic Height values in whole meters if decimeter value is 0; otherwise show the decimeter value as a subscript to the meter value.

Date shall only be shown if Existence = Reported. Date shall be shown in parentheses if available. Always show "Rep" and date on the same line.

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Hazardous Rock (BD130) *(Continued)*

Point

Notes

Priority for type placement Rules:
1-right center,
2-bottom center,
3-left center,
4-top center.
(A) Minimum distance from symbol - 1 mm.
(B) Maximum distance from symbol before choosing the next highest priority:
(#1) 4 mm measured to the West end;
(#2) 4 mm measured to the North side (top);
(#3) 4 mm measured to the East end;
(#4) 4 mm measured to the South side (bottom)

If symbols overprint each other, labels are condensed as follows:
(1.) If the labels are identical, only one is retained.;
(2,) If the labels are not identical, they shall be condensed in to one legend, e.g., "Fishhaven and Well". If multiple depths are shown, only the shallowest is retained.

Labels may be stacked, readable top to bottom to avoid overprints.

Type Preference Placement Rules:
(a.) Placed on one horizontal line centered in feature.;
(b.) Broken into multiple horizontal lines, centered in feature, with a 1 mm space between lines.;
(c.) Placed on one line outside feature. Use Priority for type placement Rules for placement, Hydrographic Depth shall always be positioned inside area.;
(d.) Placed on two lines to avoid overprints. Use Priority for type placement Rules. If present, Hydrographic Depth shall always be placed inside area.

Labels shall be placed in water area, and shall not overprint the shoreline or be placed on land or in the foreshore.

When two or more identical features fall within 10 mm of each other, one legend with plural description shall be shown, with type placed as close as possible to the center of the group without overprinting features.
Where symbols with the same feature code but different values of Hydrographic Vertical Positioning Information: Water Level Effect (i.e. Always Dry , Always Submerged, Covers and Uncovers)) are adjacent to each other, the boundary of the feature that is the highest in vertical plane shall be shown, and the boundary of the feature above water is shown whole, while the feature that covers and uncovers has a portion of its symbol perimeter that overprints the higher feature deleted.

When two or more identical symbols overprint each other, delete the portions that overlap and show as one symbol with an outline of the combined area of the symbols.

Show land tint inside the symbol. If the Hydrographic Vertical Positioning Information: Water Level Effect: [Always Dry] show land tint only.

A blue 31% tint shall not overprint other blue 31% tints. If two blue tinted symbols, or one tinted symbol and water tint overprint, only one 31% tint shall be shown in the area.

Feature Information					
Specification:	*Chart #1*	**Feature Origin:**	*Hydrographic*	**Symbol:**	K13
Symbol Criteria:	(Hydrographic Vertical Positioning Information : Water Level Effect =[Always Submerged]) AND (Dangerous to Navigation = [True]) AND (Hydrographic Depth: (Null) or [0]) AND (Structural Material Type = [Lava] or [Rock])			**Label:**	None
				Color:	N/A

Feature Information					
Specification:	*Chart #1*	**Feature Origin:**	*Hydrographic*	**Symbol:**	K14.2
Symbol Criteria:	(Hydrographic Vertical Positioning Information : Water Level Effect = [Always Submerged]) AND (Dangerous to Navigation = [True]) AND (Hydrographic Depth > 0)			**Label:**	Hydrographic Depth: 7 point lower case/ 5 point subscript Italic; "R" if symbol is not shown, 7 point Upper case Italic
				Color:	Black (58600)

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Hazardous Rock (BD130) *(Continued)*

Point

Feature Information					
Specification:	*Chart #1*	**Feature Origin:**	*Hydrographic*	**Symbol:**	K14.1
Symbol Criteria:	Hydrographic Vertical Positioning Information : Water Level Effect = [Always Submerged] AND Dangerous to Navigation = [False]			**Label:**	Hydrographic Depth: 7 point lower case/ 5 point subscript Italic; "R" if symbol is not shown, 7 point Upper case Italic
				Color:	Black (58600)

Feature Information					
Specification:	*Chart #1*	**Feature Origin:**	*Hydrographic*	**Symbol:**	K12
Symbol Criteria:	Hydrographic Vertical Positioning Information : Water Level Effect = [Awash at Chart Datum] or [Awash at Low Water]			**Label:**	None
				Color:	N/A

Feature Information					
Specification:	*Chart #1*	**Feature Origin:**	*Hydrographic*	**Symbol:**	K11
Symbol Criteria:	Hydrographic Vertical Positioning Information : Water Level Effect = [Covers and Uncovers] or [Partially Submerged]			**Label:**	"([Hydrographic Drying Height])" if available, 7 point lower case/ 5 point subscript Italic
				Color:	Black (58600)

Heating Facility (AD050)

Surface

Feature Description:	*A facility for the generation of thermal energy for heating purposes.*
Inclusion Condition:	*Area: >= 15,625 sq m.*

Notes

None

Feature Information					
Specification:	*LTDS*	**Feature Origin:**	*Topographic*	**Symbol:**	AR_Black_HeatingFacility
Symbol Criteria:	All			**Label:**	[Geographic Name Information : Full Name] if available, 6 point Condensed Mixed case; "Heating facility", 6 point Condensed Mixed case
				Color:	Black (58600)

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Heating Facility (AD050)

Point

Inclusion Condition:	*Area <15,625sq.m*

Notes

None

Feature Information					
Specification:	*LTDS*	**Feature Origin:**	*Topographic*	**Symbol:**	PT_Black_HeatingFacility
Symbol Criteria:	All			**Label:**	[Geographic Name Information : Full Name] if available, 6 point Condensed Mixed case; "Heating facility", 6 point Condensed Mixed case
				Color:	Black (58600)

Hedgerow (EA020)

Curve

Feature Description:	*A continuous growth of shrubs planted as a fence, a boundary, and/or a windbreak.*
Inclusion Condition:	*Length or Diameter >= 250 m. AND Predominant Vegetation Height >= 3 m. OR Navigation Landmark = [True]*

Notes

Area and line features will be generalized to detail compatible with scale.

Feature Information					
Specification:	*LTDS*	**Feature Origin:**	*Topographic*	**Symbol:**	LN_Green362_Hedgerow
Symbol Criteria:	All			**Label:**	None
				Color:	N/A

Helipad (GB030)

Point

Feature Description:	*A designated area, usually with a prepared surface, used for the take-off, landing, or parking of helicopters.*
Inclusion Condition:	*ALL REQUIRED*

Notes

None

Feature Information					
Specification:	*LTDS*	**Feature Origin:**	*Topographic*	**Symbol:**	PT_Blue072_Helipad
Symbol Criteria:	Helipad Associated Facility <> [Hospital]			**Label:**	[Geographic Name Information : Full Name], 7 point Condensed mixed case if available
				Color:	Blue (46351)

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Helipad (GB030) *(Continued)*

Point

Feature Information					
Specification:	*LTDS*	**Feature Origin:**	*Topographic*	**Symbol:**	PT_Blue072_Helipad_Hospital
Symbol Criteria:	Helipad Associated Facility = [Hospital]			**Label:**	[Geographic Name Information : Full Name], 7 point Condensed mixed case if available
				Color:	Blue (46351)

Heliport (GB035)

Point

Feature Description:	*An aerodrome intended to be used for the arrival, landing, takeoff or departure of vertical takeoff and landing aircraft/helicopters.*
Inclusion Condition:	*ALL REQUIRED*

Notes

None

Feature Information					
Specification:	*LTDS*	**Feature Origin:**	*Topographic*	**Symbol:**	AR_Black_Heliport
Symbol Criteria:	Helipad Associated Facility: <> [Hospital]			**Label:**	[Geographic Name Information : Full Name], 7 point Condensed mixed case if available and "(abandoned)" if Condition of Facility: [Abandoned], 7 point Condensed lower case ; "(not usable)" if Condition of Facility: [Destroyed] or [Ruined], 7 point Condensed lower case
				Color:	Blue (46351)

High Water Line (BA005)

Curve

Feature Description:	*The line dividing the backshore and foreshore along a coast or beach as defined by the National Authority.*
Inclusion Condition:	*ALL REQUIRED*

Notes

None

Feature Information					
Specification:	*LTDS*	**Feature Origin:**	*Topographic*	**Symbol:**	TBD
Symbol Criteria:	TBD			**Label:**	TBD
				Color:	TBD

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Holding Pen (AJ030)

Surface

Feature Description:	*An enclosed tract on which livestock are temporarily kept.*
Inclusion Condition:	*Area >= 15,625 sq m.*

Notes

Area and line features will be generalized to detail compatible with scale.

Label feature as per hierarchy for topo type placement to south neatline corners reading left to right:
1. Position hierarchy:
(a.) northeast (preferred position
(b.) southeast (1st alternative)
(c.) northwest (2nd alternative
(d.) southwest (3rd alternative)
(e.) top-centered (4th alternative)
(f.) bottom-centered (5th alternative)
(Hierarchy is based on type positioning so as to avoid overprinting other type or obscuring detail.)
2. Minimum space between type placement and feature symbol is 0.5 mm
3. This method of type placement shall be used for areal features when space does not permit labeling within that feature.

Names placement shall be oriented to the longest axis of the feature reading left to right and placed within the area outline and centered. If the longest axis is perpendicular to the south neatline, the type shall be placed outside of the area outline, preferred position is northeast of the feature (Rule L-3505), but may be placed at any position around the feature so as not to overprint any other feature type and reading left to right.

Type for features on land shall be positioned on land if possible. If type must be placed in the water, it shall be positioned so it does not obscure or overprint hydrographic detail. Type for features in the water shall be positioned in the water. If possible, type shall not be positioned across the shoreline.

If a clearing exists inside of the area feature, and the size of the clearing is equal to or greater than the area for the surrounding area feature, the clearing is shown as an open space inside the surround feature. If the area of the clearing is less than the area for the surrounding feature, the clearing is deleted and absorbed into the surrounding area feature.

If two area features with the same feature code do not connect at any point, and have a space between them less than 2.5 mm at map/chart scale, delete the open space that is less than 2.5 mm between the features and combine them into one area feature.

If a portion of an area feature has a minimum width of less than 2.5 mm at map/chart scale, delete that portion of the area feature that is not at least 2.5 mm wide, measured from perimeter to perimeter. If the deletion of the portion of the area based on the above criteria will reduce the area of the remaining portion of the area feature to below the minimum area inclusion, do not delete the narrow portion of the feature that is less than 2.5 mm wide. If the deletion of the portion of the area based on the above criteria will split two larger areas connected by a narrow strip into two separate areas, either of which would be below minimum area inclusion, do not delete the narrow portion of the feature that is less than 2.5 mm wide.

Feature Information

Specification:	*LTDS*	**Feature Origin:**	*Topographic*	**Symbol:**	AR_Black_HoldingPen
Symbol Criteria:	All			**Label:**	"Holding Pen", 6 point Condensed Mixed case
				Color:	Black (58600)

Holding Pen (AJ030)

Point

Inclusion Condition:	*Area < 15,625 sq m.*

Notes

None

Feature Information

Specification:	*LTDS*	**Feature Origin:**	*Topographic*	**Symbol:**	PT_Dk-Brown1815_HoldingPen
Symbol Criteria:	All			**Label:**	None
				Color:	N/A

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Hop Field (EA055)

Surface

Feature Description:	*A tract covered by the systematic planting of hop vines.*
Inclusion Condition:	*Area >= 15,625 sq m.*

Notes

Coincident similar area features will be blended to form a single feature.

Area and line features will be generalized to detail compatible with scale.

Feature will be generalized to provide a more aesthetic contoured feature (i.e., smoothed).

Symbols and associated area patterns of underpassing features (except drainage shorelines) are broken for all bridges, except footbridges. This rule does not apply to land tint on Littoral Planning Charts.

If a clearing exists inside of the area feature, and the size of the clearing is equal to or greater than the area for the surrounding area feature, the clearing is shown as an open space inside the surround feature. If the area of the clearing is less than the area for the surrounding feature, the clearing is deleted and absorbed into the surrounding area feature.

If two area features with the same feature code do not connect at any point, and have a space between them less than 2.5 mm at map/chart scale, delete the open space that is less than 2.5 mm between the features and combine them into one area feature.

If a portion of an area feature has a minimum width of less than 2.5 mm at map/chart scale, delete that portion of the area feature that is not at least 2.5 mm wide, measured from perimeter to perimeter. If the deletion of the portion of the area based on the above criteria will reduce the area of the remaining portion of the area feature to below the minimum area inclusion, do not delete the narrow portion of the feature that is less than 2.5 mm wide. If the deletion of the portion of the area based on the above criteria will split two larger areas connected by a narrow strip into two separate areas, either of which would be below minimum area inclusion, do not delete the narrow portion of the feature that is less than 2.5 mm wide.

Feature Information

Specification:	*LTDS*	**Feature Origin:**	*Topographic*	**Symbol:**	AR_Green362_HopField
Symbol Criteria:	Area >= 15,625			**Label:**	[Geographic Name Information : Full Name], 6 point Condensed Mixed case if available; and "Hops", 6 point Condensed Mixed case and "Terraces", 6 point Condensed Mixed case if Crop Information : Farming Pattern [1] = [Terraced]
				Color:	Black (58600)

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Hop Field (EA055)

Point

Inclusion Condition:	*Area < 15,625 sq m.*

Notes

Coincident similar area features will be blended to form a single feature.

Area and line features will be generalized to detail compatible with scale.

Feature will be generalized to provide a more aesthetic contoured feature (i.e., smoothed).

Symbols and associated area patterns of underpassing features (except drainage shorelines) are broken for all bridges, except footbridges. This rule does not apply to land tint on Littoral Planning Charts.

If a clearing exists inside of the area feature, and the size of the clearing is equal to or greater than the area for the surrounding area feature, the clearing is shown as an open space inside the surround feature. If the area of the clearing is less than the area for the surrounding feature, the clearing is deleted and absorbed into the surrounding area feature.

If two area features with the same feature code do not connect at any point, and have a space between them less than 2.5 mm at map/chart scale, delete the open space that is less than 2.5 mm between the features and combine them into one area feature.

If a portion of an area feature has a minimum width of less than 2.5 mm at map/chart scale, delete that portion of the area feature that is not at least 2.5 mm wide, measured from perimeter to perimeter. If the deletion of the portion of the area based on the above criteria will reduce the area of the remaining portion of the area feature to below the minimum area inclusion, do not delete the narrow portion of the feature that is less than 2.5 mm wide. If the deletion of the portion of the area based on the above criteria will split two larger areas connected by a narrow strip into two separate areas, either of which would be below minimum area inclusion, do not delete the narrow portion of the feature that is less than 2.5 mm wide.

Feature Information

Specification:	*LTDS*	**Feature Origin:**	*Topographic*	**Symbol:**	PT_Green362_HopField
Symbol Criteria:	Area < 15,625 sq m. AND Navigation Landmark = [True]			**Label:**	[Geographic Name Information : Full Name], 6 point Condensed Mixed case if available; and "Hops", 6 point Condensed Mixed case
				Color:	Black (58600)

Hopper (AF080)

Point

Feature Description:	*A top-loaded funnel-shaped structure for temporary holding of loose material which will be dispensed from its bottom.*
Inclusion Condition:	*Height Above Surface Level >= 46 m.*

Notes

None

Feature Information

Specification:	*LTDS*	**Feature Origin:**	*Topographic*	**Symbol:**	PT_Blue072_VO
Symbol Criteria:	Height Above Surface Level >= 46			**Label:**	"Hopper ", 6 point Condensed Mixed case; [Highest Elevation], 6 point Condensed Bold Mixed case; "([Height Above Surface Level])", 6 point Condensed Mixed case
				Color:	Blue (46351)

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Hulk (BD181)

Point

Feature Description:	*A vessel, either stranded aground or permanently moored, that is no longer seaworthy due to an inoperable propulsion plant or compromised vessel integrity.*
Inclusion Condition:	*Length < 200 meters OR Height Above Surface Level >= 46 m. OR Navigation Landmark = [True]*

Notes

Priority for type placement: 1-right center, 2-bottom center, 3-left center, 4-top center.
(A) Minimum distance from symbol - 1 mm.
(B) Maximum distance from symbol before choosing the next highest priority:
(#1) 4 mm measured to the West end;
(#2) 4 mm measured to the North side (top);
(#3) 4 mm measured to the East end;
(#4) 4 mm measured to the South side (bottom)

If symbols overprint each other, labels are condensed as follows:
(1.) If the labels are identical, only one is retained.;
(2,) If the labels are not identical, they shall be condensed in to one legend, e.g., "Fishhaven and Well".
If multiple depths are shown, only the shallowest is retained.

Labels may be stacked, readable top to bottom to avoid overprints

Labels shall be placed in water area, and shall not overprint the shoreline or be placed on land or in the foreshore.

When two or more identical features fall within 10 mm of each other, one legend with plural description shall be shown, with type placed as close as possible to the center of the group without overprinting features.

A blue 31% tint shall not overprint other blue 31% tints. If two blue tinted symbols, or one tinted symbol and water tint overprint, only one 31% tint shall be shown in the area.

The diameter of the circular symbol perimeter line may be reduced in congested area to avoid overprinting other symbols, but must be at least 2.0 mm diameter, and cannot come closer than 0.2 mm to any type or graphic symbol shown inside the circle. In non-congested areas the normal symbol diameter shall be shown.

Feature Information

Specification:	*LTDS*	**Feature Origin:**	*Topographic*	**Symbol:**	PT_Black_Hulk
Symbol Criteria:	Navigation Landmark = [True]			**Label:**	[Geographic Name Information : Full Name], 6 point Condensed Mixed case if available; and "Vessel", 6 point Condensed Mixed case
				Color:	Black (58600)

Feature Information

Specification:	*LTDS*	**Feature Origin:**	*Topographic*	**Symbol:**	PT_Blue072_VO
Symbol Criteria:	Height Above Surface Level >= 46			**Label:**	"Vessel" , 6 point Condensed Mixed case; Highest Elevation: 6 point Condensed Bold Mixed case; "([Height Above Surface Level])": 6 point Condensed Mixed case
				Color:	Blue (46351)

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Hummock (BH077)

Surface

Feature Description:	*A place of higher elevation within a wetland (for example: a swamp, a bog, or a marsh).*
Inclusion Condition:	*Area >= 15,625 sq m.*

Notes

Coincident similar area features will be blended to form a single feature.

Area and line features will be generalized to detail compatible with scale.

Symbols and associated area patterns of underpassing features (except drainage shorelines) are broken for all bridges, except footbridges. This rule does not apply to land tint on Littoral Planning Charts.

If a clearing exists inside of the area feature, and the size of the clearing is equal to or greater than the area for the surrounding area feature, the clearing is shown as an open space inside the surround feature. If the area of the clearing is less than the area for the surrounding feature, the clearing is deleted and absorbed into the surrounding area feature.

If two area features with the same feature code do not connect at any point, and have a space between them less than 2.5 mm at map/chart scale, delete the open space that is less than 2.5 mm between the features and combine them into one area feature.

If a portion of an area feature has a minimum width of less than 2.5 mm at map/chart scale, delete that portion of the area feature that is not at least 2.5 mm wide, measured from perimeter to perimeter. If the deletion of the portion of the area based on the above criteria will reduce the area of the remaining portion of the area feature to below the minimum area inclusion, do not delete the narrow portion of the feature that is less than 2.5 mm wide. If the deletion of the portion of the area based on the above criteria will split two larger areas connected by a narrow strip into two separate areas, either of which would be below minimum area inclusion, do not delete the narrow portion of the feature that is less than 2.5 mm wide.

Feature Information

Specification:	*LTDS*	**Feature Origin:**	*Topographic*	**Symbol:**	AR_Cyan_Hummock
Symbol Criteria:	All			**Label:**	None
				Color:	N/A

Hut (AL099)

Point

Feature Description:	*A small, simple free-standing (detached) self-contained residence usually having only a single multi-function room.*
Inclusion Condition:	*Length or Diameter >= 3 m. OR Navigation Landmark = [True]*

Notes

The feature (when height <= 46 meters or when height is not known) shall be oriented perpendicular (90 degrees) to the nearby road (AP030), cart track (AP010), trail (AP050), or railroad (AN010).

Rows of huts with common walls shall be shown with each individual hut symbol abutting together, showing one common line between each.

Feature Information

Specification:	*LTDS*	**Feature Origin:**	*Topographic*	**Symbol:**	PT_Dk-Brown1815_Hut
Symbol Criteria:	All			**Label:**	None
				Color:	N/A

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Hydrocarbon Products Facility (AC040)

Surface

Feature Description:	*A facility involved in the production or distribution of petroleum, oil and/or natural gas products.*
Inclusion Condition:	*Area: >= 15,625 sq m.*

Notes

None

Feature Information

Specification:	*LTDS*	**Feature Origin:**	*Topographic*	**Symbol:**	AR_Black_Hydrocarbon ProdFacility
Symbol Criteria:	All			**Label:**	[Geographic Name Information : Full Name] if available: 6 point Condensed Mixed case, and [Manufacturing Information : Product] "facility", 6 point Condensed Mixed case; "Hydrocarbons facility" if [Manufacturing Information : Product] = [Other] or [No information], 6 point Condensed Mixed case; "(destroyed)" if Condition of Facility: [Destroyed] or [Ruined], 6 point Condensed lower case
				Color:	Black (58600)

Hydrocarbon Products Facility (AC040)

Point

Inclusion Condition:	*Area: < 15,625 sq m.*

Notes

None

Feature Information

Specification:	*LTDS*	**Feature Origin:**	*Topographic*	**Symbol:**	PT_Black_Hydrocarbon ProdFacility
Symbol Criteria:	All			**Label:**	[Geographic Name Information : Full Name] if available: 6 point Condensed Mixed case, and [Manufacturing Information : Product] "facility", 6 point Condensed Mixed case; "Hydrocarbons facility" if [Manufacturing Information : Product] = [Other] or [No information], 6 point Condensed Mixed case; "(destroyed)" if Condition of Facility: [Destroyed] or [Ruined], 6 point Condensed lower case
				Color:	Black (58600)

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HydrocarbonsField (AA052)

Surface

Feature Description:	*An area where the presence of recoverable petroleum, oil, and/or natural gas has been identified.*
Inclusion Condition:	*All required*

Notes

None

Feature Information

Specification:	*LTDS*	**Feature Origin:**	*Topographic*	**Symbol:**	AR_Black_HydrocarbonsField
Symbol Criteria:	All			**Label:**	[Geographic Name Information : Full Name] if available: 6 point Condensed Mixed case; "Gas" if Product [1]: [Gas], or "Oil", 6 point Condensed Mixed case if Product [1]: [Petroleum] and "field", 6 point Condensed Lower case; "Hydrocarbons field", 6 point Condensed Mixed case if Product [1]: [Other] or [No information]
				Color:	Black (58600)

Ice Cliff (BJ040)

Curve

Feature Description:	*The vertical face of a glacier or ice shelf.*
Inclusion Condition:	*Length or Diameter >= 125 m. AND Predominant Feature Height >= 2 m.*

Notes

Area and line features will be generalized to detail compatible with scale.

Feature will be generalized to provide a more aesthetic contoured feature (i.e., smoothed).

When feature coincides with Coastal Shoreline or River/Stream, feature shall replace Coastal Shoreline or River/Stream at coalescence.

If an Ice Cliff is coincident with an Ice Shelf, the dashed outline of the Ice Shelf shall be dropped.

Feature Information

Specification:	*LTDS*	**Feature Origin:**	*Topographic*	**Symbol:**	LN_Cyan_IceCliff
Symbol Criteria:	All			**Label:**	None
				Color:	N/A

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Ice Peak (BJ060)

Point

Feature Description:	*A rocky peak projecting above a surrounding ice field that may be perpetually covered with ice.*
Inclusion Condition:	*Area >= 15,625 sq m. OR Height Above Surface Level >= 46 m. OR Navigation Landmark = [True]*

Notes

Like point features which coalesce in clusters of 3 or more will be thinned to form a representative pattern.

Feature Information

Specification:	*LTDS*	**Feature Origin:**	*Topographic*	**Symbol:**	PT_Dk-Brown1815_IcePeak
Symbol Criteria:	Structural Material Type [1] <> [Frozen Water]			**Label:**	None
				Color:	N/A

Feature Information

Specification:	*LTDS*	**Feature Origin:**	*Topographic*	**Symbol:**	PT_Cyan_IcePeak
Symbol Criteria:	Structural Material Type [1] = [Frozen Water]			**Label:**	None
				Color:	N/A

Ice Route (AQ075)

Curve

Feature Description:	*A route over a frozen watercourse..*
Inclusion Condition:	*Length or Diameter >= 300 m. OR (needed to complete the component network or to establish connectivity)*

Notes

None

Feature Information

Specification:	*LTDS*	**Feature Origin:**	*Topographic*	**Symbol:**	LN_Black_IceRoute
Symbol Criteria:	Transportation System Type [1] = [Road]			**Label:**	[Geographic Name Information : Full Name], 6 point Condensed Mixed case, if available; "ICE ROUTE", 6 point Light Condensed Upper case Italic
				Color:	Black (58600)

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Ice Shelf (BJ065)

Surface

Feature Description:	*A floating ice sheet of considerable thickness that is normally attached to the land along its landward edge.*
Inclusion Condition:	*Area >= 15,625 sq m.*

Notes

Coincident similar area features will be blended to form a single feature.

Area and line features will be generalized to detail compatible with scale.

Feature will be generalized to provide a more aesthetic contoured feature (i.e., smoothed).

Symbols and associated area patterns of underpassing features (except drainage shorelines) are broken for all bridges, except footbridges. This rule does not apply to land tint on Littoral Planning Charts.

If a clearing exists inside of the area feature, and the size of the clearing is equal to or greater than the area for the surrounding area feature, the clearing is shown as an open space inside the surround feature. If the area of the clearing is less than the area for the surrounding feature, the clearing is deleted and absorbed into the surrounding area feature.

If two area features with the same feature code do not connect at any point, and have a space between them less than 2.5 mm at map/chart scale, delete the open space that is less than 2.5 mm between the features and combine them into one area feature.

If a portion of an area feature has a minimum width of less than 2.5 mm at map/chart scale, delete that portion of the area feature that is not at least 2.5 mm wide, measured from perimeter to perimeter. If the deletion of the portion of the area based on the above criteria will reduce the area of the remaining portion of the area feature to below the minimum area inclusion, do not delete the narrow portion of the feature that is less than 2.5 mm wide. If the deletion of the portion of the area based on the above criteria will split two larger areas connected by a narrow strip into two separate areas, either of which would be below minimum area inclusion, do not delete the narrow portion of the feature that is less than 2.5 mm wide.

Do not show land tint inside symbol.

Feature Information

Specification:	*LTDS*	**Feature Origin:**	*Topographic*	**Symbol:**	AR_Cyan_IceShelf
Symbol Criteria:	All			**Label:**	[Geographic Name Information : Full Name] if available, 6 point Condensed Mixed case; "Limits of ice shelf" , 6 point Condensed Mixed case
				Color:	Blue (46351)

ILS Component (GA048)

Curve

Feature Description:	*An air navigation component that is essential to the establishment and execution of a precision approach based on an Instrument Landing System (ILS).*
Inclusion Condition:	*TBD*

Notes

None

Feature Information

Specification:	*Other*	**Feature Origin:**	*Aeronautical*	**Symbol:**	TBD
Symbol Criteria:	TBD			**Label:**	TBD
				Color:	TBD

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Industrial Farm (AL270)

Surface

Feature Description:	*An estate or large farm operating on the plantation economy model in which the farm operates as a single economic unit whose operations are based on agricultural mass production of a few staple crops (for example: cotton, tobacco, sugar cane, bananas, and/or rubber) that are typically not indigenous to the region.*
Inclusion Condition:	*Area >= 15,625 sq m. OR Navigation Landmark = [True]*

Notes

None

Feature Information

Specification:	*LTDS*	**Feature Origin:**	*Topographic*	**Symbol:**	AR_Green362_IndustrialFarm
Symbol Criteria:	Area >= 15,625			**Label:**	[Geographic Name Information: Full Name], 6-16 point Condensed Mixed case if available AND "([Crop Information : Crop Species])", 6-16 point Condensed Mixed case ; and "(terraces)", 6-16 point Condensed lower case if (Crop Information : Farming Pattern [1] = [Terraced]) and (Crop Information : Crop Species [1] = [Cork-Oak] or [Cacao] or [Coffee] or [Oil Crop] or [Rubber] or [Sugar Crop] or [Tea] or [Tobacco] or [Banana] or [Oil Palm] or [Coconut] or [Sugar Cane] or [Cotton] or [Bamboo] or [Timber]); "Plantation", 6-16 point Condensed Mixed case if (Crop Information : Farming Pattern [1] = [Linear] or [Regular] or [Intermingled Woods] or [Intermingled Trees] or [Treeless] or [Trellised] or [Irregular] or [Other] or [No Information]) and (Crop Information : Crop Species [1] = [Other] or [No Information]); "Plantation (terraces)", 6-16 point Condensed Mixed case if (Crop Information : Farming Pattern [1] = [Terraced]) and (Crop Information : Crop Species [1] = [Other] or [No Information])
				Color:	Black (58600)

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Industrial Farm (AL270)

Point

Inclusion Condition:	*Area < 15,625 sq m. AND Navigation Landmark = [True]*

Notes

None

Feature Information					
Specification:	*LTDS*	**Feature Origin:**	*Topographic*	**Symbol:**	PT_Green362_IndustrialFarm
Symbol Criteria:	Area < 15,625 AND Navigation Landmark = [True]			**Label:**	[Geographic Name Information: Full Name], 6 point Condensed Mixed case if available AND "([Crop Information : Crop Species])", 6 point Condensed Mixed case AND "Plantation", 6 point Condensed Mixed case if (Crop Information : Crop Species [1] = [Cork-Oak] or [Cacao] or [Coffee] or [Oil Crop] or [Rubber] or [Sugar Crop] or [Tea] or [Tobacco] or [Banana] or [Oil Palm] or [Coconut] or [Sugar Cane] or [Cotton] or [Bamboo] or [Timber]) and (Geographic Name Information: Full Name = ("UNK") or (ZI005_FNA is null)); [Geographic Name Information: Full Name], 6 point Condensed Mixed case if (Crop Information : Crop Species [1] = [Other] or [No Information]) and (Geographic Name Information: Full Name <> "UNK" or (is not null); "Plantation", 6 point Condensed Mixed case if (Crop Information : Crop Species [1] = [Other] or [No Information]) and (Geographic Name Information: Full Name = "UNK" or (is null)
				Color:	Black (58600)

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Industrial Furnace (AC060)

Surface

Feature Description:	*A structure used in material processing that employs heating to harden (for example: to fire brick or ceramic), pyrolyse (for example: convert coal to coke, or limestone to lime), burn (for example: to incinerate waste), or dry (for example: lumber).*
Inclusion Condition:	*Length or Diameter: >= 25 m.*

Notes

None

Feature Information

Specification:	*LTDS*	**Feature Origin:**	*Topographic*	**Symbol:**	AR_Black_IndustrialFurnace
Symbol Criteria:	All			**Label:**	"([Product])" if <> [Other] or [No Information], 6 point Condensed mixed case and "furnace", 6 point Condensed lower case; "Furnace", 6 point Condensed mixed case if Product if = ([Other] or [No Information])
				Color:	Black (58600)

Industrial Furnace (AC060)

Point

Inclusion Condition:	*Length or Diameter: >= 25 m.*

Notes

None

Feature Information

Specification:	*LTDS*	**Feature Origin:**	*Topographic*	**Symbol:**	PT_Black_IndustrialFurnace
Symbol Criteria:	All			**Label:**	"([Product])" if <> [Other] or [No Information], 6 point Condensed mixed case and "furnace", 6 point Condensed lower case; "Furnace", 6 point Condensed mixed case if Product if = ([Other] or [No Information])
				Color:	Black (58600)

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Inland Waterbody (BH082)

Surface

Feature Description:	*A body of water that is entirely surrounded by land.*
Inclusion Condition:	*((('Inland Waterbody Type' is either 'Lake' or 'Reservoir' or 'Landlocked Sea') AND ('Area' >= 15,625 sq m.) OR ('Navigation Landmark' is 'True')) OR (('Inland Waterbody Type' is either 'Pond' or 'Undifferentiated Water Body') AND ('Area' >= 2500 sq m.) OR ('Navigation Landmark' is 'True')) OR (('Inland Waterbody Type' is 'Holding Basin') AND ('Area' >= 625 sq m.) OR ('Navigation Landmark' is 'True')) OR (('Inland Waterbody Type' is 'Water-hole') AND ('Navigation Landmark' is 'True')))*

Notes

Coincident similar area features will be blended to form a single feature.

Area and line features will be generalized to detail compatible with scale.

Feature will be generalized to provide a more aesthetic contoured feature (i.e., smoothed).

Type sizes per are sizes at map/chart scale:
Area features only.
06 point - <= 770 mm square area and <= 14 mm width;
07 point - <= 2,296 mm square area and <= 28 mm width;
09 point - <= 5,192 mm square area and 44 mm width;
10 point - <=9,796 mm square area and <= 62 mm width;
12 point - <= 16,632 mm square area and <= 84 mm width;
14 point - <= 24,960 mm square area and <= 104 mm width;
16 point - > 24,960 mm square area.
Where area measurements are inconsistent, the larger type size shall be used. Where the full range of type sizes is not available for a particular label, the closest available type size shall be used.

Water-surface elevation values shall be shown centered within the limits of the water feature. If the feature cannot accommodate the elevation figure without coincidence with its limits, the elevation value shall be positioned entirely outside the feature's limits.

Water surface elevations shall be shown, when known, for Lakes and River/Streams, when they are >= 1.25 mm width, at map scale.

Priority for type placement: 1-right center, 2-bottom center, 3-left center, 4-top center.
(A) Minimum distance from symbol - 1 mm.
(B) Maximum distance from symbol before choosing the next highest priority:
(#1) 4 mm measured to the West end;
(#2) 4 mm measured to the North side (top);
(#3) 4 mm measured to the East end;
(#4) 4 mm measured to the South side (bottom)

Descriptive type or name shall be positioned in the following priority: (!.)Horizontal within area feature, if the type will fit entirely within the area. If type consists of more than one word, it may be split into several lines if necessary.

If water salinity is unknown, use (Fresh)

Symbols and associated area patterns of underpassing features (except drainage shorelines) are broken for all bridges, except footbridges. This rule does not apply to land tint on Littoral Planning Charts.

A small area of land >= 1.5 mm in width occurring within an intermittent lake shall be shown as a dashed island shoreline.

Do not show land tint in the symbol. Do not show land tint if (Perennial).

Feature Information

Specification:	*LTDS*	**Feature Origin:**	*Topographic*	**Symbol:**	AR_Cyan_Lake_Perennial
Symbol Criteria:	Man-made Shoreline <> [True] and Water Resource Information : Hydrologic Persistence <> [Intermittent] or [Dry]			**Label:**	[Geographic Name Information : Full Name] if available, 7-16 point Condensed Italics Upper case and if Water Resource Information : Water Type: [Mineral] or [Saline] or [Seawater] or [Brine] label as "(salt)", 7-16 point Condensed Italics Lower case; if Water Resource Information : Water Type: [Alkaline] label as "(alkaline)", 7-16 point Condensed Italics Lower case
				Color:	Blue (48253)

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Inland Waterbody (BH082) *(Continued)*

Surface

Feature Information					
Specification:	*LTDS*	**Feature Origin:**	*Topographic*	**Symbol:**	AR_Cyan_Lake_Non-perennial
Symbol Criteria:	Man-made Shoreline <> [True] and Water Resource Information : Hydrologic Persistence = [Intermittent]			**Label:**	[Geographic Name Information : Full Name] if available, 7-16 point Condensed Italics Upper case and if Water Resource Information : Water Type: [Mineral] or [Saline] or [Seawater] or [Brine] label as "(salt)", 7-16 point Condensed Italics Lower case; if Water Resource Information : Water Type: [Alkaline] label as "(alkaline)", 7-16 point Condensed Italics Lower case
				Color:	Blue (48253)

Feature Information					
Specification:	*LTDS*	**Feature Origin:**	*Topographic*	**Symbol:**	AR_Cyan_Lake_Dry
Symbol Criteria:	Man-made Shoreline <> [True] and Water Resource Information : Hydrologic Persistence = [Dry]			**Label:**	[Geographic Name Information : Full Name] if available, 7-16 point Condensed Italics Upper case and if Water Resource Information : Water Type: [Mineral] or [Saline] or [Seawater] or [Brine] label as "(salt)", 7-16 point Condensed Italics Lower case; if Water Resource Information : Water Type: [Alkaline] label as "(alkaline)", 7-16 point Condensed Italics Lower case
				Color:	Blue (48253)

Feature Information					
Specification:	*LTDS*	**Feature Origin:**	*Topographic*	**Symbol:**	AR_Cyan_Reservoir
Symbol Criteria:	(Man-made Shoreline = [True] and Water Resource Information : Hydrologic Persistence <> [Intermittent] or [Dry]) AND (Condition of Facility <> [Under Construction] or [Destroyed])			**Label:**	[Geographic Name Information : Full Name] if available, 6 point Condensed Italics Mixed case
				Color:	Black (58600)

Feature Information					
Specification:	*LTDS*	**Feature Origin:**	*Topographic*	**Symbol:**	AR_Cyan_Reservoir_ UnderConstruction
Symbol Criteria:	(Man-made Shoreline = [True] and Water Resource Information : Hydrologic Persistence = [Intermittent] or [Dry]) OR (Condition of Facility = [Under Construction] or [Destroyed])			**Label:**	[Geographic Name Information : Full Name] if available, 6 point Condensed Italics Mixed case; Label: "([Condition of Facility])" if Condition of Facility: [Under Construction] or [Destroyed], 6 point Condensed Italics Lower case
				Color:	Black (58600)

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Insubstantial Navigation Mark (BC080)

Point

Feature Description:	*A small, insubstantial navigational mark.*
Inclusion Condition:	*All Required*

Notes

None

Feature Information					
Specification:	*Chart #1*	**Feature Origin:**	*Hydrographic*	**Symbol:**	B32
Symbol Criteria:	All			**Label:**	"([Beacon Type])", 7 point Regular Mixed case
				Color:	Black (58600)

International Date Line (FA110)

Curve

Feature Description:	*A line designated as the place on the Earth where each calendar day begins.*
Inclusion Condition:	*All Required*

Notes

A Boundary marker will be aligned with the feature.

Feature must retain all cartographic detail (i.e., not thinned or smoothed).

*"*International date line (Monday*)' will be labeled on the west side, reading left to right or bottom to top, with "(Sunday)" centered under Monday. Label twice on each chart, with a 1 mm space between the type and date line.*

Boundaries shown shall be included in legend.

Feature Information					
Specification:	*Chart #1*	**Feature Origin:**	*Hydrographic*	**Symbol:**	N1.1
Symbol Criteria:	All			**Label:**	"International date line" and "(Monday)" and "(Sunday)", 8 point Mixed case
				Color:	Black (58600)

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Island (BA030)

Surface

Feature Description:	*A land mass, other than a continent, surrounded by water.*
Inclusion Condition:	*Area >= 625 sq m.*

Notes

Coincident similar area features will be blended to form a single feature.

Area and line features will be generalized to detail compatible with scale.

Feature will be generalized to provide a more aesthetic contoured feature (i.e., smoothed).

Type sizes per are sizes at map/chart scale: Area features only.
06 point - <= 770 mm square area and <= 14 mm width
07 point - <= 2,296 mm square area and <= 28 mm width
09 point - <= 5,192 mm square area and 44 mm width
10 point - <=9,796 mm square area and <= 62 mm width
12 point - <= 16,632 mm square area and <= 84 mm width
14 point - <= 24,960 mm square area and <= 104 mm width
16 point - > 24,960 mm square area.
Where area measurements are inconsistent, the larger type size shall be used. Where the full range of type sizes is not available for a particular label, the closest available type size shall be used.

Label feature as per hierarchy for topo type placement to south neatline corners reading left to right:
1. Position hierarchy:
(a.) northeast (preferred position),
(b.) southeast (1st alternative),
(c.) northwest (2nd alternative,
(d.) southwest (3rd alternative),
(e.) top-centered (4th alternative),
(f.) bottom-centered (5th alternative):
(Hierarchy is based on type positioning so as to avoid overprinting other type or obscuring detail.);
2. Minimum space between type placement and feature symbol is 0.5 mm;
3. This method of type placement shall be used for areal features when space does not permit labeling within that feature.

Names placement shall be oriented to the longest axis of the feature reading left to right and placed within the area outline and centered. If the longest axis is perpendicular to the south neatline, the type shall be placed outside of the area outline, preferred position is northeast of the feature, but may be placed at any position around the feature so as not to overprint any other feature type and reading left to right.

Possession of islands and island groups shall be shown by placing the country name in parentheses below the island name or island group name. If all of the islands in an island group belong to one country, the country name shall be placed under the island group name only. if islands within the same island group belong to different countries, the country name shall be placed under each island name, and not under the island group name. Islands administered jointly by two countries shall show both country names, separated by a dash, e.g., (UK-US). Country names shall be abbreviated in the manner approved by the Board of Geographic Names. Type size for county names shall be 2/3 the size of the island name or island group name, but shall not be less than 5 point.

Feature Information

Specification:	*LTDS*	**Feature Origin:**	*Topographic*	**Symbol:**	AR_Island
Symbol Criteria:	All			**Label:**	[Geographic Name Information : Full Name] if available, 10-16 point Condensed Upper case
				Color:	Black (58600)

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Island (BA030)

Point

Inclusion Condition:	*Area < 625 sq m.*

Notes

Label feature as per hierarchy for topo type placement to south neatline corners reading left to right:
1. Position hierarchy:
(a.) northeast (preferred position),
(b.) southeast (1st alternative),
(c.) northwest (2nd alternative,
(d.) southwest (3rd alternative),
(e.) top-centered (4th alternative),
(f.) bottom-centered (5th alternative):
(Hierarchy is based on type positioning so as to avoid overprinting other type or obscuring detail.);
2. Minimum space between type placement and feature symbol is 0.5 mm;
3. This method of type placement shall be used for areal features when space does not permit labeling within that feature.

Names placement shall be oriented to the longest axis of the feature reading left to right and placed within the area outline and centered. If the longest axis is perpendicular to the south neatline, the type shall be placed outside of the area outline, preferred position is northeast of the feature, but may be placed at any position around the feature so as not to overprint any other feature type and reading left to right.

Possession of islands and island groups shall be shown by placing the country name in parentheses below the island name or island group name. If all of the islands in an island group belong to one country, the country name shall be placed under the island group name only. if islands within the same island group belong to different countries, the country name shall be placed under each island name, and not under the island group name. Islands administered jointly by two countries shall show both country names, separated by a dash, e.g., (UK-US). Country names shall be abbreviated in the manner approved by the Board of Geographic Names. Type size for county names shall be 2/3 the size of the island name or island group name, but shall not be less than 5 point.

Islets too small to be shown in their true (scale) size, must not be reduced to a width less than the width of the coastline symbol (to avoid confusion with printing imperfections).

Feature Information

Specification:	*Chart #1*	**Feature Origin:**	*Hydrographic*	**Symbol:**	K10
Symbol Criteria:	All			**Label:**	[Geographic Name Information : Full Name] if available, 6 point Condensed Mixed case
				Color:	Black (58600)

Land Aerodrome (GB005)

Surface

Feature Description:	*An aerodrome on land intended to be used either wholly or in part for the arrival, departure and surface movement of aircraft.*
Inclusion Condition:	*ALL REQUIRED*

Notes

Coincident similar area features will be blended to form a single feature.

Area and line features will be generalized to detail compatible with scale.

Type for features on land shall be positioned on land if possible. If type must be placed in the water, it shall be positioned so it does not obscure or overprint hydrographic detail. Type for features in the water shall be positioned in the water. If possible, type shall not be positioned across the shoreline.

Descriptive terms, e.g., "Canal" shall be shown if the name is not known. If the descriptive word appears in the name, for example, "PANAMA CANAL". The descriptive type shall not be shown, i.e. do not show "Panama Canal Canal".

Limiting lines of feature are omitted if it coalesces with a road (AP030).

Symbolize apron/hardstands and buildings inside areal Aircraft Facilities (Airport), or (Seaplane)

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Land Aerodrome (GB005) *(Continued)*

Surface

Feature Information					
Specification:	*LTDS*	**Feature Origin:**	*Topographic*	**Symbol:**	AR_Black_Aerodrome
Symbol Criteria:	All			**Label:**	[Geographic Name Information : Full Name] if available, 7 point Condensed mixed case and "(abandoned)", 7 point Condensed lower case if Airfield Symbol Type = [E - Abandoned or Closed (with usable runways or landing areas)]); "(not useable)", 7 point Condensed lower case if Airfield Symbol Type = [G - Unusable for landing or take-off] or [X - Decoy]; "(construction)", 7 point Condensed lower case if Condition of Facility = [Under Construction]
				Color:	Blue (46351)

Land Area (DB000)

Surface

Feature Description:	*A geographically defined part of the land.*
Inclusion Condition:	*ALL REQUIRED*

Notes

Land area is a Yellow (SPC 57377) tint which generally covers the Land portions of the LPC except where other feature rules say not to include tint.

Feature Information					
Specification:	*LTDS*	**Feature Origin:**	*Topographic*	**Symbol:**	Yellow (SPC 57377)
Symbol Criteria:	All			**Label:**	None
				Color:	N/A

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Land Subject to Inundation (BH090)

Surface

Feature Description:	*A tract periodically covered by flood water, excluding tidal waters.*
Inclusion Condition:	*Area >= 15,625 sq m.*

Notes

Coincident similar area features will be blended to form a single feature.

Area and line features will be generalized to detail compatible with scale.

If a clearing exists inside of the area feature, and the size of the clearing is equal to or greater than the area for the surrounding area feature, the clearing is shown as an open space inside the surround feature. If the area of the clearing is less than the area for the surrounding feature, the clearing is deleted and absorbed into the surrounding area feature.

If two area features with the same feature code do not connect at any point, and have a space between them less than 2.5 mm at map/chart scale, delete the open space that is less than 2.5 mm between the features and combine them into one area feature.

If a portion of an area feature has a minimum width of less than 2.5 mm at map/chart scale, delete that portion of the area feature that is not at least 2.5 mm wide, measured from perimeter to perimeter. If the deletion of the portion of the area based on the above criteria will reduce the area of the remaining portion of the area feature to below the minimum area inclusion, do not delete the narrow portion of the feature that is less than 2.5 mm wide. If the deletion of the portion of the area based on the above criteria will split two larger areas connected by a narrow strip into two separate areas, either of which would be below minimum area inclusion, do not delete the narrow portion of the feature that is less than 2.5 mm wide.

Feature Information

Specification:	*LTDS*	**Feature Origin:**	*Topographic*	**Symbol:**	AR_Cyan_LSI_Natural
Symbol Criteria:	Controlled <> [True]			**Label:**	None
				Color:	N/A

Feature Information

Specification:	*LTDS*	**Feature Origin:**	*Topographic*	**Symbol:**	AR_Cyan_LSI_Controlled
Symbol Criteria:	Controlled = [True]			**Label:**	None
				Color:	N/A

Land Water Boundary (BA010)

Curve

Feature Description:	*The line where a land mass is in contact with a body of water.*
Inclusion Condition:	*ALL REQUIRED*

Notes

Area and line features will be generalized to detail compatible with scale.

Feature will be generalized to provide a more aesthetic contoured feature (i.e., smoothed).

No type shall cross Shoreline. Type will either be shown entirely within the Tidal Water (BA040) or entirely on land.

Mean High Water is the preferred vertical datum for shoreline portrayal. When High Water is not available, the shoreline will be delineated by whatever means possible. There may never be a segment of missing shoreline (by definition, the line where land mass is in contact with a body of open water).

Shoreline (BA010 Line Water Boundary) which is coincident with features BB190 Berthing Structure, BB230 Seawall, AP030 Road, AN010 Railway, AN050 Railway Sidetrack, and AL260 Wall are not shown.

Symbols and associated area patterns of underpassing features (except drainage shorelines) are broken for all bridges, except footbridges. This rule does not apply to land tint on Littoral Planning Charts.

Shoreline (BA010 Line Water Boundary) shall not be shown where it becomes coincident with a manmade harbor or coastal structure.

The coastal shoreline will not be shown when a swamp is adjacent to open water. The shoreline will separate the open water from the swamp symbol.

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Land Water Boundary (BA010) *(Continued)*

Curve

Notes

The water side limit Mangrove or Nipa is always shown by a dashed line. The landside limits (Mean High Water line = Land Water Boundary) is shown when known.

When Shoreline (BA010) around an island is smaller than the symbol for a point feature on the island, delete the shoreline and show the point feature symbol in the water.

If the embankment (causeway) is adjacent to a shoreline < .25 mm from a road or railroad, suppress the shoreline.

Feature Information					
Specification:	*LTDS*	**Feature Origin:**	*Topographic*	**Symbol:**	LN_Cyan_LandWater Boundary_Dry
Symbol Criteria:	Shoreline Type = ([Stony] or [Building Rubble] or [Erosion Rubble] or [Sandy] or [Shingly] or [Coral] or [Ice] or [Mud] or [Other] or [No Information])			**Label:**	None
				Color:	N/A

Feature Information					
Specification:	*LTDS*	**Feature Origin:**	*Topographic*	**Symbol:**	LN_Cyan_LandWater Boundary_Wet
Symbol Criteria:	Shoreline Type = [Mangrove] or [Marshy]			**Label:**	None
				Color:	N/A

Landing Steps (BB151)

Point

Feature Description:	*Steps at the shoreline as the connection between land and water on different levels.*
Inclusion Condition:	*ALL REQUIRED*

Notes

Label shall be placed on land, 1 mm from shoreline and actual location of boat landing (BB149), orient label perpendicular to shoreline.

Steps (BB149) shall be positioned adjacent to the shoreline on the water side, with longer edge coincident to shoreline. If the shoreline is too curved to place the steps adjacent, delete the symbol and show the legend "Steps". Type is black (58600, 6 point upper and lower case.

Feature Information					
Specification:	*Chart #1*	**Feature Origin:**	*Hydrographic*	**Symbol:**	U7
Symbol Criteria:	All			**Label:**	"Steps", 7 point regular Mixed Case
				Color:	Black (58600)

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Large Bottom Object (BD003)

Point

Feature Description:	*An object protruding above the water body bottom of which at least one component of height, width or length is a minimum of 5 metres.*
Inclusion Condition:	*Length < 200 meters*

Notes

Use the following abbreviations for Horizontal Accuracy and Existence values:
If Position Approximate, label "PA";
If Position Doubtful, label "PD";
If Existence Doubtful, label "ED";
If Reported, label "Rep".

Date shall only be shown if Existence = Reported. Date shall be shown in parentheses if available. Always show "Rep" and date on the same line.

Unless specifically indicated on source material as inaccurate or doubtful, etc., or modified by other override rules, features shall be assumed to be accurate and definite. This rule applied to accuracy and existence of unique features, and is not related to the accuracy of the source as a whole.

Following IHO practice, features whose positions are approximate are labeled "PA" on the source charts. Features whose positions are doubtful are labeled "PD" on the source charts. Features whose existence is doubtful are labeled "ED" on the source charts. Features that have only been reported are labeled "Rep" on the source charts, and usually show the date of report in parentheses.

Definitions:
PA - Position Approximate = The position has not been accurately determined, or does not remain fixed.;
PD - Position Doubtful = The feature has been reported in various positions, but not definitely determined to be in any.;
ED - Existence Doubtful = Indicates the possible existence of the feature, the actual existence of which has not been established.

Show Hydrographic Depth or Hydrographic Height values in whole meters if decimeter value is 0; otherwise show the decimeter value as a subscript to the meter value.

Date shall only be shown if Existence = Reported. Date shall be shown in parentheses if available. Always show "Rep" and date on the same line.

Priority for type placement Rules:
1-right center,
2-bottom center,
3-left center,
4-top center.
(A) Minimum distance from symbol - 1 mm.
(B) Maximum distance from symbol before choosing the next highest priority:
(#1) 4 mm measured to the West end;
(#2) 4 mm measured to the North side (top);
(#3) 4 mm measured to the East end;
(#4) 4 mm measured to the South side (bottom)

If symbols overprint each other, labels are condensed as follows:
(1.) If the labels are identical, only one is retained.;
(2,) If the labels are not identical, they shall be condensed in to one legend, e.g., "Fishhaven and Well". If multiple depths are shown, only the shallowest is retained.

Labels may be stacked, readable top to bottom to avoid overprints.

Type Preference Placement Rules:
(a.) Placed on one horizontal line centered in feature.;
(b.) Broken into multiple horizontal lines, centered in feature, with a 1 mm space between lines.;
(c.) Placed on one line outside feature. Use Priority for type placement Rules for placement, Hydrographic Depth shall always be positioned inside area.;
(d.) Placed on two lines to avoid overprints. Use Priority for type placement Rules. If present, Hydrographic Depth shall always be placed inside area.

Labels shall be placed in water area, and shall not overprint the shoreline or be placed on land or in the foreshore.

When two or more identical features fall within 10 mm of each other, one legend with plural description shall be shown, with type placed as close as possible to the center of the group without overprinting features.
Where symbols with the same feature code but different values of Hydrographic Vertical Positioning Information: Water Level Effect (i.e. Always Dry , Always Submerged, Covers and Uncovers)) are adjacent to each other, the boundary of the feature that is the highest in vertical plane shall be shown, and the boundary of the feature above water is shown whole, while the feature that covers and uncovers has a portion of its symbol perimeter that overprints the higher feature deleted.

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Large Bottom Object (BD003) *(Continued)*

Point

Notes

When two or more identical symbols overprint each other, delete the portions that overlap and show as one symbol with an outline of the combined area of the symbols.

The height of a hydrographic feature is that above the surface of the water at High Water. Hydrographic Depth/Height information is not applicable at High Water. Hydrographic Depth/Height information values (Depth Known), (Depth Known by Wire Drag), and (Depth Unknown) are used to describe knowledge about a depth of a feature when the feature is below the water surface/ submerged. Hydrographic Depth/Height information values (Uncovering Height Known) and (Uncovering Height Unknown) are used to describe knowledge about the drying height of a feature when it covers and uncovers, i.e., between high and low water.
Hydrographic depth is used to record the depth of a feature when Hydrographic Depth/Height information values are (Depth Known) and (Depth Known by Wire Drag). Hydrographic depth is not applicable when the depth is unknown or when the feature covers and uncovers. Hydrographic Drying Height records the drying height of a feature when Hydrographic Depth/Height information values are known. Hydrographic Drying Height is not applicable when the drying height is unknown, or the feature is below water.

Vertical Datum category is the sounding datum to which a hydrographic depth, drying height, or the high water datum to which a Safe Overhead Clearance is referenced. If the hydrographic depth, drying height, or Safe Overhead Clearance is unknown or not applicable, the vertical datum category is not applicable.

When the boundary of an areal symbol crosses or overprints the shoreline, the boundary overprinting the shoreline or land is deleted.

If areas where separate dangers coalesce because of scale, the danger curves (dotted lines) are deleted except for those on the outermost perimeter of the aggregate danger hazard. A smooth dotted line is drawn around the area.

Dangers with danger curves (dotted lines) around them which fall inside other danger areas with danger curves around them, shall have the inner dotted line deleted.

A blue 31% tint shall not overprint other blue 31% tints. If two blue tinted symbols, or one tinted symbol and water tint overprint, only one 31% tint shall be shown in the area.

Feature Information					
Specification:	*Chart #1*	**Feature Origin:**	*Hydrographic*	**Symbol:**	F22
Symbol Criteria:	(Bottom Object Type = [Pilings]) and (Hydrographic Vertical Positioning Information: Water Level Effect: [Always Dry] or [Awash at Chart Datum] or [Awash at Low Water] or [Covers and Uncovers] or [Partially Submerged])			**Label:**	None
				Color:	N/A

Feature Information					
Specification:	*Chart #1*	**Feature Origin:**	*Hydrographic*	**Symbol:**	K43.1
Symbol Criteria:	(Bottom Object Type = [Pilings]) and (Hydrographic Vertical Positioning Information: Water Level Effect: [Always Submerged])			**Label:**	"Obstn", 6 point Mixed case, Italic
				Color:	Black (58600)

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Launch Pad (GB040)

Point

Feature Description:	*A designated site or structure from which a rocket or missile is launched.*
Inclusion Condition:	*Length < 200 meters*

Notes

None

Feature Information

Specification:	*LTDS*	**Feature Origin:**	*Topographic*	**Symbol:**	PT_Black_LaunchPad
Symbol Criteria:	All			**Label:**	[Geographic Name Information : Full Name], 6 point Condensed Mixed case if available; "Launch pad", 6 point Condensed Mixed case if Geographic Name Information : Full Name = ("UNK") or (is null)
				Color:	Black (58600)

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Leading Line (BC100)

Curve

Feature Description:	*A line which passes through at least two clearly defined objects, along which a vessel can safely travel.*
Inclusion Condition:	*All Required*

Notes

Break line symbol in water area for overprinting point symbol with the same color. Leave space 0.5 mm on each side of the point symbol. Do not displace either the line symbol or the point symbol. Point symbols may overprint line symbols of a different color.

If feature is linear, the label hierarchy is:
(1) Label shall be placed 1 mm above feature, centered.
(2) Top of label shall be placed 1 mm below feature, centered.
(3) Label may be displaced along the feature, use two labels if feature length 150 mm or greater.
(4) Do not label across shoreline (coastal or inland)

When a leading line is also part of a recommended track, the type for the leading line shall be omitted, if it duplicates the labels on the track.

Marks should be briefly described using the Character of Light or Description of Reference Point, if there is any doubt concerning their identity on the chart:
(a.) if space is minimal, the entire legend may be deleted.
(b.) Show Bearing of Object only, if the identity of the marks is clear.
(c.) Features may be named if desirable to identify the marks, for example, a Character of Light of "2 Lts" or a Description of Reference Point of "2Bns" or "TR & Bn".
(d.) Exceptionally, the character of the light is given to avoid confusion with other nearby lights, for example a Character of Light of "2 Fl R".

The first letter in the symbol label, and any abbreviation of lights or beacons (Lt, Lts, Bn, Bn), are capitalized. Other letters are lower case.

If a clearing line or leading line is shown, show the feature(s) associated with it (usually Marine Navigation Light or Marine Navigation Beacon).

The dashed portion of the leading line shall be the end of the leading line nearest to the (Line Associated Feature).

The length of the solid and dashed portions of the leading line shall be determined by three geographic positions (the two end points of the line and the point where the dashed and solid portions meet).

Delete leading line for moire effect light when associated moire effect light is not shown.

Bearing of Object label shall include the degree symbol in 5 point Black (58600)

Feature Information

Specification:	*Chart #1*	**Feature Origin:**	*Hydrographic*	**Symbol:**	P21
Symbol Criteria:	Maritime Navigation Line Features = [Two Lights or more] or [One other than Directional Light]			**Label:**	[Bearing of Object], 7 point lower case;
				Color:	Black (58600)

Feature Information

Specification:	*Chart #1*	**Feature Origin:**	*Hydrographic*	**Symbol:**	P30.1, P30.2
Symbol Criteria:	Maritime Navigation Line Features = [Directional Light]			**Label:**	"Dir", 7 point Mixed case; [Bearing of Object] 7 point lower case
				Color:	Black (58600)

Feature Information

Specification:	*Chart #1*	**Feature Origin:**	*Hydrographic*	**Symbol:**	S11
Symbol Criteria:	Maritime Navigation Line Features = [Directional Radiobeacon]			**Label:**	"RD", 7 point Upper case; [Bearing of Object], 7 point lower case
				Color:	Black (58600)

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Leading Line (BC100) *(Continued)*

Curve

Feature Information					
Specification:	*Chart #1*	**Feature Origin:**	*Hydrographic*	**Symbol:**	P31
Symbol Criteria:	Maritime Navigation Line Features = [Moire Effect Light]			**Label:**	"Dir", 7 point Mixed case; [Bearing of Object], 7 point lower case
				Color:	Black (58600)

Feature Information					
Specification:	*Chart #1*	**Feature Origin:**	*Hydrographic*	**Symbol:**	S3.5
Symbol Criteria:	Maritime Navigation Line Features = [Leading Radio Responder Beacon]			**Label:**	[Bearing of Object], 7 point lower case
				Color:	Black (58600)

Feature Information					
Specification:	*Chart #1*	**Feature Origin:**	*Hydrographic*	**Symbol:**	Q120
Symbol Criteria:	Maritime Navigation Line Features = [Two Beacons or more] or [Two other than Two Lights or Beacons]			**Label:**	[Description of Reference Point], 7 point Mixed case if Maritime Navigation Line Features: [Two other than Two Lights or Beacons]
				Color:	Black (58600)

Light Sector (BC060)

Surface

Feature Description:	*A sector defined by bearings from seaward within which a light shows a specified character or color, or is obscured.*
Inclusion Condition:	*All Required*

Notes

In such cases, on charts where the sector limits are normally shown by fine dashed lines, those lengths of the sector limits which mark the sides of the fairway should be shown by fine continuous lines, in order to highlight the channel.

Sector limits may also be omitted where they cross the fairway.

Feature Information					
Specification:	*Chart #1*	**Feature Origin:**	*Hydrographic*	**Symbol:**	P41.1
Symbol Criteria:	All			**Label:**	[Character of Light], 7 point Mixed case
				Color:	Black (58600)

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Light Vessel (BC070)

Point

Feature Description:	*A distinctively marked vessel anchored or moored at a charted point, to serve as an aid to navigation.*
Inclusion Condition:	*All Required*

Notes

Feature name/label shall be positioned parallel to lines of latitude and readable left to right.

Label placement hierarchy:
(1.) On land, one line.
(2.) On land, two lines, word spellings not split.
(3.) In water, one line.

Name of feature shall not be translated (Into English) or changed, but shall appear in print as they appear on the original source (i.e., Banc Sud).

The word "Light" shall not be included in the name of the light.

Abbreviations for colors for the Color Code of Feature The international abbreviations for colour are:
The International Abbreviations for Navigation Mark Colours for labeling color of Navigational Aids is:
Navigation Mark Colour: [White], abbreviate "W";

Navigation Mark Colour: [Red], abbreviate "R";

Navigation Mark Colour: [Green], abbreviate "G";

Navigation Mark Colour: [Blue], abbreviate "Bu";

Navigation Mark Colour: [Violet], abbreviate "Vi";

Navigation Mark Colour: [Yellow], abbreviate "Y";

Navigation Mark Colour: [Orange], abbreviate "Or"; (may be charted as "Y")

Navigation Mark Colour: [Amber], abbreviate "Am"; (may be charted as "Y")

Navigation Mark Colour: [Black], abbreviate "B";

Colour abbreviations must be in capital letters in all cases except for the second letter of two-letter abbreviations. These abbreviations must be used for the colours of lights and structures.

Color label is place 1mm below the position circle of the feature

Feature Information

Specification:	*LTDS*	**Feature Origin:**	*Topographic*	**Symbol:**	PT_Black_LightVessel
Symbol Criteria:	All			**Label:**	[Geographic Name Information : Full Name], 6 point Regular Mixed case Italic if available; [Navigation Mark Colour], 5 point Upper case Italic
				Color:	Black (58600)

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Lighthouse (BC050)

Point

Feature Description:	*A distinctive structure on or off a coast exhibiting a major light designed to serve as an aid to navigation.*
Inclusion Condition:	*All Required*

Notes

When aids to navigation overprint hydrographic dangers, or depth curves, the following displacement criteria applies:
(a.) If the aid to navigation is a fixed aid (Electronic Beacon, Light, or Visual Beacon), and the hydrographic danger symbol is a point symbol, delete the hydrographic danger point symbol. Show both if the hydrographic danger symbol is an area symbol.
(b.) If the aid to navigation is a buoy, the hydrographic danger symbol's danger curve, i.e., the dotted perimeter line, if present, is broken for the buoy symbol (excluding type).
(c.) If a point danger symbol contains a central graphic element, such as a + or posicut 104, 110, 114, etc., the buoy symbol is displaced off the danger symbol's central graphic element. The displacement shall be as little as possible to resolve the overprint.
(d.) The above criteria refers to the graphic portion of symbols only. All type and labels are movable around the graphic elements to avoid overprints. If the hydrographic danger symbol's central hydrographic depth type must be moved, it shall be placed outside the danger circle, and enclosed in parentheses.
(e.) Depth curves are broken for aids to navigation and dangers, and the type associated with these symbols.

Feature name/label shall be positioned parallel to lines of latitude and readable left to right.

Label placement hierarchy:
(1.) On land, one line.
(2.) On land, two lines, word spellings not split.
(3.) In water, one line.

Name of feature shall not be translated (Into English) or changed, but shall appear in print as they appear on the original source (i.e., Banc Sud).

If the name of the feature is the same name as the land which it is on (i.e., Cape Dana, Calva Island, Bull Hill), and the land is labeled with its name within 10 mm of the feature, no name shall be shown on the feature.

The legends on lights within harbors and restricted channels may be shortened to reduce chart clutter and eliminate overprints. In restricted channels, the priority for condensing legends is different from those lights in more open areas, because most, if not all lights are visible when navigating close inshore, and therefore, the range is less important than the other characteristics of a light.
The following priority is used within harbors and restricted channels:
(1.) Omit (Light Visibility Range or Multiple Light Ranges) first;
(2.) Omit (Elevation of Light) second;
(3.) Omit (Period of Light) third;
(4.) Omit all detail except for flare and dot. Where the ends of numerous piers/wharves are uniformly lighted along a river or channel, a standard note covering them all may be used.

Feature Information

Specification:	*LTDS*	**Feature Origin:**	*Topographic*	**Symbol:**	PT_Black_Lighthouse
Symbol Criteria:	All			**Label:**	[Geographic Name Information : Full Name], 6 point Regular Mixed case if available
				Color:	Black (58600)

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Local Magnetic Anomaly (ZC040)

Surface

Feature Description:	*A localized anomaly in the Earth's magnetic field.*
Inclusion Condition:	*All Required*

Notes

Where the magnitude and extent of permanent local magnetic anomalies have been established to be 3° or greater, they should be shown by a limiting undulating magenta line with the value of the anomalous variation.

Within the enclosed area the magnetic variation may deviate from the normal by the value shown. Where the magnetic compass is known to be deflected either to the west alone or to the east alone, 5°W or 5°E should be quoted, rather than +5° or -5°, to reduce the ambiguity resulting from either a W or E 'normal' variation in the general area. Where the deflection may be in either direction it is expressed as ±5°. In all cases, the value quoted for the anomaly must be the deviation from the normal magnetic variation expected for the area.

Where local magnetic anomalies have not been investigated in detail, an appropriate legend should be shown in magenta, eg. Local Magnetic Anomaly (see Note) or Area of Magnetic Disturbance (see Note), with further information in an accompanying note.

Labeling areas, in order of preference:
(1.) Centered in area on one line in the area, type is horizontal, reading left to right.
(2.) Centered in area on one line in the area, oriented along the long axis of the feature, reading left to right, or bottom to top if axis is vertical.
(3.) Centered in area on two approximately equal lines, without splitting a word, type is horizontal, reading left to right.
(4.) Centered on two approximately equal lines without splitting a word, type is oriented along the long axis of the feature, reading left to right, or bottom to top if axis is vertical.

Priority for type placement: 1-right center, 2-bottom center, 3-left center, 4-top center.
(A) Minimum distance from symbol - 1 mm.
(B) Maximum distance from symbol before choosing the next highest priority:
(#1) 4 mm measured to the West end;
(#2) 4 mm measured to the North side (top);
(#3) 4 mm measured to the East end;
(#4) 4 mm measured to the South side (bottom)

Feature name/label shall be positioned parallel to lines of latitude and readable left to right.

Feature Information

Specification:	*Chart #1*	**Feature Origin:**	*Hydrographic*	**Symbol:**	B82.1
Symbol Criteria:	All			**Label:**	[Magnetic Variation Anomaly], 12 point Mixed case
				Color:	Purple (96532)

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Lock (BI030)

Surface

Feature Description:	*An enclosure with a pair or series of gates used for raising or lowering vessels as they pass from one water level to another.*
Inclusion Condition:	*Width >= 25 meters AND Length or Diameter >= 50 meters*

Notes

When 2 or more similar area features are separated by less than 0.5 mm at chart scale, the feature will be agglomerated to form an area multiple feature outline.

Area and line features will be generalized to detail compatible with scale.

Descriptive terms, e.g., "Canal" shall be shown if the name is not known. If the descriptive word appears in the name, for example, "PANAMA CANAL". The descriptive type shall not be shown, i.e. do not show "Panama Canal Canal".

Omit if not shown in conjunction with drainage feature.

The point of the Lock or Sluice Gate symbol shall be positioned pointing upstream.

Do not show land tint inside symbol.

If a feature is not associated with (touching) a river, omit the feature.

Feature Information

Specification:	*LTDS*	**Feature Origin:**	*Topographic*	**Symbol:**	AR_Black_Lock
Symbol Criteria:	Width >= 25 meters			**Label:**	[Geographic Name Information : Full Name] if available, 6 point Condensed Mixed case
				Color:	Black (58600)

Lock (BI030)

Point

Inclusion Condition:	*Width < 25 meters AND Length or Diameter < 50 meters*

Notes

Label feature as per hierarchy for topo type placement to south neatline corners reading left to right:
1. Position hierarchy: (a.) northeast (preferred position), (b.) southeast (1st alternative), (c.) northwest (2nd alternative, (d.) southwest (3rd alternative), (e.) top-centered (4th alternative), (f.) bottom-centered (5th alternative): (Hierarchy is based on type positioning so as to avoid overprinting other type or obscuring detail.);
2. Minimum space between type placement and feature symbol is 0.5 mm;
3. This method of type placement shall be used for areal features when space does not permit labeling within that feature.

Omit if not shown in conjunction with drainage feature.

The point of the Lock or Sluice Gate symbol shall be positioned pointing upstream.

If a feature is not associated with (touching) a river, omit the feature.

Feature Information

Specification:	*LTDS*	**Feature Origin:**	*Topographic*	**Symbol:**	PT_Black_Lock
Symbol Criteria:	Width < 25 meters			**Label:**	None
				Color:	N/A

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Low Air Maneuver Pylon (GB222)

Point

Feature Description:	*A vertical structure that is used to mark turns and/or provide obstacles around which aircraft may perform low level aerobatic manoeuvres (for example: during racing or training).*
Inclusion Condition:	*TBD*

Notes

None

Feature Information					
Specification:	*Other*	**Feature Origin:**	*Other*	**Symbol:**	TBD
Symbol Criteria:	TBD			**Label:**	TBD
				Color:	TBD

Low Water Line (BE021)

Curve

Feature Description:	*The line dividing foreshore and nearshore along a coast at the seaward edge of the beach to which the sea recedes at low water.*
Inclusion Condition:	*TBD*

Notes

None

Feature Information					
Specification:	*Chart #1*	**Feature Origin:**	*Hydrographic*	**Symbol:**	C1
Symbol Criteria:	All			**Label:**	None
				Color:	N/A

Magnetic Station (BK020)

Point

Feature Description:	*The geographic location at which a set of magnetic observations were taken.*
Inclusion Condition:	*TBD*

Notes

None

Feature Information					
Specification:	*Other*	**Feature Origin:**	*Other*	**Symbol:**	TBD
Symbol Criteria:	TBD			**Label:**	TBD
				Color:	TBD

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Mariculture Site (BH050)

Surface

Feature Description:	*A site where marine organisms are cultivated for food and other products in either the open ocean, in an enclosed section of the ocean, or in tanks, ponds or raceways that are filled with seawater (for example: the farming of marine fish, prawns, or oysters in saltwater ponds).*
Inclusion Condition:	*Width >= 75 meters*

Notes

None

Feature Information

Specification:	*Chart #1*	**Feature Origin:**	*Hydrographic*	**Symbol:**	K48.1
Symbol Criteria:	All			**Label:**	"Kelp Farm", 8 point Mixed case Italic if Aquaculture Facility Type = [Kelp Farm]
				Color:	Black (58600)

Mariculture Site (BH050)

Point

Inclusion Condition:	*Width < 75 meters*

Notes

None

Feature Information

Specification:	*Chart #1*	**Feature Origin:**	*Hydrographic*	**Symbol:**	K48.2
Symbol Criteria:	All			**Label:**	None
				Color:	N/A

Marine Range Centre-line (FC102)

Curve

Feature Description:	*A surveyed reference track marking the centre-line of a marine test range.*
Inclusion Condition:	*TBD*

Notes

None

Feature Information

Specification:	*Other*	**Feature Origin:**	*Other*	**Symbol:**	TBD
Symbol Criteria:	TBD			**Label:**	TBD
				Color:	TBD

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Maritime Caution Area (FC037)

Surface

Feature Description:	*Generally, an area where the mariner has to be made aware of circumstances influencing the safety of navigation.*
Inclusion Condition:	*Length >= 200 meters AND Width >= 250 meters*

Notes

Type sizes for Maritime Limits and Areas:
8 point - < 8 square cm.
10 point - >= 8 and < 12 square cm,
12 point - >= 12 and < 24 square cm.
14 point - >= 24 and < 100 square cm.
8 point - >= 100 square cm.

Type placement for areas >= 100 square cm. to < 500 square cm.
Two labels are shown on approximately opposite sides of the area, preferably top and bottom.

Type placement for areas >= 500 square cm. Labels are placed at approximately 250 mm. interval around the perimeter of the feature.

Do not place labels around sharp corners (interior angle < 135°), or along chart neatlines. Place type 1 mm away and parallel to area limit line, reading left to right or bottom to top if limit line is vertical. Type is placed on the INSIDE of the area.

Priority for Type Placement:
1-right center
2-bottom center
3-left center
4-top center
(A) Minimum distance from symbol - 1 mm.
(B) Maximum distance from symbol before choosing the next highest priority:
(#1) 4 mm measured to the West end;
(#2) 4 mm measured to the North side (top);
(#3) 4 mm measured to the East end;
(#4) 4 mm measured to the South side (bottom)

Type placement for areas < 100 square cm: Type shall be center in the area reading from left to right.

If longer of the two axes is North-South +/- 20 degrees or East-West +/- 20 degrees, type is parallel to the south neatline.
(a.) If length < width times two, type shall be placed on two approximately equal lines without splitting words;
(b.) If length >= width times two, and major axis is East-West +/- 20 degrees, type shall be placed on one line;
(c.) If length >= width times two, and major axis is North-South +/- 20 degrees, type shall be placed with each word on a separate line.
If longer of the two axis is more than 20 degrees from either North-South or East-West, and length < width times two, type shall be parallel to south neatline and on two approximately equal lines without splitting words.
If longer of two axis is more than 20 degrees from either North-South or East-West, and length >= width times two, type shall be placed parallel to the major axis on one line. Type is to be placed inside area, but if type does not fit, place type outside area using Priority for Type Placement above.

Coincident Maritime Limits (FC021), Channel (BH191) and Caution Areas (FC037):
In areas where there are Maritime Limits or Cnannel (BH191) and associated Caution Areas relating to the Maritime Limits or Channels, the symbology and label information for the Maritime Limit and Channel features shall be used and the Caution Area feature guidance on related notes shall be used.

Cable and Pipeline Rules:
Descriptive type for cables and pipelines shall be parallel to the direction of the pipeline or cable, i.e., bank to bank, rather than aligned with the body of water. Type may be shown on more than one line if necessary.

Labeling of pipeline areas: If width <10 mm, Product Category label shall be placed in the center of the area, parallel to the major axis, reading left to right, or bottom to top if major axis is vertical. Type may be moved sideways to avoid overprints. If width >= 10 mm, Product Category label shall be placed parallel to and 1 mm away from the boundary, inside the area adjacent to pipeline portion of the symbol. Both sides of the area shall be labeled. If line is > 150 mm it shall be labeled every 100 mm.

Pipelines, pipeline areas, and cable and pipeline areas shall show a label for the following Product values, using the label shown below:
If Product is Chemical, label "Chem";
If Product is Gas, label "Gas";
If Product is Petrol, label "Gasoline";
If Product is Petroleum, label "Oil";
If Product is Other and [Note : Memorandum] indicates Product is water, label "Water"; Otherwise do not label
No product label is shown if Product is [No Product] or [Radioactive Material].

Charts shall have the following caution notes shown in the margin if pipelines, pipeline areas, or cable and pipeline areas, are shown on the chart, and products are chemicals, gas, gasoline, or oil.: "CAUTION- Mariners risk prosecution if they anchor or trawl near a pipeline and so damage it. [Product category] from a damaged pipeline could cause fire or loss of a vessels buoyancy. " The product name [Product] is indicated in the text of the note. Product [Chemical] is shown in plural, i.e., "Chemicals". See notes and cautions section of product specifications for color, type size, type style, and other information regarding caution notes.

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Maritime Caution Area (FC037) *(Continued)*

Surface

Notes

If Pipelines or cables in the water meet the following criteria, they are represented as pipeline areas (FC037) or cable areas (FC037) respectively, rather than shown as separate pipelines or cables. If pipelines and cables together meet the following criteria, they are represented as "cables and pipelines area" (FC037). Criteria: (a.) more than two linear features, AND (b.) space between any two linear features is less than 8 mm at chart scale, AND (c.) space between the outermost linear features in the group is greater than 3 mm at chart scale. If more than two cables or pipelines are <= 3 mm apart at chart scale, show only the outermost linear features. If cable symbols overprint other cable symbols, show only one cable. If pipeline symbols overprint other pipeline symbols, show only one pipeline. If cable symbols and pipeline symbols overprint, show a cable and pipeline area (FC037). The outer limits of a cable or pipeline areas must enclose the area in which anchoring and certain forms of fishing are prohibited or inadvisable, ie, the limits must lie a safe distance beyond the actual lines of the outermost cables or pipelines.

In Caution Areas where both Cables and Pipelines (Maritime Caution Type: [Submarine Cable] or [Submarine Pipeline] or [Intake Area] or [Outfall Area]) exist, a label of "Cables and Pipelines", 8-14 point Mixed case Italic in Green (52813) should be added within area.

The area must be delimited by the general symbol for the limits of restricted areas (N2.1), interspersed at intervals of about 30mm with short sections of the cable symbol. The cable symbol must be repeated sufficiently to characterize the line. Individual cables within a cable area should not be shown.

Mine Danger Area Rules:
If danger from mines is significant, based on ancillary data, cautions on source charts, or other available information, show mine danger area as maintained minefield even if the field is no longer maintained.

If the boundary of an area showing alternating T shaped dashes and other graphic components joins with a similar line, for example if a boundary closes on itself, number of dashes or graphic components shown in a series (usually three) shall be reduced so that no more than four of any one kind of symbol component are shown in a row.

When the boundary of an areal symbol crosses or overprints the shoreline, the boundary overprinting the shoreline or land is deleted.

If danger to surface navigation still exists, they must be charted as minefields (N34); the legend 'Mine Danger Area' may be used instead of 'Minefield '. If danger to surface navigation is now no greater than the normal hazards of marine navigation, but there is a possible residual danger for submarine or seabed activities, the legend should be 'Former Mined Area (see Note)', or equivalent and an associated explanatory note should be added.

The following cautions shall be shown on charts showing Mine Danger Areas. If the minefield is a maintained minefield or Mine Danger Area, the following caution is shown: "CAUTION - Mariners should stay out of the area indicated because of the presence of mines. See Annual NM 1 (36)." If the area is a former Mine Danger Area no longer maintained, the following caution is shown: "CAUTION - Mariners are warned not to anchor, trawl, ground, or conduct other bottom activities because of the residual danger of mines on the bottom. See Annual NM 1 (36)." The cautions shown may have been modified to provide additional or different information, based on ancillary sources such as Sailing Directions or cautions shown on source charts.

Notifications of Works Rules:
Confirmation of completion should normally be obtained before permanent action is taken on features such as cable-laying, planning consents and harbour works, as the finished works may not be exactly as planned. Such features may be covered in the interim by (P) NM action, and/or the use of legends such as 'Under construction', 'Being reclaimed' or 'Works in progress', with an associated date.

If two work in progress areas are within 20 mm of each other and both are Operational: [True], show only one legend with the later date legend (if available for FC037 Caution Area or FC021 Maritime Limit [Note : Memorandum] attribute) centered between the two features.

If work in progress area is extending the shoreline seaward, the old shoreline is retained until the work is completed. Water tint is deleted from the area, but land tint is not extended into the area. If it is a feature under construction, the coincident shoreline is deleted, and land tint is extended into the area.

Nature Reserve Rules:

All details associated with ESSA (Environmentally Sensitive Sea Area) should be charted in green.

Limits may be shown by a symbolized line or, if such a line is not appropriate or available, limits may be charted by a general maritime limit or restricted area limit (see below), with an appropriate legend within the area of the ESSA. Where it is necessary to highlight specific restrictions, reference to a charted note may be included. Where symbols are incorporated in an ESSA limit, they must be oriented to indicate the side of the line on which the area lies. In all cases, the basic line style employed in the depiction of these limits (which may or may not be amplified by specific ESSA symbology referred to below), should follow the normal conventions for charting of unrestricted and restricted areas (see B-439.2), that is:

- *dashed line (N 1.2), the general maritime limit, in green, implying no restrictions or physical obstructions;*
- *T-shaped dashed line with down-strokes pointing inwards, in green, if legislation prohibits certain activities such as anchoring or fishing (N 2.1), or restricts entry to certain types of vessels (N 2.2).*

Where the limit of the ESSA coincides with other limits which need to be charted, for example associated protective measures such as anchoring and fishing prohibited which apply within the ESSA, they may be incorporated in the symbolized charted limit.

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Maritime Caution Area (FC037) ***(Continued)***

Surface

Notes

The limit of the Nature Reserve may be inserted using the appropriate basic line style as described above with the appropriate symbol below inserted within the area. Symbols used should be selected from the following:

a. *Bird Sanctuary or similar nature reserve (N22 Bird symbol)*
b. *Seal sanctuary (N22 Seal symbol)*
c. *Non-specific nature reserve, National Park, Marine Sanctuary, Marine Reserve, etc (Label: "MR", 8 point Upper case Italic, (Green (52813)*

If other limits which need to be charted coincide with the limit of the Nature Reserve, for example restrictions which apply within the Nature Reserve, they may be incorporated in the symbolized charted limit. If insufficient space is available, they may be charted using one of the symbols above as a point symbol.

*A legend, eg 'Marine Sanctuary (see Note)' may be inserted (in green) within the area. (Omit the reference '(see Note)' if a note is not necessary). A suitably worded note may be inserted in the title area of relevant charts; the following are examples, and will be in green: "**MARINE SANCTUARY** (...insert approximate position...) To avoid the risk of pollution and damage to the environment, this area has been designated a Marine Sanctuary. All vessels carrying dangerous or toxic cargoes, or any other vessel exceeding .... grt, should avoid the area." "**NATIONAL PARKS** (...insert approximate positions...) Entry into the national parks shown on this chart is affected by numerous restrictions and prohibitions. For further details, see ... [name of publication] .... " The exact wording of the note should be tailored to cover the specific case, ie location, the type of measures, restrictions, etc; it may be detailed or may be simply a reference which draws attention to the full details contained in a publication. Such a note may be combined with other related notes. For nature reserves on land, omit the MR from the limit. A note will not usually be necessary.*

*MARPOL are special areas that are an IMO-adopted measure designated under the International Convention for the Prevention of Pollution from Ships 1973, modified by the Protocol of 1978 (MARPOL 73/78). Given the wide extent of the area covered by individual designated Special Areas, and the fact that they are not directly related to safety of navigation, their limits should not normally be inserted on navigation charts. It is more appropriate to include details in associated publications, such as Sailing Directions, Annual Notices to Mariners or special charts depicting MARPOL 73/78 limits. If necessary, a note may be inserted (in green or magenta) on appropriate charts referring to the fact that the chart (or a specified part of it) lies within an IMO-designated Special Area: **MARPOL 73/78 SPECIAL AREA** This chart lies within a Special Area designated by IMO under MARPOL 73/78. For details, see .....[name of chart or publication].....*

***Particularly Sensitive Sea Area (PSSA)** is an IMO-designated measure, established in accordance with IMO Resolution. A suitably worded note should be inserted on the relevant charts; the following is an example (in green): "**PARTICULARLY SENSITIVE SEA AREA (PSSA)** An IMO-approved PSSA is designated in [general area or the area of this chart]. Mariners ... [insert any special requirements, procedures, etc]. For further details, see [insert name of publication]." The exact wording of the note should be tailored to cover each specific area, ie location, the type of associated protective measures, etc. It may be detailed or may be simply a reference which draws attention to the full details contained in an associated publication such as Sailing Directions. Such a note may be combined with other related notes. A simple note, providing a reference to an associated publication may be the only way in which some associated protective measures, such as special discharge restrictions, can be identified on charts.*

The legend 'Particularly Sensitive Sea Area (see Note)' should be inserted within the area of the PSSA, at appropriate positions on relevant charts. Where space is limited, the abbreviated legend 'PSSA (see Note)' may be inserted. The limit of a PSSA should be charted using a dashed line (in green), with a tint band of maximum 5 mm width and the international abbreviation 'PSSA' on the PSSA side of the dashed line limit, in green. The actual width of the band, and density of the tint, should be carefully selected so that the area is not given undue prominence compared with other areas. Use of the tint band in addition to the dashed limit serves to emphasize or differentiate the limit and provides visual continuity to the entirety of a PSSA's area. The dashed limit may be broken for any associated protective measures whose limits coincide with those of the PSSA itself. It may be appropriate to reverse this convention (ie instead of breaking the dashed limit, break the tint band and continue the dashes) where the limit coincides with a Traffic Separation Scheme tint band.

Where the limits of any Associated Protective Measures, which according to the specifications detailed above should be inserted on charts, coincide with those of the PSSA, both limits should be inserted. The Associated Protective Measure limits should be in accordance with the appropriate specifications, one component of the limit of the PSSA (ie the tint band or the dashes) being broken.

Areas to be Avoided (ATBA) Rules:
ATBA vary in size from small circular areas, which 'protect' vital buoys or major lights (M29.1), to much larger areas which protect natural features, such as large coral reefs (M 29.2).

ATBA may be established specifically to provide additional environmental protection to the areas concerned.

The limits of an ATBA must be shown by bold T-shaped green dashes (M14).

The green legend 'AREA TO BE AVOIDED (see Note)', or equivalent, should be inserted within the area of the ATBA. Where space is limited, the abbreviated legend 'ATBA (see Note)', or equivalent, should be inserted.

*A green note should be inserted explaining the reasons for establishment of the area, specifying the vessels to which it applies and stating whether the ATBA is IMO-adopted, as appropriate, eg:"**AREA TO BE AVOIDED (ATBA)** (...insert approximate position ...) To avoid the risk of pollution and damage to the environment, this area has been designated an Area to be Avoided. All vessels carrying dangerous or toxic cargoes, or any other vessel exceeding ... grt, should avoid the area. This Area is IMO-adopted." The exact wording of the note should be tailored to reflect the specific criteria for each area; it may be detailed, as in the example above, or may be simply a reference which draws attention to the full details contained in a publication.*

Historic Wreck Rule:
An explanatory note may be added, in green, eg: "HISTORIC WRECKS The sites of historic wrecks are protected from unauthorized interference."

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Maritime Caution Area (FC037) *(Continued)*

Surface

Notes

Oil and Gas Fields Rule:
Where areas of Gas and or Oil fields exist on the Chart, the following note should be shown in the marginalia in green. "OFFSHORE OIL AND GAS FIELDS. Numerous structures (usually carrying lights) pipes and submerged obstructions (sometimes marked by buoys) exist in certain areas indicated on this chart. For further details, the larger scale charts should be consulted."

Traffic Services Limit Rule:
A note should be added to the chart, listing all the VTS areas which have vessel reporting requirements if they fall wholly or partially within the chart, whether the limits can be charted or not or the reporting points cannot be charted (for example because they are time rather than position related). For example, for one VTS: "VESSEL REPORTING For details of the [Name] VTS, see [associated publication]" or for several VTS, eg: "VESSEL REPORTING For details of the following [vessel traffic services and vessel reporting systems], see [associated publication]: • London VTS • Ramsgate Port Information Service • Medway Navigation Service • Dover Strait Reporting System (CALDOVREP)". If there is no available publication, essential details may be given in the note. To assist cross referencing, it is important to use exactly the form of name used in any associated publication, noting that not all vessel traffic services or reporting systems have 'VTS' as part of the name.

Feature Information					
Specification:	*Chart #1*	**Feature Origin:**	*Hydrographic*	**Symbol:**	N34
Symbol Criteria:	Maritime Area Restriction = [Mine Danger Area] or [Former Mine Danger Area]			**Label:**	"Mine Danger Area" if (Maritime Area Restriction = [Mine Danger Area]), 8-14 point Mixed case, Italic; "Former Mined Area (see Note)" if (Maritime Area Restriction = [Former Mine Danger Area]), 8-14 point Mixed case, Italic
				Color:	Green (52813)

Feature Information					
Specification:	*Chart #1*	**Feature Origin:**	*Hydrographic*	**Symbol:**	N31
Symbol Criteria:	Maritime Navigation Restriction = [Entry Prohibited] or [Shipping Prohibited]			**Label:**	"[Maritime Navigation Restriction]", 8-14 point Mixed case, Italic
				Color:	Green (52813)

Feature Information					
Specification:	*Chart #1*	**Feature Origin:**	*Hydrographic*	**Symbol:**	N2.1
Symbol Criteria:	(Maritime Area Restriction = [Other] or [No Information]) or (Maritime Navigation Restriction = [Other] or [No Information] or [Limited Speed Area] or (Maritime Caution Type = [Other] or [No Information]))			**Label:**	Label restriction type within [Note : Memorandum] if available and if (Maritime Area Restriction = [Other] or [No Information]) or (Maritime Navigation Restriction = [Other] or [No Information]), 8-14 point Mixed case, Italic; Label speed restriction within [Note : Memorandum] if available and if (Maritime Navigation Restriction = [Limited Speed Area]) (Example: 5 kn), 8-14 point Lower case Italic
				Color:	Green (52813)

Feature Information					
Specification:	*Chart #1*	**Feature Origin:**	*Hydrographic*	**Symbol:**	L30.2
Symbol Criteria:	(Cable Type = [Telephone] or [Telegraph] or [Digital Communication Line] or [Communication Line] or [Fibre-optic] or [Other] or [No Information]) and (Maritime Caution Type = [Submarine Cable])			**Label:**	None
				Color:	N/A

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Maritime Caution Area (FC037) *(Continued)*

Surface

Feature Information					
Specification:	*Chart #1*	**Feature Origin:**	*Hydrographic*	**Symbol:**	L31.2
Symbol Criteria:	(Cable Type = [Power Line]) and (Maritime Caution Type = [Submarine Cable])			**Label:**	None
				Color:	N/A

Feature Information					
Specification:	*Chart #1*	**Feature Origin:**	*Hydrographic*	**Symbol:**	L40.2
Symbol Criteria:	(Maritime Caution Type = [Submarine Pipeline]) and (Product: [Chemical] or [Gas] or [Petrol] or [Petroleum] or [Radioactive Material] or [No Product] or [Other] or [No Information])			**Label:**	"Oil" if Product = [Petroleum], 8-14 point Mixed case, Italic; "Gas" if Product = [Gas], 8-14 point Mixed case, Italic; "Chem" if Product = [Chemical], 8-14 point Mixed case, Italic; "Water" if (Product = [Other]) and ([Note : Memorandum] indicates Product is water), 8-14 point Mixed case, Italic; "Gasoline" if (Product = [Petrol]), 8-14 point Mixed case, Italic
				Color:	Green (52813)

Feature Information					
Specification:	*Chart #1*	**Feature Origin:**	*Hydrographic*	**Symbol:**	N20
Symbol Criteria:	Maritime Navigation Restriction = [Anchoring Prohibited]			**Label:**	"No Anchoring Area (see Note)", 8-14 point Mixed case, Italic
				Color:	Green (52813)

Feature Information					
Specification:	*Chart #1*	**Feature Origin:**	*Hydrographic*	**Symbol:**	N45
Symbol Criteria:	Maritime Area Restriction = [Fishery Zone]			**Label:**	None
				Color:	N/A

Feature Information					
Specification:	*Chart #1*	**Feature Origin:**	*Hydrographic*	**Symbol:**	M14
Symbol Criteria:	((Maritime Navigation Restriction = [Avoidance Area] or (Maritime Caution Type = [Ice Advisory] or [Pollution Zone])) and (Maritime Area Restriction <> [Unrestricted Area] or [Bird Sanctuary] or [Conservation and Management Zone] or [Ecological Reserve] or [Fish Sanctuary] or [Game Preserve] or [Marine Nature Reserve] or [Nature Reserve] or [Seal Sanctuary])) or (Maritime Navigation Restriction = [Anchoring Restricted] or [Fishing Restricted] or [Trawling Prohibited] or [Trawling Restricted] or [Entry Restricted] or [Dredging Prohibited] or [Dredging Restricted] or [Diving Prohibited] or [Diving Restricted] or [Reduced Wake Area] or [Construction Prohibited] or [Discharging Prohibited] or [Discharging Restricted] or [Exploration and/or Development Prohibited] or [Exploration and/or Development Restricted] or [Drilling Prohibited] or [Drilling Restricted] or [Artifact Removal Prohibited] or [Cargo Transshipment Prohibited] or [Dragging Prohibited] or [Stopping Prohibited] or [Landing Prohibited] or [Navigational Aid Safety Zone] or [Offshore Safety Zone])			**Label:**	"AREA TO BE AVOIDED (see Note)" if (Maritime Navigation Restriction = [Avoidance Area]) or Maritime Caution Type = [Ice Advisory] or [Pollution Zone]), 8-14 point Mixed case, Italic; "[Maritime Navigation Restriction]" if (Maritime Navigation Restriction <> [Navigational Aid Safety Zone] or [Offshore Safety Zone]), 8-14 point Mixed case, Italic;
				Color:	Green (52813)

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Maritime Caution Area (FC037) *(Continued)*

Surface

Feature Information					
Specification:	*Chart #1*	**Feature Origin:**	*Hydrographic*	**Symbol:**	L41.2
Symbol Criteria:	(Maritime Caution Type = [Intake Area] or [Outfall Area] or [Submarine Sewer]) and (Product = [No Product] or [Other] or [No Information])			**Label:**	"Water" if (Product = [Other]) and ([Note : Memorandum] indicates Product is water), 8-14 point Mixed case, Italic; "Sewer" if (Maritime Caution Type = [Submarine Sewer]), 8-14 point Mixed case, Italic; "Outfall" if (Maritime Caution Type = [Outfall Area], 8-14 point Mixed case, Italic; "Intake" if (Maritime Caution Type = [Intake Area]), 8-14 point Mixed case, Italic
				Color:	Green (52813)

Feature Information					
Specification:	*Chart #1*	**Feature Origin:**	*Hydrographic*	**Symbol:**	F31
Symbol Criteria:	(Coastal Work Type = [Land Reclamation] or [Construction of Structures] or [Other]) or (Maritime Caution Type = [Works in Progress Area]) or (Maritime Area Restriction = [Reclamation Area])			**Label:**	"Being Reclaimed ([Note : Memorandum])" if ((Coastal Work Type = [Land Reclamation]) or (Maritime Area Restriction = [Reclamation Area]) and [Note : Memorandum]if available and describes a date, 8 point Mixed case; "Works in progress ([Note : Memorandum])" if (Coastal Work Type = [Construction of Structures] or [Other]) or (Maritime Caution Type = [Works in Progress Area]) and [Note : Memorandum]if available and describes a date, 8 point Mixed case
				Color:	Green (52813)

Feature Information					
Specification:	*Chart #1*	**Feature Origin:**	*Hydrographic*	**Symbol:**	N22
Symbol Criteria:	(Maritime Navigation Restriction = [Avoidance Area]) and (Maritime Area Restriction = [Bird Sanctuary] or [Conservation and Management Zone] or [Ecological Reserve] or [Fish Sanctuary] or [Game Preserve] or [Marine Nature Reserve] or [Nature Reserve] or [Seal Sanctuary] or [Maritime Pollution (MARPOL) Reporting Area])			**Label:**	"Particularly Sensitive Sea Area (see Note)", 8-14 point Mixed case, Italic;
				Color:	Green (52813)

Feature Information					
Specification:	*Chart #1*	**Feature Origin:**	*Hydrographic*	**Symbol:**	N25
Symbol Criteria:	Maritime Area Restriction = [Degaussing Range]			**Label:**	"[Maritime Area Restriction]", 8 point Mixed Case Italic
				Color:	Black (58600)

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Maritime Caution Area (FC037) ***(Continued)***

Surface

Feature Information					
Specification:	*Chart #1*	**Feature Origin:**	*Hydrographic*	**Symbol:**	N63
Symbol Criteria:	Maritime Area Restriction = [Dredging Area]			**Label:**	"Extraction Area", 8 point Mixed Case Italic
				Color:	Green (52813)

Feature Information					
Specification:	*Chart #1*	**Feature Origin:**	*Hydrographic*	**Symbol:**	M18
Symbol Criteria:	Maritime Area Restriction = [Fairway]			**Label:**	"FAIRWAY [Note : Memorandum]m (see Note)", [Note : Memorandum]if available and described a depth, 8-14 point Mixed Case Italic
				Color:	Green (52813)

Feature Information					
Specification:	*Chart #1*	**Feature Origin:**	*Hydrographic*	**Symbol:**	N26
Symbol Criteria:	Maritime Area Restriction = [Historic Wreck Area]			**Label:**	"Historic Wk", 8 Point Mixed case Italic
				Color:	Green (52813)

Feature Information					
Specification:	*Chart #1*	**Feature Origin:**	*Hydrographic*	**Symbol:**	N2.1
Symbol Criteria:	Maritime Area Restriction = [Natural Gas Field] or [Petroleum Field] or [Joint Economic Development Zone]			**Label:**	"[Geographic Name Information : Full Name] OILFIELD" if available and if (Maritime Area Restriction = [Petroleum Field], 8-14 Point Mixed case Italic; "[Geographic Name Information : Full Name] GASFIELD" if available and if (Maritime Area Restriction = [Natural Gas Field], 8-14 Point Mixed case Italic; "DEVELOPMENT AREA (see Note)" if Maritime Area Restriction = [Joint Economic Development Zone], 8-14 Point Mixed case Italic
				Color:	Green (52813)

Feature Information					
Specification:	*Chart #1*	**Feature Origin:**	*Hydrographic*	**Symbol:**	N1.1
Symbol Criteria:	Maritime Area Restriction = [Unrestricted Area]			**Label:**	None
				Color:	N/A

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Maritime Caution Area (FC037) *(Continued)*

Surface

Feature Information					
Specification:	*Chart #1*	**Feature Origin:**	*Hydrographic*	**Symbol:**	N1.2
Symbol Criteria:	(Maritime Area Restriction = [Waiting Area] or [Traffic Services Limit]) or (Maritime Caution Type = [COLREGS Demarcation Line])			**Label:**	"Waiting Area" if Maritime Area Restriction = [Waiting Area], 8-14 point Mixed case Italic; "[Geographic Name Information : Full Name] VTS (see Note)" if Maritime Area Restriction = [Traffic Services Limit], 8-14 point Mixed case Italic
				Color:	Green (52813)

Feature Information					
Specification:	*Chart #1*	**Feature Origin:**	*Hydrographic*	**Symbol:**	N47
Symbol Criteria:	Maritime Area Restriction = [Exclusive Economic Zone]			**Label:**	"[Geographic Name Information : Full Name] EEZ", 8-14 point Mixed case Italic
				Color:	Green (52813)

Feature Information					
Specification:	*Chart #1*	**Feature Origin:**	*Hydrographic*	**Symbol:**	N46
Symbol Criteria:	Maritime Area Restriction = [Continental Shelf Area] or [Extended Continental Shelf Area]			**Label:**	"[Geographic Name Information : Full Name] Continental Shelf" if Maritime Area Restriction = [Continental Shelf Area], 8-14 point Mixed case Italic; "[Geographic Name Information : Full Name] Extended Continental Shelf" if Maritime Area Restriction = [Extended Continental Shelf Area], 8-14 point Mixed case Italic;
				Color:	Green (52813)

Feature Information					
Specification:	*Chart #1*	**Feature Origin:**	*Hydrographic*	**Symbol:**	N44
Symbol Criteria:	Maritime Area Restriction = [Contiguous Zone]			**Label:**	None
				Color:	N/A

Feature Information					
Specification:	*Chart #1*	**Feature Origin:**	*Hydrographic*	**Symbol:**	N43
Symbol Criteria:	Maritime Area Restriction = [Territorial Sea Area]			**Label:**	None
				Color:	N/A

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Maritime Caution Area (FC037) *(Continued)*

Surface

Feature Information					
Specification:	*Chart #1*	**Feature Origin:**	*Hydrographic*	**Symbol:**	N41
Symbol Criteria:	Maritime Area Restriction = [Internal Waters]			**Label:**	"[Geographic Name Information : Full Name]" if available, 8-14 point Upper case Italic
				Color:	Green (52813)

Feature Information					
Specification:	*Chart #1*	**Feature Origin:**	*Hydrographic*	**Symbol:**	N64
Symbol Criteria:	Maritime Caution Type = [Cargo Transshipment]			**Label:**	"Cargo Transhipment Area", 8 point Upper case Italic
				Color:	Green (52813)

Feature Information					
Specification:	*Chart #1*	**Feature Origin:**	*Hydrographic*	**Symbol:**	N32
Symbol Criteria:	Maritime Caution Type = [Unexploded Ordnance]			**Label:**	"Unexploded Ordnance", 8 point Upper case Italic
				Color:	Green (52813)

Feature Information					
Specification:	*Chart #1*	**Feature Origin:**	*Hydrographic*	**Symbol:**	M2
Symbol Criteria:	Maritime Caution Type = [Clearing Line]			**Label:**	None
				Color:	N/A

Feature Information					
Specification:	*Chart #1*	**Feature Origin:**	*Hydrographic*	**Symbol:**	N49
Symbol Criteria:	Maritime Caution Type = [Harbour Area]			**Label:**	"Harbour Limit", 8 point Mixed Case Italic
				Color:	Green (52813)

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Maritime Caution Area (FC037)

Point

Inclusion Condition:	*Length < 200 meters*

Notes

Type sizes for Maritime Limits and Areas:
8 point - < 8 square cm.
10 point - >= 8 and < 12 square cm,
12 point - >= 12 and < 24 square cm.
14 point - >= 24 and < 100 square cm.
8 point - >= 100 square cm.

Type placement for areas >= 100 square cm. to < 500 square cm.
Two labels are shown on approximately opposite sides of the area, preferably top and bottom.

Type placement for areas >= 500 square cm. Labels are placed at approximately 250 mm. interval around the perimeter of the feature.

Do not place labels around sharp corners (interior angle < 135°), or along chart neatlines. Place type 1 mm away and parallel to area limit line, reading left to right or bottom to top if limit line is vertical. Type is placed on the INSIDE of the area.

Priority for Type Placement:
1-right center
2-bottom center
3-left center
4-top center
(A) Minimum distance from symbol - 1 mm.
(B) Maximum distance from symbol before choosing the next highest priority:
(#1) 4 mm measured to the West end;
(#2) 4 mm measured to the North side (top);
(#3) 4 mm measured to the East end;
(#4) 4 mm measured to the South side (bottom)

Coincident Maritime Limits (FC021), Channel (BH191) and Caution Areas (FC037):
In areas where there are Maritime Limits or Cnannel (BH191) and associated Caution Areas relating to the Maritime Limits or Channels, the symbology and label information for the Maritime Limit and Channel features shall be used and the Caution Area feature guidance on related notes shall be used.

Mine Danger Area Rules:
The following cautions shall be shown on charts showing Mine Danger Areas. If the minefield is a maintained minefield or Mine Danger Area, the following caution is shown: "CAUTION - Mariners should stay out of the area indicated because of the presence of mines. See Annual NM 1 (36)." If the area is a former Mine Danger Area no longer maintained, the following caution is shown: "CAUTION - Mariners are warned not to anchor, trawl, ground, or conduct other bottom activities because of the residual danger of mines on the bottom. See Annual NM 1 (36)." The cautions shown may have been modified to provide additional or different information, based on ancillary sources such as Sailing Directions or cautions shown on source charts.

Nature Reserve Rules:

All details associated with ESSA (Environmentally Sensitive Sea Area) should be charted in green.

The limit of the Nature Reserve may be inserted using the appropriate basic line style as described above with the appropriate symbol below inserted within the area. Symbols used should be selected from the following:

a. *Bird Sanctuary or similar nature reserve (N22 Bird symbol)*
b. *Seal sanctuary (N22 Seal symbol)*
c. *Non-specific nature reserve, National Park, Marine Sanctuary, Marine Reserve, etc (Label: "MR", 8 point Upper case Italic, (Green (52813)*

If other limits which need to be charted coincide with the limit of the Nature Reserve, for example restrictions which apply within the Nature Reserve, they may be incorporated in the symbolized charted limit. If insufficient space is available, they may be charted using one of the symbols above as a point symbol.

Feature Information

Specification:	*Chart #1*	**Feature Origin:**	*Hydrographic*	**Symbol:**	L3
Symbol Criteria:	Maritime Navigation Restriction = [Navigational Aid Safety Zone]			**Label:**	None
				Color:	N/A

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Maritime Caution Area (FC037) *(Continued)*

Point

Feature Information					
Specification:	*Chart #1*	**Feature Origin:**	*Hydrographic*	**Symbol:**	N23.1
Symbol Criteria:	Maritime Area Restriction = [Mine Danger Area]			**Label:**	None
				Color:	N/A

Feature Information					
Specification:	*Chart #1*	**Feature Origin:**	*Hydrographic*	**Symbol:**	N20
Symbol Criteria:	Maritime Navigation Restriction = [Anchoring Prohibited]			**Label:**	None
				Color:	N/A

Feature Information					
Specification:	*Chart #1*	**Feature Origin:**	*Hydrographic*	**Symbol:**	N21
Symbol Criteria:	Maritime Navigation Restriction = [Fishing Prohibited]			**Label:**	None
				Color:	N/A

Feature Information					
Specification:	*Chart #1*	**Feature Origin:**	*Hydrographic*	**Symbol:**	N31
Symbol Criteria:	Maritime Navigation Restriction = [Entry Prohibited] or [Shipping Prohibited]			**Label:**	None
				Color:	N/A

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Maritime Limit (FC021)

Curve

Feature Description:	*A line where on either side certain activities or factors of significance to navigation and/or operation apply.*
Inclusion Condition:	*All Required*

Notes

Type sizes for Maritime Limits and Areas:
8 point - < 8 square cm.
10 point - >= 8 and < 12 square cm,
12 point - >= 12 and < 24 square cm.
14 point - >= 24 and < 100 square cm.
8 point - >= 100 square cm.

Type placement for areas >= 100 square cm. to < 500 square cm.
Two labels are shown on approximately opposite sides of the area, preferably top and bottom.

Type placement for areas >= 500 square cm. Labels are placed at approximately 250 mm. interval around the perimeter of the feature.

Do not place labels around sharp corners (interior angle < 135°), or along chart neatlines. Place type 1 mm away and parallel to area limit line, reading left to right or bottom to top if limit line is vertical. Type is placed on the INSIDE of the area.

Priority for Type Placement:
1-right center
2-bottom center
3-left center
4-top center
(A) Minimum distance from symbol - 1 mm.
(B) Maximum distance from symbol before choosing the next highest priority:
(#1) 4 mm measured to the West end;
(#2) 4 mm measured to the North side (top);
(#3) 4 mm measured to the East end;
(#4) 4 mm measured to the South side (bottom)

Type placement for areas < 100 square cm: Type shall be center in the area reading from left to right.

If longer of the two axes is North-South +/- 20 degrees or East-West +/- 20 degrees, type is parallel to the south neatline.
(a.) If length < width times two, type shall be placed on two approximately equal lines without splitting words;
(b.) If length >= width times two, and major axis is East-West +/- 20 degrees, type shall be placed on one line;
(c.) If length >= width times two, and major axis is North-South +/- 20 degrees, type shall be placed with each word on a separate line.
If longer of the two axis is more than 20 degrees from either North-South or East-West, and length < width times two, type shall be parallel to south neatline and on two approximately equal lines without splitting words.
If longer of two axis is more than 20 degrees from either North-South or East-West, and length >= width times two, type shall be placed parallel to the major axis on one line. Type is to be placed inside area, but if type does not fit, place type outside area using Priority for Type Placement above.

Coincident Maritime Limits (FC021), Channel (BH191) and Caution Areas (FC037):
In areas where there are Maritime Limits or Cnannel (BH191) and associated Caution Areas relating to the Maritime Limits or Channels, the symbology and label information for the Maritime Limit and Channel features shall be used and the Caution Area feature guidance on related notes shall be used.

Mine Danger Area Rules:
If danger from mines is significant, based on ancillary data, cautions on source charts, or other available information, show mine danger area as maintained minefield even if the field is no longer maintained.

If the boundary of an area showing alternating T shaped dashes and other graphic components joins with a similar line, for example if a boundary closes on itself, number of dashes or graphic components shown in a series (usually three) shall be reduced so that no more than four of any one kind of symbol component are shown in a row.

When the boundary of an areal symbol crosses or overprints the shoreline, the boundary overprinting the shoreline or land is deleted.

If danger to surface navigation still exists, they must be charted as minefields (N34); the legend 'Mine Danger Area' may be used instead of 'Minefield '. If danger to surface navigation is now no greater than the normal hazards of marine navigation, but there is a possible residual danger for submarine or seabed activities, the legend should be 'Former Mined Area (see Note)', or equivalent and an associated explanatory note should be added.

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Maritime Limit (FC021) *(Continued)*

Curve

Notes

The following cautions shall be shown on charts showing Mine Danger Areas. If the minefield is a maintained minefield or Mine Danger Area, the following caution is shown: "CAUTION - Mariners should stay out of the area indicated because of the presence of mines. See Annual NM 1 (36)." If the area is a former Mine Danger Area no longer maintained, the following caution is shown: "CAUTION - Mariners are warned not to anchor, trawl, ground, or conduct other bottom activities because of the residual danger of mines on the bottom. See Annual NM 1 (36)." The cautions shown may have been modified to provide additional or different information, based on ancillary sources such as Sailing Directions or cautions shown on source charts.

Notifications of Works Rules:
Confirmation of completion should normally be obtained before permanent action is taken on features such as cable-laying, planning consents and harbour works, as the finished works may not be exactly as planned. Such features may be covered in the interim by (P) NM action, and/or the use of legends such as 'Under construction', 'Being reclaimed' or 'Works in progress', with an associated date.

If two work in progress areas are within 20 mm of each other and both are Operational: [True], show only one legend with the later date legend (if available for FC037 Caution Area or FC021 Maritime Limit [Note : Memorandum] attribute) centered between the two features.

If work in progress area is extending the shoreline seaward, the old shoreline is retained until the work is completed. Water tint is deleted from the area, but land tint is not extended into the area. If it is a feature under construction, the coincident shoreline is deleted, and land tint is extended into the area.

Nature Reserve Rules:

All details associated with ESSA (Environmentally Sensitive Sea Area) should be charted in green.

Limits may be shown by a symbolized line or, if such a line is not appropriate or available, limits may be charted by a general maritime limit or restricted area limit (see below), with an appropriate legend within the area of the ESSA. Where it is necessary to highlight specific restrictions, reference to a charted note may be included. Where symbols are incorporated in an ESSA limit, they must be oriented to indicate the side of the line on which the area lies. In all cases, the basic line style employed in the depiction of these limits (which may or may not be amplified by specific ESSA symbology referred to below), should follow the normal conventions for charting of unrestricted and restricted areas (see B-439.2), that is:
• dashed line (N 1.2), the general maritime limit, in green, implying no restrictions or physical obstructions;
• T-shaped dashed line with down-strokes pointing inwards, in green, if legislation prohibits certain activities such as anchoring or fishing (N 2.1), or restricts entry to certain types of vessels (N 2.2).

Where the limit of the ESSA coincides with other limits which need to be charted, for example associated protective measures such as anchoring and fishing prohibited which apply within the ESSA, they may be incorporated in the symbolized charted limit.

The limit of the Nature Reserve may be inserted using the appropriate basic line style as described above with the appropriate symbol below inserted within the area. Symbols used should be selected from the following:
a. Bird Sanctuary or similar nature reserve (N22 Bird symbol)
b. Seal sanctuary (N22 Seal symbol)
c. Non-specific nature reserve, National Park, Marine Sanctuary, Marine Reserve, etc (Label: "MR", 8 point Upper case Italic, (Green (52813)
If other limits which need to be charted coincide with the limit of the Nature Reserve, for example restrictions which apply within the Nature Reserve, they may be incorporated in the symbolized charted limit. If insufficient space is available, they may be charted using one of the symbols above as a point symbol.

A legend, eg 'Marine Sanctuary (see Note)' may be inserted (in green) within the area. (Omit the reference '(see Note)' if a note is not necessary). A suitably worded note may be inserted in the title area of relevant charts; the following are examples, and will be in green: "MARINE SANCTUARY (…insert approximate position…) To avoid the risk of pollution and damage to the environment, this area has been designated a Marine Sanctuary. All vessels carrying dangerous or toxic cargoes, or any other vessel exceeding …. Grt, should avoid the area." "NATIONAL PARKS (…insert approximate positions…) Entry into the national parks shown on this chart is affected by numerous restrictions and prohibitions.

For further details, see … [name of publication] …. " The exact wording of the note should be tailored to cover the specific case, ie location, the type of measures, restrictions, etc; it may be detailed or may be simply a reference which draws attention to the full details contained in a publication. Such a note may be combined with other related notes. For nature reserves on land, omit the MR from the limit. A note will not usually be necessary.

MARPOL are special areas that are an IMO-adopted measure designated under the International Convention for the Prevention of Pollution from Ships 1973, modified by the Protocol of 1978 (MARPOL 73/78). Given the wide extent of the area covered by individual designated Special Areas, and the fact that they are not directly related to safety of navigation, their limits should not normally be inserted on navigation charts. It is more appropriate to include details in associated publications, such as Sailing Directions, Annual Notices to Mariners or special charts depicting MARPOL 73/78 limits. If necessary, a note may be inserted (in green or magenta) on appropriate charts referring to the fact that the chart (or a specified part of it) lies within an IMO-designated Special Area: MARPOL 73/78 SPECIAL AREA This chart lies within a Special Area designated by IMO under MARPOL 73/78. For details, see ……[name of chart or publication]…..

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Maritime Limit (FC021) *(Continued)*

Curve

Notes

Particularly Sensitive Sea Area (PSSA) is an IMO-designated measure, established in accordance with IMO Resolution. A suitably worded note should be inserted on the relevant charts; the following is an example (in green): "PARTICULARLY SENSITIVE SEA AREA (PSSA) An IMO-approved PSSA is designated in [general area or the area of this chart]. Mariners … [insert any special requirements, procedures, etc]. For further details, see [insert name of publication]." The exact wording of the note should be tailored to cover each specific area, ie location, the type of associated protective measures, etc. It may be detailed or may be simply a reference which draws attention to the full details contained in an associated publication such as Sailing Directions. Such a note may be combined with other related notes. A simple note, providing a reference to an associated publication may be the only way in which some associated protective measures, such as special discharge restrictions, can be identified on charts.

The legend 'Particularly Sensitive Sea Area (see Note)' should be inserted within the area of the PSSA, at appropriate positions on relevant charts. Where space is limited, the abbreviated legend 'PSSA (see Note)' may be inserted. The limit of a PSSA should be charted using a dashed line (in green), with a tint band of maximum 5 mm width and the international abbreviation 'PSSA' on the PSSA side of the dashed line limit, in green. The actual width of the band, and density of the tint, should be carefully selected so that the area is not given undue prominence compared with other areas. Use of the tint band in addition to the dashed limit serves to emphasize or differentiate the limit and provides visual continuity to the entirety of a PSSA's area. The dashed limit may be broken for any associated protective measures whose limits coincide with those of the PSSA itself. It may be appropriate to reverse this convention (ie instead of breaking the dashed limit, break the tint band and continue the dashes) where the limit coincides with a Traffic Separation Scheme tint band.

Where the limits of any Associated Protective Measures, which according to the specifications detailed above should be inserted on charts, coincide with those of the PSSA, both limits should be inserted. The Associated Protective Measure limits should be in accordance with the appropriate specifications, one component of the limit of the PSSA (ie the tint band or the dashes) being broken.
Historic Wreck Rule:
An explanatory note may be added, in green, eg: "HISTORIC WRECKS The sites of historic wrecks are protected from unauthorized interference."

Oil and Gas Fields Rule:
Where areas of Gas and or Oil fields exist on the Chart, the following note should be shown in the marginalia in green. "OFFSHORE OIL AND GAS FIELDS. Numerous structures (usually carrying lights) pipes and submerged obstructions (sometimes marked by buoys) exist in certain areas indicated on this chart. For further details, the larger scale charts should be consulted."

Traffic Services Limit Rule:
A note should be added to the chart, listing all the VTS areas which have vessel reporting requirements if they fall wholly or partially within the chart, whether the limits can be charted or not or the reporting points cannot be charted (for example because they are time rather than position related). For example, for one VTS: "VESSEL REPORTING For details of the [Name] VTS, see [associated publication]" or for several VTS, eg: "VESSEL REPORTING For details of the following [vessel traffic services and vessel reporting systems], see [associated publication]: • London VTS • Ramsgate Port Information Service • Medway Navigation Service • Dover Strait Reporting System (CALDOVREP)". If there is no available publication, essential details may be given in the note. To assist cross referencing, it is important to use exactly the form of name used in any associated publication, noting that not all vessel traffic services or reporting systems have 'VTS' as part of the name.

Feature Information					
Specification:	*Chart #1*	**Feature Origin:**	*Hydrographic*	**Symbol:**	N34
Symbol Criteria:	Maritime Area Restriction = [Mine Danger Area] or [Former Mine Danger Area]			**Label:**	"Mine Danger Area" if (Maritime Area Restriction: [Mine Danger Area]), 8 point Mixed case, Italic;
				Color:	Green (52813)

Feature Information					
Specification:	*Chart #1*	**Feature Origin:**	*Hydrographic*	**Symbol:**	N31
Symbol Criteria:	Maritime Navigation Restriction = [Entry Prohibited]			**Label:**	"[Maritime Navigation Restriction]", 8 point Mixed case, Italic
				Color:	Green (52813)

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Maritime Limit (FC021) *(Continued)*

Curve

Feature Information					
Specification:	*Chart #1*	**Feature Origin:**	*Hydrographic*	**Symbol:**	N45
Symbol Criteria:	Maritime Area Restriction = [Fishery Zone]			**Label:**	None
				Color:	N/A

Feature Information					
Specification:	*Chart #1*	**Feature Origin:**	*Hydrographic*	**Symbol:**	F31
Symbol Criteria:	Maritime Area Restriction = [Reclamation Area]			**Label:**	"Being Reclaimed ([Note : Memorandum])" if (Maritime Area Restriction = [Reclamation Area]) and [Note : Memorandum]if available and describes a date, 8 point Mixed case
				Color:	Green (52813)

Feature Information					
Specification:	*Chart #1*	**Feature Origin:**	*Hydrographic*	**Symbol:**	N22
Symbol Criteria:	(Maritime Area Restriction = [Conservation and Management Zone]			**Label:**	"Particularly Sensitive Sea Area (see Note)", 8-14 point Mixed case, Italic;
				Color:	Green (52813)

Feature Information					
Specification:	*Chart #1*	**Feature Origin:**	*Hydrographic*	**Symbol:**	N25
Symbol Criteria:	Maritime Area Restriction = [Degaussing Range]			**Label:**	"[Maritime Area Restriction]", 8 point Mixed Case Italic
				Color:	Black (58600)

Feature Information					
Specification:	*Chart #1*	**Feature Origin:**	*Hydrographic*	**Symbol:**	N63
Symbol Criteria:	Maritime Area Restriction = [Dredging Area]			**Label:**	"Extraction Area", 8 point Mixed Case Italic
				Color:	Green (52813)

Feature Information					
Specification:	*Chart #1*	**Feature Origin:**	*Hydrographic*	**Symbol:**	M18
Symbol Criteria:	Maritime Area Restriction = [Fairway]			**Label:**	"FAIRWAY [Note : Memorandum]m (see Note)", [Note : Memorandum]if available and described a depth, 8-14 point Mixed Case Italic
				Color:	Green (52813)

Feature Information					
Specification:	*Chart #1*	**Feature Origin:**	*Hydrographic*	**Symbol:**	N26
Symbol Criteria:	Maritime Area Restriction = [Historic Wreck Area]			**Label:**	"Historic Wk", 8 Point Mixed case Italic
				Color:	Green (52813)

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Maritime Limit (FC021) *(Continued)*

Curve

Feature Information					
Specification:	*Chart #1*	**Feature Origin:**	*Hydrographic*	**Symbol:**	N2.1
Symbol Criteria:	Maritime Area Restriction = [Natural Gas Field] or [Petroleum Field] or [Joint Economic Development Zone]			**Label:**	"[Geographic Name Information : Full Name] OILFIELD" if available and if (Maritime Area Restriction = [Petroleum Field], 8-14 Point Mixed case Italic; "[Geographic Name Information : Full Name] GASFIELD" if available and if (Maritime Area Restriction = [Natural Gas Field], 8-14 Point Mixed case Italic; "DEVELOPMENT AREA (see Note)" if Maritime Area Restriction = [Joint Economic Development Zone], 8-14 Point Mixed case Italic
				Color:	Green (52813)

Feature Information					
Specification:	*Chart #1*	**Feature Origin:**	*Hydrographic*	**Symbol:**	N1.2
Symbol Criteria:	(Maritime Area Restriction = [Traffic Services Limit]) or (Maritime Caution Type = [COLREGS Demarcation Line])			**Label:**	"[Geographic Name Information : Full Name] VTS (see Note)" if Maritime Area Restriction = [Traffic Services Limit], 8-14 point Mixed case Italic
				Color:	Green (52813)

Feature Information					
Specification:	*Chart #1*	**Feature Origin:**	*Hydrographic*	**Symbol:**	N64
Symbol Criteria:	Maritime Caution Type = [Cargo Transshipment]			**Label:**	"Cargo Transhipment Area", 8 point Upper case Italic
				Color:	Green (52813)

Feature Information					
Specification:	*Chart #1*	**Feature Origin:**	*Hydrographic*	**Symbol:**	M2
Symbol Criteria:	Maritime Caution Type = [Clearing Line]			**Label:**	None
				Color:	N/A

Feature Information					
Specification:	*Chart #1*	**Feature Origin:**	*Hydrographic*	**Symbol:**	N49
Symbol Criteria:	Maritime Caution Type = [Harbour Area]			**Label:**	"Harbour Limit", 8 point Mixed Case Italic
				Color:	Green (52813)

Feature Information					
Specification:	*Chart #1*	**Feature Origin:**	*Hydrographic*	**Symbol:**	N30
Symbol Criteria:	Naval Firing and/or Practice Type = [Firing Danger Area]			**Label:**	"None
				Color:	N/A

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Maritime Limit (FC021) *(Continued)*

Curve

Feature Information					
Specification:	*Chart #1*	**Feature Origin:**	*Hydrographic*	**Symbol:**	N32
Symbol Criteria:	Naval Firing and/or Practice Type = [Mine Laying Practice Area]			**Label:**	None
				Color:	N/A

Feature Information					
Specification:	*Chart #1*	**Feature Origin:**	*Hydrographic*	**Symbol:**	N33
Symbol Criteria:	Naval Operations Type = [Submarine Exercise Area]			**Label:**	"SUBMARINE EXERCISE AREA (see Note)", 8 point Mixed case Italic
				Color:	Green (52813)

Feature Information					
Specification:	*Chart #1*	**Feature Origin:**	*Hydrographic*	**Symbol:**	M14
Symbol Criteria:	Maritime Caution Type = [Pollution Zone]			**Label:**	"Pollution Zone", 8 point Mixed case Italic
				Color:	Green (52813)

Feature Information					
Specification:	*Chart #1*	**Feature Origin:**	*Hydrographic*	**Symbol:**	N11.2
Symbol Criteria:	Maritime Area Restriction = [Swinging Area]			**Label:**	None
				Color:	N/A

Feature Information					
Specification:	*Chart #1*	**Feature Origin:**	*Hydrographic*	**Symbol:**	No symbology as they are prohibited by IHO.
Symbol Criteria:	Maritime Area Restriction = [Incineration area]			**Label:**	None
				Color:	N/A

Maritime Navigation Beacon (BC010)

Point

Feature Description:	*A prominent, specially constructed object forming a conspicuous mark as a fixed aid to navigation or for use in hydrographic survey.*
Inclusion Condition:	*All Required*

Notes

Beacons and additional co-located Maritime Features:

Beacons may contain additional Maritime features that co-located as the same coordinate. These additional features can be Maritime Navigation Light (BC040), Fog Signal (BC101), Sound Signal Device (BC103), and Maritime Radio Beacon (BC034). These other Maritime features in conjunction with the Buoy feature will provide all of the necessary components of a Buoy that will be located at the same coordinate, but collected as separate features. Refer to rules pertaining to those additional features.

Displacement Rules for NAVAIDS overprinting hydrographic dangers or depth curves:

When aids to navigation overprint hydrographic dangers, or depth curves, the following displacement criteria applies:

(a.) If the aid to navigation is a fixed aid (Maritime Navigation Beacon and Maritime Navigation Light), and the hydrographic danger symbol is a point symbol, delete the hydrographic danger point symbol. Show both if the hydrographic danger symbol is an area symbol.
(b.) If the aid to navigation is a beacon, the hydrographic danger symbol's danger curve, i.e., the dotted perimeter line, if present, is broken for the beacon symbol (excluding type).

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Maritime Navigation Beacon (BC010) *(Continued)*

Point

Notes

(c.) If a point danger symbol and a beacon are overprinting, refer to Cardinal Displacement Rules for hazards and displacement shall be as little as possible to resolve the overprint.
(d.) The above criteria refer to the graphic portion of symbols only. All type and labels are movable around the graphic elements to avoid overprints. If the hydrographic danger symbol's hydrographic depth type must be moved, it shall be placed outside the danger circle, and enclosed in parentheses.
(e.) Depth curves are broken for aids to navigation and dangers, and the type associated with these symbols.

Lateral Marks Guidance:

Lateral marks may exhibit red or green lights of any rhythm (but not fixed), including Long Flashing lights. A composite group flashing red or green light (eg Fl(2+1)R) is exhibited only from modified lateral (preferred channel) buoys.

IALA Region A:
- Port-hand marks are red with cylindrical topmarks (if any). Lights are red and have any rhythm except Fl(2+1)R
- Preferred channel to port Buoy will show Maritime Navigation Navigation Lights as Fl(2+1)G, if lit
- Starboard-hand marks are green with conical topmarks (if any). Lights are green and have any rhythm except Fl(2+1)G
- Preferred channel to starboard Buoy will show Maritime Navigation Navigation Lights as Fl(2+1)R, if lit
- Port-hand Edge Buoys (Buoy Type:[Port-hand Edge]) may be shown Can and Pillar and will be colored red with cylinder Topmarks.
- Starboard-hand Edge Buoys (Buoy Type:[Starboard-hand Edge]) may be shown Conical and Pillar and will be colored green with cone Topmarks

IALA Region B:
- Port-hand marks are green with cylindrical topmarks (if any). Lights are green and have any rhythm except Fl(2+1)G
- Preferred channel to port will show Maritime Navigation Navigation Lights as Fl(2+1)R, if lit
- Starboard-hand marks are red with conical topmarks (if any). Lights are red and have any rhythm except Fl(2+1)R
- Preferred channel to starboard will show Maritime Navigation Navigation Lights as Fl(2+1)G, if lit

Yellow, abbreviated "Y", shall be substituted for Orange "Or" or Amber "Am" when describing light color in the character of light.
If the only light color is white, omit color from light. e.g., "Fl 6s 12m 8M" is a white light.

The Period of Light label for a Maritime Navigation Light that is associated with a buoy may be omitted if that buoy is shown on a chart at a larger scale.

In areas of congested type, if it is necessary to abridge a Maritime Navigation Light legend due to clutter, the period may be omitted.

With the lower end of the topmark touching the feature, use the following IALA Maritime Buoyage System - Topmarks guidance.
IALA Maritime Buoyage System - Topmarks guidance

Lateral marks may have a single can (cylindrical) topmark (Topmark Shape: [Cylinder]) on the port hand and a single conical topmark (point up) (Topmark Shape: [Cone Point Upward]) on the starboard hand, coloured red or green as appropriate for Region A and Region B. Can and conical buoys indicate by their shape which is the correct side to pass, so these buoys may not be fitted with topmarks. If the buoy does not have a distinctive shape, a topmark will usually be fitted. Region A: red to port, green to starboard, Region B: green to port, red to starboard.

Cardinal marks have two black cone topmarks, one above the other. There must be a clear separation between each cone; in particular, two cones base to base must not be shown as a diamond shape.

North Cardinal - (Buoy Type: [North Cardinal]) will use Topmark Shape: [2 Cones Points Upward]
South Cardinal - (Buoy Type: [South Cardinal]) will use Topmark Shape: [2 Cones Points Downward]
East Cardinal - (Buoy Type: [East Cardinal]) will use Topmark Shape: [2 Cones Base to Base]
West Cardinal - (Buoy Type: [West Cardinal]) will use Topmark Shape: [2 Cones Point to Point]

The point of the cone reflects the position of the black band(s) on the body of the buoy (or beacon), eg for a West cardinal mark, the black band is in the middle.

Isolated danger marks, which indicate the location of an isolated danger of limited size that is completely surrounded by navigable water, have two black sphere topmarks, one above the other (Topmark Shape: [2 Spheres One over Other].

Safe water marks, as used for centre-lines of channels or as landfall marks, may have a single red sphere topmark (Topmark Shape: [Sphere]. Spherical buoys indicate by their shape that there is navigable water all around them, so such buoys may not be fitted with topmarks.

Special marks, not primarily intended to assist navigation but indicating a special area or feature, may have a single yellow diagonal 'X' shaped topmark (St Andrew's cross) (Topmark Shape: [X-shaped].

Special marks may also be used to mark Traffic Separation Schemes, or channels within channels (eg a Deep Water route within a wider navigation channel marked by standard lateral buoys) or special purpose channels (eg for small craft).

Sound Signal Device Guidance:

To avoid clutter, where a Sound Signal Device is co-located with another aid to navigation, the [Fog Signal Type] legend or the Chart #1 symbol R1 should be shown, but not both.

If a Sound Signal Device (Fog Signal Type: [No Information]) or not shown, label the sound signal type with appropriate legend and show Chart #1 (R1) symbol.

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Maritime Navigation Beacon (BC010) *(Continued)*

Point

Notes

If Fog Signal Type: [Bell], use label "Bell", Chart #1 (R14)
If Fog Signal Type: [Diaphone], use legend "Dia", Chart #1 (R11)
If Fog Signal Type: [Explosive], use legend "Explos", Chart #1 (R10)
If Fog Signal Type: [Gong], use legend "Gong", Chart #1 (R16)
If Fog Signal Type: [Horn], use legend "Horn", Chart #1 (R13)
If Fog Signal Type: [Siren], use legend "Siren", Chart #1 (R12)

A Sound Signal Device with Fog Signal Type: [Other] or [No Information] the Chart #1 (R1) symbol shall be shown and can be used alone or in conjunction with other aids to navigation such as Buoys, Maritime Navigational Lights, or Maritime Radiobeacons.

Color of Navigational Aids Guidance:

The international abbreviations for colour are:
The International Abbreviations for Navigation Mark Colours for labeling color of Navigational Aids is:
Navigation Mark Colour: [White], abbreviate "W";

Navigation Mark Colour: [Red], abbreviate "R";

Navigation Mark Colour: [Green], abbreviate "G";

Navigation Mark Colour: [Blue], abbreviate "Bu";

Navigation Mark Colour: [Violet], abbreviate "Vi";

Navigation Mark Colour: [Yellow], abbreviate "Y";

Navigation Mark Colour: [Orange], abbreviate "Or"; (may be charted as "Y")

Navigation Mark Colour: [Amber], abbreviate "Am"; (may be charted as "Y")

Navigation Mark Colour: [Black], abbreviate "B";

Colour abbreviations must be in capital letters in all cases except for the second letter of two-letter abbreviations. These abbreviations must be used for the colours of lights and structures.

Where the structure and topmark are coloured differently, the colour that best assists the mariner in identifying the beacon should be charted. If it is useful to give the colour of both topmark and structure, then the topmark colour should be given first, divided from the structure colour by a slash, eg for a red lateral topmark on a white painted beacon tower: R/W.

Multicoloured structures. Where the colours are in bands, the sequence of colour abbreviations must be from top to bottom. Where the colours are in stripes (vertical or diagonal) or the sequence of horizontal bands is not known, the darker colour must be given first. Buoys and Beacons which consist of multiple colors (Bands or Stripes) should contain multiple Navigation Mark Colour attributes collected for each color and in the order top to bottom for Bands and darker color first for Stripes.
The Navigation Mark Colour label for Buoys and Maritime Navigation Beacon shall be centered immediately below the small circle marking the position of the feature.

Within the IALA Maritime Buoyage System:
• red and green are used for lateral marks,
• black and yellow bands are used for cardinal marks,
• black and red bands are used for isolated danger marks,
• red and white stripes are used for safe-water buoys,
• yellow is used for special marks,
• blue and yellow stripes are used for emergency wreck marking buoys (on trial from 2006).

Red and green on lateral marks have different meanings in IALA Regions A and B, therefore the applicable region must be stated on charts.

Where there is insufficient space on charts for abbreviations, the topmarks alone (for cardinal buoys and beacons) or the filled and open symbols (for lateral buoys) may be considered adequate to indicate colours, without abbreviations.
Abbreviations for multiple colours on buoys (and beacons) must be shown in accordance with the following conventions:

a. Where the colours are in bands the sequence of colour abbreviations must be from top to bottom, eg in the IALA System: Chart #1 (Q4)
• a north cardinal mark (black above yellow): BY
• an east cardinal mark (black with single broad horizontal yellow band): BYB
• a south cardinal mark (yellow above black): YB
• a west cardinal mark (yellow with a single broad horizontal black band): YBY
• a preferred channel lateral mark: GRG or RGR
• an isolated danger mark (black with one or more broad horizontal red bands): BRB

Note: It helps to remember that the points of the topmark cones (for cardinal marks) reflect the position of the black band(s) on the body of the buoy (or beacon), eg for a north cardinal mark, the black band is at the top.

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Maritime Navigation Beacon (BC010) *(Continued)*

Point

Notes

b. Where the colours are in stripes (vertical or diagonal) or the sequence of horizontal bands is not known, the darker colour must be given first, eg in the IALA System: Chart #1 (Q5)
- *a safe water buoy (red and white vertical stripes): RW*
- *an emergency wreck marking buoy (blue and yellow vertical stripes): BuY*

RADAR REFLECTORS ON BEACONS Guidance:

Radar reflectors are not generally charted on buoys and buoyant beacons within the IALA System as they are commonly constructed to be radar reflective. However, beacons do not consistently have radar reflectors, therefore where it is known that a beacon has a radar reflector installed (Radar Reflector Present:[True]), it should be charted by the symbol S4, placed above the beacon if space allows.

'Radar reflectors', are appropriate for buoyant beacons (Beacon Type: [Articulated Beacon]).

If Radar Reflector Present:[False] do not show Radar Reflector (Chart #1 (S4)).

SPECIAL-PURPOSE BEACONS:

Special-purpose beacons are beacons which are not part of a cardinal or lateral system. They may conform to IALA special marks, or to a national or other standard system, eg triangles placed point up on front leading marks and point down on rear leading marks. Full details (if known and if scale permits) should be shown on charts. However, if special-purpose beacons conform to a standard shape and colour throughout a nation's waters, they may be charted in less detail where an adequate description of the standard system is in the Sailing Directions (eg colour abbreviations may be omitted, or be charted simply by a 'beacon in general' symbol, Q80).

The functions of beacons marking leading lines, cables, outfalls or measured distance will be clear from the associated line symbols, so legends such as 'Cable Beacons', 'Leading Beacons', or equivalents should not be added.

An isolated danger mark is an example of a special-purpose beacon which conforms to an international (ie the IALA) system. The following are some other examples of special-purpose beacons, the purpose of which is identifiable by reference to other charted features.
- *Leading beacons*
- *Beacons marking a clearing line*
- *Beacons marking measured distance with quoted bearings*
- *Cable landing beacon*

Miscellaneous Beacon Guidance:

Symbols and associated legends for beacons must be upright (except those on submerged or drying rocks, to help distinguish them from floating spar buoys, which are less reliable for position fixing.
Miscellaneous labels occasionally may be found in association with marine navigational aids. If a text label is shown on hydrographic source material, it should be considered significant for navigation. Examples are:
A fog detection light, label "Fog Det Lt";
A floodlit structure near navigable water, label "(Illuminated)"; (Floodlit: [True])
A daytime light, if character of light in the day is different from the character shown at night. Show daytime character, followed by "Day" in parentheses, for example: (F 37m 11M Day);
Unwatched light, with no standby or emergency arrangements, label "(U)";
A temporary light or buoy, label "(temp)". If seasonal, include months, for example: "(Apr-Oct)";
A fog light, if light is only shown in fog, or the light during fog is different from the character of light shown at other times, show character in fog, followed by "(in fog)", in parentheses. For example: Fl 5s (in fog);
Reserve fog signals shall not be shown on product.
If a channel is marked by a feature but because of the product scale individual features overprint or are closer than 3 mm, replace the features with the legend "Buoyed channel" or "Channel marked by beacons". Legend will be aligned with the channel.

Feature Information					
Specification:	*Chart #1*	**Feature Origin:**	*Hydrographic*	**Symbol:**	Q83
Symbol Criteria:	Beacon Type = [Insubstantial Navigation Mark]			**Label:**	"[Navigational Mark Colour]", 5 point Upper Case
				Color:	Black (58600)

Feature Information					
Specification:	*Chart #1*	**Feature Origin:**	*Hydrographic*	**Symbol:**	Q82
Symbol Criteria:	Beacon Type = [Pile Beacon] or [Articulated Beacon]			**Label:**	"[Navigational Mark Colour]", 5 point Upper Case
				Color:	Black (58600)

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Maritime Navigation Beacon (BC010) *(Continued)*

Point

Feature Information					
Specification:	*Chart #1*	**Feature Origin:**	*Hydrographic*	**Symbol:**	Q110
Symbol Criteria:	Beacon Type = [Beacon Tower]			**Label:**	"[Navigational Mark Colour]", 5 point Upper Case
				Color:	Black (58600)

Feature Information					
Specification:	*Chart #1*	**Feature Origin:**	*Hydrographic*	**Symbol:**	Q90
Symbol Criteria:	Beacon Type = [Withy]			**Label:**	None
				Color:	N/A

Feature Information					
Specification:	*Chart #1*	**Feature Origin:**	*Hydrographic*	**Symbol:**	Q111
Symbol Criteria:	Beacon Type = [Lattice Beacon]			**Label:**	"[Navigational Mark Colour]", 5 point Upper Case
				Color:	Black (58600)

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Maritime Navigation Light (BC040)

Curve

Feature Description:	*A luminous or lighted device intended for the purpose of aiding maritime navigation.*
Inclusion Condition:	*Length >= 10m*

Notes

Lights and additional co-located Maritime Features:

While Maritime Navigation Light may be a single feature, these lights are often co-located as a component with other Maritime features at the same coordinate such as Maritime Navigation Beacon (BC010), Buoy (BC020) Light Sector (BC060), Light Vessel (BC070), Lighthouse (BC050), and Maritime Navigation Light Support.

Displacement Rules for NAVAIDS overprinting hydrographic dangers or depth curves:

When aids to navigation overprint hydrographic dangers, or depth curves, the following displacement criteria applies:

(a.) If the aid to navigation is a fixed aid (Maritime Navigation Beacon and Maritime Navigation Light), and the hydrographic danger symbol is a point symbol, delete the hydrographic danger point symbol. Show both if the hydrographic danger symbol is an area symbol.
(b.) If the aid to navigation is a beacon, the hydrographic danger symbol's danger curve, i.e., the dotted perimeter line, if present, is broken for the beacon symbol (excluding type).
(c.) If a point danger symbol and a beacon are overprinting, refer to Cardinal Displacement Rules for hazards and displacement shall be as little as possible to resolve the overprint.
(d.) The above criteria refer to the graphic portion of symbols only. All type and labels are movable around the graphic elements to avoid overprints. If the hydrographic danger symbol's hydrographic depth type must be moved, it shall be placed outside the danger circle, and enclosed in parentheses.
(e.) Depth curves are broken for aids to navigation and dangers, and the type associated with these symbols

Lateral Marks Guidance:

Lateral marks may exhibit red or green lights of any rhythm (but not fixed), including Long Flashing lights. A composite group flashing red or green light (eg Fl(2+1)R) is exhibited only from modified lateral (preferred channel) buoys.

IALA Region A:
- *Port-hand marks are red with cylindrical topmarks (if any). Lights are red and have any rhythm except Fl(2+1)R*
- *Preferred channel to port Buoy will show Maritime Navigation Navigation Lights as Fl(2+1)G, if lit*
- *Starboard-hand marks are green with conical topmarks (if any). Lights are green and have any rhythm except Fl(2+1)G*
- *Preferred channel to starboard Buoy will show Maritime Navigation Navigation Lights as Fl(2+1)R, if lit*
- *Port-hand Edge Buoys (Buoy Type:[Port-hand Edge]) may be shown Can and Pillar and will be colored red with cylinder Topmarks.*
- *Starboard-hand Edge Buoys (Buoy Type:[Starboard-hand Edge]) may be shown Conical and Pillar and will be colored green with cone Topmarks*

IALA Region B:
- *Port-hand marks are green with cylindrical topmarks (if any). Lights are green and have any rhythm except Fl(2+1)G*
- *Preferred channel to port will show Maritime Navigation Navigation Lights as Fl(2+1)R, if lit*
- *Starboard-hand marks are red with conical topmarks (if any). Lights are red and have any rhythm except Fl(2+1)R*
- *Preferred channel to starboard will show Maritime Navigation Navigation Lights as Fl(2+1)G, if lit*

Yellow, abbreviated "Y", shall be substituted for Orange "Or" or Amber "Am" when describing light color in the character of light.
If the only light color is white, omit color from light. e.g., "Fl 6s 12m 8M" is a white light.
The Period of Light label for a Maritime Navigation Light that is associated with a buoy may be omitted if that buoy is shown on a chart at a larger scale.

In areas of congested type, if it is necessary to abridge a Maritime Navigation Light legend due to clutter, the period may be omitted.

Special marks may also be used to mark Traffic Separation Schemes, or channels within channels (eg a Deep Water route within a wider navigation channel marked by standard lateral buoys) or special purpose channels (eg for small craft).

Sound Signal Device Guidance:

To avoid clutter, where a Sound Signal Device is co-located with another aid to navigation, the [Fog Signal Type] legend or the Chart #1 symbol R1 should be shown, but not both.

If a Sound Signal Device (Fog Signal Type: [No Information]) or not shown, label the sound signal type with appropriate legend and show Chart #1 (R1) symbol.
If Fog Signal Type: [Bell], use label "Bell", Chart #1 (R14)
If Fog Signal Type: [Diaphone], use legend "Dia", Chart #1 (R11)
If Fog Signal Type: [Explosive], use legend "Explos", Chart #1 (R10)
If Fog Signal Type: [Gong], use legend "Gong", Chart #1 (R16)
If Fog Signal Type: [Horn], use legend "Horn", Chart #1 (R13)
If Fog Signal Type: [Siren], use legend "Siren", Chart #1 (R12)
A Sound Signal Device with Fog Signal Type: [Other] or [No Information] the Chart #1 (R1) symbol shall be shown and can be used alone or in conjunction with other aids to navigation such as Buoys, Maritime Navigational Lights, or Maritime Radiobeacons.

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Maritime Navigation Light (BC040) *(Continued)*

Curve

Notes

Color of Navigational Aids Guidance:
The international abbreviations for colour are:
The International Abbreviations for Navigation Mark Colours for labeling color of Navigational Aids is:
Navigation Mark Colour: [White], abbreviate "W";
Navigation Mark Colour: [Red], abbreviate "R";
Navigation Mark Colour: [Green], abbreviate "G";
Navigation Mark Colour: [Blue], abbreviate "Bu";
Navigation Mark Colour: [Violet], abbreviate "Vi";
Navigation Mark Colour: [Yellow], abbreviate "Y";
Navigation Mark Colour: [Orange], abbreviate "Or"; (may be charted as "Y")
Navigation Mark Colour: [Amber], abbreviate "Am"; (may be charted as "Y")
Navigation Mark Colour: [Black], abbreviate "B";

Colour abbreviations must be in capital letters in all cases except for the second letter of two-letter abbreviations. These abbreviations must be used for the colours of lights and structures.

Where the structure and topmark are coloured differently, the colour that best assists the mariner in identifying the beacon should be charted. If it is useful to give the colour of both topmark and structure, then the topmark colour should be given first, divided from the structure colour by a slash, eg for a red lateral topmark on a white painted beacon tower: R/W.

Multicoloured structures. Where the colours are in bands, the sequence of colour abbreviations must be from top to bottom. Where the colours are in stripes (vertical or diagonal) or the sequence of horizontal bands is not known, the darker colour must be given first. Buoys and Beacons which consist of multiple colors (Bands or Stripes) should contain multiple Navigation Mark Colour attributes collected for each color and in the order top to bottom for Bands and darker color first for Stripes.
The Navigation Mark Colour label for Buoys and Maritime Navigation Beacon shall be centered immediately below the small circle marking the position of the feature.

Within the IALA Maritime Buoyage System:
• red and green are used for lateral marks,
• black and yellow bands are used for cardinal marks,
• black and red bands are used for isolated danger marks,
• red and white stripes are used for safe-water buoys,
• yellow is used for special marks,
• blue and yellow stripes are used for emergency wreck marking buoys (on trial from 2006).

Red and green on lateral marks have different meanings in IALA Regions A and B, therefore the applicable region must be stated on charts.

Where there is insufficient space on charts for abbreviations, the topmarks alone (for cardinal buoys and beacons) or the filled and open symbols (for lateral buoys) may be considered adequate to indicate colours, without abbreviations.
Abbreviations for multiple colours on buoys (and beacons) must be shown in accordance with the following conventions:

a. Where the colours are in bands the sequence of colour abbreviations must be from top to bottom, eg in the IALA System: Chart #1 (Q4)
• a north cardinal mark (black above yellow): BY
• an east cardinal mark (black with single broad horizontal yellow band): BYB
• a south cardinal mark (yellow above black): YB
• a west cardinal mark (yellow with a single broad horizontal black band): YBY
• a preferred channel lateral mark: GRG or RGR
• an isolated danger mark (black with one or more broad horizontal red bands): BRB

Note: It helps to remember that the points of the topmark cones (for cardinal marks) reflect the position of the black band(s) on the body of the buoy (or beacon), eg for a north cardinal mark, the black band is at the top.

b. Where the colours are in stripes (vertical or diagonal) or the sequence of horizontal bands is not known, the darker colour must be given first, eg in the IALA System: Chart #1 (Q5)
• a safe water buoy (red and white vertical stripes): RW
• an emergency wreck marking buoy (blue and yellow vertical stripes): BuY

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Maritime Navigation Light (BC040) *(Continued)*

Curve

Notes

Light Flare Guidance:

The point of a light flare P11 should be about 1 mm from the charted position of the light.

The orientation of flares should be such as to avoid obscuring other detail, eg symbols, soundings, text. In the case of a leading light (see B-475.6), lights in line and direction lights, the flares should be oriented to seaward along the line, provided this does not obscure the front light star. Black lines (eg transit lines, coastline, graticule) should not be broken where they cross a light flare.

Flares must not be inserted against air obstruction lights or strip lights where light stars are usually omitted.

Names of major lights are very important. If a light has a name which is unrelated to any other charted feature, the name must be inserted against the position of the light on at least the largest scale charts, above or preceding the light description, and should be in the same style as the light description. If the name of a light is obviously that of the named feature on which the light stands, eg Saint Catherine's Point, the name of the light need not be repeated above the light description. The name must be in the style appropriate to the feature, eg a headland or a shoal, and in many cases can be sited immediately above the light description. Where a light description is unavoidably sited some distance from the light star, the name of the light should be repeated above the light description, in the same style as the light description.

Minor lights may be identifiable in the List of Lights and Fog Signals by a charted general name and a (possibly uncharted) descriptive term, eg Royal Pier. Names or descriptions of individual lights of a pair of leading lights, eg 'Rear' or 'Upper', 'Front' or 'Lower', can normally be deduced from the positions shown on the chart and, to save clutter and translation, should not be inserted on paper charts.

LIGHT DESCRIPTIONS:

The principal character of a light is its rhythm (although, strictly, fixed lights and some alternating lights are not 'rhythmic'). The basic international abbreviations including Chart #1 reference are shown below for the following [Navigation Light Characteristic] attribute:

[Alternating]: Al (P10.11) (Example: AlWR)
[Fixed]: F (P10.1)
[Fixed and Flashing]: FFl (P10.10)
[Fixed Long-flashing]: F LFl (P10.1, P10.5)
[Fixed with Alternating Flashing]: F Al (P10.1, P10.11)
[Flashing]: Fl (P10.4)
[Flashing Alternating]: Fl Al (P10.4, P10.11)
[Flashing with Long-flash]: Fl LFl (P10.4, P10.5)
[Group Quick-flashing]: Q() (P10.6) (Example: Q(3))
[Group Very Quick-flashing]: VQ() (P10.7) (Example: VQ(3))
[Interrupted Quick-flashing]: IQ (P10.6)
[Interrupted Ultra Quick-flashing]: IUQ (P10.8)
[Interrupted Very Quick-flashing]: IVQ (P10.7)
[Isophase]: Iso (P10.3)
[Long-flashing]: LFl (P10.5)
[Long-flashing Alternating]: LFl Al (P10.5, P10.11)
[Morse Code]: Mo() (P10.9) (Example: Mo(K))
[Occulting]: Oc (P10.2)
[Occulting Alternating]: Oc Al (P10.2, P10.11)
[Occulting Flashing]: Oc Fl (P10.2, P10.4)
[Quick-flashing]: Q (P10.6)
[Quick-flashing with Long-flash]: Q LFl (P10.6, P10.5)
[Ultra Quick-flashing]: UQ (P10.8)
[Ultra Quick-flashing with Long-flash]: UQ LFl (P10.8, P10.5)
[Very Quick-flashing]: VQ (P10.7)
[Very Quick-flashing with Long-flash]: VQ LFl (P10.7, P10.5)

Lights exhibited from the same structure (or charted at the same light star):

a. If more than one light is exhibited from a light structure the description of the main one (eg a light visible from all directions) should be shown on one line and the subsidiary light (eg a red sector of different character, covering a danger) on a line below.
Two short descriptions may be shown on one line linked by '&'. This also applies where two separate lights which are close together are charted by one light star, because of scale.

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Maritime Navigation Light (BC040) *(Continued)*

Curve

Notes

b. Disposition of lights. Lights exhibited from the same structure which are disposed horizontally or vertically must be charted by the abbreviation '(hor)' or '(vert)' P15, as appropriate, immediately following the colour in the light description.
Two (or more) fixed lights of the same colour disposed horizontally or vertically must be charted, eg:
• 2F.G(hor) means that two fixed green lights are disposed horizontally
• 2F.R(vert) means that two fixed red lights are disposed vertically
• 3F.R(vert) means that three fixed red lights are disposed vertically
It is possible to show lights exhibited from the same structure arranged in other ways by means of a geometric symbol, eg:
• 3F.R(Δ) means 3 fixed red lights disposed in the shape of a triangle (the appropriate way up).
Two (or more) lights of different colour disposed horizontally or vertically must be charted, eg:
• F.GR(vert) means that 2 fixed lights are disposed vertically, the uppermost being green, the lower being red.
• F.RGR(hor) means that 3 fixed lights are disposed horizontally, the middle one being green.
The '&' sign is not required, as the qualifier (vert) or (hor) clearly indicates that there is more than one light. These conventions must not be used for Traffic Signals (see B-495).
c. If a fixed light is varied at intervals by a flash of greater intensity, it is charted as FFl, P10.10.
Combining the elements of a light description must be achieved in a way that enables complex descriptions to be shown compactly. However, some spacing of the elements is needed for ease of interpretation. Full stops are specified below to ensure spacing, but the full stops may be omitted providing the spacing is retained:

a. Insert full stops (or spaces):
• at the end of the characteristic rhythm (except where there is a bracket);
• at the end of all colours (not between colours);
• after Al (Alternating) - although Al is not a rhythmic characteristic it is often juxtaposed with one.

b. Omit full stops:
• after s (seconds);
• after m (elevation);
• after M (range);
• where there is a bracket;
• at the end of the light description.

c. If more than one range is given in the light description for a single light, show as follows:
eg 15/10M P14 Light with two different ranges (use forward slash).
eg 15-7M P14 Light with three or more different ranges (use hyphen).

Colours of a light must be arranged in the same order as the ranges, with the longest range normally given first However, in the case of a FFl light, where the flash is always brighter, the ranges should be shown in the same order as the character to which they refer, eg: FFl.10/15M

LIGHT DESCRIPTIONS: ABRIDGING, OMISSION:

For paper charts, the order of omission of details in an abridged (shortened) description is given below. It is not the same for all types of lights.

Major lights (ie lights intended for use at sea, usually with a range of 15 miles or more, and in outer approaches to harbours). When reducing the detail to be charted as the chart scale decreases, the following must be the order of omission:
a. Elevation of light, eg 23m
b. Period of light, eg 10s
c. Range (visibility) eg 22M
d. Character and colour.

Where useful on some smaller scale charts, a light star, major floating light symbol, or offshore platform symbol may be shown with flare and possibly name but without light description.

Lights within harbours and in restricted channels. It may be advisable to abridge light descriptions even on the largest scale charts to eliminate details of little interest to the mariner, especially where space is very limited. The order of omission must be:
a. Range
b Elevation
c. Period
d. Character and colour.

Where numerous quays, wharves, etc, along a river channel have similar lights, the light star and flare may be retained and a standard note covering them all may be used, eg:
LIGHTS: Light stars without legends represent two fixed lights displayed vertically. They are seen as red to port [or starboard] and green to starboard [or port] when proceeding upriver.

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Maritime Navigation Light (BC040) *(Continued)*

Curve

Notes

Miscellaneous Light Guidance:

Miscellaneous labels occasionally may be found in association with marine navigational aids. If a text label is shown on hydrographic source material, it should be considered significant for navigation. Examples are:
A fog detection light, label "Fog Det Lt";
A floodlit structure near navigable water, label "(Illuminated)"; (Floodlit: [True])
A daytime light, if character of light in the day is different from the character shown at night. Show daytime character, followed by "Day" in parentheses, for example: (F 37m 11M Day);
Unwatched light, with no standby or emergency arrangements, label "(U)";
A temporary light or buoy, label "(temp)". If seasonal, include months, for example: "(Apr-Oct)";
A fog light, if light is only shown in fog, or the light during fog is different from the character of light shown at other times, show character in fog, followed by "(in fog)", in parentheses. For example: Fl 5s (in fog)

Feature Information

Specification:	*Chart #1*	**Feature Origin:**	*Hydrographic*	**Symbol:**	P64
Symbol Criteria:	Maritime Light Type = [Strip Light]			**Label:**	[Geographic Name Information: Full Name], 7 point Mixed case if available; [Character of Light], 7 point Mixed case; [Period of Light]"s", 7 point lower case; [Light Elevation]"m", 7 point lower case; [Nominal Range]"M", 7 point upper case; [Multiple Light Ranges]"M", 7 point upper case (if available)
				Color:	Black (58600)

Maritime Navigation Light (BC040)

Point

Inclusion Condition:	*Length < 10m*

Notes

Lights and additional co-located Maritime Features:

While Maritime Navigation Light may be a single feature, these lights are often co-located as a component with other Maritime features at the same coordinate such as Maritime Navigation Beacon (BC010), Buoy (BC020) Light Sector (BC060), Light Vessel (BC070), Lighthouse (BC050), and Maritime Navigation Light Support.

Displacement Rules for NAVAIDS overprinting hydrographic dangers or depth curves:

When aids to navigation overprint hydrographic dangers, or depth curves, the following displacement criteria applies:

(a.) If the aid to navigation is a fixed aid (Maritime Navigation Beacon and Maritime Navigation Light), and the hydrographic danger symbol is a point symbol, delete the hydrographic danger point symbol. Show both if the hydrographic danger symbol is an area symbol.
(b.) If the aid to navigation is a beacon, the hydrographic danger symbol's danger curve, i.e., the dotted perimeter line, if present, is broken for the beacon symbol (excluding type).
(c.) If a point danger symbol and a beacon are overprinting, refer to Cardinal Displacement Rules for hazards and displacement shall be as little as possible to resolve the overprint.
(d.) The above criteria refer to the graphic portion of symbols only. All type and labels are movable around the graphic elements to avoid overprints. If the hydrographic danger symbol's hydrographic depth type must be moved, it shall be placed outside the danger circle, and enclosed in parentheses.
(e.) Depth curves are broken for aids to navigation and dangers, and the type associated with these symbols.

Lateral Marks Guidance:

Lateral marks may exhibit red or green lights of any rhythm (but not fixed), including Long Flashing lights. A composite group flashing red or green light (eg Fl(2+1)R) is exhibited only from modified lateral (preferred channel) buoys.

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Maritime Navigation Light (BC040) ***(Continued)***

Point

Notes

IALA Region A:
- *Port-hand marks are red with cylindrical topmarks (if any). Lights are red and have any rhythm except Fl(2+1)R*
- *Preferred channel to port Buoy will show Maritime Navigation Navigation Lights as Fl(2+1)G, if lit*
- *Starboard-hand marks are green with conical topmarks (if any). Lights are green and have any rhythm except Fl(2+1)G*
- *Preferred channel to starboard Buoy will show Maritime Navigation Navigation Lights as Fl(2+1)R, if lit*
- *Port-hand Edge Buoys (Buoy Type:[Port-hand Edge]) may be shown Can and Pillar and will be colored red with cylinder Topmarks.*
- *Starboard-hand Edge Buoys (Buoy Type:[Starboard-hand Edge]) may be shown Conical and Pillar and will be colored green with cone Topmarks*

IALA Region B:
- *Port-hand marks are green with cylindrical topmarks (if any). Lights are green and have any rhythm except Fl(2+1)G*
- *Preferred channel to port will show Maritime Navigation Navigation Lights as Fl(2+1)R, if lit*
- *Starboard-hand marks are red with conical topmarks (if any). Lights are red and have any rhythm except Fl(2+1)R*
- *Preferred channel to starboard will show Maritime Navigation Navigation Lights as Fl(2+1)G, if lit*

Yellow, abbreviated "Y", shall be substituted for Orange "Or" or Amber "Am" when describing light color in the character of light.
If the only light color is white, omit color from light. e.g., "Fl 6s 12m 8M" is a white light.
The Period of Light label for a Maritime Navigation Light that is associated with a buoy may be omitted if that buoy is shown on a chart at a larger scale.

In areas of congested type, if it is necessary to abridge a Maritime Navigation Light legend due to clutter, the period may be omitted.

Special marks may also be used to mark Traffic Separation Schemes, or channels within channels (eg a Deep Water route within a wider navigation channel marked by standard lateral buoys) or special purpose channels (eg for small craft).

Sound Signal Device Guidance:

To avoid clutter, where a Sound Signal Device is co-located with another aid to navigation, the [Fog Signal Type] legend or the Chart #1 symbol R1 should be shown, but not both.

If a Sound Signal Device (Fog Signal Type: [No Information]) or not shown, label the sound signal type with appropriate legend and show Chart #1 (R1) symbol.
If Fog Signal Type: [Bell], use label "Bell", Chart #1 (R14)
If Fog Signal Type: [Diaphone], use legend "Dia", Chart #1 (R11)
If Fog Signal Type: [Explosive], use legend "Explos", Chart #1 (R10)
If Fog Signal Type: [Gong], use legend "Gong", Chart #1 (R16)
If Fog Signal Type: [Horn], use legend "Horn", Chart #1 (R13)
If Fog Signal Type: [Siren], use legend "Siren", Chart #1 (R12)
A Sound Signal Device with Fog Signal Type: [Other] or [No Information] the Chart #1 (R1) symbol shall be shown and can be used alone or in conjunction with other aids to navigation such as Buoys, Maritime Navigational Lights, or Maritime Radiobeacons.

Color of Navigational Aids Guidance:
The international abbreviations for colour are:
The International Abbreviations for Navigation Mark Colours for labeling color of Navigational Aids is:
Navigation Mark Colour: [White], abbreviate "W";
Navigation Mark Colour: [Red], abbreviate "R";
Navigation Mark Colour: [Green], abbreviate "G";
Navigation Mark Colour: [Blue], abbreviate "Bu";
Navigation Mark Colour: [Violet], abbreviate "Vi";
Navigation Mark Colour: [Yellow], abbreviate "Y";
Navigation Mark Colour: [Orange], abbreviate "Or"; (may be charted as "Y")
Navigation Mark Colour: [Amber], abbreviate "Am"; (may be charted as "Y")
Navigation Mark Colour: [Black], abbreviate "B";

Colour abbreviations must be in capital letters in all cases except for the second letter of two-letter abbreviations. These abbreviations must be used for the colours of lights and structures.

Where the structure and topmark are coloured differently, the colour that best assists the mariner in identifying the beacon should be charted. If it is useful to give the colour of both topmark and structure, then the topmark colour should be given first, divided from the structure colour by a slash, eg for a red lateral topmark on a white painted beacon tower: R/W.

Multicoloured structures. Where the colours are in bands, the sequence of colour abbreviations must be from top to bottom. Where the colours are in stripes (vertical or diagonal) or the sequence of horizontal bands is not known, the darker colour must be given first. Buoys and Beacons which consist of multiple colors (Bands or Stripes) should contain multiple Navigation Mark Colour attributes collected for each color and in the order top to bottom for Bands and darker color first for Stripes.
The Navigation Mark Colour label for Buoys and Maritime Navigation Beacon shall be centered immediately below the small circle marking the position of the feature.

Within the IALA Maritime Buoyage System:
• red and green are used for lateral marks,
• black and yellow bands are used for cardinal marks,
• black and red bands are used for isolated danger marks,

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Maritime Navigation Light (BC040) ***(Continued)***

Point

Notes

• red and white stripes are used for safe-water buoys,
• yellow is used for special marks,
• blue and yellow stripes are used for emergency wreck marking buoys (on trial from 2006).

Red and green on lateral marks have different meanings in IALA Regions A and B, therefore the applicable region must be stated on charts.

Where there is insufficient space on charts for abbreviations, the topmarks alone (for cardinal buoys and beacons) or the filled and open symbols (for lateral buoys) may be considered adequate to indicate colours, without abbreviations.
Abbreviations for multiple colours on buoys (and beacons) must be shown in accordance with the following conventions:

a. Where the colours are in bands the sequence of colour abbreviations must be from top to bottom, eg in the IALA System: Chart #1 (Q4)
• a north cardinal mark (black above yellow): BY
• an east cardinal mark (black with single broad horizontal yellow band): BYB
• a south cardinal mark (yellow above black): YB
• a west cardinal mark (yellow with a single broad horizontal black band): YBY
• a preferred channel lateral mark: GRG or RGR
• an isolated danger mark (black with one or more broad horizontal red bands): BRB

Note: It helps to remember that the points of the topmark cones (for cardinal marks) reflect the position of the black band(s) on the body of the buoy (or beacon), eg for a north cardinal mark, the black band is at the top.

b. Where the colours are in stripes (vertical or diagonal) or the sequence of horizontal bands is not known, the darker colour must be given first, eg in the IALA System: Chart #1 (Q5)
• a safe water buoy (red and white vertical stripes): RW
• an emergency wreck marking buoy (blue and yellow vertical stripes): BuY

Light Flare Guidance:

The point of a light flare P11 should be about 1 mm from the charted position of the light.

The orientation of flares should be such as to avoid obscuring other detail, eg symbols, soundings, text. In the case of a leading light (see B-475.6), lights in line and direction lights, the flares should be oriented to seaward along the line, provided this does not obscure the front light star. Black lines (eg transit lines, coastline, graticule) should not be broken where they cross a light flare.

Flares must not be inserted against air obstruction lights or strip lights where light stars are usually omitted.

Names of major lights are very important. If a light has a name which is unrelated to any other charted feature, the name must be inserted against the position of the light on at least the largest scale charts, above or preceding the light description, and should be in the same style as the light description. If the name of a light is obviously that of the named feature on which the light stands, eg Saint Catherine's Point, the name of the light need not be repeated above the light description. The name must be in the style appropriate to the feature, eg a headland or a shoal, and in many cases can be sited immediately above the light description. Where a light description is unavoidably sited some distance from the light star, the name of the light should be repeated above the light description, in the same style as the light description.

Minor lights may be identifiable in the List of Lights and Fog Signals by a charted general name and a (possibly uncharted) descriptive term, eg Royal Pier. Names or descriptions of individual lights of a pair of leading lights, eg 'Rear' or 'Upper', 'Front' or 'Lower', can normally be deduced from the positions shown on the chart and, to save clutter and translation, should not be inserted on paper charts.

LIGHT DESCRIPTIONS:

The principal character of a light is its rhythm (although, strictly, fixed lights and some alternating lights are not 'rhythmic'). The basic international abbreviations including Chart #1 reference are shown below for the following [Navigation Light Characteristic] attribute:

[Alternating]: Al (P10.11) (Example: AlWR)
[Fixed]: F (P10.1)
[Fixed and Flashing]: FFl (P10.10)
[Fixed Long-flashing]: F LFl (P10.1, P10.5)
[Fixed with Alternating Flashing]: F Al (P10.1, P10.11)
[Flashing]: Fl (P10.4)
[Flashing Alternating]: Fl Al (P10.4, P10.11)
[Flashing with Long-flash]: Fl LFl (P10.4, P10.5)
[Group Quick-flashing]: Q() (P10.6) (Example: Q(3))
[Group Very Quick-flashing]: VQ() (P10.7) (Example: VQ(3))
[Interrupted Quick-flashing]: IQ (P10.6)
[Interrupted Ultra Quick-flashing]: IUQ (P10.8)
[Interrupted Very Quick-flashing]: IVQ (P10.7)
[Isophase]: Iso (P10.3)
[Long-flashing]: LFl (P10.5)
[Long-flashing Alternating]: LFl Al (P10.5, P10.11)
[Morse Code]: Mo() (P10.9) (Example: Mo(K))
[Occulting]: Oc (P10.2)
[Occulting Alternating]: Oc Al (P10.2, P10.11)
[Occulting Flashing]: Oc Fl (P10.2, P10.4)
[Quick-flashing]: Q (P10.6)
[Quick-flashing with Long-flash]: Q LFl (P10.6, P10.5)

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Maritime Navigation Light (BC040) ***(Continued)***

Point

Notes

[Ultra Quick-flashing]: UQ (P10.8)
[Ultra Quick-flashing with Long-flash]: UQ LFl (P10.8, P10.5)
[Very Quick-flashing]: VQ (P10.7)
[Very Quick-flashing with Long-flash]: VQ LFl (P10.7, P10.5)

Lights exhibited from the same structure (or charted at the same light star):

a. If more than one light is exhibited from a light structure the description of the main one (eg a light visible from all directions) should be shown on one line and the subsidiary light (eg a red sector of different character, covering a danger) on a line below.
Two short descriptions may be shown on one line linked by '&'. This also applies where two separate lights which are close together are charted by one light star, because of scale.

b. Disposition of lights. Lights exhibited from the same structure which are disposed horizontally or vertically must be charted by the abbreviation '(hor)' or '(vert)' P15, as appropriate, immediately following the colour in the light description.
Two (or more) fixed lights of the same colour disposed horizontally or vertically must be charted, eg:
• 2F.G(hor) means that two fixed green lights are disposed horizontally
• 2F.R(vert) means that two fixed red lights are disposed vertically
• 3F.R(vert) means that three fixed red lights are disposed vertically
It is possible to show lights exhibited from the same structure arranged in other ways by means of a geometric symbol, eg:
• 3F.R(Δ) means 3 fixed red lights disposed in the shape of a triangle (the appropriate way up).
Two (or more) lights of different colour disposed horizontally or vertically must be charted, eg:
• F.GR(vert) means that 2 fixed lights are disposed vertically, the uppermost being green, the lower being red.
• F.RGR(hor) means that 3 fixed lights are disposed horizontally, the middle one being green.
The '&' sign is not required, as the qualifier (vert) or (hor) clearly indicates that there is more than one light. These conventions must not be used for Traffic Signals (see B-495).
c. If a fixed light is varied at intervals by a flash of greater intensity, it is charted as FFl, P10.10.
Combining the elements of a light description must be achieved in a way that enables complex descriptions to be shown compactly. However, some spacing of the elements is needed for ease of interpretation. Full stops are specified below to ensure spacing, but the full stops may be omitted providing the spacing is retained:

a. Insert full stops (or spaces):
• at the end of the characteristic rhythm (except where there is a bracket);
• at the end of all colours (not between colours);
• after Al (Alternating) - although Al is not a rhythmic characteristic it is often juxtaposed with one.

b. Omit full stops:
• after s (seconds);
• after m (elevation);
• after M (range);
• where there is a bracket;
• at the end of the light description.

c. If more than one range is given in the light description for a single light, show as follows:
eg 15/10M P14 Light with two different ranges (use forward slash).
eg 15-7M P14 Light with three or more different ranges (use hyphen).

Colours of a light must be arranged in the same order as the ranges, with the longest range normally given first However, in the case of a FFl light, where the flash is always brighter, the ranges should be shown in the same order as the character to which they refer, eg: FFl.10/15M

LIGHT DESCRIPTIONS: ABRIDGING, OMISSION:

For paper charts, the order of omission of details in an abridged (shortened) description is given below. It is not the same for all types of lights.

Major lights (ie lights intended for use at sea, usually with a range of 15 miles or more, and in outer approaches to harbours). When reducing the detail to be charted as the chart scale decreases, the following must be the order of omission:
a. Elevation of light, eg 23m
b. Period of light, eg 10s
c. Range (visibility) eg 22M
d. Character and colour.

Where useful on some smaller scale charts, a light star, major floating light symbol, or offshore platform symbol may be shown with flare and possibly name but without light description.

Lights within harbours and in restricted channels. It may be advisable to abridge light descriptions even on the largest scale charts to eliminate details of little interest to the mariner, especially where space is very limited. The order of omission must be:
a. Range
b Elevation
c. Period
d. Character and colour.

Where numerous quays, wharves, etc, along a river channel have similar lights, the light star and flare may be retained and a standard note covering them all may be used, eg:
LIGHTS: Light stars without legends represent two fixed lights displayed vertically. They are seen as red to port [or starboard] and green to starboard [or port] when proceeding upriver.

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Maritime Navigation Light (BC040) *(Continued)*

Point

Notes

Miscellaneous Light Guidance:

Miscellaneous labels occasionally may be found in association with marine navigational aids. If a text label is shown on hydrographic source material, it should be considered significant for navigation. Examples are:
A fog detection light, label "Fog Det Lt";
A floodlit structure near navigable water, label "(Illuminated)"; (Floodlit: [True])
A daytime light, if character of light in the day is different from the character shown at night. Show daytime character, followed by "Day" in parentheses, for example: (F 37m 11M Day);
Unwatched light, with no standby or emergency arrangements, label "(U)";
A temporary light or buoy, label "(temp)". If seasonal, include months, for example: "(Apr-Oct)";
A fog light, if light is only shown in fog, or the light during fog is different from the character of light shown at other times, show character in fog, followed by "(in fog)", in parentheses. For example: Fl 5s (in fog)

Feature Information					
Specification:	*Chart #1*	**Feature Origin:**	*Hydrographic*	**Symbol:**	P61.2
Symbol Criteria:	Maritime Light Type = [Air Obstruction Light]			**Label:**	"(R Lts)", 7 point Mixed case; "([Light Elevation]), 7 point lower case;
				Color:	Black (58600)

Feature Information					
Specification:	*Chart #1*	**Feature Origin:**	*Hydrographic*	**Symbol:**	P63
Symbol Criteria:	Maritime Light Type = [Flood Light]			**Label:**	"(illum)", 7 point Mixed case;
				Color:	Black (58600)

Feature Information					
Specification:	*Chart #1*	**Feature Origin:**	*Hydrographic*	**Symbol:**	P1
Symbol Criteria:	Maritime Light Type = [Bearing Light] or [Emergency Light] or [Fog Detector Light] or [Front] or [Horizontally Disposed] or [Leading Light] or [Lower] or [Rear] or [Spotlight] or [Subsidiary Light] or [Upper] or [Vertically Disposed]			**Label:**	[Geographic Name Information: Full Name], 7 point Mixed case if available; [Character of Light], 7 point Mixed case; [Period of Light]"s", 7 point lower case; [Light Elevation]"m", 7 point lower case; [Nominal Range]"M", 7 point upper case; [Multiple Light Ranges]"M", 7 point upper case (if available); "Fog Det Lt", 7 point Mixed case if (Maritime Light Type: [Fog Detector Light] and Fog Detector Light Present: [True]); "(hor)", 7 point lower case if Maritime Light Type: [Horizontally Disposed]; "(vert)", 7 point lower case if Maritime Light Type: [Vertically Disposed]; "Front Lt", 7 point Mixed case if Maritime Light Type: [Front] or [Lower]; "Rear Lt", 7 point Mixed case if Maritime Light Type: [Rear] or [Upper];
				Color:	Black (58600)

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Maritime Navigation Light (BC040) ***(Continued)***

Point

Feature Information					
Specification:	*Chart #1*	**Feature Origin:**	*Hydrographic*	**Symbol:**	P41.1
Symbol Criteria:	Maritime Light Type = [Sectored Light]			**Label:**	[Geographic Name Information: Full Name], 7 point Mixed case if available; [Character of Light], 7 point Mixed case; [Period of Light]"s", 7 point lower case; [Light Elevation]"m", 7 point lower case; [Nominal Range]"M", 7 point upper case; [Multiple Light Ranges]"M", 7 point upper case (if available)
				Color:	Black (58600)

Feature Information					
Specification:	*Chart #1*	**Feature Origin:**	*Hydrographic*	**Symbol:**	P31
Symbol Criteria:	Maritime Light Type = [Directional Function]			**Label:**	[Geographic Name Information: Full Name], 7 point Mixed case if available; "Dir", 7 point Mixed case; Character of Light: 7 point Upper/lower case, Light Range Nominal, Multiple Light Ranges: 7 point upper case; Name: 7 point Upper/lower case; Label: "Dir"7 point Upper/lower case; Other: 7 point Upper/lower case; Period of Light: 7 point lower case; Elevation of Light: 7 point lower case;
				Color:	Black (58600)

Feature Information					
Specification:	*Chart #1*	**Feature Origin:**	*Hydrographic*	**Symbol:**	P50
Symbol Criteria:	Maritime Light Type = [Moiré Effect Light]			**Label:**	[Geographic Name Information: Full Name], 7 point Mixed case if available; "Dir", 7 point Mixed case
				Color:	Black (58600)

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Maritime Navigation Light Structure (BC041)

Point

Feature Description:	*An otherwise unspecified fixed structure which supports a Maritime Navigation Light.*
Inclusion Condition:	*All Required*

Notes

Lights and additional co-located Maritime Features:

While Maritime Navigation Light may be a single feature, these lights are often co-located as a component with other Maritime features at the same coordinate such as Maritime Navigation Beacon (BC010), Buoy (BC020) Light Sector (BC060), Light Vessel (BC070), Lighthouse (BC050), and Maritime Navigation Light Support.

Feature Information

Specification:	*Chart #1*	**Feature Origin:**	*Hydrographic*	**Symbol:**	P1
Symbol Criteria:	All			**Label:**	None
				Color:	N/A

Maritime Navigation Marker (BC055)

Point

Feature Description:	*A coloured (usually white) mark on a vertical surface (for example: a cliff, a rocky outcrop, and/or a wall) that is a conspicuous landmark for maritime navigation.*
Inclusion Condition:	*All Required*

Notes

Priority for type placement: 1-right center, 2-bottom center, 3-left center, 4-top center.
(A) Minimum distance from symbol - 1 mm.
(B) Maximum distance from symbol before choosing the next highest priority:
(#1) 4 mm measured to the West end;
(#2) 4 mm measured to the North side (top);
(#3) 4 mm measured to the East end;
(#4) 4 mm measured to the South side (bottom)

Feature Information

Specification:	*Chart #1*	**Feature Origin:**	*Hydrographic*	**Symbol:**	Q101
Symbol Criteria:	All			**Label:**	"Mk", 6 point Mixed case
				Color:	Black (58600)

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Maritime Notice Board (BC120)

Point

Feature Description:	*A signboard used to indicate speed restrictions, cable landings, and other items of interest to maritime navigation.*
Inclusion Condition:	*All Required*

Notes

None

Feature Information					
Specification:	*Chart #1*	**Feature Origin:**	*Hydrographic*	**Symbol:**	Q126
Symbol Criteria:	All			**Label:**	None
				Color:	N/A

Maritime Radar Reference Line (FC130)

Curve

Feature Description:	*A track along which ships may be guided by coastal radar stations, especially in the event of reduced visibility.*
Inclusion Condition:	*All Required*

Notes

Label "Ra" is repeated at 150 mm interval along top of line.

Feature Information					
Specification:	*Chart #1*	**Feature Origin:**	*Hydrographic*	**Symbol:**	M32.1
Symbol Criteria:	All			**Label:**	"Ra", 6 point Mixed case, Italic; [Bearing and Reciprocal Category], 6 point lower case, Italic
				Color:	Green (52813)

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Maritime Radiobeacon (BC034)

Point

Feature Description:	*An electronic aid to maritime navigation consisting of a radio transmitter that broadcasts distinctive signals.*
Inclusion Condition:	*All Required*

Notes

Radiobeacons and additional co-located Maritime Features:
While Maritime Radiobeacon are often co-located as a component with other Maritime features at the same coordinate such as Maritime Navigation Beacon (BC010), Buoy (BC020) Light Sector (BC060), Light Vessel (BC070), Lighthouse (BC050), and Maritime Navigation Light Support.

Displacement Rules for NAVAIDS overprinting hydrographic dangers or depth curves:
When aids to navigation overprint hydrographic dangers, or depth curves, the following displacement criteria applies:

(a.) If the aid to navigation is a fixed aid (Maritime Navigation Beacon and Maritime Navigation Light), and the hydrographic danger symbol is a point symbol, delete the hydrographic danger point symbol. Show both if the hydrographic danger symbol is an area symbol.
(b.) If the aid to navigation is a beacon, the hydrographic danger symbol's danger curve, i.e., the dotted perimeter line, if present, is broken for the beacon symbol (excluding type).
(c.) If a point danger symbol and a beacon are overprinting, refer to Cardinal Displacement Rules for hazards and displacement shall be as little as possible to resolve the overprint.
(d.) The above criteria refer to the graphic portion of symbols only. All type and labels are movable around the graphic elements to avoid overprints. If the hydrographic danger symbol's hydrographic depth type must be moved, it shall be placed outside the danger circle, and enclosed in parentheses.
(e.) Depth curves are broken for aids to navigation and dangers, and the type associated with these symbols.

Miscellaneous Rules for Maritime Radiobeacons:
If an electronic beacon is within 2 mm, at chart scale, of a light, the electronic beacon shall be shown and the light shall be changed to include the electronic beacon as a part of the light. The radio aids of the electronic beacon will be added to the radio aids of the light. The electronic beacon and the light are not combined if the light is a moire effect light or a strip light. In this case, show both the light and the electronic beacon as separate symbols.

Feature Information

Specification:	*Chart #1*	**Feature Origin:**	*Hydrographic*	**Symbol:**	S17.1
Symbol Criteria:	Maritime Radiobeacon Type = [Automated Identification System (AIS)]			**Label:**	"AIS", 7 point Upper Case
				Color:	Purple (96532)

Feature Information

Specification:	*Chart #1*	**Feature Origin:**	*Hydrographic*	**Symbol:**	S11
Symbol Criteria:	Maritime Radiobeacon Type = [Directional Radiobeacon]			**Label:**	"RD", 7 point Upper Case
				Color:	Purple (96532)

Feature Information

Specification:	*Chart #1*	**Feature Origin:**	*Hydrographic*	**Symbol:**	S3.3
Symbol Criteria:	Maritime Radiobeacon Type = [Radar Responder Beacon (RACON)]			**Label:**	"Racon", 7 point Mixed Case; "([Characters Emitted])", 7 point Upper Case
				Color:	Purple (96532)

Feature Information

Specification:	*Chart #1*	**Feature Origin:**	*Hydrographic*	**Symbol:**	S12
Symbol Criteria:	Maritime Radiobeacon Type = [Rotating Pattern Radiobeacon]			**Label:**	"RW", 7 point Upper Case
				Color:	Purple (96532)

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Maritime Radiobeacon (BC034) *(Continued)*

Point

Feature Information					
Specification:	*Chart #1*	**Feature Origin:**	*Hydrographic*	**Symbol:**	S15
Symbol Criteria:	Maritime Radiobeacon Type = [QTG Station]			**Label:**	"R", 7 point Upper Case
				Color:	Purple (96532)

Feature Information					
Specification:	*Chart #1*	**Feature Origin:**	*Hydrographic*	**Symbol:**	S2
Symbol Criteria:	Maritime Radiobeacon Type = [Radar Marker (RAMARK)]			**Label:**	"Ramark", 7 point Mixed Case
				Color:	Purple (96532)

Feature Information					
Specification:	*Chart #1*	**Feature Origin:**	*Hydrographic*	**Symbol:**	S10
Symbol Criteria:	Maritime Radiobeacon Type = [Circular Radiobeacon]			**Label:**	[Geographic Name Information: Full Name], 7 point Mixed Case if available; "RC", 7 point Upper Case
				Color:	Purple (96532)

Feature Information					
Specification:	*Chart #1*	**Feature Origin:**	*Hydrographic*	**Symbol:**	S14
Symbol Criteria:	Maritime Radiobeacon Type = [Radio Direction Finding Station]			**Label:**	"RG", 7 point Upper Case
				Color:	Purple (96532)

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Maritime Route (FC165)

Surface

Feature Description:	*A track or lane established for the safe passage of vessels.*
Inclusion Condition:	*Width >= 50 meters*

Notes

Break line symbol in water area for overprinting point symbol with the same color. Leave space 0.5 mm on each side of the point symbol. Do not displace either the line symbol or the point symbol. Point symbols may overprint line symbols of a different color.

Show Hydrographic Depth or Hydrographic Height values in whole meters if decimeter value is 0; otherwise show the decimeter value as a subscript to the meter value.

If feature is linear, the label hierarchy is:
(1) Label shall be placed 1 mm above feature, centered.
(2) Top of label shall be placed 1 mm below feature, centered.
(3) Label may be displaced along the feature, use two labels if feature length 150 mm or greater.
(4) Do not label across shoreline (coastal or inland)

The (Bearing and Reciprocal) and (Bearing from Seaward) shall be printed above track, 1 mm away from and parallel to it. Type shall be read from left to right, or bottom to top if track is vertical. One attribute shall be shown for each straight line segment, centered on that segment, but can be moved sideways to avoid conflicting with arrows or other chart detail. (Bearing from Seaward) is used on one-way tracks to indicate the bearing steered by a ship following the track in the direction indicated. (Bearing and Reciprocal) is used on two-way tracks with the bearing from seaward, i.e., when preceding from seaward toward land, or in the direction of buoyage, followed by its reciprocal bearing, except as follows: When a two way route is of such length that reciprocal bearings are shown at both extremities, i.e., a straight line segment over 25 cm long, the bearing quoted first shall be the bearing followed by a ship joining the track at that extremity.

Descriptive terms, e.g., "Canal" shall be shown if the name is not known. If the descriptive word appears in the name, for example, "PANAMA CANAL". The descriptive type shall not be shown, i.e. do not show "Panama Canal Canal".

To reduce translation difficulties, a "not equal to" sign, i.e., a slash (/) overprinting an equal sign (=), or Posicut #224, shall be used in the (Description of Reference Points) to indicate any two objects in line. The (Description of Reference Point) should use this symbol in place of the words "in line", e.g., "2 Bns ≠" rather than "2 Bns in line".

If two line features of the same feature type meet end to end, and have different depths, a short line is shown centered on the point of intersection. It bisects the angle at which the line features meet (i.e., if the lines meet at 180° angle the bisecting line is perpendicular to the meeting line features). The bisecting line is 0.1 mm lineweight, length is 3.0 mm, and it is shown in the same color as the line features.

Vertical Datum category is the sounding datum to which a hydrographic depth, drying height, or the high water datum to which a Safe Overhead Clearance is referenced. If the hydrographic depth, drying height, or Safe Overhead Clearance is unknown or not applicable, the vertical datum category is not applicable.

Tracks of value only to local fishing boats or pleasure craft should not be included on nautical charts except in the Bahama Islands, Nova Scotia-Newfoundland, and Bermuda areas. This exception applies to all charts of 1:300,000 scale and larger.

Two way tracks that are not marked by fixed objects are represented by arrows pointing in opposite directions. Each pair of arrows is separated by two dashes. No arrows are shown on two way tracks that are marked by fixed objects, except for those showing depths.
If depth is known, the (Hydrographic Depth) is placed between the arrowheads and the track line is deleted between the arrow points, to avoid overprinting (Hydrographic Depth).
One way tracks are represented by a single arrow pointing in the direction of traffic flow. If depth is known, the (Hydrographic Depth) is placed 2 mm behind the point of the arrow and the track line is deleted from the arrow to 1 mm past the type.
Deep water tracks shall have a "DW" inserted before the arrow, approximately 25 mm in front of the arrowhead.
Representation of arrows on tracks:
(1.) Each segment of tracks that has a different depth shall have one arrows/(Hydrographic Depth) set as described above centered approximately in the center of the segment. Type /arrows shall be moved sideways along track to avoid being placed around sharp corners (interior angle < 135°). Additional arrows without type shall be spaced along tracks at 100 mm interval, or once for each straight line segment over 15 mm long, whichever is less.
(2.) Tracks without depths shall show arrows, or pairs of arrows, on two way tracks, once every 100 mm along a track, or once for each straight line segment over 15 mm long, whichever is less.

Feature Information

Specification:	*Chart #1*	**Feature Origin:**	*Hydrographic*	**Symbol:**	M27.2
Symbol Criteria:	Route Intended Use = [Deep Water Route]			**Label:**	"DW", 8 point Upper case, Italic; [Hydrographic Depth]"m", 8 point lower case, Italic;
				Color:	Green (52813)

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Maritime Route (FC165) *(Continued)*

Surface

Feature Information					
Specification:	*Chart #1*	**Feature Origin:**	*Hydrographic*	**Symbol:**	M28.2
Symbol Criteria:	(Route Intended Use = [Two-way Route])			**Label:**	None
				Color:	N/A

Maritime Route (FC165)

Curve

Inclusion Condition:	*Width < 50 meters*

Notes

Break line symbol in water area for overprinting point symbol with the same color. Leave space 0.5 mm on each side of the point symbol. Do not displace either the line symbol or the point symbol. Point symbols may overprint line symbols of a different color.

Show Hydrographic Depth or Hydrographic Height values in whole meters if decimeter value is 0; otherwise show the decimeter value as a subscript to the meter value.

If feature is linear, the label hierarchy is:
(1) Label shall be placed 1 mm above feature, centered.
(2) Top of label shall be placed 1 mm below feature, centered.
(3) Label may be displaced along the feature, use two labels if feature length 150 mm or greater.
(4) Do not label across shoreline (coastal or inland)

The (Bearing and Reciprocal) and (Bearing from Seaward) shall be printed above track, 1 mm away from and parallel to it. Type shall be read from left to right, or bottom to top if track is vertical. One attribute shall be shown for each straight line segment, centered on that segment, but can be moved sideways to avoid conflicting with arrows or other chart detail. (Bearing from Seaward) is used on one-way tracks to indicate the bearing steered by a ship following the track in the direction indicated. (Bearing and Reciprocal) is used on two-way tracks with the bearing from seaward, i.e., when preceding from seaward toward land, or in the direction of buoyage, followed by its reciprocal bearing, except as follows: When a two way route is of such length that reciprocal bearings are shown at both extremities, i.e., a straight line segment over 25 cm long, the bearing quoted first shall be the bearing followed by a ship joining the track at that extremity.

Descriptive terms, e.g., "Canal" shall be shown if the name is not known. If the descriptive word appears in the name, for example, "PANAMA CANAL". The descriptive type shall not be shown, i.e. do not show "Panama Canal Canal".

To reduce translation difficulties, a "not equal to" sign, i.e., a slash (/) overprinting an equal sign (=), or Posicut #224, shall be used in the (Description of Reference Points) to indicate any two objects in line. The (Description of Reference Point) should use this symbol in place of the words "in line", e.g., "2 Bns ≠" rather than "2 Bns in line".

If two line features of the same feature type meet end to end, and have different depths, a short line is shown centered on the point of intersection. It bisects the angle at which the line features meet (i.e., if the lines meet at 180° angle the bisecting line is perpendicular to the meeting line features). The bisecting line is 0.1 mm lineweight, length is 3.0 mm, and it is shown in the same color as the line features.

Vertical Datum category is the sounding datum to which a hydrographic depth, drying height, or the high water datum to which a Safe Overhead Clearance is referenced. If the hydrographic depth, drying height, or Safe Overhead Clearance is unknown or not applicable, the vertical datum category is not applicable.

Tracks of value only to local fishing boats or pleasure craft should not be included on nautical charts except in the Bahama Islands, Nova Scotia-Newfoundland, and Bermuda areas. This exception applies to all charts of 1:300,000 scale and larger.

Two way tracks that are not marked by fixed objects are represented by arrows pointing in opposite directions. Each pair of arrows is separated by two dashes. No arrows are shown on two way tracks that are marked by fixed objects, except for those showing depths.
If depth is known, the (Hydrographic Depth) is placed between the arrowheads and the track line is deleted between the arrow points, to avoid overprinting (Hydrographic Depth).
One way tracks are represented by a single arrow pointing in the direction of traffic flow. If depth is known, the (Hydrographic Depth) is placed 2 mm behind the point of the arrow and the track line is deleted from the arrow to 1 mm past the type.
Deep water tracks shall have a "DW" inserted before the arrow, approximately 25 mm in front of the arrowhead.
Representation of arrows on tracks:
(1.) Each segment of tracks that has a different depth shall have one arrows/(Hydrographic Depth) set as described above centered approximately in the center of the segment. Type /arrows shall be moved sideways along track to avoid being placed around sharp corners (interior angle < 135°). Additional arrows without type shall be spaced along tracks at 100 mm interval, or once for each straight line segment over 15 mm long, whichever is less.
(2.) Tracks without depths shall show arrows, or pairs of arrows, on two way tracks, once every 100 mm along a track, or once for each straight line segment over 15 mm long, whichever is less.

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Maritime Route (FC165) ***(Continued)***

Curve

Feature Information					
Specification:	*Chart #1*	**Feature Origin:**	*Hydrographic*	**Symbol:**	M27.3
Symbol Criteria:	(Route Intended Use = [Recommended Track for Deep Draft Vessels]) and (Bidirectional = [True])			**Label:**	"DW", 8 point Upper , Italic
				Color:	Green (52813)

Feature Information					
Specification:	*Chart #1*	**Feature Origin:**	*Hydrographic*	**Symbol:**	M28.1
Symbol Criteria:	(Route Intended Use = [Recommended Route] or [Transit Route])			**Label:**	None
				Color:	N/A

Feature Information					
Specification:	*Chart #1*	**Feature Origin:**	*Hydrographic*	**Symbol:**	M4
Symbol Criteria:	(Route Intended Use = [Recommended Track for Other Than Deep Draft Vessels]) and (Bidirectional = [True]) and (Maritime Navigation Marked = [False])			**Label:**	[Bearing and Reciprical Category], 6 point lower case; [Hydrographic Depth]"m" if available, 6 point lower case, Italic
				Color:	Green (52813)

Feature Information					
Specification:	*Chart #1*	**Feature Origin:**	*Hydrographic*	**Symbol:**	M3
Symbol Criteria:	(Route Intended Use = [Recommended Track for Other Than Deep Draft Vessels]) and (Bidirectional = [True]) and (Maritime Navigation Marked = [True])			**Label:**	[Bearing and Reciprical Category], 6 point lower case; [Hydrographic Depth]"m" if available, 6 point lower case
				Color:	Green (52813)

Feature Information					
Specification:	*Chart #1*	**Feature Origin:**	*Hydrographic*	**Symbol:**	M5.1
Symbol Criteria:	(Route Intended Use = [Recommended Track for Other Than Deep Draft Vessels]) and (Bidirectional = [False]) and (Maritime Navigation Marked = [True])			**Label:**	None
				Color:	N/A

Feature Information					
Specification:	*Chart #1*	**Feature Origin:**	*Hydrographic*	**Symbol:**	M5.1
Symbol Criteria:	(Route Intended Use = [Recommended Track for Deep Draft Vessels]) and (Bidirectional = [False]) and (Maritime Navigation Marked = [True])			**Label:**	"DW", 8 point Upper case Italic; "(see Note)", 8 point Mixed case Italic
				Color:	Green (52813)

Feature Information					
Specification:	*Chart #1*	**Feature Origin:**	*Hydrographic*	**Symbol:**	TBD
Symbol Criteria:	(Route Intended Use = [Q Route])			**Label:**	None
				Color:	N/A

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Maritime Route (FC165) *(Continued)*

Curve

Feature Information					
Specification:	*Chart #1*	**Feature Origin:**	*Hydrographic*	**Symbol:**	M27.3
Symbol Criteria:	(Route Intended Use = [Recommended Track for Deep Draft Vessels]) and (Bidirectional = [False])			**Label:**	"DW", 8 point Upper , Italic
				Color:	Green (52813)

Maritime Signal Station (BB155)

Point

Feature Description:	*A place on shore from which signals are made to vessels at sea.*
Inclusion Condition:	*All Required*

Notes

The feature shall be oriented perpendicular (90 degrees) to the nearby road (AP030), cart track (AP010), trail (AP050), or railroad (AN010).

If feature coalesce at map scale, when shown in their true positions, they shall be displaced 0.2 mm from one another.

When symbolized feature is < 0.2 mm from a line feature, displace to 0.2 mm (map scale).

Like point features which coalesce in clusters of 3 or more will be thinned to form a representative pattern.

Type for features on land shall be positioned on land if possible. If type must be placed in the water, it shall be positioned so it does not obscure or overprint hydrographic detail. Type for features in the water shall be positioned in the water. If possible, type shall not be positioned across the shoreline.

Feature Information					
Specification:	*Chart #1*	**Feature Origin:**	*Hydrographic*	**Symbol:**	T22
Symbol Criteria:	Maritime Station Type = [Traffic Signal Station]			**Label:**	"SS (Traffic)", 6 point Mixed case
				Color:	Black (58600)

Feature Information					
Specification:	*Chart #1*	**Feature Origin:**	*Hydrographic*	**Symbol:**	T34
Symbol Criteria:	Maritime Station Type = [Tidal Stream Station]			**Label:**	"SS (Stream)", 6 point Mixed case
				Color:	Black (58600)

Feature Information					
Specification:	*Chart #1*	**Feature Origin:**	*Hydrographic*	**Symbol:**	M30
Symbol Criteria:	Maritime Station Type = [Radar Surveillance Station]			**Label:**	"Radar Surveillance Station", 6 point Mixed case
				Color:	Black (58600)

Feature Information					
Specification:	*Chart #1*	**Feature Origin:**	*Hydrographic*	**Symbol:**	Tb
Symbol Criteria:	Maritime Station Type = [Signal Mast]			**Label:**	None
				Color:	N/A

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Maritime Signal Station (BB155) *(Continued)*

Point

Feature Information					
Specification:	*Chart #1*	**Feature Origin:**	*Hydrographic*	**Symbol:**	T31
Symbol Criteria:	Maritime Station Type = [Time Signal Station]			**Label:**	"SS (Time)", 6 point Mixed case
				Color:	Black (58600)

Feature Information					
Specification:	*Chart #1*	**Feature Origin:**	*Hydrographic*	**Symbol:**	T2
Symbol Criteria:	Maritime Station Type = [Pilot Lookout Station]			**Label:**	"Pilot Lookout", 6 point Mixed case
				Color:	Black (58600)

Feature Information					
Specification:	*Chart #1*	**Feature Origin:**	*Hydrographic*	**Symbol:**	T20
Symbol Criteria:	Maritime Station Type = [Signal Station] or [Fog Signal] or [Fireboat Station] or [Time Ball Station] or [Unmanned Oceanographic Station] or [Warning Signal Station] or [Water-police Station]			**Label:**	"SS", 6 point Upper case; "SS (Fog), 6 point Mixed case if Maritime Station Type = [Fog Signal]; "Fireboat Station", 6 point Mixed case if Maritime Station Type = [Fireboat Station]; "SS (Time Ball), 6 point Mixed case if Maritime Station Type = [Time Ball Station]; "Unmanned Oceanographic Station", 6 point Mixed case if Maritime Station Type = [Unmanned Oceanographic Station]; "SS (Warning), 6 point Mixed case if Maritime Station Type = [Warning Signal Station]; "Water-police Station", 6 point Mixed case if Maritime Station Type = [Water-police Station]
				Color:	Black (58600)

Feature Information					
Specification:	*Chart #1*	**Feature Origin:**	*Hydrographic*	**Symbol:**	T10
Symbol Criteria:	Maritime Station Type = [Coast Guard Station]			**Label:**	"CG", 6 point Upper case
				Color:	Black (58600)

Feature Information					
Specification:	*Chart #1*	**Feature Origin:**	*Hydrographic*	**Symbol:**	T30
Symbol Criteria:	Maritime Station Type = [Ice Signal Station]			**Label:**	"SS (Ice)", 6 point Mixed case
				Color:	Black (58600)

Feature Information					
Specification:	*Chart #1*	**Feature Origin:**	*Hydrographic*	**Symbol:**	T12
Symbol Criteria:	Maritime Station Type = [Rescue Station]			**Label:**	None
				Color:	N/A

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Maritime Signal Station (BB155) *(Continued)*

Point

Feature Information					
Specification:	*Chart #1*	**Feature Origin:**	*Hydrographic*	**Symbol:**	T23
Symbol Criteria:	Maritime Station Type = [Port Control Station]			**Label:**	"SS (Port Control)", 6 point Mixed case
				Color:	Black (58600)

Feature Information					
Specification:	*Chart #1*	**Feature Origin:**	*Hydrographic*	**Symbol:**	T3
Symbol Criteria:	Maritime Station Type = [Pilot Station]			**Label:**	"Pilots", 6 point Mixed case
				Color:	Black (58600)

Feature Information					
Specification:	*Chart #1*	**Feature Origin:**	*Hydrographic*	**Symbol:**	T28
Symbol Criteria:	Maritime Station Type = [Storm Signal Station]			**Label:**	"SS (Storm)", 6 point Mixed case
				Color:	Black (58600)

Feature Information					
Specification:	*Chart #1*	**Feature Origin:**	*Hydrographic*	**Symbol:**	T33
Symbol Criteria:	Maritime Station Type = [Tide Station]			**Label:**	"SS (Tide)", 6 point Upper/lower case
				Color:	Black (58600)

Feature Information					
Specification:	*Chart #1*	**Feature Origin:**	*Hydrographic*	**Symbol:**	T29
Symbol Criteria:	Maritime Station Type = [Weather Signal Station]			**Label:**	"SS (Weather)", 6 point Mixed case
				Color:	Black (58600)

Feature Information					
Specification:	*Chart #1*	**Feature Origin:**	*Hydrographic*	**Symbol:**	T25.1
Symbol Criteria:	Maritime Station Type = [Bridge Signal Station]			**Label:**	"SS (Bridge)", 6 point Mixed case
				Color:	Black (58600)

Feature Information					
Specification:	*Chart #1*	**Feature Origin:**	*Hydrographic*	**Symbol:**	T24
Symbol Criteria:	Maritime Station Type = [Lock Signal Station]			**Label:**	"SS (Lock)", 6 point Mixed case
				Color:	Black (58600)

Feature Information					
Specification:	*Chart #1*	**Feature Origin:**	*Hydrographic*	**Symbol:**	T21
Symbol Criteria:	Maritime Station Type = [International Port Signals Station]			**Label:**	"SS (INT)", 6 point Upper case
				Color:	Black (58600)

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Maritime Signal Station (BB155) *(Continued)*

Point

Feature Information					
Specification:	*Chart #1*	**Feature Origin:**	*Hydrographic*	**Symbol:**	T36
Symbol Criteria:	Maritime Station Type = [Military Practice Signal Station]			**Label:**	"SS (Firing)", 6 point Mixed case
				Color:	Black (58600)

Marsh (ED010)

Surface

Feature Description:	*A soft, poorly drained wetland that is characterized by the growth of only non-woody plants (for example: grasses) and often forms a transition region between a waterbody and land.*
Inclusion Condition:	*Area >= 15,625 sq m.*

Notes

Coincident similar area features will be blended to form a single feature.

Area and line features will be generalized to detail compatible with scale.

Feature will be generalized to provide a more aesthetic contoured feature (i.e., smoothed).

Symbols and associated area patterns of underpassing features (except drainage shorelines) are broken for all bridges, except footbridges. This rule does not apply to land tint on Littoral Planning Charts.

If a clearing exists inside of the area feature, and the size of the clearing is equal to or greater than the area for the surrounding area feature, the clearing is shown as an open space inside the surround feature. If the area of the clearing is less than the area for the surrounding feature, the clearing is deleted and absorbed into the surrounding area feature.

If two area features with the same feature code do not connect at any point, and have a space between them less than 2.5 mm at map/chart scale, delete the open space that is less than 2.5 mm between the features and combine them into one area feature.

If a portion of an area feature has a minimum width of less than 2.5 mm at map/chart scale, delete that portion of the area feature that is not at least 2.5 mm wide, measured from perimeter to perimeter. If the deletion of the portion of the area based on the above criteria will reduce the area of the remaining portion of the area feature to below the minimum area inclusion, do not delete the narrow portion of the feature that is less than 2.5 mm wide. If the deletion of the portion of the area based on the above criteria will split two larger areas connected by a narrow strip into two separate areas, either of which would be below minimum area inclusion, do not delete the narrow portion of the feature that is less than 2.5 mm wide.

Feature Information					
Specification:	*LTDS*	**Feature Origin:**	*Topographic*	**Symbol:**	AR_Cyan_Marsh
Symbol Criteria:	All			**Label:**	[Geographic Name Information : Full Name], 6-20 point Condensed Mixed case if available
				Color:	Blue (46351)

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Measured Distance Line (FC100)

Surface

Feature Description:	*A maritime course, the length of which has been accurately measured, whose endpoints are often established by leading lines.*
Inclusion Condition:	*All required*

Notes

Break line symbol in water area for overprinting point symbol with the same color. Leave space 0.5 mm on each side of the point symbol. Do not displace either the line symbol or the point symbol. Point symbols may overprint line symbols of a different color.

Labels are positioned 1 mm away from and parallel to the line. "Measured Distance" is placed above the line, or to the left, if the line is vertical. (Length of range), and (Bearing and Reciprocal) labels are placed below the line, or to the right, if the line is vertical. Type reads left to right, or bottom to top if line is vertical. Labels shall be centered in the middle of the line, if possible, but may be moved sideways to avoid overprints. "Measured Distance and (Length of range), and (Bearing and Reciprocal) type should be kept together if possible. If the space is limited, thin type in the following priority:
(1.) Delete "Measured Distance" first.

Feature Information

Specification:	*Chart #1*	**Feature Origin:**	*Hydrographic*	**Symbol:**	Q122
Symbol Criteria:	All			**Label:**	"Measured Distance", 6 point Mixed case; [Length of Range] "m", 6 point lower case; [Bearing and Reciprocal Category], 6 point Lower case;
				Color:	Black (58600)

Memorial Monument (AL130)

Surface

Feature Description:	*A marker erected and/or maintained as a memorial to a person and/or event.*
Inclusion Condition:	*Area >= 625 sq m.*

Notes

None

Feature Information

Specification:	*LTDS*	**Feature Origin:**	*Topographic*	**Symbol:**	AR_Black_MemorialMonument
Symbol Criteria:	(Height Above Surface Level < 46) OR (Area >= 5625 and Height Above Surface Level >= 46)			**Label:**	[Geographic Name Information: Full Name], 6 point Condensed Mixed case if available OR "Monument", 6 point Condensed Mixed case
				Color:	Black (58600)

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Memorial Monument (AL130)

Point

Inclusion Condition:	*Area < 625 sq m.*

Notes

Label feature as per hierarchy for topo type placement to south neatline corners reading left to right:
1. Position hierarchy:
(a.) northeast (preferred position
(b.) southeast (1st alternative)
(c.) northwest (2nd alternative
(d.) southwest (3rd alternative)
(e.) top-centered (4th alternative)
(f.) bottom-centered (5th alternative)
(Hierarchy is based on type positioning so as to avoid overprinting other type or obscuring detail.)
2. Minimum space between type placement and feature symbol is 0.5 mm
3. This method of type placement shall be used for areal features when space does not permit labeling within that feature.

Type for features on land shall be positioned on land if possible. If type must be placed in the water, it shall be positioned so it does not obscure or overprint hydrographic detail. Type for features in the water shall be positioned in the water. If possible, type shall not be positioned across the shoreline.

If the existence is Doubtful or Reported, label as such.

When obstructions coalesce at map scale, label with highest obstruction information.

If >= 3 equal symbols would coalesce at map scale, portray with a representative pattern.

Feature Information

Specification:	*LTDS*	**Feature Origin:**	*Topographic*	**Symbol:**	PT_Black_Monument
Symbol Criteria:	(Navigation Landmark = [True]) and (Height Above Surface Level < 46)			**Label:**	[Geographic Name Information: Full Name], 6 point Condensed Mixed case if available
				Color:	Black (58600)

Feature Information

Specification:	*LTDS*	**Feature Origin:**	*Topographic*	**Symbol:**	PT_Blue072_VO
Symbol Criteria:	Height Above Surface Level >= 46			**Label:**	"Monument", 6 point Condensed Mixed case; [Highest Elevation], 6 point Condensed Bold Mixed case; "([Height Above Surface Level])", 6 point Condensed Mixed case
				Color:	Blue (46351)

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Military Installation (SU001)

Surface

Feature Description:	*An installation designed for military use.*
Inclusion Condition:	*Area >= 15,625 sq m.*

Notes

Coincident similar area features will be blended to form a single feature.

Area and line features will be generalized to detail compatible with scale.

Type for features on land shall be positioned on land if possible. If type must be placed in the water, it shall be positioned so it does not obscure or overprint hydrographic detail. Type for features in the water shall be positioned in the water. If possible, type shall not be positioned across the shoreline.

Descriptive terms, e.g., "Canal" shall be shown if the name is not known. If the descriptive word appears in the name, for example, "PANAMA CANAL". The descriptive type shall not be shown, i.e. do not show "Panama Canal Canal".

Feature Information

Specification:	*LTDS*	**Feature Origin:**	*Topographic*	**Symbol:**	AR_Black_MilitaryInstallation
Symbol Criteria:	All			**Label:**	[Geographic Name Information : Full Name], 6 point Condensed Mixed case if available; or "Military", 6 point Condensed Mixed case; "(destroyed)" , 6 point Condensed Lower case if Condition of Facility = (Destroyed) or (Ruined) and (Geographic Name Information : Full Name <> ("UNK") and (is not null); "Military (destroyed)", 6 point Condensed Mixed case if Condition of Facility = (Destroyed) or (Ruined) and (Geographic Name Information : Full Name = ("UNK") and (is null)
				Color:	Black (58600)

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Military Installation (SU001)

Point

Inclusion Condition:	*Area < 15,625 sq m.*

Notes

The feature (when height <= 46 meters or when height is not known) shall be oriented perpendicular (90 degrees) to the nearby road (AP030), cart track (AP010), trail (AP050), or railroad (AN010).

If one symbol coalesces with another symbol for the same type feature, displace symbols to allow a minimum separation of 0.21mm.

Like point features which coalesce in clusters of 3 or more will be thinned to form a representative pattern.

Type for features on land shall be positioned on land if possible. If type must be placed in the water, it shall be positioned so it does not obscure or overprint hydrographic detail. Type for features in the water shall be positioned in the water. If possible, type shall not be positioned across the shoreline.

Descriptive terms, e.g., "Canal" shall be shown if the name is not known. If the descriptive word appears in the name, for example, "PANAMA CANAL". The descriptive type shall not be shown, i.e. do not show "Panama Canal Canal".

Feature Information

Specification:	*LTDS*	Feature Origin:	*Topographic*	Symbol:	PT_Black_MilitaryInstallation
Symbol Criteria:	All			Label:	[Geographic Name Information : Full Name], 6 point Condensed Mixed case if available; or "Military", 6 point Condensed Mixed case; "(destroyed)" , 6 point Condensed Lower case if Condition of Facility = (Destroyed) or (Ruined) and (Geographic Name Information : Full Name <> ("UNK") and (is not null); "Military (destroyed)", 6 point Condensed Mixed case if Condition of Facility = (Destroyed) or (Ruined) and (Geographic Name Information : Full Name = ("UNK") and (is null)
				Color:	Black (58600)

Mine Shaft Superstructure (AA020)

Point

Feature Description:	*A structure over a vertical mine shaft containing machinery (for example: a winding head or hoist) used to transport minerals, equipment, and/or workers between the surface and various levels within the mine.*
Inclusion Condition:	*Height Above Surface Level: >= 46 meters*

Notes

None

Feature Information

Specification:	*LTDS*	Feature Origin:	*Topographic*	Symbol:	PT_Blue072_VO
Symbol Criteria:	Height Above Surface Level >= 46 meters			Label:	"Mine superstructure", 6 point Condensed Mixed case; [Highest Elevation], 6 point Condensed lower case; "([Height Above Surface Level])": 6 point Condensed lower case
				Color:	Blue (46351)

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Minefield (AL065)

Surface

Feature Description:	*A site laid with explosive mines.*
Inclusion Condition:	*Area >= 15,625 sq m.*

Notes

None

Feature Information					
Specification:	*LTDS*	**Feature Origin:**	*Topographic*	**Symbol:**	AR_Black_Minefield
Symbol Criteria:	Area >= 15,625			**Label:**	[Geographic Name Information: Full Name], 6 point Condensed Mixed case if available; "Mine field", 6 point Condensed Mixed case
				Color:	Black (58600)

Minefield (AL065)

Point

Inclusion Condition:	*Area < 15,625 sq m.*

Notes

None

Feature Information					
Specification:	*LTDS*	**Feature Origin:**	*Topographic*	**Symbol:**	PT_Black_Minefield
Symbol Criteria:	Area < 15,625 sq m.			**Label:**	[Geographic Name Information: Full Name], 6 point Condensed Mixed case if available; "Mine field", 6 point Condensed Mixed case
				Color:	Black (58600)

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Mineral Pile (AM040)

Surface

Feature Description:	*A man-made heap of mining or quarrying products that does not contain waste materials.*
Inclusion Condition:	*Area >= 625 sq m.*

Notes

Labeling areas, in order of preference:
(1.) Centered in area on one line in the area, type is horizontal, reading left to right.
(2.) Centered in area on one line in the area, oriented along the long axis of the feature, reading left to right, or bottom to top if axis is vertical.
(3.) Centered in area on two approximately equal lines, without splitting a word, type is horizontal, reading left to right.
(4.) Centered on two approximately equal lines without splitting a word, type is oriented along the long axis of the feature, reading left to right, or bottom to top if axis is vertical.
(5.) Use priority for type placement if the area is too small to place the type inside the area.

Priority for type placement: 1-right center, 2-bottom center, 3-left center, 4-top center.
(A) Minimum distance from symbol - 1 mm.
(B) Maximum distance from symbol before choosing the next highest priority:
(#1) 4 mm measured to the West end;
(#2) 4 mm measured to the North side (top);
(#3) 4 mm measured to the East end;
(#4) 4 mm measured to the South side (bottom)

Feature name/label shall be positioned parallel to lines of latitude and readable left to right.

Feature Information

Specification:	*LTDS*	**Feature Origin:**	*Topographic*	**Symbol:**	AR_Black_MineralPile
Symbol Criteria:	All			**Label:**	"([Raw Material])", 6 point Condensed Mixed case and "pile", 6 point Condensed Mixed case if Raw Material <> ([Other] or [No information]); or "Mineral pile", 6 point Condensed Mixed case
				Color:	Black (58600)

Mineral Pile (AM040)

Point

Inclusion Condition:	*Area < 625 sq m.*

Notes

None

Feature Information

Specification:	*LTDS*	**Feature Origin:**	*Topographic*	**Symbol:**	PT_Black_MineralPile
Symbol Criteria:	Height Above Surface Level < 46			**Label:**	"([Raw Material])", 6 point Condensed Mixed case and "pile", 6 point Condensed Mixed case if Raw Material <> ([Other] or [No information]); or "Mineral pile", 6 point Condensed Mixed case
				Color:	Black (58600)

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Mineral Pile (AM040) *(Continued)*

Point

Feature Information					
Specification:	*LTDS*	**Feature Origin:**	*Topographic*	**Symbol:**	PT_Blue072_VO
Symbol Criteria:	Height Above Surface Level >= 46			**Label:**	"Mineral pile", 6 point Condensed Mixed case; [Highest Elevation], 6 point Condensed Bold Mixed case; "([Height Above Surface Level])", 6 point Condensed Mixed case
				Color:	Blue (46351)

Moat (BH100)

Surface

Feature Description:	*A trench surrounding a tract of land and serving as a barrier.*
Inclusion Condition:	*Width >= 25 m.*

Notes

None

Feature Information					
Specification:	*LTDS*	**Feature Origin:**	*Topographic*	**Symbol:**	AR_Cyan_Moat
Symbol Criteria:	Water Resource Information : Hydrologic Persistence = [Perennial]			**Label:**	"Moat", 6 point Condensed Mixed case
				Color:	Blue (46351)

Feature Information					
Specification:	*LTDS*	**Feature Origin:**	*Topographic*	**Symbol:**	AR_Black_Moat
Symbol Criteria:	Water Resource Information : Hydrologic Persistence = ([Intermittent] or [Dry])			**Label:**	"Moat", 6 point Condensed Mixed case
				Color:	Black (58600)

Moat (BH100)

Curve

Inclusion Condition:	*Width < 25 m.*

Notes

None

Feature Information					
Specification:	*LTDS*	**Feature Origin:**	*Topographic*	**Symbol:**	LN_Cyan_Moat
Symbol Criteria:	Water Resource Information : Hydrologic Persistence = [Perennial]			**Label:**	"Moat", 6 point Condensed Mixed case
				Color:	Blue (46351)

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Moat (BH100) *(Continued)*

Curve

Feature Information					
Specification:	*LTDS*	**Feature Origin:**	*Topographic*	**Symbol:**	LN_Black_Moat
Symbol Criteria:	Water Resource Information : Hydrologic Persistence: ([Intermittent] or [Dry])			**Label:**	"Moat", 6 point Condensed Mixed case
				Color:	Black (58600)

Mobile Home Park (AI021)

Surface

Feature Description:	*A site for the permanent parking of mobile homes used as dwellings and designed without a permanent foundation.*
Inclusion Condition:	*Area >= 15,625 sq m.*

Notes

Coincident similar area features will be blended to form a single feature.

Area and line features will be generalized to detail compatible with scale.

Type for features on land shall be positioned on land if possible. If type must be placed in the water, it shall be positioned so it does not obscure or overprint hydrographic detail. Type for features in the water shall be positioned in the water. If possible, type shall not be positioned across the shoreline.

Limiting lines of feature are omitted if it coalesces with a road (AP030).

If a clearing exists inside of the area feature, and the size of the clearing is equal to or greater than the area for the surrounding area feature, the clearing is shown as an open space inside the surround feature. If the area of the clearing is less than the area for the surrounding feature, the clearing is deleted and absorbed into the surrounding area feature.

If two area features with the same feature code do not connect at any point, and have a space between them less than 2.5 mm at map/chart scale, delete the open space that is less than 2.5 mm between the features and combine them into one area feature.

If a portion of an area feature has a minimum width of less than 2.5 mm at map/chart scale, delete that portion of the area feature that is not at least 2.5 mm wide, measured from perimeter to perimeter. If the deletion of the portion of the area based on the above criteria will reduce the area of the remaining portion of the area feature to below the minimum area inclusion, do not delete the narrow portion of the feature that is less than 2.5 mm wide. If the deletion of the portion of the area based on the above criteria will split two larger areas connected by a narrow strip into two separate areas, either of which would be below minimum area inclusion, do not delete the narrow portion of the feature that is less than 2.5 mm wide.

Feature Information					
Specification:	*LTDS*	**Feature Origin:**	*Topographic*	**Symbol:**	AR_Black_MobileHomePark
Symbol Criteria:	Area >= 15,625			**Label:**	[Geographic Name Information : Full Name], 6 point Condensed Mixed case; "Mobile home park", 6 point Condensed Mixed case
				Color:	Black (58600)

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Mobile Home Park (AI021)

Point

Inclusion Condition:	*Area < 15,625 sq m. and Navigation Landmark = [True]*

Notes

None

Feature Information					
Specification:	*LTDS*	**Feature Origin:**	*Topographic*	**Symbol:**	PT_Black_MobileHomePark
Symbol Criteria:	Area < 15,625 and Navigation Landmark = [True]			**Label:**	[Geographic Name Information : Full Name], 6 point Condensed Mixed case; "Mobile home park", 6 point Condensed Mixed case
				Color:	Black (58600)

Mooring Chain (BD074)

Curve

Feature Description:	*A chain or similar physical connection (for example: a cable) used to moor a vessel.*
Inclusion Condition:	*All Required*

Notes

Underwater chains, cables and anchors, if required, must be charted by the magenta symbol L18.

The anchor size should be the same as used in anchorage limits (N12), the shaft with ring being not more than 3.0mm long.

Ground tackle may be used, eg, for stabilising or fixing in position mooring buoys; fish farms; floating energy installations; for holding floating structures, eg vessels, pontoons, away from quays. If the actual chains, cables and anchors cannot be charted, an area (N1.2) with magenta legend, eg: 'Chains and anchors' may be used.

Mooring trots: Exceptionally, and on very large scale charts only, mooring berths between buoys may be shown with their numbers or letters inserted in circles, in magenta.

Feature Information					
Specification:	*Chart #1*	**Feature Origin:**	*Hydrographic*	**Symbol:**	L18
Symbol Criteria:	Mooring Type = [Buoy Mooring] or [Fore and Aft Mooring] or [Island Mooring] or [Swing Mooring]			**Label:**	"Chains and anchors", 8 point Mixed case
				Color:	Purple (96532)

Feature Information					
Specification:	*Chart #1*	**Feature Origin:**	*Hydrographic*	**Symbol:**	Q42
Symbol Criteria:	Mooring Type = [Trot Mooring]			**Label:**	None
				Color:	N/A

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Mooring Mast (AQ110)

Point

Feature Description:	*A mast used to secure an airship.*
Inclusion Condition:	*All Required*

Notes

Type for features on land shall be positioned on land if possible. If type must be placed in the water, it shall be positioned so it does not obscure or overprint hydrographic detail. Type for features in the water shall be positioned in the water. If possible, type shall not be positioned across the shoreline.

If the existence is Doubtful or Reported, label as such.

When obstructions coalesce at map scale, label with highest obstruction information.

Feature Information

Specification:	*LTDS*	**Feature Origin:**	*Topographic*	**Symbol:**	PT_Black_MooringMast
Symbol Criteria:	Height Above Surface Level < 46			**Label:**	None
				Color:	N/A

Feature Information

Specification:	*LTDS*	**Feature Origin:**	*Topographic*	**Symbol:**	PT_Blue072_VO
Symbol Criteria:	Height Above Surface Level >= 46			**Label:**	"Mooring mast", 6 point Condensed Mixed case; [Highest Elevation], 6 point Condensed Bold Mixed case; "([Height Above Surface Level])", 6 point Condensed Mixed case
				Color:	Blue (46351)

Mooring Ring (BB160)

Point

Feature Description:	*A metal ring attached to a structure and used to secure a vessel.*
Inclusion Condition:	*All Required*

Notes

None

Feature Information

Specification:	*Chart #1*	**Feature Origin:**	*Hydrographic*	**Symbol:**	TBD
Symbol Criteria:	All			**Label:**	"Mooring ring", 6 point Upper/lower Case
				Color:	Black (58600)

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Moraine (BJ020)

Surface

Feature Description:	*An accumulation of soil and stone debris deposited by a glacier.*
Inclusion Condition:	*Area >= 15,625 sq m.*

Notes

When 2 or more similar area features are separated by less than 0.5 mm at chart scale, the feature will be agglomerated.

Coincident similar area features will be blended to form a single feature.

Area and line features will be generalized to detail compatible with scale.

Feature will be generalized to provide a more aesthetic contoured feature (i.e., smoothed).

Symbols and associated area patterns of underpassing features (except drainage shorelines) are broken for all bridges, except footbridges. This rule does not apply to land tint on Littoral Planning Charts.

Feature Information

Specification:	*LTDS*	**Feature Origin:**	*Topographic*	**Symbol:**	AR_Dk-Brown1815_Moraine
Symbol Criteria:	All			**Label:**	None
				Color:	N/A

Motor Vehicle Station (AQ170)

Surface

Feature Description:	*A building at, or in which, motor vehicles are refuelled, serviced, and sometimes repaired.*
Inclusion Condition:	*Length or Diameter >= 25 m.*

Notes

None

Feature Information

Specification:	*LTDS*	**Feature Origin:**	*Topographic*	**Symbol:**	AR_Black_MotorVehicleStation
Symbol Criteria:	All			**Label:**	None
				Color:	N/A

Motor Vehicle Station (AQ170)

Point

Inclusion Condition:	*Length or Diameter < 25 m.*

Notes

None

Feature Information

Specification:	*LTDS*	**Feature Origin:**	*Topographic*	**Symbol:**	PT_Black_MotorVehicleStation
Symbol Criteria:	All			**Label:**	None
				Color:	N/A

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Mountain Pass (DB150)

Point

Feature Description:	*A narrow route through a mountainous region or over a mountain range.*
Inclusion Condition:	*ALL REQUIRED*

Notes

Like point features which coalesce in clusters of 3 or more will be thinned to form a representative pattern.

Label feature as per hierarchy for topo type placement to south neatline corners reading left to right:
1. Position hierarchy: (a.) northeast (preferred position), (b.) southeast (1st alternative), (c.) northwest (2nd alternative, (d.) southwest (3rd alternative), (e.) top-centered (4th alternative), (f.) bottom-centered (5th alternative): (Hierarchy is based on type positioning so as to avoid overprinting other type or obscuring detail.);
2. Minimum space between type placement and feature symbol is 0.5 mm;
3. This method of type placement shall be used for areal features when space does not permit labeling within that feature.

Descriptive terms, e.g., "Canal" shall be shown if the name is not known. If the descriptive word appears in the name, for example, "PANAMA CANAL". The descriptive type shall not be shown, i.e. do not show "Panama Canal Canal".

Feature Information

Specification:	*LTDS*	**Feature Origin:**	*Topographic*	**Symbol:**	PT_Black_MountainPass
Symbol Criteria:	All			**Label:**	[Geographic Name Information : Full Name], 6 point Condensed Mixed case if available and [Highest Elevation], 6 point Condensed Mixed case
				Color:	Black (58600)

Munition Storage Facility (AM065)

Surface

Feature Description:	*A facility for the storage of munitions (for example: bombs, missiles, warheads, mines or ammunition).*
Inclusion Condition:	*Area >= 15,625 sq m.*

Notes

None

Feature Information

Specification:	*LTDS*	**Feature Origin:**	*Topographic*	**Symbol:**	AR_Black_Munition StorageFacility
Symbol Criteria:	All			**Label:**	"Munition storage", 6 point Condensed Mixed case OR "Underground munition storage", 6 point Condensed Mixed case if Located Underground = [True]
				Color:	Black (58600)

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Munition Storage Facility (AM065)

Point

Inclusion Condition:	*Area < 15,625 sq m.*

Notes

None

Feature Information					
Specification:	*LTDS*	**Feature Origin:**	*Topographic*	**Symbol:**	PT_Black_Munition StorageFacility
Symbol Criteria:	All			**Label:**	"Munition storage", 6 point Condensed Mixed case OR "Underground munition storage", 6 point Condensed Mixed case if Located Underground = [True]
				Color:	Black (58600)

Named Location (ZD040)

Point

Feature Description:	*A location that normally does not appear as a specific, characterized object but that has a name that is required to be displayed in association with that location.*
Inclusion Condition:	*ALL REQUIRED*

Notes

Populated places are classified by complete up-to-date population figures, and by administrative importance. When complete up-to-date population data is not available, populated places are classified solely by administrative importance.

FIRST ORDER OF PRECEDENCE:
Populated classification for culturally developed areas:
>= 500,000, first importance, 14 point Upper case;
>= 100,000 and < 500,000, second importance, 10 point Upper case;
>= 25,000 and < 100,000, third importance, 10 point Upper/lower case;
>= 5,000 and < 25,000, fourth importance, 10 point Upper/lower case;
< 5,000, fifth importance, 8 point Upper/lower case.

SECOND ORDER OF PRECEDENCE: Population and relative importance classification for an area not as yet well culturally developed:
>= 100,000, first importance, 14 point Upper case;
>= 50,000 and < 100,000, second importance, 10 point Upper case;
>= 10,000 and < 50,000, third importance, 10 point Upper/lower case;
>= 2,000 and < 10,000, fourth importance, 10 point Upper/lower case;
< 2,000, fifth importance, 8 point upper/lower case.

THIRD ORDER OF PRECEDENCE: the categories of administrative importance may vary from region to region:
National Capital, first importance, 14 point Upper case;
Province, state, or department capital, second importance, 10 point Upper case;
County seat or chartered city, third importance, 10 point Upper/lower case;
Town, fourth importance, 10 point Upper/lower case;
Village or settlement, fifth importance, 8 point Upper/lower case.

Label feature as per hierarchy for topo type placement to south neatline corners reading left to right:
1. Position hierarchy: (a.) northeast (preferred position), (b.) southeast (1st alternative), (c.) northwest (2nd alternative, (d.) southwest (3rd alternative), (e.) top-centered (4th alternative), (f.) bottom-centered (5th alternative): (Hierarchy is based on type positioning so as to avoid overprinting other type or obscuring detail.);
2. Minimum space between type placement and feature symbol is 0.5 mm;
3. This method of type placement shall be used for areal features when space does not permit labeling within that feature.

Geographic names shall not be placed along the axis of deepest water in a confined area, such as a channel, fairway, etc.

The following features shall be named if name is known:
(a.) Non-tidal basin (an artificially enclosed area within which water can be maintained at a desired level to keep ships afloat while loading or discharging cargo, etc.) Water level is maintained by locks, or sluice gates, in IHO terminology, a "caisson". Symbolize name in italic type.
(b.) Tidal basin or tidal harbor (an enclosure in which the tide freely rises and falls, i.e., there is no lock or gate to regulate the water level. Symbolize name in italic.

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Named Location (ZD040) *(Continued)*

Point

Feature Information					
Specification:	*LTDS*	**Feature Origin:**	*Topographic*	**Symbol:**	PT_NamedLocation
Symbol Criteria:	Named Location Type <> [Other] or [No Information]			**Label:**	[Geographic Name Information : Full Name], 6-16 Light Condensed Upper case
				Color:	Black (58600)

Natural Pool (BH170)

Point

Feature Description:	*A naturally formed pool of water.*
Inclusion Condition:	*ALL REQUIRED*

Notes

Like point features which coalesce in clusters of 3 or more will be thinned to form a representative pattern.

Label feature as per hierarchy for topo type placement to south neatline corners reading left to right:
1. Position hierarchy: (a.) northeast (preferred position), (b.) southeast (1st alternative), (c.) northwest (2nd alternative, (d.) southwest (3rd alternative), (e.) top-centered (4th alternative), (f.) bottom-centered (5th alternative): (Hierarchy is based on type positioning so as to avoid overprinting other type or obscuring detail.);
2. Minimum space between type placement and feature symbol is 0.5 mm;
3. This method of type placement shall be used for areal features when space does not permit labeling within that feature.

Omit from Built-up Area.

Squiggly tail of symbol to point downhill to align with the Direction Of Flow (DOF). If DOF cannot be determined, then orient the tail to the bottom of the sheet.

Feature Information					
Specification:	*LTDS*	**Feature Origin:**	*Topographic*	**Symbol:**	PT_Cyan_NaturalPool_ Perennial
Symbol Criteria:	Water Resource Information : Hydrologic Persistence <> ([Intermittent] or [Dry])			**Label:**	[Geographic Name Information : Full Name] if available, 6 point Condensed Italics Mixed case, and if Water Resource Information : Water Type = (Mineral) or (Salty) or (Brackish) or (Seawater) or (Saltier than Seawater) label as "(salt)", 6 point Condensed Italics Lower case; if Water Resource Information : Water Type = (Alkaline), label as "(alkaline)", 6 point Condensed Italics Lower case
				Color:	Blue (48253)

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Natural Pool (BH170) *(Continued)*

Point

Feature Information					
Specification:	*LTDS*	**Feature Origin:**	*Topographic*	**Symbol:**	PT_Cyan_NaturalPool_Non-perennial
Symbol Criteria:	Water Resource Information : Hydrologic Persistence = ([Intermittent] or [Dry])			**Label:**	[Geographic Name Information : Full Name] if available, 6 point Condensed Italics Mixed case, and if Water Resource Information : Water Type = (Mineral) or (Salty) or (Brackish) or (Seawater) or (Saltier than Seawater) label as "(salt)", 6 point Condensed Italics Lower case; if Water Resource Information : Water Type = (Alkaline), label as "(alkaline)", 6 point Condensed Italics Lower case
				Color:	Blue (48253)

Naval Firing and/or Practice Area (FC050)

Surface

Feature Description:	*An area where Naval forces (for example: vessels or aircraft) conduct firing and/or munition exercises.*
Inclusion Condition:	*ALL REQUIRED*

Notes

Type sizes for Maritime Limits and areas:
8 point - < 8 square cm.,
10 point - >= 8 and < 12 square cm,,
12 point - >= 12 and < 24 square cm.,
14 point - >= 24 and < 100 square cm.,
8 point - >= 100 square cm.;
Type placement for areas >= 100 square cm. to < 500 square cm.:
Two labels are shown on approximately opposite sides of the area, preferably top and bottom.;
Type placement for areas >= 500 square cm.:
Labels are placed at approximately 250 mm. interval around the perimeter of the feature.;
Do not place labels around sharp corners (interior angle < 135°), or along chart neatlines. Place type 1 mm away and parallel to area limit line, reading left to right, or bottom to top if limit line is vertical. Type is placed on the INSIDE of the area.

Priority for type placement: 1-right center, 2-bottom center, 3-left center, 4-top center.
(A) Minimum distance from symbol - 1 mm.
(B) Maximum distance from symbol before choosing the next highest priority:
(#1) 4 mm measured to the West end;
(#2) 4 mm measured to the North side (top);
(#3) 4 mm measured to the East end;
(#4) 4 mm measured to the South side (bottom)

Type placement for areas < 100 square cm. Type shall be center in the area reading from left to right. If longer of the two axes is North-South +/- 20 degrees or East-West +/- 20 degrees, type is parallel to the south neatline.
(a.) If length < width times two, type shall be placed on two approximately equal lines without splitting words;
(b.) If length >= width times two, and major axis is East-West +/- 20 degrees, type shall be placed on one line;
(c.) If length >= width times two, and major axis is North-South +/- 20 degrees, type shall be placed with each word on a separate line.
If longer of the two axis is more than 20 degrees from either North-South or East-West, and length < width times two, type shall be parallel to south neatline and on two approximately equal lines without splitting words.
If longer of two axis is more than 20 degrees from either North-South or East-West, and length >= width times two, type shall be placed parallel to the major axis on one line. Type to be placed inside area, place type outside area.

When the boundary of an areal symbol crosses or overprints the shoreline, the boundary overprinting the shoreline or land is deleted.

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Naval Firing and/or Practice Area (FC050) ***(Continued)***

Surface

Feature Information					
Specification:	*Chart #1*	**Feature Origin:**	*Hydrographic*	**Symbol:**	N32
Symbol Criteria:	Naval Firing and/or Practice Type = [Mine Laying Practice Area]			**Label:**	[Naval Firing and/or Practice Type], 8-14 point Mixed case, Italic
				Color:	Green (52813)

Feature Information					
Specification:	*Chart #1*	**Feature Origin:**	*Hydrographic*	**Symbol:**	N30
Symbol Criteria:	Naval Firing and/or Practice Type = ([Firing Danger Area] or [Missile Firing and Practice Area])			**Label:**	[Naval Firing and/or Practice Type], 8-14 point Mixed case, Italic
				Color:	Green (52813)

Feature Information					
Specification:	*Chart #1*	**Feature Origin:**	*Hydrographic*	**Symbol:**	N34
Symbol Criteria:	Naval Firing and/or Practice Type = ([Naval Training Minefield] or [Abandoned Naval Training Minefield])			**Label:**	[Naval Firing and/or Practice Type] "(see Note)", 8-14 point Mixed case, Italic
				Color:	Green (52813)

Feature Information					
Specification:	*Chart #1*	**Feature Origin:**	*Hydrographic*	**Symbol:**	N31
Symbol Criteria:	Naval Firing and/or Practice Type = ([Bombing and/or Strafing Targets Area] or [Inwater Tracking Range] or [Naval Impact Area] or [Torpedo Practice Area] or [Weapons Range])			**Label:**	[Naval Firing and/or Practice Type], 8-14 point Mixed case, Italic
				Color:	Green (52813)

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Naval Operations Area (FC055)

Surface

Feature Description:	*An area where Naval forces conduct operations.*
Inclusion Condition:	*ALL REQUIRED*

Notes

Type sizes for Maritime Limits and areas:
8 point - < 8 square cm.,
10 point - >= 8 and < 12 square cm,,
12 point - >= 12 and < 24 square cm.,
14 point - >= 24 and < 100 square cm.,
8 point - >= 100 square cm.;
Type placement for areas >= 100 square cm. to < 500 square cm.:
Two labels are shown on approximately opposite sides of the area, preferably top and bottom.;
Type placement for areas >= 500 square cm.:
Labels are placed at approximately 250 mm. interval around the perimeter of the feature.;
Do not place labels around sharp corners (interior angle < 135°), or along chart neatlines. Place type 1 mm away and parallel to area limit line, reading left to right, or bottom to top if limit line is vertical. Type is placed on the INSIDE of the area.

Priority for type placement: 1-right center, 2-bottom center, 3-left center, 4-top center.
(A) Minimum distance from symbol - 1 mm.
(B) Maximum distance from symbol before choosing the next highest priority:
(#1) 4 mm measured to the West end;
(#2) 4 mm measured to the North side (top);
(#3) 4 mm measured to the East end;
(#4) 4 mm measured to the South side (bottom)

Type placement for areas < 100 square cm. Type shall be center in the area reading from left to right. If longer of the two axes is North-South +/- 20 degrees or East-West +/- 20 degrees, type is parallel to the south neatline.
(a.) If length < width times two, type shall be placed on two approximately equal lines without splitting words;
(b.) If length >= width times two, and major axis is East-West +/- 20 degrees, type shall be placed on one line;
(c.) If length >= width times two, and major axis is North-South +/- 20 degrees, type shall be placed with each word on a separate line.
If longer of the two axis is more than 20 degrees from either North-South or East-West, and length < width times two, type shall be parallel to south neatline and on two approximately equal lines without splitting words.
If longer of two axis is more than 20 degrees from either North-South or East-West, and length >= width times two, type shall be placed parallel to the major axis on one line. Type to be placed inside area, place type outside area.

When the boundary of an areal symbol crosses or overprints the shoreline, the boundary overprinting the shoreline or land is deleted.

Feature Information

Specification:	*Chart #1*	**Feature Origin:**	*Hydrographic*	**Symbol:**	N33
Symbol Criteria:	Naval Operations Type = ([Submarine Exercise Area or [Submarine Transit Lane (Submerged)] or [Submarine Transit Lane (Surface)])			**Label:**	[Naval Operations Type] "(see Note)", 8-14 point Mixed case, Italic
				Color:	Green (52813)

Feature Information

Specification:	*Chart #1*	**Feature Origin:**	*Hydrographic*	**Symbol:**	N34
Symbol Criteria:	Naval Operations Type = ([Naval Minefield])			**Label:**	[Naval Operations Type] "(see Note)", 8-14 point Mixed case, Italic
				Color:	Green (52813)

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Nearshore (BA021)

Surface

Feature Description:	*The part of the shore or beach which lies between the low water mark and the landward limit of the offshore region.*
Inclusion Condition:	*ALL REQUIRED*

Notes

None

Feature Information					
Specification:	*Chart #1*	**Feature Origin:**	*Hydrographic*	**Symbol:**	TBD
Symbol Criteria:	All			**Label:**	None
				Color:	N/A

Non-building Structure (AL014)

Point

Feature Description:	*A free-standing self-supporting construction (for example: a large piece of equipment) designed to support human activities (for example: agriculture, manufacturing, or mining) but not intended for human occupancy and/or habitation (for example: a house, a bank, an office, or a stadium).*
Inclusion Condition:	*Length or Diameter < 25 meters And Height Above Surface Level >= 46 meters*

Notes

None

Feature Information					
Specification:	*LTDS*	**Feature Origin:**	*Topographic*	**Symbol:**	PT_Blue072_VO
Symbol Criteria:	Height Above Surface Level >= 46			**Label:**	"Industrial structure", 6 point Condensed Mixed case; [Highest Elevation], 6 point Condensed Bold Mixed case; "([Height Above Surface Level])", 6 point Condensed Mixed case
				Color:	Blue (48253)

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Non-Water Well (AA054)

Point

Feature Description:	*A shaft sunk into the ground to reach and tap a supply of liquids and/or gases other than water intended for use in agriculture or domestic consumption.*
Inclusion Condition:	*Condition of Facility: (Fully Functional) or [No Information] OR Navigation Landmark = [True] AND Location Referenced to Shoreline = ([At Shoreline] or [Inland])*

Notes

If one symbol coalesces with another symbol for the same type feature, displace symbols to allow a minimum separation of 0.21mm.

When Product Category (No Information), omit the Product label.

Type for features on land shall be positioned on land if possible. If type must be placed in the water, it shall be positioned so it does not obscure or overprint hydrographic detail. Type for features in the water shall be positioned in the water. If possible, type shall not be positioned across the shoreline.

Descriptive terms, e.g., "Canal" shall be shown if the name is not known. If the descriptive word appears in the name, for example, "PANAMA CANAL". The descriptive type shall not be shown, i.e. do not show "Panama Canal Canal".

When project area or sheet falls within an area defined as having sparse drainage; the inclusion condition defaults to all required.

If Existence (Abandoned), use only if (Landmark).

If >= 3 equal symbols would coalesce at map scale, portray with a representative pattern.

Priority for type placement: 1-right center, 2-bottom center, 3-left center, 4-top center
(A) Minimum distance from symbol - 1 mm.
(B) Maximum distance from symbol before choosing the next highest priority:
#1 4 mm measured to the West end
#2 4 mm measured to the North side (top)
#3 4 mm measured to the East end
#4 4 mm measured to the South side (bottom)

Labels may be stacked, readable top to bottom to avoid overprints.

Unless specifically indicated on source material as inaccurate or doubtful, etc., or modified by other override rules, features shall be assumed to be accurate and definite. This rule applied to accuracy and existence of unique features, and is not related to the accuracy of the source as a whole.

Feature Information

Specification:	*LTDS*	**Feature Origin:**	*Topographic*	**Symbol:**	PT_Black_Well_Notwater
Symbol Criteria:	All			**Label:**	Label as "Gas" if "Product [1]" = [Gas] or [Liquefied Petroleum Gas (LPG]) or [Coalbed Methane] or [Natural Gas Condensate] or [Helium] , 6 point Condensed Mixed case; label as "Oil" if "Product [1]" = [Petroleum], 6 point Condensed Mixed case
				Color:	Black (58600)

Nuclear Reactor Containment (AD041)

Surface

Feature Description:	*A building-like structure intended to create a barrier against the release of radioactivity generated during nuclear power operations.*
Inclusion Condition:	*Area >= 625 sq m.*

Notes

None

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Nuclear Reactor Containment (AD041) *(Continued)*

Surface

Feature Information					
Specification:	*LTDS*	**Feature Origin:**	*Topographic*	**Symbol:**	AR_Black_NuclearReactor Containment
Symbol Criteria:	(Height Above Surface Level < 46) OR (Area >= 5625 and Height Above Surface Level >= 46)			**Label:**	[Geographic Name Information : Full Name] if avaialble, 6 point Condensed Mixed case; and "Containment structure", 6 point Condensed Mixed case
				Color:	Black (58600)

Nuclear Reactor Containment (AD041)

Point

Inclusion Condition:	*Area < 625 sq m.*

Notes

None

Feature Information					
Specification:	*LTDS*	**Feature Origin:**	*Topographic*	**Symbol:**	PT_Black_NuclearReactor Containment
Symbol Criteria:	Height Above Surface Level < 46			**Label:**	[Geographic Name Information : Full Name] if avaialble, 6 point Condensed Mixed case; and "Containment structure", 6 point Condensed Mixed case
				Color:	Black (58600)

Feature Information					
Specification:	*LTDS*	**Feature Origin:**	*Topographic*	**Symbol:**	PT_Blue072_VO
Symbol Criteria:	Height Above Surface Level >= 46			**Label:**	"Containment structure", 6 point Condensed Mixed case; [Highest Elevation], 6 point Condensed Bold Mixed case; "([Height Above Surface Level])", 6 point Condensed Mixed case
				Color:	Blue (46351)

Oasis (EC020)

Surface

Feature Description:	*A small, isolated, fertile or green area in a desert region usually having a spring or well.*
Inclusion Condition:	*Area >= 15,625*

Notes

Coincident similar area features will be blended to form a single feature.

Area and line features will be generalized to detail compatible with scale.

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Oasis (EC020) *(Continued)*

Surface

Notes

Type sizes per are sizes at map/chart scale: Area features only.
06 point - <= 770 mm square area and <= 14 mm width;
07 point - <= 2,296 mm square area and <= 28 mm width;
09 point - <= 5,192 mm square area and 44 mm width;
10 point - <=9,796 mm square area and <= 62 mm width;
12 point - <= 16,632 mm square area and <= 84 mm width;
14 point - <= 24,960 mm square area and <= 104 mm width;
16 point - > 24,960 mm square area. Where area measurements are inconsistent, the larger type size shall be used.
Where the full range of type sizes is not available for a particular label, the closest available type size shall be used.

Label feature as per hierarchy for topo type placement to south neatline corners reading left to right:
1. Position hierarchy: (a.) northeast (preferred position), (b.) southeast (1st alternative), (c.) northwest (2nd alternative, (d.) southwest (3rd alternative), (e.) top-centered (4th alternative), (f.) bottom-centered (5th alternative): (Hierarchy is based on type positioning so as to avoid overprinting other type or obscuring detail.);
2. Minimum space between type placement and feature symbol is 0.5 mm;
3. This method of type placement shall be used for areal features when space does not permit labeling within that feature.

Names placement shall be oriented to the longest axis of the feature reading left to right and placed within the area outline and centered. If the longest axis is perpendicular to the south neatline, the type shall be placed outside of the area outline, preferred position is northeast of the feature, but may be placed at any position around the feature so as not to overprint any other feature type and reading left to right.

If a clearing exists inside of the area feature, and the size of the clearing is equal to or greater than the area for the surrounding area feature, the clearing is shown as an open space inside the surround feature. If the area of the clearing is less than the area for the surrounding feature, the clearing is deleted and absorbed into the surrounding area feature.

If two area features with the same feature code do not connect at any point, and have a space between them less than 2.5 mm at map/chart scale, delete the open space that is less than 2.5 mm between the features and combine them into one area feature.

If a portion of an area feature has a minimum width of less than 2.5 mm at map/chart scale, delete that portion of the area feature that is not at least 2.5 mm wide, measured from perimeter to perimeter. If the deletion of the portion of the area based on the above criteria will reduce the area of the remaining portion of the area feature to below the minimum area inclusion, do not delete the narrow portion of the feature that is less than 2.5 mm wide. If the deletion of the portion of the area based on the above criteria will split two larger areas connected by a narrow strip into two separate areas, either of which would be below minimum area inclusion, do not delete the narrow portion of the feature that is less than 2.5 mm wide.

Feature Information

Specification:	*LTDS*	**Feature Origin:**	*Topographic*	**Symbol:**	AR_Green362_Oasis
Symbol Criteria:	All			**Label:**	[Geographic Name Information : Full Name], 6 point Condensed Mixed case if available and "Oasis", 6 point Condensed Mixed case
				Color:	Black (58600)

Oasis (EC020)

Point

Inclusion Condition:	*Area < 15,625*

Notes

Coincident similar area features will be blended to form a single feature.

Area and line features will be generalized to detail compatible with scale.

Label feature as per hierarchy for topo type placement to south neatline corners reading left to right:
1. Position hierarchy: (a.) northeast (preferred position), (b.) southeast (1st alternative), (c.) northwest (2nd alternative, (d.) southwest (3rd alternative), (e.) top-centered (4th alternative), (f.) bottom-centered (5th alternative): (Hierarchy is based on type positioning so as to avoid overprinting other type or obscuring detail.);
2. Minimum space between type placement and feature symbol is 0.5 mm;
3. This method of type placement shall be used for areal features when space does not permit labeling within that feature.
feature, but may be placed at any position around the feature so as not to overprint any other feature type and reading left to right.

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Oasis (EC020) *(Continued)*

Point

Notes

If a clearing exists inside of the area feature, and the size of the clearing is equal to or greater than the area for the surrounding area feature, the clearing is shown as an open space inside the surround feature. If the area of the clearing is less than the area for the surrounding feature, the clearing is deleted and absorbed into the surrounding area feature.

If two area features with the same feature code do not connect at any point, and have a space between them less than 2.5 mm at map/chart scale, delete the open space that is less than 2.5 mm between the features and combine them into one area feature.

If a portion of an area feature has a minimum width of less than 2.5 mm at map/chart scale, delete that portion of the area feature that is not at least 2.5 mm wide, measured from perimeter to perimeter. If the deletion of the portion of the area based on the above criteria will reduce the area of the remaining portion of the area feature to below the minimum area inclusion, do not delete the narrow portion of the feature that is less than 2.5 mm wide. If the deletion of the portion of the area based on the above criteria will split two larger areas connected by a narrow strip into two separate areas, either of which would be below minimum area inclusion, do not delete the narrow portion of the feature that is less than 2.5 mm wide.

Feature Information

Specification:	*LTDS*	**Feature Origin:**	*Topographic*	**Symbol:**	PT_Oasis
Symbol Criteria:	Area < 15,625			**Label:**	[Geographic Name Information : Full Name], 6 point Condensed Mixed case if available and "Oasis", 6 point Condensed Mixed case
				Color:	Black (58600)

Offshore Construction (BD115)

Point

Feature Description:	*An artificial structure that is located offshore.*
Inclusion Condition:	*Length < 40 meters*

Notes

Priority for type placement: 1-right center, 2-bottom center, 3-left center, 4-top center.
(A) Minimum distance from symbol - 1 mm.
(B) Maximum distance from symbol before choosing the next highest priority:
(#1) 4 mm measured to the West end;
(#2) 4 mm measured to the North side (top);
(#3) 4 mm measured to the East end;
(#4) 4 mm measured to the South side (bottom)

Labels may be stacked, readable top to bottom to avoid overprints.

Light flares (Posicut # 94) shall be oriented in order as follows:
(1) So that it does not overprint on the feature or a legend associated with that feature (except for the 7.1 mm diameter circle representing a radio aid to navigation, which the flare may overprint). ;
(2) So it does not overprint other chart data (i.e., soundings, pipelines, submarine cables, etc).;
(3) So that the wide end of the flare is pointed toward the legend of the light, or to seaward along the line, in the case of clearing lines and leading lines.

Feature Information

Specification:	*LTDS*	**Feature Origin:**	*Topographic*	**Symbol:**	PT_Black_OffshorePlatform
Symbol Criteria:	Offshore Construction Primary Structure = [Platform]			**Label:**	"Ru", 6 point Mixed case if Condition of Facility: ([Destroyed] or [Ruined]); [Geographic Name Information : Full Name] if available, 6 point Mixed case
				Color:	Black (58600)

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Offshore Construction (BD115) *(Continued)*

Point

Feature Information					
Specification:	*Chart #1*	**Feature Origin:**	*Hydrographic*	**Symbol:**	L16
Symbol Criteria:	Offshore Construction Primary Structure = [Catenary Turntable] or [Submerged Turret] OR (Offshore Positioning Method <> [Fixed])			**Label:**	None
				Color:	N/A

Feature Information					
Specification:	*Chart #1*	**Feature Origin:**	*Hydrographic*	**Symbol:**	L17
Symbol Criteria:	Offshore Construction Primary Structure = [Barge] or [Vessel]			**Label:**	"FPSO", 6 point upper case Italic
				Color:	Black (58600)

Feature Information					
Specification:	*Chart #1*	**Feature Origin:**	*Hydrographic*	**Symbol:**	L12
Symbol Criteria:	(Offshore Construction Primary Structure = [Terminal Buoy]) OR (Offshore Positioning Method = [Fixed])			**Label:**	"SPM", 6 point Upper case
				Color:	Black (58600)

Orchard (EA040)

Surface

Feature Description:	*A tract covered by systematic plantings of trees that yield fruits (including nuts).*
Inclusion Condition:	*Area >= 15,625 sq m.*

Notes

Coincident similar area features will be blended to form a single feature.

Area and line features will be generalized to detail compatible with scale.

Feature will be generalized to provide a more aesthetic contoured feature (i.e., smoothed).

Label feature as per hierarchy for topo type placement to south neatline corners reading left to right:
1. Position hierarchy: (a.) northeast (preferred position), (b.) southeast (1st alternative), (c.) northwest (2nd alternative, (d.) southwest (3rd alternative), (e.) top-centered (4th alternative), (f.) bottom-centered (5th alternative): (Hierarchy is based on type positioning so as to avoid overprinting other type or obscuring detail.);
2. Minimum space between type placement and feature symbol is 0.5 mm;
3. This method of type placement shall be used for areal features when space does not permit labeling within that feature.

Names placement shall be oriented to the longest axis of the feature reading left to right and placed within the area outline and centered. If the longest axis is perpendicular to the south neatline, the type shall be placed outside of the area outline, preferred position is northeast of the feature, but may be placed at any position around the feature so as not to overprint any other feature type and reading left to right.

Symbols and associated area patterns of underpassing features (except drainage shorelines) are broken for all bridges, except footbridges. This rule does not apply to land tint on Littoral Planning Charts.

If a clearing exists inside of the area feature, and the size of the clearing is equal to or greater than the area for the surrounding area feature, the clearing is shown as an open space inside the surround feature. If the area of the clearing is less than the area for the surrounding feature, the clearing is deleted and absorbed into the surrounding area feature.

If two area features with the same feature code do not connect at any point, and have a space between them less than 2.5 mm at map/chart scale, delete the open space that is less than 2.5 mm between the features and combine them into one area feature.

If a portion of an area feature has a minimum width of less than 2.5 mm at map/chart scale, delete that portion of the area feature that is not at least 2.5 mm wide, measured from perimeter to perimeter. If the deletion of the portion of the area based on the above criteria will reduce the area of the remaining portion of the area feature to below the minimum area inclusion, do not delete the narrow portion of the feature that is less than 2.5 mm wide. If the deletion of the portion of the area based on the above criteria will split two larger areas connected by a narrow strip into two separate areas, either of which would be below minimum area inclusion, do not delete the narrow portion of the feature that is less than 2.5 mm wide.

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Orchard (EA040) *(Continued)*

Surface

Feature Information					
Specification:	*LTDS*	**Feature Origin:**	*Topographic*	**Symbol:**	AR_Green362_Orchard
Symbol Criteria:	Area >= 15,625			**Label:**	[Geographic Name Information : Full Name], 6-16 point Condensed Mixed case if available; "(terraces)", 6-16 point Condensed Lower case if Crop Information : Farming Pattern [1] = [Terraced] and (Geographic Name Information : Full Name <> "UNK") and (is not null); "Terraces", , 6-16 point Condensed Mixed case if Crop Information : Farming Pattern [1] = [Terraced] and (Geographic Name Information : Full Name = "UNK") and (is null);
				Color:	Black (58600)

Orchard (EA040)

Point

Inclusion Condition:	*Area < 15,625 sq m. AND Navigation Landmark = [True]*

Notes

Label feature as per hierarchy for topo type placement to south neatline corners reading left to right:
1. Position hierarchy: (a.) northeast (preferred position), (b.) southeast (1st alternative), (c.) northwest (2nd alternative, (d.) southwest (3rd alternative), (e.) top-centered (4th alternative), (f.) bottom-centered (5th alternative): (Hierarchy is based on type positioning so as to avoid overprinting other type or obscuring detail.);
2. Minimum space between type placement and feature symbol is 0.5 mm;
3. This method of type placement shall be used for areal features when space does not permit labeling within that feature.

Feature Information					
Specification:	*LTDS*	**Feature Origin:**	*Topographic*	**Symbol:**	PT_Green362_Orchard
Symbol Criteria:	Area < 15,625 AND Navigation Landmark = [True]			**Label:**	[Geographic Name Information : Full Name], 6 point Condensed Mixed case if available; and "Orchard", 6 point Condensed Mixed case
				Color:	Black (58600)

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Outdoor Theatre Screen (AK080)

Point

Feature Description:	*A large outdoor screen for showing motion pictures.*
Inclusion Condition:	*Height Above Surface Level >= 46 m. OR Navigation Landmark = [True]*

Notes

None

Feature Information

Specification:	*LTDS*	**Feature Origin:**	*Topographic*	**Symbol:**	PT_Blue072_VO
Symbol Criteria:	Height Above Surface Level >= 46			**Label:**	"Outdoor screen" , 7 point Condensed Mixed case; [Highest Elevation], 6 point Condensed Bold Mixed case; "([Height Above Surface Level])", 6 point Condensed Mixed case
				Color:	Blue (46351)

Pack Ice (BJ070)

Surface

Feature Description:	*Term used in a wide sense to include any area of sea ice other than fast ice.*
Inclusion Condition:	*Area >= 15,625 sq m.*

Notes

None

Feature Information

Specification:	*Chart #1*	**Feature Origin:**	*Hydrographic*	**Symbol:**	N60.2
Symbol Criteria:	All			**Label:**	"Polar ice, 6 point Mixed case; "([Seasonal Ice Limit])", 6 point Mixed case, italic
				Color:	Black (58600)

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Park (AK120)

Surface

Feature Description:	*An area of defined limits which is set aside for human recreation and enjoyment and/or for historic preservation purposes, and is usually maintained in a natural, semi-natural, or ornamentally planted state.*
Inclusion Condition:	*Area >= 15,625 sq m.*

Notes

Label feature as per hierarchy for topo type placement to south neatline corners reading left to right:
1. Position hierarchy:
(a.) northeast (preferred position
(b.) southeast (1st alternative)
(c.) northwest (2nd alternative
(d.) southwest (3rd alternative)
(e.) top-centered (4th alternative)
(f.) bottom-centered (5th alternative)
(Hierarchy is based on type positioning so as to avoid overprinting other type or obscuring detail.)
2. Minimum space between type placement and feature symbol is 0.5 mm
3. This method of type placement shall be used for areal features when space does not permit labeling within that feature.

Names placement shall be oriented to the longest axis of the feature reading left to right and placed within the area outline and centered. If the longest axis is perpendicular to the south neatline, the type shall be placed outside of the area outline, preferred position is northeast of the feature, but may be placed at any position around the feature so as not to overprint any other feature type and reading left to right.

Type for features on land shall be positioned on land if possible. If type must be placed in the water, it shall be positioned so it does not obscure or overprint hydrographic detail. Type for features in the water shall be positioned in the water. If possible, type shall not be positioned across the shoreline.

Limiting lines of feature are omitted if it coalesces with a road (AP030).

If a clearing exists inside of the area feature, and the size of the clearing is equal to or greater than the area for the surrounding area feature, the clearing is shown as an open space inside the surround feature. If the area of the clearing is less than the area for the surrounding feature, the clearing is deleted and absorbed into the surrounding area feature.

If two area features with the same feature code do not connect at any point, and have a space between them less than 2.5 mm at map/chart scale, delete the open space that is less than 2.5 mm between the features and combine them into one area feature.

If a portion of an area feature has a minimum width of less than 2.5 mm at map/chart scale, delete that portion of the area feature that is not at least 2.5 mm wide, measured from perimeter to perimeter. If the deletion of the portion of the area based on the above criteria will reduce the area of the remaining portion of the area feature to below the minimum area inclusion, do not delete the narrow portion of the feature that is less than 2.5 mm wide. If the deletion of the portion of the area based on the above criteria will split two larger areas connected by a narrow strip into two separate areas, either of which would be below minimum area inclusion, do not delete the narrow portion of the feature that is less than 2.5 mm wide.

Feature Information

Specification:	*LTDS*	**Feature Origin:**	*Topographic*	**Symbol:**	AR_Black_Park
Symbol Criteria:	Area >= 15,625			**Label:**	[Geographic Name Information : Full Name], 6 point Condensed Mixed case; "Park", 6 point Condensed Mixed case
				Color:	Black (58600)

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Park (AK120)

Point

Inclusion Condition:	*Area < 15,625 square meters*

Notes

None

Feature Information					
Specification:	*LTDS*	**Feature Origin:**	*Topographic*	**Symbol:**	PT_Green362_Park
Symbol Criteria:	Area < 15,625 square meters AND Navigation Landmark = [True]			**Label:**	[Geographic Name Information : Full Name], 6 point Condensed Mixed case; "Park", 6 point Condensed Mixed case
				Color:	Black (58600)

Parking Garage (AQ141)

Surface

Feature Description:	*A designated, multi-level, structure used for parking and/or storing vehicles.*
Inclusion Condition:	*Area >= 625 sq m. OR Height Above Surface Level >= 46 m.*

Notes

None

Feature Information					
Specification:	*LTDS*	**Feature Origin:**	*Topographic*	**Symbol:**	AR_Black_ParkingGarage
Symbol Criteria:	(Located Underground <> [True] and (Height Above Surface Level < 46)) OR (Area >= 5625 and Height Above Surface Level >= 46)			**Label:**	"Parking garage", 6 point Condensed Mixed case
				Color:	Black (58600)

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Parking Garage (AQ141)

Point

Inclusion Condition:	*Area < 625 sq m. OR Height Above Surface Level >= 46 m.*

Notes

None

Feature Information					
Specification:	*LTDS*	**Feature Origin:**	*Topographic*	**Symbol:**	PT_Black_ParkingGarage
Symbol Criteria:	Located Underground <> [True] AND Height Above Surface Level < 46 m			**Label:**	"Parking garage". 6 point Condensed Mixed case
				Color:	Black (58600)

Feature Information					
Specification:	*LTDS*	**Feature Origin:**	*Topographic*	**Symbol:**	PT_Blue072_VO
Symbol Criteria:	Located Underground <> [True] AND Height Above Surface Level >= 46 m			**Label:**	"Parking garage", 6 point Condensed Mixed case; [Highest Elevation], 6 point Condensed Bold Mixed case; "([Height Above Surface Level])", 6 point Condensed Mixed case
				Color:	Blue (46351)

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Particle Accelerator (AL140)

Surface

Feature Description:	*An apparatus for imparting high velocities to charged particles.*
Inclusion Condition:	*Length or Diameter >= 125 m. and Width >= 25 m.*

Notes

Coincident similar area features will be blended to form a single feature.

Area and line features will be generalized to detail compatible with scale.

Label feature as per hierarchy for topo type placement to south neatline corners reading left to right:
1. Position hierarchy:
(a.) northeast (preferred position
(b.) southeast (1st alternative)
(c.) northwest (2nd alternative
(d.) southwest (3rd alternative)
(e.) top-centered (4th alternative)
(f.) bottom-centered (5th alternative)
(Hierarchy is based on type positioning so as to avoid overprinting other type or obscuring detail.)
2. Minimum space between type placement and feature symbol is 0.5 mm
3. This method of type placement shall be used for areal features when space does not permit labeling within that feature.

Type for features on land shall be positioned on land if possible. If type must be placed in the water, it shall be positioned so it does not obscure or overprint hydrographic detail. Type for features in the water shall be positioned in the water. If possible, type shall not be positioned across the shoreline.

Feature Information

Specification:	*LTDS*	**Feature Origin:**	*Topographic*	**Symbol:**	AR_Black_ParticleAccelerator
Symbol Criteria:	All			**Label:**	[Geographic Name Information: Full Name], 6 point Condensed Mixed case if available; "Particle accelerator ", 6 point Condensed Mixed case
				Color:	Black (58600)

Particle Accelerator (AL140)

Curve

Inclusion Condition:	*Length or Diameter >= 125 m. and Width < 25 m.*

Notes

None

Feature Information

Specification:	*LTDS*	**Feature Origin:**	*Topographic*	**Symbol:**	LN_Black_ParticleAccelerator
Symbol Criteria:	All			**Label:**	[Geographic Name Information: Full Name], 6 point Condensed Mixed case if available; "Particle accelerator ", 6 point Condensed Mixed case
				Color:	Black (58600)

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Penstock (BH110)

Curve

Feature Description:	*A pipeline that is used by hydroelectric plants to transport water by gravity from a reservoir to the turbine(s).*
Inclusion Condition:	*Length or Diameter >= 75 m.*

Notes

Area and line features will be generalized to detail compatible with scale.

Label shall be position above feature, reading from left to right (or to the left of the vertical feature, reading bottom to top), at a 0.5 mm distance and parallel to respective feature. Label shall preferably be positioned at the midpoint of the line segment or symbol; however, it may be displaced laterally along respective feature to avoid overprinting other symbols or labels. If space will not permit placing label parallel to feature, offset the label and use a leader line to identify its location along the feature.

Feature name, label, data information holder, and/or symbol shall be positioned, reading left to right, parallel to the tangent of the center of the southern neatline of the Map/Chart.

Feature Information

Specification:	*LTDS*	**Feature Origin:**	*Topographic*	**Symbol:**	LN_Cyan_Penstock_OnSurface
Symbol Criteria:	Vertical Relative Location = [On Surface] or [No Information]			**Label:**	"Penstock", 6 point Condensed Italics Mixed case
				Color:	Blue (48253)

Feature Information

Specification:	*LTDS*	**Feature Origin:**	*Topographic*	**Symbol:**	LN_Cyan_Penstock_Below Surface
Symbol Criteria:	Vertical Relative Location = [Below Ground Surface]			**Label:**	"Penstock", 6 point Condensed Italics Mixed case
				Color:	Blue (48253)

Feature Information

Specification:	*LTDS*	**Feature Origin:**	*Topographic*	**Symbol:**	LN_Cyan_Penstock_Above Surface
Symbol Criteria:	Vertical Relative Location = [Above Surface]			**Label:**	"Penstock", 6 point Condensed Italics Mixed case
				Color:	Blue (48253)

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Pilot Boarding Place (BB270)

Point

Feature Description:	*A location offshore where a pilot may board a vessel in preparation to piloting it through local waters.*
Inclusion Condition:	*Length < 20 mm at chart scale*

Notes

Priority for type placement: 1-right center, 2-bottom center, 3-left center, 4-top center.
(A) Minimum distance from symbol - 1 mm.
(B) Maximum distance from symbol before choosing the next highest priority:
(#1) 4 mm measured to the West end;
(#2) 4 mm measured to the North side (top);
(#3) 4 mm measured to the East end;
(#4) 4 mm measured to the South side (bottom)

Feature Information

Specification:	*Chart #1*	**Feature Origin:**	*Hydrographic*	**Symbol:**	T1.1
Symbol Criteria:	Pilot Boarding Method = [Pilot-cruising Vessel] or [Requested Vessel]			**Label:**	[Geographic Name Information : Full Name] if available, 6 point Mixed case, Italic
				Color:	Green (52813)

Feature Information

Specification:	*Chart #1*	**Feature Origin:**	*Hydrographic*	**Symbol:**	T1.4
Symbol Criteria:	Pilot Boarding Method = [Helicopter]			**Label:**	[Geographic Name Information : Full Name] if available, 6 point Mixed case, Italic; "H", 6 point Upper case, Italic;
				Color:	Green (52813)

Pipeline (AQ113)

Curve

Feature Description:	*A connected set of pipes for conveying liquids, slurries, or gases.*
Inclusion Condition:	*Length or Diameter >= 1250 m. OR (needed to complete the component network or to establish connectivity)*

Notes

Area and line features will be generalized to detail compatible with scale.

Type for features on land shall be positioned on land if possible. If type must be placed in the water, it shall be positioned so it does not obscure or overprint hydrographic detail. Type for features in the water shall be positioned in the water. If possible, type shall not be positioned across the shoreline.

Label shall be position above feature, reading from left to right (or to the left of the vertical feature, reading bottom to top), at a 0.5 mm distance and parallel to respective feature. Label shall preferably be positioned at the midpoint of the line segment or symbol; however, it may be displaced laterally along respective feature to avoid overprinting other symbols or labels. If space will not permit placing label parallel to feature, offset the label in accordance with Rule L-4261 below and use a leader line to identify its location along the feature.

Feature name, label, data information holder, and/or symbol shall be positioned, reading left to right, parallel to the tangent of the center of the southern neatline of the Map/Chart.

If feature is linear, the label hierarchy is:
(1) Label shall be placed 1 mm above feature, centered.
(2) Top of label shall be placed 1 mm below feature, centered.
(3) Label may be displaced along the feature, use two labels if feature length 150 mm or greater.
(4) Do not label across shoreline (coastal or inland)

Pipelines, pipeline areas, and cable and pipeline areas shall show a label for the following Product category values, using the label shown below: If Product Category is Chemical, label "Chem"; If Product Category is Natural Gas, label "Gas"; If Product Category is Gasoline, label "Gasoline"; If Product Category is Oil, label "Oil"; If Product Category is Water, label "Water".

Depth Below Surface Level is used when Location/origin (Below Sea Bottom). If Depth Below Surface Level is Unknown, show as Location/Origin (On Sea Bottom).

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Pipeline (AQ113) *(Continued)*

Curve

Notes

Pipelines shall not be shown within Built-up tinted areal features.

The Height Above Sea Bottom is applicable when Location/Origin (Suspended or Elevated Above Sea Bottom). If a pipeline is (Suspended or Elevated Above Sea Bottom), and the height above the bottom is unknown, a caution label (see cautions section of product specification) shall be shown near the feature, and the following caution shall be shown in the caution box:
"CAUTION - Pipelines are elevated above the sea bottom, and the clearance over them is less than the charted depth." If the feature is (Suspended or Elevated Above Sea Bottom), and the height above the bottom is known, the caution in the caution box shall be:
"CAUTION - Pipelines are elevated up to (Height above sea bottom value) meters above the sea bottom, and the clearance over them is less than the charted depth." The height above sea bottom is indicated in the text of the note.

Omit from Built-up Area.

Show Pipelines that are below ground surface to show connections to pipelines that are on ground surface or elevated, or when scars in the earth from underground feature is a Landmark

Pipelines shall not be shown when coincident with Roads and Railroads, except in desert regions or arctic regions where Landmark.

If Pipelines or cables in the water meet the following criteria, they are represented as pipeline areas (FC037)or cable areas (FC037) respectively, rather than shown as separate pipelines or cables. If pipelines and cables together meet the following criteria, they are represented as "cables and pipelines area" (FC037). Criteria: (a.) more than two linear features, AND (b.) space between any two linear features is less than 8 mm at chart scale, AND (c.) space between the outermost linear features in the group is greater than 3 mm at chart scale. If more than two cables or pipelines are <= 3 mm apart at chart scale, show only the outermost linear features. If cable symbols overprint other cable symbols, show only one cable. If pipeline symbols overprint other pipeline symbols, show only one pipeline. If cable symbols and pipeline symbols overprint, show a cable and pipeline area (FC037). The outermost limits of the cable, pipeline, or cables and pipeline area (FC037) area feature are displaced for 2 mm past the outermost cables and pipelines, so that the area within which anchoring, trawling, and dredging are prohibited or inadvisable includes a safety margin beyond the outermost cables and/or pipelines.

Charts shall have the following caution notes shown in the margin if pipelines, pipeline areas, or cable and pipeline areas, are shown on the chart, and products are chemicals, gas, gasoline, or oil.: "CAUTION- Mariners risk prosecution if they anchor or trawl near a pipeline and so damage it. (Product category) leaking from a damaged pipeline could cause fire or loss of a vessels buoyancy. " The product name (Product category) is indicated in the text of the note. Product category (Chemical) is shown in plural, i.e., "Chemicals". See notes and cautions section of product specifications for color, type size, type style, and other information regarding caution notes.

Pipelines coincident with traveled ways are not shown, except in desert areas.

Feature Information

Specification:	*LTDS*	**Feature Origin:**	*Topographic*	**Symbol:**	LN_Black_Pipeline_ NotWaterOnSurface
Symbol Criteria:	(Vertical Relative Location = [On Surface] or [No Information] and Product [1] <> [Water]) AND (Location Referenced to Shoreline = ([At Shoreline] or [Inland]))			**Label:**	"[Product]" and "pipeline" if <> [Other] or [No Information], 6 point Condensed Mixed case; "Steam" if Product = [Heating Steam and/or Water], 6 point Condensed Mixed case or "Pipeline", 6 point Condensed Mixed case
				Color:	Black (58600)

Feature Information

Specification:	*LTDS*	**Feature Origin:**	*Topographic*	**Symbol:**	LN_Cyan_Pipeline_ WaterOnSurface
Symbol Criteria:	(Vertical Relative Location = [On Surface] or [No Information] and Product [1] = [Water]) AND (Location Referenced to Shoreline = ([At Shoreline] or [Inland]))			**Label:**	"Pipeline", 6 point Condensed Mixed case
				Color:	Blue (46351)

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Pipeline (AQ113) *(Continued)*

Curve

Feature Information					
Specification:	*LTDS*	**Feature Origin:**	*Topographic*	**Symbol:**	LN_Black_Pipeline_ NotWaterBelowSurface
Symbol Criteria:	(Vertical Relative Location = [Below Ground Surface] and Product [1] <> [Water]) AND (Location Referenced to Shoreline = ([At Shoreline] or [Inland]))			**Label:**	"[Product]" and "pipeline" if <> [Other] or [No Information], 6 point Condensed Mixed case; "Steam" if Product = [Heating Steam and/or Water], 6 point Condensed Mixed case or "Pipeline", 6 point Condensed Mixed case
				Color:	Black (58600)

Feature Information					
Specification:	*LTDS*	**Feature Origin:**	*Topographic*	**Symbol:**	LN_Cyan_Pipeline_ WaterBelowSurface
Symbol Criteria:	(Vertical Relative Location = [Below Ground Surface] and Product [1] = [Water]) AND (Location Referenced to Shoreline = ([At Shoreline] or [Inland]))			**Label:**	"Pipeline", 6 point Condensed Mixed case
				Color:	Blue (46351)

Feature Information					
Specification:	*LTDS*	**Feature Origin:**	*Topographic*	**Symbol:**	LN_Black_Pipeline_ NotWaterElevated
Symbol Criteria:	(Vertical Relative Location = [Above Surface] and Product [1] <> [Water]) AND (Location Referenced to Shoreline = ([At Shoreline] or [Inland]))			**Label:**	"[Product]" and "pipeline" if <> [Other] or [No Information], 6 point Condensed Mixed case; "Steam" if Product = [Heating Steam and/or Water], 6 point Condensed Mixed case or "Pipeline", 6 point Condensed Mixed case
				Color:	Black (58600)

Feature Information					
Specification:	*LTDS*	**Feature Origin:**	*Topographic*	**Symbol:**	LN_Cyan_Pipeline_ WaterElevated
Symbol Criteria:	(Vertical Relative Location = [Above Surface] and Product [1] = [Water]) AND (Location Referenced to Shoreline = ([At Shoreline] or [Inland]))			**Label:**	"Pipeline", 6 point Condensed Mixed case
				Color:	Blue (46351)

Feature Information					
Specification:	*Chart #1*	**Feature Origin:**	*Hydrographic*	**Symbol:**	L40.1
Symbol Criteria:	(Location Referenced to Shoreline = [Offshore]) and (Condition of Facility = ([Fully Functional] or [Partially Functional] or [Under Construction])) and (Vertical Relative Location = ([Above Waterbody Bottom] or [On Waterbody Bottom])) and (Pipeline Type = [Transport Pipe])			**Label:**	[Product], 6 point Mixed case, Italic
				Color:	Black (58600)

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Pipeline (AQ113) *(Continued)*

Curve

Feature Information					
Specification:	*Chart #1*	**Feature Origin:**	*Hydrographic*	**Symbol:**	L44
Symbol Criteria:	(Location Referenced to Shoreline = [Offshore]) and (Condition of Facility = ([Abandoned] or [Damaged] or [Destroyed] or [Disused] or [Partially Destroyed])) and (Vertical Relative Location = ([Above Waterbody Bottom] or [On Waterbody Bottom]))			**Label:**	None
				Color:	N/A

Feature Information					
Specification:	*Chart #1*	**Feature Origin:**	*Hydrographic*	**Symbol:**	L42
Symbol Criteria:	(Location Referenced to Shoreline = [Offshore]) and (Condition of Facility = ([Fully Functional] or [Partially Functional] or [Under Construction])) and (Vertical Relative Location = ([Below Waterbody Bottom])) and (Pipeline Type = [Transport Pipe])			**Label:**	"Buried", 6 point Mixed case, Italic; [Depth below Surface Level], 6 point lower case, Italic; "m", 6 point lower case, Italic
				Color:	Black (58600)

Feature Information					
Specification:	*Chart #1*	**Feature Origin:**	*Hydrographic*	**Symbol:**	L41.1
Symbol Criteria:	(Location Referenced to Shoreline = [Offshore]) and (Condition of Facility = ([Fully Functional] or [Partially Functional] or [Under Construction])) and (Vertical Relative Location = ([Above Waterbody Bottom] or [On Waterbody Bottom])) and (Pipeline Type = [Intake] or [Outfall] or [Sewer])			**Label:**	[Pipeline Type], 6 point Mixed case, Italic; "Water" if Product = [Water], 6 Point Mixed case italic
				Color:	Black (58600)

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Plant Nursery (EA030)

Surface

Feature Description:	*A tract where plants (for example: shrubs, flowers, and/or trees) are grown for transplanting, seed, and/or grafting.*
Inclusion Condition:	*Area >= 15,625 sq m.*

Notes

Coincident similar area features will be blended to form a single feature.

Area and line features will be generalized to detail compatible with scale.

Feature will be generalized to provide a more aesthetic contoured feature (i.e., smoothed).

Label feature as per hierarchy for topo type placement to south neatline corners reading left to right:
1. Position hierarchy: (a.) northeast (preferred position), (b.) southeast (1st alternative), (c.) northwest (2nd alternative, (d.) southwest (3rd alternative), (e.) top-centered (4th alternative), (f.) bottom-centered (5th alternative): (Hierarchy is based on type positioning so as to avoid overprinting other type or obscuring detail.);
2. Minimum space between type placement and feature symbol is 0.5 mm;
3. This method of type placement shall be used for areal features when space does not permit labeling within that feature.

Names placement shall be oriented to the longest axis of the feature reading left to right and placed within the area outline and centered. If the longest axis is perpendicular to the south neatline, the type shall be placed outside of the area outline, preferred position is northeast of the feature, but may be placed at any position around the feature so as not to overprint any other feature type and reading left to right.

Type for features on land shall be positioned on land if possible. If type must be placed in the water, it shall be positioned so it does not obscure or overprint hydrographic detail. Type for features in the water shall be positioned in the water. If possible, type shall not be positioned across the shoreline.

Symbols and associated area patterns of underpassing features (except drainage shorelines) are broken for all bridges, except footbridges. This rule does not apply to land tint on Littoral Planning Charts.

If a clearing exists inside of the area feature, and the size of the clearing is equal to or greater than the area for the surrounding area feature, the clearing is shown as an open space inside the surround feature. If the area of the clearing is less than the area for the surrounding feature, the clearing is deleted and absorbed into the surrounding area feature.

If two area features with the same feature code do not connect at any point, and have a space between them less than 2.5 mm at map/chart scale, delete the open space that is less than 2.5 mm between the features and combine them into one area feature.

If a portion of an area feature has a minimum width of less than 2.5 mm at map/chart scale, delete that portion of the area feature that is not at least 2.5 mm wide, measured from perimeter to perimeter. If the deletion of the portion of the area based on the above criteria will reduce the area of the remaining portion of the area feature to below the minimum area inclusion, do not delete the narrow portion of the feature that is less than 2.5 mm wide. If the deletion of the portion of the area based on the above criteria will split two larger areas connected by a narrow strip into two separate areas, either of which would be below minimum area inclusion, do not delete the narrow portion of the feature that is less than 2.5 mm wide.

Feature Information

Specification:	*LTDS*	**Feature Origin:**	*Topographic*	**Symbol:**	AR_Green362_PlantNursery
Symbol Criteria:	Area >= 15,625			**Label:**	[Geographic Name Information : Full Name], 6-16 point Condensed Mixed case if available; "Nursery", 6-16 point Condensed Mixed case
				Color:	Black (58600)

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Plant Nursery (EA030)

Point

Inclusion Condition:	*Area < 15,625 sq m. AND Navigation Landmark = [True]*

Notes

Label feature as per hierarchy for topo type placement to south neatline corners reading left to right:
1. Position hierarchy: (a.) northeast (preferred position), (b.) southeast (1st alternative), (c.) northwest (2nd alternative, (d.) southwest (3rd alternative), (e.) top-centered (4th alternative), (f.) bottom-centered (5th alternative): (Hierarchy is based on type positioning so as to avoid overprinting other type or obscuring detail.);
2. Minimum space between type placement and feature symbol is 0.5 mm;
3. This method of type placement shall be used for areal features when space does not permit labeling within that feature.

Type for features on land shall be positioned on land if possible. If type must be placed in the water, it shall be positioned so it does not obscure or overprint hydrographic detail. Type for features in the water shall be positioned in the water. If possible, type shall not be positioned across the shoreline.

Feature Information

Specification:	*LTDS*	Feature Origin:	*Topographic*	Symbol:	PT_Green362_PlantNursery
Symbol Criteria:	Area < 15,625 AND Navigation Landmark = [True]			Label:	[Geographic Name Information : Full Name], 6 point Condensed Mixed case if available; "Nursery", 6 point Condensed Mixed case
				Color:	Black (58600)

Polar Ice (BJ080)

Surface

Feature Description:	*Sea ice that is more than one year old and more than 3 metres thick.*
Inclusion Condition:	*Area >= 15,625 sq m.*

Notes

None

Feature Information

Specification:	*Chart #1*	Feature Origin:	*Hydrographic*	Symbol:	N60.2
Symbol Criteria:	All			Label:	"Polar ice", 6 point Mixed case; "([Seasonal Ice Limit])", 6 point Mixed case, italic
				Color:	Black (58600)

Port (BB009)

Surface

Feature Description:	*A place provided with terminal and transfer facilities for loading and/or discharging cargo or passengers, usually located in a harbour.*
Inclusion Condition:	*Area >= 15,625 sq m.*

Notes

None

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Port (BB009) *(Continued)*

Surface

Feature Information					
Specification:	*LTDS*	**Feature Origin:**	*Topographic*	**Symbol:**	AR_Black_Port
Symbol Criteria:	Area >= 15,625			**Label:**	[Geographic Name Information : Full Name] if available, 6 point Condensed Mixed case; "Port", 6 point Condensed Mixed case
				Color:	Black (58600)

Port (BB009)

Point

Inclusion Condition:	*Area < 15,625 sq m. AND Navigation Landmark = [True]*

Notes

None

Feature Information					
Specification:	*LTDS*	**Feature Origin:**	*Topographic*	**Symbol:**	PT_Black_Port
Symbol Criteria:	Area < 15,625 AND Navigation Landmark = [True]			**Label:**	[Geographic Name Information : Full Name] if available, 6 point Condensed Mixed case; "Port", 6 point Condensed Mixed case
				Color:	Black (58600)

Power Station (AD010)

Surface

Feature Description:	*A facility including one or more buildings and equipment used for power generation.*
Inclusion Condition:	*Area >= 15,625 sq.m*

Notes

Coincident similar area features will be blended to form a single feature.

Area and line features will be generalized to detail compatible with scale.

Type for features on land shall be positioned on land if possible. If type must be placed in the water, it shall be positioned so it does not obscure or overprint hydrographic detail. Type for features in the water shall be positioned in the water. If possible, type shall not be positioned across the shoreline.

Descriptive terms, e.g., "Canal" shall be shown if the name is not known. If the descriptive word appears in the name, for example, "PANAMA CANAL". The descriptive type shall not be shown, i.e. do not show "Panama Canal Canal".

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Power Station (AD010) *(Continued)*

Surface

Feature Information					
Specification:	*LTDS*	**Feature Origin:**	*Topographic*	**Symbol:**	AR_Black_PowerStation
Symbol Criteria:	All			**Label:**	[Geographic Name Information : Full Name] if available: 6 point Condensed Mixed case and "([Power Plant Type])" if <> [Other] or [No information], 6 point Condensed Mixed case and "Power plant" if [Other] or [No information], 6 point Condensed lower case
				Color:	Black (58600)

Power Station (AD010)

Point

Inclusion Condition:	*Area < 15,625 sq.m*

Notes

None

Feature Information					
Specification:	*LTDS*	**Feature Origin:**	*Topographic*	**Symbol:**	PT_Black_PowerStation
Symbol Criteria:	All			**Label:**	[Geographic Name Information : Full Name] if available: 6 point Condensed Mixed case and "([Power Plant Type])" if <> [Other] or [No information], 6 point Condensed Mixed case and "Power plant" if [Other] or [No information], 6 point Condensed lower case
				Color:	Black (58600)

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Power Substation (AD030)

Surface

Feature Description:	*A facility, along a power transmission line, in which electric current is switched, transformed, and/or converted.*
Inclusion Condition:	*Area >= 15,625 sq m.*

Notes

When 2 or more similar area features are separated by less than 0.5 mm at chart scale, the feature will be agglomerated.

Coincident similar area features will be blended to form a single feature.

Area and line features will be generalized to detail compatible with scale.

Feature Information

Specification:	*LTDS*	**Feature Origin:**	*Topographic*	**Symbol:**	AR_Black_PowerSubstation
Symbol Criteria:	All			**Label:**	None
				Color:	N/A

Power Substation (AD030)

Point

Inclusion Condition:	*Area < 15,625sq.m*

Notes

The feature (when height <= 46 meters or when height is not known) shall be oriented perpendicular (90 degrees) to the nearby road (AP030), cart track (AP010), trail (AP050), or railroad (AN010).

If one symbol coalesces with another symbol for the same type feature, displace symbols to allow a minimum separation of 0.21mm.

If >= 3 equal symbols would coalesce at map scale, portray with a representative pattern.

Feature Information

Specification:	*LTDS*	**Feature Origin:**	*Topographic*	**Symbol:**	PT_Black_PowerSubstation
Symbol Criteria:	All			**Label:**	None
				Color:	N/A

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Public Square (AL170)

Surface

Feature Description:	*An open site that serves as a public meeting location in a built-up area.*
Inclusion Condition:	*Area >= 625 sq m.*

Notes

When 2 or more similar area features are separated by less than 0.5 mm at chart scale, the feature will be agglomerated.

Area and line features will be generalized to detail compatible with scale.

If the width of the symbolized Road is greater than the width of the Public Square, then suppress the Public Square.

Feature Information

Specification:	*LTDS*	**Feature Origin:**	*Topographic*	**Symbol:**	AR_Black_PublicSquare
Symbol Criteria:	All			**Label:**	[Geographic Name Information: Full Name], 6 point Condensed Mixed case if available; and "Plaza", 6 point Condensed Mixed case
				Color:	Black (58600)

Pumping Station (AQ116)

Surface

Feature Description:	*A facility to move solids, liquids or gases by means of pressure or suction.*
Inclusion Condition:	*Area >= 15,625 sq m.*

Notes

Area and line features will be generalized to detail compatible with scale.

When Product Category (No Information), omit the Product label.

Type for features on land shall be positioned on land if possible. If type must be placed in the water, it shall be positioned so it does not obscure or overprint hydrographic detail. Type for features in the water shall be positioned in the water. If possible, type shall not be positioned across the shoreline.

The limiting outline of the Built-up Area tint shall be dropped when it overprints linear features (Streams, Roads, and Railroads, etc), or if the space between the symbols is < 0.5mm.

Feature Information

Specification:	*LTDS*	**Feature Origin:**	*Topographic*	**Symbol:**	AR_Black_PumpingStation
Symbol Criteria:	All			**Label:**	[Product [1]], 6 point Condensed Mixed case if Product [1] <> [Other] or [No Information] and "pumping station", 6 point Condensed Mixed case; "Pumping station", 6 point Condensed Mixed case if Product [1] = [Other] or [No Information]
				Color:	Black (58600)

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Pumping Station (AQ116)

Point

Inclusion Condition:	*Area < 15,625 sq m.*

Notes

When symbolized feature is < 0.2 mm from a line feature, displace to 0.2 mm (map scale).

Like point features which coalesce in clusters of 3 or more will be thinned to form a representative pattern.

When Product Category (No Information), omit the Product label.

Type for features on land shall be positioned on land if possible. If type must be placed in the water, it shall be positioned so it does not obscure or overprint hydrographic detail. Type for features in the water shall be positioned in the water. If possible, type shall not be positioned across the shoreline.

Feature Information

Specification:	*LTDS*	**Feature Origin:**	*Topographic*	**Symbol:**	PT_Black_PumpingStation
Symbol Criteria:	All			**Label:**	[Product [1]], 6 point Condensed Mixed case if Product [1] <> [Other] or [No Information] and "pumping station", 6 point Condensed Mixed case; "Pumping station", 6 point Condensed Mixed case if Product [1] = [Other] or [No Information]
				Color:	Black (58600)

Pylon (AT042)

Point

Feature Description:	*A pylon or pole used to support one or more cables.*
Inclusion Condition:	*Height Above Surface Level >= 46 m.*

Notes

Label feature as per hierarchy for topo type placement to south neatline corners reading left to right:
1. Position hierarchy:
(a.) northeast (preferred position),
(b.) southeast (1st alternative),
(c.) northwest (2nd alternative,
(d.) southwest (3rd alternative),
(e.) top-centered (4th alternative),
(f.) bottom-centered (5th alternative):
(Hierarchy is based on type positioning so as to avoid overprinting other type or obscuring detail.);
2. Minimum space between type placement and feature symbol is 0.5 mm;
3. This method of type placement shall be used for areal features when space does not permit labeling within that feature.

Feature Information

Specification:	*LTDS*	**Feature Origin:**	*Topographic*	**Symbol:**	PT_Blue072_VO
Symbol Criteria:	Height Above Surface Level >= 46			**Label:**	"Pylon", 6 point Condensed Mixed case ; [Highest Elevation], 6 point Condensed Bold Mixed case; "([Height Above Surface Level])", 6 point Condensed Mixed case
				Color:	Blue (46351)

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Racetrack (AK130)

Curve

Feature Description:	*A ground or tract marked out for racing.*
Inclusion Condition:	*Length >= 125 meters OR Navigation Landmark: (True)*

Notes

Area and line features will be generalized to detail compatible with scale.

Label feature as per hierarchy for topo type placement to south neatline corners reading left to right:
1. Position hierarchy:
(a.) northeast (preferred position
(b.) southeast (1st alternative)
(c.) northwest (2nd alternative
(d.) southwest (3rd alternative)
(e.) top-centered (4th alternative)
(f.) bottom-centered (5th alternative)
(Hierarchy is based on type positioning so as to avoid overprinting other type or obscuring detail.)
2. Minimum space between type placement and feature symbol is 0.5 mm
3. This method of type placement shall be used for areal features when space does not permit labeling within that feature.

Type for features on land shall be positioned on land if possible. If type must be placed in the water, it shall be positioned so it does not obscure or overprint hydrographic detail. Type for features in the water shall be positioned in the water. If possible, type shall not be positioned across the shoreline.

Descriptive terms, e.g., "Canal" shall be shown if the name is not known. If the descriptive word appears in the name, for example, "PANAMA CANAL". The descriptive type shall not be shown, i.e. do not show "Panama Canal Canal".

Feature Information

Specification:	*LTDS*	**Feature Origin:**	*Topographic*	**Symbol:**	LN_Black_Racetrack
Symbol Criteria:	All			**Label:**	[Geographic Name Information : Full Name] if <> ("UNK") or (is not null), 6 point Condensed Mixed case; "race track", 6 point Condensed lower case; "Race track", 6 point Condensed Mixed case if Geographic Name Information : Full Name = ("UNK") or (is null)
				Color:	Black (58600)

Radar Range (BC033)

Surface

Feature Description:	*Indicates the coverage of a sea area by a radar surveillance station.*
Inclusion Condition:	*All Required*

Notes

None

Feature Information

Specification:	*Chart #1*	**Feature Origin:**	*Hydrographic*	**Symbol:**	M31
Symbol Criteria:	All			**Label:**	"Ra", 6 point Mixed case; "[Geographic Name Information : Full Name]", 6 point Mixed case
				Color:	Purple (96532)

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Radar Station (AT045)

Surface

Feature Description:	*A facility utilizing radar to detect and analyze objects (for example: aircraft, artificial satellites, asteroids, and/or missiles) and/or environmental phenomena (for example: tornadoes).*
Inclusion Condition:	*Area >= 15,625 sq m.*

Notes

None

Feature Information

Specification:	*LTDS*	**Feature Origin:**	*Topographic*	**Symbol:**	AR_Black_RadarStation
Symbol Criteria:	All			**Label:**	[Geographic Name Information : Full Name], 6 point Condensed Mixed case if available ; "Radar station", 6 point Condensed Mixed case
				Color:	Black (58600)

Radar Station (AT045)

Point

Inclusion Condition:	*Area < 15,625 sq m.*

Notes

None

Feature Information

Specification:	*LTDS*	**Feature Origin:**	*Topographic*	**Symbol:**	PT_Black_RadarStation
Symbol Criteria:	Height Above Surface Level < 46			**Label:**	[Geographic Name Information : Full Name], 6 point Condensed Mixed case if available ; "Radar station", 6 point Condensed Mixed case
				Color:	Black (58600)

Feature Information

Specification:	*LTDS*	**Feature Origin:**	*Topographic*	**Symbol:**	PT_Blue072_VO
Symbol Criteria:	Height Above Surface Level >= 46			**Label:**	"Radar station", 6 point Condensed Mixed case ; [Highest Elevation], 6 point Condensed Bold Mixed case; "([Height Above Surface Level])", 6 point Condensed Mixed case
				Color:	Blue (46351)

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Railway (AN010)

Curve

Feature Description:	*One or more railway tracks comprising a network that is operated for the conveyance of passengers and/or goods.*
Inclusion Condition:	*All required*

Notes

Contours will be adjusted to planimetric features.

If two Railroads are on separate roadbeds, and the symbols coalesce, the spacing between rail lines shall be 3.0 mm. When the distance between two parallel railroads is too small to plot to scale without the symbols coalescing, the distance between the center line is exaggerated to 3.0 mm.

Area and line features will be generalized to detail compatible with scale.

Type for features on land shall be positioned on land if possible. If type must be placed in the water, it shall be positioned so it does not obscure or overprint hydrographic detail. Type for features in the water shall be positioned in the water. If possible, type shall not be positioned across the shoreline.

Broad gauge Railroads shall be labeled parallel to the Railroad alignment.

The gauge label of narrow gauge Railroads with lines of varying widths shall be positioned parallel to the alignment of the gauge.

Electrified Railroads shall be labeled "ELECTRIFIED" positioned parallel to the Railroad alignment.

The label "ELECTRIFIED" shall be dropped when the Railroad name indicates the rail is electrified (Example: "OHIO ELECTRIC").

Names shall be shown and positioned parallel to the Railroad alignment.

When Location (On Ground Surface), no Location label is required.

Label shall be position above feature, reading from left to right (or to the left of the vertical feature, reading bottom to top), at a 0.5 mm distance and parallel to respective feature. Label shall preferably be positioned at the midpoint of the line segment or symbol; however, it may be displaced laterally along respective feature to avoid overprinting other symbols or labels. If space will not permit placing label parallel to feature, offset the label and use a leader line to identify its location along the feature.

Feature name, label, data information holder, and/or symbol shall be positioned, reading left to right, parallel to the tangent of the center of the southern neatline of the Map/Chart.

Railroad crosstie ticks may overlap Cut Line and Embankment symbols.

If Railroads and Piers/Wharves symbologies coalesce, only the Pier/Wharf and crossties of the Railroad shall be shown.

Railroad symbol ticks shall begin and end not less than 6.5 mm from the Bridge ticks.

Car lines, operating or non-operating, shall not be shown within Built-Up Areas.

When a Railroad Main line/Branch line enters a Railroad Yard, the Main line/Branch line shall remain at its portrayed lineweight whether or not the track terminates at, in or passes through the yard feature.

Show land tint inside the symbol. If the vertical reference category is (above surface /does not cover at High Water), show land tint only.

A blue 31% tint shall not overprint other blue 31% tints. If two blue tinted symbols, or one tinted symbol and water tint overprint, only one 31% tint shall be shown in the area.

When a Road or a Railroad coincide or coalesce at map scale when on the same bridge, the Railroad shall be suppressed to a distance of 0.25 mm back from the wing ticks at each end of the bridge.

If a Railroad Track is coincident with features (Tunnel (point)), or (Tunnel (line)), then suppress that section of the Railroad Track.

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Railway (AN010) *(Continued)*

Curve

Feature Information					
Specification:	*LTDS*	**Feature Origin:**	*Topographic*	**Symbol:**	LN_Black_Railway_ DoubleNormalGauge_Electric
Symbol Criteria:	(Condition of Facility = [Fully Functional] or [Other] or [No Information]) and (Railway Use [1] = [Logging] or [Rail Rapid Transit] or [Long-haul] or [Other] or [No Information]) and (Track Information : Railway Gauge Classification = [Broad] or [Standard] or [No Information]) and (Track Information : Railway in Road = [False] or [No Information]) and (Track Information : Railway Power Method = [Electrified Track] or [Overhead Electrified]) and (Track or Lane Count >= 2) and (Vertical Relative Location = [On Surface] or [No Information])			**Label:**	[Geographic Name Information: Full Name], 6 point Light Condensed Italics Upper case if available; "([Track or Lane Count]) TRACKS" if Track or Lane Count >2, 6 point Light Condensed Italics Upper case; "([Track Information : Railway Gauge]) m", 6 point Light Condensed Italics Upper case if Track Information : Railway Gauge Classification = [Broad] if available
				Color:	Black (58600)

Feature Information					
Specification:	*LTDS*	**Feature Origin:**	*Topographic*	**Symbol:**	LN_Black_Railway_ DoubleNarrowGauge_Electric _NotInUse
Symbol Criteria:	(Condition of Facility = [Under Construction] or [Abandoned] or [Dismantled] or [Ruined] or [Partially Functional] or [Partially Destroyed] or [Damaged] or [Disused]) and (Railway Use [1] = [Logging] or [Rail Rapid Transit] or [Long-haul] or [Other] or [No Information]) and (Track Information : Railway Gauge Classification = [Narrow]) and (Track Information : Railway in Road = [False] or [No Information]) and (Track Information : Railway Power Method = [Electrified Track] or [Overhead Electrified]) and (Track or Lane Count >= 2) and (Vertical Relative Location = [On Surface] or [No Information])			**Label:**	[Geographic Name Information: Full Name], 6 point Light Condensed Italics Upper case if available; "([Track or Lane Count]) TRACKS" if Track or Lane Count > 2, 6 point Light Condensed Italics Upper case; if Condition of Facility = [Under Construction] or [Partially Functional] label as "CONSTRUCTION", 6 point Light Condensed Italics Upper case; if Condition of Facility = [Abandoned] or [Destroyed] or [Dismantled] or [Ruined] or [Partially Destroyed] or [Damaged] or [Disused], label as "NOT USABLE", 6 point Light Condensed Italics Upper case
				Color:	Black (58600)

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Railway (AN010) *(Continued)*

Curve

Feature Information					
Specification:	*LTDS*	**Feature Origin:**	*Topographic*	**Symbol:**	LN_Black_Railway_ SingleNarrowGauge_Non-Electric_NotInUse
Symbol Criteria:	(Condition of Facility = [Under Construction] or [Abandoned] or [Dismantled] or [Ruined] or [Partially Functional] or [Partially Destroyed] or [Damaged] or [Disused]) and (Railway Use [1] = [Logging] or [Rail Rapid Transit] or [Long-haul] or [Other] or [No Information]) and (Track Information : Railway Gauge Classification = [Narrow]) and (Track Information : Railway in Road = [False] or [No Information]) and (Track Information : Railway Power Method = [Non-electrified] or [Other] or [No Information]) and (Track or Lane Count = 1) and (Vertical Relative Location = [On Surface] or [No Information])				**Label:**	[Geographic Name Information: Full Name], 6 point Light Condensed Italics Upper case if available; if Condition of Facility = [Under Construction] or [Partially Functional] label as "CONSTRUCTION", 6 point Light Condensed Italics Upper case; if Condition of Facility = [Abandoned] or [Destroyed] or [Dismantled] or [Ruined] or [Partially Destroyed] or [Damaged] or [Disused], label as "NOT USABLE", 6 point Light Condensed Italics Upper case
				Color:	Black (58600)

Feature Information					
Specification:	*LTDS*	**Feature Origin:**	*Topographic*	**Symbol:**	LN_Black_Railway_ SingleNarrowGauge_Electric _NotInUse
Symbol Criteria:	(Condition of Facility = [Under Construction] or [Abandoned] or [Dismantled] or [Ruined] or [Partially Functional] or [Partially Destroyed] or [Damaged] or [Disused]) and (Railway Use [1]= [Logging] or [Rail Rapid Transit] or [Long-haul] or [Other] or [No Information]) and (Track Information : Railway Gauge Classification = [Narrow]) and (Track Information : Railway in Road = [False] or [No Information]) and (Track Information : Railway Power Method = [Electrified Track] or [Overhead Electrified])and (Track or Lane Count = 1) and (Vertical Relative Location = [On Surface] or [No Information])				**Label:**	[Geographic Name Information: Full Name], 6 point Light Condensed Italics Upper case if available; "([Track or Lane Count]) TRACKS" if Track or Lane Count > 2, 6 point Light Condensed Italics Upper case; if Condition of Facility = [Under Construction] or [Partially Functional] label as "CONSTRUCTION", 6 point Light Condensed Italics Upper case; if Condition of Facility = [Abandoned] or [Destroyed] or [Dismantled] or [Ruined] or [Partially Destroyed] or [Damaged] or [Disused], label as "NOT USABLE", 6 point Light Condensed Italics Upper case
				Color:	Black (58600)

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Railway (AN010) *(Continued)*

Curve

Feature Information					
Specification:	*LTDS*	**Feature Origin:**	*Topographic*	**Symbol:**	LN_Black_Car-Line
Symbol Criteria:	(Condition of Facility = [Fully Functional] or [Other] or [No Information]) and (Railway Use [1] = [Carline] or [Marine Railway] or [Tramway] or [Funicular] or [Museum] or [Automated Transit System]) and (Track Information : Railway in Road = [False] or [No Information]) and (Vertical Relative Location = [On Surface] or [No Information])			**Label:**	[Geographic Name Information : Full Name], 6 point Light Condensed Italics Upper case if available; "([Railway Use [1]])", 6 point Light Condensed Italics Upper case
				Color:	Black (58600)

Feature Information					
Specification:	*LTDS*	**Feature Origin:**	*Topographic*	**Symbol:**	LN_Black_Car-Line_NotInUse
Symbol Criteria:	(Condition of Facility = [Under Construction] or [Abandoned] or [Dismantled] or [Ruined] or [Partially Functional] or [Partially Destroyed or [Damaged] or [Disused]) and (Railway Use [1] = [Carline] or [Marine Railway] or [Tramway] or [Funicular] or [Museum] or [Automated Transit System]) and (Track Information : Railway in Road = [False] or [No Information]) and (Vertical Relative Location = [On Surface] or [No Information])			**Label:**	[Geographic Name Information: Full Name], 6 point Light Condensed Italics Upper case if available; "([Railway Use [1]])", 6 point Light Condensed Italics Upper case; if Condition of Facility = [Under Construction] or [Partially Functional] label as "CONSTRUCTION", 6 point Light Condensed Italics Upper case; if Condition of Facility =[Abandoned] or [Destroyed] or [Dismantled] or [Ruined] or [Partially Destroyed] or [Damaged] or [Disused], label as "NOT USABLE", 6 point Light Condensed Italics Upper case
				Color:	Black (58600)

Feature Information					
Specification:	*LTDS*	**Feature Origin:**	*Topographic*	**Symbol:**	LN_Black_Railway_In_Road
Symbol Criteria:	(Condition of Facility = [Fully Functional] or [Other] or [No Information]) and (Track Information : Railway in Road = [True]) and (Vertical Relative Location = [On Surface] or [No Information])			**Label:**	[Geographic Name Information : Full Name], 6 point Light Condensed Italics Upper case if available
				Color:	Black (58600)

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Railway (AN010) *(Continued)*

Curve

Feature Information					
Specification:	*LTDS*	**Feature Origin:**	*Topographic*	**Symbol:**	LN_Black_Railway_ DoubleNormalGauge_Non-Electric
Symbol Criteria:	(Condition of Facility = [Fully Functional] or [Other] or [No Information]) and (Railway Use [1] = [Logging] or [Rail Rapid Transit] or [Long-haul] or [Other] or [No Information]) and (Track Information : Railway Gauge Classification = [Broad] or [Standard] or [No Information]) and (Track Information : Railway in Road = [False] or [No Information]) and (Track Information : Railway Power Method = [Non-electrified] or [Other] or [No Information]) and (Track or Lane Count >= 2) and (Vertical Relative Location = [On Surface] or [No Information])			**Label:**	[Geographic Name Information: Full Name], 6 point Light Condensed Italics Upper case if available; "([Track or Lane Count]) TRACKS" if Track or Lane Count > 2, 6 point Light Condensed Italics Upper case; "([Track Information : Railway Gauge]) m", 6 point Light Condensed Italics Upper case, ("m" in Lower case) if Track Information : Railway Gauge Classification = [Broad] if available
				Color:	Black (58600)

Feature Information					
Specification:	*LTDS*	**Feature Origin:**	*Topographic*	**Symbol:**	LN_Black_Railway_ SingleNormalGauge_Electric
Symbol Criteria:	(Condition of Facility = [Fully Functional] or [Other] or [No Information]) and (Railway Use [1] = [Logging] or [Rail Rapid Transit] or [Long-haul] or [Other] or [No Information]) and (Track Information : Railway Gauge Classification = [Broad] or [Standard] or [No Information]) and (Track Information : Railway in Road = [False] or [No Information]) and (Track Information : Railway Power Method = [Electrified Track] or [Overhead Electrified]) and (Track or Lane Count = 1) and (Vertical Relative Location = [On Surface] or [No Information])			**Label:**	[Geographic Name Information: Full Name]if available, 6 point Upper case Light condensed Italic ; "([Track Information : Railway Gauge]) m" if Track Information : Railway Gauge Classification = (Broad), 6 point Light Condensed Italics Upper case, ("m" is lower case);
				Color:	Black (58600)

Feature Information					
Specification:	*LTDS*	**Feature Origin:**	*Topographic*	**Symbol:**	LN_Black_Railway_ SingleNormalGauge_Non-Electric
Symbol Criteria:	(Condition of Facility = [Fully Functional] or [Other] or [No Information]) and (Railway Use [1] = [Logging] or [Rail Rapid Transit] or [Long-haul] or [Other] or [No Information]) and (Track Information : Railway Gauge Classification = [Broad] or [Standard] or [No Information]) and (Track Information : Railway in Road = [False] or [No Information]) and (Track Information : Railway Power Method = [Non-electrified] or [Other] or [No Information]) and (Track or Lane Count = 1) and (Vertical Relative Location = [On Surface] or [No Information])			**Label:**	[Geographic Name Information: Full Name], 6 point Light Condensed Italics Upper case if available; "([Track Information : Railway Gauge]) m", 6 point Light Condensed Italics Upper case ("m" in lower case)if Track Information : Railway Gauge Classification = [Broad] ;
				Color:	Black (58600)

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Railway (AN010) *(Continued)*

Curve

Feature Information					
Specification:	*LTDS*	**Feature Origin:**	*Topographic*	**Symbol:**	LN_Black_Railway_ DoubleNarrowGauge_Non-Electric
Symbol Criteria:	(Condition of Facility = [Fully Functional] or [Other] or [No Information]) and (Railway Use [1] = [Logging] or [Rail Rapid Transit] or [Long-haul] or [Other] or [No Information]) and (Track Information : Railway Gauge Classification = [Narrow]) and (Track Information : Railway in Road = [False] or [No Information]) and (Track Information : Railway Power Method = [Non-electrified] or [Other] or [No Information]) and (Track or Lane Count >= 2) and (Vertical Relative Location = [On Surface] or [No Information])			**Label:**	Geographic Name Information: Full Name, 6 point Light Condensed Italics Upper case if available; Label: "([Track or Lane Count]) TRACKS" if Track or Lane Count > 2, 6 point Light Condensed Italics Upper case;
				Color:	Black (58600)

Feature Information					
Specification:	*LTDS*	**Feature Origin:**	*Topographic*	**Symbol:**	LN_Black_Railway_ DoubleNarrowGauge_Electric
Symbol Criteria:	(Condition of Facility = [Fully Functional] or [Other] or [No Information]) and (Railway Use [1] = [Logging] or [Rail Rapid Transit] or [Long-haul] or [Other] or [No Information]) and (Track Information : Railway Gauge Classification = [Narrow]) and (Track Information : Railway in Road = [False] or [No Information]) and (Track Information : Railway Power Method = [Electrified Track] or [Overhead Electrified]) and (Track or Lane Count >= 2) and (Vertical Relative Location = [On Surface] or [No Information])			**Label:**	[Geographic Name Information: Full Name], 6 point Light Condensed Italics Upper case if available: "([Track or Lane Count]) TRACKS", 6 point Light Condensed Italics Upper case if [Track or Lane Count] > 2
				Color:	Black (58600)

Feature Information					
Specification:	*LTDS*	**Feature Origin:**	*Topographic*	**Symbol:**	LN_Black_Railway_ SingleNarrowGauge_Non-Electric
Symbol Criteria:	(Condition of Facility = [Fully Functional] or [Other] or [No Information]) and (Railway Use [1] = [Logging] or [Rail Rapid Transit] or [Long-haul] or [Other] or [No Information]) and (Track Information : Railway Gauge Classification = [Narrow]) and (Track Information : Railway in Road = [False] or [No Information]) and (Track Information : Railway Power Method = [Non-electrified] or [Other] or [No Information]) and (Track or Lane Count = 1) and (Vertical Relative Location = [On Surface] or [No Information])			**Label:**	[Geographic Name Information: Full Name], 6 point Light Condensed Italics Upper case if available
				Color:	Black (58600)

Feature Information					
Specification:	*LTDS*	**Feature Origin:**	*Topographic*	**Symbol:**	LN_Black_Railway_ SingleNarrowGauge_Electric
Symbol Criteria:	(Condition of Facility = [Fully Functional] or [Other] or [No Information]) and (Railway Use [1] = [Logging] or [Rail Rapid Transit] or [Long-haul] or [Other] or [No Information]) and (Track Information : Railway Gauge Classification = [Narrow]) and (Track Information : Railway in Road = [False] or [No Information]) and (Track Information : Railway Power Method = [Electrified Track] or [Overhead Electrified]) and (Track or Lane Count = 1) and (Vertical Relative Location = [On Surface] or [No Information])			**Label:**	[Geographic Name Information: Full Name], 6 point Light Condensed Italics Upper case if available;
				Color:	Black (58600)

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Railway (AN010) *(Continued)*

Curve

Feature Information					
Specification:	*LTDS*	**Feature Origin:**	*Topographic*	**Symbol:**	LN_Black_Railway_ DoubleNormalGauge_Non-Electric_NotInUse
Symbol Criteria:	(Condition of Facility = [Under Construction] or [Abandoned] or [Dismantled] or [Ruined] or [Partially Functional] or [Partially Destroyed] or [Damaged] or [Disused]) and (Railway Use [1] = [Logging] or [Rail Rapid Transit] or [Long-haul] or [Other] or [No Information]) and (Track Information : Railway Gauge Classification = [Broad] or [Standard] or [No Information]) and (Track Information : Railway in Road = [False] or [No Information]) and (Track Information : Railway Power Method = [Non-electrified] or [Other] or [No Information]) and (Track or Lane Count >= 2) and (Vertical Relative Location = [On Surface] or [No Information])			**Label:**	[Geographic Name Information: Full Name], 6 point Light Condensed Italics Upper case if available; "([Track or Lane Count]) TRACKS" if Track or Lane Count > 2, 6 point Light Condensed Italics Upper case; "([Track Information : Railway Gauge]) m", 6 point Light Condensed Italics Upper case, ("m" in lower case) if Track Information : Railway Gauge Classification = [Broad] if available; if Condition of Facility = [Under Construction] or [Partially Functional] label as "CONSTRUCTION", 6 point Light Condensed Italics Upper case; if Condition of Facility = [Abandoned] or [Destroyed] or [Dismantled] or [Ruined] or [Partially Destroyed] or [Damaged] or [Disused], label as "NOT USABLE", 6 point Light Condensed Italics Upper case
				Color:	Black (58600)

Feature Information					
Specification:	*LTDS*	**Feature Origin:**	*Topographic*	**Symbol:**	LN_Black_Railway_ DoubleNormalGauge_Electric_ NotInUse
Symbol Criteria:	(Condition of Facility = [Under Construction] or [Abandoned] or [Dismantled] or [Ruined] or [Partially Functional] or [Partially Destroyed] or [Damaged] or [Disused]) and (Railway Use [1] = [Logging] or [Rail Rapid Transit] or [Long-haul] or [Other] or [No Information]) and (Track Information : Railway Gauge Classification = [Broad] or [Standard] or [No Information]) and (Track Information : Railway in Road = [False] or [No Information]) and (Track Information : Railway Power Method = [Electrified Track] or [Overhead Electrified]) and (Track or Lane Count >= 2) and (Vertical Relative Location = [On Surface] or [No Information])			**Label:**	[Geographic Name Information: Full Name], 6 point Light Condensed Italics Upper case if available; "([Track or Lane Count]) TRACKS" if Track or Lane Count > 2, 6 point Light Condensed Italics Upper case; "([Track Information : Railway Gauge]) m", 6 point Light Condensed Italics Upper case, ("m" in Lower case) if Track Information : Railway Gauge Classification = [Broad] if available; if Condition of Facility = [Under Construction] or [Partially Functional] label as "CONSTRUCTION", 6 point Light Condensed Italics Upper case; if Condition of Facility = [Abandoned] or [Destroyed] or [Dismantled] or [Ruined] or [Partially Destroyed] or [Damaged] or [Disused], label as "NOT USABLE", 6 point Light Condensed Italics Upper case
				Color:	Black (58600)

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Railway (AN010) *(Continued)*

Curve

Feature Information					
Specification:	*LTDS*	**Feature Origin:**	*Topographic*	**Symbol:**	LN_Black_Railway_ SingleNormalGauge_Non-Electric_NotInUse
Symbol Criteria:	(Condition of Facility = [Under Construction] or [Abandoned] or [Dismantled] or [Ruined] or [Partially Functional] or [Partially Destroyed] or [Damaged] or [Disused]) and (Railway Use [1] = [Logging] or [Rail Rapid Transit] or [Long-haul] or [Other] or [No Information]) and (Track Information : Railway Gauge Classification = [Broad] or [Standard] or [No Information]) and (Track Information : Railway in Road = [False] or [No Information]) and (Track Information : Railway Power Method = [Non-electrified] or [Other] or [No Information]) and (Track or Lane Count = 1) and (Vertical Relative Location = [On Surface] or [No Information])				**Label:**	[Geographic Name Information: Full Name], 6 point Light Condensed Italics Upper case if available; "([Track Information : Railway Gauge]) m", 6 point Light Condensed Italics Upper case, ("m" in lower case) if Track Information : Railway Gauge Classification = [Broad] if available; if Condition of Facility = [Under Construction] or [Partially Functional] label as "CONSTRUCTION", 6 point Light Condensed Italics Upper case; if Condition of Facility = [Abandoned] or [Destroyed] or [Dismantled] or [Ruined] or [Partially Destroyed] or [Damaged] or [Disused], label as "NOT USABLE", 6 point Light Condensed Italics Upper case
				Color:	Black (58600)

Feature Information					
Specification:	*LTDS*	**Feature Origin:**	*Topographic*	**Symbol:**	LN_Black_Railway_ SingleNormalGauge_Electric_ NotInUse
Symbol Criteria:	(Condition of Facility = [Under Construction] or [Abandoned] or [Dismantled] or [Ruined] or [Partially Functional] or [Partially Destroyed] or [Damaged] or [Disused]) and (Railway Use [1] = [Logging] or [Rail Rapid Transit] or [Long-haul] or [Other] or [No Information]) and (Track Information : Railway Gauge Classification = [Broad] or [Standard] or [No Information]) and (Track Information : Railway in Road = [False] or [No Information]) and (Track Information : Railway Power Method = [Electrified Track] or [Overhead Electrified]) and (Track or Lane Count = 1) and (Vertical Relative Location = [On Surface] or [No Information])				**Label:**	[Geographic Name Information: Full Name], 6 point Light Condensed Italics Upper case if available; "([Track Information : Railway Gauge]) m", 6 point Light Condensed Italics Upper case, ("m" in lower case) if Track Information : Railway Gauge Classification = [Broad] if available; if Condition of Facility = [Under Construction] or [Partially Functional] label as "CONSTRUCTION", 6 point Light Condensed Italics Upper case; if Condition of Facility = [Abandoned] or [Destroyed] or [Dismantled] or [Ruined] or [Partially Destroyed] or [Damaged] or [Disused], label as "NOT USABLE", 6 point Light Condensed Italics Upper case
				Color:	Black (58600)

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Railway (AN010) *(Continued)*

Curve

Feature Information					
Specification:	*LTDS*	**Feature Origin:**	*Topographic*	**Symbol:**	LN_Black_Railway_DoubleNarrowGauge_Non-Electric_NotInUse
Symbol Criteria:	(Condition of Facility = [Under Construction] or [Abandoned] or [Dismantled] or [Ruined] or [Partially Functional] or [Partially Destroyed] or [Damaged] or [Disused]) and (Railway Use [1] = [Logging] or [Rail Rapid Transit] or [Long-haul] or [Other] or [No Information]) and (Track Information : Railway Gauge Classification = [Narrow]) and (Track Information : Railway in Road = [False] or [No Information]) and (Track Information : Railway Power Method = [Non-electrified] or [Other] or [No Information]) and (Track or Lane Count >= 2) and (Vertical Relative Location = [On Surface] or [No Information])			**Label:**	[Geographic Name Information: Full Name], 6 point Light Condensed Italics Upper case if available; "([Track or Lane Count]) TRACKS" if Track or Lane Count > 2, 6 point Light Condensed Italics Upper case; if Condition of Facility = [Under Construction] or [Partially Functional] label as "CONSTRUCTION", 6 point Light Condensed Italics Upper case; if Condition of Facility = [Abandoned] or [Destroyed] or [Dismantled] or [Ruined] or [Partially Destroyed] or [Damaged] or [Disused], label as "NOT USABLE", 6 point Light Condensed Italics Upper case
				Color:	Black (58600)

Feature Information					
Specification:	*LTDS*	**Feature Origin:**	*Topographic*	**Symbol:**	LN_Black_Railway_In_Road_NotInUse
Symbol Criteria:	(Condition of Facility = [Under Construction] or [Abandoned] or [Dismantled] or [Ruined] or [Partially Functional] or [Partially Destroyed or [Damaged] or [Disused]) and (Track Information : Railway in Road = [True]) and (Vertical Relative Location = [On Surface] or [No Information])			**Label:**	[Geographic Name Information: Full Name], 6 point Light Condensed Italics Upper case if available; if Condition of Facility = [Under Construction] or [Partially Functional] label as "CONSTRUCTION", 6 point Light Condensed Italics Upper case; if Condition of Facility = [Abandoned] or [Destroyed] or [Dismantled] or [Ruined] or [Partially Destroyed] or [Damaged] or [Disused], label as "NOT USABLE", 6 point Light Condensed Italics Upper case
				Color:	Black (58600)

Feature Information					
Specification:	*LTDS*	**Feature Origin:**	*Topographic*	**Symbol:**	LN_Black_Railway_Destroyed
Symbol Criteria:	(Condition of Facility = [Destroyed])			**Label:**	"Destroyed railroad", 6 point Light Condensed Italics Mixed case
				Color:	Black (58600)

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Railway Sidetrack (AN050)

Curve

Feature Description:	*A stretch of railway track connected to a main railway and used for temporary storage, passing, loading, and/or unloading.*
Inclusion Condition:	*All required*

Notes

Contours will be adjusted to planimetric features.

If a Railroad and Sidetrack or Spur coalesce, the Sidetrack center line shall be displaced to 3.0 mm from the Railroad center line.

Area and line features will be generalized to detail compatible with scale.

If the Railroad Siding/Spur (Sidetrack) or (Passing), the Rail Gauge, Existence and Railroad Power Source shall be equal to associated Railroad.

Spurs and Sidetracks shall not be shown in Built-up Areas when their symbology coalesces with other features.

If a feature is not associated with (touching, stacked_on, etc.) a Railroad Track, omit the feature.

Feature Information

Specification:	*LTDS*	**Feature Origin:**	*Topographic*	**Symbol:**	LN_Black_RailwaySidetrack _StandardNonElectric
Symbol Criteria:	(Condition of Facility = [Fully Functional] or [Other] or [No Information]) and (Track Information : Railway Gauge Classification = [Broad] or [Standard] or [No Information]) and (Track Information : Railway Power Method = [Non-electrified] or [Other] or [No Information]) and (Vertical Relative Location = [On Surface] or [No Information])			**Label:**	None
				Color:	N/A

Feature Information

Specification:	*LTDS*	**Feature Origin:**	*Topographic*	**Symbol:**	LN_Black_RailwaySidetrack _SingleNormalGauge_Electric
Symbol Criteria:	(Condition of Facility = [Fully Functional] or [Other] or [No Information]) and (Track Information : Railway Gauge Classification = [Broad] or [Standard] or [No Information]) and (Track Information : Railway Power Method = [Electrified Track] or [Overhead Electrified]) and (Vertical Relative Location = [On Surface] or [No Information])			**Label:**	None
				Color:	N/A

Feature Information

Specification:	*LTDS*	**Feature Origin:**	*Topographic*	**Symbol:**	LN_Black_RailwaySidetrack _SingleNarrowGauge_Non-Electric
Symbol Criteria:	(Condition of Facility = [Fully Functional] or [Other] or [No Information]) and (Track Information : Railway Gauge Classification = [Narrow]) and (Track Information : Railway Power Method = [Non-electrified] or [Other] or [No Information]) and (Vertical Relative Location = [On Surface] or [No Information])			**Label:**	None
				Color:	N/A

Feature Information

Specification:	*LTDS*	**Feature Origin:**	*Topographic*	**Symbol:**	LN_Black_RailwaySidetrack _SingleNarrowGauge_Electric
Symbol Criteria:	(Condition of Facility = [Fully Functional] or [Other] or [No Information]) and (Track Information : Railway Gauge Classification = [Narrow]) and (Track Information : Railway Power Method = [Electrified Track] or [Overhead Electrified]) and (Vertical Relative Location = [On Surface] or [No Information])			**Label:**	None
				Color:	N/A

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Railway Sidetrack (AN050) *(Continued)*

Curve

Feature Information					
Specification:	*LTDS*	**Feature Origin:**	*Topographic*	**Symbol:**	LN_Black_RailwaySidetrack _SingleNormalGauge_Electric_ NotInUse
Symbol Criteria:	(Condition of Facility = [Under Construction] or [Abandoned] or [Dismantled] or [Ruined] or [Partially Functional] or [Partially Destroyed or [Damaged] or [Disused]) and (Track Information : Railway Gauge Classification = [Broad] or [Standard] or [No Information]) and (Track Information : Railway Power Method = [Electrified Track] or [Overhead Electrified]) and (Vertical Relative Location = [On Surface] or [No Information])			**Label:**	None
				Color:	N/A

Feature Information					
Specification:	*LTDS*	**Feature Origin:**	*Topographic*	**Symbol:**	LN_Black_RailwaySidetrack _SingleNarrowGauge_Electric_ NotInUse
Symbol Criteria:	(Condition of Facility = [Under Construction] or [Abandoned] or [Dismantled] or [Ruined] or [Partially Functional] or [Partially Destroyed or [Damaged] or [Disused]) and (Track Information : Railway Gauge Classification = [Narrow]) and (Track Information : Railway Power Method = [Electrified Track] or [Overhead Electrified]) and (Vertical Relative Location = [On Surface] or [No Information])			**Label:**	None
				Color:	N/A

Feature Information					
Specification:	*LTDS*	**Feature Origin:**	*Topographic*	**Symbol:**	LN_Black_RailwaySidetrack _SingleNormalGauge_ NonElectric_ NotInUse
Symbol Criteria:	(Condition of Facility = [Under Construction] or [Abandoned] or [Dismantled] or [Ruined] or [Partially Functional] or [Partially Destroyed or [Damaged] or [Disused]) and (Track Information : Railway Gauge Classification = [Broad] or [Standard] or [No Information]) and (Track Information : Railway Power Method = [Non-electrified] or [Other] or [No Information]) and (Vertical Relative Location = [On Surface] or [No Information])			**Label:**	None
				Color:	N/A

Feature Information					
Specification:	*LTDS*	**Feature Origin:**	*Topographic*	**Symbol:**	LN_Black_RailwaySidetrack _SingleNarrowGauge_ NonElectric_NotInUse
Symbol Criteria:	(Condition of Facility = [Under Construction] or [Abandoned] or [Dismantled] or [Ruined] or [Partially Functional] or [Partially Destroyed or [Damaged] or [Disused]) and (Track Information : Railway Gauge Classification = [Narrow]) and (Track Information : Railway Power Method = [Non-electrified] or [Other] or [No Information]) and (Vertical Relative Location = [On Surface] or [No Information])			**Label:**	None
				Color:	N/A

Railway Turntable (AN075)

Point

Feature Description:	*A rotating platform with railway tracks used for turning locomotives and/or railway carriages.*
Inclusion Condition:	*All required*

Notes

Like point features which coalesce in clusters of 3 or more will be thinned to form a representative pattern.

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Railway Turntable (AN075) *(Continued)*

Point

Feature Information					
Specification:	*LTDS*	**Feature Origin:**	*Topographic*	**Symbol:**	PT_Black_RailwayTurntable
Symbol Criteria:	All			**Label:**	[Geographic Name Information: Full Name], 6 point Light Condensed Italics Upper case if available; If Condition of Facility = [Under Construction] or [Partially Functional], label as "(construction)", 6 point Light Condensed Italics lower case ; if Condition of Facility = [Abandoned] or [Destroyed] or [Partially Destroyed] or [Damaged] or [Disused], label as "(not useable)", 6 point Light Condensed Italics lower case
				Color:	Black (58600)

Railway Yard (AN060)

Surface

Feature Description:	*A system of railway tracks and associated structures that are located within defined limits and that provide for loading, unloading, and/or assembling trains.*
Inclusion Condition:	*All required*

Notes

When 2 or more similar area features are separated by less than 0.5 mm at chart scale, the feature will be agglomerated.

Coincident similar area features will be blended to form a single feature.

Area and line features will be generalized to detail compatible with scale.

If area is not large enough to place type within, move to outside.

Type for features on land shall be positioned on land if possible. If type must be placed in the water, it shall be positioned so it does not obscure or overprint hydrographic detail. Type for features in the water shall be positioned in the water. If possible, type shall not be positioned across the shoreline.

When Railroad Yard, or any part, is an area feature and does not converge on itself (open at one end), no hardline lineweight symbol shall be shown closing or connecting the feature symbol at the open end.

Interior Track alignment hall run parallel to the longest axis of the feature and conform to the true shape of the feature.

If a feature is not associated with (touching, stacked_on, etc.) a Railroad Track, omit the feature.

Feature Information					
Specification:	*LTDS*	**Feature Origin:**	*Topographic*	**Symbol:**	AR_Black_RailwayYard
Symbol Criteria:	All			**Label:**	[Geographic Name Information: Full Name], 6 point Light Condensed Italics Upper case if available; If Condition of Facility = [Under Construction], label as "Construction", 6 point Light Condensed Italics Mixed case ; if Condition of Facility = [Abandoned] or [Dismantled] or [Ruined] or [Partially Functional] or [Partially Destroyed] or [Damaged] or [Disused], label as "Not useable", 6 point Light Condensed Italics Mixed case
				Color:	Black (58600)

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Rapids (BH120)

Surface

Feature Description:	*Portions of a stream with accelerated current where it descends rapidly but without a break in the slope of the bed sufficient to form a waterfall.*
Inclusion Condition:	*Width >= 25 m.*

Notes

Area and line features will be generalized to detail compatible with scale.

Feature will be generalized to provide a more aesthetic contoured feature (i.e., smoothed).

Omit if not shown in conjunction with drainage feature.

Rapid symbols shall be shown on double-line River/Stream perpendicular to the River/Stream centerline. The Rapids Length is to be considered coincident with the River/Stream centerline.

If a feature is not associated with (touching) a river, omit the feature.

Feature Information

Specification:	*LTDS*	**Feature Origin:**	*Topographic*	**Symbol:**	AR_Cyan_Rapids
Symbol Criteria:	Length or Diameter >= 50			**Label:**	None
				Color:	N/A

Rapids (BH120)

Curve

Inclusion Condition:	*Width < 25 m.*

Notes

Area and line features will be generalized to detail compatible with scale.

Feature will be generalized to provide a more aesthetic contoured feature (i.e., smoothed).

Label feature as per hierarchy for topo type placement to south neatline corners reading left to right:
1. Position hierarchy: (a.) northeast (preferred position), (b.) southeast (1st alternative), (c.) northwest (2nd alternative, (d.) southwest (3rd alternative), (e.) top-centered (4th alternative), (f.) bottom-centered (5th alternative): (Hierarchy is based on type positioning so as to avoid overprinting other type or obscuring detail.);
2. Minimum space between type placement and feature symbol is 0.5 mm;
3. This method of type placement shall be used for areal features when space does not permit labeling within that feature.

Omit if not shown in conjunction with drainage feature.

Rapid symbols shall be shown on double-line River/Stream perpendicular to the River/Stream centerline. The Rapids Length is to be considered coincident with the River/Stream centerline.

If a feature is not associated with (touching) a river, omit the feature.

Feature Information

Specification:	*LTDS*	**Feature Origin:**	*Topographic*	**Symbol:**	LN_Cyan_Rapids
Symbol Criteria:	Length or Diameter < 50			**Label:**	"Rapids", 6 point Condensed Italics Mixed case
				Color:	Blue (48253)

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Recycling Site (AB010)

Surface

Feature Description:	*A site engaged in the wrecking, dismantling, storage, recycling, and/or disposal of discarded or scrap products.*
Inclusion Condition:	*Area: >= 15,625 sq m.*

Notes

Coincident similar area features will be blended to form a single feature.

Area and line features will be generalized to detail compatible with scale.

Type for features on land shall be positioned on land if possible. If type must be placed in the water, it shall be positioned so it does not obscure or overprint hydrographic detail. Type for features in the water shall be positioned in the water. If possible, type shall not be positioned across the shoreline.

Limiting lines of feature are omitted if it coalesces with a road (AP030).

If a clearing exists inside of the area feature, and the size of the clearing is equal to or greater than the area for the surrounding area feature, the clearing is shown as an open space inside the surround feature. If the area of the clearing is less than the area for the surrounding feature, the clearing is deleted and absorbed into the surrounding area feature.

If two area features with the same feature code do not connect at any point, and have a space between them less than 2.5 mm at map/chart scale, delete the open space that is less than 2.5 mm between the features and combine them into one area feature.

If a portion of an area feature has a minimum width of less than 2.5 mm at map/chart scale, delete that portion of the area feature that is not at least 2.5 mm wide, measured from perimeter to perimeter. If the deletion of the portion of the area based on the above criteria will reduce the area of the remaining portion of the area feature to below the minimum area inclusion, do not delete the narrow portion of the feature that is less than 2.5 mm wide. If the deletion of the portion of the area based on the above criteria will split two larger areas connected by a narrow strip into two separate areas, either of which would be below minimum area inclusion, do not delete the narrow portion of the feature that is less than 2.5 mm wide.

Feature Information

Specification:	*LTDS*	**Feature Origin:**	*Topographic*	**Symbol:**	AR_Black_RecyclingSite
Symbol Criteria:	All			**Label:**	"Recycling site", 6 point Condensed Mixed case
				Color:	Black (58600)

Reef (BD120)

Surface

Feature Description:	*A mass of rock or coral which either reaches close to the sea surface or is exposed at low tide, posing a hazard to navigation.*
Inclusion Condition:	*Area >= 625m*

Notes

Submerged coral reefs and pinnacles must be charted by the same symbols as isolated rocks, using the abbreviation for coral as appropriate as shown on symbol K16 (Chart #1).

If point symbol overprints shoreline, the Hydrographic Depth type, posicut, or graphic in the center of the symbol shall be displaced seaward until they no longer overprint the shoreline. If the danger curve (dotted perimeter line) overprints the shoreline, that portion of the dotted perimeter line falling on land shall be deleted.

Rocks and isolated coral heads within submerged reefs shall be charted using the appropriate rock symbol. When depths over selected rocks are shown, an overall depth over the reef is not required, since the depth over the reef is shown by the depth of the shallowest rock. Where it is possible to chart depth information for separate rocks, the shallowest depth over the reef shall be shown by (Hydrographic Depth) or (Hydrographic Drying Height) on reef.

Use the following abbreviations for Horizontal Accuracy and Existence values:
If Position Approximate, label "PA";
If Position Doubtful, label "PD";
If Existence Doubtful, label "ED";
If Reported, label "Rep".

Date shall only be shown if Existence = Reported. Date shall be shown in parentheses if available. Always show "Rep" and date on the same line.

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Reef (BD120) ***(Continued)***

Surface

Notes

Following IHO practice, features whose positions are approximate are labeled "PA" on the source charts. Features whose positions are doubtful are labeled "PD" on the source charts. Features whose existence is doubtful are labeled "ED" on the source charts. Features that have only been reported are labeled "Rep" on the source charts, and usually show the date of report in parentheses.

Definitions:
PA - Position Approximate = The position has not been accurately determined, or does not remain fixed.;
PD - Position Doubtful = The feature has been reported in various positions, but not definitely determined to be in any.;
ED - Existence Doubtful = Indicates the possible existence of the feature, the actual existence of which has not been established.

Show Hydrographic Depth or Hydrographic Height values in whole meters if decimeter value is 0; otherwise show the decimeter value as a subscript to the meter value.

Date shall only be shown if Existence = Reported. Date shall be shown in parentheses if available. Always show "Rep" and date on the same line.

Priority for type placement Rules:
1-right center,
2-bottom center,
3-left center,
4-top center.
(A) Minimum distance from symbol - 1 mm.
(B) Maximum distance from symbol before choosing the next highest priority:
(#1) 4 mm measured to the West end;
(#2) 4 mm measured to the North side (top);
(#3) 4 mm measured to the East end;
(#4) 4 mm measured to the South side (bottom)

If symbols overprint each other, labels are condensed as follows:
(1.) If the labels are identical, only one is retained.;
(2,) If the labels are not identical, they shall be condensed in to one legend, e.g., "Fishhaven and Well". If multiple depths are shown, only the shallowest is retained.

Labels may be stacked, readable top to bottom to avoid overprints.

Type Preference Placement Rules:
(a.) Placed on one horizontal line centered in feature.;
(b.) Broken into multiple horizontal lines, centered in feature, with a 1 mm space between lines.;
(c.) Placed on one line outside feature. Use Priority for type placement Rules for placement, Hydrographic Depth shall always be positioned inside area.;
(d.) Placed on two lines to avoid overprints. Use Priority for type placement Rules. If present, Hydrographic Depth shall always be placed inside area.
Labels shall be placed in water area, and shall not overprint the shoreline or be placed on land or in the foreshore.

When two or more identical features fall within 10 mm of each other, one legend with plural description shall be shown, with type placed as close as possible to the center of the group without overprinting features.

Where symbols with the same feature code but different values of Hydrographic Vertical Positioning Information: Water Level Effect (i.e. Always Dry , Always Submerged, Covers and Uncovers)) are adjacent to each other, the boundary of the feature that is the highest in vertical plane shall be shown, and the boundary of the feature above water is shown whole, while the feature that covers and uncovers has a portion of its symbol perimeter that overprints the higher feature deleted.

When two or more identical symbols overprint each other, delete the portions that overlap and show as one symbol with an outline of the combined area of the symbols.

Show land tint inside the symbol. If the Hydrographic Vertical Positioning Information: Water Level Effect: [Always Dry] show land tint only.

Descriptive terms, e.g., "Canal" shall be shown if the name is not known. If the descriptive word appears in the name, for example, "PANAMA CANAL". The descriptive type shall not be shown, i.e. do not show "Panama Canal Canal".

Unless specifically indicated on source material as inaccurate or doubtful, etc., or modified by other override rules, features shall be assumed to be accurate and definite. This rule applied to accuracy and existence of unique features, and is not related to the accuracy of the source as a whole.

Symbol consists of arcs and Vs along the area perimeter. If the reef edge symbol overprints the shoreline, the symbol is deleted for that section that overprints.

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Reef (BD120) ***(Continued)***

Surface

Notes

The height of a hydrographic feature is that above the surface of the water at High Water. Hydrographic Depth/Height information is not applicable at High Water. Hydrographic Depth/Height information values (Depth Known), (Depth Known by Wire Drag), and (Depth Unknown) are used to describe knowledge about a depth of a feature when the feature is below the water surface/ submerged. Hydrographic Depth/Height information values (Uncovering Height Known) and (Uncovering Height Unknown) are used to describe knowledge about the drying height of a feature when it covers and uncovers, i.e., between high and low water.
Hydrographic depth is used to record the depth of a feature when Hydrographic Depth/Height information values are (Depth Known) and (Depth Known by Wire Drag). Hydrographic depth is not applicable when the depth is unknown or when the feature covers and uncovers.

Hydrographic Drying Height records the drying height of a feature when Hydrographic Depth/Height information values are known. Hydrographic Drying Height is not applicable when the drying height is unknown, or the feature is below water.

Vertical Datum category is the sounding datum to which a hydrographic depth, drying height, or the high water datum to which a Safe Overhead Clearance is referenced. If the hydrographic depth, drying height, or Safe Overhead Clearance is unknown or not applicable, the vertical datum category is not applicable.

Where symbols with the same feature code but different values of vertical reference categories (i.e. cover/uncovers, submerged, above surface (at high water)) are adjacent to each other, the boundary of the feature that is the highest in vertical plane shall be shown, and the boundary of the feature above water is shown whole, while the feature that covers and uncovers has a portion of its symbol perimeter that overprints the higher feature deleted.

If areas where separate dangers coalesce because of scale, the danger curves (dotted lines) are deleted except for those on the outermost perimeter of the aggregate danger hazard. A smooth dotted line is drawn around the area.

The minimum size of a reef that covers and uncovers shall be 2 mm diameter. The minimum size of a reef that is under water shall be 3 mm. If the reef at chart scale is smaller than these minimum sizes, it shall be shown as a rock.

A blue 31% tint shall not overprint other blue 31% tints. If two blue tinted symbols, or one tinted symbol and water tint overprint, only one 31% tint shall be shown in the area.

If a hole exists inside of an area feature, and the width of the hole is greater than 3 mm at chart scale, the hole is shown as an open space inside the surrounding feature. If the hole is 3 mm wide or less, the hole is deleted and absorbed into the surrounding area feature.

Feature Information					
Specification:	*LTDS*	**Feature Origin:**	*Topographic*	**Symbol:**	AR_BlackCyan_Reef
Symbol Criteria:	Water Level Effect = [Always Dry] or [Awash at Chart Datum] or [Awash at Low Water] or [Covers and Uncovers] or [Partly Submerged]			**Label:**	[Geographic Name Information: Full Name] if available, 6 point Mixed Case, Italic; [Hydrographic Drying Height] if available, 7 point lower Case Italic, 5 point Subscript; [Hydrographic Depth] if available, 7 point lower Case Italic, 5 point Subscript; "Co" if Structure Material Type = [Coral] or [Coral Head], 6 point Mixed case Italic; "Rky" if Structure Material Type = [Rock], 6 point Mixed case Italic
				Color:	Black (58600)

Feature Information					
Specification:	*Chart #1*	**Feature Origin:**	*Hydrographic*	**Symbol:**	K16
Symbol Criteria:	Water Level Effect = [Always Submerged]			**Label:**	[Geographic Name Information: Full Name] if available, 6 point Mixed Case, Italic; "Co" if Structure Material Type = [Coral] or [Coral Head], 6 point Mixed case Italic
				Color:	Black (58600)

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Reef (BD120) ***(Continued)***

Curve

Inclusion Condition:	*Width < 25m*

Notes

Submerged coral reefs and pinnacles must be charted by the same symbols as isolated rocks, using the abbreviation for coral as appropriate as shown on symbol K16 (Chart #1).

If point symbol overprints shoreline, the Hydrographic Depth type, posicut, or graphic in the center of the symbol shall be displaced seaward until they no longer overprint the shoreline. If the danger curve (dotted perimeter line) overprints the shoreline, that portion of the dotted perimeter line falling on land shall be deleted.

Rocks and isolated coral heads within submerged reefs shall be charted using the appropriate rock symbol. When depths over selected rocks are shown, an overall depth over the reef is not required, since the depth over the reef is shown by the depth of the shallowest rock. Where it is possible to chart depth information for separate rocks, the shallowest depth over the reef shall be shown by (Hydrographic Depth) or (Hydrographic Drying Height) on reef.

Use the following abbreviations for Horizontal Accuracy and Existence values:
If Position Approximate, label "PA";
If Position Doubtful, label "PD";
If Existence Doubtful, label "ED";
If Reported, label "Rep".

Date shall only be shown if Existence = Reported. Date shall be shown in parentheses if available. Always show "Rep" and date on the same line.

Following IHO practice, features whose positions are approximate are labeled "PA" on the source charts. Features whose positions are doubtful are labeled "PD" on the source charts. Features whose existence is doubtful are labeled "ED" on the source charts. Features that have only been reported are labeled "Rep" on the source charts, and usually show the date of report in parentheses.

Definitions:
PA - Position Approximate = The position has not been accurately determined, or does not remain fixed.;
PD - Position Doubtful = The feature has been reported in various positions, but not definitely determined to be in any.;
ED - Existence Doubtful = Indicates the possible existence of the feature, the actual existence of which has not been established.

Show Hydrographic Depth or Hydrographic Height values in whole meters if decimeter value is 0; otherwise show the decimeter value as a subscript to the meter value.

Date shall only be shown if Existence = Reported. Date shall be shown in parentheses if available. Always show "Rep" and date on the same line.

Priority for type placement Rules:
1-right center,
2-bottom center,
3-left center,
4-top center.
(A) Minimum distance from symbol - 1 mm.
(B) Maximum distance from symbol before choosing the next highest priority:
(#1) 4 mm measured to the West end;
(#2) 4 mm measured to the North side (top);
(#3) 4 mm measured to the East end;
(#4) 4 mm measured to the South side (bottom)

If symbols overprint each other, labels are condensed as follows:
(1.) If the labels are identical, only one is retained.;
(2,) If the labels are not identical, they shall be condensed in to one legend, e.g., "Fishhaven and Well". If multiple depths are shown, only the shallowest is retained.

Labels may be stacked, readable top to bottom to avoid overprints.

Type Preference Placement Rules:
(a.) Placed on one horizontal line centered in feature.;
(b.) Broken into multiple horizontal lines, centered in feature, with a 1 mm space between lines.;
(c.) Placed on one line outside feature. Use Priority for type placement Rules for placement, Hydrographic Depth shall always be positioned inside area.;
(d.) Placed on two lines to avoid overprints. Use Priority for type placement Rules. If present, Hydrographic Depth shall always be placed inside area.
Labels shall be placed in water area, and shall not overprint the shoreline or be placed on land or in the foreshore.

When two or more identical features fall within 10 mm of each other, one legend with plural description shall be shown, with type placed as close as possible to the center of the group without overprinting features.

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Reef (BD120) ***(Continued)***

Curve

Notes

Where symbols with the same feature code but different values of Hydrographic Vertical Positioning Information: Water Level Effect (i.e. Always Dry , Always Submerged, Covers and Uncovers)) are adjacent to each other, the boundary of the feature that is the highest in vertical plane shall be shown, and the boundary of the feature above water is shown whole, while the feature that covers and uncovers has a portion of its symbol perimeter that overprints the higher feature deleted.

When two or more identical symbols overprint each other, delete the portions that overlap and show as one symbol with an outline of the combined area of the symbols.

Show land tint inside the symbol. If the Hydrographic Vertical Positioning Information: Water Level Effect: [Always Dry] show land tint only.

Descriptive terms, e.g., "Canal" shall be shown if the name is not known. If the descriptive word appears in the name, for example, "PANAMA CANAL". The descriptive type shall not be shown, i.e. do not show "Panama Canal Canal".

Unless specifically indicated on source material as inaccurate or doubtful, etc., or modified by other override rules, features shall be assumed to be accurate and definite. This rule applied to accuracy and existence of unique features, and is not related to the accuracy of the source as a whole.

Symbol consists of arcs and Vs along the area perimeter. If the reef edge symbol overprints the shoreline, the symbol is deleted for that section that overprints.

The height of a hydrographic feature is that above the surface of the water at High Water. Hydrographic Depth/Height information is not applicable at High Water. Hydrographic Depth/Height information values (Depth Known), (Depth Known by Wire Drag), and (Depth Unknown) are used to describe knowledge about a depth of a feature when the feature is below the water surface/ submerged. Hydrographic Depth/Height information values (Uncovering Height Known) and (Uncovering Height Unknown) are used to describe knowledge about the drying height of a feature when it covers and uncovers, i.e., between high and low water.
Hydrographic depth is used to record the depth of a feature when Hydrographic Depth/Height information values are (Depth Known) and (Depth Known by Wire Drag). Hydrographic depth is not applicable when the depth is unknown or when the feature covers and uncovers.

Hydrographic Drying Height records the drying height of a feature when Hydrographic Depth/Height information values are known. Hydrographic Drying Height is not applicable when the drying height is unknown, or the feature is below water.

Vertical Datum category is the sounding datum to which a hydrographic depth, drying height, or the high water datum to which a Safe Overhead Clearance is referenced. If the hydrographic depth, drying height, or Safe Overhead Clearance is unknown or not applicable, the vertical datum category is not applicable.

Where symbols with the same feature code but different values of vertical reference categories (i.e. cover/uncovers, submerged, above surface (at high water)) are adjacent to each other, the boundary of the feature that is the highest in vertical plane shall be shown, and the boundary of the feature above water is shown whole, while the feature that covers and uncovers has a portion of its symbol perimeter that overprints the higher feature deleted.

If areas where separate dangers coalesce because of scale, the danger curves (dotted lines) are deleted except for those on the outermost perimeter of the aggregate danger hazard. A smooth dotted line is drawn around the area.

The minimum size of a reef that covers and uncovers shall be 2 mm diameter. The minimum size of a reef that is under water shall be 3 mm. If the reef at chart scale is smaller than these minimum sizes, it shall be shown as a rock.

A blue 31% tint shall not overprint other blue 31% tints. If two blue tinted symbols, or one tinted symbol and water tint overprint, only one 31% tint shall be shown in the area.

If a hole exists inside of an area feature, and the width of the hole is greater than 3 mm at chart scale, the hole is shown as an open space inside the surrounding feature. If the hole is 3 mm wide or less, the hole is deleted and absorbed into the surrounding area feature.

Feature Information					
Specification:	*LTDS*	**Feature Origin:**	*Topographic*	**Symbol:**	LN_Black_Reef
Symbol Criteria:	All			**Label:**	[Geographic Name Information: Full Name] if available, 6 point Mixed Case, Italic;
				Color:	Black (58600)

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Reef (BD120)

Point

Inclusion Condition:	*Area < 625m*

Notes

If point symbol overprints shoreline, the Hydrographic Depth type, posicut, or graphic in the center of the symbol shall be displaced seaward until they no longer overprint the shoreline. If the danger curve (dotted perimeter line) overprints the shoreline, that portion of the dotted perimeter line falling on land shall be deleted.

Rocks and isolated coral heads within submerged reefs shall be charted using the appropriate rock symbol. When depths over selected rocks are shown, an overall depth over the reef is not required, since the depth over the reef is shown by the depth of the shallowest rock. Where it is possible to chart depth information for separate rocks, the shallowest depth over the reef shall be shown by (Hydrographic Depth) or (Hydrographic Drying Height) on reef.

Use the following abbreviations for Horizontal Accuracy and Existence values:
If Position Approximate, label "PA";
If Position Doubtful, label "PD";
If Existence Doubtful, label "ED";
If Reported, label "Rep".

Date shall only be shown if Existence = Reported. Date shall be shown in parentheses if available. Always show "Rep" and date on the same line.
Unless specifically indicated on source material as inaccurate or doubtful, etc., or modified by other override rules, features shall be assumed to be accurate and definite. This rule applied to accuracy and existence of unique features, and is not related to the accuracy of the source as a whole.

Following IHO practice, features whose positions are approximate are labeled "PA" on the source charts. Features whose positions are doubtful are labeled "PD" on the source charts. Features whose existence is doubtful are labeled "ED" on the source charts. Features that have only been reported are labeled "Rep" on the source charts, and usually show the date of report in parentheses.

Definitions:
PA - Position Approximate = The position has not been accurately determined, or does not remain fixed.;
PD - Position Doubtful = The feature has been reported in various positions, but not definitely determined to be in any.;
ED - Existence Doubtful = Indicates the possible existence of the feature, the actual existence of which has not been established.

Show Hydrographic Depth or Hydrographic Height values in whole meters if decimeter value is 0; otherwise show the decimeter value as a subscript to the meter value.

Date shall only be shown if Existence = Reported. Date shall be shown in parentheses if available. Always show "Rep" and date on the same line.

Priority for type placement Rules:
1-right center,
2-bottom center,
3-left center,
4-top center.
(A) Minimum distance from symbol - 1 mm.
(B) Maximum distance from symbol before choosing the next highest priority:
(#1) 4 mm measured to the West end;
(#2) 4 mm measured to the North side (top);
(#3) 4 mm measured to the East end;
(#4) 4 mm measured to the South side (bottom)

If symbols overprint each other, labels are condensed as follows:
(1.) If the labels are identical, only one is retained.;
(2,) If the labels are not identical, they shall be condensed in to one legend, e.g., "Fishhaven and Well". If multiple depths are shown, only the shallowest is retained.

Labels may be stacked, readable top to bottom to avoid overprints.

Type Preference Placement Rules:
(a.) Placed on one horizontal line centered in feature.;
(b.) Broken into multiple horizontal lines, centered in feature, with a 1 mm space between lines.;
(c.) Placed on one line outside feature. Use Priority for type placement Rules for placement, Hydrographic Depth shall always be positioned inside area.;
(d.) Placed on two lines to avoid overprints. Use Priority for type placement Rules. If present, Hydrographic Depth shall always be placed inside area.
Labels shall be placed in water area, and shall not overprint the shoreline or be placed on land or in the foreshore.

When two or more identical features fall within 10 mm of each other, one legend with plural description shall be shown, with type placed as close as possible to the center of the group without overprinting features.

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Reef (BD120) ***(Continued)***

Point

Notes

Where symbols with the same feature code but different values of Hydrographic Vertical Positioning Information: Water Level Effect (i.e. Always Dry , Always Submerged, Covers and Uncovers)) are adjacent to each other, the boundary of the feature that is the highest in vertical plane shall be shown, and the boundary of the feature above water is shown whole, while the feature that covers and uncovers has a portion of its symbol perimeter that overprints the higher feature deleted.

When two or more identical symbols overprint each other, delete the portions that overlap and show as one symbol with an outline of the combined area of the symbols.

Descriptive terms, e.g., "Canal" shall be shown if the name is not known. If the descriptive word appears in the name, for example, "PANAMA CANAL". The descriptive type shall not be shown, i.e. do not show "Panama Canal Canal".

Unless specifically indicated on source material as inaccurate or doubtful, etc., or modified by other override rules, features shall be assumed to be accurate and definite. This rule applied to accuracy and existence of unique features, and is not related to the accuracy of the source as a whole.

Symbol consists of arcs and Vs along the area perimeter. If the reef edge symbol overprints the shoreline, the symbol is deleted for that section that overprints.

The height of a hydrographic feature is that above the surface of the water at High Water. Hydrographic Depth/Height information is not applicable at High Water. Hydrographic Depth/Height information values (Depth Known), (Depth Known by Wire Drag), and (Depth Unknown) are used to describe knowledge about a depth of a feature when the feature is below the water surface/ submerged. Hydrographic Depth/Height information values (Uncovering Height Known) and (Uncovering Height Unknown) are used to describe knowledge about the drying height of a feature when it covers and uncovers, i.e., between high and low water.
Hydrographic depth is used to record the depth of a feature when Hydrographic Depth/Height information values are (Depth Known) and (Depth Known by Wire Drag). Hydrographic depth is not applicable when the depth is unknown or when the feature covers and uncovers.

Hydrographic Drying Height records the drying height of a feature when Hydrographic Depth/Height information values are known. Hydrographic Drying Height is not applicable when the drying height is unknown, or the feature is below water.

Vertical Datum category is the sounding datum to which a hydrographic depth, drying height, or the high water datum to which a Safe Overhead Clearance is referenced. If the hydrographic depth, drying height, or Safe Overhead Clearance is unknown or not applicable, the vertical datum category is not applicable.

Where symbols with the same feature code but different values of vertical reference categories (i.e. cover/uncovers, submerged, above surface (at high water)) are adjacent to each other, the boundary of the feature that is the highest in vertical plane shall be shown, and the boundary of the feature above water is shown whole, while the feature that covers and uncovers has a portion of its symbol perimeter that overprints the higher feature deleted.

The minimum size of a reef that covers and uncovers shall be 2 mm diameter. The minimum size of a reef that is under water shall be 3 mm. If the reef at chart scale is smaller than these minimum sizes, it shall be shown as a rock.

A blue 31% tint shall not overprint other blue 31% tints. If two blue tinted symbols, or one tinted symbol and water tint overprint, only one 31% tint shall be shown in the area.

Feature Information

Specification:	*LTDS*	**Feature Origin:**	*Topographic*	**Symbol:**	PT_Black_Reef
Symbol Criteria:	Area < 625m			**Label:**	"Co" if Structure Material Type = [Coral] or [Coral Head], 6 point Mixed case Italic; "Rk" if Structure Material Type = [Rock], 6 point Mixed case Italic; [Hydrographic Drying Height] if available, 7 point lower Case Italic, 5 point Subscript; [Hydrographic Depth] if available, 7 point lower Case Italic, 5 point Subscript;
				Color:	Black (58600)

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Rice Field (BH135)

Surface

Feature Description:	*A tract that is periodically covered with water and is used for growing rice.*
Inclusion Condition:	*Area >= 15,625 sq m.*

Notes

None

Feature Information					
Specification:	*LTDS*	**Feature Origin:**	*Topographic*	**Symbol:**	AR_Cyan_Rice
Symbol Criteria:	All			**Label:**	"Terraces" , 6 point Condensed Italics Mixed case if Crop Information : Farming Pattern [1] = [Terraced]
				Color:	Black (58600)

Rig (AA040)

Point

Feature Description:	*A superstructure fitted for drilling or lifting operations for extraction and/or exploitation of natural resources.*
Inclusion Condition:	*Location Referenced to Shoreline = [Offshore] OR Height Above Surface Level >= 46 m. OR Navigation Landmark = [True]*

Notes

Type for features on land shall be positioned on land if possible. If type must be placed in the water, it shall be positioned so it does not obscure or overprint hydrographic detail. Type for features in the water shall be positioned in the water. If possible, type shall not be positioned across the shoreline.

If Rigs coalesce and area is < 2.5 mm x 2.5 mm, show one Rig symbol and label "RIGS", If area >= 2.5 mm x 2.5 mm, show dashed outline and label "NUMEROUS RIGS".

When obstructions coalesce at map scale, label with highest obstruction information.

If a Rig AA040 overprints Platform BD110, delete the rig.

If a Rig AA040's coalesce and area is < 2.5 mm x 2.5 mm, show one Rig symbol in its true geographic location. If area is > 2.5 mm x 2.5 mm, show a dashed outline.

Feature Information					
Specification:	*LTDS*	**Feature Origin:**	*Topographic*	**Symbol:**	PT_Black_Rig_Inland
Symbol Criteria:	Location Referenced to Shoreline <> [Offshore]			**Label:**	"Product [1]" if = (Gas) or (Water) or (Brine), 6 point Condensed Mixed case; "Oil" if Product [1] = (Petroleum), 6 point Condensed Mixed case
				Color:	Black (58600)

Feature Information					
Specification:	*LTDS*	**Feature Origin:**	*Topographic*	**Symbol:**	PT_Blue072_VO
Symbol Criteria:	Height Above Surface Level >= 46 meters and Location Referenced to Shoreline <> [Offshore]			**Label:**	"Rig" : 6 point Condensed Mixed case; [Highest Elevation], 6 point Condensed Bold Mixed Case; "([Height Above Surface Level])": 6 point Condensed Mixed case
				Color:	Blue (46351)

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Rig (AA040) *(Continued)*

Point

Feature Information					
Specification:	*Chart #1*	**Feature Origin:**	*Hydrographic*	**Symbol:**	L10
Symbol Criteria:	Location Referenced to Shoreline = [Offshore]			**Label:**	"Oil", 6 point Condensed Mixed case if Product [1] = [Petroleum]; "Rig", 6 point Condensed Mixed case if Product [1] = [Other] or [No information]; and "([Height Above Surface Level])", 6 point Condensed Mixed case
				Color:	Black (58600)

River (BH140)

Surface

Feature Description:	*A natural flowing watercourse.*
Inclusion Condition:	*Length or Diameter >= 300 m. AND Width >= 25 m.*

Notes

Rivers, canals, and ditches will be partially collapsed when area and line delineations are supported on the product and the area feature does not meet the minimum geometric inclusion condition.

Coincident similar area features will be blended to form a single feature.

Area and line features will be generalized to detail compatible with scale.

Feature will be generalized to provide a more aesthetic contoured feature (i.e., smoothed).

Type sizes for single line features at map/chart scale.
06 point - <= 80 mm length;
07 point - <= 160 mm length;
09 point - > 160 mm length.

Label area features with upper case type within its limits and centered between the sides with a proportional size if the width of the feature will allow its inclusion. However, should the feature be too narrow, then place the type 0.5 mm above and parallel to the feature. When the feature is continuous, repeat label approximately every 30 to 40 cm for either situation, or at least two items, length permitting. In either condition (in or above the feature), curve the type necessary to the curvature of the feature. Should the feature change back and forth between the area and a line feature, the type style will change from upper case type for the area portions, to upper and lower case type for the linear portions. The repeat dimensions remain the same.

Names placement shall be oriented to the longest axis of the feature reading left to right and placed within the area outline and centered. If the longest axis is perpendicular to the south neatline, the type shall be placed outside of the area outline, preferred position is northeast of the feature, but may be placed at any position around the feature so as not to overprint any other feature type and reading left to right.

Name shall be positioned in the center of that part of the feature appearing on a chart, i.e., centered from bank to bank, and centered from mouth to neatline. Type shall run parallel to center line, reading left to right, or bottom to top if feature is vertical. Type may be moved sideways to avoid overprints or sharp bends (>= 5°).

If River/Stream is Perennial and <= 3° slope along this feature and no contours are present, then add Flow Arrow symbol to indicate direction of flow.

Rivers under the influence of the rise and fall of the tide shall have their banks delineated at the high water line. Inland of tidal influence, average water level shall be shown for perennial rivers, and flood stage shall be shown for intermittent, or dry rivers.

Do not show land tint in the symbol. Do not show land tint if (Perennial).

Feature Information					
Specification:	*LTDS*	**Feature Origin:**	*Topographic*	**Symbol:**	AR_Cyan_River_Perennial
Symbol Criteria:	Water Resource Information : Hydrologic Persistence = [Perennial] or [No Information]			**Label:**	[Geographic Name Information : Full Name] if available, 7-16 point Condensed Italics Upper case
				Color:	Blue (46351)

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River (BH140)

Surface

Feature Information					
Specification:	*LTDS*	**Feature Origin:**	*Topographic*	**Symbol:**	AR_Cyan_River_Non-perennial
Symbol Criteria:	Water Resource Information : Hydrologic Persistence = [Intermittent] or [Ephemeral] or [Dry]			**Label:**	[Geographic Name Information : Full Name] if available, 7-16 point Condensed Italics Upper case
				Color:	Blue (46351)

River (BH140)

Curve

Inclusion Condition:	*Length or Diameter >= 300 m. AND Width < 25 m.*

Notes

Area and line features will be generalized to detail compatible with scale.

Feature will be generalized to provide a more aesthetic contoured feature (i.e., smoothed).

Label shall be position above feature, reading from left to right (or to the left of the vertical feature, reading bottom to top), at a 0.5 mm distance and parallel to respective feature. Label shall preferably be positioned at the midpoint of the line segment or symbol; however, it may be displaced laterally along respective feature to avoid overprinting other symbols or labels. If space will not permit placing label parallel to feature, offset the label and use a leader line to identify its location along the feature.

Feature name, label, data information holder, and/or symbol shall be positioned, reading left to right, parallel to the tangent of the center of the southern neatline of the Map/Chart.

If River/Stream is Perennial and <= 3° slope along this feature and no contours are present, then add Flow Arrow symbol to indicate direction of flow.

Rivers under the influence of the rise and fall of the tide shall have their banks delineated at the high water line. Inland of tidal influence, average water level shall be shown for perennial rivers, and flood stage shall be shown for intermittent, or dry rivers.

Feature Information					
Specification:	*LTDS*	**Feature Origin:**	*Topographic*	**Symbol:**	LN_Cyan_River_Perennial
Symbol Criteria:	Water Resource Information : Hydrologic Persistence = [Perennial] or [No Information]			**Label:**	[Geographic Name Information : Full Name] if available, 6 point Condensed Italics Mixed case
				Color:	Blue (46351)

Feature Information					
Specification:	*LTDS*	**Feature Origin:**	*Topographic*	**Symbol:**	LN_Cyan_River_Non-perennial
Symbol Criteria:	Water Resource Information : Hydrologic Persistence = [Intermittent] or [Ephemeral] or [Dry]			**Label:**	[Geographic Name Information : Full Name] if available, 6 point Condensed Italics Mixed case
				Color:	Blue (46351)

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Road (AP030)

Curve

Feature Description:	*A route with a specially prepared surface that is intended for use by wheeled vehicles.*
Inclusion Condition:	*((Length or Diameter >= 300 m.) AND (spaced >= 300 m. apart)) OR (within a Built-up Area and street pattern required to adequately portray the density of the area and [suggested] spacing >= 200 m. apart) OR (needed to complete the component network or to establish connectivity)*

Notes

The feature which coalesces (<0.2 mm) with a railroad track or RR Sidetrack/Spur shall be displaced to a minimum of 0.2 mm apart.

Contours will be adjusted to planimetric features.

When a Road of any classification enters a "hairpin turn" condition, such as in a steep mountainous region, displace the coalescing road symbol apart 0.15 mm (symbol - edge to edge).

If feature coalesce at map scale, when shown in their true positions, they shall be displaced 0.2 mm from one another.

If a Road (Line) is coincident with features: then suppress that section of the road. (Point Tunnel) (Line Tunnel) Coalesces - to grow together, blend, mingle; Coincident - occupy the same place

Area and line features will be generalized to detail compatible with scale.

Type for features on land shall be positioned on land if possible. If type must be placed in the water, it shall be positioned so it does not obscure or overprint hydrographic detail. Type for features in the water shall be positioned in the water. If possible, type shall not be positioned across the shoreline.

Road alignments that lack adequate information for proper alignment shall be labeled "APPROXIMATE ALIGNMENT" or "APPROX. ALIGN.".

Approximate alignments less than 13 mm in length at map scale shall not be labeled.

First preferences for Road name position shall be along the upper side of the Road symbol.

When an elevated highway is >= 12.5 mm long at map scale, it shall be labeled "ELEVATED" parallel to the Road.

When Location (On Ground Surface), no Location label is required.

Label shall be position above feature, reading from left to right (or to the left of the vertical feature, reading bottom to top), at a 0.5 mm distance and parallel to respective feature. Label shall preferably be positioned at the midpoint of the line segment or symbol; however, it may be displaced laterally along respective feature to avoid overprinting other symbols or labels. If space will not permit placing label parallel to feature, offset the label and use a leader line to identify its location along the feature.

Feature name, label, data information holder, and/or symbol shall be positioned, reading left to right, parallel to the tangent of the center of the southern neatline of the Map/Chart.

For Road, Cart track, and Trail within Built-up Area; Symbolize the portion of the feature within the Built-up Area as white.

Retain any road of any classification that is < 12.5 mm at map scale when part of the main road. Example: A two lane road that changes to a 3 or 4 lane road, and back again. When this condition exists, portray at the lower road classification.

Feature under construction, to be operational by the time the map in progress is to be complete, shall be symbolized as operational.

If a Road can be classified in more than one category ((Route Weatherability, Road/Runway Surface Type. Lane/Track number, or Existence) where the total length is <= 13.0 mm at map scale, then classify this road at the lowest road classification for this condition.

A Road that can predominantly be classified in one category (>= 75% Surface Type, Route Weatherablility & Road/Runway type) within a distance of <= 13.0 mm at map scale shall be classified at that predominant road classification for this entire distance.

The Built-Up Area tint shall be cleared from all through Routes and streets.

Suppress Road when Road, Railroad, or Railroad and Road are coincident with a Dam. Label as "Road on dam" , "Railroad on dam", and "Railroad and Road on dam".

Suppress any road of any classification, cart track, or trail that intersects on side, and that is < 7.5 mm at finishing scale, and does not terminate at a cultural feature. Exception: Any road, cart track, or trail must be retained when needed to complete the network.

Do not symbolize road when outside of Built-up Area or Shanty town, and length < 300 meters, and spacing is < 300 meters, unless needed to complete the road network.

Do not symbolize road when within Built-up Area or Shanty town, and length < 300 meters, and spacing is < 300 meters, unless needed to complete the road network.

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Road (AP030) *(Continued)*

Curve

Feature Information					
Specification:	*LTDS*	**Feature Origin:**	*Topographic*	**Symbol:**	LN_Road_AllWeather_ HardSurface_Operational_ Median_MultipleLanes
Symbol Criteria:	(Centerline Spacing <= 50) and (Condition of Facility = [Abandoned] or [Fully Functional] or [Damaged] or [Disused] or [Other] or [No Information]) and ((Divided = [True]) or (Median Present = [True])) and (Route Pavement Information : Road Weather Restriction = [All-weather] or [Limited All-weather] or [Closed in Winter]) and (Route Pavement Information : Route Surface Composition = [Rigid Pavement] or [Concrete] or [Asphalt] or [Asphalt over Concrete] or [Cobble-stone] or [Brick]) and (Thoroughfare Type = [Road] or [Boulevard] or [Avenue] or [Drive] or [Circle] or [Close] or [Parkway] or [Lane] or [Terrace] or [Motorway] or [Place] or [Limited Access Motorway] or [Other] or [No Information]) and (Track or Lane Count >= 2)			**Label:**	[Geographic Name Information : Full Name] AND [Route Designation] if available, 6 point Light Condensed Italics Upper case; Label: "([Track or Lane Count]) LANES", 6 point Light Condensed Italics Upper case if >4
				Color:	Black (58600)

Feature Information					
Specification:	*LTDS*	**Feature Origin:**	*Topographic*	**Symbol:**	LN_Road_AllWeather_ HardSurface_Construction_ Median_MultipleLanes
Symbol Criteria:	(Centerline Spacing <= 50) and (Condition of Facility = [Under Construction] or [Destroyed] or [Dismantled] or [Ruined] or [Partially Functional] or [Partially Destroyed]) and ((Divided = [True]) or (Median Present = [True])) and (Route Pavement Information : Road Weather Restriction = [All-weather] or [Limited All-weather] or [Closed in Winter]) and (Route Pavement Information : Route Surface Composition = [Rigid Pavement] or [Concrete] or [Asphalt] or [Asphalt over Concrete] or [Cobble-stone] or [Brick]) and (Thoroughfare Type = [Road] or [Boulevard] or [Avenue] or [Drive] or [Circle] or [Close] or [Parkway] or [Lane] or [Terrace] or [Motorway] or [Place] or [Limited Access Motorway] or [Other] or [No Information]) and (Track or Lane Count >= 2)			**Label:**	[Geographic Name Information : Full Name] AND "([Track or Lane Count]) LANES", 6 point Light Condensed Italics Upper case if > 4; if Condition of Facility = [Under Construction] or [Partially Functional] , label as "CONSTRUCTION", 6 point Light Condensed Italics Upper case; if Condition of Facility = [Destroyed] or [Dismantled] or [Ruined] or [Partially Destroyed], label as "NOT USABLE", 6 point Light Condensed Italics Upper case
				Color:	Black (58600)

Feature Information					
Specification:	*LTDS*	**Feature Origin:**	*Topographic*	**Symbol:**	LN_Road_AllWeather_ HardSurface_Operational_ WideMedian__MultipleLanes
Symbol Criteria:	(Centerline Spacing > 50) and (Condition of Facility = [Abandoned] or [Fully Functional] or [Damaged] or [Disused] or [Other] or [No Information]) and ((Divided = [True]) or (Median Present = [True])) and (Route Pavement Information : Road Weather Restriction = [All-weather] or [Limited All-weather] or [Closed in Winter]) and (Route Pavement Information : Route Surface Composition = [Rigid Pavement] or [Concrete] or [Asphalt] or [Asphalt over Concrete] or [Cobble-stone] or [Brick]) and (Thoroughfare Type = [Road] or [Boulevard] or [Avenue] or [Drive] or [Circle] or [Close] or [Parkway] or [Lane] or [Terrace] or [Motorway] or [Place] or [Limited Access Motorway] or [Other] or [No Information]) and (Track or Lane Count >= 2)			**Label:**	[Geographic Name Information : Full Name] if available, 6 point Light Condensed Italics Upper case; "([Track or Lane Count]) LANES", 6 point Light Condensed Italics Upper case if >4
				Color:	Black (58600)

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Road (AP030) *(Continued)*

Curve

Feature Information					
Specification:	*LTDS*	**Feature Origin:**	*Topographic*	**Symbol:**	LN_Road_AllWeather_ HardSurface_Construction_ WideMedian_MultipleLanes
Symbol Criteria:	(Centerline Spacing > 50) and (Condition of Facility = [Under Construction] or [Destroyed] or [Dismantled] or [Ruined] or [Partially Functional] or [Partially Destroyed]) and ((Divided = [True]) or (Median Present = [True])) and (Route Pavement Information : Road Weather Restriction = [All-weather] or [Limited All-weather] or [Closed in Winter]) and (Route Pavement Information : Route Surface Composition = [Rigid Pavement] or [Concrete] or [Asphalt] or [Asphalt over Concrete] or [Cobble-stone] or [Brick]) and (Thoroughfare Type = [Road] or [Boulevard] or [Avenue] or [Drive] or [Circle] or [Close] or [Parkway] or [Lane] or [Terrace] or [Motorway] or [Place] or [Limited Access Motorway] or [Other] or [No Information]) and (Track or Lane Count >= 2)			**Label:**	[Geographic Name Information : Full Name] AND "([Track or Lane Count]) LANES", 6 point Light Condensed Italics Upper case if > 4; if Condition of Facility = [(Under Construction] or [Partially Functional] , label as "CONSTRUCTION", 6 point Light Condensed Italics Upper case; if Condition of Facility = [Destroyed] or [Dismantled] or [Ruined] or [Partially Destroyed], label as "NOT USABLE", 6 point Light Condensed Italics Upper case
				Color:	Black (58600)

Feature Information					
Specification:	*LTDS*	**Feature Origin:**	*Topographic*	**Symbol:**	LN_Road_AllWeather_ HardSurface_Operational_ NoMedian_MultipleLanes
Symbol Criteria:	(Condition of Facility = [Abandoned] or [Fully Functional] or [Damaged] or [Disused] or [Other] or [No Information]) and ((Divided = 1000 [False] or [No Information]) or (Median Present = [False] or [No Information])) and (Route Pavement Information : Road Weather Restriction = [All-weather] or [Limited All-weather] or [Closed in Winter]) and (Route Pavement Information : Route Surface Composition = [Rigid Pavement] or [Concrete] or [Asphalt] or [Asphalt over Concrete] or [Cobble-stone] or [Brick]) and (Thoroughfare Type = [Road] or [Boulevard] or [Avenue] or [Drive] or [Circle] or [Close] or [Parkway] or [Lane] or [Terrace] or [Motorway] or [Place] or [Limited Access Motorway] or [Other] or [No Information]) and (Track or Lane Count >= 2)			**Label:**	[Geographic Name Information : Full Name] if available, 6 point Light Condensed Italics Upper case; Label: "([Track or Lane Count]) LANES", 6 point Light Condensed Italics Upper case if >2
				Color:	Black (58600)

Feature Information					
Specification:	*LTDS*	**Feature Origin:**	*Topographic*	**Symbol:**	LN_Road_AllWeather_ HardSurface_Construction_ NoMedian_MultipleLanes
Symbol Criteria:	(Condition of Facility = [Under Construction] or [Destroyed] or [Dismantled] or [Ruined] or [Partially Functional] or [Partially Destroyed]) and ((Divided = 1000 [False] or [No Information]) or (Median Present = [False] or [No Information])) and (Route Pavement Information : Road Weather Restriction = [All-weather] or [Limited All-weather] or [Closed in Winter]) and (Route Pavement Information : Route Surface Composition = [Rigid Pavement] or [Concrete] or [Asphalt] or [Asphalt over Concrete] or [Cobble-stone] or [Brick]) and (Thoroughfare Type = [Road] or [Boulevard] or [Avenue] or [Drive] or [Circle] or [Close] or [Parkway] or [Lane] or [Terrace] or [Motorway] or [Place] or [Limited Access Motorway] or [Other] or [No Information]) and (Track or Lane Count >= 2)			**Label:**	[Geographic Name Information : Full Name] AND "([Track or Lane Count]) LANES", 6 point Light Condensed Italics Upper case if > 4; if Condition of Facility = [Under Construction] or [Partially Functional] , label as "CONSTRUCTION", 6 point Light Condensed Italics Upper case; if Condition of Facility = [Destroyed] or [Dismantled] or [Ruined] or [Partially Destroyed], label as "NOT USABLE", 6 point Light Condensed Italics Upper case
				Color:	Black (58600)

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Road (AP030) *(Continued)*

Curve

Feature Information					
Specification:	*LTDS*	**Feature Origin:**	*Topographic*	**Symbol:**	LN_Road_AllWeather_ HardSurface_Operational_ NoMedian_1Lane
Symbol Criteria:	(Condition of Facility = [Abandoned] or [Fully Functional] or [Damaged] or [Disused] or [Other] or [No Information]) and ((Divided = [False] or [No Information]) or (Median Present = [False] or [No Information])) and (Route Pavement Information : Road Weather Restriction = [All-weather] or [Limited All-weather] or [Closed in Winter]) and (Route Pavement Information : Route Surface Composition = [Rigid Pavement] or [Concrete] or [Asphalt] or [Asphalt over Concrete] or [Cobble-stone] or [Brick]) and (Thoroughfare Type = [Road] or [Boulevard] or [Avenue] or [Drive] or [Circle] or [Close] or [Parkway] or [Lane] or [Terrace] or [Motorway] or [Place] or [Limited Access Motorway] or [Other] or [No Information]) and (Track or Lane Count = 1)			**Label:**	[Geographic Name Information : Full Name] if available, 6 point Light Condensed Italics Upper case
				Color:	Black (58600)

Feature Information					
Specification:	*LTDS*	**Feature Origin:**	*Topographic*	**Symbol:**	LN_Road_AllWeather_ HardSurface_Construction_ NoMedian_1Lane
Symbol Criteria:	(Condition of Facility = [Under Construction] or [Destroyed] or [Dismantled] or [Ruined] or [Partially Functional] or [Partially Destroyed]) and ((Divided = [False] or [No Information]) or (Median Present = [False] or [No Information])) and (Route Pavement Information : Road Weather Restriction = [All-weather] or [Limited All-weather] or [Closed in Winter]) and (Route Pavement Information : Route Surface Composition = [Rigid Pavement] or [Concrete] or [Asphalt] or [Asphalt over Concrete] or [Cobble-stone] or [Brick]) and (Thoroughfare Type = [Road] or [Boulevard] or [Avenue] or [Drive] or [Circle] or [Close] or [Parkway] or [Lane] or [Terrace] or [Motorway] or [Place] or [Limited Access Motorway] or [Other] or [No Information]) and (Track or Lane Count = 1)			**Label:**	[Geographic Name Information : Full Name] AND "([Track or Lane Count]) LANES", 6 point Light Condensed Italics Upper case if > 4; if Condition of Facility = [Under Construction] or [Partially Functional] , label as "CONSTRUCTION", 6 point Light Condensed Italics Upper case; if Condition of Facility = [Destroyed] or [Dismantled] or [Ruined] or [Partially Destroyed], label as "NOT USABLE", 6 point Light Condensed Italics Upper case
				Color:	Black (58600)

Feature Information					
Specification:	*LTDS*	**Feature Origin:**	*Topographic*	**Symbol:**	LN_Road_AllWeather_ LooseSurface_Operational_ NoMedian_MultipleLanes
Symbol Criteria:	(Condition of Facility = [Abandoned] or [Fully Functional] or [Damaged] or [Disused] or [Other] or [No Information]) and ((Divided = [False] or [No Information]) or (Median Present = [False] or [No Information])) and (Route Pavement Information : Road Weather Restriction = [All-weather] or [Limited All-weather] or [Closed in Winter]) and (Route Pavement Information : Route Surface Composition = [Flexible Pavement] or [Aggregate] or [Macadam] or [Bound Surface] or [Metal] or [Wood] or [Corduroy] or [Wood Plank]) and (Thoroughfare Type = [Road] or [Boulevard] or [Avenue] or [Drive] or [Circle] or [Close] or [Parkway] or [Lane] or [Terrace] or [Motorway] or [Place] or [Limited Access Motorway] or [Other] or [No Information]) and (Track or Lane Count >= 2)			**Label:**	[Geographic Name Information : Full Name] if available, 6 point Light Condensed Italics Upper case; Label: "([Track or Lane Count]) LANES", 6 point Light Condensed Italics Upper case if >2
				Color:	Black (58600)

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Road (AP030) *(Continued)*

Curve

Feature Information					
Specification:	*LTDS*	**Feature Origin:**	*Topographic*	**Symbol:**	LN_Road_AllWeather_ LooseSurface_Construction_ NoMedian_MultipleLanes
Symbol Criteria:	(Condition of Facility = [Under Construction] or [Destroyed] or [Dismantled] or [Ruined] or [Partially Functional] or [Partially Destroyed]) and ((Divided = [False] or [No Information]) or (Median Present = [False] or [No Information])) and (Route Pavement Information : Road Weather Restriction = [All-weather] or [Limited All-weather] or [Closed in Winter]) and (Route Pavement Information : Route Surface Composition = [Flexible Pavement] or [Aggregate] or [Macadam] or [Bound Surface] or [Metal] or [Wood] or [Corduroy] or [Wood Plank]) and (Thoroughfare Type = [Road] or [Boulevard] or [Avenue] or [Drive] or [Circle] or [Close] or [Parkway] or [Lane] or [Terrace] or [Motorway] or [Place] or [Limited Access Motorway] or [Other] or [No Information]) and (Track or Lane Count >= 2)			**Label:**	[Geographic Name Information : Full Name] AND "([Track or Lane Count]) LANES", 6 point Light Condensed Italics Upper case if > 4; if Condition of Facility = [Under Construction] or [Partially Functional] , label as "CONSTRUCTION", 6 point Light Condensed Italics Upper case; if Condition of Facility = [Destroyed] or [Dismantled] or [Ruined] or [Partially Destroyed], label as "NOT USABLE", 6 point Light Condensed Italics Upper case
				Color:	Black (58600)

Feature Information					
Specification:	*LTDS*	**Feature Origin:**	*Topographic*	**Symbol:**	LN_Road_AllWeather_ LooseSurface_Operational_ NoMedian_1Lane
Symbol Criteria:	(Condition of Facility = [Abandoned] or [Fully Functional] or [Damaged] or [Disused] or [Other] or [No Information]) and ((Divided = [False] or [No Information]) or (Median Present = [False] or [No Information])) and (Route Pavement Information : Road Weather Restriction = [All-weather] or [Limited All-weather] or [Closed in Winter]) and (Route Pavement Information : Route Surface Composition = [Flexible Pavement] or [Aggregate] or [Macadam] or [Bound Surface] or [Metal] or [Wood] or [Corduroy] or [Wood Plank]) and (Thoroughfare Type = [Road] or [Boulevard] or [Avenue] or [Drive] or [Circle] or [Close] or [Parkway] or [Lane] or [Terrace] or [Motorway] or [Place] or [Limited Access Motorway] or [Other] or [No Information]) and (Track or Lane Count = 1)			**Label:**	[Geographic Name Information : Full Name] if available, 6 point Light Condensed Italics Upper case
				Color:	Black (58600)

Feature Information					
Specification:	*LTDS*	**Feature Origin:**	*Topographic*	**Symbol:**	LN_Road_AllWeather_ LooseSurface_Construction_ NoMedian_1Lane
Symbol Criteria:	(Condition of Facility = [Under Construction] or [Destroyed] or [Dismantled] or [Ruined] or [Partially Functional] or [Partially Destroyed]) and ((Divided = [False] or [No Information]) or (Median Present = [False] or [No Information])) and (Route Pavement Information : Road Weather Restriction = [All-weather] or [Limited All-weather] or [Closed in Winter]) and (Route Pavement Information : Route Surface Composition = [Flexible Pavement] or [Aggregate] or [Macadam] or [Bound Surface] or [Metal] or [Wood] or [Corduroy] or [Wood Plank]) and (Thoroughfare Type = [Road] or [Boulevard] or [Avenue] or [Drive] or [Circle] or [Close] or [Parkway] or [Lane] or [Terrace] or [Motorway] or [Place] or [Limited Access Motorway] or [Other] or [No Information]) and (Track or Lane Count = 1)			**Label:**	[Geographic Name Information : Full Name]; if Condition of Facility = [Under Construction] or [Partially Functional] , label as "CONSTRUCTION", 6 point Light Condensed Italics Upper case; if Condition of Facility = [Destroyed] or [Dismantled] or [Ruined] or [Partially Destroyed], label as "NOT USABLE", 6 point Light Condensed Italics Upper case
				Color:	Black (58600)

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Road (AP030) *(Continued)*

Curve

Feature Information					
Specification:	*LTDS*	**Feature Origin:**	*Topographic*	**Symbol:**	LN_Road_FairWeather_ Operational
Symbol Criteria:	(Condition of Facility = [Abandoned] or [Fully Functional] or [Damaged] or [Disused] or [Other] or [No Information]) and (Route Pavement Information : Road Weather Restriction = [Fair-weather] or [Winter Only] or [Closed in Winter] or [Other] or [No Information]) and (Route Pavement Information : Route Surface Composition = [Unimproved] or [Stabilized Earth] or [Other] or [No Information]) and (Thoroughfare Type = [Road] or [Boulevard] or [Avenue] or [Drive] or [Circle] or [Close] or [Parkway] or [Lane] or [Terrace] or [Motorway] or [Place] or [Limited Access Motorway] or [Other] or [No Information])			**Label:**	[Geographic Name Information : Full Name] if available, 6 point Light Condensed Italics Upper case
				Color:	Black (58600)

Feature Information					
Specification:	*LTDS*	**Feature Origin:**	*Topographic*	**Symbol:**	LN_Road_FairWeather_ Construction
Symbol Criteria:	(Condition of Facility = [Under Construction] or [Destroyed] or [Dismantled] or [Ruined] or [Partially Functional] or [Partially Destroyed]) and (Route Pavement Information : Road Weather Restriction = [Fair-weather] or [Winter Only] or [Closed in Winter] or [Other] or [No Information]) and (Route Pavement Information : Route Surface Composition = [Unimproved] or [Stabilized Earth] or [Other] or [No Information]) and (Thoroughfare Type = [Road] or [Boulevard] or [Avenue] or [Drive] or [Circle] or [Close] or [Parkway] or [Lane] or [Terrace] or [Motorway] or [Place] or [Limited Access Motorway] or [Other] or [No Information])			**Label:**	[Geographic Name Information : Full Name]; if Condition of Facility = [Under Construction] or [Partially Functional] , label as "CONSTRUCTION", 6 point Light Condensed Italics Upper case; if Condition of Facility = [Destroyed] or [Dismantled] or [Ruined] or [Partially Destroyed], label as "NOT USABLE", 6 point Light Condensed Italics Upper case
				Color:	Black (58600)

Feature Information					
Specification:	*LTDS*	**Feature Origin:**	*Topographic*	**Symbol:**	LN_Road_Street_Functional
Symbol Criteria:	(Condition of Facility = [Abandoned] or [Fully Functional] or [Damaged] or [Disused] or [Other] or [No Information]) and (Thoroughfare Type = [Street]) and (Through Route = [False] or [No Information])			**Label:**	None
				Color:	N/A

Feature Information					
Specification:	*LTDS*	**Feature Origin:**	*Topographic*	**Symbol:**	LN_Road_Street_Construction
Symbol Criteria:	(Condition of Facility = [Under Construction] or [Destroyed] or [Dismantled] or [Ruined] or [Partially Functional] or [Partially Destroyed]) and (Thoroughfare Type = [Street]) and (Through Route = [False] or [No Information])			**Label:**	if Condition of Facility = [Under Construction] or [Partially Functional] , label as "CONSTRUCTION", 6 point Light Condensed Italics Upper case; if Condition of Facility = [Destroyed] or [Dismantled] or [Ruined] or [Partially Destroyed], label as "NOT USABLE", 6 point Light Condensed Italics Upper case
				Color:	Black (58600)

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Road (AP030) *(Continued)*

Curve

Feature Information					
Specification:	*LTDS*	**Feature Origin:**	*Topographic*	**Symbol:**	LN_Road_Planned
Symbol Criteria:	Condition of Facility = [Proposed] or [Planned]			**Label:**	[Geographic Name Information : Full Name] and "([Condition of Facility])" , 6 point Light Condensed Italics Mixed case
				Color:	Black (58600)

Road (AP030)

Point

Inclusion Condition:	*((Length or Diameter >= 300 m.) AND (spaced >= 300 m. apart)) OR (within a Built-up Area and street pattern required to adequately portray the density of the area and [suggested] spacing >= 200 m. apart) OR (needed to complete the component network or to establish connectivity)*

Notes

None

Feature Information					
Specification:	*LTDS*	**Feature Origin:**	*Topographic*	**Symbol:**	PT_Black_RouteMarker_ International
Symbol Criteria:	(Road (AP030) feature (curve) with attribute Thoroughfare Class = [No Information] and [Note : Memorandum] = "International route" and [Route Designation] is known)			**Label:**	[Route Designation], 6 point Condensed Mixed case
				Color:	Black (58600)

Feature Information					
Specification:	*LTDS*	**Feature Origin:**	*Topographic*	**Symbol:**	PT_Black_RouteMarker_ National
Symbol Criteria:	(Road (AP030) feature (curve) with attribute Thoroughfare Class = [Primary Route] and [Route Designation] is known)			**Label:**	[Route Designation], 6 point Condensed Mixed case
				Color:	Black (58600)

Feature Information					
Specification:	*LTDS*	**Feature Origin:**	*Topographic*	**Symbol:**	PT_Black_RouteMarker_State
Symbol Criteria:	(Road (AP030) feature (curve) with attribute Thoroughfare Class = [Secondary Route] or [Local Route] and {Route Designation] is known)			**Label:**	[Route Designation], 6 point Condensed Mixed case
				Color:	Black (58600)

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Roadside Rest Area (AQ135)

Surface

Feature Description:	*A roadside place usually having facilities for people and/or vehicles.*
Inclusion Condition:	*Area >= 15,625 sq m.*

Notes

Area and line features will be generalized to detail compatible with scale.

Label feature as per hierarchy for topo type placement to south neatline corners reading left to right:
1. Position hierarchy:
(a.) northeast (preferred position),
(b.) southeast (1st alternative),
(c.) northwest (2nd alternative,
(d.) southwest (3rd alternative),
(e.) top-centered (4th alternative),
(f.) bottom-centered (5th alternative):
(Hierarchy is based on type positioning so as to avoid overprinting other type or obscuring detail.);
2. Minimum space between type placement and feature symbol is 0.5 mm;
3. This method of type placement shall be used for areal features when space does not permit labeling within that feature.

Names placement shall be oriented to the longest axis of the feature reading left to right and placed within the area outline and centered. If the longest axis is perpendicular to the south neatline, the type shall be placed outside of the area outline, preferred position is northeast of the feature, but may be placed at any position around the feature so as not to overprint any other feature type and reading left to right.

Type for features on land shall be positioned on land if possible. If type must be placed in the water, it shall be positioned so it does not obscure or overprint hydrographic detail. Type for features in the water shall be positioned in the water. If possible, type shall not be positioned across the shoreline.

Omit from Built-up Area.

Limiting lines of feature are omitted if it coalesces with a road (AP030).

Feature Information

Specification:	*LTDS*	**Feature Origin:**	*Topographic*	**Symbol:**	AR_Black_RestArea
Symbol Criteria:	All			**Label:**	"Rest area", 6 point Condensed Mixed case
				Color:	Black (58600)

Roadside Rest Area (AQ135)

Point

Inclusion Condition:	*Area < 15,625 sq m. AND Navigation Landmark = [True]*

Notes

None

Feature Information

Specification:	*LTDS*	**Feature Origin:**	*Topographic*	**Symbol:**	PT_Black_RestArea
Symbol Criteria:	Area < 15,625 AND Navigation Landmark = [True]			**Label:**	"Rest area", 6 point Condensed Mixed case
				Color:	Black (58600)

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Roadstead (BB015)

Surface

Feature Description:	*An area near the shore where vessels can anchor in safety, usually a shallow indentation in the coast.*
Inclusion Condition:	*Area >= 15,625*

Notes

None

Feature Information

Specification:	*Other*	**Feature Origin:**	*Hydrographic*	**Symbol:**	No Symbology Needed
Symbol Criteria:	All			**Label:**	None
				Color:	N/A

Rock Formation (DB160)

Surface

Feature Description:	*A significant outcropping of exposed bedrock.*
Inclusion Condition:	*('Rock Formation Structure' is either 'Columnar' or 'Fossilized Forest')*

Notes

When 2 or more similar area features are separated by less than 0.5 mm at chart scale, the feature will be agglomerated.

Coincident similar area features will be blended to form a single feature.

Area and line features will be generalized to detail compatible with scale.

Feature will be generalized to provide a more aesthetic contoured feature (i.e., smoothed).

Feature Information

Specification:	*LTDS*	**Feature Origin:**	*Topographic*	**Symbol:**	AR_Dk-Brown1815_ColumnarRock_Outline
Symbol Criteria:	All			**Label:**	[Geographic Name Information : Full Name], 6 point Condensed Mixed case if available
				Color:	Black (58600)

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Rock Formation (DB160)

Point

Inclusion Condition:	*('Rock Formation Structure' is either 'Pinnacle' or 'Needle')*

Notes

None

Feature Information					
Specification:	*LTDS*	**Feature Origin:**	*Topographic*	**Symbol:**	PT_Dk-Brown1815_RockFormation
Symbol Criteria:	All			**Label:**	[Geographic Name Information : Full Name], 6 point Condensed Mixed case if available
				Color:	Black (58600)

Roundhouse (AN076)

Surface

Feature Description:	*A circular or semicircular building, with a railway turntable in the centre, used for storing and/or repairing railway locomotives.*
Inclusion Condition:	*Length or Diameter >= 25 m.*

Notes

None

Feature Information					
Specification:	*LTDS*	**Feature Origin:**	*Topographic*	**Symbol:**	AR_Black_Roundhouse
Symbol Criteria:	All			**Label:**	"Roundhouse", 6 point Condensed Mixed case
				Color:	Black (58600)

Roundhouse (AN076)

Point

Inclusion Condition:	*Length or Diameter < 25 m.*

Notes

None

Feature Information					
Specification:	*LTDS*	**Feature Origin:**	*Topographic*	**Symbol:**	PT_Black_Roundhouse
Symbol Criteria:	All			**Label:**	"Roundhouse", 6 point Condensed Mixed case
				Color:	Black (58600)

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Ruins (AL200)

Surface

Feature Description:	*The deteriorated remains of an unspecified structure.*
Inclusion Condition:	*Area >= 15,625 sq m.*

Notes

When 2 or more similar area features are separated by less than 0.5 mm at chart scale, the feature will be agglomerated.

Area and line features will be generalized to detail compatible with scale.

Label feature as per hierarchy for topo type placement to south neatline corners reading left to right:
1. Position hierarchy:
(a.) northeast (preferred position
(b.) southeast (1st alternative)
(c.) northwest (2nd alternative
(d.) southwest (3rd alternative)
(e.) top-centered (4th alternative)
(f.) bottom-centered (5th alternative)
(Hierarchy is based on type positioning so as to avoid overprinting other type or obscuring detail.)
2. Minimum space between type placement and feature symbol is 0.5 mm
3. This method of type placement shall be used for areal features when space does not permit labeling within that feature.

Names placement shall be oriented to the longest axis of the feature reading left to right and placed within the area outline and centered. If the longest axis is perpendicular to the south neatline, the type shall be placed outside of the area outline, preferred position is northeast of the feature, but may be placed at any position around the feature so as not to overprint any other feature type and reading left to right.

Type for features on land shall be positioned on land if possible. If type must be placed in the water, it shall be positioned so it does not obscure or overprint hydrographic detail. Type for features in the water shall be positioned in the water. If possible, type shall not be positioned across the shoreline.

Show Hydrographic Depth or Hydrographic Drying Height values in whole meters if decimeter value is 0; otherwise show the decimeter value as a subscript to the meter value.

Labeling areas, in order of preference:
(1.) Centered in area on one line in the area, type is horizontal, reading left to right.
(2.) Centered in area on one line in the area, oriented along the long axis of the feature, reading left to right, or bottom to top if axis is vertical.
(3.) Centered in area on two approximately equal lines, without splitting a word, type is horizontal, reading left to right.
(4.) Centered on two approximately equal lines without splitting a word, type is oriented along the long axis of the feature, reading left to right, or bottom to top if axis is vertical.

Priority for type placement: 1-right center, 2-bottom center, 3-left center, 4-top center.
(A) Minimum distance from symbol - 1 mm.
(B) Maximum distance from symbol before choosing the next highest priority:
(#1) 4 mm measured to the West end;
(#2) 4 mm measured to the North side (top);
(#3) 4 mm measured to the East end;

If symbols overprint each other, labels are condensed as follows: (1.) If the labels are identical, only one is retained.; (2,) If the labels are not identical, they shall be condensed in to one legend, e.g., "Fishhaven and Well". If multiple depths are shown, only the shallowest is retained.

Descriptive terms, e.g., "Canal" shall be shown if the name is not known. If the descriptive word appears in the name, for example, "PANAMA CANAL". The descriptive type shall not be shown, i.e. do not show "Panama Canal Canal".

The height of a hydrographic feature is that above the surface of the water at High Water. Hydrographic Depth/Height information is not applicable at High Water. Hydrographic Depth/Height information values (Depth Known), (Depth Known by Wire Drag), and (Depth Unknown) are used to describe knowledge about a depth of a feature when the feature is below the water surface/ submerged. Hydrographic Depth/Height information values (Uncovering Height Known) and (Uncovering Height Unknown) are used to describe knowledge about the drying height of a feature when it covers and uncovers, i.e., between high and low water. Hydrographic depth is used to record the depth of a feature when Hydrographic Depth/Height information values are (Depth Known) and (Depth Known by Wire Drag). Hydrographic depth is not applicable when the depth is unknown or when the feature covers and uncovers. Hydrographic Drying Height records the drying height of a feature when Hydrographic Depth/Height information values are known. Hydrographic Drying Height is not applicable when the drying height is unknown, or the feature is below water.

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Ruins (AL200) *(Continued)*

Surface

Notes

Vertical Datum category is the sounding datum to which a hydrographic depth, drying height, or the high water datum to which a Safe Overhead Clearance is referenced. If the hydrographic depth, drying height, or Safe Overhead Clearance is unknown or not applicable, the vertical datum category is not applicable.

The limiting outline of the Built-up Area tint shall be dropped when it overprints linear features (Streams, Roads, and Railroads, etc), or if the space between the symbols is < 0.5mm.

When the boundary of an areal symbol crosses or overprints the shoreline, the boundary overprinting the shoreline or land is deleted.

If areas where separate dangers coalesce because of scale, the danger curves (dotted lines) are deleted except for those on the outermost perimeter of the aggregate danger hazard. A smooth dotted line is drawn around the area.

Show land tint inside the symbol. If the vertical reference category is (above surface /does not cover at High Water), show land tint only.

A blue 31% tint shall not overprint other blue 31% tints. If two blue tinted symbols, or one tinted symbol and water tint overprint, only one 31% tint shall be shown in the area.

Feature Information					
Specification:	*LTDS*	**Feature Origin:**	*Topographic*	**Symbol:**	AR_Black_Ruins
Symbol Criteria:	Location Referenced to Shoreline = [At Shoreline] or [Inland]			**Label:**	[Geographic Name Information: Full Name], 6 point Condensed Mixed case if available; "Ruins ", 6 point Condensed Mixed case
				Color:	Black (58600)

Feature Information					
Specification:	*Chart #1*	**Feature Origin:**	*Hydrographic*	**Symbol:**	F33
Symbol Criteria:	(Location Referenced to Shoreline = [Offshore]) and (Hydrographic Vertical Positioning Information : Water Level Effect = [Always Submerged] or [Awash at Chart Datum] or [Awash at Low Water] or [Covers and Uncovers] or [Partly Submerged]			**Label:**	"Ru", 6 point Mixed case; [Hydrographic Depth], 7 point lower case/ 5 point subscript, Italic if available; [Hydrographic Drying Height], 7 point lower case/ 5 point subscript, Italic if available
				Color:	Black (58600)

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Ruins (AL200)

Point

Inclusion Condition:	*Area < 15,625 sq m.*

Notes

The feature (when height <= 46 meters or when height is not known) shall be oriented perpendicular (90 degrees) to the nearby road (AP030), cart track (AP010), trail (AP050), or railroad (AN010).

When symbolized feature is < 0.2 mm from a line feature, displace to 0.2 mm (map scale).

If point symbol overprints shoreline, the Hydrographic Depth type, posicut, or graphic in the center of the symbol shall be displaced seaward until they no longer overprint the shoreline. If the danger curve (dotted perimeter line) overprints the shoreline, that portion of the dotted perimeter line falling on land shall be deleted.

Label feature as per hierarchy for topo type placement to south neatline corners reading left to right:
1. Position hierarchy:
(a.) northeast (preferred position
(b.) southeast (1st alternative)
(c.) northwest (2nd alternative
(d.) southwest (3rd alternative)
(e.) top-centered (4th alternative)
(f.) bottom-centered (5th alternative)
(Hierarchy is based on type positioning so as to avoid overprinting other type or obscuring detail.)
2. Minimum space between type placement and feature symbol is 0.5 mm
3. This method of type placement shall be used for areal features when space does not permit labeling within that feature.

Type for features on land shall be positioned on land if possible. If type must be placed in the water, it shall be positioned so it does not obscure or overprint hydrographic detail. Type for features in the water shall be positioned in the water. If possible, type shall not be positioned across the shoreline.

Show Hydrographic Depth or Hydrographic Height values in whole meters if decimeter value is 0; otherwise show the decimeter value as a subscript to the meter value.

Priority for type placement: 1-right center, 2-bottom center, 3-left center, 4-top center.
(A) Minimum distance from symbol - 1 mm.
(B) Maximum distance from symbol before choosing the next highest priority:
(#1) 4 mm measured to the West end;
(#2) 4 mm measured to the North side (top);
(#3) 4 mm measured to the East end;
(#4) 4 mm measured to the South side (bottom)

If symbols overprint each other, labels are condensed as follows: (1.) If the labels are identical, only one is retained.; (2,) If the labels are not identical, they shall be condensed in to one legend, e.g., "Fishhaven and Well". If multiple depths are shown, only the shallowest is retained.

Variable type size for Hydrographic Depth values enclosed by danger curves (dotted circles): If Hydrographic Depth < 10, (a single digit principle digit), apply 7 point type to the principle digit, and 5 point type to the subscript, if there is one. If Hydrographic Depth >= 10 (a double digit principle digit), apply 6 point type to the principle digit, and 5 point type to the subscript, if there is one.

Vertical Datum category is the sounding datum to which a hydrographic depth, drying height, or the high water datum to which a Safe Overhead Clearance is referenced. If the hydrographic depth, drying height, or Safe Overhead Clearance is unknown or not applicable, the vertical datum category is not applicable.

If areas where separate dangers coalesce because of scale, the danger curves (dotted lines) are deleted except for those on the outermost perimeter of the aggregate danger hazard. A smooth dotted line is drawn around the area.

Show land tint inside the symbol. If the vertical reference category is (above surface /does not cover at High Water), show land tint only.

A blue 31% tint shall not overprint other blue 31% tints. If two blue tinted symbols, or one tinted symbol and water tint overprint, only one 31% tint shall be shown in the area.

The diameter of the circular symbol perimeter line may be reduced in congested area to avoid overprinting other symbols, but must be at least 2.0 mm diameter, and cannot come closer than 0.2 mm to any type or graphic symbol shown inside the circle. In non-congested areas the normal symbol diameter shall be shown.

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Ruins (AL200) *(Continued)*

Point

Feature Information					
Specification:	*LTDS*	**Feature Origin:**	*Topographic*	**Symbol:**	PT_Black_Ruins
Symbol Criteria:	Height Above Surface Level < 46 meters			**Label:**	[Geographic Name Information: Full Name], 6 point Condensed Mixed case if available; [Hydrographic Depth], 7 point lower case/ 5 point subscript, Italic if available; [Hydrographic Drying Height], 7 point lower case/ 5 point subscript, Italic if available
				Color:	Black (58600)

Feature Information					
Specification:	*LTDS*	**Feature Origin:**	*Topographic*	**Symbol:**	PT_Blue072_VO
Symbol Criteria:	Height Above Surface Level >= 46 meters and Location Referenced to Shoreline: [At Shoreline] or [Inland]			**Label:**	"Ruins", 6 point Condensed Mixed case; [Highest Elevation], 6 point Condensed Bold Mixed case; "([Height Above Surface Level])", 6 point Condensed Mixed case
				Color:	Blue (46351)

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Runway (GB055)

Surface

Feature Description:	*A defined rectangular area on a land aerodrome prepared for the landing and take-off of aircraft.*
Inclusion Condition:	*ALL REQUIRED*

Notes

Contours will be adjusted to planimetric features.

Area and line features will be generalized to detail compatible with scale.

Type for features on land shall be positioned on land if possible. If type must be placed in the water, it shall be positioned so it does not obscure or overprint hydrographic detail. Type for features in the water shall be positioned in the water. If possible, type shall not be positioned across the shoreline.

Feature Information

Specification:	*LTDS*	**Feature Origin:**	*Topographic*	**Symbol:**	AR_Blue072_Runway
Symbol Criteria:	All			**Label:**	"Hard surface", 7 point Condensed Mixed case if Aerodrome Pavement Information : Aerodrome Movement Area Surface Composition [1] = [Asphalt] or [Bituminous Mix] or [Brick] or [Concrete] or [Ice] or [Landing Mat] or [Pierced Steel Planking] or [Wood] or [Asphalt Over Concrete] ; "Soft surface", 7 point Condensed Mixed case if Aerodrome Pavement Information : Aerodrome Movement Area Surface Composition [1] = [Clay] or [Coral] or [Earthen] or [Gravel] or [Laterite] or [Macadam] or [Membrane] or [Non-bituminous Mix] or [Sand] or [Snow] or [Stone] ; "Surface unknown", 7 point Condensed Mixed case if Aerodrome Pavement Information : Aerodrome Movement Area Surface Composition [1] = [No Information]; and [Highest Elevation], 7 point Condensed Upper case
				Color:	Blue (46351)

Sabkha (BH160)

Surface

Feature Description:	*A natural depression in arid or semi-arid regions whose bed is covered with salt encrusted clayey soil.*
Inclusion Condition:	*Area >= 15,625 sq m. OR Navigation Landmark: (True)*

Notes

Coincident similar area features will be blended to form a single feature.

Area and line features will be generalized to detail compatible with scale.

Feature will be generalized to provide a more aesthetic contoured feature (i.e., smoothed).

If a clearing exists inside of the area feature, and the size of the clearing is equal to or greater than the area for the surrounding area feature, the clearing is shown as an open space inside the surround feature. If the area of the clearing is less than the area for the surrounding feature, the clearing is deleted and absorbed into the surrounding area feature.

If two area features with the same feature code do not connect at any point, and have a space between them less than 2.5 mm at map/chart scale, delete the open space that is less than 2.5 mm between the features and combine them into one area feature.

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Sabkha (BH160) *(Continued)*

Surface

Notes

If a portion of an area feature has a minimum width of less than 2.5 mm at map/chart scale, delete that portion of the area feature that is not at least 2.5 mm wide, measured from perimeter to perimeter. If the deletion of the portion of the area based on the above criteria will reduce the area of the remaining portion of the area feature to below the minimum area inclusion, do not delete the narrow portion of the feature that is less than 2.5 mm wide. If the deletion of the portion of the area based on the above criteria will split two larger areas connected by a narrow strip into two separate areas, either of which would be below minimum area inclusion, do not delete the narrow portion of the feature that is less than 2.5 mm wide.

Feature Information					
Specification:	*LTDS*	**Feature Origin:**	*Topographic*	**Symbol:**	AR_Cyan_Sabkha
Symbol Criteria:	All			**Label:**	[Geographic Name Information : Full Name] if available, 6-20 point Condensed Italics Mixed case
				Color:	Blue (46351)

Salt Evaporator (BH155)

Surface

Feature Description:	*Shallow pools, normally man-made, used for the natural evaporation of water for the collection of salt.*
Inclusion Condition:	*Area >= 15,625 sq m.*

Notes

Coincident similar area features will be blended to form a single feature.

Area and line features will be generalized to detail compatible with scale.

Feature will be generalized to provide a more aesthetic contoured feature (i.e., smoothed).

Label feature as per hierarchy for topo type placement to south neatline corners reading left to right:
1. Position hierarchy: (a.) northeast (preferred position), (b.) southeast (1st alternative), (c.) northwest (2nd alternative, (d.) southwest (3rd alternative), (e.) top-centered (4th alternative), (f.) bottom-centered (5th alternative): (Hierarchy is based on type positioning so as to avoid overprinting other type or obscuring detail.);
2. Minimum space between type placement and feature symbol is 0.5 mm;
3. This method of type placement shall be used for areal features when space does not permit labeling within that feature.

Names placement shall be oriented to the longest axis of the feature reading left to right and placed within the area outline and centered. If the longest axis is perpendicular to the south neatline, the type shall be placed outside of the area outline, preferred position is northeast of the feature, but may be placed at any position around the feature so as not to overprint any other feature type and reading left to right.

If a clearing exists inside of the area feature, and the size of the clearing is equal to or greater than the area for the surrounding area feature, the clearing is shown as an open space inside the surround feature. If the area of the clearing is less than the area for the surrounding feature, the clearing is deleted and absorbed into the surrounding area feature.

If two area features with the same feature code do not connect at any point, and have a space between them less than 2.5 mm at map/chart scale, delete the open space that is less than 2.5 mm between the features and combine them into one area feature.

If a portion of an area feature has a minimum width of less than 2.5 mm at map/chart scale, delete that portion of the area feature that is not at least 2.5 mm wide, measured from perimeter to perimeter. If the deletion of the portion of the area based on the above criteria will reduce the area of the remaining portion of the area feature to below the minimum area inclusion, do not delete the narrow portion of the feature that is less than 2.5 mm wide. If the deletion of the portion of the area based on the above criteria will split two larger areas connected by a narrow strip into two separate areas, either of which would be below minimum area inclusion, do not delete the narrow portion of the feature that is less than 2.5 mm wide.

Feature Information					
Specification:	*LTDS*	**Feature Origin:**	*Topographic*	**Symbol:**	AR_Cyan_SaltEvaporator
Symbol Criteria:	Area >= 15,625			**Label:**	None
				Color:	N/A

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Salt Evaporator (BH155)

Point

Inclusion Condition:	*Area < 15,625 sq m. AND Navigation Landmark = [True]*

Notes

Coincident similar area features will be blended to form a single feature.

Area and line features will be generalized to detail compatible with scale.

Feature will be generalized to provide a more aesthetic contoured feature (i.e., smoothed).

Label feature as per hierarchy for topo type placement to south neatline corners reading left to right:
1. Position hierarchy: (a.) northeast (preferred position), (b.) southeast (1st alternative), (c.) northwest (2nd alternative, (d.) southwest (3rd alternative), (e.) top-centered (4th alternative), (f.) bottom-centered (5th alternative): (Hierarchy is based on type positioning so as to avoid overprinting other type or obscuring detail.);
2. Minimum space between type placement and feature symbol is 0.5 mm;
3. This method of type placement shall be used for areal features when space does not permit labeling within that feature.

Names placement shall be oriented to the longest axis of the feature reading left to right and placed within the area outline and centered. If the longest axis is perpendicular to the south neatline, the type shall be placed outside of the area outline, preferred position is northeast of the feature, but may be placed at any position around the feature so as not to overprint any other feature type and reading left to right.

If a clearing exists inside of the area feature, and the size of the clearing is equal to or greater than the area for the surrounding area feature, the clearing is shown as an open space inside the surround feature. If the area of the clearing is less than the area for the surrounding feature, the clearing is deleted and absorbed into the surrounding area feature.

If two area features with the same feature code do not connect at any point, and have a space between them less than 2.5 mm at map/chart scale, delete the open space that is less than 2.5 mm between the features and combine them into one area feature.

If a portion of an area feature has a minimum width of less than 2.5 mm at map/chart scale, delete that portion of the area feature that is not at least 2.5 mm wide, measured from perimeter to perimeter. If the deletion of the portion of the area based on the above criteria will reduce the area of the remaining portion of the area feature to below the minimum area inclusion, do not delete the narrow portion of the feature that is less than 2.5 mm wide. If the deletion of the portion of the area based on the above criteria will split two larger areas connected by a narrow strip into two separate areas, either of which would be below minimum area inclusion, do not delete the narrow portion of the feature that is less than 2.5 mm wide.

Feature Information

Specification:	*LTDS*	**Feature Origin:**	*Topographic*	**Symbol:**	PT_Cyan_SaltEvaporator
Symbol Criteria:	Area < 15,625 AND Navigation Landmark = [True]			**Label:**	"Salt evaporator", 6 point Condensed Italics Mixed case
				Color:	Blue (48253)

Salt Flat (BH150)

Surface

Feature Description:	*A flat area of natural surface salt deposits.*
Inclusion Condition:	*Area >= 15,625 sq m.*

Notes

Coincident similar area features will be blended to form a single feature.

Area and line features will be generalized to detail compatible with scale.

Feature will be generalized to provide a more aesthetic contoured feature (i.e., smoothed).

Type sizes per are sizes at map/chart scale:
Area features only.
06 point - <= 770 mm square area and <= 14 mm width;
07 point - <= 2,296 mm square area and <= 28 mm width;
09 point - <= 5,192 mm square area and 44 mm width;
10 point - <=9,796 mm square area and <= 62 mm width;
12 point - <= 16,632 mm square area and <= 84 mm width;
14 point - <= 24,960 mm square area and <= 104 mm width;
16 point - > 24,960 mm square area. Where area measurements are inconsistent, the larger type size shall be used.
Where the full range of type sizes is not available for a particular label, the closest available type size shall be used.

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Salt Flat (BH150) *(Continued)*

Surface

Notes

Label feature as per hierarchy for topo type placement to south neatline corners reading left to right:
1. Position hierarchy: (a.) northeast (preferred position), (b.) southeast (1st alternative), (c.) northwest (2nd alternative, (d.) southwest (3rd alternative), (e.) top-centered (4th alternative), (f.) bottom-centered (5th alternative): (Hierarchy is based on type positioning so as to avoid overprinting other type or obscuring detail.);
2. Minimum space between type placement and feature symbol is 0.5 mm;
3. This method of type placement shall be used for areal features when space does not permit labeling within that feature.

Names placement shall be oriented to the longest axis of the feature reading left to right and placed within the area outline and centered. If the longest axis is perpendicular to the south neatline, the type shall be placed outside of the area outline, preferred position is northeast of the feature, but may be placed at any position around the feature so as not to overprint any other feature type and reading left to right.

Symbols and associated area patterns of underpassing features (except drainage shorelines) are broken for all bridges, except footbridges. This rule does not apply to land tint on Littoral Planning Charts.

If a clearing exists inside of the area feature, and the size of the clearing is equal to or greater than the area for the surrounding area feature, the clearing is shown as an open space inside the surround feature. If the area of the clearing is less than the area for the surrounding feature, the clearing is deleted and absorbed into the surrounding area feature.

If two area features with the same feature code do not connect at any point, and have a space between them less than 2.5 mm at map/chart scale, delete the open space that is less than 2.5 mm between the features and combine them into one area feature.

If a portion of an area feature has a minimum width of less than 2.5 mm at map/chart scale, delete that portion of the area feature that is not at least 2.5 mm wide, measured from perimeter to perimeter. If the deletion of the portion of the area based on the above criteria will reduce the area of the remaining portion of the area feature to below the minimum area inclusion, do not delete the narrow portion of the feature that is less than 2.5 mm wide. If the deletion of the portion of the area based on the above criteria will split two larger areas connected by a narrow strip into two separate areas, either of which would be below minimum area inclusion, do not delete the narrow portion of the feature that is less than 2.5 mm wide.

Feature Information

Specification:	*LTDS*	**Feature Origin:**	*Topographic*	**Symbol:**	AR_Cyan_SaltFlat
Symbol Criteria:	All			**Label:**	[Geographic Name Information : Full Name] if available, 6-16 point Condensed Italics Mixed case; "Salt flat", 6-16 point Condensed Italics Mixed case
				Color:	Blue (48253)

Sand Dunes (DB170)

Surface

Feature Description:	*One or more mounds or ridges of sand usually formed by the wind.*
Inclusion Condition:	*Area >= 15,625 sq m.*

Notes

Coincident similar area features will be blended to form a single feature.

Area and line features will be generalized to detail compatible with scale.

Feature will be generalized to provide a more aesthetic contoured feature (i.e., smoothed).

If the type of Sand Dunes is unknown, the label "Dunes" is positioned at 100.0 mm intervals to the overall extent of the area.

Use the structure shape (symbol) that most closely approximates the configuration of the dunes.

Symbols and associated area patterns of underpassing features (except drainage shorelines) are broken for all bridges, except footbridges. This rule does not apply to land tint on Littoral Planning Charts.

Sand Dune patterns shall be positioned according to Sand Dune orientation, to the nearest 15° increment, to indicate their orientation relative to the prevailing winds.

If two area features with the same feature code do not connect at any point, and have a space between them less than 2.5 mm at map/chart scale, delete the open space that is less than 2.5 mm between the features and combine them into one area feature.

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Sand Dunes (DB170) *(Continued)*

Surface

Notes

If a portion of an area feature has a minimum width of less than 2.5 mm at map/chart scale, delete that portion of the area feature that is not at least 2.5 mm wide, measured from perimeter to perimeter. If the deletion of the portion of the area based on the above criteria will reduce the area of the remaining portion of the area feature to below the minimum area inclusion, do not delete the narrow portion of the feature that is less than 2.5 mm wide. If the deletion of the portion of the area based on the above criteria will split two larger areas connected by a narrow strip into two separate areas, either of which would be below minimum area inclusion, do not delete the narrow portion of the feature that is less than 2.5 mm wide.

Feature Information					
Specification:	*LTDS*	**Feature Origin:**	*Topographic*	**Symbol:**	AR_Dk-Brown1815_ SandDunes_Unknown
Symbol Criteria:	Sand Dune Type = [Other] or [No information]			**Label:**	[Geographic Name Information : Full Name], 7-10 point Regular Mixed case if available; "Dunes", 7-10 point Regular Mixed case
				Color:	Black (58600)

Feature Information					
Specification:	*LTDS*	**Feature Origin:**	*Topographic*	**Symbol:**	AR_Dk-Brown1815_ SandDunes_Crescent
Symbol Criteria:	Sand Dune Type = [Crescent] or [Parabolic]			**Label:**	[Geographic Name Information : Full Name], 7-10 point Regular Mixed case if available
				Color:	Black (58600)

Feature Information					
Specification:	*LTDS*	**Feature Origin:**	*Topographic*	**Symbol:**	AR_Dk-Brown1815_ SandDunes_Dome
Symbol Criteria:	Sand Dune Type = [Dome]			**Label:**	[Geographic Name Information : Full Name], 7-10 point Regular Mixed case if available
				Color:	Black (58600)

Feature Information					
Specification:	*LTDS*	**Feature Origin:**	*Topographic*	**Symbol:**	AR_Dk-Brown1815_ SandDunes_Transverse
Symbol Criteria:	Sand Dune Type = [Transverse] or [Dome and Transverse]			**Label:**	[Geographic Name Information : Full Name], 7-10 point Regular Mixed case if available
				Color:	Black (58600)

Feature Information					
Specification:	*LTDS*	**Feature Origin:**	*Topographic*	**Symbol:**	AR_Dk-Brown1815_ SandDunes_Linear
Symbol Criteria:	Sand Dune Type = [Linear]			**Label:**	[Geographic Name Information : Full Name], 7-10 point Regular Mixed case if available
				Color:	Black (58600)

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Sand Dunes (DB170) *(Continued)*

Surface

Feature Information					
Specification:	*LTDS*	**Feature Origin:**	*Topographic*	**Symbol:**	AR_Dk-Brown1815_ SandDunes_Ripple
Symbol Criteria:	Sand Dune Type = [Ripple]			**Label:**	[Geographic Name Information : Full Name], 7-10 point Regular Mixed case if available
				Color:	Black (58600)

Feature Information					
Specification:	*LTDS*	**Feature Origin:**	*Topographic*	**Symbol:**	AR_Dk-Brown1815_ SandDunes_Star
Symbol Criteria:	Sand Dune Type = [Star]			**Label:**	[Geographic Name Information : Full Name], 7-10 point Regular Mixed case if available
				Color:	Black (58600)

Scoreboard (AK161)

Point

Feature Description:	*A large outdoor board for publicly displaying the score in an athletic event.*
Inclusion Condition:	*Height Above Surface Level: >= 46 m.*

Notes

None

Feature Information					
Specification:	*LTDS*	**Feature Origin:**	*Topographic*	**Symbol:**	PT_Blue072_VO
Symbol Criteria:	Height Above Surface >= 46			**Label:**	"Scoreboard" , 7 point Condensed Mixed case; [Highest Elevation], 6 point Condensed Bold Mixed case; "([Height Above Surface Level])", 6 point Condensed Mixed case
				Color:	Blue (46351)

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Seaplane Run (GB070)

Surface

Feature Description:	*A designated portion of water that is outlined by visual surface markings and used by seaplanes to land and take-off.*
Inclusion Condition:	*All Required*

Notes

None

Feature Information					
Specification:	*Chart #1*	**Feature Origin:**	*Hydrographic*	**Symbol:**	N13
Symbol Criteria:	All			**Label:**	None
				Color:	N/A

Settlement (AL105)

Surface

Feature Description:	*A continuously occupied concentration of tents or lightweight fixed structures (for example: huts) serving as residences.*
Inclusion Condition:	*Area >= 15,625 sq m.*

Notes

Label feature as per hierarchy for topo type placement to south neatline corners reading left to right:
1. Position hierarchy:
(a.) northeast (preferred position
(b.) southeast (1st alternative)
(c.) northwest (2nd alternative
(d.) southwest (3rd alternative)
(e.) top-centered (4th alternative)
(f.) bottom-centered (5th alternative)
(Hierarchy is based on type positioning so as to avoid overprinting other type or obscuring detail.)
2. Minimum space between type placement and feature symbol is 0.5 mm
3. This method of type placement shall be used for areal features when space does not permit labeling within that feature.

Feature Information					
Specification:	*LTDS*	**Feature Origin:**	*Topographic*	**Symbol:**	AR_Black_Settlement
Symbol Criteria:	All			**Label:**	[Geographic Name Information: Full Name], 6 point Condensed Mixed case if available
				Color:	Black (58600)

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Settling Pond (AC030)

Surface

Feature Description:	*A small reservoir where solid matter is precipitated from a liquid by evaporating or settling.*
Inclusion Condition:	*Area >= 625 sq m.*

Notes

When 2 or more similar area features are separated by less than 0.5 mm at chart scale, the feature will be agglomerated.

Area and line features will be generalized to detail compatible with scale.

Label feature as per hierarchy for topo type placement to south neatline corners reading left to right:
1. Position hierarchy:
(a.) northeast (preferred position
(b.) southeast (1st alternative)
(c.) northwest (2nd alternative
(d.) southwest (3rd alternative)
(e.) top-centered (4th alternative)
(f.) bottom-centered (5th alternative)
(Hierarchy is based on type positioning so as to avoid overprinting other type or obscuring detail.)
2. Minimum space between type placement and feature symbol is 0.5 mm
3. This method of type placement shall be used for areal features when space does not permit labeling within that feature.

Type for features on land shall be positioned on land if possible. If type must be placed in the water, it shall be positioned so it does not obscure or overprint hydrographic detail. Type for features in the water shall be positioned in the water. If possible, type shall not be positioned across the shoreline.

Limiting lines of feature are omitted if it coalesces with a road (AP030).

Feature Information

Specification:	*LTDS*	**Feature Origin:**	*Topographic*	**Symbol:**	AR_Cyan_SettlingPond
Symbol Criteria:	All			**Label:**	"Settling pond", 6 point Condensed Mixed case
				Color:	Black (58600)

Sewage Treatment Plant (AC507)

Surface

Feature Description:	*An operational area with buildings and other facilities for the purification of wastewater.*
Inclusion Condition:	*Area >= 15,625 sq m.*

Notes

None

Feature Information

Specification:	*LTDS*	**Feature Origin:**	*Topographic*	**Symbol:**	AR_Black_SewageTreatment Plant
Symbol Criteria:	All			**Label:**	[Geographic Name Information : Full Name], 6 point Condensed Mixed case; "Sewage plant", 6 point Condensed Mixed case
				Color:	Black (58600)

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Sewage Treatment Plant (AC507)

Point

Inclusion Condition:	*Area: < 15,625 sq m.*

Notes

None

Feature Information					
Specification:	*LTDS*	**Feature Origin:**	*Topographic*	**Symbol:**	PT_Black_SewageTreatment Plant
Symbol Criteria:	All			**Label:**	[Geographic Name Information : Full Name], 6 point Condensed Mixed case; "Sewage plant", 6 point Condensed Mixed case
				Color:	Black (58600)

Shallow Water Area (BH250)

Surface

Feature Description:	*That part of an inland body of water where the depth is relatively slight (less than 1 metre) and thus likely to limit the passage of watercraft.*
Inclusion Condition:	*Area >= 15,625 sq m.*

Notes

A solid blue tint must be shown on all charts to emphasize shallow water.

Feature Information					
Specification:	*Chart #1*	**Feature Origin:**	*Hydrographic*	**Symbol:**	I30
Symbol Criteria:	All			**Label:**	"Shallow water area", 7 Point Mixed case, Italic
				Color:	Black (58600)

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Shanty Town (AL208)

Surface

Feature Description:	*A section of a built-up area consisting chiefly of densely packed shacks and having few, if any, streets and no public facilities.*
Inclusion Condition:	*Area >= 15,625 sq m.*

Notes

Coincident similar area features will be blended to form a single feature.

Area and line features will be generalized to detail compatible with scale.

Type sizes per are sizes at map/chart scale: Area features only.
06 point - <= 770 mm square area and <= 14 mm width
07 point - <= 2,296 mm square area and <= 28 mm width
09 point - <= 5,192 mm square area and 44 mm width
10 point - <=9,796 mm square area and <= 62 mm width
12 point - <= 16,632 mm square area and <= 84 mm width
Where area measurements are inconsistent, the larger type size shall be used. Where the full range of type sizes is not available for a particular label, the closest available type size shall be used.

When a Wall symbol coalesces with Built-Up Area outline, or Shantytown outline, omit Built-Up Area outline or Shantytown outline, and show Wall with Built-Up Area tint only.

Feature Information

Specification:	*LTDS*	**Feature Origin:**	*Topographic*	**Symbol:**	AR_Black_BUA_ShantyTown
Symbol Criteria:	All			**Label:**	[Geographic Name Information: Full Name], 6-12 point Condensed Mixed case if available
				Color:	Black (58600)

Shed (AL019)

Surface

Feature Description:	*A small building, generally of light construction, that usually has one or more open sides.*
Inclusion Condition:	*Length or Diameter >= 25 meters*

Notes

None

Feature Information

Specification:	*LTDS*	**Feature Origin:**	*Topographic*	**Symbol:**	AR_Black_Shed
Symbol Criteria:	All			**Label:**	None
				Color:	N/A

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Shed (AL019)

Point

Inclusion Condition:	*Length or Diameter < 25 meters*

Notes

None

Feature Information					
Specification:	*LTDS*	**Feature Origin:**	*Topographic*	**Symbol:**	PT_Black_Shed
Symbol Criteria:	All			**Label:**	None
				Color:	N/A

Ship Elevator (BI006)

Curve

Feature Description:	*A device used to raise ships vertically between water bodies with different elevations.*
Inclusion Condition:	*Length or Diameter >= 125 m.*

Notes

None

Feature Information					
Specification:	*LTDS*	**Feature Origin:**	*Topographic*	**Symbol:**	LN_Black_ShipElevator
Symbol Criteria:	All			**Label:**	"Ship elevator", 6 point Condensed Mixed case
				Color:	Black (58600)

Ship Elevator (BI006)

Point

Inclusion Condition:	*Length or Diameter < 125 m.*

Notes

None

Feature Information					
Specification:	*LTDS*	**Feature Origin:**	*Topographic*	**Symbol:**	PT_Black_ShipElevator
Symbol Criteria:	Height Above Surface Level < 46			**Label:**	"Ship elevator", 6 point Condensed Mixed case
				Color:	Black (58600)

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Shipyard (BB241)

Surface

Feature Description:	*A large enclosed area adjoining the sea or a major river, including facilities in which ships are built or repaired.*
Inclusion Condition:	*Area >= 15,625 sq m.*

Notes

None

Feature Information

Specification:	*LTDS*	**Feature Origin:**	*Topographic*	**Symbol:**	AR_Black_Shipyard
Symbol Criteria:	Area >= 15,625			**Label:**	[Geographic Name Information: Full Name]. 6 point Regular Mixed case if available; "Shipyard", 6 point Regular Mixed case
				Color:	Black (58600)

Shipyard (BB241)

Point

Inclusion Condition:	*Area < 15,625 sq m. AND Navigation Landmark = [True]*

Notes

None

Feature Information

Specification:	*LTDS*	**Feature Origin:**	*Topographic*	**Symbol:**	PT_Black_Shipyard
Symbol Criteria:	Area < 15,625 sq m. AND Navigation Landmark = [True]			**Label:**	[Geographic Name Information: Full Name]. 6 point Regular Mixed case if available; "Shipyard", 6 point Regular Mixed case
				Color:	Black (58600)

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Shopping Complex (AG030)

Surface

Feature Description:	*A commercial facility tenanted by multiple retail stores (for example: household goods, clothing, or books), restaurants (for example: a food court), entertainment venues (for example: a movie theatre or electronic games arcade), and/or other businesses (for example: professional services), that is completely enclosed, climate controlled, has a common public arcade (for example: furnished with decorative plantings, benches, and/or fountains) and a common vehicle parking area (for example: a parking lot or parking garage).*
Inclusion Condition:	*(Area >= 15,625 and Height Above Surface Level < 46) OR (Area >= 5,625 and Height Above Surface Level >= 46)*

Notes

None

Feature Information

Specification:	*LTDS*	**Feature Origin:**	*Topographic*	**Symbol:**	AR_Black_ShoppingComplex
Symbol Criteria:	(Area >= 15,625 and Height Above Surface Level < 46) OR (Area >= 5,625 and Height Above Surface Level >= 46)			**Label:**	"Shopping mall", 6 point Condensed Mixed case
				Color:	Black (58600)

Shopping Complex (AG030)

Point

Inclusion Condition:	*(Area < 15,625) and (Navigation Landmark = [True]) and (Height Above Surface Level < 46)*

Notes

None

Feature Information

Specification:	*LTDS*	**Feature Origin:**	*Topographic*	**Symbol:**	PT_Black_ShoppingComplex
Symbol Criteria:	(Area < 15,625) and (Navigation Landmark = [True]) and (Height Above Surface Level < 46)			**Label:**	"Shopping mall", 6 point Condensed Mixed case
				Color:	Black (58600)

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Shoreline (BA024)

Curve

Feature Description:	*A line drawn along the normal limit of wave action above the higher high water line as defined by the National Authority.*
Inclusion Condition:	*All required*

Notes

Area and line features will be generalized to detail compatible with scale.

Feature will be generalized to provide a more aesthetic contoured feature (i.e., smoothed).

No type shall cross Shoreline. Type will either be shown entirely within the Tidal Water (BA040) or entirely on land.

Mean High Water is the preferred vertical datum for shoreline portrayal. When High Water is not available, the shoreline will be delineated by whatever means possible. There may never be a segment of missing shoreline (by definition, the line where land mass is in contact with a body of open water).

Shoreline (BA010 Line Water Boundary) which is coincident with features BB190 Berthing Structure, BB230 Seawall, AP030 Road, AN010 Railway, AN050 Railway Sidetrack, and AL260 Wall are not shown.

Symbols and associated area patterns of underpassing features (except drainage shorelines) are broken for all bridges, except footbridges. This rule does not apply to land tint on Littoral Planning Charts.

Shoreline (BA010 Line Water Boundary) shall not be shown where it becomes coincident with a manmade harbor or coastal structure.

The coastal shoreline will not be shown when a swamp is adjacent to open water. The shoreline will separate the open water from the swamp symbol.

The water side limit Mangrove or Nipa is always shown by a dashed line. The landside limits (Mean High Water line = Land Water Boundary) is shown when known.

When Shoreline (BA010) around an island is smaller than the symbol for a point feature on the island, delete the shoreline and show the point feature symbol in the water.

If the embankment (causeway) is adjacent to a shoreline < .25 mm from a road or railroad, suppress the shoreline.

Feature Information

Specification:	*Chart #1*	**Feature Origin:**	*Hydrographic*	**Symbol:**	C1
Symbol Criteria:	Shoreline Delineated = [True]			**Label:**	None
				Color:	N/A

Feature Information

Specification:	*Chart #1*	**Feature Origin:**	*Hydrographic*	**Symbol:**	C2
Symbol Criteria:	Shoreline Delineated = [False]			**Label:**	None
				Color:	N/A

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Shoreline Construction (BB081)

Surface

Feature Description:	*An artificial structure attached to land bordering a body of water and fixed in position.*
Inclusion Condition:	*Length or Diameter >= 125 m. AND Width >= 20 m.*

Notes

Breakwater Rules:
If feature is (Below surface) or (Covers and uncovers), add label "Breakwater". Type shall be 6 point Mixed case italic. If feature length <=13 mm at chart scale, abbreviate Breakwater as "Bkw". Type placement for Breakwater or Bkw shall be in the water, parallel to the feature, readable from left to right or bottom to top.

On very large scales, the above-water nature of the structure may often be represented; an indication that ships do not go alongside may be given by showing the sloping sides. Examples of such symbols are the use of hachures, small irregular circles indicating loose boulders, or the seawall symbol indicating a slope of masonry or concrete.
If there is a possibility of misinterpretation by the mariner, it is recommended that a dotted danger line is drawn parallel to and 1 mm away from the edge of the structure to indicate the danger. All labeling and the danger curve coincident with the edge of the shoreline construction feature are deleted.

General Rules:
A shoreline construction feature (BB081) is generally not intended for berthing, even on the sheltered side. A "mole" is a term used to describe a breakwater alongside which vessels may lie on the sheltered side only. In this case, it should be shown as a wharf for the side that is used for berthing, and as a breakwater on the unsheltered side. In some cases, a mole may lie entirely within an artificial harbor, permitting vessels to lie along both sides. In this case, it should be shown as an offshore loading facility.

Where symbols with the same feature code but different values of vertical reference categories (i.e. cover/uncovers, submerged, above surface (at high water)) are adjacent to each other, the boundary of the feature that is the highest in vertical plane shall be shown, and the boundary of the feature above water is shown whole, while the feature that covers and uncovers has a portion of its symbol perimeter that overprints the higher feature deleted.

When the area symbol boundary overprints the shoreline, and the vertical reference category is above high water, the coincident feature boundary and coincident shoreline are deleted, and the land tint extends to fill the feature.

Show land tint inside the symbol. If the vertical reference category is (above surface /does not cover at High Water), show land tint only.

A blue 31% tint shall not overprint other blue 31% tints. If two blue tinted symbols, or one tinted symbol and water tint overprint, only one 31% tint shall be shown in the area.

When the area symbol or cased line symbol overprints the shoreline, shoreline is deleted.

Feature Information

Specification:	*LTDS*	**Feature Origin:**	*Topographic*	**Symbol:**	AR_Black_ShorelineConstruction_Dry
Symbol Criteria:	(Shoreline Construction Type = [Pier] or [Wharf] or [Quay] or [Recreational Pier] or [Seawall] or (Promenade] or [Other] or [No Information]) or (Shoreline Construction Type = [Breakwater] or [Groin] or [Mole] or [Training Wall]) and (Water Level Effect = [Always Dry] or [Other] or [No Information])			**Label:**	[Geographic Name Information : Full Name] and "Pier" if Shoreline Construction Type = [Pier] or [Recreational Pier], 6 Point Regular Mixed case; [Geographic Name Information : Full Name] and "Wharf" if Shoreline Construction Type = [Wharf] or [Quay], 6 Point Regular Mixed case; "Seawall" if Shoreline Construction Type = [Seawall] or [Promenade], 6 Point Regular Mixed case; "Breakwater" if Shoreline Construction Type = [Breakwater] or [Mole], 6 Point Regular Mixed case; "Groin" if Shoreline Construction Type = [Groin], 6 Point Regular Mixed case; "Training Wall" if Shoreline Construction Type = [Training Wall], 6 Point Regular Mixed case
				Color:	Black (58600)

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Shoreline Construction (BB081) ***(Continued)***

Surface

Feature Information					
Specification:	*LTDS*	**Feature Origin:**	*Topographic*	**Symbol:**	AR_BlackCyan_Shoreline Construction_NotDry
Symbol Criteria:	Shoreline Construction Type = [Breakwater] or [Groin] or [Training Wall] AND Water Level Effect = [Partly Submerged] or [Always Submerged] or [Covers and Uncovers] or [Awash at Low Water] or [Awash at Chart Datum]			**Label:**	"Breakwater" if Shoreline Construction Type = [Breakwater], 6 Point Mixed case Italic; "Groin" if Shoreline Construction Type = [Groin], 6 Point Mixed case Italic; "Training Wall" if Shoreline Construction Type: [Training Wall], 6 Point Mixed case Italic
				Color:	Black (58600)

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Shoreline Construction (BB081)

Curve

Inclusion Condition:	*Length or Diameter >= 125 m. AND Width < 20 m.*

Notes

Breakwater Rules:
If feature is (Below surface) or (Covers and uncovers), add label "Breakwater". Type shall be 6 point Mixed case italic. If feature length <=13 mm at chart scale, abbreviate Breakwater as "Bkw". Type placement for Breakwater or Bkw shall be in the water, parallel to the feature, readable from left to right or bottom to top.

On very large scales, the above-water nature of the structure may often be represented; an indication that ships do not go alongside may be given by showing the sloping sides. Examples of such symbols are the use of hachures, small irregular circles indicating loose boulders, or the seawall symbol indicating a slope of masonry or concrete.
If there is a possibility of misinterpretation by the mariner, it is recommended that a dotted danger line is drawn parallel to and 1 mm away from the edge of the structure to indicate the danger. All labeling and the danger curve coincident with the edge of the shoreline construction feature are deleted.

General Rules:
A shoreline construction feature (BB081) is generally not intended for berthing, even on the sheltered side. A "mole" is a term used to describe a breakwater alongside which vessels may lie on the sheltered side only. In this case, it should be shown as a wharf for the side that is used for berthing, and as a breakwater on the unsheltered side. In some cases, a mole may lie entirely within an artificial harbor, permitting vessels to lie along both sides. In this case, it should be shown as an offshore loading facility.

Where symbols with the same feature code but different values of vertical reference categories (i.e. cover/uncovers, submerged, above surface (at high water)) are adjacent to each other, the boundary of the feature that is the highest in vertical plane shall be shown, and the boundary of the feature above water is shown whole, while the feature that covers and uncovers has a portion of its symbol perimeter that overprints the higher feature deleted.

Feature Information

Specification:	*LTDS*	Feature Origin:	*Topographic*	Symbol:	LN_Black_Shoreline Construction_Dry
Symbol Criteria:	(Shoreline Construction Type = [Pier] or [Wharf] or [Quay] or [Recreational Pier] or [Seawall] or (Promenade] or [Other] or [No Information]) or (Shoreline Construction Type = [Breakwater] or [Groin] or [Mole] or [Training Wall]) and (Water Level Effect = [Always Dry] or [Other] or [No Information])			Label:	[Geographic Name Information : Full Name] and "Pier" if Shoreline Construction Type = [Pier] or [Recreational Pier], 6 Point Regular Mixed case; [Geographic Name Information : Full Name] and "Wharf" if Shoreline Construction Type = [Wharf] or [Quay], 6 Point Regular Mixed case; "Seawall" if Shoreline Construction Type = [Seawall] or [Promenade], 6 Point Regular Mixed case; "Breakwater" if Shoreline Construction Type = [Breakwater] or [Mole], 6 Point Regular Mixed case; "Groin" if Shoreline Construction Type = [Groin], 6 Point Regular Mixed case; "Training Wall" if Shoreline Construction Type = [Training Wall], 6 Point Regular Mixed case
				Color:	Black (58600)

Feature Information

Specification:	*LTDS*	Feature Origin:	*Topographic*	Symbol:	LN_Black_Shoreline Construction_NotDry
Symbol Criteria:	Shoreline Construction Type = [Breakwater] or [Groin] or [Training Wall] AND Water Level Effect = [Partly Submerged] or [Always Submerged] or [Covers and Uncovers] or [Awash at Low Water] or [Awash at Chart Datum]			Label:	"Breakwater" if Shoreline Construction Type = [Breakwater], 6 Point Mixed case Italic; "Groin" if Shoreline Construction Type = [Groin], 6 Point Mixed case Italic; "Training Wall" if Shoreline Construction Type = [Training Wall], 6 Point Mixed case Italic
				Color:	Black (58600)

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Shoreline Ramp (BB082)

Surface

Feature Description:	*A ramp-like structure on a shoreline that is intended to facilitate the movement of vessels and/or materials (for example: logs) into or out of the water.*
Inclusion Condition:	*Width >= 20 m.*

Notes

Label shall be on land, parallel to length of feature, 2 mm separation, readable left to right or bottom to top if feature is vertical.

Where symbols with the same feature code but different values of vertical reference categories (i.e. cover/uncovers, submerged, above surface (at high water)) are adjacent to each other, the boundary of the feature that is the highest in vertical plane shall be shown, and the boundary of the feature above water is shown whole, while the feature that covers and uncovers has a portion of its symbol perimeter that overprints the higher feature deleted.

When the area symbol boundary overprints the shoreline, and the vertical reference category is above high water, the coincident feature boundary and coincident shoreline are deleted, and the land tint extends to fill the feature.

A blue 31% tint shall not overprint other blue 31% tints. If two blue tinted symbols, or one tinted symbol and water tint overprint, only one 31% tint shall be shown in the area.

Feature Information

Specification:	*LTDS*	**Feature Origin:**	*Topographic*	**Symbol:**	AR_Black_ShorelineRamp_Dry
Symbol Criteria:	Water Level Effect = [Always Dry] or [Other] or [No Information]			**Label:**	"Ramp", 6 point Condensed Mixed case
				Color:	Black (58600)

Feature Information

Specification:	*LTDS*	**Feature Origin:**	*Topographic*	**Symbol:**	AR_BlackCyan_ShorelineRamp_NotDry
Symbol Criteria:	Water Level Effect = [Partly Submerged] or [Always Submerged] or [Covers and Uncovers] or [Awash at Low Water] or [Awash at Chart Datum]			**Label:**	"Ramp", 6 point Condensed Mixed case
				Color:	Black (58600)

Shoreline Ramp (BB082)

Curve

Inclusion Condition:	*Width < 20 m.*

Notes

Label shall be on land, parallel to length of feature, 2 mm separation, readable left to right or bottom to top if feature is vertical.

Feature Information

Specification:	*LTDS*	**Feature Origin:**	*Topographic*	**Symbol:**	LN_Black_ShorelineRamp_Dry
Symbol Criteria:	Water Level Effect = [Always Dry] or [Other] or [No Information]			**Label:**	"Ramp", 6 point Condensed Mixed case
				Color:	Black (58600)

Feature Information

Specification:	*LTDS*	**Feature Origin:**	*Topographic*	**Symbol:**	LN_Black_ShorelineRamp_NotDry
Symbol Criteria:	Water Level Effect = [Partly Submerged] or [Always Submerged] or [Covers and Uncovers] or [Awash at Low Water] or [Awash at Chart Datum]			**Label:**	"Ramp", 6 point Condensed Mixed case
				Color:	Black (58600)

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Side Scan Sonar Coverage (FC179)

Surface

Feature Description:	*An area of horizontal coverage recorded by side scan SONAR survey.*
Inclusion Condition:	*All Required*

Notes

None

Feature Information					
Specification:	*Other*	**Feature Origin:**	*Hydrographic*	**Symbol:**	TBD; no Chart #1 symbology
Symbol Criteria:	All			**Label:**	TBD
				Color:	TBD

Ski-jump (AK150)

Curve

Feature Description:	*A man-made structure consisting of a steep ramp levelling off at the end and built on a natural slope, used in ski-jumping.*
Inclusion Condition:	*Length or Diameter >= 125 m.*

Notes

Area and line features will be generalized to detail compatible with scale.

Label feature as per hierarchy for topo type placement to south neatline corners reading left to right:
1. Position hierarchy:
(a.) northeast (preferred position
(b.) southeast (1st alternative)
(c.) northwest (2nd alternative
(d.) southwest (3rd alternative)
(e.) top-centered (4th alternative)
(f.) bottom-centered (5th alternative)
(Hierarchy is based on type positioning so as to avoid overprinting other type or obscuring detail.)
2. Minimum space between type placement and feature symbol is 0.5 mm
3. This method of type placement shall be used for areal features when space does not permit labeling within that feature.

Type for features on land shall be positioned on land if possible. If type must be placed in the water, it shall be positioned so it does not obscure or overprint hydrographic detail. Type for features in the water shall be positioned in the water. If possible, type shall not be positioned across the shoreline.

Feature Information					
Specification:	*LTDS*	**Feature Origin:**	*Topographic*	**Symbol:**	LN_Black_SkiJump
Symbol Criteria:	All			**Label:**	"Ski jump", 6 point Condensed Mixed case
				Color:	Black (58600)

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Ski-jump (AK150)

Point

Inclusion Condition:	*Length or Diameter < 125 m.*

Notes

Area and line features will be generalized to detail compatible with scale.

Label feature as per hierarchy for topo type placement to south neatline corners reading left to right:
1. Position hierarchy:
(a.) northeast (preferred position
(b.) southeast (1st alternative)
(c.) northwest (2nd alternative
(d.) southwest (3rd alternative)
(e.) top-centered (4th alternative)
(f.) bottom-centered (5th alternative)
(Hierarchy is based on type positioning so as to avoid overprinting other type or obscuring detail.)
2. Minimum space between type placement and feature symbol is 0.5 mm
3. This method of type placement shall be used for areal features when space does not permit labeling within that feature.

Type for features on land shall be positioned on land if possible. If type must be placed in the water, it shall be positioned so it does not obscure or overprint hydrographic detail. Type for features in the water shall be positioned in the water. If possible, type shall not be positioned across the shoreline.

When obstructions coalesce at map scale, label with highest obstruction information.

Feature Information

Specification:	*LTDS*	**Feature Origin:**	*Topographic*	**Symbol:**	PT_Black_SkiJump
Symbol Criteria:	Height Above Surface < 46			**Label:**	"Ski jump", 6 point Condensed Mixed case
				Color:	Black (58600)

Feature Information

Specification:	*LTDS*	**Feature Origin:**	*Topographic*	**Symbol:**	PT_Blue072_VO
Symbol Criteria:	Height Above Surface >= 46			**Label:**	"Ski jump" , 7 point Condensed Mixed case; [Highest Elevation], 6 point Condensed Bold Mixed case; "([Height Above Surface Level])", 6 point Condensed Mixed case
				Color:	Blue (46351)

Sluice Gate (BI040)

Curve

Feature Description:	*A gate used to regulate the flow or level of water in a watercourse (for example: stream, irrigation ditch, or sluice).*
Inclusion Condition:	*Length or Diameter >= 25 m.*

Notes

Area and line features will be generalized to detail compatible with scale.

Label feature as per hierarchy for topo type placement to south neatline corners reading left to right:
1. Position hierarchy: (a.) northeast (preferred position), (b.) southeast (1st alternative), (c.) northwest (2nd alternative, (d.) southwest (3rd alternative), (e.) top-centered (4th alternative), (f.) bottom-centered (5th alternative): (Hierarchy is based on type positioning so as to avoid overprinting other type or obscuring detail.);
2. Minimum space between type placement and feature symbol is 0.5 mm;
3. This method of type placement shall be used for areal features when space does not permit labeling within that feature.

Omit if not shown in conjunction with drainage feature.

The point of the Lock or Sluice Gate symbol shall be positioned pointing upstream.

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Sluice Gate (BI040) *(Continued)*

Curve

Feature Information					
Specification:	*LTDS*	**Feature Origin:**	*Topographic*	**Symbol:**	LN_Black_SluiceGate
Symbol Criteria:	All			**Label:**	"Sluice gate", 6 point Condensed Mixed case
				Color:	Black (58600)

Sluice Gate (BI040)

Point

Inclusion Condition:	*Length or Diameter < 25 m.*

Notes

Label feature as per hierarchy for topo type placement to south neatline corners reading left to right:
1. Position hierarchy: (a.) northeast (preferred position), (b.) southeast (1st alternative), (c.) northwest (2nd alternative, (d.) southwest (3rd alternative), (e.) top-centered (4th alternative), (f.) bottom-centered (5th alternative): (Hierarchy is based on type positioning so as to avoid overprinting other type or obscuring detail.);
2. Minimum space between type placement and feature symbol is 0.5 mm;
3. This method of type placement shall be used for areal features when space does not permit labeling within that feature.

Omit if not shown in conjunction with drainage feature.

The point of the Lock or Sluice Gate symbol shall be positioned pointing upstream.

Feature Information					
Specification:	*LTDS*	**Feature Origin:**	*Topographic*	**Symbol:**	PT_Black_SluiceGate
Symbol Criteria:	All			**Label:**	None
				Color:	N/A

Small Bottom Object (BD002)

Point

Feature Description:	*An object protruding above the water body bottom whose greatest dimension is less than five metres.*
Inclusion Condition:	*All Required*

Notes

None

Feature Information					
Specification:	*Other*	**Feature Origin:**	*Hydrographic*	**Symbol:**	TBD; no Chart #1 symbology
Symbol Criteria:	All			**Label:**	TBD
				Color:	TBD

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Small Craft Facility (BB201)

Point

Feature Description:	*A place at which a service generally of interest to small craft or pleasure boats is available.*
Inclusion Condition:	*Length < 150 meters*

Notes

None

Feature Information					
Specification:	*Chart #1*	**Feature Origin:**	*Hydrographic*	**Symbol:**	U1.1
Symbol Criteria:	All			**Label:**	None
				Color:	N/A

Smokestack (AF010)

Point

Feature Description:	*A vertical structure containing a passage or flue for discharging smoke and gases of combustion.*
Inclusion Condition:	*All Required*

Notes

If one symbol coalesces with another symbol for the same type feature, displace symbols to allow a minimum separation of 0.21mm.

Type for features on land shall be positioned on land if possible. If type must be placed in the water, it shall be positioned so it does not obscure or overprint hydrographic detail. Type for features in the water shall be positioned in the water. If possible, type shall not be positioned across the shoreline.

When obstructions coalesce at map scale, label with highest obstruction information.

If >= 3 equal symbols would coalesce at map scale, portray with a representative pattern.

Feature Information					
Specification:	*LTDS*	**Feature Origin:**	*Topographic*	**Symbol:**	PT_Black_Smokestack
Symbol Criteria:	Height Above Surface Level < 46 meters			**Label:**	None
				Color:	N/A

Feature Information					
Specification:	*LTDS*	**Feature Origin:**	*Topographic*	**Symbol:**	PT_Blue072_VO
Symbol Criteria:	Height Above Surface Level >= 46 meters			**Label:**	"Smokestack", 6 point Condensed Mixed case; [Highest Elevation], 6 point Condensed Bold Mixed case; "([Height Above Surface Level])", 6 point Condensed Mixed case
				Color:	Blue (46351)

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Snag (BD140)

Surface

Feature Description:	*A tree, branch or broken pile embedded in the ocean floor, river or lake bottom and may not be visible on the surface, forming thereby a hazard to vessels.*
Inclusion Condition:	*Length >= 150 meters*

Notes

Use the following abbreviations for Horizontal Accuracy and Existence values:
If Position Approximate, label "PA";
If Position Doubtful, label "PD";
If Existence Doubtful, label "ED";
If Reported, label "Rep".

Date shall only be shown if Existence = Reported. Date shall be shown in parentheses if available. Always show "Rep" and date on the same line.
Unless specifically indicated on source material as inaccurate or doubtful, etc., or modified by other override rules, features shall be assumed to be accurate and definite. This rule applied to accuracy and existence of unique features, and is not related to the accuracy of the source as a whole.
Following IHO practice, features whose positions are approximate are labeled "PA" on the source charts. Features whose positions are doubtful are labeled "PD" on the source charts. Features whose existence is doubtful are labeled "ED" on the source charts. Features that have only been reported are labeled "Rep" on the source charts, and usually show the date of report in parentheses.

Definitions:
PA - Position Approximate = The position has not been accurately determined, or does not remain fixed.;
PD - Position Doubtful = The feature has been reported in various positions, but not definitely determined to be in any.;
ED - Existence Doubtful = Indicates the possible existence of the feature, the actual existence of which has not been established.

Date shall only be shown if Existence = Reported. Date shall be shown in parentheses if available. Always show "Rep" and date on the same line.

Labels shall be placed in water area, and shall not overprint the shoreline or be placed on land or in the foreshore.

When two or more identical features fall within 10 mm of each other, one legend with plural description shall be shown, with type placed as close as possible to the center of the group without overprinting features.
Where symbols with the same feature code but different values of Hydrographic Vertical Positioning Information: Water Level Effect (i.e. Always Dry , Always Submerged, Covers and Uncovers)) are adjacent to each other, the boundary of the feature that is the highest in vertical plane shall be shown, and the boundary of the feature above water is shown whole, while the feature that covers and uncovers has a portion of its symbol perimeter that overprints the higher feature deleted.

When two or more identical symbols overprint each other, delete the portions that overlap and show as one symbol with an outline of the combined area of the symbols.

A blue 31% tint shall not overprint other blue 31% tints. If two blue tinted symbols, or one tinted symbol and water tint overprint, only one 31% tint shall be shown in the area.

Unless specifically indicated on source material as inaccurate or doubtful, etc., or modified by other override rules, features shall be assumed to be accurate and definite. This rule applied to accuracy and existence of unique features, and is not related to the accuracy of the source as a whole.

When the boundary of an areal symbol crosses or overprints the shoreline, the boundary overprinting the shoreline or land is deleted.

If areas where separate dangers coalesce because of scale, the danger curves (dotted lines) are deleted except for those on the outermost perimeter of the aggregate danger hazard. A smooth dotted line is drawn around the area.

Feature Information

Specification:	*Chart #1*	**Feature Origin:**	*Hydrographic*	**Symbol:**	K40
Symbol Criteria:	Hydrographic Vertical Positioning Information : Water Level Effect = [Always Submerged]			**Label:**	"Obstn", 6 point Mixed case, Italic; [Hydrographic Depth] if available, 7 point lower case/ 5 subscript, Italic
				Color:	Black (58600)

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Snag (BD140)

Point

Inclusion Condition:	*Length < 150 meters*

Notes

Use the following abbreviations for Horizontal Accuracy and Existence values:
If Position Approximate, label "PA";
If Position Doubtful, label "PD";
If Existence Doubtful, label "ED";
If Reported, label "Rep".

Date shall only be shown if Existence = Reported. Date shall be shown in parentheses if available. Always show "Rep" and date on the same line.
Unless specifically indicated on source material as inaccurate or doubtful, etc., or modified by other override rules, features shall be assumed to be accurate and definite. This rule applied to accuracy and existence of unique features, and is not related to the accuracy of the source as a whole.
Following IHO practice, features whose positions are approximate are labeled "PA" on the source charts. Features whose positions are doubtful are labeled "PD" on the source charts. Features whose existence is doubtful are labeled "ED" on the source charts. Features that have only been reported are labeled "Rep" on the source charts, and usually show the date of report in parentheses.

Definitions:
PA - Position Approximate = The position has not been accurately determined, or does not remain fixed.;
PD - Position Doubtful = The feature has been reported in various positions, but not definitely determined to be in any.;
ED - Existence Doubtful = Indicates the possible existence of the feature, the actual existence of which has not been established.

Date shall only be shown if Existence = Reported. Date shall be shown in parentheses if available. Always show "Rep" and date on the same line.

Labels shall be placed in water area, and shall not overprint the shoreline or be placed on land or in the foreshore.

When two or more identical features fall within 10 mm of each other, one legend with plural description shall be shown, with type placed as close as possible to the center of the group without overprinting features.
Where symbols with the same feature code but different values of Hydrographic Vertical Positioning Information: Water Level Effect (i.e. Always Dry , Always Submerged, Covers and Uncovers)) are adjacent to each other, the boundary of the feature that is the highest in vertical plane shall be shown, and the boundary of the feature above water is shown whole, while the feature that covers and uncovers has a portion of its symbol perimeter that overprints the higher feature deleted.

When two or more identical symbols overprint each other, delete the portions that overlap and show as one symbol with an outline of the combined area of the symbols.

A blue 31% tint shall not overprint other blue 31% tints. If two blue tinted symbols, or one tinted symbol and water tint overprint, only one 31% tint shall be shown in the area.

Unless specifically indicated on source material as inaccurate or doubtful, etc., or modified by other override rules, features shall be assumed to be accurate and definite. This rule applied to accuracy and existence of unique features, and is not related to the accuracy of the source as a whole.

When considered advisable to show the exact position of the object, a small circle shall be added at the base of the sloping stroke (K43.2).

Feature Information					
Specification:	*Chart #1*	**Feature Origin:**	*Hydrographic*	**Symbol:**	K43.1
Symbol Criteria:	Hydrographic Vertical Positioning Information : Water Level Effect = [Always Submerged]			**Label:**	"Obstn", 6 point Mixed case, Italic; [Hydrographic Depth] if available, 7 point lower case/ 5 subscript, Italic
				Color:	Black (58600)

Feature Information					
Specification:	*Chart #1*	**Feature Origin:**	*Hydrographic*	**Symbol:**	K43.1, K43.2
Symbol Criteria:	Hydrographic Vertical Positioning Information : Water Level Effect = [Always Dry] or [Awash at Chart Datum] or [Awash at Low Water] or [Covers and Uncovers] or [Partly Submerged]			**Label:**	None
				Color:	N/A

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Snow Field and/or Ice-field (BJ100)

Surface

Feature Description:	*A large area permanently covered by snow and/or ice.*
Inclusion Condition:	*Area >= 15,625 sq m.*

Notes

Coincident similar area features will be blended to form a single feature.

Area and line features will be generalized to detail compatible with scale.

Feature will be generalized to provide a more aesthetic contoured feature (i.e., smoothed).

Type sizes per are sizes at map/chart scale: Area features only.
06 point - <= 770 mm square area and <= 14 mm width;
07 point - <= 2,296 mm square area and <= 28 mm width;
09 point - <= 5,192 mm square area and 44 mm width;
10 point - <=9,796 mm square area and <= 62 mm width;
12 point - <= 16,632 mm square area and <= 84 mm width;
14 point - <= 24,960 mm square area and <= 104 mm width;
16 point - > 24,960 mm square area.
Where area measurements are inconsistent, the larger type size shall be used. Where the full range of type sizes is not available for a particular label, the closest available type size shall be used.

Label feature as per hierarchy for topo type placement to south neatline corners reading left to right:
1. Position hierarchy: (a.) northeast (preferred position), (b.) southeast (1st alternative), (c.) northwest (2nd alternative, (d.) southwest (3rd alternative), (e.) top-centered (4th alternative), (f.) bottom-centered (5th alternative): (Hierarchy is based on type positioning so as to avoid overprinting other type or obscuring detail.);
2. Minimum space between type placement and feature symbol is 0.5 mm;
3. This method of type placement shall be used for areal features when space does not permit labeling within that feature.

Names placement shall be oriented to the longest axis of the feature reading left to right and placed within the area outline and centered. If the longest axis is perpendicular to the south neatline, the type shall be placed outside of the area outline, preferred position is northeast of the feature, but may be placed at any position around the feature so as not to overprint any other feature type and reading left to right.

Symbols and associated area patterns of underpassing features (except drainage shorelines) are broken for all bridges, except footbridges. This rule does not apply to land tint on Littoral Planning Charts.

If a clearing exists inside of the area feature, and the size of the clearing is equal to or greater than the area for the surrounding area feature, the clearing is shown as an open space inside the surround feature. If the area of the clearing is less than the area for the surrounding feature, the clearing is deleted and absorbed into the surrounding area feature.

If two area features with the same feature code do not connect at any point, and have a space between them less than 2.5 mm at map/chart scale, delete the open space that is less than 2.5 mm between the features and combine them into one area feature.

If a portion of an area feature has a minimum width of less than 2.5 mm at map/chart scale, delete that portion of the area feature that is not at least 2.5 mm wide, measured from perimeter to perimeter. If the deletion of the portion of the area based on the above criteria will reduce the area of the remaining portion of the area feature to below the minimum area inclusion, do not delete the narrow portion of the feature that is less than 2.5 mm wide. If the deletion of the portion of the area based on the above criteria will split two larger areas connected by a narrow strip into two separate areas, either of which would be below minimum area inclusion, do not delete the narrow portion of the feature that is less than 2.5 mm wide.

Do not show land tint inside symbol.

Feature Information

Specification:	*LTDS*	**Feature Origin:**	*Topographic*	**Symbol:**	AR_Cyan_SnowIceField
Symbol Criteria:	All			**Label:**	[Geographic Name Information : Full Name] if available, 6-16 point Condensed Mixed case; "Snow", 6-16 point Condensed Mixed case if Frozen Cover Type = [Snowfield]; "Ice", 6-16 point Condensed Mixed case if Frozen Cover Type = [Ice-field]
				Color:	Blue (46351)

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Soil Surface Region (DA010)

Surface

Feature Description:	*A region of the land that is homogeneous with respect to a soil characteristic.*
Inclusion Condition:	*Area >= 15,625 sq m.*

Notes

Coincident similar area features will be blended to form a single feature.

Area and line features will be generalized to detail compatible with scale.

Feature will be generalized to provide a more aesthetic contoured feature (i.e., smoothed).

Type sizes per are sizes at map/chart scale: Area features only.
06 point - <= 770 mm square area and <= 14 mm width;
07 point - <= 2,296 mm square area and <= 28 mm width;
09 point - <= 5,192 mm square area and 44 mm width;
10 point - <=9,796 mm square area and <= 62 mm width;
12 point - <= 16,632 mm square area and <= 84 mm width;
14 point - <= 24,960 mm square area and <= 104 mm width;
16 point - > 24,960 mm square area.
Where area measurements are inconsistent, the larger type size shall be used. Where the full range of type sizes is not available for a particular label, the closest available type size shall be used.

Symbols and associated area patterns of underpassing features (except drainage shorelines) are broken for all bridges, except footbridges. This rule does not apply to land tint on Littoral Planning Charts.

Karst areas >=25.4 mm square at map scale shall not be symbolized with the area pattern. Standard contouring shall depict the area and the description label KARST shall be added throughout such areas.

If a clearing exists inside of an area feature, and the size of the clearing is equal to or greater than the area inclusion condition for the surrounding feature, the clearing is shown as an open space inside the surrounding feature. If the area of the clearing is less than the area inclusion condition for the surrounding feature, the clearing is deleted and absorbed into the surrounding area feature.

If two area features with the same feature code do not connect at any point, and have a space between them of less than 2.5 mm at map/chart scale, delete the open space that is less than 2.5 mm wide between the features and combine them into one area feature.

If a portion of an area feature has a minimum width of less than 2.5 mm at map/chart scale, delete that portion of the area feature that is not at least 2.5 mm wide, measured from perimeter to perimeter. If the deletion of the portion of the area based on the above criteria will reduce the area of the remaining portion of the area feature to below the minimum Area Inclusion, do not delete the narrow portion of the feature that is less than 2.5 mm wide. If the deletion of the portion of the area based on the above criteria will split two larger areas connected by a narrow strip into two separate areas, either of which would be below minimum Area inclusion, do not delete the narrow portion of the feature that is less than 2.5 mm wide.

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Soil Surface Region (DA010) ***(Continued)***

Surface

Feature Information					
Specification:	*LTDS*	**Feature Origin:**	*Topographic*	**Symbol:**	AR_Dk-Brown1815_SoilSurface_DistortedSurface
Symbol Criteria:	(Area < 360000) and (Terrain Morphology = [Numerous Cobbles and Boulders] or [Stony Areas] or [Stony Soil with Surface Rock] or [Stony Soil with scattered Boulders] or [Stony Soil with numerous Boulders] or [Numerous Boulders] or [Numerous Rock Outcrops] or [Scattered Boulders] or [Talus] or [Boulder Field] or [Highly Fractured Rock] or [Weathered Lava] or [Unweathered Lava] or [Stony Soil with numerous Rock Outcrops] or [Irregular with deep Foliation Fractures] or [Rugged with numerous Rock Outcrops] or [Rugged Bedrock] or [Highly Distorted with sharp Rocky Ridges] or [Stony Soil with numerous Gullies] or [Moderately Dissected] or [Moderately Dissected with scattered Rock Outcrops] or [Dissected Floodplain] or [Highly Dissected] or [Deep Erosional Gullies] or [Steep Rugged Dissected with narrow Gullies] or [Karst with numerous Sinkholes and Solution Valleys] or [Karst with numerous Sinkholes] or [Hummocky Karst with Large Hills] or [Hummocky Karst with Low Mounds]) or (Terrain Surface Material [1] = [Loess])			**Label:**	"Boulders", 6 point Light Condensed Italics Mixed case if Terrain Morphology = [Numerous Cobbles and Boulders] or [Stony Soil with scattered Boulders] or [Stony Soil with numerous Boulders] or [Numerous Boulders] or [Scattered Boulders] or [Boulder Field]; "Rocky", 6 point Light Condensed Italics Mixed case if Terrain Morphology = [Stony Areas] or [Stony Soil with Surface Rock] or [Numerous Rock Outcrops] or [Talus] or [Highly Fractured Rock] or [Stony Soil with numerous Rock Outcrops] or [Irregular with deep Foliation Fractures] or [Rugged with numerous Rock Outcrops] or [Rugged Bedrock] or [Highly Distorted with sharp Rocky Ridges]; "Lava", 6 point Light Condensed Italics Mixed case if Terrain Morphology = [Weathered Lava] or [Unweathered Lava]; "Dissected terrain", 6 point Light Condensed Italics Mixed case if Terrain Morphology = [Stony Soil with numerous Gullies] or [Moderately Dissected] or [Moderately Dissected with scattered Rock Outcrops] or [Dissected Floodplain] or [Highly Dissected] or [Deep Erosional Gullies] or [Steep Rugged Dissected with narrow Gullies]; "Karst", 6 point Light Condensed Italics Mixed case if Terrain Morphology = [Karst with numerous Sinkholes and Solution Valleys] or [Karst with numerous Sinkholes] or [Hummocky Karst with Large Hills] or [Hummocky Karst with Low Mounds]; "Loess", 6 point Light Condensed Italics Mixed case if Terrain Surface Material [1] = [Loess]
				Color:	Black (58600)

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Soil Surface Region (DA010) ***(Continued)***

Surface

Feature Information					
Specification:	*LTDS*	**Feature Origin:**	*Topographic*	**Symbol:**	AR_SoilSurface_ DistortedSurface-Large
Symbol Criteria:	(Area >= 360000) and (Terrain Morphology = [Numerous Cobbles and Boulders] or [Stony Areas] or [Stony Soil with Surface Rock] or [Stony Soil with scattered Boulders] or [Stony Soil with numerous Boulders] or [Numerous Boulders] or [Numerous Rock Outcrops] or [Scattered Boulders] or [Talus] or [Boulder Field] or [Highly Fractured Rock] or [Weathered Lava] or [Unweathered Lava] or [Stony Soil with numerous Rock Outcrops] or [Irregular with deep Foliation Fractures] or [Rugged with numerous Rock Outcrops] or [Rugged Bedrock] or [Highly Distorted with sharp Rocky Ridges] or [Stony Soil with numerous Gullies] or [Moderately Dissected] or [Moderately Dissected with scattered Rock Outcrops] or [Dissected Floodplain] or [Highly Dissected] or [Deep Erosional Gullies] or [Steep Rugged Dissected with narrow Gullies] or [Karst with numerous Sinkholes and Solution Valleys] or [Karst with numerous Sinkholes] or [Hummocky Karst with Large Hills] or [Hummocky Karst with Low Mounds]) or (Terrain Surface Material [1] = [Loess])			**Label:**	"Boulders", 6 point Light Condensed Italics Mixed case if Terrain Morphology = [Numerous Cobbles and Boulders] or [Stony Soil with scattered Boulders] or [Stony Soil with numerous Boulders] or [Numerous Boulders] or [Scattered Boulders] or [Boulder Field]; "Rocky", 6 point Light Condensed Italics Mixed case if Terrain Morphology = [Stony Areas] or [Stony Soil with Surface Rock] or [Numerous Rock Outcrops] or [Talus] or [Highly Fractured Rock] or [Stony Soil with numerous Rock Outcrops] or [Irregular with deep Foliation Fractures] or [Rugged with numerous Rock Outcrops] or [Rugged Bedrock] or [Highly Distorted with sharp Rocky Ridges]; "Lava", 6 point Light Condensed Italics Mixed case if Terrain Morphology = [Weathered Lava] or [Unweathered Lava]; "Dissected terrain", 6 point Light Condensed Italics Mixed case if Terrain Morphology = [Stony Soil with numerous Gullies] or [Moderately Dissected] or [Moderately Dissected with scattered Rock Outcrops] or [Dissected Floodplain] or [Highly Dissected] or [Deep Erosional Gullies] or [Steep Rugged Dissected with narrow Gullies]; "Karst", 6 point Light Condensed Italics Mixed case if Terrain Morphology = [Karst with numerous Sinkholes and Solution Valleys] or [Karst with numerous Sinkholes] or [Hummocky Karst with Large Hills] or [Hummocky Karst with Low Mounds]; "Loess", 6 point Light Condensed Italics Mixed case if Terrain Surface Material [1] = [Loess]
				Color:	Black (58600)

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Soil Surface Region (DA010) ***(Continued)***

Surface

Feature Information					
Specification:	*LTDS*	**Feature Origin:**	*Topographic*	**Symbol:**	AR_Dk-Brown1815_SoilSurface_SandandGravel
Symbol Criteria:	(Terrain Surface Material [1] = [Gravel]) or ([Sand] and [Gravel]) or ([Gravel] and [Sand])			**Label:**	None
				Color:	N/A

Feature Information					
Specification:	*LTDS*	**Feature Origin:**	*Topographic*	**Symbol:**	AR_Dk-Brown1815_SoilSurface_Sand
Symbol Criteria:	(Terrain Surface Material [1] = [Sand]) or ([Sand] and [Mud]) or ([Sand] and [Boulders]) or ([Boulders] and [Sand]) or ([Mud] and [Sand])			**Label:**	"Sand and mud", 6 point Light Condensed Italics Mixed case if (Terrain Surface Material [1] = [Sand] or [Mud]) and (Terrain Surface Material [2] = [Sand] or [Mud]); "Sand and boulders", 6 point Light Condensed Italics Mixed case if (Terrain Surface Material [1] = [Sand] or [Boulders]) and (Terrain Surface Material [2] = [Sand] or [Boulders])
				Color:	Black (58600)

Feature Information					
Specification:	*LTDS*	**Feature Origin:**	*Topographic*	**Symbol:**	AR_SoilSurface_Other
Symbol Criteria:	(Terrain Morphology = [Numerous Small Lakes and Ponds] or [Numerous Crevasses] or [Numerous Terraces] or [Mine Tailing(s)] or [Numerous Dykes] or [Numerous Dyked Fields] or [Numerous Fences] or [Numerous Stone Walls] or [Numerous Man-made Drainage] or [Numerous Terraced Fields] or [Parallel Earthen Rows] or [Numerous Hedgerows] or [Other])			**Label:**	[Terrain Morphology] , 6 point Light Condensed Italics Mixed if Terrain Morphology = [Numerous Small Lakes and Ponds] or [Numerous Crevasses] or [Numerous Terraces] or [Mine Tailing(s)] or [Numerous Dykes] or [Numerous Dyked Fields] or [Numerous Fences] or [Numerous Stone Walls] or [Numerous Man-made Drainage] or [Numerous Terraced Fields] or [Parallel Earthen Rows] or [Numerous Hedgerows]; [Note : Memorandum], 6 point Light Condensed Italics Mixed if (Terrain Morphology = [Other]) and (Note : Memorandum <> "UNK") and (Note : Memorandum is not null)
				Color:	Black (58600)

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Solar Farm (AD025)

Surface

Feature Description:	*An extensive collection of solar panels that are collocated and serve a common purpose (for example: the generation of electricity or the generation of heating steam and/or water).*
Inclusion Condition:	*Area: >= 15,625 sq m.*

Notes

None

Feature Information					
Specification:	*LTDS*	**Feature Origin:**	*Topographic*	**Symbol:**	AR_Black_SolarFarm
Symbol Criteria:	Area >= 15,625			**Label:**	[Geographic Name Information : Full Name] if available: 6 point Condensed Mixed case; "Solar Farm", 6 point Condensed Mixed case
				Color:	Black (58600)

Solar Farm (AD025)

Point

Inclusion Condition:	Area: < 15,625 sq m. AND Navigation Landmark = [True]

Notes

None

Feature Information					
Specification:	*LTDS*	**Feature Origin:**	*Topographic*	**Symbol:**	PT_Black_SolarFarm
Symbol Criteria:	Area < 15,625 sq m. AND Navigation Landmark = [True]			**Label:**	[Geographic Name Information : Full Name] if available: 6 point Condensed Mixed case; "Solar Farm", 6 point Condensed Mixed case
				Color:	Black (58600)

Sound Signal Device (BC103)

Point

Feature Description:	*An acoustic aid to navigation consisting of a device that generates distinctive audible sounds.*
Inclusion Condition:	*All Required*

Notes

If it is required to include the type of fog signal, the labels must be used. To avoid clutter, where a fog signal is co-located with another aid to navigation, the fog signal label or the symbol R 1 should be shown, but not both.

FOG SIGNALS: RHYTHM AND PERIOD:

A single sound (blast) repeated at intervals should be shown by '(1)' following the type of signal.

Multiple sounds (blasts) (other than Morse or composite signals) repeated at intervals must be shown by '(2)', '(3)', etc, following the type of signal.

Morse code rhythms must be shown by 'Mo' followed by the Morse letter in brackets. Note: there is no R1 symbol included as the type of fog signal is shown.

Composite rhythms (other than Morse) where groups of blasts are sounded must be shown as the full sequence.

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Sound Signal Device (BC103) *(Continued)*

Point

Notes

The period of a fog signal is the time taken for a complete sequence of sound emissions. Where space permits, it must be charted for major signals (and on the largest scale charts for minor signals where considered useful) following the number of blasts. The period must be given in seconds, even for periods of one minute or longer.

FOG SIGNALS ON BUOYS:

The existence of fog signals on buoys must be indicated, on appropriate scales, by legends such as 'Bell', 'Horn' or 'Gong' or the symbol R1 if there is no requirement to define the type of fog signal.

Wave-actuated fog signals have no regular rhythm and must be charted by a legend indicating the type of signal eg 'Bell', 'Horn', 'Gong' against the buoy symbol.

Fog signals operated automatically should be charted by a legend which includes the number of blasts (or strokes) and the period.

Wave-actuated signals in conjunction with automatic signals should be charted as in the following example: Horn(1)15s Whis. Note: there is no R1 symbol included as the type of fog signal is shown.

Feature Information					
Specification:	*Chart #1*	**Feature Origin:**	*Hydrographic*	**Symbol:**	R14
Symbol Criteria:	Fog Signal Type = [Bell] or [Submarine Bell]			**Label:**	"Bell", 7 point Mixed case if Submarine Acoustic Signal Type = [Shore Sound Signal]; "Bell", 7 point Mixed case Italic if Submarine Acoustic Signal Type = [Non-shore Sound Signal]
				Color:	Black (58600)

Feature Information					
Specification:	*Chart #1*	**Feature Origin:**	*Hydrographic*	**Symbol:**	R11
Symbol Criteria:	Fog Signal Type = [Diaphone]			**Label:**	"Dia", 7 point Mixed case if Submarine Acoustic Signal Type = [Shore Sound Signal]; "Dia", 7 point Mixed case Italic if Submarine Acoustic Signal Type = [Non-shore Sound Signal]
				Color:	Black (58600)

Feature Information					
Specification:	*Chart #1*	**Feature Origin:**	*Hydrographic*	**Symbol:**	R10
Symbol Criteria:	Fog Signal Type = [Explosive]			**Label:**	"Explos", 7 point Mixed case if Submarine Acoustic Signal Type = [Shore Sound Signal]; "Explos", 7 point Mixed case Italic if Submarine Acoustic Signal Type = [Non-shore Sound Signal]
				Color:	Black (58600)

Feature Information					
Specification:	*Chart #1*	**Feature Origin:**	*Hydrographic*	**Symbol:**	R16
Symbol Criteria:	Fog Signal Type = [Gong]			**Label:**	"Gong", 7 point Mixed case if Submarine Acoustic Signal Type = [Shore Sound Signal]; "Gong", 7 point Mixed case Italic if Submarine Acoustic Signal Type = [Non-shore Sound Signal]
				Color:	Black (58600)

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Sound Signal Device (BC103) *(Continued)*

Point

Feature Information					
Specification:	*Chart #1*	**Feature Origin:**	*Hydrographic*	**Symbol:**	R13
Symbol Criteria:	Fog Signal Type = [Horn] or [Nautophone]			**Label:**	"Horn", 7 point Mixed case if Submarine Acoustic Signal Type = [Shore Sound Signal]; "Horn", 7 point Mixed case Italic if Submarine Acoustic Signal Type = [Non-shore Sound Signal]
				Color:	Black (58600)

Feature Information					
Specification:	*Chart #1*	**Feature Origin:**	*Hydrographic*	**Symbol:**	R12
Symbol Criteria:	Fog Signal Type = [Siren]			**Label:**	"Siren", 7 point Mixed case if Submarine Acoustic Signal Type = [Shore Sound Signal]; "Siren", 7 point Mixed case Italic if Submarine Acoustic Signal Type = [Non-shore Sound Signal]
				Color:	Black (58600)

Sounding (BE020)

Point

Feature Description:	*A spot depth or drying height that has been referenced to a vertical datum.*
Inclusion Condition:	*All Required*

Notes

Soundings shall normally be plotted in their true positions. If a selected sounding overprints other important detail, such as aids to navigation or dangers, a different sounding is selected, if possible. If the selected sounding is the shallower than any other sounding around it, it must be shown. In this case, it is shown as an "out of position" sounding and a leader line is used to show the true position of the sounding. Leader line shall be 3-25 mm in length.

Soundings shall be displaced seaward when they overprint the shoreline until they no longer overprint.

If a channel (deep area between two shallow areas) is too narrow to place a sounding in, and the sounding is the shallowest depth in the channel between the two shallow areas, place the sounding alongside the channel, in parentheses.

Use the following abbreviations for Accuracy and Existence values:
If Accuracy value = Position Approximate, label "PA";
If Accuracy value = Position Doubtful, label "PD";
If Existence value = Existence Doubtful, label "ED";
If Existence value = Reported or Sounding Metadata : Bathymetric Measurement Quality Category = [Reported Depth], label "Rep", 6 Point Mixed case Italic, Black (58600).

Use the following abbreviations for Sounding Metadata : Bathymetric Measurement Quality Category:

If Bathymetric Measurement Quality Category = [Depth Doubtful], label "SD", 6 Point Upper case Italic, Black (58600).

Show Hydrographic Depth or Hydrographic Height values in whole meters if decimeter value is 0; otherwise show the decimeter value as a subscript to the meter value.

Date shall only be shown if Existence = Reported. Date shall be shown in parentheses. If date is unknown, delete label and parentheses from the symbol. Always show "Rep" and date on the same line.

If soundings shown on an NGA product chart are taken from graphic source chart/survey that has been enlarged by factor greater than two, for example, soundings on a 1:50,000 source chart shown on an NGA chart at a scale larger than 1:25,000, then those soundings shall be shown as vertical (non-Italic) soundings, including sounding categories as Drying Height (Approximate) soundings, No Bottom Found (Approximate) soundings, Approximate soundings, and Doubtful (Approximate) soundings.
When this override occurs, the chart shall show the following note, wording as appropriate, in the margin of the chart. See "Notes and Cautions" section of product specification for information regarding note portrayal.

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Sounding (BE020) ***(Continued)***

Point

Notes

"NOTE" to be placed centered on and stacked on top of "Soundings in vertical figures are from smaller scale charts."

Some foreign charting agencies (and the IHO standard) use vertical (upright hairline) type for questionable soundings and slant type for normal soundings. NGA has also adopted this practice and should be consistent with IHO in displaying the various sounding types. Sounding types that are questionable shall be shown in upright vertical type, where accurate (normal) soundings shall be shown in italic (slanted) type. When this occurs, the chart shall show the following note, wording as appropriate, in the margin of the chart. See" Notes and Cautions" section of product specifications for information regarding note portrayal. SOUNDINGS - In compliance with IHO specifications, soundings on this chart that are reliable are represented by sloping or italicized figures. Soundings from less reliable sources are represented by vertical figures. The label shall be centered and stacked above the note.

Sounding data that is unreliable, based on notes or cautions on the source material, or some other information known to the compiler, shall be shown as upright or vertical type (Approximate), Drying Height (Approximate), No Bottom (Approximate), Doubtful (Approximate)), depending on the type of sounding. A note explaining the reason for the upright or vertical soundings shall be given in the margin or land area. See "Notes and Cautions" section of product specifications.

Unless specifically indicated on source material as inaccurate or doubtful, etc., or modified by other override rules, features shall be assumed to be accurate and definite. This rule applied to accuracy and existence of unique features, and is not related to the accuracy of the source as a whole.

Following IHO practice, features whose positions are approximate are labeled "PA" on the source charts. Features whose positions are doubtful are labeled "PD" on the source charts. Features whose existence is doubtful are labeled "ED" on the source charts. Features that have only been reported are labeled "Rep" on the source charts, and usually show the date of report in parentheses.

Definitions:
PA - Position Approximate = The position has not been accurately determined, or does not remain fixed.;
PD - Position Doubtful = The feature has been reported in various positions, but not definitely determined to be in any.;
ED - Existence Doubtful = Indicates the possible existence of the feature, the actual existence of which has not been established.

If a chart shows a mixture of soundings from different sources that utilize different vertical (sounding/hydrographic) datums, the sounding datum quoted in the margin of the chart shall be the highest of the datum used.
Sounding density shall be greatest (6 mm - 15 mm average spacing) between the shoreline and the 10 meter depth curve.

Soundings that are 200 meters deep or deeper shall be corrected for sound velocity using NP-139 Tables. Sound velocity measurements shall be used in place of NP-139 Tables if they are considered more reliable than the averaged values shown in the NP-139 Tables. Soundings that are less than 200 meters deep shall be corrected for sound velocity using sound velocity measurements if data is available. If it is not possible to correct soundings for sound velocity, and assumed speeds of the sound are used, uncorrected soundings are identified in the source diagram.

Vertical Datum category is the sounding datum to which a hydrographic depth, drying height, or the high water datum to which a Safe Overhead Clearance is referenced. If the hydrographic depth, drying height, or Safe Overhead Clearance is unknown or not applicable, the vertical datum category is not applicable.

Hydrographic Drying Height is used when sounding type is (Drying Height), or (Drying Height (Approximate)). Hydrographic Depth is used for all other values (types) of soundings.

The rounding off of decimeters between 21 and 30.5 shall be as follows:
Decimeters between 3 and 7 shall be shown as 5, e.g., depths from 21.3 through 21.7 shall be shown as 21.5. Decimeters 1, 2, 8, and 9, shall be shown as 21, and 21.8 and 21.9 shall be shown as 22.

Areas with soundings shallower than depth range of maximum density that are not contiguous to the shoreline shall show a sounding density of <= 10 mm average spacing.

In well surveyed areas, where sounding density on the source is <= 5 mm average spacing, sounding density shall be > 20 mm average spacing and depth curves relied on to portray the bottom topography.

The position of the sounding on an NGA or NOS chart is the center of mass of the principle digit, excluding the subscript. Soundings on foreign source material do not necessarily show the center of mass of the principal digit as the position of the sounding. The standard practice of the charting authority that produced the source shall be followed to determine the position of the sounding on the foreign source.

CONTROLLING DEPTHS OF CHANNELS: A sounding shall be shown to indicate the controlling depth of a natural channel. The controlling depth of the channel is the least depth in the shallowest part of the natural channel, analogous to the highest point in a pass between two mountains.

DEEPEST PATH ALONG A NATURAL CHANNEL: A line of soundings is shown to indicate the deepest water through a natural channel, analogous to the lowest part of a valley floor.

SOUNDINGS ALONG TRACKS AND ROUTES: A line of soundings should be shown along tracks that ships must follow, such as a leading line, radar guided track, or route. If no soundings exist directly along the track, the closest one shall be shown.

DEEPS: Deep soundings should be shown. Deep are local lows; soundings that are deeper than surrounding soundings. Soundings that are approximately 20% or more deeper than the surrounding soundings are considered important. Soundings between 10% and 20% deeper than surrounding soundings may be important, depending on the characteristic of the bottom, for example, in flat areas.

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Sounding (BE020) ***(Continued)***

Point

Notes

SOUNDINGS AT CHANGES IN SLOPE: Soundings shall be shown to indicate significant changes of slope of the bottom. Soundings that are more than 5% shallower than the surface interpolated from surrounding shoals, deeps, depth curves, and other soundings, should be shown. Soundings that are more than 10% deeper than the surface interpolated from surrounding shoals, deeps, depth curves and other soundings should also be shown.

SOUNDINGS NEAR DEPTH CURVES: Soundings shown close to depth curves, i.e., closer than 25 mm, serve to support the depth curve, especially when there is no depth curve label nearby, or for short depth curves that do not have a label. Depth curves showing long, narrow extensions of deeper water into shallower water from the depth curve's normal smoothed curve should be supported by soundings along or near the end of the extension, if there is space, without soundings overprinting depth curves.

SOUNDINGS IN SLIPS AND AROUND PIERS: Where space is available, soundings should be shown around piers and wharves. The soundings must be far enough off the face of the pier or wharf to provide the depth of water at the keel lines of the vessels that are expected to use the structures. If only one sounding can be shown, the shallowest one should be selected.

SOUNDINGS IN DANGEROUS AREAS: As scale is reduced from the source to the product chart, soundings may be omitted between groups of rocks or reefs, when there is no well defined passage between them, or if detail has been generalized in the area. Where there is a well defined passage through the dangerous area, soundings should be shown.

SOUNDINGS CLOSE TO AND THE SAME AS A DEPTH CURVE: Soundings that are the same value of the depth curve on the shallower side of the curve, and soundings that are only one unit (fathom or meter) deeper than the curve and shown on the deeper side of the depth curve, should not be shown if they are closer than 3 mm to the depth curve, because they do not contribute any useful information to the mariner.

NO BOTTOM SOUNDINGS: No bottom soundings should be shown only if no other sounding data is available within 30 mm at chart scale.

Fill soundings shall be shown in flat or deep areas between shoals. Fill soundings are shown in a somewhat regular pattern of less dense (15 to 30 mm spacing) soundings that do not have significant changes in slope.

Soundings that are the least depths in proximity to known or potential navigational routes are placed very close together to increase the amount of detail presented to the chart user. They should not generally be placed closer than about 6 mm at chart scale.

Soundings in shoal areas, natural channels and hazardous areas should be sufficiently close together so these areas are highlighted by a dense pattern of soundings. Sounding spacing should be 10 to 15 mm. Soundings around a shoal should be less than 10 mm spacing.

In areas where depth curves are less than 10 mm apart, the number of soundings should be reduced, because the function of showing the shape of the bottom has been taken over by the depth curves. Significant deviations (5% higher or 10% lower) from the slope indicated by the depth curves must still be shown by soundings.

A least depth sounding must be shown for each shoal on the chart. When selecting soundings from larger scale source for inclusion on a smaller scale product, it may become necessary to generalize a series of shoals into one shoal. When this is required, the shallowest sounding from the group is selected to represent the least depth over the generalized shoal.

If two adjacent shoal soundings have the same depth, the one shown first is the one closest to the nearest or most prominent navigational route.

For any group of soundings with equal depth values, the most seaward one is shown. The most seaward sounding is the one closest to the deeper depth curve, or closest to the next deeper sounding.

If two shoal soundings of equal depth are found in an isolated shoal area (shallow area surrounded by a depth curve that closes on itself), the farthest seaward of equal shoal soundings must be shown.

If a shoal sounding is at the same depth as a depth curve, the depth curve is shown around the sounding. If two or more soundings have the same depth as the depth curve, the curve is shown around all of them. Additional deeper soundings at 10 or 15 mm spacing are shown outside the depth curve to indicate the slope of the sea bottom around the shoal.

Deep soundings that are shown, but not surrounded by a depth curve, should be surrounded by a group of irregularly spaced soundings. Soundings shown around deeps should be spaced at a greater distance than for a comparably sized shoal, i.e., 15 to 20 mm.

The structure of natural channels should be shown by a pattern of soundings with enough density to delineate both the width and the depth of the navigable portion. Selected soundings must be the least depth in the immediate area they are to represent.

When soundings from a recent survey reveal that a satisfactory junction between the new data and existing data cannot be made, a blank band approximately 5 mm wide at chart scale should be left around the limits of the more recent survey. A note should be shown stating that hydrography is from an older survey. Example: "Hydrography to eastward is from surveys in 1934"

All hydrographic detail may be removed from certain areas undergoing continual and rapid change, such as ocean inlets and openings between barrier islands if showing soundings is considered to present an unsafe representation between chart editions. The area shall be tinted with blue tint. A note should be shown stating that hydrography is under continual change: for example: "Area of continuous change"

SHOALS: All shoal soundings must be shown. A shoal sounding is a local high: a sounding that is shallower than other soundings around it. Shoal soundings may be placed very close together, but generally not less than 6 mm spacing. The density of soundings shown around shoals should be increased to less than 15 mm spacing, so the increased density of soundings draw attention to the presence of the shoal.

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Sounding (BE020) *(Continued)*

Point

Notes

Soundings are thinned according to the following hierarchy. Those soundings at the top of the list are deleted last, and those soundings at the bottom of the list are deleted first. See the referenced representation rules for more information about each category of sounding.
(1.) Controlling depths;
(2.) The deepest path along a navigable channel;
(3.) Soundings along tracks and routes;
(4.) Deeps;
(5.) Soundings at changes of slopes;
(6.) Soundings supporting depth curves;
(7.) Soundings in slips and around piers;
(8.) Soundings other than (1.)-(7.) above;
(9.) Soundings inside dangerous areas;
(10.) Soundings close to and the same value as a depth curve;
(11.) No bottom soundings

Feature Information					
Specification:	*Chart #1*	**Feature Origin:**	*Hydrographic*	**Symbol:**	I15
Symbol Criteria:	(Hydrographic Drying Height > 0) and (Sounding Metadata : Bathymetric Measurement Quality Category = [Depth Unknown]			**Label:**	[Hydrographic Drying Height], 7 point lower case, Italic/ 5 point subscript;
				Color:	Black (58600)

Feature Information					
Specification:	*Chart #1*	**Feature Origin:**	*Hydrographic*	**Symbol:**	I14
Symbol Criteria:	(Hydrographic Depth > 0) and (Sounding Metadata : Bathymetric Measurement Quality Category = [Depth Unreliable]			**Label:**	[Hydrographic Depth], 7 point lower case, 5 point subscript;
				Color:	Black (58600)

Feature Information					
Specification:	*Chart #1*	**Feature Origin:**	*Hydrographic*	**Symbol:**	I10
Symbol Criteria:	(Hydrographic Depth > 0) and (Sounding Metadata : Bathymetric Measurement Quality Category = [Depth Known] or [Least Depth] or [Maintained Depth])			**Label:**	[Hydrographic Depth], 7 point lower case, Italic/ 5 point subscript;
				Color:	Black (58600)

Feature Information					
Specification:	*Chart #1*	**Feature Origin:**	*Hydrographic*	**Symbol:**	I13
Symbol Criteria:	(Hydrographic Depth > 0) and (Sounding Metadata : Bathymetric Measurement Quality Category = [No Bottom] or [Safe Clearance Known])			**Label:**	[Hydrographic Depth], 7 point lower case, Italic
				Color:	Black (58600)

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Space Facility (AL351)

Surface

Feature Description:	*A facility that is used to support space related activities (for example: services such as assembly, launching or recovery of spacecraft or managing flight operations).*
Inclusion Condition:	*Area >= 15,625 sq m.*

Notes

None

Feature Information

Specification:	*LTDS*	**Feature Origin:**	*Topographic*	**Symbol:**	AR_Black_SpaceFacility
Symbol Criteria:	Area >= 15,625			**Label:**	[Geographic Name Information: Full Name], 6 point Condensed Mixed case if available or "Space facility", 6 point Condensed Mixed case
				Color:	Black (58600)

Space Facility (AL351)

Point

Inclusion Condition:	*Area < 15,625 sq m. AND Navigation Landmark = [True]*

Notes

None

Feature Information

Specification:	*LTDS*	**Feature Origin:**	*Topographic*	**Symbol:**	PT_Black_SpaceFacility
Symbol Criteria:	Area < 15,625 AND Navigation Landmark = [True]			**Label:**	[Geographic Name Information: Full Name], 6 point Condensed Mixed case if available or "Space facility", 6 point Condensed Mixed case
				Color:	Black (58600)

Spillway (BH165)

Surface

Feature Description:	*A passage for surplus water to run over or around a dam.*
Inclusion Condition:	*Width >= 25*

Notes

None

Feature Information

Specification:	*LTDS*	**Feature Origin:**	*Topographic*	**Symbol:**	AR_Black_Spillway
Symbol Criteria:	Width >= 25			**Label:**	"Spillway", 6 point Condensed Mixed case
				Color:	Black (58600)

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Spillway (BH165)

Curve

Inclusion Condition:	*Width < 25*

Notes

None

Feature Information					
Specification:	*LTDS*	**Feature Origin:**	*Topographic*	**Symbol:**	LN_Black_Spillway
Symbol Criteria:	Width < 25			**Label:**	"Spillway", 6 point Condensed Mixed case
				Color:	Black (58600)

Sports Ground (AK040)

Surface

Feature Description:	*An open area where sporting events, exercises, and/or games occur.*
Inclusion Condition:	*Area >= 8,125 square meters*

Notes

When 2 or more similar area features are separated by less than 0.5 mm at chart scale, the feature will be agglomerated.

Area and line features will be generalized to detail compatible with scale.

Type for features on land shall be positioned on land if possible. If type must be placed in the water, it shall be positioned so it does not obscure or overprint hydrographic detail. Type for features in the water shall be positioned in the water. If possible, type shall not be positioned across the shoreline.

Descriptive terms, e.g., "Canal" shall be shown if the name is not known. If the descriptive word appears in the name, for example, "PANAMA CANAL". The descriptive type shall not be shown, i.e. do not show "Panama Canal Canal".

Feature Information					
Specification:	*LTDS*	**Feature Origin:**	*Topographic*	**Symbol:**	AR_Black_SportsGround
Symbol Criteria:	Area >= 8125			**Label:**	[Geographic Name Information : Full Name], 6 point Condensed Mixed case; "Sports ground ", 6 point Condensed Mixed case
				Color:	Black (58600)

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Sports Ground (AK040)

Point

Inclusion Condition:	*Area < 8,125 square meters AND Navigation Landmark = [True]*

Notes

None

Feature Information					
Specification:	*LTDS*	**Feature Origin:**	*Topographic*	**Symbol:**	PT_Black_SportsGround
Symbol Criteria:	Area < 8,125 AND Navigation Landmark = [True]			**Label:**	[Geographic Name Information : Full Name], 6 point Condensed Mixed case; "Sports ground ", 6 point Condensed Mixed case
				Color:	Black (58600)

Spot Elevation (CA030)

Point

Feature Description:	*A designated location with an elevation value relative to a vertical datum.*
Inclusion Condition:	*ALL REQUIRED*

Notes

Spot elevation values are placed to avoid obscuring features of importance to the map user, such as small tops, ridges, and saddles. The order of precedence for placement is as follows:
Preferred: The bottom line of the value is aligned to the right side of the dot, with the horizontal center of the symbol referenced (dot).
Second: The top line of the value is aligned to the bottom left with the horizontal center of the symbol referenced (dot).
Third: Value is centered directly over the top of the symbol referenced (dot).
Fourth: Value is centered directly under the symbol referenced (dot).

When a referenced spot elevation is located at a line feature intersection, the placement of the value is as follows:
Preferred: Bottom right quadrant;
Second: Top right quadrant;
Third: Top left quadrant;
Fourth: Bottom left quadrant.

When an island is too small to accommodate the symbol referenced value (spot elevation), the value is placed adjacent to the island and aligned.

Type for a spot elevation placed in the water shall be enclosed in parentheses, and print blue (SPC 48253).

If an island (area enclosed by shoreline) cannot accommodate a spot elevation value without overprinting its shoreline, the elevation figure shall be positioned adjacent to the spot and entirely in the open water area. If the island is identified with a proper name, the elevation value shall be centered below the name.

Each 30 minute X 15 minute area on the map, as defined by the latitude and longitude grid, should contain approximately 1-3 trig stations and/or bench marks when known, and supplemented with 3-5 additional normal spot elevations. In the absence of any trig stations or bench marks, show 6-8 normal spot elevations.

When an elevation is identified with intersections of Roads, Railroads, Streams, or any crossing combination of above, also to include Island shorelines without contours, the value shall be placed adjacent to the feature. No dot is shown.

Whenever possible, Spot Elevations are shown for selected readily identifiable ground features, listed:
-Railroad junctions,
-Railroad gate crossings,
-high points on grades of railroads and roads,
-extensive flat areas,
-rims and bottoms of Depressions with diameter > 125 meters,
-Stream junctions.
-Spot elevations are also needed to support of the relief presentation:
-on the sides of slopes,
-the highest elevation on the map sheet,
-the top of prominent natural features such as hilltops, isolated summits, mountain tops, mountain passes, saddles, and other high points.

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Spot Elevation (CA030) *(Continued)*

Point

Notes
Spot elevation values, when known, shall be shown for hilltops, knolls, isolated summits, mountain tops, mountain passes, saddles, road junctions, railroad crossings, high points on grades of highways and railroads, areas >= 150 mm x 150 mm without contour feature and % of slope is < 5, rims and bottoms of Depressions >= 25 mm x 25 mm, water surfaces of lakes and Ponds, and Streams junctions. Type size is 8 point, color #58600 Black-Solid

Feature Information					
Specification:	*LTDS*	**Feature Origin:**	*Topographic*	**Symbol:**	PT_Black_SpotElevation
Symbol Criteria:	Elevation Surface Category = ([Land])			**Label:**	"[Highest Elevation]" if Elevation Surface Category = ([Land])/(Highest on sheet)), 10 point Regular Bold lower case; "[Highest Elevation]" if Elevation Surface Category = ([Land]), 8 point Regular lower case
				Color:	Black (58600)

Feature Information					
Specification:	*LTDS*	**Feature Origin:**	*Topographic*	**Symbol:**	PT_Black_SpotElevation
Symbol Criteria:	Elevation Surface Category = ([Snow Field and/or Ice-field])			**Label:**	"[Highest Elevation]" if Elevation Surface Category = ([Snow Field and/or Ice-field])/(Highest on sheet)), 10 point Regular Bold lower case; "[Highest Elevation]" if Elevation Surface Category = ([Snow Field and/or Ice-field]), 8 point Regular lower case
				Color:	Blue (48253)

Feature Information					
Specification:	*LTDS*	**Feature Origin:**	*Topographic*	**Symbol:**	PT_SpotElevation_Water
Symbol Criteria:	Elevation Surface Category = ([Water] or [Inland Water])			**Label:**	"[Highest Elevation]", 8 point Regular lower case
				Color:	Blue (48253)

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Stadium (AK160)

Surface

Feature Description:	*A field and/or stage partly or completely surrounded by a structure designed to allow spectators to stand or sit while viewing an event.*
Inclusion Condition:	*Area >= 625 square meters*

Notes

Area and line features will be generalized to detail compatible with scale.

Type for features on land shall be positioned on land if possible. If type must be placed in the water, it shall be positioned so it does not obscure or overprint hydrographic detail. Type for features in the water shall be positioned in the water. If possible, type shall not be positioned across the shoreline.

Descriptive terms, e.g., "Canal" shall be shown if the name is not known. If the descriptive word appears in the name, for example, "PANAMA CANAL". The descriptive type shall not be shown, i.e. do not show "Panama Canal Canal".

Omit feature < 46 meters in Built-up Area, unless (Landmark)

Limiting lines of feature are omitted if it coalesces with a road (AP030).

If a clearing exists inside of the area feature, and the size of the clearing is equal to or greater than the area for the surrounding area feature, the clearing is shown as an open space inside the surround feature. If the area of the clearing is less than the area for the surrounding feature, the clearing is deleted and absorbed into the surrounding area feature.

If two area features with the same feature code do not connect at any point, and have a space between them less than 2.5 mm at map/chart scale, delete the open space that is less than 2.5 mm between the features and combine them into one area feature.

If a portion of an area feature has a minimum width of less than 2.5 mm at map/chart scale, delete that portion of the area feature that is not at least 2.5 mm wide, measured from perimeter to perimeter. If the deletion of the portion of the area based on the above criteria will reduce the area of the remaining portion of the area feature to below the minimum area inclusion, do not delete the narrow portion of the feature that is less than 2.5 mm wide. If the deletion of the portion of the area based on the above criteria will split two larger areas connected by a narrow strip into two separate areas, either of which would be below minimum area inclusion, do not delete the narrow portion of the feature that is less than 2.5 mm wide.

Feature Information

Specification:	*LTDS*	**Feature Origin:**	*Topographic*	**Symbol:**	AR_Black_Stadium
Symbol Criteria:	Area >= 625			**Label:**	[Geographic Name Information : Full Name], 6 point Condensed Mixed case; "Stadium", 6 point Condensed Mixed case
				Color:	Black (58600)

Stadium (AK160)

Point

Inclusion Condition:	*Area < 625 square meters and Navigation Landmark = [True]*

Notes

Type for features on land shall be positioned on land if possible. If type must be placed in the water, it shall be positioned so it does not obscure or overprint hydrographic detail. Type for features in the water shall be positioned in the water. If possible, type shall not be positioned across the shoreline.

When obstructions coalesce at map scale, label with highest obstruction information.

Feature Information

Specification:	*LTDS*	**Feature Origin:**	*Topographic*	**Symbol:**	PT_Black_Stadium
Symbol Criteria:	(Area < 625) and (Height Above Surface Level < 46) and (Navigation Landmark = [True])			**Label:**	[Geographic Name Information : Full Name], 6 point Condensed Mixed case; "Stadium", 6 point Condensed Mixed case
				Color:	Black (58600)

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Stadium (AK160) *(Continued)*

Point

Feature Information					
Specification:	*LTDS*	**Feature Origin:**	*Topographic*	**Symbol:**	PT_Blue072_VO
Symbol Criteria:	Height above Surface Level >= 46 meters			**Label:**	"Stadium" , 7 point Condensed Mixed case; [Highest Elevation], 6 point Condensed Bold Mixed case; "([Height Above Surface Level])", 6 point Condensed Mixed case
				Color:	Blue (46351)

Steep Terrain Face (DB010)

Curve

Feature Description:	*A steep, vertical, or overhanging face of rock and/or soil.*
Inclusion Condition:	*Length or Diameter >= 125 m. AND Predominant Feature Height >= 2 m. AND Surface Slope >= 100%*

Notes

Area and line features will be generalized to detail compatible with scale.

Feature will be generalized to provide a more aesthetic contoured feature (i.e., smoothed).

If a Steep Terrain Face (Bluff/Cliff, Escarpment) height is less than the contour interval, the Steep Terrain face symbol shall be omitted, unless it is an obstacle to cross country movement.

If a Steep Terrain Face (Bluff/Cliff, Escarpment) is greater in height than one contour interval and the contours coalesce, the contours shall be dropped the entire length of the Steep Terrain Face (Bluff/Cliff, Escarpment).

Feature Information					
Specification:	*LTDS*	**Feature Origin:**	*Topographic*	**Symbol:**	LN_Dk-Brown1815_Steep TerrainFace_EqualOrGreater ThanContour
Symbol Criteria:	Predominant Feature Height >= contour interval			**Label:**	[Geographic Name Information : Full Name], 6 point Condensed Mixed case Italic if available
				Color:	Black (58600)

Feature Information					
Specification:	*LTDS*	**Feature Origin:**	*Topographic*	**Symbol:**	LN_Dk-Brown1815_Steep TerrainFace_LessThanContour
Symbol Criteria:	Predominant Feature Height < contour interval			**Label:**	[Geographic Name Information : Full Name], 6 point Condensed Mixed case Italic if available
				Color:	Black (58600)

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Stopway (GB045)

Surface

Feature Description:	*A defined rectangular area on the ground at the end of the take-off run available that has been prepared as a suitable area in which an aircraft can be stopped in the case of an abandoned take-off.*
Inclusion Condition:	*ALL REQUIRED*

Notes

Area and line features will be generalized to detail compatible with scale.

Feature Information					
Specification:	*LTDS*	**Feature Origin:**	*Topographic*	**Symbol:**	AR_Blue072_Stopway
Symbol Criteria:	All			**Label:**	None
				Color:	N/A

Storage Depot (AM010)

Surface

Feature Description:	*A tract used for the storage of products and/or supplies.*
Inclusion Condition:	*Area >= 15,625 sq m.*

Notes

When 2 or more similar area features are separated by less than 0.5 mm at chart scale, the feature will be agglomerated.

Area and line features will be generalized to detail compatible with scale.

Type for features on land shall be positioned on land if possible. If type must be placed in the water, it shall be positioned so it does not obscure or overprint hydrographic detail. Type for features in the water shall be positioned in the water. If possible, type shall not be positioned across the shoreline.

Limiting lines of feature are omitted if it coalesces with a road (AP030).

Feature Information					
Specification:	*LTDS*	**Feature Origin:**	*Topographic*	**Symbol:**	AR_Black_StorageDepot
Symbol Criteria:	Area >= 15,625			**Label:**	"Storage depot", 6 point Condensed Mixed case OR "Underground storage depot", 6 point Condensed Mixed case if Located Underground = [True]
				Color:	Black (58600)

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Storage Depot (AM010)

Point

Inclusion Condition:	*Area < 15,625 sq m. AND Navigation Landmark = [True]*

Notes

None

Feature Information					
Specification:	*LTDS*	**Feature Origin:**	*Topographic*	**Symbol:**	PT_Black_StorageDepot
Symbol Criteria:	Area < 15,625 AND Navigation Landmark = [True]			**Label:**	"Storage depot", 6 point Condensed Mixed case OR "Underground storage depot", 6 point Condensed Mixed case if Located Underground = [True]
				Color:	Black (58600)

Storage Tank (AM070)

Surface

Feature Description:	*A container used for the storage of liquids and/or gases that is not supported by a tower.*
Inclusion Condition:	*Area >= 625 sq m.*

Notes

Area and line features will be generalized to detail compatible with scale.

Type for features on land shall be positioned on land if possible. If type must be placed in the water, it shall be positioned so it does not obscure or overprint hydrographic detail. Type for features in the water shall be positioned in the water. If possible, type shall not be positioned across the shoreline.

If Tank symbols coalesce and there are less than 4, show one symbol and label "Tanks". If there are 4 or more, and area is >= 2.5 mm, show areal symbol as dashed outline and label "Numerous tanks". The predominant shall be applied to the labeling.

Feature Information					
Specification:	*LTDS*	**Feature Origin:**	*Topographic*	**Symbol:**	AR_Black_StorageTank
Symbol Criteria:	(Height Above Surface Level < 46) or (Area >= 5625 and Height Above Surface Level >= 46)			**Label:**	"([Product [1]])", 6 point Condensed Mixed case if Product [1]: <> [No information] or [Other]
				Color:	Black (58600)

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Storage Tank (AM070)

Point

Inclusion Condition:	*Area < 625 sq m. OR Height Above Surface Level >= 46 m. OR Navigation Landmark = [True]*

Notes

Label feature as per hierarchy for topo type placement to south neatline corners reading left to right:
1. Position hierarchy:
(a.) northeast (preferred position
(b.) southeast (1st alternative)
(c.) northwest (2nd alternative
(d.) southwest (3rd alternative)
(e.) top-centered (4th alternative)
(f.) bottom-centered (5th alternative)
(Hierarchy is based on type positioning so as to avoid overprinting other type or obscuring detail.)
2. Minimum space between type placement and feature symbol is 0.5 mm
3. This method of type placement shall be used for areal features when space does not permit labeling within that feature.

Type for features on land shall be positioned on land if possible. If type must be placed in the water, it shall be positioned so it does not obscure or overprint hydrographic detail. Type for features in the water shall be positioned in the water. If possible, type shall not be positioned across the shoreline.

When obstructions coalesce at map scale, label with highest obstruction information.

If Tank symbols coalesce and there are less than 4, show one symbol and label "Tanks". If there are 4 or more, and area is >= 2.5 mm, show areal symbol as dashed outline and label "Numerous tanks". The predominant shall be applied to the labeling.

Feature Information

Specification:	*LTDS*	Feature Origin:	*Topographic*	Symbol:	PT_Black_StorageTank
Symbol Criteria:	(Height Above Surface Level < 46) and ((Product [1] = [Water]) and (Structure Shape = [Horizontal Capped Cylindrical] or Vertical Capped Cylindrical] or [Pyramidal] or [Spherical] or [Cylindrical with Framework] or [Cylindrical with Flat Top] or [Cylindrical with Domed Top] or [Cylindrical with Conical Top] or [Domed] or [Vertical Cylindrical] or [Cubic] or [Board-like Prism] or [Horizontal Cylindrical] or [Rectangular Prism] or [Square Prism] or [Other] or [No Information])) or ((Product [1] = [Chemical] or [Gas] or [Petrol] or [Liquefied Natural Gas (LNG)] or [Liquefied Petroleum Gas (LPG)] or [Petroleum] or [Radioactive Material] or [Sewage] or [Biodiesel] or [Other] or [No Information]))			Label:	"(Product [1])", 6 point Condensed Mixed case if Product [1]: <> (No information) or (Other)
				Color:	Black (58600)

Feature Information

Specification:	*LTDS*	Feature Origin:	*Topographic*	Symbol:	PT_Cyan_StorageTank_Water
Symbol Criteria:	(Height Above Surface Level < 46) and (Product = [Water]) and (Structure Shape = [Spherical on Column] or [Cylindrical on Tower])			Label:	None
				Color:	N/A

Feature Information

Specification:	*LTDS*	Feature Origin:	*Topographic*	Symbol:	PT_Blue072_VO
Symbol Criteria:	Height Above Surface Level >= 46			Label:	[Product [1]], 6 point Condensed Mixed case if Product [1]: <> [No information] or [Other] AND "tank", 6 point Condensed Mixed case AND [Highest Elevation], 6 point Condensed Bold Mixed case; "([Height Above Surface Level])), 6 point Condensed Mixed case; OR "Tank", 6 point Condensed Mixed case if Product [1] = [No information] or [Other] AND [Highest Elevation], 6 point Condensed Bold Mixed case; "([Height Above Surface Level])", 6 point Condensed Mixed case
				Color:	Blue (46351)

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Structural Pile (BD100)

Surface

Feature Description:	*A long pile (for example: a heavy timber or section of steel, wood, or concrete) forced into the earth that may serve as a support (for example: for a pier) or as a free standing pole within a marine environment.*
Inclusion Condition:	*Area >= 15,625 sq m.*

Notes

Use the following abbreviations for Horizontal Accuracy and Existence values:
If Position Approximate, label "PA";
If Position Doubtful, label "PD";
If Existence Doubtful, label "ED";
If Reported, label "Rep".

Date shall only be shown if Existence = Reported. Date shall be shown in parentheses if available. Always show "Rep" and date on the same line.

Unless specifically indicated on source material as inaccurate or doubtful, etc., or modified by other override rules, features shall be assumed to be accurate and definite. This rule applied to accuracy and existence of unique features, and is not related to the accuracy of the source as a whole.

Following IHO practice, features whose positions are approximate are labeled "PA" on the source charts. Features whose positions are doubtful are labeled "PD" on the source charts. Features whose existence is doubtful are labeled "ED" on the source charts. Features that have only been reported are labeled "Rep" on the source charts, and usually show the date of report in parentheses.

Definitions:
PA - Position Approximate = The position has not been accurately determined, or does not remain fixed.;
PD - Position Doubtful = The feature has been reported in various positions, but not definitely determined to be in any.;
ED - Existence Doubtful = Indicates the possible existence of the feature, the actual existence of which has not been established.

Date shall only be shown if Existence = Reported. Date shall be shown in parentheses if available. Always show "Rep" and date on the same line.

Priority for type placement: 1-right center, 2-bottom center, 3-left center, 4-top center.
(A) Minimum distance from symbol - 1 mm.
(B) Maximum distance from symbol before choosing the next highest priority:
(#1) 4 mm measured to the West end;
(#2) 4 mm measured to the North side (top);
(#3) 4 mm measured to the East end;
(#4) 4 mm measured to the South side (bottom)

Labels may be stacked, readable top to bottom to avoid overprints.

Labels may be stacked, readable top to bottom to avoid overprints.

Type shall be placed in the following preference:
(a.) Placed on one horizontal line centered in feature.;
(b.) Broken into multiple horizontal lines, centered in feature, with a 1 mm space between lines.;
(c.) Placed on one line outside feature. ;
(d.) Placed on two lines to avoid overprints. If present, Hydrographic Depth shall always be placed inside area.

Labels shall be placed in water area, and shall not overprint the shoreline or be placed on land or in the foreshore.

Unless specifically indicated on source material as inaccurate or doubtful, etc., or modified by other override rules, features shall be assumed to be accurate and definite. This rule applied to accuracy and existence of unique features, and is not related to the accuracy of the source as a whole.

When the boundary of an areal symbol crosses or overprints the shoreline, the boundary overprinting the shoreline or land is deleted.

Features shown as points on cartographic source material shall be shown as points, except as follows: If more than five point symbols fall inside an area of 1 square centimeter at product scale, show as an area symbol, labeled, and with a smooth danger curve (dotted line) around them. If any of the point features extend above High Water, show area symbol as above surface. If no point features are above the surface at High water, show area symbol as submerged.

A blue 31% tint shall not overprint other blue 31% tints. If two blue tinted symbols, or one tinted symbol and water tint overprint, only one 31% tint shall be shown in the area.

Feature Information

Specification:	*LTDS*	**Feature Origin:**	*Topographic*	**Symbol:**	AR_Cyan-31_StructuralPiles
Symbol Criteria:	All			**Label:**	"Piles", 6 point Condensed Mixed case
				Color:	Black (58600)

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Structural Pile (BD100)

Point

Inclusion Condition:	*Area < 15,625 sq m. and Navigation Landmark = [True]*

Notes

Use the following abbreviations for Horizontal Accuracy and Existence values:
If Position Approximate, label "PA";
If Position Doubtful, label "PD";
If Existence Doubtful, label "ED";
If Reported, label "Rep".

Date shall only be shown if Existence = Reported. Date shall be shown in parentheses if available. Always show "Rep" and date on the same line.

Unless specifically indicated on source material as inaccurate or doubtful, etc., or modified by other override rules, features shall be assumed to be accurate and definite. This rule applied to accuracy and existence of unique features, and is not related to the accuracy of the source as a whole.

Following IHO practice, features whose positions are approximate are labeled "PA" on the source charts. Features whose positions are doubtful are labeled "PD" on the source charts. Features whose existence is doubtful are labeled "ED" on the source charts. Features that have only been reported are labeled "Rep" on the source charts, and usually show the date of report in parentheses.

Definitions:
PA - Position Approximate = The position has not been accurately determined, or does not remain fixed.;
PD - Position Doubtful = The feature has been reported in various positions, but not definitely determined to be in any.;
ED - Existence Doubtful = Indicates the possible existence of the feature, the actual existence of which has not been established.

Date shall only be shown if Existence = Reported. Date shall be shown in parentheses if available. Always show "Rep" and date on the same line.

Priority for type placement Rules:
1-right center,
2-bottom center,
3-left center,
4-top center.
(A) Minimum distance from symbol - 1 mm.
(B) Maximum distance from symbol before choosing the next highest priority:
(#1) 4 mm measured to the West end;
(#2) 4 mm measured to the North side (top);
(#3) 4 mm measured to the East end;
(#4) 4 mm measured to the South side (bottom)

Labels may be stacked, readable top to bottom to avoid overprints.

Labels shall be placed in water area, and shall not overprint the shoreline or be placed on land or in the foreshore.

When two or more identical features fall within 10 mm of each other, one legend with plural description shall be shown, with type placed as close as possible to the center of the group without overprinting features.

Unless specifically indicated on source material as inaccurate or doubtful, etc., or modified by other override rules, features shall be assumed to be accurate and definite. This rule applied to accuracy and existence of unique features, and is not related to the accuracy of the source as a whole.

Features shown as points on cartographic source material shall be shown as points, except as follows: If more than five point symbols fall inside an area of 1 square centimeter at product scale, show as an area symbol, labeled, and with a smooth danger curve (dotted line) around them. If any of the point features extend above High Water, show area symbol as above surface. If no point features are above the surface at High water, show area symbol as submerged.

Structural Pile (BD100) *(Continued)*

Point

Feature Information					
Specification:	*Chart #1*	**Feature Origin:**	*Hydrographic*	**Symbol:**	F22
Symbol Criteria:	All			**Label:**	None
				Color:	N/A

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Submerged Volcano (DB181)

Surface

Feature Description:	*An underwater hill or mountain situated over an opening or openings in the Earth's crust through which lava, steam, and/or gases, are or have been expelled.*
Inclusion Condition:	*Area >= 15,625 sq m.*

Notes

Use the following abbreviations for Horizontal Accuracy and Existence values:
If Position Approximate, label "PA";
If Position Doubtful, label "PD";
If Existence Doubtful, label "ED";
If Reported, label "Rep".

Date shall only be shown if Existence = Reported. Date shall be shown in parentheses if available. Always show "Rep" and date on the same line.
Unless specifically indicated on source material as inaccurate or doubtful, etc., or modified by other override rules, features shall be assumed to be accurate and definite. This rule applied to accuracy and existence of unique features, and is not related to the accuracy of the source as a whole.
Following IHO practice, features whose positions are approximate are labeled "PA" on the source charts. Features whose positions are doubtful are labeled "PD" on the source charts. Features whose existence is doubtful are labeled "ED" on the source charts. Features that have only been reported are labeled "Rep" on the source charts, and usually show the date of report in parentheses.

Definitions:
PA - Position Approximate = The position has not been accurately determined, or does not remain fixed.;
PD - Position Doubtful = The feature has been reported in various positions, but not definitely determined to be in any.;
ED - Existence Doubtful = Indicates the possible existence of the feature, the actual existence of which has not been established.

Date shall only be shown if Existence = Reported. Date shall be shown in parentheses if available. Always show "Rep" and date on the same line.

Priority for type placement Rules:
1-right center,
2-bottom center,
3-left center,
4-top center.
(A) Minimum distance from symbol - 1 mm.
(B) Maximum distance from symbol before choosing the next highest priority:
(#1) 4 mm measured to the West end;
(#2) 4 mm measured to the North side (top);
(#3) 4 mm measured to the East end;
(#4) 4 mm measured to the South side (bottom)

If symbols overprint each other, labels are condensed as follows:
(1.) If the labels are identical, only one is retained.;
(2,) If the labels are not identical, they shall be condensed in to one legend, e.g., "Fishhaven and Well". If multiple depths are shown, only the shallowest is retained.

Labels may be stacked, readable top to bottom to avoid overprints.

Type Preference Placement Rules:
(a.) Placed on one horizontal line centered in feature.;
(b.) Broken into multiple horizontal lines, centered in feature, with a 1 mm space between lines.;
(c.) Placed on one line outside feature. Use Priority for type placement Rules for placement, Hydrographic Depth shall always be positioned inside area.;
(d.) Placed on two lines to avoid overprints. Use Priority for type placement Rules. If present, Hydrographic Depth shall always be placed inside area.

Labels shall be placed in water area, and shall not overprint the shoreline or be placed on land or in the foreshore.

Feature Information

Specification:	*Chart #1*	**Feature Origin:**	*Hydrographic*	**Symbol:**	K40
Symbol Criteria:	All			**Label:**	"Sub vol", 6 point Mixed case Italic
				Color:	Black (58600)

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Submerged Volcano (DB181)

Point

Inclusion Condition:	*Area < 15,625 sq m.*

Notes

Use the following abbreviations for Horizontal Accuracy and Existence values:
If Position Approximate, label "PA";
If Position Doubtful, label "PD";
If Existence Doubtful, label "ED";
If Reported, label "Rep".

Date shall only be shown if Existence = Reported. Date shall be shown in parentheses if available. Always show "Rep" and date on the same line.
Unless specifically indicated on source material as inaccurate or doubtful, etc., or modified by other override rules, features shall be assumed to be accurate and definite. This rule applied to accuracy and existence of unique features, and is not related to the accuracy of the source as a whole.
Following IHO practice, features whose positions are approximate are labeled "PA" on the source charts. Features whose positions are doubtful are labeled "PD" on the source charts. Features whose existence is doubtful are labeled "ED" on the source charts. Features that have only been reported are labeled "Rep" on the source charts, and usually show the date of report in parentheses.

Definitions:
PA - Position Approximate = The position has not been accurately determined, or does not remain fixed.;
PD - Position Doubtful = The feature has been reported in various positions, but not definitely determined to be in any.;
ED - Existence Doubtful = Indicates the possible existence of the feature, the actual existence of which has not been established.

Date shall only be shown if Existence = Reported. Date shall be shown in parentheses if available. Always show "Rep" and date on the same line.

Priority for type placement Rules:
1-right center,
2-bottom center,
3-left center,
4-top center.
(A) Minimum distance from symbol - 1 mm.
(B) Maximum distance from symbol before choosing the next highest priority:
(#1) 4 mm measured to the West end;
(#2) 4 mm measured to the North side (top);
(#3) 4 mm measured to the East end;
(#4) 4 mm measured to the South side (bottom)

If symbols overprint each other, labels are condensed as follows:
(1.) If the labels are identical, only one is retained.;
(2,) If the labels are not identical, they shall be condensed in to one legend, e.g., "Fishhaven and Well". If multiple depths are shown, only the shallowest is retained.

Labels may be stacked, readable top to bottom to avoid overprints.

Type Preference Placement Rules:
(a.) Placed on one horizontal line centered in feature.;
(b.) Broken into multiple horizontal lines, centered in feature, with a 1 mm space between lines.;
(c.) Placed on one line outside feature. Use Priority for type placement Rules for placement, Hydrographic Depth shall always be positioned inside area.;
(d.) Placed on two lines to avoid overprints. Use Priority for type placement Rules. If present, Hydrographic Depth shall always be placed inside area.

Labels shall be placed in water area, and shall not overprint the shoreline or be placed on land or in the foreshore.

Feature Information					
Specification:	*Chart #1*	**Feature Origin:**	*Hydrographic*	**Symbol:**	K40
Symbol Criteria:	All			**Label:**	"Sub vol", 6 point Mixed case Italic
				Color:	Black (58600)

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Surface Bunker (AM060)

Surface

Feature Description:	*A surface structure that may be covered and/or surrounded with earth and is resistant to ordnance.*
Inclusion Condition:	*Area >= 625 sq m.*

Notes

When 2 or more similar area features are separated by less than 0.5 mm at chart scale, the feature will be agglomerated.

Area and line features will be generalized to detail compatible with scale.

Label feature as per hierarchy for topo type placement to south neatline corners reading left to right:
1. Position hierarchy:
(a.) northeast (preferred position
(b.) southeast (1st alternative)
(c.) northwest (2nd alternative
(d.) southwest (3rd alternative)
(e.) top-centered (4th alternative)
(f.) bottom-centered (5th alternative)
(Hierarchy is based on type positioning so as to avoid overprinting other type or obscuring detail.)
2. Minimum space between type placement and feature symbol is 0.5 mm
3. This method of type placement shall be used for areal features when space does not permit labeling within that feature.

Names placement shall be oriented to the longest axis of the feature reading left to right and placed within the area outline and centered. If the longest axis is perpendicular to the south neatline, the type shall be placed outside of the area outline, preferred position is northeast of the feature, but may be placed at any position around the feature so as not to overprint any other feature type and reading left to right.

Type for features on land shall be positioned on land if possible. If type must be placed in the water, it shall be positioned so it does not obscure or overprint hydrographic detail. Type for features in the water shall be positioned in the water. If possible, type shall not be positioned across the shoreline.

Feature Information

Specification:	*LTDS*	**Feature Origin:**	*Topographic*	**Symbol:**	AR_Black_SurfaceBunker
Symbol Criteria:	All			**Label:**	"[Product [1]]", 6 point Condensed Mixed case if Product: <> [Other] or [No information] and "bunker", 6 point Condensed lower case; or "Bunker", 6 point Condensed Mixed case if Product = [Other] or [No information]
				Color:	Black (58600)

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Surface Bunker (AM060)

Point

Inclusion Condition:	*Area < 625 sq m.*

Notes

A cluster of 3 or more coalescing similar point features will be aggregated when an area delineation is supported by the product.

When Product Category (No Information), omit the Product label.

Type for features on land shall be positioned on land if possible. If type must be placed in the water, it shall be positioned so it does not obscure or overprint hydrographic detail. Type for features in the water shall be positioned in the water. If possible, type shall not be positioned across the shoreline.

Feature Information

Specification:	*LTDS*	Feature Origin:	*Topographic*	Symbol:	PT_Dk-Brown1815_SurfaceBunker
Symbol Criteria:	All			Label:	"[Product [1]]", 6 point Condensed Mixed case if Product: <> [Other] or [No information]
				Color:	Black (58600)

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Swamp (ED020)

Surface

Feature Description:	*A seasonally flooded, poorly drained wetland with more woody plants than a marsh and better drainage than a bog..*
Inclusion Condition:	*Area >= 15,625 sq m.*

Notes

Type sizes per are sizes at map/chart scale: Area features only.
06 point - <= 770 mm square area and <= 14 mm width;
07 point - <= 2,296 mm square area and <= 28 mm width;
09 point - <= 5,192 mm square area and 44 mm width;
10 point - <=9,796 mm square area and <= 62 mm width;
12 point - <= 16,632 mm square area and <= 84 mm width;
14 point - <= 24,960 mm square area and <= 104 mm width;
16 point - > 24,960 mm square area.
Where area measurements are inconsistent, the larger type size shall be used. Where the full range of type sizes is not available for a particular label, the closest available type size shall be used.

Label feature as per hierarchy for topo type placement to south neatline corners reading left to right:
1. Position hierarchy: (a.) northeast (preferred position), (b.) southeast (1st alternative), (c.) northwest (2nd alternative, (d.) southwest (3rd alternative), (e.) top-centered (4th alternative), (f.) bottom-centered (5th alternative): (Hierarchy is based on type positioning so as to avoid overprinting other type or obscuring detail.);
2. Minimum space between type placement and feature symbol is 0.5 mm;
3. This method of type placement shall be used for areal features when space does not permit labeling within that feature.

Names placement shall be oriented to the longest axis of the feature reading left to right and placed within the area outline and centered. If the longest axis is perpendicular to the south neatline, the type shall be placed outside of the area outline, preferred position is northeast of the feature, but may be placed at any position around the feature so as not to overprint any other feature type and reading left to right.

Coincident similar area features will be blended to form a single feature.

Area and line features will be generalized to detail compatible with scale.

Feature will be generalized to provide a more aesthetic contoured feature (i.e., smoothed).

Symbols and associated area patterns of underpassing features (except drainage shorelines) are broken for all bridges, except footbridges. This rule does not apply to land tint on Littoral Planning Charts.

If a clearing exists inside of the area feature, and the size of the clearing is equal to or greater than the area for the surrounding area feature, the clearing is shown as an open space inside the surround feature. If the area of the clearing is less than the area for the surrounding feature, the clearing is deleted and absorbed into the surrounding area feature.

If two area features with the same feature code do not connect at any point, and have a space between them less than 2.5 mm at map/chart scale, delete the open space that is less than 2.5 mm between the features and combine them into one area feature.

If a portion of an area feature has a minimum width of less than 2.5 mm at map/chart scale, delete that portion of the area feature that is not at least 2.5 mm wide, measured from perimeter to perimeter. If the deletion of the portion of the area based on the above criteria will reduce the area of the remaining portion of the area feature to below the minimum area inclusion, do not delete the narrow portion of the feature that is less than 2.5 mm wide. If the deletion of the portion of the area based on the above criteria will split two larger areas connected by a narrow strip into two separate areas, either of which would be below minimum area inclusion, do not delete the narrow portion of the feature that is less than 2.5 mm wide.

Feature Information

Specification:	*LTDS*	**Feature Origin:**	*Topographic*	**Symbol:**	AR_Cyan_Swamp
Symbol Criteria:	Vegetation Species [1] = [Cypress] or [Other] or [No Information			**Label:**	[Geographic Name Information : Full Name], 7-16 point Condensed Upper case Italic if available
				Color:	Blue (46351)

Feature Information

Specification:	*LTDS*	**Feature Origin:**	*Topographic*	**Symbol:**	AR_Green362_NipaPalm
Symbol Criteria:	Vegetation Species [1] = [Nipa]			**Label:**	[Geographic Name Information : Full Name], 7-16 point Condensed Upper case Italic if available
				Color:	Black (58600)

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Swamp (ED020) *(Continued)*

Surface

Feature Information					
Specification:	*LTDS*	**Feature Origin:**	*Topographic*	**Symbol:**	AR_Green362_Mangrove
Symbol Criteria:	Vegetation Species [1] = [Mangrove]			**Label:**	[Geographic Name Information : Full Name], 7-16 point Condensed Upper case Italic if available
				Color:	Black (58600)

Swept Area (FC177)

Surface

Feature Description:	*A region that has been determined to be clear of navigational dangers to a specified depth.*
Inclusion Condition:	*All required*

Notes

Show Hydrographic Depth or Hydrographic Height values in whole meters if decimeter value is 0; otherwise show the decimeter value as a subscript to the meter value.

Type sizes for Swept Areas: (Type Size:)
(If Width is:):
8 point <= 8 mm;
10 point > 8 mm <= 10 mm;
12 point > 10 mm <= 18 mm;
14 point > 18 mm.
Large areas 30 x 60 mm wide, with irregular shape, < 60% of a minimum bounding rectangle covered by area, shall be labeled in several places so it is clear to the user what the depth of the area is.
Areas that are too small to be labeled with 8 point type without overprinting area limit lines shall be aggregated into larger adjoining areas swept to a lesser depth than the small area. The larger adjoining area chosen shall be the one with the closest shallower depth value.
If the small area is shallower than the surrounding areas, the swept depth label shall be placed outside the area with a Leading Line used to indicate which area the depth value refers to.

Vertical Datum category is the sounding datum to which a hydrographic depth, drying height, or the high water datum to which a Safe Overhead Clearance is referenced. If the hydrographic depth, drying height, or Safe Overhead Clearance is unknown or not applicable, the vertical datum category is not applicable.

If two area symbols of the same type share a common boundary, the common boundary shall be shown with the lineweight reduced to 0.25 mm, dash lengths reduced to 2.5 mm, and dash spaces reduced to 0.6 mm. Color remains the same.

If a swept area falls on a chart, show this note in the margin or an open water area. Do not combine with other notes.
"NOTE- The area tinted green has been swept in (Date)-(Date) to a depth indicated thus: 40". Place date of the wire drag in the note. The "40" in the notes above is an example only, and a depth within the range of depths shown on the feature should be selected. The wire drag underline, as shown on the feature symbol, should also be placed under this number. Color is Green SPC-52813. Type for the note is 12 and 10 point. type for the swept depth number is 10 point, Italic.

Feature Information					
Specification:	*Chart #1*	**Feature Origin:**	*Hydrographic*	**Symbol:**	I24
Symbol Criteria:	All			**Label:**	"([Currency Date and Time])", 8-14 point lower case, Italic; [Hydrographic Depth], 8-14 point lower case, Italic, (Type size subscript: 6 to 12 point (Two points smaller than principle digit.))
				Color:	Green (52813)

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Swimming Pool (AK170)

Surface

Feature Description:	*A man-made pool used for swimming outdoors.*
Inclusion Condition:	*Length or Diameter >= 50m*

Notes

Area and line features will be generalized to detail compatible with scale.

Type for features on land shall be positioned on land if possible. If type must be placed in the water, it shall be positioned so it does not obscure or overprint hydrographic detail. Type for features in the water shall be positioned in the water. If possible, type shall not be positioned across the shoreline.

Symbolize feature (at map scale) 2.5 mm length, and 1.3 mm width when the feature is less than this size at its ground equivalent.

Do not show land tint inside symbol.

Feature Information

Specification:	*LTDS*	**Feature Origin:**	*Topographic*	**Symbol:**	AR_Cyan-31_SwimmingPool
Symbol Criteria:	Length or Diameter >= 50			**Label:**	"Swimming pool", 6 point Condensed Mixed case
				Color:	Black (58600)

Tank Farm (AM071)

Surface

Feature Description:	*A tract of land occupied by large-capacity tanks in which petroleum, natural gas, or liquid petrochemicals are stored.*
Inclusion Condition:	*Area >= 15,625 sq m.*

Notes

None

Feature Information

Specification:	*LTDS*	**Feature Origin:**	*Topographic*	**Symbol:**	AR_Black_TankFarm
Symbol Criteria:	All			**Label:**	[Geographic Name Information: Full Name], 6 point Condensed Mixed case if available; and "Tank farm", 6 point Condensed Mixed case
				Color:	Black (58600)

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Tank Farm (AM071)

Point

Inclusion Condition:	*Area < 15,625 sq m.*

Notes

None

Feature Information					
Specification:	*LTDS*	**Feature Origin:**	*Topographic*	**Symbol:**	PT_Black_TankFarm
Symbol Criteria:	All			**Label:**	[Geographic Name Information: Full Name], 6 point Condensed Mixed case if available; and "Tank farm", 6 point Condensed Mixed case
				Color:	Black (58600)

Taxiway (GB075)

Surface

Feature Description:	*A defined path at an aerodrome established for the taxiing of aircraft and intended to provide a ground movement link between one part of the aerodrome and another.*
Inclusion Condition:	*ALL REQUIRED*

Notes

Contours will be adjusted to planimetric features.

Area and line features will be generalized to detail compatible with scale.

Feature Information					
Specification:	*LTDS*	**Feature Origin:**	*Topographic*	**Symbol:**	AR_Blue072_Taxiway
Symbol Criteria:	All			**Label:**	None
				Color:	N/A

Test Site (FA100)

Surface

Feature Description:	*A site for the testing of technical products and equipment.*
Inclusion Condition:	*Area >= 15,625 sq m.*

Notes

None

Feature Information					
Specification:	*LTDS*	**Feature Origin:**	*Topographic*	**Symbol:**	AR_Black_TestSite
Symbol Criteria:	All			**Label:**	[Geographic Name Information : Full Name] if available and "Test site", 6 point Condensed Mixed case
				Color:	Black (58600)

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Tethered Balloon (AL510)

Point

Feature Description:	*A balloon that is tethered to the ground at an anchoring device (for example: a ring and pad) or substantial structure (for example: a mooring tower).*
Inclusion Condition:	*Height Above Surface Level >= 46*

Notes

None

Feature Information

Specification:	*LTDS*	**Feature Origin:**	*Topographic*	**Symbol:**	PT_Blue072_VO
Symbol Criteria:	*Height Above Surface Level >= 46*			**Label:**	"Tethered balloon" and 7 point Condensed Mixed case; [Highest Elevation], 6 point Condensed Bold Mixed case; "([Height Above Surface Level])", 6 point Condensed Mixed case
				Color:	Blue (46351)

Theodolite Reference Line (FC101)

Curve

Feature Description:	*A line of known direction from a fixed location that is used to determine accurate positions of a vessel on certain test ranges.*
Inclusion Condition:	*All Required*

Notes

None

Feature Information

Specification:	*Other*	**Feature Origin:**	*Other*	**Symbol:**	TBD
Symbol Criteria:	All			**Label:**	TBD
				Color:	TBD

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Thicket (EB020)

Surface

Feature Description:	*A tract covered mainly by low-growing, uncultured, woody plants that are thickly tangled together.*
Inclusion Condition:	*Area >= 15,625 sq m.*

Notes

Coincident similar area features will be blended to form a single feature.

Area and line features will be generalized to detail compatible with scale.

Feature will be generalized to provide a more aesthetic contoured feature (i.e., smoothed).

Symbols and associated area patterns of underpassing features (except drainage shorelines) are broken for all bridges, except footbridges. This rule does not apply to land tint on Littoral Planning Charts.

If a clearing exists inside of the area feature, and the size of the clearing is equal to or greater than the area for the surrounding area feature, the clearing is shown as an open space inside the surround feature. If the area of the clearing is less than the area for the surrounding feature, the clearing is deleted and absorbed into the surrounding area feature.

If two area features with the same feature code do not connect at any point, and have a space between them less than 2.5 mm at map/chart scale, delete the open space that is less than 2.5 mm between the features and combine them into one area feature.

If a portion of an area feature has a minimum width of less than 2.5 mm at map/chart scale, delete that portion of the area feature that is not at least 2.5 mm wide, measured from perimeter to perimeter. If the deletion of the portion of the area based on the above criteria will reduce the area of the remaining portion of the area feature to below the minimum area inclusion, do not delete the narrow portion of the feature that is less than 2.5 mm wide. If the deletion of the portion of the area based on the above criteria will split two larger areas connected by a narrow strip into two separate areas, either of which would be below minimum area inclusion, do not delete the narrow portion of the feature that is less than 2.5 mm wide.

Feature Information

Specification:	*LTDS*	**Feature Origin:**	*Topographic*	**Symbol:**	AR_Green362_Thicket
Symbol Criteria:	ALL			**Label:**	None
				Color:	N/A

Tidal Stream Observation Station (BG030)

Point

Feature Description:	*The location in a tidally influenced stream where observations are taken.*
Inclusion Condition:	*All Required.*

Notes

None

Feature Information

Specification:	*Chart #1*	**Feature Origin:**	*Hydrographic*	**Symbol:**	H46
Symbol Criteria:	All			**Label:**	TBD
				Color:	TBD

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Tidal Water (BA040)

Surface

Feature Description:	*Any water the level of which changes periodically due to tidal action.*
Inclusion Condition:	*All Required.*

Notes

Symbols and associated area patterns of underpassing features (except drainage shorelines) are broken for all bridges, except footbridges. This rule does not apply to land tint on Littoral Planning Charts.

Show water tint (Blue SPC-48253), 31% screen, at 45°) from the shoreline (BA010), to the 10 meter depth curve (BE010) and all offshore areas shallower than 10 meters (inside the 10 meter curve). Blue tint is deleted from inland hydrographic features (BH), in those areas that are deeper than 10 meters (outside the 10 meter depth curve).

A blue 31% tint shall not overprint other blue 31% tints. If two blue tinted symbols, or one tinted symbol and water tint overprint, only one 31% tint shall be shown in the area.

Feature Information

Specification:	*LTDS*	**Feature Origin:**	*Topographic*	**Symbol:**	AR_Cyan-31_TidalWater
Symbol Criteria:	All			**Label:**	[Geographic Name Information : Full Name] if available and Large area, 8-18 point Condensed Upper case Italic , 7-10 point Mixed case Italic if Small area
				Color:	Blue (48253)

Tide Gauge Station (BG020)

Point

Feature Description:	*A station containing an instrument for measuring the height of the tide.*
Inclusion Condition:	*All Required.*

Notes

None

Feature Information

Specification:	*Chart #1*	**Feature Origin:**	*Hydrographic*	**Symbol:**	T32.2
Symbol Criteria:	All			**Label:**	"Tide gauge", 6 point Mixed case
				Color:	Black (58600)

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Tomb (AL036)

Surface

Feature Description:	*A structure within which a corpse is entombed.*
Inclusion Condition:	*Length or Diameter >= 25 meters*

Notes

None

Feature Information					
Specification:	*LTDS*	**Feature Origin:**	*Topographic*	**Symbol:**	AR_Black_Tomb
Symbol Criteria:	All			**Label:**	[Geographic Name Information: Full Name], 6 point Condensed Mixed case if available; and "Tomb", 6 point Condensed Mixed case
				Color:	Black (58600)

Tomb (AL036)

Point

Inclusion Condition:	*Length or Diameter < 25 meters*

Notes

None

Feature Information					
Specification:	*LTDS*	**Feature Origin:**	*Topographic*	**Symbol:**	PT_Black_Tomb
Symbol Criteria:	All			**Label:**	[Geographic Name Information: Full Name], 6 point Condensed Mixed case if available; and "Tomb", 6 point Condensed Mixed case
				Color:	Black (58600)

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Tower (AL241)

Point

Feature Description:	*A relatively tall, narrow structure that may either stand alone or may form part of another structure.*
Inclusion Condition:	*Height Above Surface Level >= 46 m. OR Navigation Landmark = [True]*

Notes

Label feature as per hierarchy for topo type placement to south neatline corners reading left to right:
1. Position hierarchy:
(a.) northeast (preferred position
(b.) southeast (1st alternative)
(c.) northwest (2nd alternative
(d.) southwest (3rd alternative)
(e.) top-centered (4th alternative)
(f.) bottom-centered (5th alternative)
(Hierarchy is based on type positioning so as to avoid overprinting other type or obscuring detail.)
2. Minimum space between type placement and feature symbol is 0.5 mm
3. This method of type placement shall be used for areal features when space does not permit labeling within that feature.

Type for features on land shall be positioned on land if possible. If type must be placed in the water, it shall be positioned so it does not obscure or overprint hydrographic detail. Type for features in the water shall be positioned in the water. If possible, type shall not be positioned across the shoreline.

If coalescing features being thinned are a mix of heights, with some < 46 meters and some >= 46 meters, then only the obstruction symbol shall be shown.

When obstructions coalesce at map scale, label with highest obstruction information.

Feature Information

Specification:	*LTDS*	**Feature Origin:**	*Topographic*	**Symbol:**	PT_Black_Tower_ Communication
Symbol Criteria:	Tower Type [1] = [Telecommunication Tower] and Height Above Surface Level < 46 m			**Label:**	None
				Color:	N/A

Feature Information

Specification:	*LTDS*	**Feature Origin:**	*Topographic*	**Symbol:**	PT_Black_Tower
Symbol Criteria:	Tower Type [1] <> [Telecommunication Tower]			**Label:**	None
				Color:	N/A

Feature Information

Specification:	*LTDS*	**Feature Origin:**	*Topographic*	**Symbol:**	PT_Blue072_VO
Symbol Criteria:	Height Above Surface Level >= 46 and Location Referenced to Shoreline = [At Shoreline] or [Inland]			**Label:**	"Tower", 6 point Condensed Mixed case; [Highest Elevation], 6 point Condensed Bold Mixed case; "([Height Above Surface Level])", 6 point Condensed Mixed case
				Color:	Blue (48253)

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Traffic Separation Scheme (FC041)

Surface

Feature Description:	*A maritime routeing measure that separates opposing streams of traffic by appropriate means.*
Inclusion Condition:	*All required*

Notes

Type sizes for Maritime Limits and areas:
8 point - < 8 square cm.,
10 point - >= 8 and < 12 square cm,,
12 point - >= 12 and < 24 square cm.,
14 point - >= 24 and < 100 square cm.,
8 point - >= 100 square cm.; Type placement for areas >= 100 square cm. to < 500 square cm.:
Two labels are shown on approximately opposite sides of the area, preferably top and bottom.;
Type placement for areas >= 500 square cm.: Labels are placed at approximately 250 mm. interval around the perimeter of the feature.;
Do not place labels around sharp corners (interior angle < 135°), or along chart neatlines. Place type 1 mm away and parallel to area limit line, reading left to right, or bottom to top if limit line is vertical. Type is placed on the INSIDE of the area.

Priority for type placement: 1-right center, 2-bottom center, 3-left center, 4-top center.
(A) Minimum distance from symbol - 1 mm.
(B) Maximum distance from symbol before choosing the next highest priority:
(#1) 4 mm measured to the West end;
(#2) 4 mm measured to the North side (top);
(#3) 4 mm measured to the East end;
(#4) 4 mm measured to the South side (bottom)

Label for (Abandoned) shall be "Disused", in Mixed case italic type, and enclosed in parentheses. It shall be centered under the label for that area.

Type placement for areas < 100 square cm. Type shall be center in the area reading from left to right. If longer of the two axes is North-South +/- 20 degrees or East-West +/- 20 degrees, type is parallel to the south neatline.
(a.) If length < width times two, type shall be placed on two approximately equal lines without splitting words;
(b.) If length >= width times two, and major axis is East-West +/- 20 degrees, type shall be placed on one line;
(c.) If length >= width times two, and major axis is North-South +/- 20 degrees, type shall be placed with each word on a separate line.
If longer of the two axis is more than 20 degrees from either North-South or East-West, and length < width times two, type shall be parallel to south neatline and on two approximately equal lines without splitting words.
If longer of two axis is more than 20 degrees from either North-South or East-West, and length >= width times two, type shall be placed parallel to the major axis on one line.

When the boundary of an areal symbol crosses or overprints the shoreline, the boundary overprinting the shoreline or land is deleted.

When a traffic separation scheme is portrayed on a nautical chart, a note is shown in black indicating whether or not it has been adopted by the International Maritime Organization. The following note is shown if the:
(1.) If the traffic separation scheme has been adopted by the IMO. "NOTE- The Traffic Separation Scheme(s) on this chart is adopted by the International Maritime Organization (IMO)." ;
(2.) if the traffic separation scheme has not been adopted by the IMO. "NOTE- The Traffic Separation Scheme(s) on this chart is not adopted by the International Maritime Organization (IMO)."

Break outer boundary of the roundabout where traffic separation schemes enter and exit the roundabout. Arrows proceed in a clockwise direction and may be moved to avoid overprints. At least two arrows shall be shown, with more arrows shown as the circumference of the travelled lane increases. If the width of the traffic lane of the roundabout is less than 4 mm, delete arrows.

Minimum width for maritime limit symbols, other than pilot boarding areas, shall be 4 mm at chart scale.

Feature Information

Specification:	*Chart #1*	**Feature Origin:**	*Hydrographic*	**Symbol:**	M23
Symbol Criteria:	Maritime Traffic Separation Scheme Component = [Crossing]			**Label:**	None
				Color:	N/A

Feature Information

Specification:	*Chart #1*	**Feature Origin:**	*Hydrographic*	**Symbol:**	M16
Symbol Criteria:	Maritime Traffic Separation Scheme Component = [Precautionary Area]			**Label:**	"Precautionary Area", 8-14 point Mixed case, Italic
				Color:	Green (52813)

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Traffic Separation Scheme (FC041) *(Continued)*

Surface

Feature Information					
Specification:	*Chart #1*	**Feature Origin:**	*Hydrographic*	**Symbol:**	M21
Symbol Criteria:	Maritime Traffic Separation Scheme Component = [Roundabout] or [Point]			**Label:**	None
				Color:	N/A

Feature Information					
Specification:	*Chart #1*	**Feature Origin:**	*Hydrographic*	**Symbol:**	M14
Symbol Criteria:	Maritime Traffic Separation Scheme Component = [Inshore Traffic Zone]			**Label:**	"Inshore Traffic Zone", 8-14 point Mixed case, Italic
				Color:	Green (52813)

Feature Information					
Specification:	*Chart #1*	**Feature Origin:**	*Hydrographic*	**Symbol:**	M13
Symbol Criteria:	Maritime Traffic Separation Scheme Component = [Separation Zone] and Width >= 50			**Label:**	None
				Color:	N/A

Feature Information					
Specification:	*Chart #1*	**Feature Origin:**	*Hydrographic*	**Symbol:**	M15
Symbol Criteria:	Maritime Traffic Separation Scheme Component = [Boundary] or [Inbound Lane] or [Lane Part] or [Outbound Lane]			**Label:**	None
				Color:	N/A

Traffic Separation Scheme (FC041)

Curve

Inclusion Condition:	*All required*

Notes

When a traffic separation scheme is portrayed on a nautical chart, a note is shown in black indicating whether or not it has been adopted by the International Maritime Organization. The following note is shown if the:
(1.) If the traffic separation scheme has been adopted by the IMO. "NOTE- The Traffic Separation Scheme(s) on this chart is adopted by the International Maritime Organization (IMO)." ;
(2.) if the traffic separation scheme has not been adopted by the IMO. "NOTE- The Traffic Separation Scheme(s) on this chart is not adopted by the International Maritime Organization (IMO)."

Feature Information					
Specification:	*Chart #1*	**Feature Origin:**	*Hydrographic*	**Symbol:**	M12
Symbol Criteria:	Maritime Traffic Separation Scheme Component = [Line] and Width < 50			**Label:**	None
				Color:	N/A

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Trail (AP050)

Curve

Feature Description:	*A path worn by the passage of people or animals.*
Inclusion Condition:	*('Road Weather Restriction' is either 'Fair Weather' or 'Winter Only') AND ('Length or Diameter' > 300 m.) AND ((in moderate-to-dense 'Area's are spaced >= 1000 m. apart) OR (in sparse-to-moderate 'Area's are spaced >= 300 m. apart) OR (needed to complete the component network or to establish connectivity)))*

Notes

The feature which coalesces (<0.2 mm) with a railroad track or RR Sidetrack/Spur shall be displaced to a minimum of 0.2 mm apart.

If feature coalesce at map scale, when shown in their true positions, they shall be displaced 0.2 mm from one another.

Area and line features will be generalized to detail compatible with scale.

For Road, Cart track, and Trail within Built-up Area; Symbolize the portion of the feature within the Built-up Area as white.

Thin Cart Tracks and Trails in moderate to dense areas to a Length <= 1500 meters and a spacing of >= 6000 meters, and for sparse to moderate areas to a Length <= 1500 meters and a spacing of >= 1250 meters, unless needed to complete the road network. Two exceptions to the above rule for these features if they do not connect with another "road like" feature: (1.) If Length is less than 1500 meters and has a cultural feature at its terminus, retain this short segment to this feature. (2.) Delete this feature if area is moderate to dense and there is no cultural feature at its terminus.

Feature Information

Specification:	*LTDS*	**Feature Origin:**	*Topographic*	**Symbol:**	LN_Black_Trail
Symbol Criteria:	All			**Label:**	"Winter only", 6 point Light Condensed Italics Mixed case if Route Pavement Information : Road Weather Restriction = [Winter Only]
				Color:	Black (58600)

Training Site (FA165)

Surface

Feature Description:	*A site reserved for training.*
Inclusion Condition:	*Area >= 15,625 sq m.*

Notes

None

Feature Information

Specification:	*LTDS*	**Feature Origin:**	*Topographic*	**Symbol:**	AR_Black_TrainingSite
Symbol Criteria:	All			**Label:**	[Geographic Name Information : Full Name] if available and "Training site", 6 point Condensed Mixed case
				Color:	Black (58600)

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Training Site (FA165)

Point

Inclusion Condition:	*Area < 15,625 sq m..*

Notes

None

Feature Information					
Specification:	*LTDS*	**Feature Origin:**	*Topographic*	**Symbol:**	PT_Black_TrainingSite
Symbol Criteria:	All			**Label:**	[Geographic Name Information : Full Name] if available and "Training site", 6 point Condensed Mixed case
				Color:	Black (58600)

Transportation Route Characteristic Change (AQ059)

Point

Feature Description:	*A section of a transportation route where its size (for example: narrowing or expansion in width) and/or operating speed is significantly different than in the adjacent sections of the route.*
Inclusion Condition:	*All Required*

Notes

None

Feature Information					
Specification:	*LTDS*	**Feature Origin:**	*Topographic*	**Symbol:**	PT_Black_TransRoute CharacterChange
Symbol Criteria:	Transportation System Type [1] = [Pedestrian] or [Railway] or [Road] or [Other] or [No Information]			**Label:**	None
				Color:	N/A

Feature Information					
Specification:	*LTDS*	**Feature Origin:**	*Topographic*	**Symbol:**	PT_Cyan_TransRoute CharacterChange
Symbol Criteria:	Transportation System Type [1] = [Canal] or [Inland Waterway]			**Label:**	None
				Color:	N/A

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Transportation Route Protection Structure (AL211)

Surface

Feature Description:	*A structure built over and/or along a transportation route designed to prevent damage to, or blockage of, the route from rock slides, snow slides and/or weather phenomena.*
Inclusion Condition:	*Length or Diameter >= 125 m.*

Notes

None

Feature Information					
Specification:	*LTDS*	**Feature Origin:**	*Topographic*	**Symbol:**	AR_Black_TransRoute ProtectStruct
Symbol Criteria:	All			**Label:**	"Protection shed", 6 point Condensed Mixed case
				Color:	Black (58600)

Transportation Route Protection Structure (AL211)

Point

Inclusion Condition:	*Length or Diameter < 125 m.*

Notes

The feature symbology shall be positioned such that the longest axis of the symbol is aligned coincident with the centerline of the associated road, railroad track, or RR siding/RR spur feature.

Like point features which coalesce in clusters of 3 or more will be thinned to form a representative pattern.

Type for features on land shall be positioned on land if possible. If type must be placed in the water, it shall be positioned so it does not obscure or overprint hydrographic detail. Type for features in the water shall be positioned in the water. If possible, type shall not be positioned across the shoreline.

If a Transportation Route Protection Structure falls on more than one sheet, it will be labeled on both.

If a feature is not associated with (touching) a road (AP030) or railroad track (AN010), omit the feature.

Feature Information					
Specification:	*LTDS*	**Feature Origin:**	*Topographic*	**Symbol:**	PT_Black_TransRoute ProtectStruct
Symbol Criteria:	All			**Label:**	"Protection shed", 6 point Condensed Mixed case
				Color:	Black (58600)

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Transportation Station (AQ125)

Surface

Feature Description:	*A station that serves as a stopping place along a transportation route.*
Inclusion Condition:	*Length or Diameter >= 25 m.*

Notes

None

Feature Information

Specification:	*LTDS*	**Feature Origin:**	*Topographic*	**Symbol:**	AR_Black_TransportationStation
Symbol Criteria:	(Feature Function [1] = [Station] or [Railway Transport] or [Railway Passenger Transport]) and (Transportation System Type [1] = [Bus] or [Railway]) and (Height Above Surface Level < 46) or (Area >= 5625 and Height Above Surface Level >= 46)			**Label:**	[Geographic Name Information : Full Name], 6 point Condensed Mixed case if available and ("Station", 6 point Condensed Mixed case if Transportation System Type [1] = [Railway]) or ("Bus station" if Transportation System Type [1] = [Bus]) ; and "(construction)", 6 point Condensed Lower case if Condition of Facility: [Under Construction], "(not usable)" , 6 point Condensed Lower case if Condition of Facility: [Abandoned] or [Destroyed] or [Ruined]
				Color:	Black (58600)

Transportation Station (AQ125)

Point

Inclusion Condition:	*Length or Diameter >= 25 m.*

Notes

None

Feature Information

Specification:	*LTDS*	**Feature Origin:**	*Topographic*	**Symbol:**	PT_Black_TransportationStation
Symbol Criteria:	(Feature Function [1] = [Station] or [Railway Transport] or [Railway Passenger Transport]) and Transportation System Type [1] = [Railway]) and (Height Above Surface Level < 46)			**Label:**	[Geographic Name Information : Full Name], 6 point Condensed Mixed case if available and "(construction)", 6 point Condensed Lower case if Condition of Facility: [Under Construction], "(not usable)" , 6 point Condensed Lower case if Condition of Facility: [Abandoned] or [Destroyed] or [Ruined]
				Color:	Black (58600)

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Tree (EC005)

Point

Feature Description:	*An individual woody perennial plant, typically having a single stem or trunk growing to a considerable height and bearing lateral branches at some distance from the ground.*
Inclusion Condition:	*Navigation Landmark = [True]*

Notes

None

Feature Information					
Specification:	*LTDS*	**Feature Origin:**	*Topographic*	**Symbol:**	PT_Green362_Wood
Symbol Criteria:	Navigation Landmark = [True]			**Label:**	None
				Color:	N/A

Tundra (BJ110)

Surface

Feature Description:	*A vast, nearly level, treeless arctic region usually with a marshy surface and underlying permafrost.*
Inclusion Condition:	*Area >= 15,625 sq m.*

Notes

Coincident similar area features will be blended to form a single feature.

Area and line features will be generalized to detail compatible with scale.

Feature will be generalized to provide a more aesthetic contoured feature (i.e., smoothed).

Type sizes per are sizes at map/chart scale: Area features only.
06 point - <= 770 mm square area and <= 14 mm width;
07 point - <= 2,296 mm square area and <= 28 mm width;
09 point - <= 5,192 mm square area and 44 mm width;
10 point - <=9,796 mm square area and <= 62 mm width;
12 point - <= 16,632 mm square area and <= 84 mm width;
14 point - <= 24,960 mm square area and <= 104 mm width;
16 point - > 24,960 mm square area.

Where area measurements are inconsistent, the larger type size shall be used. Where the full range of type sizes is not available for a particular label, the closest available type size shall be used.

Symbols and associated area patterns of underpassing features (except drainage shorelines) are broken for all bridges, except footbridges. This rule does not apply to land tint on Littoral Planning Charts.

If a clearing exists inside of the area feature, and the size of the clearing is equal to or greater than the area for the surrounding area feature, the clearing is shown as an open space inside the surround feature. If the area of the clearing is less than the area for the surrounding feature, the clearing is deleted and absorbed into the surrounding area feature.

If two area features with the same feature code do not connect at any point, and have a space between them less than 2.5 mm at map/chart scale, delete the open space that is less than 2.5 mm between the features and combine them into one area feature.

If a portion of an area feature has a minimum width of less than 2.5 mm at map/chart scale, delete that portion of the area feature that is not at least 2.5 mm wide, measured from perimeter to perimeter. If the deletion of the portion of the area based on the above criteria will reduce the area of the remaining portion of the area feature to below the minimum area inclusion, do not delete the narrow portion of the feature that is less than 2.5 mm wide. If the deletion of the portion of the area based on the above criteria will split two larger areas connected by a narrow strip into two separate areas, either of which would be below minimum area inclusion, do not delete the narrow portion of the feature that is less than 2.5 mm wide.

Feature Information					
Specification:	*LTDS*	**Feature Origin:**	*Topographic*	**Symbol:**	AR_Tundra
Symbol Criteria:	All			**Label:**	"TUNDRA", 7-16 point Condensed Italics Upper case
				Color:	Black (58600)

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Tunnel (AQ130)

Curve

Feature Description:	*An underground passage that is open at both ends and usually contains a land transportation route (for example: a road and/or a railway).*
Inclusion Condition:	*Length >= 100*

Notes

Area and line features will be generalized to detail compatible with scale.

Label shall be position above feature, reading from left to right (or to the left of the vertical feature, reading bottom to top), at a 0.5 mm distance and parallel to respective feature. Label shall preferably be positioned at the midpoint of the line segment or symbol; however, it may be displaced laterally along respective feature to avoid overprinting other symbols or labels. If space will not permit placing label parallel to feature, offset the label and use a leader line to identify its location along the feature.

Feature name, label, data information holder, and/or symbol shall be positioned, reading left to right, parallel to the tangent of the center of the southern neatline of the Map/Chart.

Descriptive terms, e.g., "Canal" shall be shown if the name is not known. If the descriptive word appears in the name, for example, "PANAMA CANAL". The descriptive type shall not be shown, i.e. do not show "Panama Canal Canal".

Within Built-up Areas, include only tunnels relating to through routes.

If the alignment of a Railroad is approximate and the Railroad enter/exits a Tunnel, the dashed line representing the Tunnel symbol shall not be shown. Only the wing ticks and "headwall" at both ends of the Tunnel shall be shown.

If a feature is not associated with (touching) a road (AP030) or railroad track (AN010), omit the feature.

Feature Information

Specification:	*LTDS*	**Feature Origin:**	*Topographic*	**Symbol:**	LN_Black_Tunnel_Rail
Symbol Criteria:	(Transportation System Type [1] = [Pedestrian] or [Railway]) or (Transportation System Type [1] = [Railway] and Transportation System Type [2] = [Road]) or (Transportation System Type [1] = [Road] and Transportation System Type [2] = [Railway) and (Length >= 100)			**Label:**	[Geographic Name Information : Full Name], 6 point Condensed Mixed case if available and "Tunnel", 6 point Condensed Mixed case
				Color:	Black (58600)

Feature Information

Specification:	*LTDS*	**Feature Origin:**	*Topographic*	**Symbol:**	LN_Cyan_Tunnel_Aqueduct
Symbol Criteria:	(Transportation System Type [1] = [Aqueduct]) and (Length >= 100)			**Label:**	[Geographic Name Information : Full Name], 6 point Condensed Mixed case if available and "Tunnel", 6 point Condensed Mixed case
				Color:	Blue (48253)

Feature Information

Specification:	*LTDS*	**Feature Origin:**	*Topographic*	**Symbol:**	LN_Cyan_Tunnel_Canal
Symbol Criteria:	Transportation System Type [1] = [Canal] or [Inland Waterway]) and (Length >= 100)			**Label:**	[Geographic Name Information : Full Name], 6 point Condensed Mixed case if available and "Tunnel", 6 point Condensed Mixed case
				Color:	Blue (48253)

Feature Information

Specification:	*LTDS*	**Feature Origin:**	*Topographic*	**Symbol:**	LN_Black_Tunnel_Road
Symbol Criteria:	(Transportation System Type [1] = [Road]) and (Length >= 100)			**Label:**	[Geographic Name Information : Full Name], 6 point Condensed Mixed case if available and "Tunnel", 6 point Condensed Mixed case
				Color:	Black (58600)

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Tunnel (AQ130)

Point

Inclusion Condition:	*Length < 100*

Notes

Area and line features will be generalized to detail compatible with scale.

Label shall be position above feature, reading from left to right (or to the left of the vertical feature, reading bottom to top), at a 0.5 mm distance and parallel to respective feature. Label shall preferably be positioned at the midpoint of the line segment or symbol; however, it may be displaced laterally along respective feature to avoid overprinting other symbols or labels. If space will not permit placing label parallel to feature, offset the label and use a leader line to identify its location along the feature.

Feature name, label, data information holder, and/or symbol shall be positioned, reading left to right, parallel to the tangent of the center of the southern neatline of the Map/Chart.

Descriptive terms, e.g., "Canal" shall be shown if the name is not known. If the descriptive word appears in the name, for example, "PANAMA CANAL". The descriptive type shall not be shown, i.e. do not show "Panama Canal Canal".

Within Built-up Areas, include only tunnels relating to through routes.

If the alignment of a Railroad is approximate and the Railroad enter/exits a Tunnel, the dashed line representing the Tunnel symbol shall not be shown. Only the wing ticks and "headwall" at both ends of the Tunnel shall be shown.

If a feature is not associated with (touching) a road (AP030) or railroad track (AN010), omit the feature.

Feature Information

Specification:	*LTDS*	**Feature Origin:**	*Topographic*	**Symbol:**	PT_Black_Tunnel
Symbol Criteria:	(Transportation System Type [1] = [Pedestrian] or [Railway] or [Road]) or (Transportation System Type [1] = [Railway] and Transportation System Type [2] = [Road]) or (Transportation System Type [1] = [Road] and Transportation System Type [2] = [Railway) and (Length < 100)			**Label:**	[Geographic Name Information : Full Name], 6 point Condensed Mixed case if available and "Tunnel", 6 point Condensed Mixed case
				Color:	Black (58600)

Feature Information

Specification:	*LTDS*	**Feature Origin:**	*Topographic*	**Symbol:**	PT_Cyan_Tunnel
Symbol Criteria:	Transportation System Type [1] = [Aqueduct] or [Canal] or [Inland Waterway]) and (Length < 100)			**Label:**	[Geographic Name Information : Full Name], 6 point Condensed Mixed case if available and "Tunnel", 6 point Condensed Mixed case
				Color:	Blue (48253)

Underground Bunker (AH060)

Point

Feature Description:	*A reinforced underground shelter.*
Inclusion Condition:	*All required*

Notes

None

Feature Information

Specification:	*LTDS*	**Feature Origin:**	*Topographic*	**Symbol:**	PT_Black_UndergroundBunker
Symbol Criteria:	All			**Label:**	"Underground bunker", 6 point Condensed Mixed case
				Color:	Black (58600)

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Underground Dwelling (AL250)

Point

Feature Description:	*Underground living quarters.*
Inclusion Condition:	*Navigation Landmark = [True]*

Notes

Label feature as per hierarchy for topo type placement to south neatline corners reading left to right:
1. Position hierarchy:
(a.) northeast (preferred position
(b.) southeast (1st alternative)
(c.) northwest (2nd alternative
(d.) southwest (3rd alternative)
(e.) top-centered (4th alternative)
(f.) bottom-centered (5th alternative)
(Hierarchy is based on type positioning so as to avoid overprinting other type or obscuring detail.)
2. Minimum space between type placement and feature symbol is 0.5 mm
3. This method of type placement shall be used for areal features when space does not permit labeling within that feature.

Feature Information

Specification:	*LTDS*	**Feature Origin:**	*Topographic*	**Symbol:**	PT_Black_UndergroundDwelling
Symbol Criteria:	All			**Label:**	"Underground dwelling", 6 point Condensed Mixed case
				Color:	Black (58600)

Vanishing Point (BH145)

Point

Feature Description:	*The location at which a watercourse disappears into the terrain.*
Inclusion Condition:	*ALL REQUIRED*

Notes

Like point features which coalesce in clusters of 3 or more will be thinned to form a representative pattern.

Omit if not shown in conjunction with drainage feature.

The apex of feature to point uphill, to align with direction of flow (DOF).

Feature Information

Specification:	*LTDS*	**Feature Origin:**	*Topographic*	**Symbol:**	PT_Cyan_VanishingPoint _Disappearing
Symbol Criteria:	Watercourse Sink Type = [Disappearing] or [Sinkhole] or [Hole])			**Label:**	None
				Color:	N/A

Feature Information

Specification:	*LTDS*	**Feature Origin:**	*Topographic*	**Symbol:**	PT_Cyan_VanishingPoint _Dissipating
Symbol Criteria:	Watercourse Sink Type = [Dissipating] or [Other] or [No Information			**Label:**	None
				Color:	N/A

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Vehicle Lot (AQ140)

Surface

Feature Description:	*A tract used for storing and/or parking vehicles (for example: recreational vehicles) and/or vessels.*
Inclusion Condition:	*Area >= 15,625 sq m.*

Notes

Area and line features will be generalized to detail compatible with scale.

Label feature as per hierarchy for topo type placement to south neatline corners reading left to right:
1. Position hierarchy:
(a.) northeast (preferred position),
(b.) southeast (1st alternative),
(c.) northwest (2nd alternative,
(d.) southwest (3rd alternative),
(e.) top-centered (4th alternative),
(f.) bottom-centered (5th alternative):
(Hierarchy is based on type positioning so as to avoid overprinting other type or obscuring detail.);
2. Minimum space between type placement and feature symbol is 0.5 mm;
3. This method of type placement shall be used for areal features when space does not permit labeling within that feature.

Names placement shall be oriented to the longest axis of the feature reading left to right and placed within the area outline and centered. If the longest axis is perpendicular to the south neatline, the type shall be placed outside of the area outline, preferred position is northeast of the feature, but may be placed at any position around the feature so as not to overprint any other feature type and reading left to right.

Limiting lines of feature are omitted if it coalesces with a road (AP030).

If a clearing exists inside of the area feature, and the size of the clearing is equal to or greater than the area for the surrounding area feature, the clearing is shown as an open space inside the surround feature. If the area of the clearing is less than the area for the surrounding feature, the clearing is deleted and absorbed into the surrounding area feature.

If two area features with the same feature code do not connect at any point, and have a space between them less than 2.5 mm at map/chart scale, delete the open space that is less than 2.5 mm between the features and combine them into one area feature.

If a portion of an area feature has a minimum width of less than 2.5 mm at map/chart scale, delete that portion of the area feature that is not at least 2.5 mm wide, measured from perimeter to perimeter. If the deletion of the portion of the area based on the above criteria will reduce the area of the remaining portion of the area feature to below the minimum area inclusion, do not delete the narrow portion of the feature that is less than 2.5 mm wide. If the deletion of the portion of the area based on the above criteria will split two larger areas connected by a narrow strip into two separate areas, either of which would be below minimum area inclusion, do not delete the narrow portion of the feature that is less than 2.5 mm wide.

Feature Information

Specification:	*LTDS*	**Feature Origin:**	*Topographic*	**Symbol:**	AR_Black_VehicleLot
Symbol Criteria:	Area >= 15,625			**Label:**	"Parking area", 6 point Condensed Mixed case
				Color:	Black (58600)

Vineyard (EA050)

Surface

Feature Description:	*A tract covered by the systematic planting of grape vines.*
Inclusion Condition:	*Area >= 15,625 sq m. and Navigation Landmark = [True]*

Notes

Coincident similar area features will be blended to form a single feature.

Area and line features will be generalized to detail compatible with scale.

Feature will be generalized to provide a more aesthetic contoured feature (i.e., smoothed).

Symbols and associated area patterns of underpassing features (except drainage shorelines) are broken for all bridges, except footbridges. This rule does not apply to land tint on Littoral Planning Charts.

If a clearing exists inside of the area feature, and the size of the clearing is equal to or greater than the area for the surrounding area feature, the clearing is shown as an open space inside the surround feature. If the area of the clearing is less than the area for the surrounding feature, the clearing is deleted and absorbed into the surrounding area feature.

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Vineyard (EA050) *(Continued)*

Surface

Notes

If a portion of an area feature has a minimum width of less than 2.5 mm at map/chart scale, delete that portion of the area feature that is not at least 2.5 mm wide, measured from perimeter to perimeter. If the deletion of the portion of the area based on the above criteria will reduce the area of the remaining portion of the area feature to below the minimum area inclusion, do not delete the narrow portion of the feature that is less than 2.5 mm wide. If the deletion of the portion of the area based on the above criteria will split two larger areas connected by a narrow strip into two separate areas, either of which would be below minimum area inclusion, do not delete the narrow portion of the feature that is less than 2.5 mm wide.

Feature Information					
Specification:	*LTDS*	**Feature Origin:**	*Topographic*	**Symbol:**	AR_Green362_Vineyard
Symbol Criteria:	Area >= 15,625 and Navigation Landmark = [True]			**Label:**	[Geographic Name Information : Full Name], 6 point Condensed Mixed case if available; and "Vineyard", 6 point Condensed Mixed case and "Terraces", 6 point Condensed Mixed case if Crop Information : Farming Pattern [1] = [Terraced]
				Color:	Black (58600)

Vineyard (EA050)

Point

Inclusion Condition:	*Area < 15,625 sq m. and Navigation Landmark = [True]*

Notes

Coincident similar area features will be blended to form a single feature.

Area and line features will be generalized to detail compatible with scale.

Feature will be generalized to provide a more aesthetic contoured feature (i.e., smoothed).

Symbols and associated area patterns of underpassing features (except drainage shorelines) are broken for all bridges, except footbridges. This rule does not apply to land tint on Littoral Planning Charts.

If a clearing exists inside of the area feature, and the size of the clearing is equal to or greater than the area for the surrounding area feature, the clearing is shown as an open space inside the surround feature. If the area of the clearing is less than the area for the surrounding feature, the clearing is deleted and absorbed into the surrounding area feature.

If two area features with the same feature code do not connect at any point, and have a space between them less than 2.5 mm at map/chart scale, delete the open space that is less than 2.5 mm between the features and combine them into one area feature.

If a portion of an area feature has a minimum width of less than 2.5 mm at map/chart scale, delete that portion of the area feature that is not at least 2.5 mm wide, measured from perimeter to perimeter. If the deletion of the portion of the area based on the above criteria will reduce the area of the remaining portion of the area feature to below the minimum area inclusion, do not delete the narrow portion of the feature that is less than 2.5 mm wide. If the deletion of the portion of the area based on the above criteria will split two larger areas connected by a narrow strip into two separate areas, either of which would be below minimum area inclusion, do not delete the narrow portion of the feature that is less than 2.5 mm wide.

Feature Information					
Specification:	*LTDS*	**Feature Origin:**	*Topographic*	**Symbol:**	PT_Green362_Vineyard
Symbol Criteria:	Area < 15,625 and Navigation Landmark = [True]			**Label:**	"Vineyard", 6 point Condensed Mixed cas
				Color:	Black (58600)

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Void Collection Area (ZD020)

Surface

Feature Description:	*A data collection region lacking suitable source coverage and/or where data is not required.*
Inclusion Condition:	*ALL REQUIRED*

Notes

Feature must retain all cartographic detail (i.e., not thinned or smoothed).

Type sizes per are sizes at map/chart scale:
Area features only.
06 point - <= 770 mm square area and <= 14 mm width;
07 point - <= 2,296 mm square area and <= 28 mm width;
09 point - <= 5,192 mm square area and 44 mm width;
10 point - <=9,796 mm square area and <= 62 mm width;
12 point - <= 16,632 mm square area and <= 84 mm width;
14 point - <= 24,960 mm square area and <= 104 mm width;
16 point - > 24,960 mm square area. Where area measurements are inconsistent, the larger type size shall be used.
Where the full range of type sizes is not available for a particular label, the closest available type size shall be used.

Label feature as per hierarchy for topo type placement to south neatline corners reading left to right:
1. Position hierarchy: (a.) northeast (preferred position), (b.) southeast (1st alternative), (c.) northwest (2nd alternative, (d.) southwest (3rd alternative), (e.) top-centered (4th alternative), (f.) bottom-centered (5th alternative): (Hierarchy is based on type positioning so as to avoid overprinting other type or obscuring detail.);
2. Minimum space between type placement and feature symbol is 0.5 mm;
3. This method of type placement shall be used for areal features when space does not permit labeling within that feature.

Names placement shall be oriented to the longest axis of the feature reading left to right and placed within the area outline and centered. If the longest axis is perpendicular to the south neatline, the type shall be placed outside of the area outline, preferred position is northeast of the feature, but may be placed at any position around the feature so as not to overprint any other feature type and reading left to right.

An area void of Contours or form lines due to lack of, or poor quality source data, shall be labeled "RELIEF DATA INCOMPLETE". An area void of relief which is greater than 75 mm X 75 mm at map scale shall carry the additional note "Limits of Reliable Relief Information" repeated along the perimeter of the contoured area.

Feature Information

Specification:	*LTDS*	**Feature Origin:**	*Topographic*	**Symbol:**	AR_Black_VoidCollectionArea
Symbol Criteria:	Area >= 1,000,000 AND Void Collection Type [1] = [Hypsography] or [Other])			**Label:**	"DATA INCOMPLETE", 6-16 Condensed Upper case
				Color:	Black (58600)

Volcanic Dyke (DB190)

Curve

Feature Description:	*A sheet of rock filling a fissure that sometimes shows as a terrain ridge.*
Inclusion Condition:	*Length or Diameter >= 250 m*

Notes

None

Feature Information

Specification:	*LTDS*	**Feature Origin:**	*Topographic*	**Symbol:**	LN_Black_VolcanicDyke
Symbol Criteria:	Length or Diameter >= 250			**Label:**	"Volcanic dyke", 6 point Regular Mixed case
				Color:	Black (58600)

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Volcano (DB180)

Point

Feature Description:	*A hill or mountain situated over an opening or openings in the Earth's crust through which lava, cinders, steam, and/or gases, are or have been expelled.*
Inclusion Condition:	*ALL REQUIRED*

Notes

Label feature as per hierarchy for topo type placement to south neatline corners reading left to right:
1. Position hierarchy: (a.) northeast (preferred position), (b.) southeast (1st alternative), (c.) northwest (2nd alternative, (d.) southwest (3rd alternative), (e.) top-centered (4th alternative), (f.) bottom-centered (5th alternative): (Hierarchy is based on type positioning so as to avoid overprinting other type or obscuring detail.);
2. Minimum space between type placement and feature symbol is 0.5 mm;
3. This method of type placement shall be used for areal features when space does not permit labeling within that feature.

Feature Information

Specification:	*LTDS*	**Feature Origin:**	*Topographic*	**Symbol:**	PT_Dk-Brown1815_Volcano
Symbol Criteria:	(Volcano Shape = [Cone] or [Cinder Cone])			**Label:**	[Geographic Name Information : Full Name], 7-10 point Regular Mixed case if available
				Color:	Black (58600)

Feature Information

Specification:	*LTDS*	**Feature Origin:**	*Topographic*	**Symbol:**	PT_Volcano
Symbol Criteria:	(Volcano Shape = [Shield] or [Caldera] or [Composite] or [No Information])			**Label:**	[Geographic Name Information : Full Name], 7-10 point Regular Mixed case if available; "Volcano" if Volcano Shape = [Shield] or [Composite] or [No Information]; "Caldera" if Volcano Shape = [Caldera], 7-10 point Regular Mixed case
				Color:	Black (58600)

Wall (AL260)

Curve

Feature Description:	*A solid man-made barrier of generally heavy material used as an enclosure, boundary, or for protection.*
Inclusion Condition:	*Length or Diameter >= 125 m. AND Predominant Feature Height: >= 2 m.*

Notes

Area and line features will be generalized to detail compatible with scale.

Omit feature when coincident with another unlike line feature.

Walls or Fences which enclose the following areal features shall not be shown: Mobile Home park, Amusement Park, Athletic Field, Campground, Drive-In Theater, Fairgrounds, Golf Course, Stadium, Zoo, and Cemetery.

Feature Information

Specification:	*LTDS*	**Feature Origin:**	*Topographic*	**Symbol:**	LN_Black_Wall
Symbol Criteria:	All			**Label:**	[Geographic Name Information: Full Name] if available, 6 point Condensed Mixed case
				Color:	Black (58600)

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Water Aerodrome (GB065)

Surface

Feature Description:	*An aerodrome intended to be used either wholly or in part for the arrival, departure and surface movement of aircraft on water.*
Inclusion Condition:	*ALL REQUIRED*

Notes

Coincident similar area features will be blended to form a single feature.

Area and line features will be generalized to detail compatible with scale.

Type for features on land shall be positioned on land if possible. If type must be placed in the water, it shall be positioned so it does not obscure or overprint hydrographic detail. Type for features in the water shall be positioned in the water. If possible, type shall not be positioned across the shoreline.

Coincident similar area features will be blended to form a single feature.

Feature Information

Specification:	*Chart #1*	**Feature Origin:**	*Hydrographic*	**Symbol:**	N13
Symbol Criteria:	All			**Label:**	[Geographic Name Information : Full Name], 7 point Condensed Mixed case if available; and "Seaplane base" , 7 point Condensed Mixed case and; "abandoned", 7 point Condensed Lower case if Airfield Symbol Type = [Abandoned or Closed (with usable runways or landing areas)] or [Decoy]; "not useable", 7 point Condensed Lower case if Airfield Symbol Type = [Unusable for landing or take-off]; "construction", 7 point Condensed Lower case if Airfield Symbol Type = [with less than the minimum facilities)]and "([Highest Elevation])", 7 point Condensed Lower case
				Color:	Purple (96532)

Water Current Region (BG013)

Surface

Feature Description:	*A waterbody region in which a water current persists.*
Inclusion Condition:	*ALL REQUIRED*

Notes

None

Feature Information

Specification:	*Other*	**Feature Origin:**	*Other*	**Symbol:**	TBD
Symbol Criteria:	All			**Label:**	None
				Color:	N/A

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Water Intake Tower (BI050)

Surface

Feature Description:	*A tower-like structure associated with a dam or water source and used for the intake of water.*
Inclusion Condition:	*Area >= 625 sq m.*

Notes

When 2 or more similar area features are separated by less than 0.5 mm at chart scale, the feature will be agglomerated to form an area multiple feature outline.

Area and line features will be generalized to detail compatible with scale.

Omit if not shown in conjunction with drainage feature.

Feature Information

Specification:	*LTDS*	**Feature Origin:**	*Topographic*	**Symbol:**	AR_Black_WaterIntakeTower
Symbol Criteria:	(Height Above Surface Level < 46) or (Area >= 5625 and Height Above Surface Level >= 46)			**Label:**	"Water intake tower", 6 point Condensed Mixed case
				Color:	Black (58600)

Water Intake Tower (BI050)

Point

Inclusion Condition:	*Area < 625 sq m*

Notes

A cluster of 3 or more coalescing similar point features will be aggregated to form an area multiple feature outline.

Omit if not shown in conjunction with drainage feature.

Feature Information

Specification:	*LTDS*	**Feature Origin:**	*Topographic*	**Symbol:**	PT_Black_WaterIntakeTower
Symbol Criteria:	Height Above Surface Level < 46			**Label:**	None
				Color:	N/A

Water Mill (AJ055)

Point

Feature Description:	*A structure that uses a water wheel or turbine to drive a mechanical process (for example: grinding grain into flour, cutting timber into lumber, or stripping bark from trees for use in tanning).*
Inclusion Condition:	*Length or Diameter >= 3 m. and (Height Above Surface level < 46)*

Notes

None

Feature Information

Specification:	*LTDS*	**Feature Origin:**	*Topographic*	**Symbol:**	PT_Black_WaterMill
Symbol Criteria:	(Height Above Surface level < 46)			**Label:**	None
				Color:	N/A

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Water Movement Data Location (BG010)

Point

Feature Description:	*A location at which currents, tides, and/or other types of water movement are specified.*
Inclusion Condition:	*All Required*

Notes

The feature shall be aligned with a river/stream, canal, or ditch.

Current speed Current Rate Minimum or Current Rate Maximum shall be printed 1 mm above the line reading from left to right, or from bottom to top if arrow is vertical. When Current Rate Minimum and Current Rate Maximum are shown together, they are separated by a hyphen, e.g., 2.2 - 4.0 kn.

A Flow Arrow shall be positioned within the limits of a double-line River/Stream without coincidence. The arrow shall be positioned parallel center within the River/Stream symbol.

Flow Arrow shall be centered between shoreline symbols on area features with arrow pointing downstream or 0.25 mm away from linear feature. Repeat use of symbol every 25.0 mm.

Arrow shall point in the direction of the flow.

Current Arrows shall not be shown closer than 40 mm apart, unless the difference in Direction Of Flow is greater than 45°, or difference in Current Rate Maximum is greater than 2 knots. If arrows are thinned, the one with the greatest Current Rate Maximum shall be retained.

Feature Information

Specification:	*Chart #1*	**Feature Origin:**	*Hydrographic*	**Symbol:**	H42
Symbol Criteria:	Current Type Category = [River Flow] or [Tidal Flow] or [Underwater River Flow]			**Label:**	None
				Color:	N/A

Feature Information

Specification:	*Chart #1*	**Feature Origin:**	*Hydrographic*	**Symbol:**	H43
Symbol Criteria:	Current Type Category = [Ocean Current]			**Label:**	None
				Color:	N/A

Feature Information

Specification:	*Chart #1*	**Feature Origin:**	*Hydrographic*	**Symbol:**	H40
Symbol Criteria:	Current Type Category = [Flood Stream]			**Label:**	[Current Rate (Speed)] "kn", 6 point lower case, Italic
				Color:	Black (58600)

Feature Information

Specification:	*Chart #1*	**Feature Origin:**	*Hydrographic*	**Symbol:**	H41
Symbol Criteria:	Current Type Category = [Ebb Stream]			**Label:**	[Current Rate (Speed)] "kn", 6 point lower case, Italic
				Color:	Black (58600)

Feature Information

Specification:	*Chart #1*	**Feature Origin:**	*Hydrographic*	**Symbol:**	H44
Symbol Criteria:	Current Type Category = [Longshore] or [Rip]			**Label:**	None
				Color:	N/A

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Water Race (BH065)

Curve

Feature Description:	*An artificial channel for a current of water, especially one built to provide water for industrial or agricultural purposes or for transporting water-borne materials.*
Inclusion Condition:	*Length or Diameter >= 75 m.*

Notes

Area and line features will be generalized to detail compatible with scale.

Label shall be position above feature, reading from left to right (or to the left of the vertical feature, reading bottom to top), at a 0.5 mm distance and parallel to respective feature. Label shall preferably be positioned at the midpoint of the line segment or symbol; however, it may be displaced laterally along respective feature to avoid overprinting other symbols or labels. If space will not permit placing label parallel to feature, offset the label and use a leader line to identify its location along the feature.

Feature name, label, data information holder, and/or symbol shall be positioned, reading left to right, parallel to the tangent of the center of the southern neatline of the Map/Chart.

Omit from Built-up Area.

Feature Information

Specification:	*LTDS*	**Feature Origin:**	*Topographic*	**Symbol:**	LN_Cyan_WaterRace_ OnGround
Symbol Criteria:	(Vertical Relative Location = [Level] or [Depressed] or [No Information])			**Label:**	"Water race", 6 point Condensed Italics Mixed case
				Color:	Blue (48253)

Feature Information

Specification:	*LTDS*	**Feature Origin:**	*Topographic*	**Symbol:**	LN_Cyan_WaterRace_Elevated
Symbol Criteria:	(Vertical Relative Location = [Raised])			**Label:**	"Water race", 6 point Condensed Italics Mixed case
				Color:	Blue (48253)

Water Tower (AM080)

Point

Feature Description:	*A tower supporting an elevated storage tank of water.*
Inclusion Condition:	*Height Above Surface Level >= 46 m. OR Navigation Landmark = [True]*

Notes

Label feature as per hierarchy for topo type placement to south neatline corners reading left to right:
1. Position hierarchy:
(a.) northeast (preferred position
(b.) southeast (1st alternative)
(c.) northwest (2nd alternative
(d.) southwest (3rd alternative)
(e.) top-centered (4th alternative)
(f.) bottom-centered (5th alternative)
(Hierarchy is based on type positioning so as to avoid overprinting other type or obscuring detail.)
2. Minimum space between type placement and feature symbol is 0.5 mm
3. This method of type placement shall be used for areal features when space does not permit labeling within that feature.

Type for features on land shall be positioned on land if possible. If type must be placed in the water, it shall be positioned so it does not obscure or overprint hydrographic detail. Type for features in the water shall be positioned in the water. If possible, type shall not be positioned across the shoreline.

When obstructions coalesce at map scale, label with highest obstruction information.

Omit feature < 46 meters in Built-up Area, unless (Landmark)

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Water Tower (AM080) *(Continued)*

Point

Feature Information					
Specification:	*LTDS*	**Feature Origin:**	*Topographic*	**Symbol:**	PT_Cyan_WaterTower
Symbol Criteria:	Height Above Surface Level < 46			**Label:**	None
				Color:	N/A

Feature Information					
Specification:	*LTDS*	**Feature Origin:**	*Topographic*	**Symbol:**	PT_Blue072_VO
Symbol Criteria:	Height Above Surface Level >= 46			**Label:**	"Water tower", 6 point Condensed Mixed case; [Highest Elevation], 6 point Condensed Bold Mixed case; "([Height Above Surface Level])", 6 point Condensed Mixed case
				Color:	Blue (46351)

Water Treatment Bed (BH040)

Surface

Feature Description:	*A tract for the treatment of water that consists of a bed of material where water is aerated or filtered.*
Inclusion Condition:	*Area >= 625 sq m. OR Navigation Landmark = [True]*

Notes

Area and line features will be generalized to detail compatible with scale.

Label feature as per hierarchy for topo type placement to south neatline corners reading left to right:
1. Position hierarchy: (a.) northeast (preferred position), (b.) southeast (1st alternative), (c.) northwest (2nd alternative, (d.) southwest (3rd alternative), (e.) top-centered (4th alternative), (f.) bottom-centered (5th alternative): (Hierarchy is based on type positioning so as to avoid overprinting other type or obscuring detail.);
2. Minimum space between type placement and feature symbol is 0.5 mm;
3. This method of type placement shall be used for areal features when space does not permit labeling within that feature.

Names placement shall be oriented to the longest axis of the feature reading left to right and placed within the area outline and centered. If the longest axis is perpendicular to the south neatline, the type shall be placed outside of the area outline, preferred position is northeast of the feature, but may be placed at any position around the feature so as not to overprint any other feature type and reading left to right.

Feature Information					
Specification:	*LTDS*	**Feature Origin:**	*Topographic*	**Symbol:**	AR_Cyan_WaterTreatmentBed
Symbol Criteria:	All			**Label:**	"Filtration bed", 6 point Condensed Italics Mixed case
				Color:	Blue (46351)

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Water Turbulence (BG012)

Surface

Feature Description:	*A disturbance of water caused by the interaction of any combination of waves, currents, eddies, tidal streams, wind, shoals, and obstructions.*
Inclusion Condition:	*Area >= 250 Sq m.*

Notes

Point features, or individual posicuts of an area symbol, may be displaced <= 5 mm, at chart scale, to avoid overprints.

When the boundary of an areal symbol crosses or overprints the shoreline, the boundary overprinting the shoreline or land is deleted.

When two or more identical symbols overprint each other, delete the portions that overlap and show as one symbol with an outline of the combined area of the symbols.

Features with areas greater than 100 square cm shall be represented by a legend, e.g., Kelp, Eddies, Overfalls, rather than by the graphic symbol. Type is 6 point, Black SPC-58600, and label is scattered over area at approximately 50 mm intervals. Position is horizontal and shall not overprint other features.

Feature Information

Specification:	*Chart #1*	**Feature Origin:**	*Hydrographic*	**Symbol:**	K17
Symbol Criteria:	Water Turbulence Type = [Breakers] or [Plunging Breakers] or [Spilling Breakers] or [Surging Breakers]			**Label:**	None
				Color:	N/A

Feature Information

Specification:	*Chart #1*	**Feature Origin:**	*Hydrographic*	**Symbol:**	H44
Symbol Criteria:	Water Turbulence Type = [Overfalls] or [Tide Rips] or [Confused Seas]			**Label:**	None
				Color:	N/A

Feature Information

Specification:	*Chart #1*	**Feature Origin:**	*Hydrographic*	**Symbol:**	H45
Symbol Criteria:	Water Turbulence Type = [Eddies] or [Whirlpool]			**Label:**	None
				Color:	N/A

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Water Turbulence (BG012)

Point

Inclusion Condition:	*Area < 250 Sq m.*

Notes

Point features, or individual posicuts of an area symbol, may be displaced <= 5 mm, at chart scale, to avoid overprints.

Use the following abbreviations for Accuracy and Existence values:
If Accuracy value = Position Approximate, label "PA";
If Accuracy value = Position Doubtful, label "PD";
If Existence value = Existence Doubtful, label "ED";
If Existence value = Reported, label "Rep".

Date shall only be shown if Existence = Reported. Date shall be shown in parentheses. If date is unknown, delete label and parentheses from the symbol. Always show "Rep" and date on the same line.

Priority for type placement: 1-right center, 2-bottom center, 3-left center, 4-top center.
(A) Minimum distance from symbol - 1 mm.
(B) Maximum distance from symbol before choosing the next highest priority:
(#1) 4 mm measured to the West end;
(#2) 4 mm measured to the North side (top);
(#3) 4 mm measured to the East end;
(#4) 4 mm measured to the South side (bottom)

Labels may be stacked, readable top to bottom to avoid overprints.

Labels shall be placed in water area, and shall not overprint the shoreline or be placed on land or in the foreshore.

Unless specifically indicated on source material as inaccurate or doubtful, etc., or modified by other override rules, features shall be assumed to be accurate and definite. This rule applied to accuracy and existence of unique features, and is not related to the accuracy of the source as a whole.

Following IHO practice, features whose positions are approximate are labeled "PA" on the source charts. Features whose positions are doubtful are labeled "PD" on the source charts. Features whose existence is doubtful are labeled "ED" on the source charts. Features that have only been reported are labeled "Rep" on the source charts, and usually show the date of report in parentheses.

Definitions:
PA - Position Approximate = The position has not been accurately determined, or does not remain fixed.;
PD - Position Doubtful = The feature has been reported in various positions, but not definitely determined to be in any.;
ED - Existence Doubtful = Indicates the possible existence of the feature, the actual existence of which has not been established.

If two or more point breakers fall within 15 mm of each other, show one symbol in the center of the group.

Feature Information					
Specification:	*Chart #1*	**Feature Origin:**	*Hydrographic*	**Symbol:**	K17
Symbol Criteria:	Water Turbulence Type = [Breakers] or [Plunging Breakers] or [Spilling Breakers] or [Surging Breakers]			**Label:**	"Br", 6 point Mixed case, Italic
				Color:	N/A

Feature Information					
Specification:	*Chart #1*	**Feature Origin:**	*Hydrographic*	**Symbol:**	H44
Symbol Criteria:	Water Turbulence Type = [Overfalls] or [Tide Rips] or [Confused Seas]			**Label:**	None
				Color:	N/A

Feature Information					
Specification:	*Chart #1*	**Feature Origin:**	*Hydrographic*	**Symbol:**	H45
Symbol Criteria:	Water Turbulence Type = [Eddies] or [Whirlpool]			**Label:**	None
				Color:	N/A

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Water Well (BH230)

Point

Feature Description:	*A shaft sunk into the ground to reach and tap a supply of water.*
Inclusion Condition:	*(Condition of Facility = (Fully Functional) or (No Information)) OR (Navigation Landmark = ([True])*

Notes

None

Feature Information

Specification:	*LTDS*	**Feature Origin:**	*Topographic*	**Symbol:**	PT_Cyan_Well_Water Intermittent
Symbol Criteria:	Water Resource Information : Hydrologic Persistence = [Intermittent] or [Dry]			**Label:**	[Geographic Name Information : Full Name] if available, 6 point Condensed Italics Mixed case, and if Water Resource Information : Water Type = [Mineral] or [Salty] or [Brackish] or [Seawater] or [Saltier than Seawater] label as "Salt", 6 point Condensed Italics Mixed case; if Water Resource Information : Water Type = [Alkaline], label as "Alkaline", 6 point Condensed Italics Mixed case; and "(abandoned)", 6 point Condensed Italics Lower case if Condition of Facility = [Abandoned]
				Color:	Blue (46351)

Feature Information

Specification:	*LTDS*	**Feature Origin:**	*Topographic*	**Symbol:**	PT_Cyan_Well_WaterPerennial
Symbol Criteria:	Water Resource Information : Hydrologic Persistence = [No information] or [Perennial]			**Label:**	[Geographic Name Information : Full Name] if available, 6 point Condensed Italics Mixed case, and if Water Resource Information : Water Type = [Mineral] or [Salty] or [Brackish] or [Seawater] or [Saltier than Seawater] label as "Salt", 6 point Condensed Italics Mixed case; if Water Resource Information : Water Type = [Alkaline], label as "Alkaline", 6 point Condensed Italics Mixed case; and "(abandoned)", 6 point Condensed Italics Lower case if Condition of Facility = [Abandoned]
				Color:	Blue (46351)

Waterbody Area (FC046)

Surface

Feature Description:	*A geographically defined part of a waterbody (for example: the sea or other navigable waters).*
Inclusion Condition:	*All Required*

Notes

None

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Waterbody Area (FC046) *(Continued)*

Surface

Feature Information					
Specification:	*Other*	**Feature Origin:**	*Other*	**Symbol:**	No Chart #1 symbology
Symbol Criteria:	All			**Label:**	[Geographic Name Information : Full Name] if available, 6 point Condensed Italics Mixed case,
				Color:	Blue (46351)

Waterbody Bar (BA070)

Surface

Feature Description:	*A shallow ridge or mound of unconsolidated mineral material that is located in a stream channel, at the mouth of a stream, in an estuary, at a lagoon, in the wave-break zone along a coast or at the sea bottom.*
Inclusion Condition:	*TBD*

Notes

None

Feature Information					
Specification:	*Other*	**Feature Origin:**	*Other*	**Symbol:**	TBD
Symbol Criteria:	TBD			**Label:**	TBD
				Color:	TBD

Waterbody Morphology Area (FC047)

Surface

Feature Description:	*A region of a waterbody, either surface or subsurface, that is homogeneous with respect to form.*
Inclusion Condition:	*All Required*

Notes

None

Feature Information					
Specification:	*Other*	**Feature Origin:**	*Other*	**Symbol:**	No Chart #1 symbology
Symbol Criteria:	All			**Label:**	[Geographic Name Information : Full Name] if available, 6 point Condensed Italics Mixed case,
				Color:	Blue (46351)

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Waterfall (BH180)

Curve

Feature Description:	*A vertically descending part of a watercourse where it falls from a height (for example: over a rock or a precipice).*
Inclusion Condition:	*Length or Diameter >= 25 m.*

Notes

Area and line features will be generalized to detail compatible with scale.

Feature will be generalized to provide a more aesthetic contoured feature (i.e., smoothed).

Label feature as per hierarchy for topo type placement to south neatline corners reading left to right:
1. Position hierarchy: (a.) northeast (preferred position), (b.) southeast (1st alternative), (c.) northwest (2nd alternative, (d.) southwest (3rd alternative), (e.) top-centered (4th alternative), (f.) bottom-centered (5th alternative): (Hierarchy is based on type positioning so as to avoid overprinting other type or obscuring detail.);
2. Minimum space between type placement and feature symbol is 0.5 mm; 3. This method of type placement shall be used for areal features when space does not permit labeling within that feature.

Descriptive terms, e.g., "Canal" shall be shown if the name is not known. If the descriptive word appears in the name, for example, "PANAMA CANAL". The descriptive type shall not be shown, i.e. do not show "Panama Canal Canal".

Omit if not shown in conjunction with drainage feature.

If a feature is not associated with (touching) a river, omit the feature.

Feature Information

Specification:	*LTDS*	**Feature Origin:**	*Topographic*	**Symbol:**	LN_Cyan_Waterfall
Symbol Criteria:	All			**Label:**	[Geographic Name Information : Full Name] if available, 6 point Condensed Italics Mixed case and "Falls", 6 point Condensed Italics Mixed case
				Color:	Blue (48253)

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Waterfall (BH180)

Point

Inclusion Condition:	*Length or Diameter < 25 m.*

Notes

A cluster of 3 or more coalescing similar point features will be aggregated when an area delineation is supported by the product.

Like point features which coalesce in clusters of 3 or more will be thinned to form a representative pattern.

Label feature as per hierarchy for topo type placement to south neatline corners reading left to right:
1. Position hierarchy: (a.) northeast (preferred position), (b.) southeast (1st alternative), (c.) northwest (2nd alternative, (d.) southwest (3rd alternative), (e.) top-centered (4th alternative), (f.) bottom-centered (5th alternative): (Hierarchy is based on type positioning so as to avoid overprinting other type or obscuring detail.);
2. Minimum space between type placement and feature symbol is 0.5 mm; 3. This method of type placement shall be used for areal features when space does not permit labeling within that feature.

Descriptive terms, e.g., "Canal" shall be shown if the name is not known. If the descriptive word appears in the name, for example, "PANAMA CANAL". The descriptive type shall not be shown, i.e. do not show "Panama Canal Canal".

Omit if not shown in conjunction with drainage feature.

If a feature is not associated with (touching) a river, omit the feature.

Feature Information

Specification:	*LTDS*	Feature Origin:	*Topographic*	Symbol:	PT_Cyan_Waterfall
Symbol Criteria:	All			Label:	[Geographic Name Information : Full Name] if available, 6 point Condensed Italics Mixed case and "Falls", 6 point Condensed Italics Mixed case
				Color:	Blue (48253)

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Waterwork (BH220)

Surface

Feature Description:	*An establishment for storing, purifying, and supplying an area or town with water.*
Inclusion Condition:	*Area >= 15,625 sq m.*

Notes

None

Feature Information

Specification:	*LTDS*	**Feature Origin:**	*Topographic*	**Symbol:**	AR_Black_Waterwork
Symbol Criteria:	All			**Label:**	[Geographic Name Information : Full Name] if available, 6 point Condensed Mixed case; and "Waterworks", and "(destroyed)" if Condition of Facility = [Destroyed]
				Color:	Black (58600)

Waterwork (BH220)

Point

Inclusion Condition:	*Area < 15,625 sq m.*

Notes

None

Feature Information

Specification:	*LTDS*	**Feature Origin:**	*Topographic*	**Symbol:**	PT_Black_Waterwork
Symbol Criteria:	All			**Label:**	[Geographic Name Information : Full Name] if available, 6 point Condensed Mixed case; and "Waterworks", and "(destroyed)" if Condition of Facility = [Destroyed]
				Color:	Black (58600)

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Weigh Station (AQ180)

Surface

Feature Description:	*A building and associated equipment that is used to inspect and weigh motor vehicles.*
Inclusion Condition:	*Length or Diameter >= 25 m.*

Notes

None

Feature Information					
Specification:	*LTDS*	**Feature Origin:**	*Topographic*	**Symbol:**	AR_Black_Weighstation
Symbol Criteria:	All			**Label:**	[Geographic Name Information : Full Name], 6 point Condensed Mixed case if available and "Weigh station", 6 point Condensed Mixed case
				Color:	Black (58600)

Weigh Station (AQ180)

Point

Inclusion Condition:	*Length or Diameter < 25 m.*

Notes

None

Feature Information					
Specification:	*LTDS*	**Feature Origin:**	*Topographic*	**Symbol:**	PT_Black_Weighstation
Symbol Criteria:	All			**Label:**	[Geographic Name Information : Full Name], 6 point Condensed Mixed case if available and "Weigh station", 6 point Condensed Mixed case
				Color:	Black (58600)

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Well (AA050)

Point

Feature Description:	*An excavation drilled or dug into the ground (for example: the sea bed) for the extraction of liquids and/or gases.*
Inclusion Condition:	*All Required*

Notes

Use the following abbreviations for Horizontal Accuracy and Existence values:
If Position Approximate, label "PA";
If Position Doubtful, label "PD";
If Existence Doubtful, label "ED";
If Reported, label "Rep".

Date shall only be shown if Existence = Reported. Date shall be shown in parentheses if available. Always show "Rep" and date on the same line.
Unless specifically indicated on source material as inaccurate or doubtful, etc., or modified by other override rules, features shall be assumed to be accurate and definite. This rule applied to accuracy and existence of unique features, and is not related to the accuracy of the source as a whole.
Following IHO practice, features whose positions are approximate are labeled "PA" on the source charts. Features whose positions are doubtful are labeled "PD" on the source charts. Features whose existence is doubtful are labeled "ED" on the source charts. Features that have only been reported are labeled "Rep" on the source charts, and usually show the date of report in parentheses.

Definitions:
PA - Position Approximate = The position has not been accurately determined, or does not remain fixed.;
PD - Position Doubtful = The feature has been reported in various positions, but not definitely determined to be in any.;
ED - Existence Doubtful = Indicates the possible existence of the feature, the actual existence of which has not been established.

Date shall only be shown if Existence = Reported. Date shall be shown in parentheses if available. Always show "Rep" and date on the same line.

Priority for type placement Rules:
1-right center,
2-bottom center,
3-left center,
4-top center.
(A) Minimum distance from symbol - 1 mm.
(B) Maximum distance from symbol before choosing the next highest priority:
(#1) 4 mm measured to the West end;
(#2) 4 mm measured to the North side (top);
(#3) 4 mm measured to the East end;
(#4) 4 mm measured to the South side (bottom)

If symbols overprint each other, labels are condensed as follows:
(1.) If the labels are identical, only one is retained.;
(2,) If the labels are not identical, they shall be condensed in to one legend, e.g., "Fishhaven and Well". If multiple depths are shown, only the shallowest is retained.

Labels may be stacked, readable top to bottom to avoid overprints.

Type Preference Placement Rules:
(a.) Placed on one horizontal line centered in feature.;
(b.) Broken into multiple horizontal lines, centered in feature, with a 1 mm space between lines.;
(c.) Placed on one line outside feature. Use Priority for type placement Rules for placement, Hydrographic Depth shall always be positioned inside area.;
(d.) Placed on two lines to avoid overprints. Use Priority for type placement Rules. If present, Hydrographic Depth shall always be placed inside area.

Labels shall be placed in water area, and shall not overprint the shoreline or be placed on land or in the foreshore.

When two or more identical features fall within 10 mm of each other, one legend with plural description shall be shown, with type placed as close as possible to the center of the group without overprinting features.

Where symbols with the same feature code but different values of Hydrographic Vertical Positioning Information: Water Level Effect (i.e. Always Dry , Always Submerged, Covers and Uncovers)) are adjacent to each other, the boundary of the feature that is the highest in vertical plane shall be shown, and the boundary of the feature above water is shown whole, while the feature that covers and uncovers has a portion of its symbol perimeter that overprints the higher feature deleted.

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Well (AA050) *(Continued)*

Point

Notes

When two or more identical symbols overprint each other, delete the portions that overlap and show as one symbol with an outline of the combined area of the symbols.

Show land tint inside the symbol. If the Hydrographic Vertical Positioning Information: Water Level Effect: [Always Dry] show land tint only.

A blue 31% tint shall not overprint other blue 31% tints. If two blue tinted symbols, or one tinted symbol and water tint overprint, only one 31% tint shall be shown in the area.

Unless specifically indicated on source material as inaccurate or doubtful, etc., or modified by other override rules, features shall be assumed to be accurate and definite. This rule applied to accuracy and existence of unique features, and is not related to the accuracy of the source as a whole.

When the boundary of an areal symbol crosses or overprints the shoreline, the boundary overprinting the shoreline or land is deleted.

If areas where separate dangers coalesce because of scale, the danger curves (dotted lines) are deleted except for those on the outermost perimeter of the aggregate danger hazard. A smooth dotted line is drawn around the area.

If one symbol coalesces with another symbol for the same type feature, displace symbols to allow a minimum separation of 0.21mm.

Type for features on land shall be positioned on land if possible. If type must be placed in the water, it shall be positioned so it does not obscure or overprint hydrographic detail. Type for features in the water shall be positioned in the water. If possible, type shall not be positioned across the shoreline.

Descriptive terms, e.g., "Canal" shall be shown if the name is not known. If the descriptive word appears in the name, for example, "PANAMA CANAL". The descriptive type shall not be shown, i.e. do not show "Panama Canal Canal".

If Well symbol AA050 coalesce, and there is less than four individual sells, show on symbol and label "Wells". If there are four or more, and the area is >= 2.5 mm x 2.5 mm, show a representative pattern and label "Numerous wells".

If point symbol overprints shoreline, the Hydrographic Depth type, posicut, or graphic in the center of the symbol shall be displaced seaward until they no longer overprint the shoreline. If the danger curve (dotted perimeter line) overprints the shoreline, that portion of the dotted perimeter line falling on land shall be deleted.

Hydrographic Depth label shall be centered in the circle.

Variable type size for Hydrographic Depth values enclosed by danger curves (dotted circles): If Hydrographic Depth < 10, (a single digit principle digit), apply 7 point type to the principle digit, and 5 point type to the subscript, if there is one. If Hydrographic Depth >= 10 (a double digit principle digit), apply 6 point type to the principle digit, and 5 point type to the subscript, if there is one.

The height of a hydrographic feature is that above the surface of the water at High Water. Hydrographic Depth/Height information is not applicable at High Water. Hydrographic Depth/Height information values (Depth Known), (Depth Known by Wire Drag), and (Depth Unknown) are used to describe knowledge about a depth of a feature when the feature is below the water surface/ submerged. Hydrographic Depth/Height information values (Uncovering Height Known) and (Uncovering Height Unknown) are used to describe knowledge about the drying height of a feature when it covers and uncovers, i.e., between high and low water.
Hydrographic depth is used to record the depth of a feature when Hydrographic Depth/Height information values are (Depth Known) and (Depth Known by Wire Drag). Hydrographic depth is not applicable when the depth is unknown or when the feature covers and uncovers.
Hydrographic Drying Height records the drying height of a feature when Hydrographic Depth/Height information values are known. Hydrographic Drying Height is not applicable when the drying height is unknown, or the feature is below water.

Vertical Datum category is the sounding datum to which a hydrographic depth, drying height, or the high water datum to which a Safe Overhead Clearance is referenced. If the hydrographic depth, drying height, or Safe Overhead Clearance is unknown or not applicable, the vertical datum category is not applicable.

If areas where separate dangers coalesce because of scale, the danger curves (dotted lines) are deleted except for those on the outermost perimeter of the aggregate danger hazard. A smooth dotted line is drawn around the area.

Dangers with danger curves (dotted lines) around them which fall inside other danger areas with danger curves around them, shall have the inner dotted line deleted.

The diameter of the circular symbol perimeter line may be reduced in congested area to avoid overprinting other symbols, but must be at least 2.0 mm diameter, and cannot come closer than 0.2 mm to any type or graphic symbol shown inside the circle. In non-congested areas the normal symbol diameter shall be shown.

When two or more point obstruction symbols, (dotted lines) overprint, and the attribution values are identical, one symbol shall be placed in the center of the group and shall be labeled with the number of obstructions in the group, e.g., 2 Obstr's, etc. Type is 6 point italic, in color Black SPC-58600.

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Well (AA050) *(Continued)*

Point

Feature Information					
Specification:	*Chart #1*	**Feature Origin:**	*Hydrographic*	**Symbol:**	L21.1
Symbol Criteria:	(Conspicuous Sea Category = [Sonar]) and (Depth Below Surface Level <= 0)			**Label:**	"Well",7 point Mixed case, Italic
				Color:	Black (58600)

Feature Information					
Specification:	*Chart #1*	**Feature Origin:**	*Hydrographic*	**Symbol:**	L21.2
Symbol Criteria:	(Conspicuous Sea Category = [Sonar]) and (Depth Below Surface Level > 0)			**Label:**	"Well", 7 point Mixed case, Italic; [Hydrographic Depth], 6 point lower case/ 5 point subscript, Italic
				Color:	Black (58600)

Feature Information					
Specification:	*Chart #1*	**Feature Origin:**	*Hydrographic*	**Symbol:**	L23
Symbol Criteria:	(Conspicuous Sea Category = [Visual] or [Visual and Sonar])			**Label:**	"Pipe", 7 point Mixed case, Italic; "([Height Above Surface Level])" if ([Height Above Surface Level] > 0), 6 point lower case/ 5 point subscript, Italic
				Color:	Black (58600)

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Wind Farm (AD060)

Surface

Feature Description:	*A collection of windmotors that are collocated and are organized as a single power generation unit.*
Inclusion Condition:	*Area >= 15,625 sq m.*

Notes

None

Feature Information

Specification:	*LTDS*	**Feature Origin:**	*Topographic*	**Symbol:**	AR_Black_WindFarm
Symbol Criteria:	(Area >= 15,625) and (Location inland) and (not restricted for navigation)			**Label:**	[Geographic Name Information : Full Name] if available: 6 point Condensed Mixed case and "Wind farm", 6 point Condensed Mixed case
				Color:	Black (58600)

Feature Information

Specification:	*LTDS*	**Feature Origin:**	*Topographic*	**Symbol:**	L5.2
Symbol Criteria:	(Located Offshore) and (restricted for navigation)			**Label:**	[Geographic Name Information : Full Name] if available: 6 point Condensed Mixed case and "Wind farm", 6 point Condensed Mixed case
				Color:	Purple (96532)

Wind Farm (AD060)

Point

Inclusion Condition:	*Area < 15,625 sq m. AND Navigation Landmark = [True]*

Notes

None

Feature Information

Specification:	*LTDS*	**Feature Origin:**	*Topographic*	**Symbol:**	PT_Black_WindFarm
Symbol Criteria:	Area: < 15,625 AND Navigation Landmark = [True]			**Label:**	None
				Color:	N/A

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Wind Turbine (AJ051)

Point

Feature Description:	*A tower and associated equipment that generates electrical power from wind.*
Inclusion Condition:	*All Required*

Notes

None

Feature Information					
Specification:	*LTDS*	**Feature Origin:**	*Topographic*	**Symbol:**	PT_Black_Windturbine_ OnShore
Symbol Criteria:	Location Referenced to Shoreline <> [Offshore] and Height Above Surface Level < 46			**Label:**	None
				Color:	N/A

Feature Information					
Specification:	*LTDS*	**Feature Origin:**	*Topographic*	**Symbol:**	PT_Blue072_VO
Symbol Criteria:	Location Referenced to Shoreline <> [Offshore] and Height Above Surface Level >= 46			**Label:**	"Wind turbine", 6 point Condensed Mixed; [Highest Elevation], 6 point Condensed Mixed; "([Height Above Surface Level])", 6 Point Bold Condensed Mixed case
				Color:	Blue (46351)

Feature Information					
Specification:	*Chart #1*	**Feature Origin:**	*Hydrographic*	**Symbol:**	L5.1
Symbol Criteria:	Location Referenced to Shoreline = [Offshore]			**Label:**	"([Height Above Surface Level])" if available, 7 Point Upper case
				Color:	Black (58600)

Windmill (AJ050)

Point

Feature Description:	*A system of vanes attached to a tower and driven by wind (excluding wind turbines).*
Inclusion Condition:	*Height Above Surface Level >= 46 m. OR Navigation Landmark = [True]*

Notes

If one symbol coalesces with another symbol for the same type feature, displace symbols to allow a minimum separation of 0.21mm.

Type for features on land shall be positioned on land if possible. If type must be placed in the water, it shall be positioned so it does not obscure or overprint hydrographic detail. Type for features in the water shall be positioned in the water. If possible, type shall not be positioned across the shoreline.

When obstructions coalesce at map scale, label with highest obstruction information.

If >= 3 equal symbols would coalesce at map scale, portray with a representative pattern.

Feature Information					
Specification:	*LTDS*	**Feature Origin:**	*Topographic*	**Symbol:**	PT_Black_Windmill
Symbol Criteria:	Height Above Surface Level < 46			**Label:**	None
				Color:	N/A

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Windmill (AJ050) *(Continued)*

Point

Feature Information					
Specification:	*LTDS*	**Feature Origin:**	*Topographic*	**Symbol:**	PT_Blue072_VO
Symbol Criteria:	Height Above Surface Level >= 46			**Label:**	"Windmill", 6 point Condensed Mixed case; Highest Elevation: 6 point Condensed Bold Mixed case; Height Above Surface Level: 6 point Condensed Mixed case
				Color:	Blue (46351)

Wreck (BD180)

Surface

Feature Description:	*The ruined remains of a stranded or sunken vessel that has been rendered useless.*
Inclusion Condition:	*Length >= 200 meters*

Notes

Show Hydrographic Depth or Hydrographic Height values in whole meters if decimeter value is 0; otherwise show the decimeter value as a subscript to the meter value.

Priority for type placement: 1-right center, 2-bottom center, 3-left center, 4-top center.
(A) Minimum distance from symbol - 1 mm.
(B) Maximum distance from symbol before choosing the next highest priority:
(#1) 4 mm measured to the West end;
(#2) 4 mm measured to the North side (top);
(#3) 4 mm measured to the East end;
(#4) 4 mm measured to the South side (bottom)

If symbols overprint each other, labels are condensed as follows:
(1.) If the labels are identical, only one is retained.;
(2,) If the labels are not identical, they shall be condensed in to one legend, e.g., "Fishhaven and Well".
If multiple depths are shown, only the shallowest is retained.

Labels may be stacked, readable top to bottom to avoid overprints

Labels shall be placed in water area, and shall not overprint the shoreline or be placed on land or in the foreshore.

Label placement for area wreck symbols. The preferred position for type is for the "Wk" to be centered in the area symbol and reading horizontally. If depth is shown, it shall be centered in area reading horizontally, and the "Wk" shall be positioned to the right, if possible; otherwise placed inside the wreck where it will fit, or outside to the right of the wreck.

If the wreck at chart scale is too small to place type inside, the legend "Wk" shall be placed outside the wreck in the following priority:
(a.) To the right or below wreck, centered along major axis.
(b.) To the left or above the wreck, centered along major axis.
(c.) Any other available location.

Type is horizontal, and if a depth is shown, it is in parentheses to the right of the "Wk" label.

The height of a hydrographic feature is that above the surface of the water at High Water. Hydrographic Depth/Height information is not applicable at High Water. Hydrographic Depth/Height information values (Depth Known), (Depth Known by Wire Drag), and (Depth Unknown) are used to describe knowledge about a depth of a feature when the feature is below the water surface/ submerged. Hydrographic Depth/Height information values (Uncovering Height Known) and (Uncovering Height Unknown) are used to describe knowledge about the drying height of a feature when it covers and uncovers, i.e., between high and low water.
Hydrographic depth is used to record the depth of a feature when Hydrographic Depth/Height information values are (Depth Known) and (Depth Known by Wire Drag). Hydrographic depth is not applicable when the depth is unknown or when the feature covers and uncovers.
Hydrographic Drying Height records the drying height of a feature when Hydrographic Depth/Height information values are known. Hydrographic Drying Height is not applicable when the drying height is unknown, or the feature is below water.

Vertical Datum category is the sounding datum to which a hydrographic depth, drying height, or the high water datum to which a Safe Overhead Clearance is referenced. If the hydrographic depth, drying height, or Safe Overhead Clearance is unknown or not applicable, the vertical datum category is not applicable.

Where symbols with the same feature code but different values of vertical reference categories (i.e. cover/uncovers, submerged, above surface (at high water)) are adjacent to each other, the boundary of the feature that is the highest in vertical plane shall be shown, and the boundary of the feature above water is shown whole, while the feature that covers and uncovers has a portion of its symbol perimeter that overprints the higher feature deleted.

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Wreck (BD180) *(Continued)*

Surface

Feature Information					
Specification:	*Chart #1*	**Feature Origin:**	*Hydrographic*	**Symbol:**	K21
Symbol Criteria:	Water Level Effect = [Covers and Uncovers] or [Partly Submerged] or [Awash at Low Water]			**Label:**	"Wk", 6 point Mixed case, Italic; "Mast", 6 point Mixed case, Italic if Wreck or Hulk Exposure = [Masts and Funnel Showing] or [Masts Showing]; [Hydrographic Drying Height] if exists: 7 point lower case/ 5 point subscript;
				Color:	Black (58600)

Feature Information					
Specification:	*Chart #1*	**Feature Origin:**	*Hydrographic*	**Symbol:**	K20
Symbol Criteria:	Water Level Effect = [Always Dry]			**Label:**	"Wk", 6 point Mixed case, Italic; "Mast" if Wreck or Hulk Exposure = [Masts and Funnel Showing] or [Masts Showing]
				Color:	Black (58600)

Feature Information					
Specification:	*Chart #1*	**Feature Origin:**	*Hydrographic*	**Symbol:**	K22
Symbol Criteria:	Water Level Effect = [Always Submerged] and Bathymetric Measurement Quality Category = [Depth Known]			**Label:**	"Wk", 6 point Mixed case, Italic; [Hydrographic Depth] if exists: 7 point lower case/ 5 point subscript;
				Color:	Black (58600)

Feature Information					
Specification:	*Chart #1*	**Feature Origin:**	*Hydrographic*	**Symbol:**	K23
Symbol Criteria:	Water Level Effect = [Always Submerged] and Bathymetric Measurement Quality Category = [Depth Unknown]			**Label:**	"Wk", 6 point Mixed case, Italic; [Hydrographic Depth] if exists: 7 point lower case/ 5 point subscript;
				Color:	Black (58600)

Wreck (BD180)

Point

Inclusion Condition:	*Length < 200 meters*

Notes

Use the following abbreviations for Horizontal Accuracy and Existence values:
If Position Approximate, label "PA";
If Position Doubtful, label "PD";
If Existence Doubtful, label "ED";
If Reported, label "Rep".

Date shall only be shown if Existence = Reported. Date shall be shown in parentheses if available. Always show "Rep" and date on the same line.
Unless specifically indicated on source material as inaccurate or doubtful, etc., or modified by other override rules, features shall be assumed to be accurate and definite. This rule applied to accuracy and existence of unique features, and is not related to the accuracy of the source as a whole.
Following IHO practice, features whose positions are approximate are labeled "PA" on the source charts. Features whose positions are doubtful are labeled "PD" on the source charts. Features whose existence is doubtful are labeled "ED" on the source charts. Features that have only been reported are labeled "Rep" on the source charts, and usually show the date of report in parentheses.

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Wreck (BD180) ***(Continued)***

Point

Notes

Definitions:
PA - Position Approximate = The position has not been accurately determined, or does not remain fixed.;
PD - Position Doubtful = The feature has been reported in various positions, but not definitely determined to be in any.;
ED - Existence Doubtful = Indicates the possible existence of the feature, the actual existence of which has not been established.

Date shall only be shown if Existence = Reported. Date shall be shown in parentheses if available. Always show "Rep" and date on the same line.

Priority for type placement Rules:
1-right center,
2-bottom center,
3-left center,
4-top center.
(A) Minimum distance from symbol - 1 mm.
(B) Maximum distance from symbol before choosing the next highest priority:
(#1) 4 mm measured to the West end;
(#2) 4 mm measured to the North side (top);
(#3) 4 mm measured to the East end;
(#4) 4 mm measured to the South side (bottom)

If symbols overprint each other, labels are condensed as follows:
(1.) If the labels are identical, only one is retained.;
(2,) If the labels are not identical, they shall be condensed in to one legend, e.g., "Fishhaven and Well". If multiple depths are shown, only the shallowest is retained.

Labels may be stacked, readable top to bottom to avoid overprints.

Type Preference Placement Rules:
(a.) Placed on one horizontal line centered in feature.;
(b.) Broken into multiple horizontal lines, centered in feature, with a 1 mm space between lines.;
(c.) Placed on one line outside feature. Use Priority for type placement Rules for placement, Hydrographic Depth shall always be positioned inside area.;
(d.) Placed on two lines to avoid overprints. Use Priority for type placement Rules. If present, Hydrographic Depth shall always be placed inside area.

Labels shall be placed in water area, and shall not overprint the shoreline or be placed on land or in the foreshore.

When two or more identical features fall within 10 mm of each other, one legend with plural description shall be shown, with type placed as close as possible to the center of the group without overprinting features.

Where symbols with the same feature code but different values of Hydrographic Vertical Positioning Information: Water Level Effect (i.e. Always Dry , Always Submerged, Covers and Uncovers)) are adjacent to each other, the boundary of the feature that is the highest in vertical plane shall be shown, and the boundary of the feature above water is shown whole, while the feature that covers and uncovers has a portion of its symbol perimeter that overprints the higher feature deleted.

When two or more identical symbols overprint each other, delete the portions that overlap and show as one symbol with an outline of the combined area of the symbols.

Show land tint inside the symbol. If the Hydrographic Vertical Positioning Information: Water Level Effect: [Always Dry] show land tint only.

A blue 31% tint shall not overprint other blue 31% tints. If two blue tinted symbols, or one tinted symbol and water tint overprint, only one 31% tint shall be shown in the area.

Unless specifically indicated on source material as inaccurate or doubtful, etc., or modified by other override rules, features shall be assumed to be accurate and definite. This rule applied to accuracy and existence of unique features, and is not related to the accuracy of the source as a whole.

When the boundary of an areal symbol crosses or overprints the shoreline, the boundary overprinting the shoreline or land is deleted.

If areas where separate dangers coalesce because of scale, the danger curves (dotted lines) are deleted except for those on the outermost perimeter of the aggregate danger hazard. A smooth dotted line is drawn around the area.

If one symbol coalesces with another symbol for the same type feature, displace symbols to allow a minimum separation of 0.21mm.

Type for features on land shall be positioned on land if possible. If type must be placed in the water, it shall be positioned so it does not obscure or overprint hydrographic detail. Type for features in the water shall be positioned in the water. If possible, type shall not be positioned across the shoreline.

Descriptive terms, e.g., "Canal" shall be shown if the name is not known. If the descriptive word appears in the name, for example, "PANAMA CANAL". The descriptive type shall not be shown, i.e. do not show "Panama Canal Canal".

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Wreck (BD180) *(Continued)*

Point

Notes

If point symbol overprints shoreline, the Hydrographic Depth type, posicut, or graphic in the center of the symbol shall be displaced seaward until they no longer overprint the shoreline. If the danger curve (dotted perimeter line) overprints the shoreline, that portion of the dotted perimeter line falling on land shall be deleted.

Hydrographic Depth label shall be centered in the circle.

Variable type size for Hydrographic Depth values enclosed by danger curves (dotted circles): If Hydrographic Depth < 10, (a single digit principle digit), apply 7 point type to the principle digit, and 5 point type to the subscript, if there is one. If Hydrographic Depth >= 10 (a double digit principle digit), apply 6 point type to the principle digit, and 5 point type to the subscript, if there is one.

The height of a hydrographic feature is that above the surface of the water at High Water. Hydrographic Depth/Height information is not applicable at High Water. Hydrographic Depth/Height information values (Depth Known), (Depth Known by Wire Drag), and (Depth Unknown) are used to describe knowledge about a depth of a feature when the feature is below the water surface/ submerged. Hydrographic Depth/Height information values (Uncovering Height Known) and (Uncovering Height Unknown) are used to describe knowledge about the drying height of a feature when it covers and uncovers, i.e., between high and low water.

Hydrographic depth is used to record the depth of a feature when Hydrographic Depth/Height information values are (Depth Known) and (Depth Known by Wire Drag). Hydrographic depth is not applicable when the depth is unknown or when the feature covers and uncovers.
Hydrographic Drying Height records the drying height of a feature when Hydrographic Depth/Height information values are known. Hydrographic Drying Height is not applicable when the drying height is unknown, or the feature is below water.

Vertical Datum category is the sounding datum to which a hydrographic depth, drying height, or the high water datum to which a Safe Overhead Clearance is referenced. If the hydrographic depth, drying height, or Safe Overhead Clearance is unknown or not applicable, the vertical datum category is not applicable.

If areas where separate dangers coalesce because of scale, the danger curves (dotted lines) are deleted except for those on the outermost perimeter of the aggregate danger hazard. A smooth dotted line is drawn around the area.

Dangers with danger curves (dotted lines) around them which fall inside other danger areas with danger curves around them, shall have the inner dotted line deleted.

The diameter of the circular symbol perimeter line may be reduced in congested area to avoid overprinting other symbols, but must be at least 2.0 mm diameter, and cannot come closer than 0.2 mm to any type or graphic symbol shown inside the circle. In non-congested areas the normal symbol diameter shall be shown.

When two or more point obstruction symbols, (dotted lines) overprint, and the attribution values are identical, one symbol shall be placed in the center of the group and shall be labeled with the number of obstructions in the group, e.g., 2 Obstr's, etc. Type is 6 point italic, in color Black SPC-58600.

When two or more stranded wreck symbols (Wreck or Hulk Exposure = (any)) overlap, one symbol shall be placed in the center of the group and shall be labeled with the number of stranded wrecks in the group, e.g., "2 Wrecks", "3 Wrecks", etc. Type shall be 6 point, in color Black SPC-58600.

If more than five wrecks, other than stranded (Water Level Effect = (Always Dry) or (Covers and Uncovers)), fall within an area less than 20 mm X 20 mm, individual wreck symbols are not shown. Instead, a generalized danger line (dotted line) shall be shown surrounding the area, and the area shall be labeled as follows: "Numerous wrecks", "Numerous Wks", "Wks". Condense label as necessary to place it inside the danger line. Type is 6 point upper and lower case italic, in color SPC-58600.

Where two or more wrecks, except stranded wrecks (Water Level Effect = (Always Dry) or (Covers and Uncovers)), overprint each other: (1.) If only the danger lines (dotted lines) overprint, delete the danger lines that are inside the outer perimeter danger lines. (2.) If two Hydrographic Depths overprint, retain the shallowest depth and the danger line surrounding the wrecks.

Feature Information					
Specification:	*Chart #1*	**Feature Origin:**	*Hydrographic*	**Symbol:**	K24
Symbol Criteria:	Water Level Effect = [Always Dry] or [Partly Submerged] or {Covers and Uncovers] and Dangerous to Navigation = [True]			**Label:**	None
				Color:	N/A

Feature Information					
Specification:	*Chart #1*	**Feature Origin:**	*Hydrographic*	**Symbol:**	K25
Symbol Criteria:	(Water Level Effect = [Always Submerged]) and (Dangerous to Navigation = [True]) and (Wreck or Hulk Exposure = [Masts and Funnel Showing] or [Masts Showing])			**Label:**	"Wk", 6 point Mixed case, Italic; "Masts", 6 point Mixed case, Italic; [Hydrographic Drying Height] if exists: 7 point lower case/ 5 point subscript;
				Color:	Black (58600)

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Wreck (BD180) ***(Continued)***

Point

Feature Information					
Specification:	*Chart #1*	**Feature Origin:**	*Hydrographic*	**Symbol:**	K27
Symbol Criteria:	(Water Level Effect = [Always Submerged]) and (Bathymetric Measurement Quality Category = [Depth Known]) and (Bathymetric Measurement Technique = [Area Swept by Side-scan Sonar] or [Area Swept by Vertical Acoustic System] or [Area Swept by Wire-drag])			**Label:**	"Wk", 6 point Mixed case, Italic; [Hydrographic Depth], 7 point lower case/ 5 point subscript
				Color:	Black (58600)

Feature Information					
Specification:	*Chart #1*	**Feature Origin:**	*Hydrographic*	**Symbol:**	K26
Symbol Criteria:	(Water Level Effect = [Always Submerged]) and (Bathymetric Measurement Quality Category = [Depth Known])			**Label:**	"Wk", 6 point Mixed case, Italic; [Hydrographic Depth], 7 point lower case/ 5 point subscript
				Color:	Black (58600)

Feature Information					
Specification:	*Chart #1*	**Feature Origin:**	*Hydrographic*	**Symbol:**	K3,K30
Symbol Criteria:	(Water Level Effect = [Always Submerged]) and (Bathymetric Measurement Quality Category = [Safe Clearance Known])			**Label:**	Hydrographic Depth: 7 point lower case/ 5 point subscript; Other: 6 point Upper/lower case, Italic
				Color:	Black (58600)

Feature Information					
Specification:	*Chart #1*	**Feature Origin:**	*Hydrographic*	**Symbol:**	K29
Symbol Criteria:	(Water Level Effect = [Always Submerged]) and (Dangerous to Navigation = [False])			**Label:**	None
				Color:	N/A

Feature Information					
Specification:	*Chart #1*	**Feature Origin:**	*Hydrographic*	**Symbol:**	K28
Symbol Criteria:	(Water Level Effect = [Always Submerged]) and (Dangerous to Navigation = [True])			**Label:**	None
				Color:	N/A

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Zoo (AK180)

Surface

Feature Description:	*A site where wild animals are kept for exhibition to the public, that may also support breeding and/or study.*
Inclusion Condition:	*Area >= 15,625 square meters*

Notes

Coincident similar area features will be blended to form a single feature.

Area and line features will be generalized to detail compatible with scale.

Type for features on land shall be positioned on land if possible. If type must be placed in the water, it shall be positioned so it does not obscure or overprint hydrographic detail. Type for features in the water shall be positioned in the water. If possible, type shall not be positioned across the shoreline.

Descriptive terms, e.g., "Canal" shall be shown if the name is not known. If the descriptive word appears in the name, for example, "PANAMA CANAL". The descriptive type shall not be shown, i.e. do not show "Panama Canal Canal".

Limiting lines of feature are omitted if it coalesces with a road (AP030).

If a clearing exists inside of the area feature, and the size of the clearing is equal to or greater than the area for the surrounding area feature, the clearing is shown as an open space inside the surround feature. If the area of the clearing is less than the area for the surrounding feature, the clearing is deleted and absorbed into the surrounding area feature.

If two area features with the same feature code do not connect at any point, and have a space between them less than 2.5 mm at map/chart scale, delete the open space that is less than 2.5 mm between the features and combine them into one area feature.

If a portion of an area feature has a minimum width of less than 2.5 mm at map/chart scale, delete that portion of the area feature that is not at least 2.5 mm wide, measured from perimeter to perimeter. If the deletion of the portion of the area based on the above criteria will reduce the area of the remaining portion of the area feature to below the minimum area inclusion, do not delete the narrow portion of the feature that is less than 2.5 mm wide. If the deletion of the portion of the area based on the above criteria will split two larger areas connected by a narrow strip into two separate areas, either of which would be below minimum area inclusion, do not delete the narrow portion of the feature that is less than 2.5 mm wide.

Feature Information

Specification:	*LTDS*	**Feature Origin:**	*Topographic*	**Symbol:**	AR_Black_Zoo
Symbol Criteria:	Area >= 15,625			**Label:**	[Geographic Name Information : Full Name], 6 point Condensed Mixed case; "Zoo", 6 point Condensed Mixed case
				Color:	Black (58600)

Zoo (AK180)

Point

Inclusion Condition:	*Area < 15,625 square meters AND Navigation Landmark = [True]*

Notes

None

Feature Information

Specification:	*LTDS*	**Feature Origin:**	*Topographic*	**Symbol:**	PT_Black_Zoo
Symbol Criteria:	Area < 15,625 AND Navigation Landmark = [True]			**Label:**	[Geographic Name Information : Full Name], 6 point Condensed Mixed case; "Zoo", 6 point Condensed Mixed case
				Color:	Black (58600)

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TABLE II. Feature origin

LPC Feature Code	LPC Feature Name	Feature Source Origin	Feature Geometry	Feature Symbol
FA005	Access Zone	Other	Surface	TBD
BK010	Acoustic Station	Other	Point	TBD
FA000	Administrative Boundary	Topographic (LTDS)	Curve	LN_Dk-Brown1815_AdminBoundary_International
FA000	Administrative Boundary	Topographic (LTDS)	Curve	LN_Dk-Brown1815_AdminBoundary_Armistice
FA000	Administrative Boundary	Topographic (LTDS)	Curve	LN_Black_AdminBoundary_CeaseFire
FA000	Administrative Boundary	Topographic (LTDS)	Curve	LN_AdminBoundary_NoDefinedBoundary
FA000	Administrative Boundary	Hydrographic (Chart #1)	Curve	N41 (Chart #1)
FA000	Administrative Boundary	Hydrographic (Chart #1)	Curve	N43 (Chart #1)
FA000	Administrative Boundary	Hydrographic (Chart #1)	Curve	N43 (Chart #1)
FA000	Administrative Boundary	Hydrographic (Chart #1)	Curve	N48 (Chart #1)
FA000	Administrative Boundary	Hydrographic (Chart #1)	Curve	N44 (Chart #1)
FA000	Administrative Boundary	Hydrographic (Chart #1)	Curve	N47 (Chart #1)
FA000	Administrative Boundary	Hydrographic (Chart #1)	Curve	N46 (Chart #1)
FA000	Administrative Boundary	Hydrographic (Chart #1)	Curve	N46 (Chart #1)
FA000	Administrative Boundary	Hydrographic (Chart #1)	Curve	N46 (Chart #1)
FA000	Administrative Boundary	Topographic (LTDS)	Curve	LN_Dk-Brown1815_AdminBoundary_OtherLineOfSeparation
FA003	Administrative Division	Topographic (LTDS)	Surface	LN_Black_AdminDivision_Primary
FA003	Administrative Division	Topographic (LTDS)	Surface	LN_Black_AdminDivision_Secondary
FA003	Administrative Division	Topographic (LTDS)	Surface	LN_Black_AdminDivision_Tertiary
AT011	Aerial	Topographic (LTDS)	Point	PT_Black_Aerial
AT011	Aerial	Topographic (LTDS)	Point	PT_Blue072_VO

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TABLE II. Feature origin - Continued

LPC Feature Code	LPC Feature Name	Feature Source Origin	Feature Geometry	Feature Symbol
AT012	Aerial Farm	Topographic (LTDS)	Surface	AR_Black_AerialFarm
AT012	Aerial Farm	Topographic (LTDS)	Point	PT_Black_AerialFarm
GB013	Aerodrome Beacon	Hydrographic (Chart #1)	Point	P60 (Chart #1)
GB007	Aerodrome Boundary	Aeronautical	Curve	TBD
GA033	Aeronautical Radio Navigation Installation	Hydrographic (Chart #1)	Point	S16 (Chart #1)
GB230	Aircraft Hangar	Topographic (LTDS)	Surface	AR_Black_AircraftHangar
GB230	Aircraft Hangar	Topographic (LTDS)	Point	PT_Black_AircraftHangar
GB050	Aircraft Revetment	Topographic (LTDS)	Curve	LN_Blue072_AircraftRevetment
GB016	Aircraft Stand	Aeronautical	Point	TBD
AK164	Amphitheatre	Topographic (LTDS)	Surface	AR_Black_Amphitheatre
AK164	Amphitheatre	Topographic (LTDS)	Point	PT_Black_Amphitheatre
AK030	Amusement Park	Topographic (LTDS)	Surface	AR_Black_AmusementPark
AK030	Amusement Park	Topographic (LTDS)	Point	PT_Black_AmusementPark
AK020	Amusement Park Attraction	Topographic (LTDS)	Point	PT_Black_ParkAttraction
AK020	Amusement Park Attraction	Topographic (LTDS)	Point	PT_Blue072_VO
BB019	Anchor	Hydrographic (Chart #1)	Point	L18 (Chart #1)
BB010	Anchorage	Hydrographic (Chart #1)	Surface	N12.1 (Chart #1)
BB010	Anchorage	Hydrographic (Chart #1)	Surface	N11.2, N12.2 (Chart #1)
BB010	Anchorage	Hydrographic (Chart #1)	Surface	N12.4 (Chart #1)
BB010	Anchorage	Hydrographic (Chart #1)	Surface	N12.7 (Chart #1)
BB010	Anchorage	Hydrographic (Chart #1)	Surface	N12.8 (Chart #1)
BB010	Anchorage	Hydrographic (Chart #1)	Surface	N12.9 (Chart #1)

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TABLE II. Feature origin - Continued

LPC Feature Code	LPC Feature Name	Feature Source Origin	Feature Geometry	Feature Symbol
BB010	Anchorage	Hydrographic (Chart #1)	Surface	N12.5 (Chart #1)
BB010	Anchorage	Hydrographic (Chart #1)	Surface	N12.6 (Chart #1)
BB010	Anchorage	Hydrographic (Chart #1)	Surface	N13 (Chart #1)
BB010	Anchorage	Hydrographic (Chart #1)	Point	N12.1 (Chart #1)
BB010	Anchorage	Hydrographic (Chart #1)	Point	N12.4 (Chart #1)
BB010	Anchorage	Hydrographic (Chart #1)	Point	N12.7 (Chart #1)
BB010	Anchorage	Hydrographic (Chart #1)	Point	N12.8 (Chart #1)
BB010	Anchorage	Hydrographic (Chart #1)	Point	N12.9 (Chart #1)
BB010	Anchorage	Hydrographic (Chart #1)	Point	N12.5 (Chart #1)
BB010	Anchorage	Hydrographic (Chart #1)	Point	N12.6 (Chart #1)
BB010	Anchorage	Hydrographic (Chart #1)	Point	N13, N14 (Chart #1)
BB010	Anchorage	Hydrographic (Chart #1)	Point	N11.1 (Chart #1)
ZD045	Annotated Location	Topographic (LTDS)	Surface	AR_AnnotatedLocation_Culture
ZD045	Annotated Location	Hydrographic (Chart #1)	Surface	AR_AnnotatedLocation_Hydro
ZD045	Annotated Location	Topographic (LTDS)	Curve	LN_AnnotatedLocation_Culture
ZD045	Annotated Location	Hydrographic (Chart #1)	Curve	LN_AnnotatedLocation_Hydro
ZD045	Annotated Location	Topographic (LTDS)	Point	PT_AnnotatedLocation_Culture
ZD045	Annotated Location	Hydrographic (Chart #1)	Point	PT_AnnotatedLocation_Hydro
ZV010	Anti-shipping Activity	Hydrographic (Chart #1)	Point	TBD
GB485	Approach Lighting System	Aeronautical	Surface	TBD
GB485	Approach Lighting System	Aeronautical	Curve	TBD
GB015	Apron	Topographic (LTDS)	Surface	AR_Blue072_Apron

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TABLE II. Feature origin - Continued

LPC Feature Code	LPC Feature Name	Feature Source Origin	Feature Geometry	Feature Symbol
BD061	Aquatic Vegetation	Hydrographic (Chart #1)	Surface	J13.1, J13.2 (Chart #1)
BD061	Aquatic Vegetation	Hydrographic (Chart #1)	Point	J13.2 (Chart #1)
BH010	Aqueduct	Topographic (LTDS)	Surface	AR_Cyan_Aqueduct
BH010	Aqueduct	Topographic (LTDS)	Surface	AR_Cyan_Aqueduct_UnderConstruction_Destroyed
BH010	Aqueduct	Topographic (LTDS)	Curve	LN_Cyan_Aqueduct_OnSurface
BH010	Aqueduct	Topographic (LTDS)	Curve	LN_Cyan_Aqueduct_AboveSurface
BH010	Aqueduct	Topographic (LTDS)	Curve	LN_Cyan_Aqueduct_BelowGround
BH010	Aqueduct	Topographic (LTDS)	Curve	LN_Cyan_Aqueduct_WithQanats
AL012	Archeological Site	Topographic (LTDS)	Surface	AR_Black_ArcheologicalSite
AL012	Archeological Site	Topographic (LTDS)	Point	PT_Black_ArcheologicalSite
AL012	Archeological Site	Topographic (LTDS)	Point	PT_Blue072_VO
DA005	Asphalt Lake	Topographic (LTDS)	Surface	AR_Black_AsphaltLake
AL142	Astronomical Observatory	Topographic (LTDS)	Surface	AR_Black_AstronomicalObservatory
AL142	Astronomical Observatory	Topographic (LTDS)	Point	PT_Black_AstronomicalObservatory
AL142	Astronomical Observatory	Topographic (LTDS)	Point	PT_Blue072_VO
BA022	Backshore	Other	Surface	TBD
EE060	Bank Vegetation Zone	Hydrographic (Chart #1)	Surface	C33 (Chart #1)
BI045	Basin Gate	Topographic (LTDS)	Curve	Not symbolized
BI045	Basin Gate	Topographic (LTDS)	Point	Not symbolized
BA050	Beach	Other	Surface	TBD
BA050	Beach	Other	Curve	TBD
BA050	Beach	Other	Point	TBD
BB150	Beach Landing Site	Hydrographic (Chart #1)	Surface	TBD
BB150	Beach Landing Site	Hydrographic (Chart #1)	Point	TBD

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TABLE II. Feature origin - Continued

LPC Feature Code	LPC Feature Name	Feature Source Origin	Feature Geometry	Feature Symbol
BE050	Beach Profile	Hydrographic (Chart #1)	Curve	TBD
BB020	Berth	Hydrographic (Chart #1)	Point	F19 (Chart #1)
AG050	Billboard	Topographic (LTDS)	Point	PT_Blue072_VO
GB025	Blast Barrier	Aeronautical	Curve	TBD
GB025	Blast Barrier	Aeronautical	Point	TBD
AC010	Blast-furnace	Topographic (LTDS)	Point	PT_Blue072_VO
BB149	Boat Landing	Hydrographic (Chart #1)	Surface	F17 (Chart #1)
BB149	Boat Landing	Hydrographic (Chart #1)	Point	Not Symbolized
BH015	Bog	Topographic (LTDS)	Surface	AR_Cyan_Bog
BB030	Bollard	Hydrographic (Chart #1)	Point	F22 (Chart #1)
BD076	Booming Ground	Hydrographic (Chart #1)	Surface	F29.1 (Chart #1)
BF010	Bottom Characteristic Region	Hydrographic (Chart #1)	Surface	J12.1, J12.2 (Chart #1)
BF010	Bottom Characteristic Region	Hydrographic (Chart #1)	Point	J12.1, J12.2 (Chart #1)
BF010	Bottom Characteristic Region	Hydrographic (Chart #1)	Point	J14 (Chart #1)
BF010	Bottom Characteristic Region	Hydrographic (Chart #1)	Point	J15 (Chart #1)
ZB030	Boundary Monument	Topographic (LTDS)	Point	PT_Black_BoundaryMonument
AQ040	Bridge	Topographic (LTDS)	Curve	LN_Black_Bridge_DrawLift
AQ040	Bridge	Topographic (LTDS)	Curve	LN_Black_Bridge
AQ040	Bridge	Topographic (LTDS)	Curve	LN_Black_Bridge_Pedestrian
AQ040	Bridge	Topographic (LTDS)	Point	PT_Black_Bridge
AQ040	Bridge	Topographic (LTDS)	Point	PT_Black_Bridge_Pedestrian
AQ040	Bridge	Topographic (LTDS)	Point	PT_Black_Bridge_DrawLift

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TABLE II. Feature origin - Continued

LPC Feature Code	LPC Feature Name	Feature Source Origin	Feature Geometry	Feature Symbol
AQ045	Bridge Span		Curve	No Symbology needed
AQ050	Bridge Superstructure	Topographic (LTDS)	Point	PT_Blue072_LinearObstruction
AQ055	Bridge Tower	Topographic (LTDS)	Point	PT_Blue072_VO
EB070	Brush	Topographic (LTDS)	Surface	AR_Green362_Brush
AF021	Bucket Elevator	Topographic (LTDS)	Point	PT_Blue072_VO
AL013	Building	Topographic (LTDS)	Surface	AR_Black_Building_Important
AL013	Building	Topographic (LTDS)	Surface	AR_Black_Building_Destroyed
AL013	Building	Topographic (LTDS)	Surface	AR_Black_Building_Diplomatic
AL013	Building	Topographic (LTDS)	Surface	AR_Black_Building_Hospital
AL013	Building	Topographic (LTDS)	Surface	AR_Black_School
AL013	Building	Topographic (LTDS)	Surface	AR_Black_Mosque
AL013	Building	Topographic (LTDS)	Surface	AR_Black_Synagogue
AL013	Building	Topographic (LTDS)	Surface	AR_Black_Church
AL013	Building	Topographic (LTDS)	Surface	AR_Black_Pagoda
AL013	Building	Topographic (LTDS)	Surface	AR_Black_Temple
AL013	Building	Topographic (LTDS)	Surface	AR_Black_Stupa
AL013	Building	Topographic (LTDS)	Surface	AR_Black_Building_Generic
AL013	Building	Topographic (LTDS)	Surface	AR_Black_Building_LargeAreaVO
AL013	Building	Topographic (LTDS)	Surface	AR_Black_Building_HealthOffice
AL013	Building	Topographic (LTDS)	Point	PT_Black_Building_Generic
AL013	Building	Topographic (LTDS)	Point	PT_Black_Building_Diplomatic
AL013	Building	Topographic (LTDS)	Point	PT_Black_Hospital
AL013	Building	Topographic (LTDS)	Point	PT_Black_School

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TABLE II. Feature origin - Continued

LPC Feature Code	LPC Feature Name	Feature Source Origin	Feature Geometry	Feature Symbol
AL013	Building	Topographic (LTDS)	Point	PT_Black_Marabout_MuslimShrine
AL013	Building	Topographic (LTDS)	Point	PT_Black_Synagogue
AL013	Building	Topographic (LTDS)	Point	PT_Black_Church
AL013	Building	Topographic (LTDS)	Point	PT_Black_Pagoda
AL013	Building	Topographic (LTDS)	Point	PT_Black_Temple
AL013	Building	Topographic (LTDS)	Point	PT_Black_Stupa
AL013	Building	Topographic (LTDS)	Point	PT_Black_Minaret
AL013	Building	Topographic (LTDS)	Point	PT_Black_Mosque
AL013	Building	Topographic (LTDS)	Point	PT_Black_Customs_Checkpoint
AL013	Building	Topographic (LTDS)	Point	PT_Black_ChristianShrine
AL013	Building	Topographic (LTDS)	Point	PT_Black_Health_Office
AL013	Building	Topographic (LTDS)	Point	PT_Black_Building_Important_Landmark
AL013	Building	Topographic (LTDS)	Point	PT_Black_TransportationStation
AL013	Building	Topographic (LTDS)	Surface	AR_Black_Building_HouseOfWorship_Generic
AL013	Building	Topographic (LTDS)	Point	PT_Black_Building_HouseOfWorship_Generic
AL013	Building	Topographic (LTDS)	Point	PT_Black_Building_Landmark_Destroyed
AL013	Building	Topographic (LTDS)	Point	PT_Black_Shrine_Generic
AL013	Building	Topographic (LTDS)	Point	PT_Black_Building_Important_Non-landmark
AL013	Building	Topographic (LTDS)	Surface	PT_Blue072_VO
AL013	Building	Topographic (LTDS)	Point	PT_Blue072_VO
AL018	Building Superstructure	Topographic (LTDS)	Surface	PT_Blue072_VO
AL018	Building Superstructure	Topographic (LTDS)	Point	PT_Blue072_VO
AL020	Built-up Area	Topographic (LTDS)	Surface	AR_Dk-Brown1815-42_BUA_Dense

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TABLE II. Feature origin - Continued

LPC Feature Code	LPC Feature Name	Feature Source Origin	Feature Geometry	Feature Symbol
AL020	Built-up Area	Topographic (LTDS)	Surface	AR_Dk-Brown1815-12_BUA_Sparse
AL020	Built-up Area	Topographic (LTDS)	Surface	AR_Dk-Brown1815_BUA_Destroyed
BC020	Buoy	Hydrographic (Chart #1)	Point	Q23 (Chart #1)
BC020	Buoy	Hydrographic (Chart #1)	Point	Q25 (Chart #1)
BC020	Buoy	Hydrographic (Chart #1)	Point	Q20 (Chart #1)
BC020	Buoy	Hydrographic (Chart #1)	Point	Q21 (Chart #1)
BC020	Buoy	Hydrographic (Chart #1)	Point	Q24 (Chart #1)
BC020	Buoy	Hydrographic (Chart #1)	Point	Q22 (Chart #1)
BC020	Buoy	Hydrographic (Chart #1)	Point	Q26 (Chart #1)
BC020	Buoy	Hydrographic (Chart #1)	Point	Q58 (Chart #1)
BC020	Buoy	Hydrographic (Chart #1)	Point	Q40 (Chart #1)
BC020	Buoy	Hydrographic (Chart #1)	Point	P6 (Chart #1)
BC020	Buoy	Hydrographic (Chart #1)	Point	P5 (Chart #1)
BC020	Buoy	Hydrographic (Chart #1)	Point	Q30 (Chart #1)
BC020	Buoy	Hydrographic (Chart #1)	Point	Q130.4 (Chart #1)
BC020	Buoy	Hydrographic (Chart #1)	Point	Q130.5 (Chart #1)
BC020	Buoy	Hydrographic (Chart #1)	Point	Q130.6 (Chart #1)
BC020	Buoy	Hydrographic (Chart #1)	Point	Q130.1 (Chart #1)
BC020	Buoy	Hydrographic (Chart #1)	Point	Q43 (Chart #1)
BC020	Buoy	Hydrographic (Chart #1)	Point	Q50 (Chart #1)
BC020	Buoy	Hydrographic (Chart #1)	Point	Q56(Chart #1)
BC020	Buoy	Hydrographic (Chart #1)	Point	Q42(Chart #1)
BC020	Buoy	Hydrographic (Chart #1)	Point	Q130.3 (Chart #1)

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TABLE II. Feature origin - Continued

LPC Feature Code	LPC Feature Name	Feature Source Origin	Feature Geometry	Feature Symbol
BC020	Buoy	Hydrographic (Chart #1)	Point	Q130.3 (Chart #1)
BC020	Buoy	Hydrographic (Chart #1)	Point	Q130.3 (Chart #1)
BC020	Buoy	Hydrographic (Chart #1)	Point	Q130.3 (Chart #1)
BC020	Buoy	Hydrographic (Chart #1)	Point	Q54 (Chart #1)
BC020	Buoy	Hydrographic (Chart #1)	Point	L16 (Chart #1)
BC020	Buoy	Hydrographic (Chart #1)	Point	Q130.1 (Chart #1)
BC020	Buoy	Hydrographic (Chart #1)	Point	Q130.1 (Chart #1)
BC020	Buoy	Hydrographic (Chart #1)	Point	Q55 (Chart #1)
BC020	Buoy	Hydrographic (Chart #1)	Point	Q57 (Chart #1)
AT005	Cable	Topographic (LTDS)	Curve	LN_Blue072_Cable_Powerline
AT005	Cable	Topographic (LTDS)	Curve	LN_Black_Cable_CommunicationLine
AT005	Cable	Topographic (LTDS)	Curve	LN_Blue072_Cable_Obstruction
AT005	Cable	Hydrographic (Chart #1)	Curve	L30.1 (Chart #1)
AT005	Cable	Hydrographic (Chart #1)	Curve	L31.1 (Chart#1)
AT005	Cable	Hydrographic (Chart #1)	Curve	L32(Chart#1)
AT005	Cable	Topographic (LTDS)	Point	PT_Blue072_CableVO
AT041	Cableway	Topographic (LTDS)	Curve	LN_Black_Cableway
AT041	Cableway	Topographic (LTDS)	Curve	LN_Blue072_Cableway_Obstruction
AT041	Cableway	Topographic (LTDS)	Point	PT_Blue072_CablewayVO
ID010	Cadastral Control Point	Topographic (LTDS)	Point	TBD
AL025	Cairn	Topographic (LTDS)	Point	PT_Black_Cairn
BB050	Calling-in Point	Hydrographic (Chart #1)	Point	M40.1 (Chart #1)
AI030	Camp	Topographic (LTDS)	Surface	AR_Black_Camp

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TABLE II. Feature origin - Continued

LPC Feature Code	LPC Feature Name	Feature Source Origin	Feature Geometry	Feature Symbol
AI030	Camp	Topographic (LTDS)	Point	PT_Black_Camp
AK060	Camp-site	Topographic (LTDS)	Surface	AR_Black_CampSite
AK060	Camp-site	Topographic (LTDS)	Point	PT_Black_CampSite
BH020	Canal	Topographic (LTDS)	Surface	AR_Cyan_Canal_Perennial
BH020	Canal	Topographic (LTDS)	Surface	AR_Cyan_Canal_Non-perennial
BH020	Canal	Topographic (LTDS)	Surface	AR_Cyan_Canal_UnderConstruction_Destroyed
BH020	Canal	Topographic (LTDS)	Curve	LN_Cyan_Canal_Perennial
BH020	Canal	Topographic (LTDS)	Curve	LN_Cyan_Canal_Dry
EC010	Cane	Topographic (LTDS)	Surface	AR_Green362_Cane
AI020	Caravan Park	Topographic (LTDS)	Surface	AR_Black_CaravanPark
AI020	Caravan Park	Topographic (LTDS)	Point	PT_Black_CaravanPark
AP010	Cart Track	Topographic (LTDS)	Curve	LN_Black_CartTrack
AL375	Castle	Topographic (LTDS)	Surface	AR_Black_Castle
AL375	Castle	Topographic (LTDS)	Point	PT_Black_Castle
AL375	Castle	Topographic (LTDS)	Point	PT_Blue072_VO
AL376	Castle Complex	Topographic (LTDS)	Surface	AR_Black_CastleComplex
AL376	Castle Complex	Topographic (LTDS)	Point	PT_Black_CastleComplex
AC020	Catalytic Cracker	Topographic (LTDS)	Point	PT_Black_CatalyticCracker
AC020	Catalytic Cracker	Topographic (LTDS)	Point	PT_Blue072_VO
AQ063	Causeway Structure	Topographic (LTDS)	Surface	AR_Cyan_CausewayStructure_Dry
AQ063	Causeway Structure	Topographic (LTDS)	Curve	LN_Causeway
AQ063	Causeway Structure	Topographic (LTDS)	Surface	AR_BlackCyan_CausewayStructure_NotDry
DB029	Cave Mouth	Topographic (LTDS)	Point	PT_Black_CaveMouth

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TABLE II. Feature origin - Continued

LPC Feature Code	LPC Feature Name	Feature Source Origin	Feature Geometry	Feature Symbol
AL030	Cemetery	Topographic (LTDS)	Surface	AR_Black_Cemetery
AL030	Cemetery	Topographic (LTDS)	Surface	AR_Black_Cemetery_Christian
AL030	Cemetery	Topographic (LTDS)	Surface	AR_Black_Cemetery_Buddhism
AL030	Cemetery	Topographic (LTDS)	Surface	AR_Black_Cemetery_Islam
AL030	Cemetery	Topographic (LTDS)	Surface	AR_Black_Cemetery_Judaism
AL030	Cemetery	Topographic (LTDS)	Surface	AR_Black_Cemetery_Shinto
AL030	Cemetery	Topographic (LTDS)	Point	PT_Black_Cemetery_Unknown
AL030	Cemetery	Topographic (LTDS)	Point	PT_Black_Cemetery_Christian
AL030	Cemetery	Topographic (LTDS)	Point	PT_Black_Cemetery_Buddhism
AL030	Cemetery	Topographic (LTDS)	Point	PT_Black_Cemetery_Islam
AL030	Cemetery	Topographic (LTDS)	Point	PT_Black_Cemetery_Judaism
AL030	Cemetery	Topographic (LTDS)	Point	PT_Black_Cemetery_Shinto
BH191	Channel	Hydrographic (Chart #1)	Surface	C21 (Chart #1)
BH191	Channel	Hydrographic (Chart #1)	Surface	M18 (Chart #1)
BH191	Channel	Hydrographic (Chart #1)	Curve	C21 (Chart #1)
BH191	Channel	Hydrographic (Chart #1)	Surface	C20 (Chart #1)
BH191	Channel	Hydrographic (Chart #1)	Curve	C20 (Chart #1)
AH070	Checkpoint	Topographic (LTDS)	Point	PT_Black_Checkpoint
BI010	Cistern	Topographic (LTDS)	Point	PT_Cyan_Cistern
EC040	Cleared Way	Topographic (LTDS)	Surface	No Symbology needed
FA210	Conservation Area	Topographic (LTDS)	Surface	AR_Black_ConservationArea
FA041	Contact Zone	Other	Surface	TBD
FA012	Contaminated Region	Other	Surface	TBD

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TABLE II. Feature origin - Continued

LPC Feature Code	LPC Feature Name	Feature Source Origin	Feature Geometry	Feature Symbol
FA012	Contaminated Region	Other	Point	TBD
ZB035	Control Point	Topographic (LTDS)	Point	PT_Black_ControlPoint_Horizontal
ZB035	Control Point	Topographic (LTDS)	Point	PT_Black_ControlPoint_Vertical
AQ060	Control Tower	Topographic (LTDS)	Surface	AR_Black_ControlTower
AQ060	Control Tower	Topographic (LTDS)	Point	PT_Black_ControlTower
AQ060	Control Tower	Topographic (LTDS)	Point	PT_Blue072_VO
AF020	Conveyor	Topographic (LTDS)	Curve	LN_Black_Conveyor
AD055	Cooling Facility	Topographic (LTDS)	Surface	AR_Black_CoolingFacility
AD055	Cooling Facility	Topographic (LTDS)	Point	PT_Black_CoolingFacility
AF030	Cooling Tower	Topographic (LTDS)	Point	PT_Black_CoolingTower
AF030	Cooling Tower	Topographic (LTDS)	Point	PT_Blue072_VO
AF040	Crane	Topographic (LTDS)	Point	PT_Black_Crane
AF040	Crane	Topographic (LTDS)	Point	PT_Black_Crane_Offshore
AF040	Crane	Topographic (LTDS)	Point	PT_Blue072_VO
BJ031	Crevasse	Topographic (LTDS)	Surface	AR_Cyan_Crevasse
BJ031	Crevasse	Topographic (LTDS)	Curve	LN_Cyan_Crevasse
DB061	Crevice	Topographic (LTDS)	Surface	AR_Dk-Brown1815_Crevice
DB061	Crevice	Topographic (LTDS)	Curve	LN_Dk-Brown1815_Crevice
BD020	Crib	Hydrographic (Chart #1)	Surface	K40 (Chart #1)
BD020	Crib	Hydrographic (Chart #1)	Point	K41 (Chart #1)
BD020	Crib	Hydrographic (Chart #1)	Point	K40 (Chart #1)
BD020	Crib	Hydrographic (Chart #1)	Point	K2, K42 (Chart #1)
BD020	Crib	Hydrographic (Chart #1)	Point	K3 (Chart #1)

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TABLE II. Feature origin - Continued

LPC Feature Code	LPC Feature Name	Feature Source Origin	Feature Geometry	Feature Symbol
EA010	Crop Land	Topographic (LTDS)	Surface	AR_Dk-Brown1815-7_Cropland
AQ065	Culvert	Topographic (LTDS)	Curve	LN_Black_Culvert
DB070	Cut	Topographic (LTDS)	Curve	No Symbology
DB071	Cut Line	Topographic (LTDS)	Curve	LN_Dk-Brown1815_Cutline_GreaterthanContour
DB071	Cut Line	Topographic (LTDS)	Curve	LN_Dk-Brown1815_Cutline_LessthanContour
BI020	Dam	Topographic (LTDS)	Surface	AR_Black_Dam
BI020	Dam	Topographic (LTDS)	Surface	AR_Dk-Brown1815_Dam
BI020	Dam	Topographic (LTDS)	Curve	LN_Black_Dam
BI020	Dam	Topographic (LTDS)	Curve	LN_Dk-Brown1815_Dam
BI020	Dam	Topographic (LTDS)	Curve	LN_Black_Dam_CarryingTransportation
BI020	Dam	Topographic (LTDS)	Curve	LN_Dk-Brown1815_Dam_CarryingTransportation
BI020	Dam	Topographic (LTDS)	Point	PT_Black_Dam
BI020	Dam	Topographic (LTDS)	Point	PT_Dk-Brown1815_Dam
BI020	Dam	Topographic (LTDS)	Point	PT_Blue072_VO
BE019	Depth Area	Hydrographic (Chart #1)	Surface	I30 (Chart #1)
BE010	Depth Curve	Hydrographic (Chart #1)	Curve	I30 (Chart #1)
FC010	Direction of Buoyage Indicator	Hydrographic (Chart #1)	Point	Q130.2 (Chart #1)
BD030	Discoloured Water	Hydrographic (Chart #1)	Surface	K40 (Chart #1)
BD030	Discoloured Water	Hydrographic (Chart #1)	Point	K40 (Chart #1)
AT010	Dish Aerial	Topographic (LTDS)	Point	PT_Black_DishAerial
AT010	Dish Aerial	Topographic (LTDS)	Point	PT_Blue072_VO

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TABLE II. Feature origin - Continued

LPC Feature Code	LPC Feature Name	Feature Source Origin	Feature Geometry	Feature Symbol
AL050	Display Sign	Hydrographic (Chart #1)	Point	Q126 (Chart #1)
AB000	Disposal Site	Topographic (LTDS)	Surface	AR_Black_DisposalSite_Refuse
AB000	Disposal Site	Topographic (LTDS)	Point	PT_Blue072_VO
ZB036	Distance Mark	Hydrographic (Chart #1)	Point	B25 (Chart #1)
BH030	Ditch	Topographic (LTDS)	Surface	AR_Cyan_Ditch_Perennial
BH030	Ditch	Topographic (LTDS)	Surface	AR_Cyan_Ditch_Intermittent
BH030	Ditch	Topographic (LTDS)	Curve	LN_Cyan_Ditch_Perennial
BH030	Ditch	Topographic (LTDS)	Curve	LN_Cyan_Ditch_Intermittent
BB095	Dock	Hydrographic (Chart #1)	Surface	F27 (Chart #1)
BB080	Dolphin	Hydrographic (Chart #1)	Point	F20 (Chart #1)
BB080	Dolphin	Hydrographic (Chart #1)	Point	F21 (Chart #1)
AL060	Dragon's Teeth	Topographic (LTDS)	Surface	AR_Black_DragonsTeeth
AL060	Dragon's Teeth	Topographic (LTDS)	Curve	LN_Black_DragonsTeeth
FC034	Dredged Area	Hydrographic (Chart #1)	Surface	I22 (Chart #1)
FC034	Dredged Area	Hydrographic (Chart #1)	Surface	I23 (Chart #1)
FC034	Dredged Area	Hydrographic (Chart #1)	Curve	I20 (Chart #1)
AK070	Drive-in Theatre	Topographic (LTDS)	Surface	AR_Black_DriveInTheatre
AK070	Drive-in Theatre	Topographic (LTDS)	Point	PT_Black_DriveInTheatre
BB090	Dry Dock	Topographic (LTDS)	Surface	AR_Black_DryDock
FC033	Dumping Ground	Hydrographic (Chart #1)	Surface	N23.1 (Chart #1)
FC033	Dumping Ground	Hydrographic (Chart #1)	Surface	N23.2 (Chart #1)
FC033	Dumping Ground	Hydrographic (Chart #1)	Surface	N24 (Chart #1)
FC033	Dumping Ground	Hydrographic (Chart #1)	Point	N23.1 (Chart #1)

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TABLE II. Feature origin - Continued

LPC Feature Code	LPC Feature Name	Feature Source Origin	Feature Geometry	Feature Symbol
FC033	Dumping Ground	Hydrographic (Chart #1)	Surface	N2.1 (Chart #1)
CA010	Elevation Contour	Topographic (LTDS)	Curve	LN_Cyan_Contour_SnowIceHalfAuxillary
CA010	Elevation Contour	Topographic (LTDS)	Curve	LN_Dk-Brown1815_Contour_LandFormLine
CA010	Elevation Contour	Topographic (LTDS)	Curve	LN_Cyan_Contour_SnowIceFormLine
CA010	Elevation Contour	Topographic (LTDS)	Curve	LN_Dk-Brown1815_Contour_Land_DepressionIndex
CA010	Elevation Contour	Topographic (LTDS)	Curve	LN_Cyan_Contour_SnowIce_DepressionIndex
CA010	Elevation Contour	Topographic (LTDS)	Curve	LN_Dk-Brown1815_Contour_Land_DepressionIntermediate
CA010	Elevation Contour	Topographic (LTDS)	Curve	LN_Dk-Brown1815_Contour_LandIndexAccurate
CA010	Elevation Contour	Topographic (LTDS)	Curve	LN_Dk-Brown1815_Contour_LandIndexApproximate
CA010	Elevation Contour	Topographic (LTDS)	Curve	LN_Cyan_Contour_SnowIceIndexAccurate
CA010	Elevation Contour	Topographic (LTDS)	Curve	LN_Cyan_Contour_SnowIceIndexApproximate
CA010	Elevation Contour	Topographic (LTDS)	Curve	LN_Dk-Brown1815_Contour_LandIntermediateAccurate
CA010	Elevation Contour	Topographic (LTDS)	Curve	LN_Dk-Brown1815_Contour_LandIntermediateApproximate
CA010	Elevation Contour	Topographic (LTDS)	Curve	LN_Cyan_Contour_SnowIceIntermediateAccurate
CA010	Elevation Contour	Topographic (LTDS)	Curve	LN_Cyan_Contour_SnowIceIntermediateApproximate
CA010	Elevation Contour	Topographic (LTDS)	Curve	LN_Dk-Brown1815_Contour_LandHalfAuxillary

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TABLE II. Feature origin - Continued

LPC Feature Code	LPC Feature Name	Feature Source Origin	Feature Geometry	Feature Symbol
CA010	Elevation Contour	Topographic (LTDS)	Curve	LN_Cyan_Contour_SnowIce_DepressionIntermediate
CA010	Elevation Contour	Topographic (LTDS)	Curve	LN_Dk-Brown1815_Contour_Land_MoundIndex
CA010	Elevation Contour	Topographic (LTDS)	Curve	LN_Dk-Brown1815_Contour_Land_MoundIntermediate
CA010	Elevation Contour	Topographic (LTDS)	Curve	LN_Dk-Brown1815_Contour_Land_QuarterAuxillary
CA010	Elevation Contour	Topographic (LTDS)	Curve	LN_Cyan_Contour_SnowIce_QuarterAuxillary
CA010	Elevation Contour	Topographic (LTDS)	Curve	LN_Dk-Brown1815_Contour_Land_DepressionAuxillary
CA010	Elevation Contour	Topographic (LTDS)	Curve	LN_Cyan_Contour_SnowIce_DepressionAuxillary
DB090	Embankment	Topographic (LTDS)	Surface	AR_Dk-Brown1815_Embankment_GreaterthanContourInterval
DB090	Embankment	Topographic (LTDS)	Surface	AR_Dk-Brown1815_Embankment_LessthanContourInterval
DB090	Embankment	Topographic (LTDS)	Curve	LN_Dk-Brown1815_Embankment_GreaterthanContourInterval
DB090	Embankment	Topographic (LTDS)	Curve	LN_Dk-Brown1815_Embankment_LessthanContourInterval
DB090	Embankment	Topographic (LTDS)	Curve	LN_Dk-Brown1815_Embankment_DykeWithTransportation
DB090	Embankment	Topographic (LTDS)	Curve	LN_Dk-Brown1815_Embankment_DykeWithoutTransportation

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TABLE II. Feature origin - Continued

LPC Feature Code	LPC Feature Name	Feature Source Origin	Feature Geometry	Feature Symbol
AF060	Engine Test Cell	Topographic (LTDS)	Surface	AR_Black_EngineTestCell
AF060	Engine Test Cell	Topographic (LTDS)	Point	PT_Black_EngineTestCell
AF060	Engine Test Cell	Topographic (LTDS)	Point	PT_Blue072_VO
AH025	Engineered Earthwork	Topographic (LTDS)	Surface	AR_Black_EngineeredEarthwork
AH025	Engineered Earthwork	Topographic (LTDS)	Curve	LN_Black_EngineeredEarthwork
DB100	Esker	Topographic (LTDS)	Curve	LN_Dk-Brown1815_Esker
AF050	Excavating Machine	Topographic (LTDS)	Curve	PT_Blue072_VO
BD001	Explosive Naval Mine	Hydrographic (Chart #1)	Point	N23.1 (Chart #1)
AA010	Extraction Mine	Topographic (LTDS)	Surface	AR_Black_Mine_Operational
AA010	Extraction Mine	Topographic (LTDS)	Surface	AR_Black_Mine_Abandoned
AA010	Extraction Mine	Topographic (LTDS)	Surface	AR_Black_Mine_PeatCuttings
AA010	Extraction Mine	Topographic (LTDS)	Point	PT_Black_Mine_Operational
AA010	Extraction Mine	Topographic (LTDS)	Point	PT_Black_Mine_Abandoned
AA010	Extraction Mine	Topographic (LTDS)	Point	PT_Black_Mine_PeatCutting
AL010	Facility	Topographic (LTDS)	Surface	AR_Black_Facility
AL010	Facility	Topographic (LTDS)	Surface	AR_Black_Facility
AL010	Facility	Topographic (LTDS)	Surface	AR_Black_Facility
AL010	Facility	Topographic (LTDS)	Surface	AR_Black_Facility
AL010	Facility	Topographic (LTDS)	Surface	AR_Black_Facility
AL010	Facility	Topographic (LTDS)	Point	PT_Black_Facility
AL010	Facility	Topographic (LTDS)	Point	PT_Black_Facility
AL010	Facility	Topographic (LTDS)	Point	PT_Black_Facility
AL010	Facility	Topographic (LTDS)	Point	PT_Black_Facility

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TABLE II. Feature origin - Continued

LPC Feature Code	LPC Feature Name	Feature Source Origin	Feature Geometry	Feature Symbol
AL010	Facility	Topographic (LTDS)	Point	PT_Black_Facility
AL010	Facility	Topographic (LTDS)	Point	PT_Black_Facility_Destroyed
AK090	Fairground	Topographic (LTDS)	Surface	AR_Black_Fairground
AK090	Fairground	Topographic (LTDS)	Point	PT_Black_Fairground
AL070	Fence	Topographic (LTDS)	Curve	LN_Black_Fence
AQ070	Ferry Crossing	Hydrographic (Chart #1)	Curve	M51 (Chart #1)
AQ070	Ferry Crossing	Hydrographic (Chart #1)	Curve	M50 (Chart #1)
AQ080	Ferry Station	Topographic (LTDS)	Surface	AR_Black_FerryStation
AQ080	Ferry Station	Topographic (LTDS)	Point	PT_Black_FerryStation
FA015	Firing Range	Topographic (LTDS)	Surface	AR_Black_FiringRange
BH051	Fish Farm Facility	Topographic (LTDS)	Surface	AR_Cyan-31_MarineFarm
BH051	Fish Farm Facility	Topographic (LTDS)	Point	PT_Cyan_FishFarm
BD125	Fish Haven	Hydrographic (Chart #1)	Surface	K46.1 (Chart #1)
BD125	Fish Haven	Hydrographic (Chart #1)	Surface	K46.2 (Chart #1)
BD125	Fish Haven	Hydrographic (Chart #1)	Point	K46.1 (Chart #1)
BD125	Fish Haven	Hydrographic (Chart #1)	Point	K46.2 (Chart #1)
BB112	Fish Trap Area	Hydrographic (Chart #1)	Point	K44.2 (Chart #1)
BB112	Fish Trap Area	Hydrographic (Chart #1)	Surface	K45 (Chart #1)
BB110	Fish Weir	Hydrographic (Chart #1)	Point	K44.2 (Chart #1)
BB110	Fish Weir	Hydrographic (Chart #1)	Surface	K45 (Chart #1)
BB100	Fishing Stakes	Hydrographic (Chart #1)	Curve	K44.1 (Chart #1)
AL073	Flagpole	Topographic (LTDS)	Point	PT_Blue072_VO
AF070	Flare Pipe	Topographic (LTDS)	Point	PT_Black_FlarePipe

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TABLE II. Feature origin - Continued

LPC Feature Code	LPC Feature Name	Feature Source Origin	Feature Geometry	Feature Symbol
AF070	Flare Pipe	Hydrographic (Chart #1)	Point	L11 (Chart #1)
AF070	Flare Pipe	Topographic (LTDS)	Point	PT_Blue072_VO
BD123	Floating Barrier	Hydrographic (Chart #1)	Curve	F29.1 (Chart #1)
BD123	Floating Barrier	Hydrographic (Chart #1)	Curve	N61 (Chart #1)
BB199	Floating Dry Dock	Topographic (LTDS)	Surface	AR_Black_FloatingDryDock
BI044	Flood Control Structure	Topographic (LTDS)	Surface	AR_Black_FloodControlStructure
BI044	Flood Control Structure	Topographic (LTDS)	Curve	LN_Black_FloodControlStructure
BC101	Fog Signal	Hydrographic (Chart #1)	Point	R1 (Chart #1)
BH070	Ford	Topographic (LTDS)	Curve	LN_Black_Ford
BA023	Foreshore	Topographic (LTDS)	Surface	AR_BlackCyan_Foreshore
EC015	Forest	Topographic (LTDS)	Surface	AR_Green362-31_Forest_Evergreen
EC015	Forest	Topographic (LTDS)	Surface	AR_Green362-31_Forest_Deciduous
EC015	Forest	Topographic (LTDS)	Surface	AR_Green362-31_Forest_Mixed
EC015	Forest	Topographic (LTDS)	Surface	AR_Green362_Forest_ScatteredTrees
EC015	Forest	Topographic (LTDS)	Surface	AR_Green362-31_Forest_Unknown
EC015	Forest	Topographic (LTDS)	Curve	LN_Green362_RowOfTrees
EC015	Forest	Topographic (LTDS)	Point	PT_Green362_Tree
AH055	Fortified Building	Topographic (LTDS)	Surface	AR_Black_Fortification
AH055	Fortified Building	Topographic (LTDS)	Point	PT_Black_Fortification
AH055	Fortified Building	Topographic (LTDS)	Point	PT_Blue072_VO
BD050	Foul Ground	Hydrographic (Chart #1)	Surface	K31 (Chart #1)
BD050	Foul Ground	Hydrographic (Chart #1)	Point	K31, L22 (Chart #1)
BH075	Fountain	Topographic (LTDS)	Point	PT_Black_Fountain

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TABLE II. Feature origin - Continued

LPC Feature Code	LPC Feature Name	Feature Source Origin	Feature Geometry	Feature Symbol
AM075	Fuel Storage Facility	Topographic (LTDS)	Surface	AR_Black_FuelStorageFacility
AM075	Fuel Storage Facility	Topographic (LTDS)	Point	PT_Black_FuelStorageFacility
AL080	Gantry	Topographic (LTDS)	Point	PT_Blue072_VO
AP040	Gate	Topographic (LTDS)	Curve	LN_Black_Gate
AP040	Gate	Topographic (LTDS)	Point	PT_Black_Gate
BI070	Gauging Station	Other	Surface	TBD
BI070	Gauging Station	Other	Point	TBD
ZB060	Geodetic Point	Other	Point	TBD
ZD012	Geographic Information Location	Hydrographic (Chart #1)	Surface	No symbology; no Chart #1 symbology
ZD012	Geographic Information Location	Hydrographic (Chart #1)	Curve	No symbology; no Chart #1 symbology
ZD012	Geographic Information Location	Hydrographic (Chart #1)	Point	No symbology; no Chart #1 symbology
DB110	Geologic Fault	Topographic (LTDS)	Curve	LN_Dk-Brown1815_GeologicFault
FA002	Geopolitical Entity	Topographic (LTDS)	Surface	AR_Black_GeopoliticalEntity_DMZ
FA002	Geopolitical Entity	Topographic (LTDS)	Surface	AR_Black_GeopoliticalEntity_ZoneOfOccupation
DB115	Geothermal Outlet	Topographic (LTDS)	Surface	AR_Dk-Brown1815_GeothermalOutlet
DB115	Geothermal Outlet	Topographic (LTDS)	Point	PT_Dk-Brown1815_GeothermalOutlet
BJ030	Glacier	Topographic (LTDS)	Surface	AR_Cyan_Glacier
AK100	Golf Course	Topographic (LTDS)	Surface	AR_Black_GolfCourse
AK100	Golf Course	Topographic (LTDS)	Point	PT_Green362_GolfCourse
AM030	Grain Elevator	Topographic (LTDS)	Surface	AR_Black_GrainElevator

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TABLE II. Feature origin - Continued

LPC Feature Code	LPC Feature Name	Feature Source Origin	Feature Geometry	Feature Symbol
AM030	Grain Elevator	Topographic (LTDS)	Point	PT_Black_GrainElevator
AM030	Grain Elevator	Topographic (LTDS)	Point	PT_Blue072_VO
AM020	Grain Storage Structure	Topographic (LTDS)	Surface	AR_Black_GrainStorage
AM020	Grain Storage Structure	Topographic (LTDS)	Point	PT_Black_GrainStorage
AM020	Grain Storage Structure	Topographic (LTDS)	Point	PT_Blue072_VO
AK110	Grandstand	Topographic (LTDS)	Surface	AR_Black_Grandstand
AK110	Grandstand	Topographic (LTDS)	Point	PT_Blue072_VO
EB010	Grassland	Topographic (LTDS)	Surface	AR_Green362_Grassland
AJ110	Greenhouse	Topographic (LTDS)	Surface	AR_Green362_Greenhouse
AJ110	Greenhouse	Topographic (LTDS)	Surface	PT_Green362_Greenhouse
AJ110	Greenhouse	Topographic (LTDS)	Point	PT_Green362_Greenhouse
BB115	Gridiron	Hydrographic (Chart #1)	Surface	F24 (Chart #1)
BB005	Harbour	Hydrographic (Chart #1)	Surface	No Symbol needed
GB250	Hardened Aircraft Shelter	Topographic (LTDS)	Surface	AR_Black_HardenedAircraftShelter
GB250	Hardened Aircraft Shelter	Topographic (LTDS)	Point	PT_Black_HardenedAircraftShelter
BD130	Hazardous Rock	Hydrographic (Chart #1)	Point	K13 (Chart #1)
BD130	Hazardous Rock	Hydrographic (Chart #1)	Surface	K16 (Chart #1)
BD130	Hazardous Rock	Hydrographic (Chart #1)	Point	K14.2 (Chart #1)
BD130	Hazardous Rock	Hydrographic (Chart #1)	Point	K14.1 (Chart #1)
BD130	Hazardous Rock	Hydrographic (Chart #1)	Point	K12 (Chart #1)
BD130	Hazardous Rock	Hydrographic (Chart #1)	Point	K11 (Chart #1)
BD130	Hazardous Rock	Hydrographic (Chart #1)	Surface	K10 (Chart #1)
AD050	Heating Facility	Topographic (LTDS)	Surface	AR_Black_HeatingFacility

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TABLE II. Feature origin - Continued

LPC Feature Code	LPC Feature Name	Feature Source Origin	Feature Geometry	Feature Symbol
AD050	Heating Facility	Topographic (LTDS)	Point	PT_Black_HeatingFacility
EA020	Hedgerow	Topographic (LTDS)	Curve	LN_Green362_Hedgerow
GB030	Helipad	Topographic (LTDS)	Point	PT_Blue072_Helipad
GB030	Helipad	Topographic (LTDS)	Point	PT_Blue072_Helipad_Hospital
GB035	Heliport	Topographic (LTDS)	Surface	AR_Black_Heliport
BA005	High Water Line	Other	Curve	TBD
AJ030	Holding Pen	Topographic (LTDS)	Surface	AR_Black_HoldingPen
AJ030	Holding Pen	Topographic (LTDS)	Point	PT_Dk-Brown1815_HoldingPen
EA055	Hop Field	Topographic (LTDS)	Surface	AR_Green362_HopField
EA055	Hop Field	Topographic (LTDS)	Point	PT_Green362_HopField
AF080	Hopper	Topographic (LTDS)	Point	PT_Blue072_VO
BD181	Hulk	Topographic (LTDS)	Point	PT_Black_Hulk
BD181	Hulk	Topographic (LTDS)	Point	PT_Blue072_VO
BH077	Hummock	Topographic (LTDS)	Surface	AR_Cyan_Hummock
AL099	Hut	Topographic (LTDS)	Point	PT_Dk-Brown1815_Hut
AC040	Hydrocarbon Products Facility	Topographic (LTDS)	Surface	AR_Black_HydrocarbonProdFacility
AC040	Hydrocarbon Products Facility	Topographic (LTDS)	Point	PT_Black_HydrocarbonProdFacility
AA052	HydrocarbonsField	Topographic (LTDS)	Surface	AR_Black_HydrocarbonsField
BJ040	Ice Cliff	Topographic (LTDS)	Curve	LN_Cyan_IceCliff
BJ060	Ice Peak	Topographic (LTDS)	Point	PT_Dk-Brown1815_IcePeak
BJ060	Ice Peak	Topographic (LTDS)	Point	PT_Cyan_IcePeak
AQ075	Ice Route	Topographic (LTDS)	Curve	LN_Black_IceRoute

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TABLE II. Feature origin - Continued

LPC Feature Code	LPC Feature Name	Feature Source Origin	Feature Geometry	Feature Symbol
BJ065	Ice Shelf	Topographic (LTDS)	Surface	AR_Cyan_IceShelf
GA048	ILS Component	Aeronautical	Curve	TBD; no Chart #1 symbology
GA048	ILS Component	Aeronautical	Point	TBD; no Chart #1 symbology
AL270	Industrial Farm	Topographic (LTDS)	Surface	AR_Green362_IndustrialFarm
AL270	Industrial Farm	Topographic (LTDS)	Point	PT_Green362_IndustrialFarm
AC060	Industrial Furnace	Topographic (LTDS)	Surface	AR_Black_IndustrialFurnace
AC060	Industrial Furnace	Topographic (LTDS)	Point	PT_Black_IndustrialFurnace
BH082	Inland Waterbody	Topographic (LTDS)	Surface	AR_Cyan_Lake_Perennial
BH082	Inland Waterbody	Topographic (LTDS)	Surface	AR_Cyan_Lake_Non-perennial
BH082	Inland Waterbody	Topographic (LTDS)	Surface	AR_Cyan_Lake_Dry
BH082	Inland Waterbody	Topographic (LTDS)	Surface	AR_Cyan_Reservoir
BH082	Inland Waterbody	Topographic (LTDS)	Surface	AR_Cyan_Reservoir_UnderConstruction
BC080	Insubstantial Navigation Mark	Hydrographic (Chart #1)	Point	B32 (Chart #1)
FA110	International Date Line	Hydrographic (Chart #1)	Curve	N1.1 (Chart #1)
BA030	Island	Topographic (LTDS)	Surface	AR_Island
BA030	Island	Hydrographic (Chart #1)	Point	K10 (Chart #1)
GB005	Land Aerodrome	Topographic (LTDS)	Surface	AR_Black_Aerodrome
DB000	Land Area	Topographic (LTDS)	Surface	Yellow (SPC 57377)
BH090	Land Subject to Inundation	Topographic (LTDS)	Surface	AR_Cyan_LSI_Natural
BH090	Land Subject to Inundation	Topographic (LTDS)	Surface	AR_Cyan_LSI_Controlled
BA010	Land Water Boundary	Topographic (LTDS)	Curve	LN_Cyan_LandWaterBoundary_Dry
BA010	Land Water Boundary	Topographic (LTDS)	Curve	LN_Cyan_LandWaterBoundary_Wet
BB151	Landing Steps	Other	Point	U7 (Chart #1)

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TABLE II. Feature origin - Continued

LPC Feature Code	LPC Feature Name	Feature Source Origin	Feature Geometry	Feature Symbol
BD003	Large Bottom Object	Hydrographic (Chart #1)	Surface	K40 (Chart #1)
BD003	Large Bottom Object	Hydrographic (Chart #1)	Surface	K41 (Chart #1)
BD003	Large Bottom Object	Hydrographic (Chart #1)	Surface	K42 (Chart #1)
BD003	Large Bottom Object	Hydrographic (Chart #1)	Point	K40 (Chart #1)
BD003	Large Bottom Object	Hydrographic (Chart #1)	Point	K41 (Chart #1)
BD003	Large Bottom Object	Hydrographic (Chart #1)	Point	K42 (Chart #1)
BD003	Large Bottom Object	Hydrographic (Chart #1)	Point	L24 (Chart #1)
BD003	Large Bottom Object	Hydrographic (Chart #1)	Point	F22 (Chart #1)
BD003	Large Bottom Object	Hydrographic (Chart #1)	Point	K43.1 (Chart #1)
GB040	Launch Pad	Topographic (LTDS)	Point	PT_Black_LaunchPad
BC100	Leading Line	Hydrographic (Chart #1)	Curve	P21 (Chart #1)
BC100	Leading Line	Hydrographic (Chart #1)	Curve	P30.1, P30.2 (Chart #1)
BC100	Leading Line	Hydrographic (Chart #1)	Curve	S11 (Chart #1)
BC100	Leading Line	Hydrographic (Chart #1)	Curve	P31 (Chart #1)
BC100	Leading Line	Hydrographic (Chart #1)	Curve	S3.5 (Chart #1)
BC100	Leading Line	Hydrographic (Chart #1)	Curve	Q120 (Chart #1)
BC060	Light Sector	Hydrographic (Chart #1)	Surface	P41.1 (Chart #1)
BC070	Light Vessel	Topographic (LTDS)	Point	PT_Black_LightVessel
BC050	Lighthouse	Topographic (LTDS)	Point	PT_Black_Lighthouse
ZC040	Local Magnetic Anomaly	Hydrographic (Chart #1)	Surface	B82.1 (Chart #1)
BI030	Lock	Topographic (LTDS)	Surface	AR_Black_Lock
BI030	Lock	Topographic (LTDS)	Point	PT_Black_Lock
GB222	Low Air Maneuver Pylon	Other	Point	TBD

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TABLE II. Feature origin - Continued

LPC Feature Code	LPC Feature Name	Feature Source Origin	Feature Geometry	Feature Symbol
BE021	Low Water Line	Hydrographic (Chart #1)	Curve	C1 (Chart #1)
BK020	Magnetic Station	Other	Point	TBD
BH050	Mariculture Site	Hydrographic (Chart #1)	Surface	K48.1 (Chart #1)
BH050	Mariculture Site	Hydrographic (Chart #1)	Point	K48.2 (Chart #1)
FC102	Marine Range Centre-line	Other	Curve	TBD; no Chart #1 symbology
FC037	Maritime Caution Area	Hydrographic (Chart #1)	Surface	N34 (Chart #1)
FC037	Maritime Caution Area	Hydrographic (Chart #1)	Surface	N31 (Chart #1)
FC037	Maritime Caution Area	Hydrographic (Chart #1)	Surface	N2.1 (Chart #1)
FC037	Maritime Caution Area	Hydrographic (Chart #1)	Surface	L30.2 (Chart #1)
FC037	Maritime Caution Area	Hydrographic (Chart #1)	Surface	L31.2 (Chart #1)
FC037	Maritime Caution Area	Hydrographic (Chart #1)	Surface	L40.2 (Chart #1)
FC037	Maritime Caution Area	Hydrographic (Chart #1)	Surface	N20 (Chart #1)
FC037	Maritime Caution Area	Hydrographic (Chart #1)	Surface	N45 (Chart #1)
FC037	Maritime Caution Area	Hydrographic (Chart #1)	Surface	M14 (Chart #1)
FC037	Maritime Caution Area	Hydrographic (Chart #1)	Surface	L41.2 (Chart #1)
FC037	Maritime Caution Area	Hydrographic (Chart #1)	Point	L3 (Chart #1)
FC037	Maritime Caution Area	Hydrographic (Chart #1)	Surface	F31 (Chart #1)
FC037	Maritime Caution Area	Hydrographic (Chart #1)	Point	N23.1 (Chart #1)
FC037	Maritime Caution Area	Hydrographic (Chart #1)	Point	N20 (Chart #1)
FC037	Maritime Caution Area	Hydrographic (Chart #1)	Point	N21 (Chart #1)
FC037	Maritime Caution Area	Hydrographic (Chart #1)	Point	N31 (Chart #1)
FC037	Maritime Caution Area	Hydrographic (Chart #1)	Surface	N22 (Chart #1)
FC037	Maritime Caution Area	Hydrographic (Chart #1)	Surface	N25 (Chart #1)

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TABLE II. Feature origin - Continued

LPC Feature Code	LPC Feature Name	Feature Source Origin	Feature Geometry	Feature Symbol
FC037	Maritime Caution Area	Hydrographic (Chart #1)	Surface	N63 (Chart #1)
FC037	Maritime Caution Area	Hydrographic (Chart #1)	Surface	M18 (Chart #1)
FC037	Maritime Caution Area	Hydrographic (Chart #1)	Surface	N26 (Chart #1)
FC037	Maritime Caution Area	Hydrographic (Chart #1)	Surface	N2.1 (Chart #1)
FC037	Maritime Caution Area	Hydrographic (Chart #1)	Surface	N1.1 (Chart #1)
FC037	Maritime Caution Area	Hydrographic (Chart #1)	Surface	N1.2 (Chart #1)
FC037	Maritime Caution Area	Hydrographic (Chart #1)	Surface	N47 (Chart #1)
FC037	Maritime Caution Area	Hydrographic (Chart #1)	Surface	N46 (Chart #1)
FC037	Maritime Caution Area	Hydrographic (Chart #1)	Surface	N44 (Chart #1)
FC037	Maritime Caution Area	Hydrographic (Chart #1)	Surface	N43 (Chart #1)
FC037	Maritime Caution Area	Hydrographic (Chart #1)	Surface	N41 (Chart #1)
FC037	Maritime Caution Area	Hydrographic (Chart #1)	Surface	N64 (Chart #1)
FC037	Maritime Caution Area	Hydrographic (Chart #1)	Surface	N32 (Chart #1)
FC037	Maritime Caution Area	Hydrographic (Chart #1)	Surface	M2 (Chart #1)
FC037	Maritime Caution Area	Hydrographic (Chart #1)	Surface	N49 (Chart #1)
FC021	Maritime Limit	Hydrographic (Chart #1)	Curve	N11.2 (Chart #1)
FC021	Maritime Limit	Hydrographic (Chart #1)	Curve	No symbology as they are prohibited by IHO.
FC021	Maritime Limit	Hydrographic (Chart #1)	Curve	N34 (Chart #1)
FC021	Maritime Limit	Hydrographic (Chart #1)	Curve	N31 (Chart #1)
FC021	Maritime Limit	Hydrographic (Chart #1)	Curve	N45 (Chart #1)
FC021	Maritime Limit	Hydrographic (Chart #1)	Curve	F31 (Chart #1)
FC021	Maritime Limit	Hydrographic (Chart #1)	Curve	N22 (Chart #1)
FC021	Maritime Limit	Hydrographic (Chart #1)	Curve	N25 (Chart #1)

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TABLE II. Feature origin - Continued

LPC Feature Code	LPC Feature Name	Feature Source Origin	Feature Geometry	Feature Symbol
FC021	Maritime Limit	Hydrographic (Chart #1)	Curve	N63 (Chart #1)
FC021	Maritime Limit	Hydrographic (Chart #1)	Curve	M18 (Chart #1)
FC021	Maritime Limit	Hydrographic (Chart #1)	Curve	N26 (Chart #1)
FC021	Maritime Limit	Hydrographic (Chart #1)	Curve	N2.1 (Chart #1)
FC021	Maritime Limit	Hydrographic (Chart #1)	Curve	N1.2 (Chart #1)
FC021	Maritime Limit	Hydrographic (Chart #1)	Curve	N64 (Chart #1)
FC021	Maritime Limit	Hydrographic (Chart #1)	Curve	M2 (Chart #1)
FC021	Maritime Limit	Hydrographic (Chart #1)	Curve	N49 (Chart #1)
FC021	Maritime Limit	Hydrographic (Chart #1)	Curve	N30 (Chart #1)
FC021	Maritime Limit	Hydrographic (Chart #1)	Curve	N32 (Chart #1)
FC021	Maritime Limit	Hydrographic (Chart #1)	Curve	N33 (Chart #1)
FC021	Maritime Limit	Hydrographic (Chart #1)	Curve	M14 (Chart #1)
BC010	Maritime Navigation Beacon	Hydrographic (Chart #1)	Point	Q83 (Chart #1)
BC010	Maritime Navigation Beacon	Hydrographic (Chart #1)	Point	Q82 (Chart #1)
BC010	Maritime Navigation Beacon	Hydrographic (Chart #1)	Point	Q110 (Chart #1)
BC010	Maritime Navigation Beacon	Hydrographic (Chart #1)	Point	Q90 (Chart #1)
BC010	Maritime Navigation Beacon	Hydrographic (Chart #1)	Point	Q111 (Chart #1)
BC040	Maritime Navigation Light	Hydrographic (Chart #1)	Curve	P64 (Chart #1)
BC040	Maritime Navigation Light	Hydrographic (Chart #1)	Point	P1 (Chart #1)

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TABLE II. Feature origin - Continued

LPC Feature Code	LPC Feature Name	Feature Source Origin	Feature Geometry	Feature Symbol
BC040	Maritime Navigation Light	Hydrographic (Chart #1)	Point	P41.1 (Chart #1)
BC040	Maritime Navigation Light	Hydrographic (Chart #1)	Point	P31 (Chart #1)
BC040	Maritime Navigation Light	Hydrographic (Chart #1)	Point	P50 (Chart #1)
BC040	Maritime Navigation Light	Hydrographic (Chart #1)	Point	P61.2 (Chart #1)
BC040	Maritime Navigation Light	Hydrographic (Chart #1)	Point	P63 (Chart #1)
BC041	Maritime Navigation Light Support	Hydrographic (Chart #1)	Point	P1 (Chart #1)
BC055	Maritime Navigation Marker	Hydrographic (Chart #1)	Point	Q101 (Chart #1)
BC120	Maritime Notice Board	Hydrographic (Chart #1)	Point	Q126 (Chart #1)
FC130	Maritime Radar Reference Line	Hydrographic (Chart #1)	Curve	M32.1 (Chart #1)
BC034	Maritime Radiobeacon	Hydrographic (Chart #1)	Point	S11 (Chart #1)
BC034	Maritime Radiobeacon	Hydrographic (Chart #1)	Point	S3.3 (Chart #1)
BC034	Maritime Radiobeacon	Hydrographic (Chart #1)	Point	S12 (Chart #1)
BC034	Maritime Radiobeacon	Hydrographic (Chart #1)	Point	S15 (Chart #1)
BC034	Maritime Radiobeacon	Hydrographic (Chart #1)	Point	S2 (Chart #1)
BC034	Maritime Radiobeacon	Hydrographic (Chart #1)	Point	S10 (Chart #1)
BC034	Maritime Radiobeacon	Hydrographic (Chart #1)	Point	S14 (Chart #1)
BC034	Maritime Radiobeacon	Hydrographic (Chart #1)	Point	S17.1 (Chart #1)
FC165	Maritime Route	Hydrographic (Chart #1)	Surface	M27.2 (Chart #1)
FC165	Maritime Route	Hydrographic (Chart #1)	Curve	M27.3 (Chart #1)

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TABLE II. Feature origin - Continued

LPC Feature Code	LPC Feature Name	Feature Source Origin	Feature Geometry	Feature Symbol
FC165	Maritime Route	Hydrographic (Chart #1)	Curve	M27.3 (Chart #1)
FC165	Maritime Route	Hydrographic (Chart #1)	Curve	M28.1 (Chart #1)
FC165	Maritime Route	Hydrographic (Chart #1)	Curve	M4 (Chart #1)
FC165	Maritime Route	Hydrographic (Chart #1)	Curve	M3 (Chart #1)
FC165	Maritime Route	Hydrographic (Chart #1)	Curve	M5.1 (Chart #1)
FC165	Maritime Route	Hydrographic (Chart #1)	Curve	M5.1 (Chart #1)
FC165	Maritime Route	Hydrographic (Chart #1)	Surface	M28.2 (Chart #1)
FC165	Maritime Route	Hydrographic (Chart #1)	Curve	TBD (Chart #1)
BB155	Maritime Signal Station	Hydrographic (Chart #1)	Point	T20 (Chart #1)
BB155	Maritime Signal Station	Hydrographic (Chart #1)	Point	T10 (Chart #1)
BB155	Maritime Signal Station	Hydrographic (Chart #1)	Point	T30 (Chart #1)
BB155	Maritime Signal Station	Hydrographic (Chart #1)	Point	T12 (Chart #1)
BB155	Maritime Signal Station	Hydrographic (Chart #1)	Point	T23 (Chart #1)
BB155	Maritime Signal Station	Hydrographic (Chart #1)	Point	T3 (Chart #1)
BB155	Maritime Signal Station	Hydrographic (Chart #1)	Point	T28 (Chart #1)
BB155	Maritime Signal Station	Hydrographic (Chart #1)	Point	T33 (Chart #1)
BB155	Maritime Signal Station	Hydrographic (Chart #1)	Point	T29 (Chart #1)
BB155	Maritime Signal Station	Hydrographic (Chart #1)	Point	T25.1 (Chart #1)
BB155	Maritime Signal Station	Hydrographic (Chart #1)	Point	T24 (Chart #1)
BB155	Maritime Signal Station	Hydrographic (Chart #1)	Point	T21 (Chart #1)
BB155	Maritime Signal Station	Hydrographic (Chart #1)	Point	T36 (Chart #1)
BB155	Maritime Signal Station	Hydrographic (Chart #1)	Point	T22 (Chart #1)
BB155	Maritime Signal Station	Hydrographic (Chart #1)	Point	T34 (Chart #1)

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TABLE II. Feature origin - Continued

LPC Feature Code	LPC Feature Name	Feature Source Origin	Feature Geometry	Feature Symbol
BB155	Maritime Signal Station	Hydrographic (Chart #1)	Point	M30 (Chart #1)
BB155	Maritime Signal Station	Hydrographic (Chart #1)	Point	Tb (Chart #1)
BB155	Maritime Signal Station	Hydrographic (Chart #1)	Point	T31 (Chart #1)
BB155	Maritime Signal Station	Hydrographic (Chart #1)	Point	T2 (Chart #1)
ED010	Marsh	Topographic (LTDS)	Surface	AR_Cyan_Marsh
FC100	Measured Distance Line	Hydrographic (Chart #1)	Curve	Q122 (Chart #1)
AL130	Memorial Monument	Topographic (LTDS)	Surface	AR_Black_MemorialMonument
AL130	Memorial Monument	Topographic (LTDS)	Point	PT_Black_Monument
AL130	Memorial Monument	Topographic (LTDS)	Point	PT_Blue072_VO
SU001	Military Installation	Topographic (LTDS)	Surface	AR_Black_MilitaryInstallation
SU001	Military Installation	Topographic (LTDS)	Point	PT_Black_MilitaryInstallation
AA020	Mine Shaft Superstructure	Topographic (LTDS)	Point	PT_Blue072_VO
AL065	Minefield	Topographic (LTDS)	Surface	AR_Black_Minefield
AL065	Minefield	Topographic (LTDS)	Point	PT_Black_Minefield
AM040	Mineral Pile	Topographic (LTDS)	Surface	AR_Black_MineralPile
AM040	Mineral Pile	Topographic (LTDS)	Point	PT_Black_MineralPile
AM040	Mineral Pile	Topographic (LTDS)	Point	PT_Blue072_VO
BH100	Moat	Topographic (LTDS)	Surface	AR_Cyan_Moat
BH100	Moat	Topographic (LTDS)	Curve	LN_Cyan_Moat
BH100	Moat	Topographic (LTDS)	Surface	AR_Black_Moat
BH100	Moat	Topographic (LTDS)	Curve	LN_Black_Moat
AI021	Mobile Home Park	Topographic (LTDS)	Surface	AR_Black_MobileHomePark
AI021	Mobile Home Park	Topographic (LTDS)	Point	PT_Black_MobileHomePark

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TABLE II. Feature origin - Continued

LPC Feature Code	LPC Feature Name	Feature Source Origin	Feature Geometry	Feature Symbol
BD074	Mooring Chain	Hydrographic (Chart #1)	Curve	L18 (Chart #1)
BD074	Mooring Chain	Hydrographic (Chart #1)	Curve	Q42 (Chart #1)
AQ110	Mooring Mast	Topographic (LTDS)	Point	PT_Black_MooringMast
AQ110	Mooring Mast	Topographic (LTDS)	Point	PT_Blue072_VO
BB160	Mooring Ring	Hydrographic (Chart #1)	Point	TBD (Chart #1)
BJ020	Moraine	Topographic (LTDS)	Surface	AR_Dk-Brown1815_Moraine
AQ170	Motor Vehicle Station	Topographic (LTDS)	Surface	AR_Black_MotorVehicleStation
AQ170	Motor Vehicle Station	Topographic (LTDS)	Point	PT_Black_MotorVehicleStation
DB150	Mountain Pass	Topographic (LTDS)	Point	PT_Black_MountainPass
AM065	Munition Storage Facility	Topographic (LTDS)	Surface	AR_Black_MunitionStorageFacility
AM065	Munition Storage Facility	Topographic (LTDS)	Point	PT_Black_MunitionStorageFacility
ZD040	Named Location	Topographic (LTDS)	Point	PT_NamedLocation
BH170	Natural Pool	Topographic (LTDS)	Point	PT_Cyan_NaturalPool_Perennial
BH170	Natural Pool	Topographic (LTDS)	Point	PT_Cyan_NaturalPool_Non-perennial
FC050	Naval Firing and/or Practice Area	Hydrographic (Chart #1)	Surface	N32 (Chart #1)
FC050	Naval Firing and/or Practice Area	Hydrographic (Chart #1)	Surface	N30 (Chart #1)
FC050	Naval Firing and/or Practice Area	Hydrographic (Chart #1)	Surface	N34 (Chart #1)
FC050	Naval Firing and/or Practice Area	Hydrographic (Chart #1)	Surface	N31 (Chart #1)
FC055	Naval Operations Area	Hydrographic (Chart #1)	Surface	N33 (Chart #1)
FC055	Naval Operations Area	Hydrographic (Chart #1)	Surface	N34 (Chart #1)

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TABLE II. Feature origin - Continued

LPC Feature Code	LPC Feature Name	Feature Source Origin	Feature Geometry	Feature Symbol
BA021	Nearshore	Other	Surface	TBD
AL014	Non-building Structure	Topographic (LTDS)	Point	PT_Blue072_VO
AA054	Non-Water Well	Topographic (LTDS)	Point	PT_Black_Well_Notwater
AD041	Nuclear Reactor Containment	Topographic (LTDS)	Surface	AR_Black_NuclearReactorContainment
AD041	Nuclear Reactor Containment	Topographic (LTDS)	Point	PT_Black_NuclearReactorContainment
AD041	Nuclear Reactor Containment	Topographic (LTDS)	Point	PT_Blue072_VO
EC020	Oasis	Topographic (LTDS)	Surface	AR_Green362_Oasis
EC020	Oasis	Topographic (LTDS)	Point	PT_Oasis
BD115	Offshore Construction	Topographic (LTDS)	Point	PT_Black_OffshorePlatform
BD115	Offshore Construction	Hydrographic (Chart #1)	Point	L16 (Chart #1)
BD115	Offshore Construction	Hydrographic (Chart #1)	Point	L17 (Chart #1)
BD115	Offshore Construction	Hydrographic (Chart #1)	Point	L12 (Chart #1)
EA040	Orchard	Topographic (LTDS)	Surface	AR_Green362_Orchard
EA040	Orchard	Topographic (LTDS)	Point	PT_Green362_Orchard
AK080	Outdoor Theatre Screen	Topographic (LTDS)	Point	PT_Blue072_VO
BJ070	Pack Ice	Hydrographic (Chart #1)	Surface	N60.2 (Chart #1)
AK120	Park	Topographic (LTDS)	Surface	AR_Black_Park
AK120	Park	Topographic (LTDS)	Point	PT_Green362_Park
AQ141	Parking Garage	Topographic (LTDS)	Surface	AR_Black_ParkingGarage
AQ141	Parking Garage	Topographic (LTDS)	Point	PT_Black_ParkingGarage
AQ141	Parking Garage	Topographic (LTDS)	Point	PT_Blue072_VO
AL140	Particle Accelerator	Topographic (LTDS)	Surface	AR_Black_ParticleAccelerator

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TABLE II. Feature origin - Continued

LPC Feature Code	LPC Feature Name	Feature Source Origin	Feature Geometry	Feature Symbol
AL140	Particle Accelerator	Topographic (LTDS)	Curve	LN_Black_ParticleAccelerator
BH110	Penstock	Topographic (LTDS)	Curve	LN_Cyan_Penstock_OnSurface
BH110	Penstock	Topographic (LTDS)	Curve	LN_Cyan_Penstock_BelowSurface
BH110	Penstock	Topographic (LTDS)	Curve	LN_Cyan_Penstock_AboveSurface
BB270	Pilot Boarding Place	Hydrographic (Chart #1)	Point	T1.1 (Chart #1)
BB270	Pilot Boarding Place	Hydrographic (Chart #1)	Point	T1.4 (Chart #1)
AQ113	Pipeline	Topographic (LTDS)	Curve	LN_Black_Pipeline_NotWaterOnSurface
AQ113	Pipeline	Topographic (LTDS)	Curve	LN_Cyan_Pipeline_WaterOnSurface
AQ113	Pipeline	Topographic (LTDS)	Curve	LN_Black_Pipeline_NotWaterBelowSurface
AQ113	Pipeline	Topographic (LTDS)	Curve	LN_Cyan_Pipeline_WaterBelowSurface
AQ113	Pipeline	Topographic (LTDS)	Curve	LN_Black_Pipeline_NotWaterElevated
AQ113	Pipeline	Topographic (LTDS)	Curve	LN_Cyan_Pipeline_WaterElevated
AQ113	Pipeline	Hydrographic (Chart #1)	Curve	L40.1 (Chart#1)
AQ113	Pipeline	Hydrographic (Chart #1)	Curve	L44 (Chart#1)
AQ113	Pipeline	Hydrographic (Chart #1)	Curve	L42 (Chart #1)
AQ113	Pipeline	Hydrographic (Chart #1)	Curve	L41.1 (Chart #1)
EA030	Plant Nursery	Topographic (LTDS)	Surface	AR_Green362_PlantNursery
EA030	Plant Nursery	Topographic (LTDS)	Point	PT_Green362_PlantNursery
BJ080	Polar Ice	Hydrographic (Chart #1)	Surface	N60.2 (Chart #1)
BB009	Port	Topographic (LTDS)	Surface	AR_Black_Port
BB009	Port	Topographic (LTDS)	Point	PT_Black_Port
AD010	Power Station	Topographic (LTDS)	Surface	AR_Black_PowerStation
AD010	Power Station	Topographic (LTDS)	Point	PT_Black_PowerStation

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TABLE II. Feature origin - Continued

LPC Feature Code	LPC Feature Name	Feature Source Origin	Feature Geometry	Feature Symbol
AD030	Power Substation	Topographic (LTDS)	Surface	AR_Black_PowerSubstation
AD030	Power Substation	Topographic (LTDS)	Point	PT_Black_PowerSubstation
AL170	Public Square	Topographic (LTDS)	Surface	AR_Black_PublicSquare
AQ116	Pumping Station	Topographic (LTDS)	Surface	AR_Black_PumpingStation
AQ116	Pumping Station	Topographic (LTDS)	Point	PT_Black_PumpingStation
AT042	Pylon	Topographic (LTDS)	Point	PT_Blue072_VO
AK130	Racetrack	Topographic (LTDS)	Curve	LN_Black_Racetrack
BC033	Radar Range	Hydrographic (Chart #1)	Surface	M31 (Chart#1)
AT045	Radar Station	Topographic (LTDS)	Surface	AR_Black_RadarStation
AT045	Radar Station	Topographic (LTDS)	Point	PT_Black_RadarStation
AT045	Radar Station	Topographic (LTDS)	Point	PT_Blue072_VO
AN010	Railway	Topographic (LTDS)	Curve	LN_Black_Railway_DoubleNarrowGauge_Electric_NotInUse
AN010	Railway	Topographic (LTDS)	Curve	LN_Black_Railway_SingleNarrowGauge_Non-Electric_NotInUse
AN010	Railway	Topographic (LTDS)	Curve	LN_Black_Railway_SingleNarrowGauge_Electric_NotInUse
AN010	Railway	Topographic (LTDS)	Curve	LN_Black_Car-Line
AN010	Railway	Topographic (LTDS)	Curve	LN_Black_Car-Line_NotInUse
AN010	Railway	Topographic (LTDS)	Curve	LN_Black_Railway_In_Road
AN010	Railway	Topographic (LTDS)	Curve	LN_Black_Railway_DoubleNormalGauge_Non-Electric
AN010	Railway	Topographic (LTDS)	Curve	LN_Black_Railway_SingleNormalGauge_Electric
AN010	Railway	Topographic (LTDS)	Curve	LN_Black_Railway_SingleNormalGauge_Non-Electric

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TABLE II. Feature origin - Continued

LPC Feature Code	LPC Feature Name	Feature Source Origin	Feature Geometry	Feature Symbol
AN010	Railway	Topographic (LTDS)	Curve	LN_Black_Railway_DoubleNarrowGauge_Non-Electric
AN010	Railway	Topographic (LTDS)	Curve	LN_Black_Railway_DoubleNarrowGauge_Electric
AN010	Railway	Topographic (LTDS)	Curve	LN_Black_Railway_SingleNarrowGauge_Non-Electric
AN010	Railway	Topographic (LTDS)	Curve	LN_Black_Railway_SingleNarrowGauge_Electric
AN010	Railway	Topographic (LTDS)	Curve	LN_Black_Railway_DoubleNormalGauge_Non-Electric_NotInUse
AN010	Railway	Topographic (LTDS)	Curve	LN_Black_Railway_DoubleNormalGauge_Electric_NotInUse
AN010	Railway	Topographic (LTDS)	Curve	LN_Black_Railway_SingleNormalGauge_Non-Electric_NotInUse
AN010	Railway	Topographic (LTDS)	Curve	LN_Black_Railway_SingleNormalGauge_Electric_NotInUse
AN010	Railway	Topographic (LTDS)	Curve	LN_Black_Railway_DoubleNarrowGauge_Non-Electric_NotInUse
AN010	Railway	Topographic (LTDS)	Curve	LN_Black_Railway_In_Road_NotInUse
AN010	Railway	Topographic (LTDS)	Curve	LN_Black_Railway_Destroyed
AN010	Railway	Topographic (LTDS)	Curve	LN_Black_Railway_DoubleNormalGauge_Electric
AN050	Railway Sidetrack	Topographic (LTDS)	Curve	LN_Black_RailwaySidetrack_StandardNonElectric
AN050	Railway Sidetrack	Topographic (LTDS)	Curve	LN_Black_RailwaySidetrack_SingleNormalGauge_Electric
AN050	Railway Sidetrack	Topographic (LTDS)	Curve	LN_Black_RailwaySidetrack_SingleNarrowGauge_Non-Electric
AN050	Railway Sidetrack	Topographic (LTDS)	Curve	LN_Black_RailwaySidetrack_SingleNarrowGauge_Electric

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TABLE II. Feature origin - Continued

LPC Feature Code	LPC Feature Name	Feature Source Origin	Feature Geometry	Feature Symbol
AN050	Railway Sidetrack	Topographic (LTDS)	Curve	LN_Black_RailwaySidetrack_SingleNormalGauge_Electric_NotInUse
AN050	Railway Sidetrack	Topographic (LTDS)	Curve	LN_Black_RailwaySidetrack_SingleNarrowGauge_Electric_NotInUse
AN050	Railway Sidetrack	Topographic (LTDS)	Curve	LN_Black_RailwaySidetrack_SingleNormalGauge_NonElectric_NotInUse
AN050	Railway Sidetrack	Topographic (LTDS)	Curve	LN_Black_RailwaySidetrack_SingleNarrowGauge_NonElectric_NotInUse
AN075	Railway Turntable	Topographic (LTDS)	Point	PT_Black_RailwayTurntable
AN060	Railway Yard	Topographic (LTDS)	Surface	AR_Black_RailwayYard
BH120	Rapids	Topographic (LTDS)	Surface	AR_Cyan_Rapids
BH120	Rapids	Topographic (LTDS)	Curve	LN_Cyan_Rapids
AB010	Recycling Site	Topographic (LTDS)	Surface	AR_Black_RecyclingSite
BD120	Reef	Topographic (LTDS)	Surface	AR_BlackCyan_Reef
BD120	Reef	Hydrographic (Chart #1)	Surface	K16 (Chart #1)
BD120	Reef	Topographic (LTDS)	Point	PT_Black_Reef
BD120	Reef	Topographic (LTDS)	Curve	LN_Black_Reef
BH135	Rice Field	Topographic (LTDS)	Surface	AR_Cyan_Rice
AA040	Rig	Topographic (LTDS)	Point	PT_Black_Rig_Inland
AA040	Rig	Topographic (LTDS)	Point	PT_Blue072_VO
AA040	Rig	Hydrographic (Chart #1)	Point	L10 (Chart #1)
BH140	River	Topographic (LTDS)	Surface	AR_Cyan_River_Perennial
BH140	River	Topographic (LTDS)	Surface	AR_Cyan_River_Non-perennial
BH140	River	Topographic (LTDS)	Curve	LN_Cyan_River_Perennial

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TABLE II. Feature origin - Continued

LPC Feature Code	LPC Feature Name	Feature Source Origin	Feature Geometry	Feature Symbol
BH140	River	Topographic (LTDS)	Curve	LN_Cyan_River_Non-perennial
AP030	Road	Topographic (LTDS)	Curve	LN_Road_AllWeather_HardSurface_Operational_Median_MultipleLanes
AP030	Road	Topographic (LTDS)	Curve	LN_Road_AllWeather_HardSurface_Construction_Median_MultipleLanes
AP030	Road	Topographic (LTDS)	Curve	LN_Road_AllWeather_HardSurface_Operational_WideMedian__MultipleLanes
AP030	Road	Topographic (LTDS)	Curve	LN_Road_AllWeather_HardSurface_Construction_WideMedian_MultipleLanes
AP030	Road	Topographic (LTDS)	Curve	LN_Road_AllWeather_HardSurface_Operational_NoMedian_MultipleLanes
AP030	Road	Topographic (LTDS)	Curve	LN_Road_AllWeather_HardSurface_Construction_NoMedian_MultipleLanes
AP030	Road	Topographic (LTDS)	Curve	LN_Road_AllWeather_HardSurface_Operational_NoMedian_1Lane
AP030	Road	Topographic (LTDS)	Curve	LN_Road_AllWeather_HardSurface_Construction_NoMedian_1Lane
AP030	Road	Topographic (LTDS)	Curve	LN_Road_AllWeather_LooseSurface_Operational_NoMedian_MultipleLanes
AP030	Road	Topographic (LTDS)	Curve	LN_Road_AllWeather_LooseSurface_Construction_NoMedian_MultipleLanes
AP030	Road	Topographic (LTDS)	Curve	LN_Road_AllWeather_LooseSurface_Operational_NoMedian_1Lane
AP030	Road	Topographic (LTDS)	Curve	LN_Road_AllWeather_LooseSurface_Construction_NoMedian_1Lane

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TABLE II. Feature origin - Continued

LPC Feature Code	LPC Feature Name	Feature Source Origin	Feature Geometry	Feature Symbol
AP030	Road	Topographic (LTDS)	Curve	LN_Road_FairWeather_Operational
AP030	Road	Topographic (LTDS)	Curve	LN_Road_FairWeather_Construction
AP030	Road	Topographic (LTDS)	Curve	LN_Road_Street_Functional
AP030	Road	Topographic (LTDS)	Curve	LN_Road_Street_Construction
AP030	Road	Topographic (LTDS)	Curve	LN_Road_Planned
AP030	Road	Topographic (LTDS)	Point	PT_Black_RouteMarker_International
AP030	Road	Topographic (LTDS)	Point	PT_Black_RouteMarker_National
AP030	Road	Topographic (LTDS)	Point	PT_Black_RouteMarker_State
AQ135	Roadside Rest Area	Topographic (LTDS)	Surface	AR_Black_RestArea
AQ135	Roadside Rest Area	Topographic (LTDS)	Point	PT_Black_RestArea
BB015	Roadstead	Hydrographic (Chart #1)	Surface	No Symbology Needed
DB160	Rock Formation	Topographic (LTDS)	Surface	AR_Dk-Brown1815_ColumnarRock_Outline
DB160	Rock Formation	Topographic (LTDS)	Point	PT_Dk-Brown1815_RockFormation
AN076	Roundhouse	Topographic (LTDS)	Surface	AR_Black_Roundhouse
AN076	Roundhouse	Topographic (LTDS)	Point	PT_Black_Roundhouse
AL200	Ruins	Topographic (LTDS)	Surface	AR_Black_Ruins
AL200	Ruins	Hydrographic (Chart #1)	Surface	F33 (Chart #1)
AL200	Ruins	Topographic (LTDS)	Point	PT_Black_Ruins
AL200	Ruins	Topographic (LTDS)	Point	PT_Blue072_VO
GB055	Runway	Topographic (LTDS)	Surface	AR_Blue072_Runway
BH160	Sabkha	Topographic (LTDS)	Surface	AR_Cyan_Sabkha
BH155	Salt Evaporator	Topographic (LTDS)	Surface	AR_Cyan_SaltEvaporator

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TABLE II. Feature origin - Continued

LPC Feature Code	LPC Feature Name	Feature Source Origin	Feature Geometry	Feature Symbol
BH155	Salt Evaporator	Topographic (LTDS)	Point	PT_Cyan_SaltEvaporator
BH150	Salt Flat	Topographic (LTDS)	Surface	AR_Cyan_SaltFlat
DB170	Sand Dunes	Topographic (LTDS)	Surface	AR_Dk-Brown1815_SandDunes_Unknown
DB170	Sand Dunes	Topographic (LTDS)	Surface	AR_Dk-Brown1815_SandDunes_Crescent
DB170	Sand Dunes	Topographic (LTDS)	Surface	AR_Dk-Brown1815_SandDunes_Dome
DB170	Sand Dunes	Topographic (LTDS)	Surface	AR_Dk-Brown1815_SandDunes_Transverse
DB170	Sand Dunes	Topographic (LTDS)	Surface	AR_Dk-Brown1815_SandDunes_Linear
DB170	Sand Dunes	Topographic (LTDS)	Surface	AR_Dk-Brown1815_SandDunes_Ripple
DB170	Sand Dunes	Topographic (LTDS)	Surface	AR_Dk-Brown1815_SandDunes_Star
AK161	Scoreboard	Topographic (LTDS)	Point	PT_Blue072_VO
GB070	Seaplane Run	Hydrographic (Chart #1)	Surface	N13 (Chart #1)
AL105	Settlement	Topographic (LTDS)	Surface	AR_Black_Settlement
AC030	Settling Pond	Topographic (LTDS)	Surface	AR_Cyan_SettlingPond
AC507	Sewage Treatment Plant	Topographic (LTDS)	Surface	AR_Black_SewageTreatmentPlant
AC507	Sewage Treatment Plant	Topographic (LTDS)	Point	PT_Black_SewageTreatmentPlant
BH250	Shallow Water Area	Hydrographic (Chart #1)	Surface	I30 (Chart #1)
AL208	Shanty Town	Topographic (LTDS)	Surface	AR_Black_BUA_ShantyTown
AL019	Shed	Topographic (LTDS)	Surface	AR_Black_Shed
AL019	Shed	Topographic (LTDS)	Point	PT_Black_Shed

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TABLE II. Feature origin - Continued

LPC Feature Code	LPC Feature Name	Feature Source Origin	Feature Geometry	Feature Symbol
BI006	Ship Elevator	Topographic (LTDS)	Curve	LN_Black_ShipElevator
BI006	Ship Elevator	Topographic (LTDS)	Point	PT_Black_ShipElevator
BB241	Shipyard	Topographic (LTDS)	Surface	AR_Black_Shipyard
BB241	Shipyard	Topographic (LTDS)	Point	PT_Black_Shipyard
AG030	Shopping Complex	Topographic (LTDS)	Surface	AR_Black_ShoppingComplex
AG030	Shopping Complex	Topographic (LTDS)	Point	PT_Black_ShoppingComplex
BA024	Shoreline	Hydrographic (Chart #1)	Curve	C1 (Chart #1)
BA024	Shoreline	Hydrographic (Chart #1)	Curve	C2 (Chart #1)
BB081	Shoreline Construction	Topographic (LTDS)	Surface	AR_Black_ShorelineConstruction_Dry
BB081	Shoreline Construction	Topographic (LTDS)	Surface	AR_BlackCyan_ShorelineConstruction_NotDry
BB081	Shoreline Construction	Topographic (LTDS)	Curve	LN_Black_ShorelineConstruction_Dry
BB081	Shoreline Construction	Topographic (LTDS)	Curve	LN_Black_ShorelineConstruction_NotDry
BB082	Shoreline Ramp	Topographic (LTDS)	Surface	AR_Black_ShorelineRamp_Dry
BB082	Shoreline Ramp	Topographic (LTDS)	Surface	AR_BlackCyan_ShorelineRamp_NotDry
BB082	Shoreline Ramp	Topographic (LTDS)	Curve	LN_Black_ShorelineRamp_Dry
BB082	Shoreline Ramp	Topographic (LTDS)	Curve	LN_Black_ShorelineRamp_NotDry
FC179	Side Scan Sonar Coverage	Hydrographic (Chart #1)	Surface	TBD; no Chart #1 symbology
AK150	Ski-jump	Topographic (LTDS)	Curve	LN_Black_SkiJump
AK150	Ski-jump	Topographic (LTDS)	Point	PT_Black_SkiJump
AK150	Ski-jump	Topographic (LTDS)	Point	PT_Blue072_VO
BI040	Sluice Gate	Topographic (LTDS)	Curve	LN_Black_SluiceGate
BI040	Sluice Gate	Topographic (LTDS)	Point	PT_Black_SluiceGate
BD002	Small Bottom Object	Hydrographic (Chart #1)	Point	TBD (No Chart #1 Symbology)

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TABLE II. Feature origin - Continued

LPC Feature Code	LPC Feature Name	Feature Source Origin	Feature Geometry	Feature Symbol
BB201	Small Craft Facility	Hydrographic (Chart #1)	Point	U1.1 (Chart #1)
AF010	Smokestack	Topographic (LTDS)	Point	PT_Black_Smokestack
AF010	Smokestack	Topographic (LTDS)	Point	PT_Blue072_VO
BD140	Snag	Hydrographic (Chart #1)	Point	K43.1, K43.2 (Chart #1)
BD140	Snag	Hydrographic (Chart #1)	Point	K43.1 (Chart #1)
BD140	Snag	Hydrographic (Chart #1)	Surface	K40 (Chart #1)
BJ100	Snow Field and/or Ice-field	Topographic (LTDS)	Surface	AR_Cyan_SnowIceField
DA010	Soil Surface Region	Topographic (LTDS)	Surface	AR_Dk-Brown1815_SoilSurface_DistortedSurface
DA010	Soil Surface Region	Topographic (LTDS)	Surface	AR_SoilSurface_DistortedSurface-Large
DA010	Soil Surface Region	Topographic (LTDS)	Surface	AR_Dk-Brown1815_SoilSurface_SandandGravel
DA010	Soil Surface Region	Topographic (LTDS)	Surface	AR_Dk-Brown1815_SoilSurface_Sand
DA010	Soil Surface Region	Topographic (LTDS)	Surface	AR_SoilSurface_Other
AD025	Solar Farm	Topographic (LTDS)	Surface	AR_Black_SolarFarm
AD025	Solar Farm	Topographic (LTDS)	Point	PT_Black_SolarFarm
BC103	Sound Signal Device	Hydrographic (Chart #1)	Point	R14 (Chart#1)
BC103	Sound Signal Device	Hydrographic (Chart #1)	Point	R11 (Chart#1)
BC103	Sound Signal Device	Hydrographic (Chart #1)	Point	R10 (Chart#1)
BC103	Sound Signal Device	Hydrographic (Chart #1)	Point	R16 (Chart#1)
BC103	Sound Signal Device	Hydrographic (Chart #1)	Point	R13 (Chart#1)
BC103	Sound Signal Device	Hydrographic (Chart #1)	Point	R12 (Chart#1)
BE020	Sounding	Hydrographic (Chart #1)	Point	I15 (Chart#1)

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TABLE II. Feature origin - Continued

LPC Feature Code	LPC Feature Name	Feature Source Origin	Feature Geometry	Feature Symbol
BE020	Sounding	Hydrographic (Chart #1)	Point	I14 (Chart#1)
BE020	Sounding	Hydrographic (Chart #1)	Point	I10 (Chart #1)
BE020	Sounding	Hydrographic (Chart #1)	Point	I13 (Chart #1)
AL351	Space Facility	Topographic (LTDS)	Surface	AR_Black_SpaceFacility
AL351	Space Facility	Topographic (LTDS)	Point	PT_Black_SpaceFacility
BH165	Spillway	Topographic (LTDS)	Surface	AR_Black_Spillway
BH165	Spillway	Topographic (LTDS)	Curve	LN_Black_Spillway
AK040	Sports Ground	Topographic (LTDS)	Surface	AR_Black_SportsGround
AK040	Sports Ground	Topographic (LTDS)	Point	PT_Black_SportsGround
CA030	Spot Elevation	Topographic (LTDS)	Point	PT_Black_SpotElevation
CA030	Spot Elevation	Topographic (LTDS)	Point	PT_SpotElevation_Water
CA030	Spot Elevation	Topographic (LTDS)	Point	PT_Black_SpotElevation
AK160	Stadium	Topographic (LTDS)	Surface	AR_Black_Stadium
AK160	Stadium	Topographic (LTDS)	Point	PT_Black_Stadium
AK160	Stadium	Topographic (LTDS)	Point	PT_Blue072_VO
DB010	Steep Terrain Face	Topographic (LTDS)	Curve	LN_Dk-Brown1815_SteepTerrainFace_EqualOrGreaterThanContour
DB010	Steep Terrain Face	Topographic (LTDS)	Curve	LN_Dk-Brown1815_SteepTerrainFace_LessThanContour
GB045	Stopway	Topographic (LTDS)	Surface	AR_Blue072_Stopway
AM010	Storage Depot	Topographic (LTDS)	Surface	AR_Black_StorageDepot
AM010	Storage Depot	Topographic (LTDS)	Point	PT_Black_StorageDepot
AM070	Storage Tank	Topographic (LTDS)	Surface	AR_Black_StorageTank

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TABLE II. Feature origin - Continued

LPC Feature Code	LPC Feature Name	Feature Source Origin	Feature Geometry	Feature Symbol
AM070	Storage Tank	Topographic (LTDS)	Point	PT_Black_StorageTank
AM070	Storage Tank	Topographic (LTDS)	Point	PT_Cyan_StorageTank_Water
AM070	Storage Tank	Topographic (LTDS)	Point	PT_Blue072_VO
BD100	Structural Pile	Topographic (LTDS)	Surface	AR_Cyan-31_StructuralPiles
BD100	Structural Pile	Hydrographic (Chart #1)	Point	F22 (Chart #1)
DB181	Submerged Volcano	Hydrographic (Chart #1)	Surface	K40 (Chart #1)
DB181	Submerged Volcano	Hydrographic (Chart #1)	Point	K40 (Chart #1)
AM060	Surface Bunker	Topographic (LTDS)	Surface	AR_Black_SurfaceBunker
AM060	Surface Bunker	Topographic (LTDS)	Point	PT_Dk-Brown1815_SurfaceBunker
ED020	Swamp	Topographic (LTDS)	Surface	AR_Cyan_Swamp
ED020	Swamp	Topographic (LTDS)	Surface	AR_Green362_NipaPalm
ED020	Swamp	Topographic (LTDS)	Surface	AR_Green362_Mangrove
FC177	Swept Area	Hydrographic (Chart #1)	Surface	I24 (Chart #1)
AK170	Swimming Pool	Topographic (LTDS)	Surface	AR_Cyan-31_SwimmingPool
AM071	Tank Farm	Topographic (LTDS)	Surface	AR_Black_TankFarm
AM071	Tank Farm	Topographic (LTDS)	Point	PT_Black_TankFarm
GB075	Taxiway	Topographic (LTDS)	Surface	AR_Blue072_Taxiway
FA100	Test Site	Topographic (LTDS)	Surface	AR_Black_TestSite
AL510	Tethered Balloon	Topographic (LTDS)	Point	PT_Blue072_VO
FC101	Theodolite Reference Line	Other	Curve	TBD; no Chart #1 symbology
EB020	Thicket	Topographic (LTDS)	Surface	AR_Green362_Thicket
BG030	Tidal Stream Observation Station	Hydrographic (Chart #1)	Point	H46 (Chart #1)

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TABLE II. Feature origin - Continued

LPC Feature Code	LPC Feature Name	Feature Source Origin	Feature Geometry	Feature Symbol
BA040	Tidal Water	Topographic (LTDS)	Surface	AR_Cyan-31_TidalWater
BG020	Tide Gauge Station	Hydrographic (Chart #1)	Point	T32.2 (Chart #1)
AL036	Tomb	Topographic (LTDS)	Surface	AR_Black_Tomb
AL036	Tomb	Topographic (LTDS)	Point	PT_Black_Tomb
AL241	Tower	Topographic (LTDS)	Point	PT_Black_Tower
AL241	Tower	Topographic (LTDS)	Point	PT_Black_Tower_Communication
AL241	Tower	Topographic (LTDS)	Point	PT_Blue072_VO
FC041	Traffic Separation Scheme	Hydrographic (Chart #1)	Surface	M16 (Chart #1)
FC041	Traffic Separation Scheme	Hydrographic (Chart #1)	Surface	M21 (Chart #1)
FC041	Traffic Separation Scheme	Hydrographic (Chart #1)	Surface	M14 (Chart #1)
FC041	Traffic Separation Scheme	Hydrographic (Chart #1)	Surface	M13 (Chart #1)
FC041	Traffic Separation Scheme	Hydrographic (Chart #1)	Surface	M15 (Chart #1)
FC041	Traffic Separation Scheme	Hydrographic (Chart #1)	Curve	M12 (Chart #1)
FC041	Traffic Separation Scheme	Hydrographic (Chart #1)	Surface	M23 (Chart #1)
AP050	Trail	Topographic (LTDS)	Curve	LN_Black_Trail
FA165	Training Site	Topographic (LTDS)	Surface	AR_Black_TrainingSite
FA165	Training Site	Topographic (LTDS)	Point	PT_Black_TrainingSite
AQ059	Transportation Route Characteristic Change	Topographic (LTDS)	Point	PT_Black_TransRouteCharacterChange

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TABLE II. Feature origin - Continued

LPC Feature Code	LPC Feature Name	Feature Source Origin	Feature Geometry	Feature Symbol
AQ059	Transportation Route Characteristic Change	Topographic (LTDS)	Point	PT_Cyan_TransRouteCharacterChange
AL211	Transportation Route Protection Structure	Topographic (LTDS)	Surface	AR_Black_TransRouteProtectStruct
AL211	Transportation Route Protection Structure	Topographic (LTDS)	Point	PT_Black_TransRouteProtectStruct
AQ125	Transportation Station	Topographic (LTDS)	Surface	AR_Black_TransportationStation
AQ125	Transportation Station	Topographic (LTDS)	Point	PT_Black_TransportationStation
EC005	Tree	Topographic (LTDS)	Point	PT_Green362_Wood
BJ110	Tundra	Topographic (LTDS)	Surface	AR_Tundra
AQ130	Tunnel	Topographic (LTDS)	Curve	LN_Black_Tunnel_Rail
AQ130	Tunnel	Topographic (LTDS)	Curve	LN_Cyan_Tunnel_Aqueduct
AQ130	Tunnel	Topographic (LTDS)	Curve	LN_Cyan_Tunnel_Canal
AQ130	Tunnel	Topographic (LTDS)	Curve	LN_Black_Tunnel_Road
AQ130	Tunnel	Topographic (LTDS)	Point	PT_Black_Tunnel
AQ130	Tunnel	Topographic (LTDS)	Point	PT_Cyan_Tunnel
AH060	Underground Bunker	Topographic (LTDS)	Point	PT_Black_UndergroundBunker
AL250	Underground Dwelling	Topographic (LTDS)	Point	PT_Black_UndergroundDwelling
BH145	Vanishing Point	Topographic (LTDS)	Point	PT_Cyan_VanishingPoint_Disappearing
BH145	Vanishing Point	Topographic (LTDS)	Point	PT_Cyan_VanishingPoint_Dissipating
AQ140	Vehicle Lot	Topographic (LTDS)	Surface	AR_Black_VehicleLot
EA050	Vineyard	Topographic (LTDS)	Surface	AR_Green362_Vineyard
EA050	Vineyard	Topographic (LTDS)	Point	PT_Green362_Vineyard

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TABLE II. Feature origin - Continued

LPC Feature Code	LPC Feature Name	Feature Source Origin	Feature Geometry	Feature Symbol
ZD020	Void Collection Area	Topographic (LTDS)	Surface	AR_Black_VoidCollectionArea
DB190	Volcanic Dyke	Topographic (LTDS)	Curve	LN_Black_VolcanicDyke
DB180	Volcano	Topographic (LTDS)	Point	PT_Dk-Brown1815_Volcano
DB180	Volcano	Topographic (LTDS)	Point	PT_Volcano
AL260	Wall	Topographic (LTDS)	Curve	LN_Black_Wall
GB065	Water Aerodrome	Hydrographic (Chart #1)	Surface	N13 (Chart #1)
BG013	Water Current Region	Other	Surface	TBD; no Chart #1 symbology
BI050	Water Intake Tower	Topographic (LTDS)	Surface	AR_Black_WaterIntakeTower
BI050	Water Intake Tower	Topographic (LTDS)	Point	PT_Black_WaterIntakeTower
AJ055	Water Mill	Topographic (LTDS)	Point	PT_Black_WaterMill
BG010	Water Movement Data Location	Hydrographic (Chart #1)	Point	H42 (Chart #1)
BG010	Water Movement Data Location	Hydrographic (Chart #1)	Point	H43 (Chart #1)
BG010	Water Movement Data Location	Hydrographic (Chart #1)	Point	H40 (Chart #1)
BG010	Water Movement Data Location	Hydrographic (Chart #1)	Point	H41 (Chart #1)
BG010	Water Movement Data Location	Hydrographic (Chart #1)	Point	H44 (Chart #1)
BH065	Water Race	Topographic (LTDS)	Curve	LN_Cyan_WaterRace_OnGround
BH065	Water Race	Topographic (LTDS)	Curve	LN_Cyan_WaterRace_Elevated
AM080	Water Tower	Topographic (LTDS)	Point	PT_Cyan_WaterTower
AM080	Water Tower	Topographic (LTDS)	Point	PT_Blue072_VO
BH040	Water Treatment Bed	Topographic (LTDS)	Surface	AR_Cyan_WaterTreatmentBed
BG012	Water Turbulence	Hydrographic (Chart #1)	Surface	K17 (Chart #1)

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TABLE II. Feature origin - Continued

LPC Feature Code	LPC Feature Name	Feature Source Origin	Feature Geometry	Feature Symbol
BG012	Water Turbulence	Hydrographic (Chart #1)	Point	H44 (Chart #1)
BG012	Water Turbulence	Hydrographic (Chart #1)	Point	H45 (Chart #1)
BG012	Water Turbulence	Hydrographic (Chart #1)	Surface	H45 (Chart #1)
BG012	Water Turbulence	Hydrographic (Chart #1)	Surface	H44 (Chart #1)
BG012	Water Turbulence	Hydrographic (Chart #1)	Point	K17 (Chart #1)
BH230	Water Well	Topographic (LTDS)	Point	PT_Cyan_Well_WaterPerennial
BH230	Water Well	Topographic (LTDS)	Point	PT_Cyan_Well_WaterIntermittent
FC046	Waterbody Area	Other	Surface	no Chart #1 symbology
BA070	Waterbody Bar	Other	Surface	TBD; no Chart #1 symbology
FC047	Waterbody Morphology Area	Other	Surface	No Chart #1 symbology
BH180	Waterfall	Topographic (LTDS)	Curve	LN_Cyan_Waterfall
BH180	Waterfall	Topographic (LTDS)	Point	PT_Cyan_Waterfall
BH220	Waterwork	Topographic (LTDS)	Surface	AR_Black_Waterwork
BH220	Waterwork	Topographic (LTDS)	Point	PT_Black_Waterwork
AQ180	Weigh Station	Topographic (LTDS)	Surface	AR_Black_Weighstation
AQ180	Weigh Station	Topographic (LTDS)	Point	PT_Black_Weighstation
AA050	Well	Hydrographic (Chart #1)	Point	L21.1 (Chart #1)
AA050	Well	Hydrographic (Chart #1)	Point	L21.2 (Chart #1)
AA050	Well	Hydrographic (Chart #1)	Point	L23 (Chart #1)
AD060	Wind Farm	Topographic (LTDS)	Surface	AR_Black_WindFarm
AD060	Wind Farm	Topographic (LTDS)	Point	PT_Black_WindFarm
AD060	Wind Farm	Topographic (LTDS)	Surface	L5.2 (Chart #1)
AJ051	Wind Turbine	Topographic (LTDS)	Point	PT_Black_Windturbine_OnShore

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TABLE II. Feature origin - Continued

LPC Feature Code	LPC Feature Name	Feature Source Origin	Feature Geometry	Feature Symbol
AJ051	Wind Turbine	Hydrographic (Chart #1)	Point	L5.1 (Chart #1)
AJ051	Wind Turbine	Topographic (LTDS)	Point	PT_Blue072_VO
AJ050	Windmill	Topographic (LTDS)	Point	PT_Black_Windmill
AJ050	Windmill	Topographic (LTDS)	Point	PT_Blue072_VO
BD180	Wreck	Hydrographic (Chart #1)	Surface	K22 (Chart #1)
BD180	Wreck	Hydrographic (Chart #1)	Surface	K20 (Chart #1)
BD180	Wreck	Hydrographic (Chart #1)	Surface	K21 (Chart #1)
BD180	Wreck	Hydrographic (Chart #1)	Point	K28 (Chart#1)
BD180	Wreck	Hydrographic (Chart #1)	Point	K29 (Chart #1)
BD180	Wreck	Hydrographic (Chart #1)	Point	K3,K30 (Chart #1)
BD180	Wreck	Hydrographic (Chart #1)	Point	K26 (Chart#1)
BD180	Wreck	Hydrographic (Chart #1)	Point	K27 (Chart#1)
BD180	Wreck	Hydrographic (Chart #1)	Point	K25 (Chart #1)
BD180	Wreck	Hydrographic (Chart #1)	Point	K24 (Chart #1)
BD180	Wreck	Hydrographic (Chart #1)	Surface	K23(Chart #1)
AK180	Zoo	Topographic (LTDS)	Surface	AR_Black_Zoo
AK180	Zoo	Topographic (LTDS)	Point	PT_Black_Zoo

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APPENDIX B
Littoral Planning Chart Style Sheet (East-West)

B.1 SCOPE.

B.1.1 Scope. This Appendix is a graphic illustration of the design, composition, and location of the margin data of charts oriented with their long dimensions in an east-west direction. This Appendix is a mandatory part of the specification. The information contained herein is intended for guidance only.

B.2 APPLICABLE DOCUMENTS. (This section is not applicable to this appendix.)

B.3 Littoral Planning Chart Style Sheet.

B.3.1 Style Sheet. Appendix B East-West Style Sheet LPC_EW_Stylesheet.pdf has been attached to this document.

B.3.2 Order of precedence. In the event of a conflict between type styles/sizes illustrated on this style sheet and the type styles/sizes specifications, the specifications shall take precedence.

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APPENDIX C
Littoral Planning Chart Style Sheet (North-South)

C.1 SCOPE.

C.1.1 Scope. This Appendix is a graphic illustration of the design, composition, and location of the margin data of charts oriented with their long dimensions in a north-south direction. This Appendix is a mandatory part of the specification. The information contained herein is intended for guidance only.

C.2 APPLICABLE DOCUMENTS. (This section is not applicable to this appendix.)

C.3 Littoral Planning Chart Style Sheet.

C.3.1 Style Sheet. Appendix C North-South Style Sheet LPC_NS_Stylesheet.pdf has been attached to this document.

C.3.2 Order of precedence. In the event of a conflict between type styles/sizes illustrated on this style sheet and the type styles/sizes specifications, the specifications shall take precedence.

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